Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						(b6) (1)	TB12-5	Splice
						L41K22	Splice	DS26
						L92J22	TB12-4	J48-F
						L92K22	P48-F	J263
						L41D22	Tb12-4	J38-F
						L41E22	J38-F	J218
						L41F22	TB12-4	P45-W
						L41G22	J45-W	J204
						(c6) (1)	TB14-2	DS14
						L18C22	TB14-1	DS14
						L18D22	TB14-1	DS14
						L18E22	TB14-1	DS14
						L18F22	TB14-2	DS14
						L18G22	TB14-2	DS14
	(7) Individual ground wire from terminal block to lights	Open circuit	Loss of individual lights	Loss of individual lights	Loss of individual lights	L91A22 (a7) (1)	TB3-6	
						L35B22N	TB3-4	
						L155A22 N		
		Short to ground	None	None	None	L35C22N	TB3-6	Ground
						L35D22	TB3-4	DS14
						(b7)	TB14-3	Instru-ment
						NONE	TB14-3	Sec-ondary
						(C7)		Lights
						L91B20N		
						L91A22		
Pilot utility (cockpit) light	Power wire	Open circuit	Pilot utility light inoperative	None	Pilot utility light inoperative	L30A20	CB5	TB6-1
							TB6-1	DS10
								Pilot Cockpit Light
		Short to ground	Circuit breaker trips	Circuit breaker trips	Pilot utility light inoperative	L30A20	CB5	TB6-1
							TB6-1	DS10
								Pilot Cockpit Light

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L30B22	CB5	J34-a
						L30C22	P34-a	8R15-3
	Ground return	Open circuit	Pilot utility light inoperative	None	Pilot utility light inoperative	Chassis	DS10	Ground
Pilot utility (map light)	Power wire	Open circuit	Pilot utility light inoperative	None	Pilot utility light inoperative	L30B22	CB5	8R15-3
						L30C22	P34-a	8R15-3
						L806A22	8R15-4	8R15-7
						L808A22	8R15-6	P34-Z
						L808B22	J34-Z	Splice
							Splice	8DS67
								Pilot Map Light
		Short to ground	Pilot utility light inoperative	Circuit breaker trips	Pilot utility light inoperative	L30B22	CB5	J34-a
						L30C22	P34-a	8R15-3
						L30A20	CB5	TB6-1
							TB6-1	DS10
								Pilot Cockpit Light
						Chassis	8DS67	Ground
Copilot utility (cockpit) light	Power wire	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	L31A20	CB5	TB7-1
						PIGTAIL	TB7-1	DS9
								Copilot Cockpit Light
		Short to ground	Circuit breaker trips	Circuit breaker trips	Copilot utility light inoperative	L31A20	CB5	TB7-1
							TB7-1	DS9
								Copilot Cockpit Light
						L31B22	CB5	J33-j
						L31C22	P33-j	8R16-3
	Ground return	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	Chassis	DS9	Ground

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
Cockpit utility (map light)	Power wire	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	L31B22	CB5	J33-j
						L31C22	P33-j	8R16-3
						L807A22	8R16-4	8R16-7
						L809A22	8R16-6	P33-k
						L809B22	J33-k	Splice
							Splice	8DS68
								Copilot Map Light
		Short to ground	Copilot utility light inoperative	Circuit breaker trips	Copilot utility light inoperative	L31B22	CB5	J33-j
						L31C22	P33-j	8R16-3
						L31A20	CB5 T	TB7-1
							B7-1	9DS9
								Copilot Cockpit Light
	Ground wire	Open circuit	Copilot utility light inoperative	Non	Copilot utility light inoperative	Chassis	8DS68	Ground
MASTER CAUTION PANEL								
Legend capsules that require ground fault signal	Legend capsules 1, 2, 7 through 12, 15, 16, 19, 20, 22 through 33, and 35 through 38							
	ENG 1 OIL PRESSURE (Capsule 1)							
	Ground wire	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q1A22	P43-B	J99-T
						Q1B20	P99-T	P73-A
						Q2A20	P73-B	P99-U
						Q2B22N	J99-U	Ground
		Short to ground	False indication	Caution light	False indication	Q1A22	P43-B	J99-T
						Q1B22	P99-T	P73-A
	ENG 2 OIL PRESSURE (Capsule 2)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q3B20	P100-R	P74-A
						Q3A22	P42-B	J100-R
						Q4A20	P74-B	P100-S

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	Q4B22N	J100-S	Ground
						Q3A22	P42-B	J100-R
						Q3B20	P100-S	P74-A
	ENG 1 FUEL BOOST (Capsule 7)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q79A22	P43-D	P45-X
						Q79B22	J45-X	TB4-8
						Q67B22	TB4-7	J45-Y
						Q67A22	P45-Y	TB11-3
						Q29A20N	TB11-5	Ground
		Short to ground	False indication	Caution light	False indication	Q79A22	P43-D	P45-X
						Q79B22	J45-X	TB4-8
						Q67B22	TB4-7	J45-Y
						Q67A22	P45-Y	TB11-3
	ENG 2 FUEL BOOST (Capsule 8)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q80A22	P42-D	P45-c
						Q80B22	J45-c	TB4-4
						Q68B22	TB4-5	J45-Z
						Q68A22	P45-Z	TB10-3
		Short to ground	False indication	Caution light	False indication	Q37A20N	TB10-5	Ground
						Q80A22	P42-D	P45-c
						Q80B22	J45-c	TB4-4
						Q68B22	TB4-5	J45-Z
						Q68A22	P45-Z	TB10-3
	ENG 1 FUEL FILTER (Capsule 9)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q5A22	P43-E	J59-H
						Q5B18	P59-H	P182-B
						Q6A18	P182-C	P59-J
		Short to ground	False indication	Caution light	False indication	Q6A22N	J59-J	Ground
						Q5A22	P43-E	J59-H
						Q5B18	P59-H	P182-B
	ENGINE 2 FUEL FILTER (Capsule 10)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q7A22	P42-E	J60-H
						Q7B18	P60-H	P183-B
						Q8A18	P183-C	P60-J
						Q8B22N	J60-J	Ground
		Short to ground	False indication	Caution light	False indication	Q7A22	P42-E	J60-H
						Q7B18	P60-H	P183-B

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 1 FUEL LOW (Capsule 11)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	E39A22	P43-F	TB11-4
						Q29A20N	TB11-5	Ground
		Short to ground	False indication	Caution light	False indication	E39A22	P43-F	TB11-4
	ENG 2 FUEL LOW (Capsule 12)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	E40A22	P42-F	TB10-4
						Q37A20N	TB10-5	Ground
		Short to ground	False indication	Caution light	False indication	E40A22	P42-F	TB10-4
	ENG 1 CHIP (Capsule 15)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W51A22	P43-H	TB15-5
		Short to ground	False indication	Caution light.	False indication	W51A22	P43-H	TB15-5
						W51B22	TB15-5	J59-W
						W51C18	P59-W	P83-B
						W51D22	TB15-5	J99-L
						W51E20	P99-L	P113-A
	ENG 2 CHIP (Capsule 16)	Open circuit	Loss of particular fault indication	None.	Loss of particular fault indiction circuit	W54A22	P42-H	TB15-4
		Short to ground	False indication	Caution light.	False indication	W54A22	P42-H	TB15-4
						W54B22	TB15-4	J60-W
						W54C18	P60-W	P84-B
						W54D22	TB15-4	J100-L
						W54E20	P100-L	P114-A
	ENG 1 GEN OVHT (Capsule 19)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W151B22	P43-K	J3043-A
						W151A20	P3043-A	S209
						W149A20	S209	P3043-B
						W149B22N	J3043-B	Ground
		Short to ground	False indication	Caution light	False indication	W151B22	P43-K	J3043-A
						W151A20	P3043-A	S209
	ENG 2 GEN OVHT (Capsule 20)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W150B22	P42-K	J60-E
						W150A28	P60-E	S208
						W148A18	S208	P60-F
						W148B2N	J60-F	Ground

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light.	False indication	W150	P42-K	J60-E
						B22	P60-E	S208
						W150A18		
	HYDRAULIC (Capsule 22)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	C30F22	P42-L	9TB4-P(N)
		Short to ground	False indication	Caution light	False indication	C30F22	P42-L	9TB4-P(N)
						C20J22	9TB4-P(T)	J58-E
						C30G22	P58-E	Splice
						C30B22	Splice	P299-A
						C30C22	Splice	P105-B
						C30D22	Splice	P106-B
						C30H22	9TB4-R(A)	J58-H
						C30A22	P58-H	P300-A
						C30K22	9TB4-P(S)	9TB4-R(G)
						C30L22	9TB4-R(D)	9XK1-X-2
	XMSN OIL PRESS (Capsule 23)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	D23A22	P43-M	J41-V
						D23B20	P41-V	P36-A
						D24A20N	P36-B	Ground
		Short to ground	False indication	Caution light	False indication	D23A22	P43-M	J41-V
						D23B20	P41-V	P36-A
	CHIP C BOX (Capsule 24)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W57A22	P43-N	P99-N
						W57B20	P99-N	P115-A
						W58A20	P115-B	P99-P
						W58B22N	J99-P	Ground
		Short to ground	False indication	Caution light	False indication	W57A22	P43-N	J99-N
						W57B20	P99-N	P115-A
	C BOX OIL PRESSURE (Capsule 25)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q69A22	P42-M	J100-N
						Q69B20	P100-N	P162-A
						Q70A20	P162-B	P100-P
						Q70B22N	J100-P	Ground
		Short to ground	False indication	Caution light	False indication	Q69A22	P42-M	J100-N
						Q69B20	P100-N	P162-A

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	INVERTER NO. 1 (Capsule 26)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W79A22	P43-P	3K1 P1-H
						W601A2N	3K1 P1-G	Ground
		Short to ground	False indication	Caution light	False indication	W79A22	P43-P	3K1 P1-H
	CHIP XMSN (Capsule 27)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W175A22	P43-R	TB80-3
		Short to ground	False indication	Caution light	False indication	W175A22	P43-R	TB80-3
						W179A22	TB80-5	J41-J
						W179B20	P41-J	E17
						W178A22	TB80-4	J41-K
						W178B20	P41-K	E16
						W177A22	TB80-2	J41-L
						W177B20	P41-L	E1
						BLK	TB80-5	DS 52
						BLK	TB80-4	DS 51
						BLK	TB80-2	DS 50
	INVERTER NO. 2 (Capsule 28)	Open circuit	Loss of particular fault indication	None.	Loss of particular indication circuit	W80A22	P42-N	3K2 P1-L
						W802A2N	3K2 P1-K	Ground
		Short to ground	False indication	Caution light	False indication	W80A22	P42-N	3K2 P1-L
	XMSN OIL TEMP (Capsule 29)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	D22A22	P43-S	J41-S
						D22B20	P41-S	S26
		Short to ground	False indication	Caution light	False indication	D22A22	P43-S	J41-S
						D22B20	P41-S	S26
	CHIP 42/90 BOX (Capsule 30)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W60A22	P43-T	Splice
		Short to ground	False indication	Caution light	False indication	W60A22	P43-T	Splice
						W59A22	Splice	P117-H
						W59B22	J117-H	E7
						W60A22	Splice	P117-G
						W60B22	J117-G	E8

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	C BOX OIL TEMP (Capsule 31)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	E90B22	P42-P	J99-a
						E90A20	P99-a	P184-A
						E91A20	P184-B	P99-b
						E91B22N	J99-b	Ground
		Short to ground	False indication	Caution light	False indication	E90B22	P42-P	J99-a
						E90A20	P99-a	P184-A
	EXTERNAL POWER (Capsule 32)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W85A22	P43-U	S105-COM
						W86A22N	S10 5-NC	Ground
		Short to ground	False indication	Caution light	False indication	W85A22	P43-U	S105-COM
	BATTERY TEMP (Capsule 33)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W161A22	P43-V	Splice
		Short to ground	False indication	Caution light	False indication	W61A22	P43-V	Splice
						W161B22	Splice	P281-B
						W161C22	Splice	P281-E
	DOOR LOCK (Capsule 35)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W65A22	P43-W	Splice
		Short to ground	False indication	Caution light	False indication	W65A22	P43-W	Splice
						W65A22	Splice	Splice
						W65B22	Splice	Splice
						W65C22	Splice	P117-E
						W65D22	J117-E	S82-NC
	HEATER AIR LINE (Capsule 37)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W132A22	P42-S	Splice
		Short circuit	False indication	Caution light	False indication	W132A22	P42-S	Splice
						H37A22	Splice	S89-6
						W112A22	Splice	Splice
						H26C22	Splice	P223-C
						H26B22	Splice	J168-J
						H26A22	P168-J	Splice
	BATTERY (Capsule 38)	Open circuit	Loss of particular fault indiction	None	Loss of particular fault indication circuit	W144A22	P43-Y	K10-14

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	W144A22	P43-Y	K10-14
						P185A22	K10-12	K57
Legend capsules that require application of 28 VDC for fault signal	Legend capsules 13 and 14							
	ENG 1 GOV MANUAL (Capsule 13)							
	Power wire	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q10B22	P43-G	P45-R
						Q10A22	J45-R	S13-2
						Q9B22	S13-3	J45-N
						Q9A22	P45-N	CB5
		Short to ground	Loss of particular fault indication	Circuit breaker trips	Loss of particular fault indication circuit	Q10B22	P43-G	P45-R
						Q10A22	J45-R	S13-2
						Q9B22	S13-3	J45-N
						Q9A22	P45-N	CB5
						Q10C22	S13-2	J45-P
						Q10D22	P45-P	J59-U
						Q10E18	P59-U	P87-B
	ENG 2 GOV MANUAL (Capsule 14)	Open circuit	Loss of particular fault indication	None	Loss of particular indication circuit	Q13B22	P42-G	P44-P
						Q12B22	S14-3	J44-N
						Q12A22	P44-N	CB5
		Short to ground	Loss of particular fault indication	Circuit breaker trips	Loss of particular fault indication circuit	Q13B22	P42-G	P44-P
						Q12B22	S14-3	J44-N
						Q12A22	P44-N	CB5
						Q13C22	S14-2	J44-R
						Q13D22	P44-R	J60-U
						Q13E18	P60-U	P88-B
						Q13A22	J44-P	S14-2
Legend capsules that require removal of 28 VDC for fault signal	Legend capsules 3 through 6, 17, 18, 34 and 40							

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	AFCS (Capsule 40)							
	Power wire	Open circuit	False indication	Caution light	False indication	C100A24	P42-T	9A3 P1-a
						C143A2N	9A3 P1-c	Ground
		Short to ground	False indication	Caution light	False indication	C100A24	P42-T	9A3 P1-a
	ENG 1 DC GENERATOR (Capsule 3)	Open circuit	False indication	Caution light	False indication	W77A22	P43-A	K3-12
		Short to ground	False indication	Caution light	False indication	W77A22	P43-A	K3-12
						W77B22	K3-12	K33
						W77C22	K3-12	2TB1-1(C)
						P39A22	K3-11	K7-A3
						P39B22	K3-11	K5-12
	ENG 2 DC GENERATOR (Capsule 4)	Open circuit	False indication	Caution light	False indication	W78A22	P42-A	K4-12
		Short to ground.	False indication.	Caution light.	False indication.	W78A22	P42-A	K4-12
						W78B22	K4-12	K34
						W78C22	K4-12	2TB1-2(C)
	ENG 1 PART SEP OFF (Capsule 5)	Open circuit	False indication	Caution light	False indiction	K25B22	P43-C	J59-h
						K25A20	P59-h	P93-D
						K13A20	P93-E	J59-g
						K13B22N	J59-g	Ground
		Short to ground	False indication	Caution light	False indication	K25B22	P43-C	J59-h
						K25A20	P59-h	P93-D
	ENG 2 PART SEP OFF (Capsule 6)	Open circuit	False indication	Caution light	False indication	K27B22	P42-C	J60-h
						K27A20	P60-h	P94-D
						K18A20	P94-E	P60-g
						K18B22N	J60-g	Ground
		Short to ground	False indication	Caution light	False indication	K27B22	P42-C	P60-h
						K27A20	P60-h	P94-D

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 1 FUEL VALVE (Capsule 17)	Open circuit	False indication	Caution light	False indication	Q93B22	P43-J	P45-h
						Q93A22	J45-h	S17-2(C)
						Q89C22	S17-(NO)	J45-j
						Q89B22	P45-j	J57-F
						Q89A22	P57-F	P121-C
						Q91C22	S17-2(NC)	J45-g
						Q91B22	P45-g	J57-R
						Q91A22	P57-R	P121-D
		Short to ground	False indication	Caution light	False indication	Q93B22	P43-J	P45-h
						Q93A22	J45-h	S17-2(C)
	ENG 2 FUEL VALVE (Capsule 18)	Open circuit	False indication	Caution light	False indiction	Q94B22	P42-J	P44-h
						Q94A22	J44-h	S18-2(C)
						Q88C22	S182(NO)	J44-j
						Q88B22	P44-j	J58-G
						Q88A22	P58-G	P122-C
						QA92C22	S18-2(NC)	J44-g
						Q92B22	P44-g	J58-R
						Q92A22	P58-R	P122-D
		Short to ground	False indication	Caution light	False indication	Q94B22	P42-J	P44-h
						Q94A22	J44-h	S18-2(C)
	FUEL XFEED (Capsule 34)	Open circuit	False indication	Caution light	False indication	Q96B22	P42-R	P44-a
						Q96A22	J44-a	S16-2(C)
		Short to ground	False indication	Caution light	False indication	Q96B22	P42-R	P44-a
						Q96A22	P44-a	S16-2(C)
						Q96C22	S16-2(C)	S16-1(C)
	CAUTION PANEL (Capsule 21)	Illuminates if applied voltage to all capsules drops below 75% rated value.						
	Ground wire	Open circuit	Loss of particular fault indication	Non	Loss of particular fault indication circuit	W158A2N	P43-L	Ground

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	Power wire	Open circuit	False indication	Caution light	False indication	L88A22	P42-Z	Splice
						L88A22	Splice	S210-2
						L152A22	S210-3	CB5
		Short to ground	False indication	Caution light	False indication	L88A22	P42-Z	Splice
						L88A22	Splice	S210-2
						L152A22	S210-3	CB5
						L88C22	Splice	P292-B
BLEED AIR HEATER SYSTEM								
Heater system	Power wire from cabin heater circuit breaker to heater switch and aft outlet switch	Open circuit	Heater inoperative	Heater stops operating	Heater inoperative	H1A22	CB5	S31-C
						H17A22	CB5	S189-2
	Power wire from heater switch through heater relay through emergency fire pull handle switches through particle separator relays to bleed air check valve and heater valves	Open circuit	Heater inoperative	Heater stops operating	Heater inoperative	H2A22	S189-3	Splice
						H2D22	Splice	P239-F
						H2C22	Splice	P239-D
						H2B22	Splice	P239-K
						H25D22	P239-G	Splice
						H25C22	P239-B	Splice
						H25A22	Splice	J34-T
						H25B22	P34-T	S2-A(NC)
						K41A22	S2-A(C)	S1-A(C)
						K37A22	S1-A(NC)	P33-Z
						K37B22	J33-Z	K27-B3
						H46A22	K27-B2	K26-B3
						H15A22	K26-B2	Splice
			Left bleed air valve inoperative	None	Left bleed air valve inoperative	H15A22	Splice	J95-C
						H15B18	P95-C	P177-A
						H14A18	P177-D	P95-D
						H14B22N	J95-D	Ground
			Right bleed air valve inoperative	None	Right bleed air valve inoperative	H15C22	Splice	J96-C
						H15D18	P96-C	P178-A
						H3A18	P178-D	P96-D
						H3B22N	J96-D	Ground

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	Power wire from heater switch to winterization kit control valve, heater valve, cabin temperature controller servo and overheat relay	Short to ground	Heater inoperative	Circuit breaker trips	Heater inoperative	H27B22	S189-3	Splice
						H27F22	Splice	P350-G
						H27D22	Splice	P222-A
						H27C22	Splice	J168-A
						H27E22	P168-A	P169-A
						H2A22	S189-3	Splice
						H2D22	Splice	P239-F
						H2C22	Splice	P239-D
						H2B22	Splice	P239-K
Heater relay	Ground wire from overheat relay to overheat	Open circuit	Heater will not shut off automatically in the event of an overheat condition	Caution light signal from other overheat switch if actuated	None. Switch may be shut off	H10A22	P239-E	J168-C
						H10B22	P168-C	S81
		Short to ground	Heater inoperative	Heater shuts off and cabin heater circuit breaker trips	Heater inoperative			
Caution light	Ground wire from light to overheat switch	Open circuit	Caution light will not illuminate if overheat switch actuates	None	None	H26A22	S187	P168-J
					Heater would shut off on signal from primary overheat switch	H26B22	J168-J	Splice
						W112A22	Splice	Splice
						W132A22	Splice	P42-S
		Short to ground	Caution light	Caution light	None	H26A22	S187	P168-J
					Heater can be shut off by heater switch	H26B22	J168-J	Splice
						H26C22	Splice	P223-C
						W112A22	Splice	Splice
						H37A22	Splice	S189-6
						W132A22	Splice	P42-S
Vent blower	Power wire from circuit breaker to switch	Open circuit	Inability to operate vent blowers	Vent blowers will not operate	Vent blowers inoperative	H12A22	CB5	S190-2
		Short to ground.	Heater inoperative.	Circuit breaker trips.	Heater inoperative.	H12A22	CB5	S190-2
Vent switch	Power wire from switch to right vent blower	Open circuit	Right vent blower inoperative	No airflow from right heat ducts	None	H13A22	S190-3	B20

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
	Power wire from switch to left vent blower	Open circuit	Left vent blower inoperative	No airflow from left heat ducts	None	H13B22 S190-3 B21

NOTE:

1 The letter indicates the Circuit column and the number indicates the Location column.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
			DC POWER SYSTEM			
Battery	Power wire from battery connector to battery bus	Open circuit	No battery power to battery bus	Zero voltage reading on voltmeters	No battery power available	Wire P2A1 (2BT1P1 + (Pos) to bus bar) loose, disconnected, or broken.
		Short to ground	Battery switch opens	Battery switch in OFF position	No battery power on buses	Wire P2A1 broken, touching structure or ground.
	Ground wire from battery connector to helicopter structure	Open circuit	No battery power to battery bus	Zero voltage reading on voltmeters	No battery power available	Wire P1A1N (2BT1P1 (Neg) to ground) loose, disconnected, or broken.
		Short to ground	None	None	None	Wire P1A1N broken, touching structure or ground.
	Wire from terminal A1 of No. 1 bus/bat control relay to terminal A1 of No. 1 generator relay	Open circuit	No battery power on No. 1 buses	None	No battery power to No. 1 buses. Engine No. 1 cannot be started	Wire P5A1 (2K7-A1 to 2K3-A1) No. 1 bus/bat control relay loose, disconnected, or broken.
		Short to ground	No battery power on battery bus	Zero voltage reading on voltmeters	No battery power available	Wires P5A1, P5B1 (2K3-A1 to 2K11-A1) non-essential bus relay, P5C1 (2K11-A1 to main DC bus No. 1) and wire P5D22 (2K7-A1 to 2Z1P1-E) No. 1 DC control unit connector broken, touching structure or ground, shorted to ground at disconnect.
	Wire from BUS 1 BATT switch circuit breaker to terminal 3 of BATTERY switch	Open circuit	Battery relay No. 1 will not close	None, if BATTERY BUS switch No. 2 is closed (normal operation)	Battery cannot be connected to No. 1 DC buses by BATTERY BUS 1 switch. Cannot start engine No. 1.	Wires P224A12 (2K8-A1 to 2CB15) EMERG BUS 1 circuit breaker, P114A12 (2CB15 to 22P1-C), P114C12 (22J1-C to 2CB17) No. 1 BUS BAT circuit breaker P138A20 (2CB17 to 22P9-A), P138D20 (22J9-A to 22J13-D), P138E20 (22P13-D to 2S3-3) BATTERY BUS 1 switch broken, loose, or disconnected. No. 1 BUS BAT circuit breaker open.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
		Short to ground	No. 1 BUS BAT circuit breaker trips	No. 1 BUS BAT circuit breaker tripped	Battery cannot be connected to No. 1 buses by BATTERY BUS 1 switch. Cannot start engine No. 1.	Wires P224A12, P114A12, P114C12, P114B12 (2CB15 to 22P1-D), P114D12 (22J1-D to 22J11-C), P114E12 (22P11-C to 2S8-12) NORMAL/EMERG LOAD switch, P59C22 (2S3-1 to 2TB1-2C), P242A22 (2TB1-2H to 2K15J1-5) emergency bus interlock relay, P212C22 (2K15J1-4 to splice), P212A22 (splice to 2K13J1-6) emergency bus interlock relay, P212B22 (splice to 2K13J1-20), P242B22 (2TB1-2G to 2K14J1-19) emergency bus interlock relay, P138A20, P138D20, P138E20, P138B22 (2CB17 to 22J5-C), P138C22 (22P5-C to 2XK9-6) non-essential bus relay socket broken, touching structure or ground.
	Wire from terminal A2 of No. 2 bus batt control relay to terminal A2 of No. 2 generator relay	Open circuit	No battery power on No. 2 buses	None	No battery power on No. 2 buses. Engine No. 2 cannot be started from battery.	Wire P30A1 (2K8-A2 to 2K4-A2) No. 2 generator relay, loose, or disconnected.
		Short to ground	No. 2 generator trips off line	ENG 2 DC GENERATOR caution light	Loss of generator No. 2	Wires P30A1, P30B1 (2K4-A2 to 2K2-A2) non-essential bus 2 relay, P30C1 (2K2-A2 to main DC bus No. 2) and P30D22 (2K8-A2 to 2Z2P1-E) No. 2 DC control unit broken, touching structure or ground, or shorted to ground at disconnect.
	Wire from No. 2 BUS BAT circuit breaker to terminal 3 of BATTERY BUS 2 switch	Open circuit	Battery relay No. 2 will not close	BATTERY caution light on unless BATTERY BUS 1 switch is closed	Battery cannot be connected to No. 2 DC buses by BATTERY BUS 2 switch. Cannot start No. 2 engine from battery.	Wire P118C12 (2CB16, No. 2 BUS BAT circuit breaker to 22J2-C), P118A12 (22P2-C to 2CB14) EMERGENCY BUS 2 circuit breaker, P139A20 (2CB16 to 22J12-Z), P139B20 (22P12-Z to 2S4-3) BATTERY BUS 2 switch broken, loose, or disconnected. NO. 2 BUS BAT circuit breaker open.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
		Short to ground	No. 2 BUS BAT circuit breaker trips	NO 2 BUS BAT circuit breaker tripped	Battery cannot be connected to No. 2 DC buses by BATTERY BUS 2 switch. Cannot start No. 2 engine from battery	Wires P118C12, P118A12, P139A20, P139B20, P118B12 (2CB14 to 22P2-D), P118D12 (22J2-D to 2S8-9) NORMAL/EMERG LOAD switch, P110C22 (2S4-1 to 2TB1-2B), P242B22 (2TB1-2G to 2K14J1-19) emergency bus interlock relay, P242A22 (2TB1-2H) to 2K15J1-5) emergency bus interlock relay, P212C22 (2K15J1-4 to splice, P212A22 (splice to 2K13J1-6) emergency bus interlock relay, P212B22) (splice to 2K13J1-20, broken, touching structure or ground.
Generator No. 1	Terminal A	Open circuit	No generator No. 1 output	ENG 1 DC GENERATOR caution light	Generator No. 1 fails to supply power	Wires P54A16 (2G1-A to 1P1-C), P54B16 (1J1-C to 22P13-N), P54C16 (22J13-N to 22J3-C), P54D16 (22P3-C to 1K1-C1) No. 1 start control relay P54E16 (1K1-C to 2Z1P1-B) No. 1 DC control unit broken, loose, or disconnected.
		Short to ground	Generator No. 1 drops off line	ENG 1 DC GENERATOR caution light	Generator No. 1 fails to supply power	Wires P54A16, P54B16, P54C16, P54D16, and P54E16 broken, touching structure or ground, shorted to ground at disconnect, or grounded with shield.
	Terminal B	Open circuit	No generator No 1 output	ENG 1 DC GENERATOR caution light	Loss of power from generator No. 1	Wires P72A1 (2G1-B to 2P3-B), P72B1 (2J3-B to 2K5-B2) No. 1 start-gen relay, P71A1 (2K5-A1 to 2K3-A2) No. 1 generator relay, P71B16 (2K3-A2 to 22P13-F), P71D16 (22J13-F to 22J3-A), P71E16 (22P3-A to 2Z1P1-D) No. 1 DC control unit connector loose, disconnected, or broken. Bus bar at 2K5 missing.
		Short to ground	Generator drops off line	ENG 1 DC GENERATOR caution light	Loss of power from generator No. 1	Wires P72A1, P72B1, P71A1, P71B16, P71D16, P71E16, P71C16 (2K3-A2 to 2CB9) GEN 1 FIELD CB, P129A16 (2CB9 to 22P13-K), P129B16 (22J13-K to 22J3-E), P129C16 (22P3-E to 1K1-D3) No. 1 start control relay, P38B16 (1K1-D2 to 2Z1P1-A), P38A22 (1K1-D2 to 2Z1P1-S) broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
	Terminal C	Open circuit	Starter No. 1 inoperative	None	Starter inoperative	Wires P95A1 (2G1-C to 2P3-D), P95B1 (2J3-D to 2K5-A2) No. 1 start-gen relay, P71A1 (2K5-A1 to 2K3-A2) No. 1 generator relay, P71B16 (2K3-A2 to 22P13-F), P71D16 (22J13-F to 22J3-A), P71E16 (22P3-A to 2Z1P1-D) No. 1 DC control unit connector, loose, disconnected, or broken 2K5 contacts A1 or A2 bent or broken.
		Short to ground	Generator drops off line	ENG 1 DC GENERATOR caution light	Loss of power from generator No. 1	Wires P95A1, P95B1, P71A1, P71B16, P71D16, P71E16, P71C16 (2K3-A2 to 2CB9) GEN 1 FIELD CB, P129A16 (2CB9 to 22P13-K), P129B16 (22J13-K to 22J3-E), starter control 1 relay, broken: P129C16 (22P3-E to 1K1-D3) touching structure to ground.
	Terminal D	Open circuit	Generator paralleling circuitry inoperative	None	Generators not paralleled	Wires P51A16 (2G1-D to 1P1-A), P51B22 (1J1-A to 22P15-1), P51C22 (22J15-1 to 22J5-F), P51D22 (22P5-F to 1K1-C2) No. 1 start control relay, P51E22 (1K1-C2 to 2Z1P1-G) No. 1 DC control unit connector, loose, disconnected, or broken.
		Short to ground	No immediately discernible effect, but paralleling circuit is affected	None	Generator paralleling affected	Wires P51A16, P51B22, P51C22, P51D22, P51E22 broken, touching structure or ground, or shorted to ground at disconnects.
	Terminal E	Open circuit	Generator No. 1 inoperative	ENG 1 DC GENERATOR caution light	Loss of power from generator No. 1	Wires P73A1 (2G1-E to 2P3-F), P73B1 (2J3-F to 2R1) No. 1 shunt, P74A1N (No. 1 shunt to ground), loose, disconnected or broken. No. 1 shunt burnt, disconnected, or broken.
		Short to ground	No effect	None	None	Wires P73A1, P73B1 broken, touching structure or ground. Shorted to ground at disconnects or grounded with shield.
Generator No. 2	Terminal A	Open circuit	No generator No. 2 output	ENG 2 DC GENERATOR caution light	Generator No. 2 fails to supply power	Wires P57A16 (2G2-A to 1P2-C), P57B16 (1J2-C to 22J4-C), P57C16 (22P4-C to 1K2-C1) No. 2 start control relay, P57D16 (1K2-C1 to 2Z2P1-B) No. 2 DC control unit, loose, disconnected, or broken.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC generator caution light	Generator No. 2 fails to supply power	Wires P57A16, P57B16, P57C16, P57D16 broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
	Terminal B	Open circuit	No generator No. 2 output	ENG 2 DC GENERATOR caution light	Loss of power from generator No. 2	Wires P85A1 (2G2-B to 2P4-B), P85B1 (2J4-B to 2K6-B2) No. 2 start-gen relay, P84A1 (2K6-A1 to 2K4-A1) No. 2 generator relay, P91A22N (2K4-X2 to ground), P90B22 (2K4-X1 to 22J6-G), P90A22 (22P6-G to 1K2-A2) No. 2 start control relay, P22A22 (1K2-A3 to 22P6-F), P22B22 (22J6-F to 2K4-11), P22C22 (2K4-11 to 2K6-12), P88A22 (2K6-13 to 22J6-N), P88B22 (22P6-N to 2Z2P1-H) No. 2 DC control unit connector, loose, disconnected, or broken. Bus bar at 2K6 missing.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC GENERATOR caution light	Loss of power from generator No. 2	Wires P85A1, P85B1, P84A1, P30B1 (2K4-A2 to 2K2-A2) non-essential bus 2 relay, P30C1 (2K2-A2 to No. 2 main DC bus), P84C16 (2K4-A1 to 2CB10) GEN 2 FIELD CB, P84B16 (2K4-A1 to 22J4-A), P84D16 (22P4-A to 2Z2P1-D) broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
	Terminal C	Open circuit	Starter No. 2 circuit inoperative	None	Starter No. 2 inoperative	Wire P98A1 (2G2-C to 2P4-D), P98B1 (2J4-D to 2K6-A2) loose or disconnected.
		Short to ground	Generator No. 2 drops off line	EGN 2 DC GENERATOR caution light	Loss of power from generator No. 2	Wires P98A1, P98B1 broken, touching structure or ground.
	Terminal D	Open circuit	Generator paralleling circuitry inoperative	None	Generators not paralleled	Wires P35A16 (2G2-D to 1P2-A), P35B22 (1J2-A to 22J6-E), P35C22 (22P6-E to 1K2-C2) No. 2 start control relay, P35D22 (1K2-C2 to 2Z2P1-G) No. 2 DC control unit connector, loose, disconnected, or broken.
		Short to ground	No immediate discernible effect, but paralleling circuit is affected	None	Generator paralleling affected	Wires P35A16, P35B22, P35C22, P35D2 broken, touching structure or ground, or shorted to ground at disconnects.
	Terminal E	Open circuit	Generator No. 2 inoperative	ENG 2 DC GENERATOR caution light	Loss of power from generator No. 2	Wires P86A1 (2G2-E to 2P4-F), P86B1 (2J4-F to 2R2) No. 2 shunt, P87A1N (2R2 to ground), loose, disconnected, or broken. No. 2 shunt burnt, disconnected, or broken.
		Short to ground	No effect	None	None	Wires P86A1, P86B1 broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
No. 1 Main DC bus	Feeder wire at bus	Open circuit	No 28 VDC power on No. 1 main DC bus	CAUTION PANEL caution light, no voltage indicated on voltmeter	No 28 VDC power supplied to buses by generator No. 1	Wires P5A1 (2K7-A1 to 2K3-A1) No. 1 generator relay, P5B1 (2K3-A1 to 2K11-A1) No. 1 non-essential bus relay, P5C1 (2K11-A1 to No. 1 main DC bus) loose, disconnected or broken. 2K7 contacts A1 or A2 broken.
		Short to ground	Generator No. 1 drops off line	ENG 1 DC GENERATOR caution light	Loss of power from generator No. 1	Wires P5A1, P5B1, P5C1, P5D22 (2K7-A1 to 2Z1P1-E) No. 1 DC control unit connector, broken, touching structure or ground.
No. 2 Main DC bus	Feeder wire at bus	Open circuit	No 28 VDC power on No. 2 main DC bus	No voltage indication on voltmeter	No. 28 VDC power supplied to buses by generator No. 2	Wires P30A1 (2K1-A1 to 2K4-A2) No. 2 generator relay, P30B1 (2K4-A2 to 2K2-A2) non-essential bus 2 relay, P30C1 (2K2-A2 to No. 2 main DC bus) loose, disconnected, or broken.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC GENERATOR caution light	Loss of power from generator No. 2	Wires P30A1, P30B1, P30C1, P30D22 (2K8-A2 to 2Z2P1-E) No. 2 DC control unit connector broken, touching structure or ground.
No. 1 Essential DC bus		Short to ground	No. 1 feeder bus circuit breakers trip (after 1 or 2 seconds when BATTERY BUS 2 switch is ON, BATTERY BUS 1 switch is OFF; immediately when BATTERY BUS 2 switch is OFF, and BATTERY BUS 1 switch is ON)	No. 1 feeder circuit breakers trip	No power on No. 1 essential bus from generator No. 1. Generator No. 2 supplies power to No. 1 essential bus. Heavy bus loads may also cause No. 2 feeder circuit breakers to trip. Inverters are not affected.	Wires P247A16 (No. 1 essential DC bus to 10CB11) WINDSHIELD WIPER CPLT CB, P247B16 (10CB11 to No. 1 essential DC bus) broken, touching structure or ground.
		Short to ground	No. 2 feeder bus circuit breakers trip (Immediately if BATTERY BUS 2 switch is ON and BATTERY BUS 1 switch is OFF; after several seconds if BATTERY BUS 1 switch is ON and BATTERY	No. 2 feeder circuit breakers trip	No power on No. 1 essential bus from generator No. 2, generator No. 2 supplies power to No. 1 essential bus. Heavy bus loads may caution No. 1 feeder breakers to trip.	Wires P15A10 (2CB2 to No. 1 essential DC bus), P16A10 (2CB4 to No. 1 essential DC bus) broken, touching structure or ground, or diodes shorted to ground.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
			BUS 2 switch is OFF).			
No. 2 Essential DC bus		Short to ground	No. 2 feeder bus circuit breakers trip (Immediately when BATTERY BUS 2 switch is ON; after several seconds when BATTERY BUS 1 switch is ON and BATTERY BUS 2 switch is OFF).	No. 2 feeder circuit breakers tripped	No power on No. 2 essential bus from generator No. 2 Generator No. 1 supplies power to No. 2 essential bus. (Heavy bus loads may cause No. 1 feeder breakers to trip). Inverters are not affected.	Wires P241B16 (No. 2 essential DC bus to 10CB10) WINDSHIELD WIPER PILOT CB, P241C16 (10CB10 to No. 2 essential DC bus), broken, touching structure or ground, or diodes shorted to ground.
Non-essential DC bus	No. 1 non-essential bus feeder wire at 2TB2	Open circuit	No DC power on non-essential bus from generator No. 1	None	Non-essential bus receives no power from generator No. 1. Generator No. 2 continues to supply power to non-essential bus. (Heavy load on non-essential bus will cause No. 2 overload sensor to trip and remove all power from non-essential bus)	Wires P174A6 (2TB2 to diodes), P148A6 (Diodes to 2S5-L2) feeder power sensor, P147A6 (2S5-L1 to 2K11-A2) non-essential bus relay, loose, disconnected, or broken. Non-essential bus 1 sensor loose or disconnected, diodes open, non-essential bus 1 sensor open or not energized.
	No. 2 non-essential bus feeder wire at 2TB2	Open circuit	No DC power on non-essential bus from generator No. 2	None	Non-essential bus receives no power from generator No. 2. Generator No. 1 generator continues to supply power to non-essential bus. (Heavy load on non-essential bus will cause No. 1 overload relay to trip and remove all power from	Wires P175A6 (2TB2 to diodes), P11A6 (Diodes to 2S6-L2) feeder power sensor, loose, disconnected, or broken. Feeder power sensor loose or disconnected, diodes open, feeder power sensor open or not energized.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
					non-essential bus)	
	No. 1 and No. 2 non-essential bus feeders at 2TB2	Short to ground	Non-essential bus trips off	Non-essential equipment inoperative	No power on Non-essential bus	Wires P174A6, P175A6 broken, touching structure or ground.
			AC POWER SYSTEM			
Inverter No. 1	Pins A and B	Open circuit	Inverter No. 1 inoperative. Inverter No. 2 automatically supplies all AC buses	NO. 1 INVERTER caution light	None Inverter No. 2 supplies all AC buses	Wires X21A16N (3PS1P1-B to ground) V3A12N (3PS1P1-A to ground) loose, disconnected, or broken.
	Pin C	Open circuit	Inverter No. 1 inoperative. Inverter No. 2 automatically supplies all AC buses	No. 1 INVERTER caution light	None Inverter No. 2 supplies all AC buses	Wires V15A16 (3PS1P1-C to 22P3-I), V15B16 (22J3-I to 22J13-S), V15C16 (22P13-S to 3S1-5) INV 1 switch, V14A16N (3S1-6 to ground), loose, disconnected, or broken. Contacts 5 and 6 of 3S1 open.
		Short to ground	Inverter will not turn off with INV 1 switch 3S1	None	None	Wires V15A16, V15B16, V15C16 to contacts 5 and 6 of 3S1 shorted to ground.
	Pin D	Open circuit	No 28 VDC power to inverter No. 1	NO. 1 INVERTER caution light	None Inverter No. 2 will supply all AC buses	Wires V1C12 (3PS1P1-D to 22P1-A), V1B12 (22J1-A to 22J11-A), V1A12 (22P11-A to 3CB7) INV 1 PWR CB, loose, disconnected, or broken. Circuit breaker 3CB7 inoperative or wire P187A12 loose, disconnected, or broken.
		Short to ground	No. 1 inverter inoperative; INV 1 PWR circuit breaker opens	INV 1 PWR circuit breaker (3CB7) open. NO. 1 INVERTER caution light	None Inverter No. 2 can supply all AC buses	Wires V1C12, V1B12, V1A12 shorted to structure or ground.
Non-essential DC bus	Pin E	Open circuit	26 VAC not supplied to 26 VAC buses No. 1 and No. 2 by inverter No. 1	Equipment requiring 26 V AC power becomes inoperative. Instrument flags come into view	No. 26 VAC power is available on 26 VAC buses No. 1 and No. 2	Wires X6A18 (3PS1P1-E to splice), X6C18 (splice to 22P3-G), X6D18 (22J3-G to 22J13-T), X6E18 (22P13-T), X6D18, X6E18 (22P13-T to 3K3-B1) AC bus control relay, broken, loose, or disconnected.
		Short to ground	Inverter No. 1 drops off line	NO. 1 INVERTER caution light. No. 2 buses	None. Inverter No. 2 can supply power to No. 1 and No. 2 buses	Wires X6A18, X6C18, X6D18, X6E18, X6B18 (splice to 3K2P1-P) No. 2 AC voltage sensor relay, 3K3-B1, or 3K2P1-P touching structure or ground.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
Inverter	Pin G	Open circuit	115 VAC not supplied to 115 VAC buses No. 1 and No. 2	NO. 1 INVERTER caution light	None Inverter No. 2 can supply No. 1, No. 2 and No. 3 115 VAC buses	Wires X2A18 (3PS1P1-G to splice), X2C22 (splice to 3K1P1-C), X2B22 (splice to 3K1P1-A) broken, loose or disconnected. Relay 3K1 contacts C and D open, wire X4A18 (3K1P1-D) to splice) broken, loose, or disconnected.
		Short to ground	Inverter No. 1 drops off line immediately. Inverter No. 2 automatically switches to supply all AC buses.	NO. 1 INVERTER caution light	None Inverter No. 2 supplies power to all AC buses	Wire X2A18, (X2B22, X2C22, X4A18 X4C18, X4B18, X4D18, X4E18 touching structure or ground.
Inverter No. 2	Pins A and B	Open circuit	Inverter No. 2 inoperative. Inverter No. 1 automatically supplies bus No. 3, if required.	NO. 2 INVERTER caution light	None Inverter No. 1 supplies all AC buses	Wires X22A16N (3PS2P1-B to ground), V11A12N (3PS2P1-A to ground) loose, disconnected or broken.
	Pin C	Open circuit	Inverter No. 2 inoperative. Inverter No. 1 automatically supplies all AC buses.	NO. 2 INVERTER caution light	None Inverter No. 1 can supply all AC buses	Wires V13A16 (3PS2P1-C to 22P4-I), V13B16 (22J4-I to 3S2-5) INV 2 switch, V12A16N (3S2-6 to ground), broken, loose, or disconnected. Contacts 5 and 6 of 3S2 open.
		Short to ground	Inverter will not turn off with INV 2 switch 3S2	None	None	Wires V13A16, V13B16, or contacts 5 and 6 of 3S2 shorted to ground.
	Pin D	Open circuit	No 28 VDC power to inverter No. 2	NO. 2 INVERTER caution light	None Inverter No. 1 will supply all AC buses	Wires V9B12 (3PS2P1-D to 22P2-A), V9A12 (22J2-A to 3CB8) INV 2 PWR CB loose, disconnected, or broken. Circuit breaker 3CB8 inoperative, wire P188A12 loose, disconnected, or broken.
		Short to ground	Inverter No. 2 inoperative. INV 2 PWR circuit breaker opens.	INV 1 PWR circuit breaker (3CB8) open. NO. 2 INVERTER caution light.	None Inverter No. 1 will supply all AC buses	Wires V9B12, V9A12, shorted to structure or ground.
	Pin E No. 2	Open circuit	26 VAC not supplied to 26 VAC Bus 3 by inverter No. 2	Equipment requiring power from 26 VAC bus 3 inoperative	No 26 VAC power available on 26 VAC bus 3 (AFCS)	Wires X10A18 (3PS2P1-E to splice), X10C18 (SPLICE TO 22P4-G), X10D18 (22J4-G TO 3CB10) bus 3 feeders 26 VAC CB, broken, loose, or disconnected.

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
		Short to ground	Inverter No. 2 drops off line	NO. 2 INVERTER caution light	None. Inverter No. 1 can supply power to 26 VAC bus 3, as well as 26 VAC buses 1 and 2	Wires X10A18, X10C18, X10D18, X10B18 (splice to 3K2P1-F) No. 2 AC voltage sensor relay, X11A18 (3K2P1-G to 22P4-L), X11B18 (22J4-L to 3K3-D3) AC bus control relay, shorted to structure or ground.
	Pin G	Open circuit	115 VAC not supplied to 115 VAC buses No. 1 and No. 2	NO. 2 inverter caution light	None Inverter No. 1 can supply power to 115 VAC buses No. 1 and No. 2	Wires X8A18 (3PS2P1-G to splice), X8C18 (splice to 22P4-K), X8D18 (22J4-K, to 3CB9) BUS 3 FEEDER 115V AC CB, X8B22 (splice to (3K2P1-A) No. 2 AC voltage sensor relay, X8E18 (3CB9 to 3K3-C3) AC bus control relay, broken, loose, or disconnected.
		Short to ground	Inverter No. 2 drops off line immediately. Inverter No. 1 automatically switched to supply all AC buses.	NO. 2 INVERTER caution light	None Inverter No. 1 supplies power to all AC buses	Wires X8A18, X8B22, X8C18, X8D18, X8E18 broken and touching structure or ground.

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						WIRE NO.	FROM	TO
EXTERIOR LIGHTING								
Searchlight	Power wire from circuit breaker to searchlight	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L77A12	8CB24	8DS12-3
							8CB24	Inoperative
		Short to ground	Circuit breaker trips	Circuit breaker trips	Searchlight inoperative	L77A12	8CB24	8DS12
	Ground wire. (searchlight to ground)	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L106A12N	8DS12	Ground
	Control wire	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L73A22	8CB26	22P14-79
						L73B22	22J14-79	4A2J1-Z
								S21-2
							4A2P1-Z	S25-G
							S21-2	Inoperative
							8CB26	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	Searchlight inoperative	Circuit breaker trips	Searchlight inoperative	L73A22 L73B22	8CB26 22J1 4-79 4A2 P1-Z S21-2	22P14-79 4A2J1-Z S21-2 S25-G
	Left rotation circuit	Open circuit	Loss of left control	None	Loss of left control	L64A22 L64B22 L64C22	S25-F 4A2 J1-W 22J1 0-10	4A2P1-W 22P10-10 8DS12-5
	Right rotation circuit		Loss of right control		Loss of right control	L63A22 L63B22 L63C22	S25-D 4A2 J1-X 22J1 0-9	4A2P1-X 22P10-9 8DS12-4
	Left rotation circuit	Short to ground	Control breaker trips	Circuit breaker trips	Searchlight inoperative	L64A22 L64B22 L64C22	S-25F 4A2 J1-W 22J1 0-10	4A2P1-W 22P10-10 8DS12-5
	Right rotation circuit					L63A22 L63B22 L63C22	S-25D 22J1 0-9 22J1 0-11	4A2P1-X 22P10-9 8DS12-4
	Extend circuit	Open circuit	Loss of extend control	None	Loss of extend control	L65A22 L65B22 L65C22	S25-C 4A2 J1-V 22J1 0-11	4A2P1-V 22P10-11 8DS12-6
	Retract circuit		Loss of retract control	Circuit breaker trips	Loss of retract control	L76A22 L76C22 L76D22	S25-E 4A2 J1-Y 22J1 0-12	4A2P1-Y 22P10-12 8DS12-7
Landing light	Power wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L69A12 L69B12	8CB25 22J1 1-E 8CB25	22P11-E 8DS13P1-H Inoperative

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L69A12	8CB25	22P11-E
						L69B12	22J11-E	8DS13P1-H
	Ground wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L70A12N	8DS13P1-I	Ground
	Control wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L78A22	8CB27	22P15-95
						L78B22	22J15-95	22J7-36
						L78C22	22P7-36	4A2J1-c
							S22 2-2	S20-2
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L78A22	8CB27	22P15-95
						L78B22	22J15-95	22J7-36
						L78C22	4A2J1-c	22P7-36
							S22-2	S20-2
							S22-2	S22-3
						L105B22	4A2J21-e	22P7-35
						L105C22	22J7-35	8DS13P1-C
	Extend wire	Open circuit	Unable to extend	None	Loss of extend control	L82C22	8DS13P1-A	22J7-33
						L82B22	22P7-33	4A2J1-b
							S20-3	4A2P1-b
		Short to ground	Unable to retract	Circuit breaker trips	Loss of retract control	L83B22	S20-1	4A2P1-d
						L83C22	4A2J1-d	22P7-34
							22J7-34	8DS13P1-B
Lower anti-collision lights	Power wire	Open circuit	Lower anti-collision lights inoperative	None	Lower anti-collision lights inoperative	L11B20	Splice	22J13-d
						L11A20	22P13-d	8S22-3
	Ground wire					L12A20N	8DS3P1-C	Ground.
Upper anti-collision lights	Power wire		Upper anti-collision light inoperative	None	Upper anti-collision light inoperative	L8A20	8S22-3	1J1-E
						L8B20	1P1-E	8J5-A
						L8C20	8P5-A	Splice

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	Ground wire					L9A20 L9B20 L9C20N	8DS8 P1-C 8J5-B 1J1-F	8P5-B 1P1-F Ground
Position lights	Power wire	Open circuit	Light/lights inoperative	None	Light/lights inoperative	L1A20 8S21-2 L5A20	8CB23 8S2 1-3 8S2 1-3	8S21-2 8TB4-6
			Lower left position light inoperative		Lower left position light inoperative	L5E20 L5J20	8TB-4-6 22J1 3-C	22P13-C 8DS17
			Upper left position light inoperative		Upper left position light inoperative	L2C20	8TB-4-5	8DS15
			Upper right position light inoperative		Upper right position light inoperative	L2B20	8TB-4-5	8DS14
			Lower right position light inoperative		Lower right position light inoperative	L5B20 L5C20 L5D20	8TB-4-6 22J1 2-B 22J8-T	22P12-B 22P8-T 8DS16
			Right tail position light inoperative		Right tail position light inoperative	L5F20 L5G20 L5H20 (WHT)	8TB-4-6 22J1 3-B 8J6-A	22P13-B 8P6-A 8DS28
			Left tail position light inoperative		Left tail position light inoperative	L2D20 L2E20 L2F20	8TB-4-5 22J1 3-A 8J6-C	22P13-A 8P6-C 8DS27
	Ground wire		Left tail position light inoperative		Left tail position light inoperative	L3B20N L2F20 (BLU)	8P6-D 8J6-D	Ground 8DS27
			Right tail position light inoperative		Right tail position light inoperative	L6B20N L5H20 (BLU)	8P6-B 8J6-B	Ground 8DS28

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
LH and RH utility lights	Power wire-circuit breaker to switch	Open circuit	Utility lights inoperative	None	Utility lights inoperative	L124A22 L125A22 L125B22	8CB18 8S2 3-2 8S2 3-3 22J1 5-93	8S23-2 8S23-3 22P15-93 Splice
L utility light			L utility light inoperative		L utility light inoperative	L125C22	Splice	8DS29
R utility light			Right utility light inoperative		Right utility light inoperative	L125D22 L125E22 L125F22	Splice 22 P7-8 22J1 0-4	22J7-8 22P10-4 8DS30
R or L utility light	Ground wire	Open circuit	L or R utility light inoperative	None	L or R utility light inoperative	Chassis	Gro-und	
INTERIOR LIGHTING								
5-VOLT INSTRUMENT LIGHTS								
a) Pilot instrument lights	(1) Wire from circuit breaker to control switch	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a1) L47A22 (b1) L54A22 (c1) L42A22	8CB14 8CB15 8CB17	8R1-2 8R2-2 8R3-2
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of instrument lights			
b) Copilot instrument lights	(2) Wire from control switch to power supply	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a2) L136A22 L136B22 L136C22	8R1-1 22J1 4-91 8TB-2-LR	22P14-91 8TB2-LR 8PS3P1-A

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of instrument lights	(b2)	8R2-1	22P15-97
						L55A22	22J1 5-97	22J7-4
						L55B22		8TB3-LS
						L55C22	22 P7-4	8PS1P1-A
						L55D22	8TB-3-LR	22P14-84
						(c2)		8TB2-RC
						L43D22	8R3-1	8PS4P1-A
						L43C22	22J1 4-84	
						L43A22	8TB-2-RA	
(c) Engine instrument		Short to power	Lights on all the time	Lights on all the time	Cannot extinguish lights		R3-1	R3-2
	(3) Wires from power supply to rheostat	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a3)	8R1-4	22J14-86
						L49A22	22P1 4-86	8PS3P1-D
						L49B22		8R2-4
						(b3)	22P1 5-96	22J7-5
						L56A22	22J1 5-96	8PS1P1-D
						L56B22		22P14-85
						L56C22	22 P7-5	8TB2-RG
						(c3)	8R3-4	8PS4P1-D
						L44D22	22J1 4-85	
						L44C22		
						L44A22	8TB-2-RE	
		Short to ground	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a3)	8PS3 P1-D	22J14-86
						L49B22		8R1-4
						L49A22	22P1 4-86	8R1-5
							8R1-3	8R1-5
							8R1-4	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						(b3)	8PS1 P1-D	22P7-5
						L56C22	22J7-5	22J15-96
						L56B22	22P1 5-96	8R2-4
						L56A22	8R2-3	8R2-5
							8R2-4	8R2-5
						(c3)	8PS2 P1-D	8TB2-RE
						L44A22	8TB-2-RF	8PS4P1-D
						L44B22	8TB-2-RG	22J14-85
						L44C22	22P1 4-85	8R3-3
						L44D22	8R3-4	8R3-5
							8R3-3	8R3-5
		Short to power	Lights on bright all the time	Lights on bright all the time	Lights on bright all the time	(a3)	8R1-4	Lowest resistance tap of 8R1-5
						(b3)	8R2-4	Lowest resistance tap of 8R2-5
						(c3)	8R3-4	Lowest resistance tap of 8R3-5
	(4) Power supply ground return	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a4)	22W-T10-B	8PS3P1-B
						L114A20N	22 WT3-K	8PS1P1-B
						(b4)		
						L112A20N		
						(c4)		
						L141A20N		
						L113A20N		
						(c4)	22WT-10C	8PS4P1-B
						L141A20N	22W-T10-A	8PS2P1-B
						L113A20N		

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to power	Loss of circuit supplying power	Circuit breaker trips	Loss of circuit supplying power	(a4) (b4) (c4)	8PS3 P1-D 8PS1 P1-D 8PS2 P1-D 8PS4 P1-D	8PS3P1-B 8PS1P1-B 8PS2P1-B 8PS4P1-B
	(5) Wire from rheostat to ground	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a5) L35H22 (b5) L57A22 (c5) L35G22	8R1-5 8R2-5 8R3-5	8TB7-5 8TB6-6 8TB6-5
		Short to power	Loss of circuit suppling power	Circuit breaker trips	Loss of circuit	(a5) (b5) (c5)	8R1-5 8R2-5 8R3-5	8R1-2 8R2-2 8R3-2
	(6) Power supply output wire to terminal board	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a6) None (b6) None (c6) L44A22 L44B22	8PS2 P1-D 8PS4 P1-D	8TB2-RE 8TB2-RF
		Short to ground	Loss of instrument lights, circuit breaker trips	Loss of instrument lights, circuit breaker trips	Loss of instrument lights			
		Short to power	Lights on bright all the time	Lights burn bright or bulbs burn out	Loss of instrument light control or loss of lights			
	(7) Power supply output wire from 8TB2 to instruments	Open circuit	Loss of individual instrument light	Loss of individual instrument light	Loss of individual instrument light	(a7) L51M20	8TB-2-NF	4M10P1-G Pilot Attitude Indicator
		Short to ground	Lights dim all the time	Lights dim all the time	Lights dim all the time	L51L20 L51F20	8TB-2-ML 22P12-X	22J12-X 4M6 Pilot Standby Compass

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to power	Lights bright all the time	Lights bright all the time	Lights bright all the time	L51B20	8TB-2-MB	4M8P2-G Pilot Dual Torque Indicator
						L51E20	8TB-2-ME	4M3P1-A Pilot Altimeter
						L51J20	8TB-2-MJ	4M11P1-A Pilot Airspeed Indicator
						L51D20	8TB-2-MD	4M8P1-G Pilot Dual Torque Indicator
						L51K20	8TB-2-MK	4M7P1-E Pilot Triple Tachometer
						L51H20	8TB-2-MH	4M5P1-2 Pilot Clock
						L51G20	8TB-2-MG	4M4P1-2 Pilot Vertical Speed Indicator
						(b7)		
						(c7)		
	(8) Power supply output wire from 8TB3 to instruments	Open circuit	Loss of individual instrument light	Loss of individual instrument light	Loss of individual instrument light	(a8) None (b8)	8TB-3-NJ	5M8P1-G Copilot Dual Torque Indicator
						L58N20		
		Short to ground	Lights dim all the time	Lights dim all the time	Lights dim all the time	L58F20	8TB-3-MF	5M8P2-G Copilot Dual Torque Indicator
		Short to power	Lights bright all the time	Lights bright all the time	Lights bright all the time	L58K20	8TB-3-MK	5M9P1-g Copilot Attitude Indicator
						L58J20	8TB-3-MJ	5M1P1-A Copilot Airspeed Indicator

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L58G20	8TB-3-MG	5M4P1-2 Copilot Rate of Climb Indicator
						L58C20	8TB-3-MC	5M7P1-E Copilot Triple Tachometer
						L58H20	8TB-3-MH	5M3P1-A Copilot Altimeter
						L58D20	8TB-3-MD	22P9-i
						L58P20	22J9-i	22J13-e
						L58R20	22P1 3-e	5M6P1 Copilot Standby Compass
						L58B20	8TB-3-MB	5M5P1-2 Copilot Clock
						(c8)		
						None		
	(9) Power supply output wire from 8TB1 to instruments	Open circuit	Loss of individual instrument light	Loss of individual instrument light	Loss of individual instrument light	(a9)		
						None		
						(b9)		
						None		
						(c9)		
		Short to ground	Lights dim all the time	Light dim all the time	Lights dim all the time	L46C20	8TB-1-RB	5M2P1-A Dual Voltmeter 1 Indicator
		Short to power	Lights bright all the time	Lights bright all the time	Lights bright all the time	L46H20	8TB-1-RG	1M8P1-A Engine 2 Fuel Pressure Indicator
						L46M20	8TB-1-RL	1M7P1-A Fuel Quantity Indicator

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L46T20	8TB-1-SE	9M1P1-A No. 1 Hydraulic Pressure and Temperature Indicator
						L46E20	8TB-1-RD	2M1P1-E Dual Ammeter
						L46V20	8TB-1-SC	4M2P1-A Dual Voltmeter 2 Indicator
						L46N20	8TB-1-SD	9M2P1-A No. 2 Hydraulic Pressure and Temperature Indicator
						L46J20	8TB-1-RH	1M1P1-A Engine 1 Oil Temperature and Pressure Indicator
						L46R20	8TB-1-SA	1M4P1-D Engine 2 Gas Producer Tachometer
						L46G20	8TB-1-RM	1M9P1-A Engine 1 Fuel Pressure Indicator
						L46D20	8TB-1-RC	1M11P1-A Engine 1 Interturbine Temperature Indicator
						L46F20	8TB-1-RE	1M10P1-A Engine 2 Interturbine Temperature Indicator

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
						L46L20	8TB-1-RK	1M5P1-A Transmission Oil Temperature and Pressure Indicator
						L46Q20	8TB-1-SF	1M6P1-A Combining Gearbox Oil Pressure and Temperature Indicator
						L46K20	8TB-1-RJ	1M2P1-A Engine 2 Oil Temperature and Pressure Indicator
						L46S20	8TB-1-SB	1M3P1-D Engine 1 Gas Producer Tachometer
	(10) Individual ground wire from instrument	Open circuit	Loss of individual light	Loss of individual light	Loss of individual light	(a10) L95A20N	22 WT2-A	4M10P1-h Pilot Attitude Indicator
		Short to power	Loss of circuit supplying power	Circuit breaker trips	Loss of circuit supplying power	L95G20N	22WT-2-G	4M8P2-H Pilot Dual Torque Indicator
						L95L20N	22 WT2-L	4M8P1-H Pilot Dual Torque Indicator
						L95F20N	22 WT2-F	4M4P1-1 Pilot Vertical Speed Indicator
						L95D20N	22 WT2-D	4M5P1 Pilot Clock
						L95C20N	22 WT2-C	4M7P1-F Pilot Triple Tachometer

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L95J20N	22 WT2-J	4M11P1-B Pilot Airspeed Indicator
						L95K20N	22 WT2-K	4M3P1-B Pilot Altimeter
						(b10)		
						L97M20N	22 WT3-H	5M8P1-H Copilot Dual Torque Indicator
						L97L20N	22 WT3-I	5M8P2-H Copilot Dual Torque Indicator
						L97F20N	22WT-3-G	5M9P1-H Copilot Attitude Indicator
						L97A20N	22 WT3-A	5M5P1-1 Copilot Clock
						L97H20N	22 WT3-F	5M3P1-B Copilot Altimeter
						L97E20N	22 WT3-E	5M7P1-F Copilot Triple Tachometer
						L97C20N	22 WT3-C	5M4P1-1 Copilot Rate of Climb Indicator
						L97D20N	22 WT3-D	5M1P1-B Copilot Airspeed Indicator
						(c10)		
						L96D20N	22 WT4-D	5M2P1-J Dual Voltmeter 1 Indicator

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L96N20N	22W-T10-E	1M8P1-E Engine 2 Fuel Pressure Indicator
						L96M20N	22WT-4-M	1M7P1-J Fuel Quantity Indicator
						L96R20N	22W-T10-M	9M1P1-E No. 1 Hydraulic Pressure and Temperature Indicator
						L96H20N	22 WT4-H	2M1P1-F Dual Ammeter
						L96P20N	22W-T10-L	4M2P1-J Dual Voltmeter 2
						L96Q20N	22W-T10-D	9M2P1-E No. 2 Hydraulic Pressure and Temperature Indicator
						L96G20N	22WT-4-G	1M9P1-E Engine 1 Fuel Pressure Indicator
						L96F20N	22 WT4-F	1M4P1-E Engine 2 Gas Producer Tachometer
						L96J20N	22 WT4-J	1M1P1-E Engine Oil Temperature and Pressure Indicator

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L96A20N	22 WT4-A	1M11P1-J Engine 1 Interturbine Temperature Indicator
						L96B20N	22 WT4-B	1M10P1-J Engine 2 Interturbine Temperature Indicator
						L96K20N	22 WT4-K	1M5P1-E Transmis-sion Oil Tem-perature
						L96C20N	22 WT4-C	1M6P1-E Combining Gearbox Oil Pressure and Temperature Indicator
						L96E20N	22 WT4-E	1M2P1-E Engine 2 Oil Temperature and Pressure Indicator
						L96L20N	22 WT4-L	1M3P1-E Engine 1 Gas Producer Tachometer
	(11) Wire from PILOT INSTR switch to caution panel	Open circuit	None	None	Cannot dim caution panel	L107A22	8R1-7	22P14-9
						L107B22	22J1 4-9	8A1P2-a
INTERIOR LIGHTING								
28-VOLT INSTRUMENT LIGHTS								
a) Overhead console lights	(1) Wire from circuit breaker to switch	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a1)	8CB21	8R5-2
						L32A22	8CB21	Inoperative
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of lights	(b1)	8CB19	8R6-2
						L111A22		
						(c1)	8CB16	8R4-2
						L59A22		
b1) Pedestal lights.								

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
c) Secondary instrument lights	(2) Wire from switch to transistor collector	Open circuit	Loss of lights.	Loss of lights.	Loss of lights	(a2)	8R5-1	8Q2-C
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of lights	L33B22	8Q1-C	8Q1-C
						L33C22	8R6-1	8Q3-C
						(b2)		
						L38B22		
		Short to ground	Lighting circuit energized continuously	None	None	(c2)	8R4-1	8Q4-C
						L60B22		
	(3) Wire from switch to transistor base	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a3)	8R5-4	8Q1-B
		Short to ground	Loss of lights	Loss of lights	Loss of lights	L34A22	8Q1-B	8Q2-B
						L34B22		
						(b3)	8R6-4	8Q3-B
						L39A22	8R4-4	8Q4-B
						(c3)		
						L61A22		
	(4) Wire from switch to ground	Open circuit	None	None	Loss of circuit supplying power	(a4)	8R5-5	8TB7-6
						L35A22		
		Short to ground	None	None	Loss of dimming control	(b4)	8R6-5	8TB7-6
						L35E22		
						(c4)	8R4-5	8TB6-6
						L57D22		
	(5) Wire from transistor emitter to terminal board	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a5)	8Q1-E	8TB7-3
		Short to ground	Loss of lights	Loss of lights	Loss of lights	L162A22	8Q2-E	8TB6-3
						L163A22		
		Short to power	None	None	Lights bright all the time	(b5)	8Q3-E	22P15-98
						L41A22	22J1 5-98	22J7-3
						L41B22		8TB1-NA
						L41C22	22P1 7-3	
						(c5)	8Q4-E	22P14-83
						L36A22	22J1 4-83	22J17-B
						L36B22		TB14-2
						L36C22	22P1 7-B	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	(6) Individual wires from terminal board to lights	Open circuit	Loss of individual light	Loss of individual light	Loss of individual light	(a6)	24A5 J1	8TB7-2
		Short to ground	Loss of lights	Loss of lights	Loss of lights	L154A22		8TB7-1
		Short to power	None	None	Lights on bright all the time	L123G22	8A3J1	8TB7-1
						L123D22	8A5J1	8TB7-1
						L123C22	24A4 J1	8TB6-1
						L123E20	24A3 J1	8TB6-1
						L123F22		8TB6-2
						L134A22	8A4J1	
							8A2J1	
						(b6)	8TB-1-NB	24A1P1-F
						L41D22		24A1J3
						L41E22	24A1 J1-F	1A1P2-Y
						L41F22	8TB-1-NF	1A1J3
						L41G22		4A2J1-f
						L41H22	1A1 J2-Y	J263
						L92K22	8TB-1-NJ	22P10-6
						L41K22		Splice
						L41L22	4A2 P1-f	10DS2
							8TB-1-NK	10DS3
							22J1 0-6	
							Splice	
							Splice	
						C85A24	8TB-1-NE	9A3P2-d
						C145A24		9A11P1-K
						C145B22	8TB-1-N	9A11J2
						(c6)	9A11 J1-K	DS28
						L18D22		DS29
						L18E22	TB1 4-1	DS30
						L18F22	TB1 4-1	DS27
						L18C22		DS44
						L18G22	TB1 4-1	DS44
						L91A22	TB1 4-2	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
							TB1 4-2	
							TB1 4-3	
	(7) Individual ground wire from terminal block to lights	Open circuit	Loss of individual lights	Loss of individual lights	Loss of individual lights	(a7) NONE		
		Short to ground	None	None	None			
						(b7)	TB1 4-3	Ground
						NONE		DS44
						c7)	TB1 4-3	Instrument Secondary Lights
						L91B20N		
						L91A22		
Pilot utility (cockpit) light	Power wire	Open circuit	Pilot utility light inoperative	None	Pilot utility light inoperative	L124A22	8CB18	8S23-2
						L125B22	22J1 5-93	Splice
						L125D22		22J7-8
						L125E22	Splice	22P10-4
						L125F22	22 P7-8	8DS30
							22J1 0-4	Right Utility Light
		Short to ground	Circuit breaker trips	Circuit breaker trips	Pilot utility light inoperative	L124A22	8CB18	8S23-2
						L125A22	8S2 3-3	22P15-93
						L125B22		Splice
						L125D22	22J1 5-93	22J7-8
						L125E22	Splice	22P10-4
						L125F22	22 P7-8	8DS30
							22J1 0-4	Right Utility Light
						L125C22	Splice	8DS29
								Left Utility Light
	Ground return	Open circuit	Pilot utility light inoperative	None	Pilot utility light inoperative	Chassis	8DS30	Ground

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
Pilot utility (map light)	Power wire	Open circuit	Pilot utility light inoperative	None	Pilot utility light inoperative	L30A20	8CB12	4TB1
							4TB1	8DS10
								Pilot Utility Light
						L30B22	8CB12	22P14-73
						L30C22	22J14-73	22J16-a
						L30D22		8R20-3
						L806A22	22P16-a	8R20-7
						L808A20	8R20-4	22P16-Z
						L808B20		Splice
							8R20-6	
							22J16-Z	
		Short to ground	Pilot utility lights inoperative	Circuit breaker trips	Pilot and copilot utility light inoperative		Splice	8DS37
								Pilot Map Light
							L30A20	8CB12
								8DS10
							4TB1	Pilot Utility Light
						L30B22	8CB12	22P14-73
						L30C22	22J14-73	22J16-a
						L30D22		8R20-3
							22P16-a	
Copilot utility (cockpit) light	Power wire	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	L125C22	Splice	8DS29
								Left Utility Light Cockpit Light
	Ground return	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	Chassis	8DS29	Ground

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
Copilot utility (map light)	Power wire	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	L31B22	8CB13	22P15-94
						L31C22	22J15-94	22J7-6
						L31D22		22J17-Y
						L31E22	22 P7-6	8R21-3
						L807A22	22P17-Y	8R21-7
						L809A22		22P17-Q
						L809B22	8R21-4	Splice
							8R21-6	8DS38
							22J17-Q	Copilot Map Light
							Splice	
						L31A22	8CB13	5TB1-1
							5TB-1-1	8DS9
								Copilot Map Light
		Short to ground	Copilot utility light inoperative	Circuit breaker trips	Copilot utility light inoperative	L31B22	8CB13	22P15-94
						L31C22	22J15-94	22J7-6
						L31D22		22J17-Y
						L31E22	22 P7-6	8R21-3
						L807A22	22P17-Y	8R21-7
							8R21-4	
	Ground wire	Open circuit	Copilot utility light inoperative	None	Copilot utility light inoperative	Chassis	8DS97	Ground
						L201A22N	8DS38	Chassis
MASTER CAUTION PANEL								
Legend capsules requiring a negative (ground) fault fault signal	Legend capsules 1, 2, 7, through 12, 15, 16, 19 through 33, and 35 through 39							

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 1 oil pressure (Capsule 1) Ground wire	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q1C22 Q1B22 Q1A20 Q2A20 Q2B22N	8A1 P1-B 22J7-9 1P5-T 8S5 P1-B 1J5-U	22P7-9 1J5-T 8S5P1-A 1P5-U Ground
		Short to ground	False indication	Caution light	False indication	Q1C22 Q1B22 Q1A20	8A1 P1-B 22J7-9 1P5-T	22P7-9 1J5-T 8S5P1-A
	ENG 2 OIL PRESSURE (Capsule 2)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q3C22 Q3B22 Q3A20 Q4A20 Q4B22N	8A1 P2-B 22J1 0-15 1P6-R 8S6 P1-B 1J6-S	22P10-15 1J6-R 8S6P1-A 1P6-S Ground
		Short to ground	False indication	Caution light	False indication	Q3C22 Q3B22 Q3A20	8A1 P2-B 22J1 0-15 1P6-R	22P10-15 1J6-R 8S6P1-A
	ENG 1 FUEL BOOST (Capsule 7)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q79A22 Q79B22 Q67A22 Q67B22 Q67C22 Q6A22N	8A1 P1-D 1A1 J1-X 1A1T-B1-KL 1A1 P1-Y 22J1 9-20 1S5 P1-B	1A1P1-X 1A1TB1-KE 1A1J1-Y 22P19-20 1S5P1-A Ground
		Short to ground	False indication	Caution light	False indication	Q79A22 Q79B22 Q67A22	8A1 P1-D 1A1 J1-X 1A1T-B1-KL	1A1P1-X 1A1TB1-KE 1A1J1-Y

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						Q67B22	1A1 P1-Y	22P19-20
						Q67C22	22J1 9-20	1S5P1-A
	ENG 2 FUEL BOOST (Capsule 8)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q80A22	8A1 P2-D	1A1P1-C
						Q80B22	1A1 J1-C	1A1TB1-KC
						Q68A22	1A1T-B1-KH	1A1J2-S
						Q68B22	1A1 P2-S	22P18-1
						Q68C22	22J1 8-1	1S6P1-A
						Q45A22 N	1S6 P1-B	Ground
		Short to ground	False indication	Caution light	False indication	Q80A22	8A1 P2-D	1A1P1-C
						Q80B22	1A1 J1-C	1A1TB1-KC
						Q68A22	1A1T-B1-KH	1A1J2-S
						Q68B22	1A1 P2-S	22P18-1
						Q68C22	22J1 8-1	1S6P1-A
	ENG 1 FUEL FILTER (Capsule 9)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q5D22	8A1 P1-E	22P7-10
						Q5B22	22P1 5-92	1J1-H
						Q5A18	1P1-H	8S9P1-B
						Q6A18	8S9 P1-C	1P1-J
						Q6B22N	1J21-J	Ground
		Short to ground	False indication	Caution light	False indication	Q5D22	8A1 P1-E	22P7-10
						Q5C22	22J7-10	22J15-92
						Q5B22	22P1 5-92	1J1-H
						Q5A18	1P1-H	8S9P1-B

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 2 FUEL FILTER (Capsule 10)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q7C22 Q7B22 Q7A18 Q8A18 Q8B22N	8A1 P2-E 22P1 4-92 1P2-H 8S10 P1-C 1J2-J	22J14-92 1J2-H 8S10P1-B 1P2-J Ground
		Short to ground	False indication	Caution light	False indication	Q7C22 Q7B22 Q7A18	8A1 P2-E 22P1 4-92 1P2-H	22J14-92 1J2-H 8S10P1-B
	ENG 1 FUEL LOW (Capsule 11)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	E39A22 E39B22 Q29A20N	8A1 P1-F 22J7-98 1Z15 TB1-4 1Z15 S2 1Z15 TB1-5	22P7-98 1Z15TB1-4 1Z15S2 1Z15TB1-5 Ground
		Short to ground	False indication	Caution light	False indication	E39A22 E39B22	8A1 P1-F 22J7-98 1Z15 TB1-4	22P7-98 1Z15TB1-4 1Z15S2
	ENG 2 FUEL LOW (Capsule 12)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	E40B22 Q37A20N	8A1 P2-F 1Z14 TB1-4 1Z14 S2 1Z15 TB1-5	22P10-18 1Z14S2 1Z14TB1-5 Ground
		Short to ground	False indication	Caution light	False indication	E40B22 E40A22	8A1 P2-F 22J10-18 1Z14 TB1-4	22P10-18 1Z14TB1-4 1Z14S2

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 1 ENGINE CHIP (Capsule 15)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W51A22	8A1 P1-H	8TB3-LA
						W51B22		22P7-20
						W51C22	8TB-3-LE	22J15-75
						W51D22	22J7-20	1J1-W
						W51E18		8E3P1-B
						W52A18	22P15-75	1P1-X
						W52B22N	1P1-W	Ground
						W51E22	8E3 P1-A	22P7-21
						W51F22	1J1-X	1J5-L
						W51G20	8TB-3-LF	8E5P1-A
						W53A20		1P5-M
						W53B22N	22J7-21	Ground
							1P5-L	
							8E5 P1-B	
							1J5-M	
		Short to ground	False indication	Caution light	False indication	W51A22	8A1 P-H	8TB3-LA
						W51B22		22P7-20
						W51C22	8TB-3-LE	22J15-75
						W51D22	22J7-20	1J1-W
						W51E18		8E3P1-B
						W51E22	22P15-75	22P7-21
						W51F22	1P1-W	1J5-L
						W51G20	8TB-3-LF	8E5P1-A
							22J7-21	
							1P5-L	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 2 ENGINE CHIP (Capsule 16)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W54A22 W54B22 W54C22 W54D18 W55A18 W55B22N W54E22 W54F22 W54G20 W56A20 W56B22N	8A1 P2-H 8TB-2-LG 22P1 4-76 1P2-W 8E4 P1-A 1J2-X 8TB-2-LH 22J1 0-13 1P6-L 8E6 P1-B 1J6-M	8TB2-LC 22J14-76 1J2-W 8E4P1-B 1P2-X Ground 22P10-13 1J6-L 8E6P1-A 1P6-M Ground
		Short to ground	False indication	Caution light	False indication	W54A22 W54B22 W54C22 W54D18 W54E22 W54F22 W54G20	8A1 P2-H 8TB-2-LG 22P1 4-76 1P2-W 8TB-2-LH 22J1 0-13 1P6-L	8TB2-LC 22J14-76 1J2-W 8E4P1-B 22P10-13 1J6-L 8E6P1-A
	ENG 1 GEN OVHT (Capsule 19)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W151D22 W151C22 W151B22 W151A20 W149A20	8A1 P1-K 22J7-27 22P1 5-82 8P7-A 8S17	22P7-27 22J15-82 8J7-A 8S17 8P7-B Ground

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						W149B22N	8J7-B	
		Short to ground	False indication	Caution light	False indication	W151D22	8A1P1-K	22P7-27
								22J15-82
						W151C22	22J7-27	8J7-A
						W151B22	22P15-82	8S17
						W151A20	8P7-A	
	ENG 2 GEN OVHT (Capsule 20)	Open circuit indication	Loss of particular fault indication	None	Loss of particular fault indication circuit	W150C22	8A1P2-K	22J12-A
								1J2-E
						W150B22	22P12-A	8S18
						W150A18	1P2-E	1P2-F
							8S18	Ground
						W148A18	1J2-F	
						W148B22N		
		Short to ground	False indication	Caution light	False indication	W150C22	8A1P2-K	22J12-A
								1J2-E
						W150B22	22P12-A	8S18
						W150A18	1P2-E	
	CAUTION PANEL (Capsule 21)	Illuminates if applied voltage to all capsules drops below 75% rated value.						
	Ground wire	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	L79C22N	8A1P2-Y	22WT7-Z
						W158A2N		22WT5-L
							8A1P1-L	
	Power wire	Open circuit	False indication	Caution light	False indication	L88C22	8A1P2-Z	8TB3-RH
						L88B22		22J14-89
						L88A22	8TB-3-Z	8S20-2
						L152A22	22P14-89	8CB5
							8S20-3	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	L88C22	8A1 P2-Z	8TB3-RH
						L88B22		22J14-89
						L88A22	8TB-3-RH	8S20-2
						L152A22	22P1 4-89	8CB5
							8S2 0-3	
	HYDRAULIC (Capsule 22)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	C30G22	8A1 P2-L	9TB4-PN
						C30F22		22J14-98
						C30E22	9TB-4-PT	1J8-E
						C30D22	22P1 4-98	Splice
							1P8-E	
		Short to ground	False indication	Caution light	False indication	C30G22	8A1 P2-L	9TB4-PN
						C30F22		22J14-98
						C30E22	9TB-4-PT	1J8-E
						C30D22	22P1 4-98	Splice
						C30B22	1P8-E	8S16P1-A
						C30C22	Splice	8S14P1-B
						C30A22	Splice	8S13P1-B
						C47A22	Splice	1P8-H
						C47B22	8S15 P1-A	22P14-71
						C47C22	1J8-H	9TB4-RA
							22J1 4-71	
	XMSN OIL PRESSURE (Capsule 23)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	D23C22	8A1 P1-M	22J14-81
						D23B22		1J9-V
							22P1 4-81	8S2P1-A
							1P9-V	Ground
							8S2 P1-B	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	D23C22 D23B22	8A1 P1-M 22P1 4-81 1P9-V	22J14-81 1J9-V 8S2P1-A
	CHIP C'BOX (Capsule 24)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W57C22 W57B22 W57A20 W58A20 W58B22 N	8A1 P1-N 22J7-22 1P5-N 8E2 P1-B 1J5-P	22P7-22 1J5-N 8E2P1-A 1P5-P Ground
		Short to ground	False indication	Caution light	False indication	W57C22 W57B22 W57A20	8A1 P1-N 22J7-22 1P5-N	22P7-22 1J5-N 8E2P1-A
	C'BOX OIL PRESSURE (Capsule 25)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication	Q69C22 Q69B22 Q69A20 Q70A20 Q70B22 N	8A1 P2-M 22J10-14 1P6-N 8S8 P1-B 1J6-P	22P10-14 1J6-N 8S8P1-A 1P6-P Ground
		Short to ground	False indication	Caution light	False indication	Q69C22 Q69B22 Q69A20	8A1 P2-M 22J10-14 1P6-N	22P10-14 1J6-N 8S8P1-A
	INVERTER -1 (Capsule 26)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W79B22 W79A22 W801A2 N	8A1 P1-P 22 P5-C 3XK-1-G	22J5-C 3XK1-H Ground
		Short to ground	False indication	Caution light	False indication	W79B22 W79A22	8A1 P1-P 22 P5-C	22J5-C 3XK1-H

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	CHIP XMSN (Capsule 27)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W175A22	8A1 P1-R	8TB8-3
								22J14-98
						W177C22	8TB-8-2	1J9-L
						W177B22	22P1 4-98	8E1
								Ground
							1P9-L	
							8E1	
						W178C22	8TB-8-4	22J14-99
								1J9-K
						W178B22	22P1 4-99	8E9
							1P9-K	Ground
							8E9	
						W179C22	8TB-8-5	22J14-100
								1J9-J
						W179B22	22P1 4-100	8E10
							1P9-J	Ground
							8E10	
		Short to ground	False indication	Caution light	False indication	W175A22	8A1 P1-R	8TB8-3
								22J14-98
						W177C22	8TB-8-2	1J9-L
						W177B22	22P1 4-98	8E1
							1P9-L	
						W178C22	8TB-8-4	22J14-99
								1J9-K
						W178B22	22P1 4-99	8E9
							1P9-K	
						W179C22	8TB-8-5	22J14-100
								1J9-J
						W179B22	22P1 4-100	8E10
							1P9-J	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	INVERTER -2 (Capsule 28)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W80B22	8A1 P2-N	22J6-c
						W80A22	22 P6-C	3K2P1-L
						W802A2 N	3K2 P1-K	Ground
		Short to ground	False indication	Caution light	False indication	W80B22	8A1 P2-N	2J6-C
						W80A22	22 P6-C	3K2P1-L
	XMSN OIL TEMP (Capsule 29)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	D22C22	8A1 P1-S	22J14-80
						D22B22	22P1 4-80	1J9-S
							1P9-S	8S1
							8S1	Ground
		Short to ground	False indication	Caution light	False indication	D22C22	8A1 P1-S	22J14-80
						D22B22	22P1 4-80	1J9-S
							1P9-S	8S1
	CHIP 42/90 BOX (Capsule 30)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W60A22	8A1 P1-T	8TB3-RJ
						W60B22	8TB-3-RL	22P7-11
						W60C22	22J7-11	8P6-G
						W60D22	8J6-G	8E8
							8E8	Ground
						W60E22	8TB-3-RM	22P7-23
						W60F22	22J7-23	8P6-H
						W60G22	8J6-H	8E7
							8E7	Ground

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	W60A22 W60B22 W60C22 W60D22 W60E22 W60F22 W60G22	8A1 P1-T 8TB-3-RL 22J7-11 8J6-G 8TB-3-RM 22J7-23 8J6-H	8TB3-RJ 22P7-11 8P6-G 8E8 22P7-23 8P6-H 8E7
	C'BOX OIL TEMP	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	E90C22 E90B22 E90A20 E91A20 E91B22N	8A1 P2-P 22J7-1 1P5-A 8S11 P1-B 1J5-B	22P7-1 1J5-A 8S11P1-A 1P5-B Ground
		Short to ground	False indication	Caution light	False indication	E90C22 E90B22 E90A20	8A1 P2-P 22J7-1 1P5-A	22P7-1 1J5-A 8S11P1-A
	EXTERNAL POWER (Capsule 32)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W85A22 W86A22N	8A1 P1-U 8S1 2-NC	8S12-COM Ground
		Short to ground	False indication	Caution light	False indication	W85A22	8A1 P1-U	8S12-COM
	BATTERY TEMP (Capsule 33)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W161A22 W161D22 W161B22 W162A22N W161C22 W162B22N	8A1 P1-V 22 P5-Y Splice 8Z10 P1-C Splice 8Z10 P1-F	22J5-Y Splice 8Z10P1-B Ground 8Z10P1-E Ground

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	W161A22	8A1 P1-V	22J5-Y
								Splice
						W161D22	22 P5-Y	8Z10P1-B
						W161B22	Splice	8Z10P1-E
								Splice
						W161C22		
	DOOR LOCK (Capsule 35)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W65A22	8A1 P1-W	22P7-24
						W65B22		Splice
						W65C22	22J7-24	22J15-84
						W65D22	Splice	Splice
							22P15-84	8S3-C
								Splice
							Splice	
							8S3-N-O	
						W66A22N	Splice	Ground
							Splice	Splice
						W65E22	Splice	8S4-C
							8S4-N-O	Splice
						W67A22N	Splice	Ground
							Splice	8P6-E
						W65F22	8J6-E	8S7-NC
							8S7-C	Ground
		Short to ground	False indication	Caution light	False indication	W65A22	8A1 P1-W	22P7-24
						W65B22		Splice
						W65C22	22J7-24	22J15-84
						W65D22	Splice	Splice
							22P15-84	8S3-C
								Splice
						W65E22	Splice	Splice
						W65F22	Splice	8S4-C
							Splice	8P6-E
							8J6-E	8S7-NC

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	HEATER AIR LINE (Capsule 37)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W112A22	8A1 P2-S	Splice
		Short to ground	False indication	Caution light	False indication	W112A22	8A1 P2-S	Splice
						H37B22	Splice	22J12-e
						H37A22	22P1 2- 10 S8-5	10S8-6
						H35A22		22P12-d
						H35B22		22P10-38
						H35C22		10S4P1-B
						W112B22	22J1 2-d	22P10-34
						W112C22	22J1 0-38	Splice
						H26A22	Splice	10S2
							22J1 0-34	10S4P1-C
							Splice	
							Splice	
	BATTERY (Capsule 38)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W144A22	8A1 P1-Y	22J6-P
						W144B22	22 P6-P	2K8-14
						W145A22	2K8-13	2K7-14
						W146A22	2K8-11	2K7-12
						W147A2N	2K7-13	2K7-11
						W147B2N	2K7-11	Ground
		Short to ground	False indication	Caution light	False indication	W144A22	8A1 P1-Y	22J6-P
						W144B22	22 P6-P	2K8-14
						W145A22	2K8-13	2K7-14
						W146A22	2K8-11	2K7-12
	Legend capsules 3 through 6, 17, 18, 34 and 40.							

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 1 DC GENERA-TOR (Cap-sule 3)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W77F22 W77D22 W77A22 P153B2N	8A1 P1-A 22J9-F 22P1 5-2 2K3-1 3	2P9-F 22J15-2 2K3-14 Ground
		Short to ground	False indication	Caution light	False indication	W77F22 W77D22 W77A22	8A1 P1-A 22J9-F 22P1 5-2	22P9-F 22J15-2 2K3-14
	ENG 2 DC GENERA-TOR (Cap-sule 4)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	W78D22 W78A22 P91B22N	8A1 P2-A 22P1 4-15 2K4-1 3	22J14-15 2K4-15 Ground
		Short to ground	False indication	Caution light	False indication	W78D22 W78A22	8A1 P2-A 22P1 4-15	22J14-15 2K4-14
	ENG 1 PART SEP OFF (Capsule 5) Power wire.	Open circuit	False indication	Caution light	False indication	K25D22 K25C22 K25B22 K25A20	8A1 P1-C 22J7-1 9 22P1 5-20 1P1-h	22P7-19 22J15-20 1J1-h 1B15P1-D
		Short to ground	False indication	Caution light	False indication	K25D22 K25C22 K25B22 K25A20	8A1 P1-C 22J7-1 9 22P1 5-20 1P1-h	22P7-19 22J15-20 1J1-h 1B15P1-D
	ENG 2 PART SEP OFF (Capsule 6)	Open circuit	False indication.	Caution light.	False indication.	K27C22 K27B22 K27A20	8A1 P2-C 22P1 4-52 1P2-h	22J14-52 1J2-h 1B16P1-D

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground.	False indication	Caution light	False indication	K27C22	8A1 P2-C	22J14-52
						K27B22		1J2-h
						K27A20	22P1 4-52	1B16P1-D
							1P2-H	
	ENG 1 FUEL VALVE (Capsule 17)	Open circuit	False indication	Caution light	False indication	Q93B22	8A1 P1-J	1A1P1-h
						Q93A22		1A1S1-8
						Q91E22	1A1 J1-h	1A1S1-8
						Q91E22	1A1 S1-9	1A1J1-g
						Q91D22		22P7-74
						Q91C22	1A1 S1-9	22J15-38
						Q91B22	1A1 P1-g	1J7-R
						Q91A22	22J7-74	1B13P1-D
							22P1 5-38	
							1P7-R	
		Short to ground	False indication	Caution light	False indication	Q93B22	8A1 P1-J	1A1P1-h
						Q93A22		1A1S1-8
						Q93E22	1A1 J1-h	1A1J1-g
						Q91D22	1A1 S1-9	22P7-74
						Q91C22		22J15-38
						Q91B22	1A1 P1-g	1J7-R
						Q91A22	22J7-74	1B13P1-D
							22P1 5-38	
							1P7-R	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 2 FUEL VALVE (Capsule 18)	Open circuit	False indication	Caution light	False indication	Q94B22	8A1 P2-J	1A1P2-h
						Q94A22		1A1S2-8
						Q92D22	1A1 J2-h	1A1J2-g
						Q92C22	1A1 S2-9	22J14-65
						Q92B22		1J8-R
						Q92A22	1A1 P2-s	1B14P1-D
							22P1 4-65	
							1P8-R	
		Short to ground	False indication	Caution light	False indication	Q94B22	8A1 P2-J	1A1P2-h
						Q94A22		1A1S2-8
						Q92D22	1A1 J2-h	1A1J2-g
						Q92C22	1A1 S2-9	22J14-65
						Q92B22		1J8-R
						Q92A22	1A1 P2-tg	1B14P1-D
							22P1 4-65	
							1P8-R	
	FUEL XFEED (Capsule 34)	Open circuit	False indication	Caution light	False indication	Q96B22	8A1 P2-R	1A1P1-f
						Q96A22		1A1S8-C2
						Q95E22	1A1 J1-f	1A1J1-Z
						Q95D22	1A1 S8-C(NO)	22P7-68
						Q95C22		22J15-33
						Q95B22	1A1 P1-Z	1J7-P
						Q95A22		1B10P1-D
							22J7-68	
							22P1 5-33	
							1P7-P	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	False indication	Caution light	False indication	Q96B22	8A1 P2-R	1A1P1-f
						Q96A22		1A1S8-C2
						Q95E22	1A1 J1-f	1A1J1-Z
						Q95D22	1A1 S8-C(NO)	2P7-68
						Q95C22		22J15-33
						Q95B22	1A1 P1-Z	1J7-P
						Q95A22		1B10P1-D
							22J7-68	
							22P15-33	
							1P7-P	
	AFCS (Capsule 40)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q97A24	8A1 P2-T	8XK3-1
						C101B2N		22WT14-K
							8XK-3-2	
		Short to ground	False indication	Caution light	False indication	C97A24	8A1 P2-T	8XK3-1
	Legend capsules requiring application of 28V DC for fault signal		Legend capsules 13 and 14					
	ENG 1 GOV MANUAL (Capsule 13)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q10B22	8A1 P1-G	1A1P1-R
						Q7A22		1A1XK3-3
						Q85B22	1A1 J1-R	1A1TB1-MK
						Q85D22N	1A1 XK3-1	Ground
							1A1 TB-1-MM	
		Short to ground	False indication	Caution light	False indication	Q10B22	8A1 P1-G	1A1P1-R
						Q7A22		1A1XK3-3
							1A1 J1-R	

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	ENG 2 GOV MANUAL (Capsule 14)	Open circuit	Loss of particular fault indication	None	Loss of particular fault indication circuit	Q13B22	8A1 P2-G	1A1P2-P
						Q8A22		1A1XK2-3
						Q84B22	1A1 J2-P	1A1TB1-MG
						Q84D22 N	1A1 XK2-1	Ground
							1A1 TB1-MJ	
		Short to ground	False indication	Caution light	False indication	Q13B22	8A1 P2-G	1A1P2-P
						Q3A22		1A1XK2-3
							1A1 J2-P	
BLEED AIR HEATER SYSTEM								
Heater system	Power wire from cabin heater circuit breaker to heater switch and aft outlet switch.	Open circuit.	Heater inoperative.	Heater stops operating.	Heater inoperative.	W17A22	10CB9	10S8-2
						H1A22	10CB9	22P12-g
						H1B22	22J1 2-g	10S1-C
		Short to ground	Heater inoperative	Circuit breaker trips	Heater inoperative			
	Power wire from heater switch through overheat relay through emergency fire pull handle switches through particle separator relays to bleed air valves.	Open circuit	Heater inoperative	Heater stops	Heater inoperative	H2A22	10 S8-3	Splice
						H2D22		10K1P1-F
						H25A22	Splice	Splice
						H2C22	10K1 P1-B	10K1P1-D
						H25B22	Splice	Splice
						H25C22	10K1 P1-G	22P12-r
						H25D22		22P16-T
						H25E22	Splice	1A4S1-ANC
						K41A22	22J1 2-r	1A3S1-AC
						K37A22	22J1 6-T	22P17-M
						K37B22		1K5-B3
						K46A22	1A4 S1-AC	1K6-B3
						H15A22	1A3 S1-ANC	8TB2-SE
						H15B22		22P10-32
						H15D22	22J1 7-M	1J4-C
						H15G18		10L4P1-A
						H15C22	1K5-B-2	22P7-95

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						H15E22	1K6-B-2	1J3-C
						H15F18	8TB-2-SF	10L3P1-A
							22J1 0-32	
							1P4-C	
							8TB-2-SG	
							22J7-95	
							1P3-C	
			Left bleed air valve inoperative	None	Left bleed air valve inoperative	H15C22	8TB-2-SG	22P7-95
						H15E22	22J7-95	1J3-C
						H15F18	1P3-C	10L3P1-A
						H14A18	10L3 P1-D	1P3-D
						H14B22N	1J3-D	Ground
			Right bleed air valve inoperative	None	Right bleed air valve inoperative	H15B22	8TB-2-SF	22P10-32
						H15D22	22J1 0-32	1J4-C
						H15G18	1P4-C	10L4P1-A
						H3A18	10L4 P1-D	1P4-D
						H3D22N	1J4-D	Ground
	Power wire from heater switch to winterization kit control valve, variable mix valve cabin temperature controller servo and overheat relay	Short to ground	Heater inoperative	Circuit breaker trips	Heater inoperative	H27A22	10 S8-2	22P12-c
						H27B22	22J1 2-c	22P10-37
						H27C22	22J1 0-37	Splice
						H27E22	Splice	10L6P1-A
						H27D22	Splice	10L1P1-A
						H27F22	Splice	10Z1P1-G
						H2A22	Splice	Splice
						H2D22	10 S8-2	10K1P1-F
						H2C22	Splice	10K1P1-D
						H2B22		10K1P1-K

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
							Splice	
							Splice	
Heater relay	Ground wire from overheat relay to overheat switch (aft)	Open circuit	Heater will not shut off automatically in the event of an overheat condition	Caution light signal from other overheat switch if actuated	None Switch may be shut off	H10A22	10K1 P1-E	22P12-K
						H10B22		22P10-35
						H10C22	22J1 2-K	Splice
						H11A22N	22J1 0-35	10S3
							Splice	Splice
							Splice	Ground
							10S3	
							Splice	
		Short to ground	Heater inoperative	Heater shuts off and cabin heater circuit breaker trips	Heater inoperative	H10A22	10K1 P1-E	22P12-K
						H10B22		22P10-35
						H10C22	22J1 2-K	Splice
							22J1 0-35	10S3
							Splice	
Caution light	Ground wire from light to overheat switch (forward)	Open circuit	Caution light will not illuminate if overheat switch actuates	None	None Heater would shut off on signal from primary overheat switch	W112A22	8A1 P2-S	Splice
						W112B22		22P10-34
						W112C22	Splice	Splice
						H31A22N	22J1 0-34	10S2
							Splice	Splice
							10S2	Ground
							Splice	
		Short to ground	Caution light illuminates	Caution light	None Heater can be shut off by heater switch	W112A22	8A1 P2-S	Splice
						W112B22		22P10-34
						W112C22	Splice	Splice
							22J1 0-34	10S2
							Splice	
Vent blower	Power wire from circuit breaker to heat vent blower switch	Open circuit	Inability to operate vent blowers	Vent blowers will not operate	Vent blowers inoperative	H12A22	10CB12	10S5-2
		Short to ground	Heater inoperative	Circuit breaker trips	Heater inoperative	H12A22	10CB12	10S5-2

Table 2. Electrical Fault Analysis (S/N 33101 through 33107) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
Vent switch	Power wire from heat vent blower switch to right vent blower	Open circuit	Right vent blower inoperative	No airflow from right heat ducts	None	H13C22	Splice	10B4
		Open circuit	Left vent blower inoperative	No airflow from left heat ducts	None	H13D22	Splice	10B3

NOTE:

The alpha-numeric references in the Wire No. column refer to the Circuit column (alpha portion) and the Location column (numeric portion).

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
			DC POWER SYSTEM			
Battery	Power wire from battery connector to battery bus	Open circuit	No battery power to battery bus	Zero voltage reading on voltmeters	No battery power available	Wire P2A1 (2BT1P1 + to bus bar) loose, disconnected, or broken.
		Short to ground	Battery switch opens	Battery switch in OFF position	No battery power on buses	Wire P2A1 broken, touching structure or ground.
	Ground wire from battery connector to aircraft structure	Open circuit	No battery power to battery bus	Zero voltage reading on voltmeters	No battery power available	Wire P1A1N (2BT1P1 to ground) loose, disconnected, or broken.
		Short to ground	None	None	None	Wire P1A1N broken, touching structure or ground.
	Wire from terminal A1 of No. 1 BUS battery relay to terminal A1 of No. 1 generator relay	Open circuit	No battery power on No. 1 buses	None	No battery power to No. 1 buses. Engine No. 1 cannot be started	Wire P5A1 (2K7-A1 to 2K3-A1) batt bus No. 1 relay to No. 1 relay loose, disconnected, or broken.
		Short to ground	No battery power on battery bus	Zero voltage reading on voltmeters	No battery power available	Wires P5A1, P5B1 (2K3-A1 to 2K11-A1) No. 1 non-essential bus relay, P5C1 (2K11-A1 to main DC bus No. 1) and wire P5D22 (2K7-A1 to 2Z1P1-E) No. 1 DC control unit connector, broken, touching structure or ground, shorted to ground at disconnect.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
	Wire from No. 1 BUS BAT circuit breaker to terminal 3 of BATTERY BUS 1 switch	Open circuit	Battery relay No. 1 will not close	None, if BATTERY BUS No. 2 switch is closed (normal operation)	Battery cannot be connected in No. 1 DC buses by BATTERY BUS 1. Cannot start engine No. 1.	Wires P224A10 (2K7-A2 to 2CB15) EMERG BUS 1 circuit breaker, P114A10 (2CB15 to 22P1-C), P114C10 (22J1-C to 2CB17) No. 1 BUS BAT circuit breaker, P138A20 (2CB17 to 22P9-A), P138D20 (22J9-A to 22J13-D), P138D20 (22J9-A to 22J13-D), P138E20 (22P13-D to 2S3-3) BATTERY BUS 1 switch broken, loose, or disconnected. No. 1 BUS BAT circuit breaker open.
		Short to ground	NO 1 BUS BAT circuit breaker trips	NO. 1 BUS BAT circuit breaker tripped	Battery cannot be connected to No. 1 buses by BATTERY BUS 1 switch. Cannot start engine No. 1.	Wires P224A10, P114A10, P114C10, P114B10 (2CB15 to 2K17-A1), emergency bus 1 power relay, P138A20, P138D20, P138E20, P138B22 (2CB17 to 22J5-C), P138C22 (22P5-C to 2XK9-6) non-essential bus relay socket broken, touching structure, or ground.
	Wire from terminal A2 of No. 2 bus battery relay to terminal A2 of No. 2 generator relay	Open circuit	No battery power on No. 2 buses	None	No battery power on No. 2 buses. Engine No. 2 cannot be started from battery.	Wire P30A1 (2K8-A2 to 2K4-A2) No. 2 generator relay loose or disconnected.
		Short to ground	No. 2 generator trips off line	ENG 2 DC GENERA-TOR caution light	Loss of generator No. 2	Wires P30A1, P30B1 (2K4-A2 to 2K2-A2) No. 2 non-essential bus relay, P30C1 (2K2-A2 to main DC bus No. 2) and P30D22 (2K8-A2 to 2Z2P1-E) No. 2 DC control unit broken, touching structure or ground, or shorted to ground at disconnect.
	Wire from No. 2 BUS BAT circuit breaker to terminal 3 of BATTERY BUS 2 switch	Open circuit	Battery relay No. 2 will not close	BATTERY caution light on unless battery bus 1 switch is closed	Battery cannot be connected to No. 2 DC buses by BATTERY BUS 2 switch. Cannot start No. 2 engine from battery.	Wires P118C10 (2CB16, No. 2 BUS BAT circuit battery to 22J2-C), P118A10 (22P2-C to 2CB14) EMERG BUS 2 circuit breaker, P139A20 (2CB16 to 22J12-Z), P139B20 (2CB16 to 22J12-Z),P139B20 (22P12-Z to 2S4-3) BATTERY BUS 2 switch broken, loose, or disconnected. NO. 2 BUS BAT circuit breaker open.
		Short to ground	NO. 2 BUS BAT circuit breaker trips	NO. 2 BUS BAT circuit breaker tripped	Battery cannot be connected to No. 2 DC buses by battery bus 2 switch. Cannot start No. 2 engine from battery.	Wires P118C10, P118A10, P139A20, P139B20, P118B10 (2CB14 to 2K16-A1) emergency bus 2 power relay broken, touching structure or ground.
Generator No. 1	Terminal A	Open circuit	No generator No. 1 output	ENG 1 DC GENERA-TOR caution light	Generator No. 1 fails to supply power	Wires P54A16 (2G1-A to 1P1-C), P54B16 (1J1-C to 22P13-N), P54C16 (22J13-N to 22J3-C), P54D16 (22P3-C to 1K1-C1) starter control 1 relay, P54E16 (1K1-C1 to 2Z1P1-B) No. 1 DC control unit broken, loose, or disconnected.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
		Short to ground	Generator No. 1 drops off line	ENG 1 DC GENERA-TOR caution light	Generator No. 1 fails to supply power	Wires P54A16, P54B16, P54C16, P54D16, and P54E16 broken, touching structure or ground, shorted to ground at disconnect, or grounded with shield.
	Terminal B	Open circuit	No generator No. 1 output	ENG 1 DC GENERA-TOR caution light	Loss of power from generator No. 1	Wires P72A1 (2G1-B to 2P3-B), P72B1 (2J3-B to 2K5-B2) No. 1 starter relay, P71A1 (2K5-A1 to 2K3-A2) No. 1 starter-generator relay, P71B16 (2K3-A2 to 22P13-F), P71D16 (22J13-F to 22J3-A), P71E16 (22P3-A to 2Z1P1-D) No. 1 DC control unit connector loose, disconnected, or broken. Bus bar at 2K5 missing.
		Short to ground	Generator drops off line	ENG 1 DC GENERA-TOR caution light	Loss of power from generator No. 1	Wires P72A1, P72B1, P71A1, P71B16, P71D16, P71E16, P71C16 (2K3-A2 to 2CB9) GEN 1 FIELD CB, P129A16 (2CB9 to 22P13-K), P129B16 (22J13-K to 22J3-E), P129C16 (22P3-E to 1K1-D3) starter control 1 relay, P38B16 (1K1-D2 to 2Z1P1-A), P38A22 (1K1-D2 to 2Z1P1-S) broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
	Terminal C	Open circuit	Starter No. 1 inoperative	None	Starter inoperative	Wires P95A1 (2G1-C to 2P3-D), P95B1 (2J3-D to 2K5-A2) No. 1 starter relay, P71A1 (2K5-A1 to 2K3-A2) No. 1 starter-generator relay, P71B16 (2K3-A2 to 22P13-F), P71D16 (22J13-F to 22J3-A), P71E16 (22P3-A to 2Z1P1-D) No. 1 DC control unit connector loose, disconnected, or broken. 2K5 contacts A1 to A2 bent or broken.
		Short to ground	Generator drops off line	ENG 1 DC GENERA-TOR caution light	Loss of power from generator No. 1	Wires P95A1, P95B1, P71A1, P71B16, P71D16, P71E16, P71C16 (2K3-A2 to 2CB9) GEN 1 FIELD CB, P129A16 (2CB9 to 22P13-K), P129B16 (22P3-E to 1K1-D3) starter control 1 relay broken, touching structure or ground.
	Terminal D	Open circuit	Generator paralleling circuitry inoperative	None	Generators not paralleled	Wires P51A16 (2G1-D to 1P1-A), P51B22 (1J1-A to 22P15-1), P51C22 (22J15-1 to 22J5-F), P51D22 (22P5-F to 1K1-C2) starter control 1 relay, P51E22 (1K1-C2 to 2Z1P1-G) No. 1 DC control unit connector loose, disconnected, or broken.
		Short to ground	No immediate discernible effect, but paralleling circuit is affected	None	Generator paralleling affected	Wires P51A16, P51B22, P51C22, P51D22, P51E22 broken, touching structure or ground, or shorted to ground at disconnects.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
	Terminal E	Open circuit	Generator No. 1 inoperative	ENG 1 DC GENERA-TOR caution light	Loss of power from generator No. 1	Wires P73A1 (2G1-E to 2P3-F), P73B1 (2J3-F to 2R1) No. 1 shunt, P74A1N (No. 1 shunt to ground) loose, disconnected or broken. No. 1 shunt burnt, disconnected, or broken.
		Short to ground	No effect	None	None	Wires P73A1, P73B1 broken, touching structure or ground. Shorted to ground at disconnects or grounded with shield.
Genera-torNo. 2	Terminal A	Open circuit	No generator No. 2 output	ENG 2 DC GENERA-TOR caution light	Generator No. 2 fails to supply power	Wires P75A16 (2G2-A to 1P2-C), P57B16 (1J2-C to 22J4-C), P57C16 (22P4-C to 1K2-C1) starter control 2 relay, P57D16 (1K2-C1 to 2Z2P1-B) No. 2 DC control unit loose, disconnected, or broken.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC GENERA-TOR caution light	Generator No. 2 fails to supply power	Wires P57A16, P57B16, P57C16, P57D16 broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
	Terminal B	Open circuit	No generator No. 2 output	ENG 2 DC GENERA-TOR caution light	Loss of power from generator No. 2	Wires P85A1 (2G2-B to 2P4-B), P85B1 (2J4-B to 2K6-B2) No. 2 starter relay, P84A1 (2K6-A1 to 2K4-A1) No. 2 starter-generator relay, P91A22N (2K4-X2 to ground), P90B22 (2K4-X2 to ground), P90B22 (2K4-X1 to 22J6-G), P90A22 (22P6-G to 1K2-A2) starter control 2 relay, P22A22 (1K2-A3 to 22P6-F), P22B22 (22J6-F to 2K4-11), P22C22 (2K4-11 to 2K6-12), P88A22 (2K6-13 to 22J6-N), P88B22 (22P6-N to 2Z2P1-H) No. 2 DC control unit connector loose, disconnected, or broken. Bus bar at 2K6 missing.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC GENERA-TOR caution light	Loss of power from generator No. 2	Wires P85A1, P85B1, P84A1, P30B1 (2K4-A2 to 2K2-A2) non-essential bus 2 relay, P30C1 (2K2-A2 to No. 2 main DC bus), P84C16 (2K4-A1 to 2CB10) GEN 2 FIELD CB, P84B16 (2K4-A1 to 22J4-A), P84D16 (22P4-A to 2Z2P1-D) broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
	Terminal C	Open circuit	Starter No. 2 circuit inoperative	None	Starter No. 2 inoperative	Wire P98A1 (2G2-C to 2P4-D), P98B1 (2J4-D to 2K6-A2) loose or disconnected.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC GENERA-TOR caution light	Loss of power from generator No. 2	Wires P98A1, P98B1 broken, touching structure or ground.
	Terminal D	Open circuit	Generator paralleling circuitry inoperative	None	Generators not paralleled	Wires P35A16 (2G2-D to 1P2-A), P35B22 (1J2-A to 22J6-E), P35C22 (22P6-E to 1K2-C2) starter control 2 relay, P35D22 (1K2-C2 to 2Z2P1-G) No. 2 DC control unit connector loose, disconnected, or broken.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
		Short to ground	No immediate discernible effect, but paralleling circuit is affected	None	Generator paralleling affected	Wires P35A16, P35B22, P35C22, P35D22 broken, touching structure or ground, or shorted to ground at disconnects.
	Terminal E	Open circuit	Generator No. 2 inoperative	ENG 2 DC GENERA-TOR caution light	Loss of power from generator No. 2	Wires P86A1 (2G2-E to 2P4-F), P86B1 (2J4-F to 2R2) No. 2 shunt, P87A1N (2R2 to ground) loose, disconnected, or broken. No. 2 shunt burnt, disconnected, or broken.
		Short to ground	No effect	None	None	Wires P86A1, P86B1 broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
No. 1 Main DC bus	Feeder wire at bus	Open circuit	No 28 VDC power on No. 1 main DC bus	No voltage indicated on voltmeter	No 28 VDC power supplied to buses by generator No. 1	Wires P5A1 (2K7-A1 to 2K3-A1) No. 1 starter-generator relay, P5B1 (2K3-A1 to 2K11-A1) No. 1 non-essential bus relay, P5C1 (2K11-A1 to No. 1 main DC bus) loose, disconnected, or broken. 2K7 contracts A1 or A2 broken.
		Short to ground	Generator No. 1 drops off line	ENG 1 DC GENERA-TOR caution light	Loss of power from generator No. 1	Wires P5A1, P5B1, P5C1, P5D22 (2K7-A1 to 2Z1P1-E) No. 1 DC control unit connector broken, touching structure or ground.
No. 2 Main DC bus	Feeder wire at bus	Open circuit	No 28 VDC power on No. 2 main DC bus	No voltage indicated on voltmeter	No 28 VDC power supplies to buses by generator No. 2	Wires P30A1 (2K1-A1 to 2K4-A2) No. 2 starter-generator relay, P30B1 (2K4-A2 to 2K2-A2) non-essential bus 2 relay, P30C1 (2K2-A2 to No. 2 main DC bus) loose, disconnected, or broken.
		Short to ground	Generator No. 2 drops off line	ENG 2 DC GENERA-TOR caution light	Loss of power from generator No. 2	Wires P30A1, P30B1, P30C1, P30D22 (2K8-A2 to 2Z2P1-E) No. 2 DC control unit connector broken, touching structure to ground.
No. 1 Essential DC bus		Short to ground	No. 1 feeder bus circuit breakers trip (after 1 or 2 seconds when BATTERY BUS 2 switch is ON, BATTERY BUS 1 switch is OFF; they trip immediately when BATTERY BUS 2 switch is OFF and BATTERY BUS 1 switch is ON).	No. 1 feeder circuit breakers trips	No power on No. 1 essential bus from generator No. 1. Generator No. 2 supplies power to No. 1 essential bus. Heavy bus loads may also cause No. 2 feeder circuit breakers to trip. Inverters are not affected.	Wires P247A16 (No. 1 essential DC bus to 10CB11) WINDSHIELD WIPER CPLT circuit breaker, P247B16 (10CB11 to No. 1 essential DC bus) broken, touching structure or ground.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
		Short to ground	No. 2 feeder bus circuit breakers trip (Immediately if BATTERY BUS 2 switch is ON and BATTERY BUS 1 switch is OFF; after several seconds if BATTERY BUS 1 switch is ON and BATTERY BUS 2 switch is OFF).	No. 2 feeder circuit breakers trip	No power on No. 1 essential bus from generator No. 2. Generator No. 2 supplies power to No. 1 essential bus. Heavy bus loads may cause No. 1 feeder circuit breakers to trip.	Wires P15A10 (2CB2 to No. 1 essential DC bus), P16A10 (2CB4 to No. 1 essential DC bus) broken, touching structure or ground, or diodes shorted to ground.
No. 2 Essential DC bus		Short to ground	No. 2 feeder bus circuit breakers trip. (Immediately when BATTERY BUS 2 switch is ON; after several seconds when BATTERY BUS 1 switch is ON and BATTERY BUS 2 switch is OFF).	No. 2 feeder circuit breaker tripped	No power on No. 2 essential bus from generator No. 2. Generator No. 1 supplies power to No. 2 essential bus. Heavy bus loads may cause No. 1 feeder breakers to trip. Inverters are not affected.	Wires P241A16 (No. 2 essential DC bus to 10CB10) WINDSHIELD WIPER PILOT circuit breaker, P241B16 (10CB10 to 9CB8) FLT DIR circuit breaker broken, touching structure or ground, or diodes shorted to ground.
Non-essential DC bus	No. 1 non-essential bus feeder wire at 2TB2	Open circuit	No DC power on non-essential bus from generator No. 1	None	Non-essential bus receives no power from generator No. 1. Generator No. 2 continues to supply power to non-essential bus. Heavy load on non-essential bus will cause No.2 overload sensor to trip and remove all power from non-essential bus.	Wires P174A6 (2TB2 to diodes), P148A6 (Diodes to 2S5-L2) non-essential bus 1 sensor, P147A6 (2S5-L1 to 2K11-A2) non-essential bus relay loose, disconnected, or broken. Non-essential bus 1 sensor loose or disconnected, diodes open, non-essential bus 1 sensor open or not energized.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
	No. 2 non-essential bus feeder wire at 2TB2	Open circuit	No DC power on non-essential bus from generator No. 2	None	Non-essential bus receives no power from generator No. 2. Generator No. 1 generator continues to supply power to non-essential bus. Heavy load on non-essential bus will cause No. 1 overload relay to trip and remove all power from non-essential bus.	Wires P175A6 (2TB2 to diodes), P11A6 (diodes to 2S6-L2) non-essential bus relay loo se, disconnected, or broken, Non-essential bus 2 sensor loose or disconnected, diodes open, non-essential bus 2 sensor open or not energized.
	No. 1 and No. 2 non-essential bus feeder at 2TB2.	Short to ground	Non-essential bus trips off	Non-essential equipment inoperative	No power on non-essential bus	Wires P174A6, P175A6 broken, touching structure or ground.
			AC POWER SYSTEM			
Inverter No. 1	Pins A and B	Open circuit	Inverter No. 1 inoperative. Inverter No. 2 automatically supplies all AC buses.	NO. 1 INVERTER caution light	None Inverter No. 2 supplies all AC buses	Wires X6A16N (3PS1P1-B to ground) V5A12N (3PS1P1-A to ground) loose, disconnected, or broken.
	Pin C	Open circuit	Inverter No. 1 inoperative. Inverter No. 2 automatically supplies all AC buses.	No. 1 INVERTER caution light	None Inverter No. 2 supplies all AC buses	Wires V4A22 (3PS1P1-C to 22P5-C), V4B22 (22P15-17 to 3S1-2) INV 1 switch, V7A22N (3S1-3 to ground) loose, disconnected, or broken. Contacts 2 and 3 of 3S1 open.
		Short to ground	Inverter will not turn off with INV 1 switch 3S1	None	None	Wires V4A22, V4B22, V4C22, or 3S1 shorted to ground.
	Pin D	Open circuit	No 28 VDC power to inverter No. 1	No. 1 INVERTER caution light	None Inverter No. 2 will supply all AC buses	Wires V1C12 (3PS1P1-D to 22P1-A), V1B12 (22J1-A to 22J11-A), V1A12 (22P11-A to 3CB7) INV 1 PWR circuit breaker loose, disconnected, or broken. Circuit breaker 3CB7 inoperative or wire P230H12 loose, disconnected, or broken.
		Short to ground	No. 1 inverter inoperative. INV 1 PWR circuit breaker opens	INV 1 PWR circuit breaker (3CB7) open. No. 1 INVERTER caution light	None Inverter No. 2 can supply all AC buses	Wires V1C12, V1B12, V1A12 broken, touching structure or ground.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
	Pin E	Open circuit	26 VAC not supplied to 26 VAC buses No. 1 and No. 2 by inverter No. 1	Equipment requiring 26 VAC power becomes inoperative. Instrument flags come into view	No 26 VAC power is available on 26 VAC buses No. 1 and No. 2	Wires X4A18 (3PS1P1-E to 22P3-L), X4B18 (22J3-L to 22J13-S), X4C18 (22P13-S to 3K1P1-F) AC voltage sensor No. 1 relay connector, loose, disconnected, or broken.
		Short to ground	Inverter No. 1 drops off line	No. 1 INVERTER caution light. No. 2 buses	None Inverter No. 2 can supply power to No. 1 and No. 2 buses	Wires X4A18, X4B18, X4C18, contact 3K1-F, broken, touching structure or ground.
	Pin G	Open circuit	115 VAC not supplied to 115 VAC buses No. 1 and No. 2	No. 1 INVERTER caution light	None Inverter No. 2 can supply No. 1, No. 2 and No. 3 115 VAC buses	Wires X2A18 (3PS1P1-G to 22P5-H), X2B18 (22J5-H to 22J13-R), X2C18 (22P13-R to 2TB1-MB) emergency bus interlock TB, X2D18 (2TB1-MD to 3K1P1-C) AC voltage sensor No. 1 relay connector, X2E18 (2TB1-MF to 3K1P1-A), X27A22N (3K1P1-B to ground), loose, dis-connected, or broken. Relay 3K1 contacts C and D open.
		Short to ground	Inverter No. 1 drops off line immediately. Inverter No. 2 automatically switched to supply all AC buses.	No. 1 INVERTER caution light	None Inverter No. 2 supplies power to all AC buses	Wire X2A18, X2B18, X2C18, X2D18 X2E18 broken, touching structure or ground.
Inverter No. 2	Pins A and B	Open circuit	Inverter No. 2 inoperative. Inverter No. 1 automatically supplies bus No. 3, if required.	No. 2 INVERTER caution light	None Inverter No. 1 supplies all AC buses	Wires V6A12N (3PS2P1-A to ground), X5A16 (3PSP1-B to ground) loose, disconnected, or broken.
	Pin C	Open circuit	Inverter No. 2 inoperative. Inverter No. 1 automatically supplies all AC buses.	No. 2 INVERTER caution light	None Inverter No. 1 can supply all AC buses	Wires V9A22 (3PS2P1-C to 22P6-H), V9B22 (22J6-H to 3XK4-2) inverter 2 interlock relay socket, V3A22 (3XK4-A3 to 3S2-3 to ground) loose, disconnected, or broken. Contacts 2 and 3 of 3S2 open.
		Short to ground	Inverter will not turn off with INV 2 switch 3S2.	None	None	Wires V9A22, V9B22, V3A22, or contacts 2 and 3 of 3S2 shorted to ground.

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE
	Pin D	Open circuit	No 28 VDC power to inverter No. 2	No. 2 INVERTER caution light	None Inverter No. 1 will supply all AC buses	Wires V2B12 (3PS2P1-D to 22P2-A), V2A12 (22J2-A to 3CB8) INV 2 PWR circuit breaker loose, disconnected, or broken. Circuit breaker 3CB8 inoperative or wire P232C12 loose, disconnected, or broken.
		Short to ground	Inverter No. 2 inoperative. INV 2 PWR circuit breaker opens.	INV 2 PWR circuit breaker (3CB8) open. No. 2 INVERTER caution light.	None Inverter No. 1 will supply all AC buses	Wires V2B12, V2A12 broken, touching structure or ground.
	Pin E	Open circuit	26 VAC not supplied to 26 VAC Bus 3 by inverter No. 2	Equipment requiring power from 26 VAC bus 3 inoperative	No 26 VAC power is available on 26 VAC bus 3 (AFCS)	Wires X3A18 (3PS2P1-E to 22P4-G), X3B18 (22J4-G to 2TB1-MA) emergency bus interlock TB, X3C18 (2TB1-MC to 3K2P1-F) AC voltage sensor No. 1 relay connector, X10A18 (3K2P1-G to 3CB10) AC FEEDERS BUS 3 26V circuit breaker loose, disconnected, or broken.
		Short to ground	Inverter No. 2 drops off line	No. 2 INVERTER caution light	None Inverter No. 1 can supply power to 26 VAC	Wires X3A18, X3B18, X3C18, X10A18 broken, touching structure or ground.
	Pin G	Open circuit	115 VAC not supplied to 115 VAC buses No. 1 and No. 2	No. 2 inverter caution light	None Inverter No. 1 can supply power to 115 VAC buses No. 1 and No. 2	Wires X1A18 (3PS2P1-G to 22P4-H), X1B18 (22J4-H to 2TB1-LA) emergency bus interlock TB, X1C18 (2TB1-LD to 3K2P1-A) AC voltage sensor No. 2 relay connector, X1D18 (2TB1-LC to 3K2P1-C), X1E18 (2TB1-LB to 3K1P1-E) AC voltage sensor No. 1 relay connector. X8A18 (3K2P1-D to 3CB9) AC FEEDERS BUS 3 115V circuit breaker X7A18 (3K1P1-D to 3CB18) AC FEEDERS EMERG BUS 115 V circuit breaker loose, disconnected, or broken.
		Short to ground	Inverter No. 2 drops off line immediately. Inverter No. 1 automatically switched to supply all AC buses.	No. 2 INVERTER caution light	None Inverter No. 1 supplies power to all AC buses	Wires X1A18, X1B18, X1C18, X1D18, X1E18, X8A18, X7A18, broken, touching structure or ground.

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
						WIRE NO.	FROM	TO
EXTERIOR LIGHTING								
Searchlight	Power wire from circuit breaker to searchlight	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L77A12	8CB24 8CB24	8DS12-3 Inoperative

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
		Short to ground	Circuit breaker trips	Circuit breaker trips	Searchlight inoperative	L77A12	8CB24	8DS12
	Ground wire (searchlight to ground)	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L106A12N	8DS12	Ground
	Control wire	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L73A22	8CB26	22P14-79
						L73B22	22J14-79	S21-2
							S21-2	S25-G
							8CB26	Inoperative
		Short to ground	Searchlight inoperative	Circuit breaker trips	Searchlight inoperative	L73A22	8CB26	22P14-79
						L73B22	22J14-79	S21-2
							S21-2	S25-G
	Left rotation circuit	Open circuit	Loss of left control	None	Loss of left control	L64B22	S25-F	22P10-10
						L64C22	22J10-10	8DS12-5
	Right rotation circuit		Loss of right control		Loss of right control	L63B22	S25-D	22P10-9
						L63C22	22J10-9	8DS12-4
	Left rotation circuit	Short to ground	Control breaker trips	Circuit breaker trips	Searchlight inoperative	L64B22	S25-F	22P10-10
						22J10-10	8DS12-5	
	Right rotation circuit					L63B22	S25-D	22P10-9
						L63C22	22J10-9	8DS12-4
	Extend circuit	Open circuit	Loss of extend control	None	Loss of extend control	L65B22	S25-C	22P10-11
						L65C22	22J10-11	8DS12-6
	Retract circuit		Loss of retract control	Circuit breaker trips	Loss of retract control	L76C22	S25-E	22P10-12
						L76D22	22J10-12	8DS12-7
Landing light	Power wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L69A12	8CB25	22P11-E
						L69B12	22J11-E	8DS13 P1-H
							8CB25	Inoperative
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L69A12	8CB25	22P11-E
						L69B12	22J11-E	8DS13 P1-H
	Ground wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L70A12N	8DS13 P1-I	Ground
	Control wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L78A22	8CB27	22P15-95
						L78B22	22J15-95	22J7-36
						L78C22	22P7-36	S22-2
							S20-2	S22-2

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L105B22	S22-3	22P7-35
						L105C22	22J7-35	8DS13 P1-C
	Extend wire	Open circuit	Unable to extend	None	Loss of extend control	L82B22	S20-3	22P7-33
						L82C22	22J7-33	8DS13 P1-A
		Short to ground	Unable to retract	Circuit breaker trips	Loss of retract control	L83B22	S20-1	22P7-34
						L83C22	22J7-34	8DS12 P1-B
Lower anti-collision lights	Power wire	Open circuit	Lower anti-collision light inoperative	None	Lower anti-collision light inoperative	L11A20	8S22-3	22P13-D
						L11B20	22J13-D	Splice
						L11C20	Splice	8DS3P1-A
						L11D20	Splice	8DS3P1-B
						L11F20	Splice	8P6-J
	Ground wire					L12A20N	8DS3 P1-C	Ground
Upper anti-collision lights	Power wire		Upper anti-collision light inoperative	None	Upper anti-collision light inoperative	L8A20	8S22-3	1J1-E
						L8B20	1P1-E	8J5-A
						L8C20	8P5-A	Splice
						L8D20	Splice	8DS8P1-B
						L8E20	Splice	8DSP1-A
	Ground wire					L9A20	8DS8 P1-C	8P5-B
						L9B20	8J5-B	1P1-B
						L9C20N	1J1-F	Ground
Position lights	Power wire	Open circuit	Light/lights inoperative	None	Light/lights inoperative	L1A20	8CB23	8S21-2
						L5A20	8S21-3	8TB4-6
			Lower left position light inoperative		Lower left position light inoperative	L5E20	8TB4-6	22P13-C
						L5J20	22J13-C	8DS17
			Upper left position light inoperative		Upper left position light inoperative	L2C20	8TB4-5	8DS15
			Upper right position light inoperative		Upper right position light inoperative	L2B20	8TB4-5	8DS14
			Lower right position light inoperative		Lower right position light inoperative	L5B20	8TB4-6	22P12-B
						L5C20	22J12-B	22P8-T
						L5D20	22J8-T	8DS16

Table 3. Electrical Fault Analysis (33108 through 33213 and 36001 through 36019) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
			Right tail position light inoperative		Right tail position light inoperative	L5F20	8TB4-6	22P13-B
						L5G20	22J13-B	8P6-A
						RED	Right	
			Left tail position light inoperative		Left tail position light inoperative	L2D20	8TB4-5	22P13-A
						L2E20	22J13-A	8P6-C
						RED	Left	
	Ground wire		Left tail position light inoperative		Left tail position light inoperative	L3B20N	8P6-D	Ground
						BLU	Left	
			Right tail position light inoperative		Right tail position light inoperative	L6B20N	8P6-B	Ground
						BLU	Right	
LH and RH utility lights	Power wire-circuit breaker to switch	Open circuit	Utility lights inoperative	None	Utility lights inoperative	L124A22	8CB18	8S23-2
						L125A22	8S23-2	8S23-3
						L125B22	8S23-3	22P15-93
							22J15-93	Splice
L Utility light			L utility light inoperative		L Utility light inoperative	L125C22	Splice	8DS29
R Utility light			Right utility light inoperative		Right utility light inoperative	L125D22	Splice	22J7-8
						L125E22	22P7-8	22P10-4
						L125F22	22J10-4	8DS30
R or L Utility light	Ground wire	Open circuit	L or R Utility light inoperative	None	L or R Utility light inoperative	Chassis	Ground	
INTERIOR LIGHTING								
5-VOLT INSTRUMENT LIGHTS								
a) Pilot instrument lights(1) Wire from circuit breaker to control switchOpen circuitLoss of instrument lightsLoss of instrument lightsLoss of instrument lights(a1)L47A22(b1)L54A22(c1)L42A228CB148CB158CB178R1-28R2-28R3-2								

CHAPTER 97
AVIONICS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Avionics Systems - Description	DMC-412-A-97-00-00-00A-028A-A / 02236	2023-05-31	ALL
Airframe Components - Description	DMC-412-A-97-00-00-00A-042A-A / 02237	2023-05-31	ALL
Energizing Power Buses - Functional Tests	DMC-412-A-97-00-00-00A-340A-A / 02238	2023-05-31	SN: 33001-33107
Energizing Power Buses - Functional Tests	DMC-412-A-97-00-00-00A-340B-A / 02239	2023-05-31	SN: 33108-Subsequent
Avionics Systems - Illustrations - Description	DMC-412-A-97-00-00-00A-042B-A / 02240	2023-05-31	ALL
Avionics System Wiring Diagrams - Description	DMC-412-A-97-29-00-00A-028A-A / 02453	2023-05-31	ALL
Intercommunication System (ICS) - Description	DMC-412-A-97-01-00-00A-028A-A / 02241	2023-05-31	ALL
Intercom Control Unit - Switch Functions - Description	DMC-412-A-97-01-00-00A-042A-A / 02242	2023-05-31	SN: 33001-36637
Intercom Control Unit - Switch Functions - Description	DMC-412-A-97-01-00-00A-042B-A / 02243	2023-05-31	SN: 36638-Subsequent
Intercom System - Operational Check	DMC-412-A-97-01-00-00A-320A-A / 02244	2023-05-31	SN: 33001-36637
Intercom System - Preliminary Check Procedures - Operational Check	DMC-412-A-97-01-00-01A-320A-A / 02245	2023-05-31	SN: 33001-36637
Intercom System - Pilot and Copilot Stations - Operational Check	DMC-412-A-97-01-00-02A-320A-A / 02246	2023-05-31	SN: 33001-36637
Intercom System - Aft ICS Stations (When Installed) - Operational Check	DMC-412-A-97-01-00-03A-320A-A / 02247	2023-05-31	SN: 33001-33107
Intercom System - Aft ICS Stations (When Installed) - Operational Check	DMC-412-A-97-01-00-04A-320A-A / 02248	2023-05-31	SN: 33108-36637
Intercom System - Receiver Mixing Switch Selections - Operational Check	DMC-412-A-97-01-00-05A-320A-A / 02249	2023-05-31	SN: 33001-36637
Intercom System - Receiver/Transmitter Selections - Operational Check	DMC-412-A-97-01-00-06A-320A-A / 02250	2023-05-31	SN: 33001-36637
Intercom System - Emergency Keying/Comm - Operational Check	DMC-412-A-97-01-00-07A-320A-A / 02251	2023-05-31	SN: 33001-36637
Intercom System - Post Operational Check Procedures	DMC-412-A-97-01-00-08A-320A-A / 02252	2023-05-31	SN: 33001-36637
Intercom System - Operational Check	DMC-412-A-97-01-01-00A-320A-A / 02253	2023-05-31	SN: 36638-Subsequent
Intercom System - Preliminary Check Procedures - Operational Check	DMC-412-A-97-01-01-01A-320A-A / 02254	2023-05-31	SN: 36638-Subsequent
Crew Intercommunication System - Operational Check	DMC-412-A-97-01-01-02A-320A-A / 02255	2023-05-31	SN: 36638-Subsequent
Crew VOX Sensitivity - Operational Check	DMC-412-A-97-01-01-03A-320A-A / 02256	2023-05-31	SN: 36638-Subsequent
COM1 - Operational Check	DMC-412-A-97-01-01-04A-320A-A / 02257	2023-05-31	SN: 36638-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
COM2 - Operational Check	DMC-412-A-97-01-01-05A-320A-A / 02258	2023-05-31	SN: 36638-Subsequent
Emergency Communication Function - Operational Check	DMC-412-A-97-01-01-06A-320A-A / 02259	2023-05-31	SN: 36638-Subsequent
Navigation Systems Audio - Operational Check	DMC-412-A-97-01-01-07A-320A-A / 02260	2023-05-31	SN: 36638-Subsequent
Radar Altimeter Voice Warning - Operational Check	DMC-412-A-97-01-01-08A-320A-A / 02261	2023-05-31	SN: 36638-Subsequent
Crew/Cabin Intercommunication System - Operational Check	DMC-412-A-97-01-01-09A-320A-A / 02262	2023-05-31	SN: 36638-Subsequent
Aft Cabin VOX Sensitivity - Operational Check	DMC-412-A-97-01-01-10A-320A-A / 02263	2023-05-31	SN: 36638-Subsequent
PVT (Private) Net - Operational Check	DMC-412-A-97-01-01-11A-320A-A / 02264	2024-10-16	SN: 36638-Subsequent
Intercom System - Completion of Ground Based - Operational Checks	DMC-412-A-97-01-01-12A-320A-A / 02265	2023-05-31	SN: 36638-Subsequent
Pre-flight - RPM Audio Warning - Operational Check	DMC-412-A-97-01-01-13A-320A-A / 02266	2023-05-31	SN: 36638-Subsequent
RAD ALT Voice Warning - Operational Flight Check	DMC-412-A-97-01-01-14A-320A-A / 02267	2023-05-31	SN: 36638-Subsequent
Intercommunication System - Troubleshooting	DMC-412-A-97-01-00-00A-422A-A / 02268	2023-05-31	ALL
Intercom Control Unit - Removal	DMC-412-A-97-01-00-01A-520A-A / 02269	2023-05-31	ALL
Foot Switches - Removal	DMC-412-A-97-01-00-02A-520A-A / 02270	2023-05-31	ALL
Cyclic Stick Switches - Removal	DMC-412-A-97-01-00-03A-520A-A / 02271	2023-05-31	ALL
Emergency Keying/Comm Panel - Removal	DMC-412-A-97-01-00-04A-520A-A / 02272	2023-05-31	SN: 33001-36637
Intercom Control Unit - Installation	DMC-412-A-97-01-00-01A-720A-A / 02273	2023-05-31	ALL
Foot Switches - Installation	DMC-412-A-97-01-00-02A-720A-A / 02274	2023-05-31	ALL
Cyclic Stick Switches - Installation	DMC-412-A-97-01-00-03A-720A-A / 02275	2023-05-31	ALL
Emergency Keying/Comm Panel - Installation	DMC-412-A-97-01-00-04A-720A-A / 02276	2023-05-31	SN: 33001-36637
Intercom System - Inspection Procedures	DMC-412-A-97-01-00-00A-280A-A / 02277	2023-05-31	ALL
Intercom System - Cleaning Procedures	DMC-412-A-97-01-00-00A-251A-A / 02278	2023-05-31	ALL
Intercom System - Repair or Replacement	DMC-412-A-97-01-00-00A-685A-A / 02279	2023-05-31	ALL
VHF/AM Communication System - Description	DMC-412-A-97-20-00-00A-028A-A / 02280	2023-05-31	ALL
VHF/AM Communication System - Functional Test	DMC-412-A-97-20-00-00A-340A-A / 02281	2023-05-31	ALL
VHF/AM Communication System - Troubleshooting	DMC-412-A-97-20-00-00A-422A-A / 02282	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
VHF COMM Transceiver (No. 1 or No. 2) - Removal	DMC-412-A-97-20-00-01A-520A-A / 02283	2023-05-31	ALL
VHF COMM Control Unit (No. 1 or No. 2) - Removal	DMC-412-A-97-20-00-02A-520A-A / 02284	2023-05-31	SN: 33001-33100
VHF COMM Control Unit (No. 1 or No. 2) - Removal	DMC-412-A-97-20-00-03A-520A-A / 02285	2023-05-31	SN: 33101-Subsequent
VHF Communication Antenna No. 1 - Removal	DMC-412-A-97-20-00-04A-520A-A / 02286	2023-05-31	ALL
VHF Communication Antenna No. 2 - Removal	DMC-412-A-97-20-00-05A-520A-A / 02287	2023-05-31	ALL
VHF COMM Transceiver (No. 1 and No. 2) - Installation	DMC-412-A-97-20-00-01A-720A-A / 02288	2023-05-31	ALL
VHF COMM Control Unit (No. 1 and No. 2) - Installation	DMC-412-A-97-20-00-02A-720A-A / 02289	2023-05-31	SN: 33001-33100
VHF COMM Control Unit (No. 1 and No. 2) - Installation	DMC-412-A-97-20-00-03A-720A-A / 02290	2023-05-31	SN: 33101-Subsequent
VHF Communication Antenna No. 1 - Installation	DMC-412-A-97-20-00-04A-720A-A / 02291	2023-05-31	ALL
VHF Communication Antenna No. 2 - Installation	DMC-412-A-97-20-00-05A-720A-A / 02292	2023-05-31	ALL
VHF/AM Communication System - Inspection Procedures	DMC-412-A-97-20-00-00A-280A-A / 02293	2023-05-31	ALL
VHF/AM Communication System - Cleaning Procedures	DMC-412-A-97-20-00-00A-251A-A / 02294	2023-05-31	ALL
VHF/AM Communication System - Repair or Replacement	DMC-412-A-97-20-00-00A-685A-A / 02295	2023-05-31	ALL
Gyromagnetic Compass System(s) - Description	DMC-412-A-97-21-00-00A-028A-A / 02296	2023-05-31	SN: 33001-33100
Gyromagnetic Compass System(s) - Description	DMC-412-A-97-21-00-00A-028B-A / 02297	2023-05-31	SN: 33101-36247
Gyromagnetic Compass System(s) - Functional Test	DMC-412-A-97-21-00-00A-340A-A / 02298	2023-05-31	SN: 33001-36247
Gyromagnetic Compass System(s) - Preliminary Operational Test	DMC-412-A-97-21-00-00A-320A-A / 02299	2023-05-31	SN: 33001-36247
Compass Swing - Initial Setup - Operational Test	DMC-412-A-97-21-00-01A-320A-A / 02300	2023-05-31	SN: 33001-36247
Compass Swing - Index Error Check - Operational Test	DMC-412-A-97-21-00-01A-320B-A / 02301	2023-05-31	SN: 33001-36247
Compass Swing - One-Cycle Error Check - Operational Test	DMC-412-A-97-21-00-01A-320C-A / 02302	2023-05-31	SN: 33001-36247
Compass Swing - Final Heading Check - Operational Test	DMC-412-A-97-21-00-01A-320D-A / 02303	2023-05-31	SN: 33001-36247
Standby Magnetic Compass - Swing Procedure	DMC-412-A-97-21-00-00A-913A-A / 02304	2023-05-31	ALL
Gyromagnetic Compass System - Troubleshooting	DMC-412-A-97-21-00-00A-422A-A / 02305	2023-05-31	ALL
3-Axis Reference Sensor - Removal	DMC-412-A-97-21-00-01A-520A-A / 02306	2023-05-31	ALL
Transmitter Flux Valve - Removal	DMC-412-A-97-21-00-02A-520A-A / 02307	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Dual Remote Compass Compensator - Removal	DMC-412-A-97-21-00-03A-520A-A / 02308	2023-05-31	ALL
Compass Control Panel - Removal	DMC-412-A-97-21-00-04A-520A-A / 02309	2023-05-31	ALL
RMI Adapter - Removal	DMC-412-A-97-21-00-05A-520A-A / 02310	2023-05-31	SN: 33001-33100
3-Axis Reference Sensor - Installation	DMC-412-A-97-21-00-01A-720A-A / 02311	2023-05-31	ALL
Transmitter Flux Valve - Installation	DMC-412-A-97-21-00-02A-720A-A / 02312	2023-05-31	ALL
Dual Remote Compass Compensator - Installation	DMC-412-A-97-21-00-03A-720A-A / 02313	2023-05-31	ALL
Compass Control Panel - Installation	DMC-412-A-97-21-00-04A-720A-A / 02314	2023-05-31	ALL
RMI Adapter - Installation	DMC-412-A-97-21-00-05A-720A-A / 02315	2023-05-31	SN: 33001-33100
Gyromagnetic Compass System(s) - Inspection	DMC-412-A-97-21-00-00A-280A-A / 02316	2023-05-31	ALL
Gyromagnetic Compass System(s) - Cleaning	DMC-412-A-97-21-00-00A-251A-A / 02317	2023-05-31	ALL
Gyromagnetic Compass System(s) - Repair or Replacement	DMC-412-A-97-21-00-00A-685A-A / 02318	2023-05-31	ALL
Attitude and Heading Reference System(s) - Description	DMC-412-A-97-22-00-00A-042A-A / 02319	2023-05-31	SN: 36248-Subsequent
AHRS Components - Description	DMC-412-A-97-22-00-00A-042B-A / 02320	2023-05-31	ALL
AHRS - Functional Test	DMC-412-A-97-22-00-00A-340A-A / 02321	2023-05-31	ALL
Heading System - Functional Test	DMC-412-A-97-22-00-00A-340B-A / 02322	2023-05-31	ALL
Attitude System - Functional Test	DMC-412-A-97-22-00-00A-340C-A / 02323	2023-05-31	ALL
Helicopters Equipped With Flight Director - Functional Test	DMC-412-A-97-22-00-01A-340A-A / 02324	2023-05-31	ALL
Helicopter Not Equipped With Flight Director - Functional Test	DMC-412-A-97-22-00-01A-340B-A / 02325	2023-05-31	ALL
AHRS - Calibration	DMC-412-A-97-22-00-00A-273A-A / 02326	2023-05-31	ALL
AHRS - Troubleshooting	DMC-412-A-97-22-00-00A-422A-A / 02327	2023-05-31	ALL
AHRS Level 1 Test Set - Description	DMC-412-A-97-22-00-00A-042C-A / 02328	2023-05-31	ALL
AHRU Module - Removal	DMC-412-A-97-22-00-01A-520A-A / 02329	2023-05-31	ALL
MSU Cal Prom - Removal	DMC-412-A-97-22-00-02A-520A-A / 02330	2023-05-31	ALL
AHRU Mounting Tray - Removal	DMC-412-A-97-22-00-03A-520A-A / 02331	2023-05-31	ALL
AHRS Control Module - Removal	DMC-412-A-97-22-00-04A-520A-A / 02332	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Flux Valve - Removal	DMC-412-A-97-22-00-05A-520A-A / 02333	2023-05-31	ALL
AHRS Compartment Cooling Fan - Removal	DMC-412-A-97-22-00-06A-520A-A / 02334	2023-05-31	ALL
AHRU Mounting Tray - Installation	DMC-412-A-97-22-00-01A-720A-A / 02335	2023-05-31	ALL
MSU Cal Prom - Installation	DMC-412-A-97-22-00-02A-720A-A / 02336	2023-05-31	ALL
AHRU Module - Installation	DMC-412-A-97-22-00-03A-720A-A / 02337	2023-05-31	ALL
AHRU Control Module - Installation	DMC-412-A-97-22-00-04A-720A-A / 02338	2023-05-31	ALL
Flux Valve - Installation	DMC-412-A-97-22-00-05A-720A-A / 02339	2023-05-31	ALL
AHRS Compartment Cooling Fan - Installation	DMC-412-A-97-22-00-06A-720A-A / 02340	2023-05-31	ALL
AHRS - Inspection	DMC-412-A-97-22-00-00A-280A-A / 02341	2023-05-31	ALL
AHRS - Cleaning	DMC-412-A-97-22-00-00A-251A-A / 02342	2023-05-31	ALL
AHRS - Repair or Replacement	DMC-412-A-97-22-00-00A-685A-A / 02343	2023-05-31	ALL
ATC Transponder System - (When Installed) - Description	DMC-412-A-97-23-00-00A-028A-A / 02344	2023-05-31	SN: 33001-33100
ATC Transponder System - (When Installed) - Description	DMC-412-A-97-23-00-00A-028B-A / 02345	2023-05-31	SN: 33101-33213 or 36001-Subsequent
KSP 756 Mode C Transponder System - Description	DMC-412-A-97-23-00-01A-028A-A / 02346	2023-05-31	SN: 33101-33213 or 36001-36272 or 36274-36275
MST 67A Mode S Transponder System - Description	DMC-412-A-97-23-00-01A-042A-A / 02347	2023-05-31	SN: 36273 or 36276-36461
MST 67A Mode S (ELS) Transponder System - Description	DMC-412-A-97-23-00-01A-042B-A / 02348	2023-05-31	SN: 36462-Subsequent
Mode C Transponder System - Functional Check	DMC-412-A-97-23-00-02A-340A-A / 02349	2023-05-31	SN: 33001-33213 or 36001-36272 or 36274-36275
MST 67A Mode S Transponder - Functional Check	DMC-412-A-97-23-00-03A-340A-A / 02350	2023-05-31	SN: 36273 or 36276-Subsequent
Mode C Transponder System - Troubleshooting	DMC-412-A-97-23-00-02A-422A-A / 02351	2023-05-31	ALL
Transponder - Removal	DMC-412-A-97-23-00-04A-520B-A / 02352	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Transponder Control - Removal	DMC-412-A-97-23-00-04A-520C-A / 02353	2023-05-31	SN: 33001-33100
Transponder Control - Removal	DMC-412-A-97-23-00-04A-520D-A / 02354	2023-05-31	SN: 33101-Subsequent
Transponder Antenna - Removal	DMC-412-A-97-23-00-04A-520E-A / 02355	2023-05-31	ALL
Transponder - Installation	DMC-412-A-97-23-00-04A-720B-A / 02356	2023-05-31	ALL
Transponder Control - Installation	DMC-412-A-97-23-00-04A-720C-A / 02357	2023-05-31	SN: 33001-33100
Transponder Control - Installation	DMC-412-A-97-23-00-04A-720D-A / 02358	2023-05-31	SN: 33101-Subsequent
Transponder Antenna - Installation	DMC-412-A-97-23-00-04A-720E-A / 02359	2023-05-31	ALL
Transponder System - Inspection	DMC-412-A-97-23-00-04A-280A-A / 02360	2023-05-31	ALL
Transponder System - Cleaning	DMC-412-A-97-23-00-04A-251A-A / 02361	2023-05-31	ALL
Transponder System - Repair or Replacement	DMC-412-A-97-23-00-04A-685A-A / 02362	2023-05-31	ALL
VHF Navigation (VOR/LOC/GS) System - Description	DMC-412-A-97-24-00-00A-028A-A / 02363	2023-05-31	SN: 33001-33100
VHF Navigation (VOR/ILS and VOR/LOC/GS) System - Description	DMC-412-A-97-24-00-00A-028B-A / 02364	2023-05-31	SN: 33101-Subsequent
Preliminary Test Procedures - Functional Test	DMC-412-A-97-24-00-00A-340B-A / 02365	2023-05-31	ALL
Single Receiver System - Functional Test	DMC-412-A-97-24-00-00A-340C-A / 02366	2023-05-31	ALL
Dual Receiver (Single Pilot) System - Functional Test	DMC-412-A-97-24-00-00A-340D-A / 02367	2023-05-31	ALL
Dual Receiver (Dual Pilot) System - Functional Test	DMC-412-A-97-24-00-00A-340E-A / 02368	2023-05-31	ALL
Localizer - Functional Test	DMC-412-A-97-24-00-00A-340F-A / 02369	2023-05-31	ALL
Glideslope (GS) - Functional Test	DMC-412-A-97-24-00-00A-340G-A / 02370	2023-05-31	ALL
VHF Navigation System - Troubleshooting	DMC-412-A-97-24-00-00A-422A-A / 02371	2023-05-31	ALL
VHF Navigation Receiver (No. 1 or No. 2) - Removal	DMC-412-A-97-24-00-01A-520A-A / 02372	2023-05-31	ALL
NAV Control Panel (No. 1 or No. 2) - Removal	DMC-412-A-97-24-00-02A-520A-A / 02373	2023-05-31	SN: 33001-33100
NAV Control Panel (No. 1 or No. 2) - Removal	DMC-412-A-97-24-00-02A-520B-A / 02374	2023-05-31	SN: 33101-Subsequent
CSI (Dual VHF Navigation Systems) - Removal	DMC-412-A-97-24-00-03A-520A-A / 02375	2023-05-31	ALL
NAV Antenna Coupler (Installed with No. 2 VHF Navigation System) - Removal	DMC-412-A-97-24-00-04A-520A-A / 02376	2023-05-31	ALL
Omni Antenna (Left or Right Half Element) - Removal	DMC-412-A-97-24-00-05A-520A-A / 02377	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Glideslope Antenna Coupler - Removal	DMC-412-A-97-24-00-06A-520A-A / 02378	2023-05-31	ALL
Glideslope Antenna - Removal	DMC-412-A-97-24-00-07A-520A-A / 02379	2023-05-31	ALL
VHF Navigation Receiver (No. 1 or No. 2) - Installation	DMC-412-A-97-24-00-01A-720A-A / 02380	2023-05-31	ALL
NAV/DME Control Unit (No. 1 or No. 2) - Installation	DMC-412-A-97-24-00-02A-720A-A / 02381	2023-05-31	SN: 33001-33100
NAV/DME Control Unit (No. 1 or No. 2) - Installation	DMC-412-A-97-24-00-02A-720B-A / 02382	2023-05-31	SN: 33101-Subsequent
CSI (Dual VOR/LOC/GS or VOR/LOC Systems) - Installation	DMC-412-A-97-24-00-03A-720A-A / 02383	2023-05-31	ALL
NAV Antenna Coupler (Installed with No. 2 VHF Navigation System) - Installation	DMC-412-A-97-24-00-04A-720A-A / 02384	2023-05-31	ALL
Omni Antenna (Left or Right Half Element) - Installation	DMC-412-A-97-24-00-05A-720A-A / 02385	2023-05-31	ALL
Glideslope Antenna Coupler - Installation	DMC-412-A-97-24-00-06A-720A-A / 02386	2023-05-31	ALL
Glideslope Antenna - Installation	DMC-412-A-97-24-00-07A-720A-A / 02387	2023-05-31	ALL
VHF Navigation System - Inspection Procedures	DMC-412-A-97-24-00-00A-280A-A / 02388	2023-05-31	ALL
VHF Navigation System - Cleaning Procedures	DMC-412-A-97-24-00-00A-251A-A / 02389	2023-05-31	ALL
VHF Navigation System - Repair or Replacement	DMC-412-A-97-24-00-00A-685A-A / 02390	2023-05-31	ALL
ADF Navigation System (When Installed) - Description	DMC-412-A-97-25-00-00A-028A-A / 02391	2023-05-31	SN: 33001-33100
ADF Navigation System (When Installed) - Description	DMC-412-A-97-25-00-00A-028B-A / 02392	2023-05-31	SN: 33101-Subsequent
ADF Navigation System - Functional Test	DMC-412-A-97-25-00-00A-340A-A / 02393	2023-05-31	ALL
ADF Navigation System - Troubleshooting	DMC-412-A-97-25-00-00A-422A-A / 02394	2023-05-31	ALL
ADF Receiver - Removal	DMC-412-A-97-25-00-01A-520A-A / 02395	2023-05-31	ALL
ADF Control Panel - Removal	DMC-412-A-97-25-00-02A-520A-A / 02396	2023-05-31	SN: 33001-33100
ADF Control Panel - Removal	DMC-412-A-97-25-00-02A-520B-A / 02397	2023-05-31	SN: 33101-Subsequent
ADF Loop Antenna - Removal	DMC-412-A-97-25-00-03A-520A-A / 02398	2023-05-31	SN: 33001-33100
ADF Loop/Sense Antenna - Removal	DMC-412-A-97-25-00-04A-520A-A / 02399	2023-05-31	SN: 33101-Subsequent
ADF Sense Antenna (Whip) - Removal	DMC-412-A-97-25-00-05A-520A-A / 02400	2023-05-31	SN: 33001-33100
ADF Receiver - Installation	DMC-412-A-97-25-00-01A-720A-A / 02401	2023-05-31	ALL
ADF Control Panel - Installation	DMC-412-A-97-25-00-02A-720A-A / 02402	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
ADF Control Panel - Installation	DMC-412-A-97-25-00-02A-720B-A / 02403	2023-05-31	SN: 33101-Subsequent
ADF Loop Antenna - Installation	DMC-412-A-97-25-00-03A-720A-A / 02404	2023-05-31	SN: 33001-33100
ADF Loop/Sense Antenna - Installation	DMC-412-A-97-25-00-04A-720A-A / 02405	2023-05-31	SN: 33101-Subsequent
ADF Sense Antenna (Whip) - Installation	DMC-412-A-97-25-00-05A-720A-A / 02406	2023-05-31	SN: 33001-33100
ADF Navigation System - Inspection Procedures	DMC-412-A-97-25-00-00A-280A-A / 02407	2023-05-31	ALL
ADF Navigation System - Cleaning Procedures	DMC-412-A-97-25-00-00A-251A-A / 02408	2023-05-31	ALL
ADF Navigation System - Repair or Replacement	DMC-412-A-97-25-00-00A-685A-A / 02409	2023-05-31	ALL
Marker Beacon System (When Installed) - Description	DMC-412-A-97-26-00-00A-028A-A / 02410	2023-05-31	SN: 33001-33100
Marker Beacon System (When installed) - Description	DMC-412-A-97-26-00-00A-028B-A / 02411	2023-05-31	SN: 33101-Subsequent
Marker Beacon System - Functional Test	DMC-412-A-97-26-00-00A-340A-A / 02412	2023-05-31	ALL
Marker Beacon System - Troubleshooting	DMC-412-A-97-26-00-00A-422A-A / 02413	2023-05-31	ALL
Marker Beacon Receiver - Removal	DMC-412-A-97-26-00-01A-520A-A / 02414	2023-05-31	SN: 33001-33100
VHF NAV Receiver No. 1 - Removal	DMC-412-A-97-26-00-02A-520A-A / 02415	2023-05-31	SN: 33101-Subsequent
NAV AUDIO Control Panel - Removal	DMC-412-A-97-26-00-03A-520A-A / 02416	2023-05-31	ALL
Marker Beacon Antenna - Removal	DMC-412-A-97-26-00-04A-520A-A / 02417	2023-05-31	ALL
Marker Beacon Receiver - Installation	DMC-412-A-97-26-00-01A-720A-A / 02418	2023-05-31	SN: 33001-33100
VHF NAV Receiver No. 1 - Installation	DMC-412-A-97-26-00-02A-720A-A / 02419	2023-05-31	SN: 33101-Subsequent
NAV AUDIO Control Panel - Installation	DMC-412-A-97-26-00-03A-720A-A / 02420	2023-05-31	ALL
Marker Beacon Antenna - Installation	DMC-412-A-97-26-00-04A-720A-A / 02421	2023-05-31	ALL
Marker Beacon System - Inspection Procedures	DMC-412-A-97-26-00-00A-280A-A / 02422	2023-05-31	ALL
Marker Beacon System - Cleaning Procedures	DMC-412-A-97-26-00-00A-251A-A / 02423	2023-05-31	ALL
Marker Beacon System - Repair or Replacement	DMC-412-A-97-26-00-00A-685A-A / 02424	2023-05-31	ALL
Radar Altimeter Navigation System - Description	DMC-412-A-97-27-00-00A-028A-A / 02425	2023-05-31	ALL
Radar Altimeter Navigation System - Functional Test	DMC-412-A-97-27-00-00A-340A-A / 02426	2023-05-31	ALL
Radar Altimeter Navigation System - Troubleshooting	DMC-412-A-97-27-00-00A-422A-A / 02427	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Radar Altimeter Receiver-Transmitter - Removal	DMC-412-A-97-27-00-01A-520A-A / 02428	2023-05-31	ALL
Radar Altimeter Indicator - Removal	DMC-412-A-97-27-00-02A-520A-A / 02429	2023-05-31	ALL
Radar Altimeter Antenna - Removal	DMC-412-A-97-27-00-03A-520A-A / 02430	2023-05-31	ALL
Radar Altimeter Receiver-Transmitter - Installation	DMC-412-A-97-27-00-01A-720A-A / 02431	2023-05-31	ALL
Radar Altimeter Indicator - Installation	DMC-412-A-97-27-00-02A-720A-A / 02432	2023-05-31	ALL
Radar Altimeter Antenna - Installation	DMC-412-A-97-27-00-03A-720A-A / 02433	2023-05-31	ALL
Radar Altimeter Navigation System - Inspection Procedures	DMC-412-A-97-27-00-00A-280A-A / 02434	2023-05-31	ALL
Radar Altimeter Navigation System - Cleaning Procedures	DMC-412-A-97-27-00-00A-251A-A / 02435	2023-05-31	ALL
Radar Altimeter Navigation System - Repair or Replacement	DMC-412-A-97-27-00-00A-685A-A / 02436	2023-05-31	ALL
DME Navigation System (When Installed) - Description	DMC-412-A-97-28-00-00A-028A-A / 02437	2023-05-31	ALL
DME Navigation System - Functional Test	DMC-412-A-97-28-00-00A-340A-A / 02438	2023-05-31	ALL
DME Navigation System - Troubleshooting	DMC-412-A-97-28-00-00A-422A-A / 02439	2023-05-31	ALL
DME Receiver-Transmitter - Removal	DMC-412-A-97-28-00-01A-520A-A / 02440	2023-05-31	ALL
NAV Control Panel - Removal	DMC-412-A-97-28-00-02A-520A-A / 02441	2023-05-31	SN: 33001-33100
NAV Control Panel - Removal	DMC-412-A-97-28-00-02A-520B-A / 02442	2023-05-31	SN: 33101-Subsequent
NAV AUDIO Control Panel - Removal	DMC-412-A-97-28-00-03A-520A-A / 02443	2023-05-31	ALL
DME Antenna - Removal	DMC-412-A-97-28-00-04A-520A-A / 02444	2023-05-31	ALL
DME Receiver-Transmitter - Installation	DMC-412-A-97-28-00-01A-720A-A / 02445	2023-05-31	ALL
NAV Control Panel - Installation	DMC-412-A-97-28-00-02A-720A-A / 02446	2023-05-31	SN: 33001-33100
NAV Control Panel - Installation	DMC-412-A-97-28-00-02A-720B-A / 02447	2023-05-31	SN: 33101-Subsequent
NAV AUDIO Control Panel - Installation	DMC-412-A-97-28-00-03A-720A-A / 02448	2023-05-31	ALL
DME Antenna - Installation	DMC-412-A-97-28-00-04A-720A-A / 02449	2023-05-31	ALL
DME Navigation System - Inspection Procedures	DMC-412-A-97-28-00-00A-280A-A / 02450	2023-05-31	ALL
DME Navigation System - Cleaning Procedures	DMC-412-A-97-28-00-00A-251A-A / 02451	2023-05-31	ALL
DME Navigation System - Repair or Replacement	DMC-412-A-97-28-00-00A-685A-A / 02452	2023-05-31	ALL

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Avionics Systems - Description

Serial numbers are included in this chapter to show effectivity differences. The absence of serial number effectivities indicates the information applies to all serial number effectivities.

The avionics communication and navigation systems installed in helicopter models 412/412EP consists of the pilot/copilot intercommunication system, VHF/AM communication system, and pilot/copilot gyromagnetic compass system. For access locations to avionics components, refer to (412-A-52-00-00-00A-028A-A / 00775).

The following optional avionics auxiliary equipment are kits and are available from, or can be installed by the manufacturer. Provisions have been incorporated in the helicopter to accept these kits; cable assemblies and wiring have been provided, and wires and connectors have been stowed in the helicopter at locations where units and components are to be installed. Complete maintenance instructions are presented in this chapter pertaining to the following system kits:

- **SN: 33101-36086 or 36087-Subsequent:** VOR/ILS Navigation System No. 1 Kit
- **SN: 33001-33100:** VOR/LOC/GS Navigation System No. 1 Kit
- VOR/LOC/GS Navigation System No. 2 Kit
- VHF AM Communication System No. 2 Kit
- VOR/LOC Navigation System No. 1 Kit
- DME Navigation System Kit
- ATC Transponder System Kit
- **SN: 33001-33100:** Marker Beacon Navigation System Kit
- Aft Crew 2 Station ICS Kit
- ADF Navigation System Kit
- Radar Altimeter Navigation System Kit

Applicable: SN: 33001-33100:

A copilot (No. 2) Three Axis Gyro Equipment kit is available from the manufacturer and may be installed in helicopter. Complete installation instructions are supplied with the kit. Complete maintenance instructions are presented in the gyromagnetic compass system section of this chapter. Provisions have been incorporated in the helicopter to accept this kit. Cable assemblies and wiring have been provided and wires and connectors have been stowed in the helicopter at the locations where the units and components are installed.

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Airframe Components - Description

Components that are a part of the avionics configuration and are termed airframe components, include the following avionics/electrical items:

AC and DC power busses	Panel lights
Circuit board assemblies	Plugs
Circuit breakers	Receptacles
Connectors	Replays
Diode assemblies	Resistors
Interconnecting wiring	Switches
Electrical leads	Terminal boards

For maintenance information of airframe components, refer to (412-A-96-00-01-02A-913A-A / 01875).

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Energizing Power Buses - Functional Tests

Procedure

1. The following is a general procedure for energizing the AC and DC buses when using an external power source during functional testing of the avionics systems and sets.

1.1. Close the following circuit breakers:

- MASTER CAUTION
- COPLT 115 VAC BUS FEEDER
- PILOT 115 VAC BUS FEEDER
- BUS 3 FEEDER 115 VAC
- COPLT 26 VAC BUS FEEDER
- PILOT 26 VAC BUS FEEDER
- BUS FEEDER 26 VAC
- NO. 1 ESNTL BUS FEEDER (4)
- NO. 2 ESNTL BUS FEEDER (4)
- GEN 1 FIELD
- GEN 2 FIELD
- INV 1 PWR
- INV 2 PWR
- EMERG BUS INTCON
- BUS 1 VM
- BUS 2 VM

1.2. Set the INV 1 and INV 2 switches to ON.

1.3. Close all the remaining AC circuit breakers to load the inverters.

2. To energize the avionics systems or sets, refer to the functional test procedures for the individual systems or sets.

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Energizing Power Buses - Functional Tests

Procedure

1. The following is a general procedure for energizing AC and DC buses when using an external power source during functional testing of the avionics systems and sets.

1.1. Close the following circuit breakers:

- MASTER CAUTION
- AC FEEDERS BUS 1 115V
- AC FEEDERS BUS 2 115V
- AC FEEDERS BUS 3 115V
- AC FEEDERS EMERG BUS 115V
- AC FEEDERS BUS 1 26V
- AC FEEDERS BUS 2 26V
- AC FEEDERS BUS 3 26V
- AC FEEDERS EMERG BUS 26V
- NO. 1 ESNTL BUS FEEDERS (4)
- NO. 2 ESNTL BUS FEEDERS (4)
- GEN 1 FIELD
- GEN 2 FIELD
- INV 1 PWR
- INV 2 PWR
- EMERG BUS INTCT
- BUS 1 VM
- BUS 2 VM

1.2. Set INV 1 switch and INV 2 switch to ON.

1.3. Close all remaining AC circuit breakers to load the inverters.

2. To energize avionics systems or sets, refer to functional test procedures for individual systems or sets.

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Avionics Systems - Illustrations - Description

For avionics systems, sets, or components installed in basic configuration helicopter, refer to following illustrations.

The avionics equipment location (Figure 1 and Figure 2) shows the physical location of the avionics and airframe components and indicates access. For code designation correlation, refer to the avionics equipment list (Table 1 and Table 2).

The control unit illustrations (Figure 3 and Figure 4), intercom control unit and block diagram, illustrate the control selectors and switches for use as a guide for settings when performing functional testing of respective systems.

Avionics wiring diagrams identify all the conductors by wire numbers, all shielding, components, and equipment in the installed avionics configuration to use as a reference when replacing components, equipment, and wiring, and to troubleshoot the respective system.

For avionics systems kits (Omni Navigation System Kit, ADF Navigation System Kit, and/or Avionics 2 Station ICS System Kit), refer to the avionics kits wiring diagram for replacement, reference of components, equipment and wiring, and for troubleshooting respective kits (when installed).

For instrument flight rules configuration or instrument meteorological conditions configuration modification kits, refer to the appropriate Service Instruction, as applicable, for equipment location and avionics wiring diagrams.

Table 1. Equipment List (S/N 33001 through 33100)

REF DES	NOMENCLATURE
J46	CONNECTOR, PILOT CYCLIC STICK
J47	CONNECTOR, COPILOT CYCLIC STICK
J308L	JACK, COPILOT HEADSET
J308R	JACK, PILOT HEADSET
J319L	DISCONNECT, LEFT AFT CREW HEADSET
J319R	DISCONNECT, AFT RIGHT CREW HEADSET
P46	CONNECTOR, PILOT CYCLIC STICK
P47	CONNECTOR, COPILOT CYCLIC STICK
P319L	DISCONNECT, AFT LEFT CREW HEADSET
P319R	DISCONNECT, AFT RIGHT CREW HEADSET
TB22	TERMINAL BLOCK, COPILOT HEADSET
TB23	TERMINAL BLOCK, PILOT HEADSET
2300DS1	ANNUNCIATOR, RIGHT AFT INT/CALL
2300DS2	ANNUNCIATOR, LEFT AFT INT/CALL
2300J5	DISCONNECT, DOORPOST
2300P5	DISCONNECT, DOORPOST
2301A1	PANEL, EMERGENCY KEY
2301A2	PANEL, INTERFACE
2301A2J1	CONNECTOR, INTERFACE PANEL
2301A2P1	CONNECTOR, INTERFACE PANEL
2301CP1	CONTROL UNIT, PILOT INTERCOMM
2301CP1P1	CONNECTOR, PILOT INTERCOMM CONTROL UNIT
2301CP2	CONTROL UNIT, COPILOT INTERCOMM
2301CP2P1	CONNECTOR, COPILOT INTERCOMM CONTROL UNIT
2301CP3	CONTROL UNIT, RIGHT AFT CREW INTERCOMM
2301CP3P1	CONNECTOR, RIGHT AFT CREW INTERCOMM CONTROL UNIT
2301CP4	CONTROL UNIT, LEFT AFT CREW INTERCOMM

Table 1. Equipment List (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
2301CP4P1	CONNECTOR, LEFT AFT CREW INTERCOMM CONTROL UNIT
2301S3	SWITCH, PILOT FOOT
2301S4	SWITCH, COPILOT FOOT
2301TB1	TERMINAL BLOCK, INTERCOMM
2301TB2	TERMINAL BLOCK, INTERCOMM
2301TB3	TERMINAL BLOCK, INTERCOMM
2303CP1	CONTROL, VHF COMM NO. 1
2303CP1P1	CONNECTOR, VHF COMM NO. 1 CONTROL
2303CP2	CONTROL, VHF COMM NO. 2
2303CP2P1	CONNECTOR, VHF COMM NO. 2 CONTROL
2303E1	ANTENNA, VHF COMM NO. 1
2303E1P1	CONNECTOR, VHF COMM NO. 1 ANTENNA
2303E2	ANTENNA, VHF COMM NO. 2
2303E2P1	CONNECTOR, VHF COMM NO. 2 ANTENNA
2303J1	DISCONNECT, VHF COMM NO. 1, NOSE
2303J2	DISCONNECT, VHF COMM NO. 2, NOSE
2303P1	DISCONNECT, VHF COMM NO. 1, NOSE
2303P2	DISCONNECT, VHF COMM NO. 2, NOSE
2303P3	DISCONNECT, VHF COMM NO. 1
2303TR1	TRANSCEIVER, VHF COMM NO. 1
2303TR1P1	CONNECTOR, VHF COMM NO. 1 TRANSCEIVER
2303TR1P2	CONNECTOR, VHF COMM NO. 2 TRANSCEIVER
2303TR2	TRANSCEIVER, VHF COMM NO. 2
2303TR2P1	CONNECTOR, VHF COMM NO. 2 TRANSCEIVER
2303TR2P2	CONNECTOR, VHF COMM NO. 2 TRANSCEIVER
3400CP1	CONTROL PANEL, NAV AUDIO
3400CP1P1	CONNECTOR, NAV AUDIO CONTROL PANEL
3400J1	DISCONNECT, INSTRUMENT PANEL
3400J8	DISCONNECT, WINDSHIELD POST
3400P1	DISCONNECT, INSTRUMENT PANEL
3400P8	DISCONNECT, WINDSHIELD POST
3400TB1	TERMINAL BLOCK, COMM, NOSE
3400TB2	TERMINAL BLOCK, COMM, NOSE
3400TB3	TERMINAL BLOCK, AC RETURN, NOSE
3400TB4	TERMINAL BLOCK, GLIDESLOPE, NOSE
3405AR1	ADAPTER, RMI NO. 1
3405AR1P1	CONNECTOR, RMI NO. 1 ADAPTER
3405AR2	ADAPTER, RMI NO. 2
3405AR2P1	CONNECTOR, RMI NO. 2 ADAPTER
3405CP1	CONTROL PANEL, NAV NO. 1
3405CP1P1	CONNECTOR, NAV NO. 1 CONTROL PANEL
3405CP2	CONTROL PANEL, NAV NO. 2

Table 1. Equipment List (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
3405CP2P1	CONNECTOR, NAV NO. 2 CONTROL PANEL
3405CP3	COUPLER, NAV ANTENNA
3405CP3P1	CONNECTOR, NAV ANTENNA COUPLER
3405CP3P2	CONNECTOR, NAV ANTENNA COUPLER
3405CP3P3	CONNECTOR, NAV ANTENNA COUPLER
3405CP4	CONTROL PANEL, PILOT NAV BEARING POINTER
3405CP4P1	CONNECTOR, PILOT NAV BEARING POINTER SWITCH
3405CP5	CONTROL PANEL, COPILOT NAV BEARING POINTER
3405CP5P1	CONNECTOR, COPILOT NAV BEARING POINTER SWITCH
3405DS1	INDICATOR, PILOT HSI
3405DS1P1	CONNECTOR, PILOT HSI
3405DS2	INDICATOR, COPILOT HSI
3405DS2P1	CONNECTOR, COPILOT HSI
3405DS3	INDICATOR, COURSE SELECT
3405DS3P1	CONNECTOR, COURSE SELECT INDICATOR
3405E1	ANTENNA, LEFT OMNI
3405E1P1	CONNECTOR, LEFT OMNI ANTENNA
3405E2	ANTENNA, RIGHT OMNI
3405E2P1	CONNECTOR, RIGHT OMNI ANTENNA
3405J1	DISCONNECT, NOSE
3405J2	DISCONNECT, NOSE
3405J3	DISCONNECT, NOSE
3405J4	DISCONNECT, NOSE
3405J5	DISCONNECT, TAILBOOM
3405J6	DISCONNECT, TAILBOOM ANTENNA
3405P1	DISCONNECT, NOSE
3405P2	DISCONNECT, NOSE
3405P3	DISCONNECT, NOSE
3405P4	DISCONNECT, NOSE
3405P5	DISCONNECT, TAILBOOM
3405P6	DISCONNECT, TAILBOOM ANTENNA
3405RE1	RECEIVER, NAV NO. 1
3405RE1P1	CONNECTOR, NAV RECEIVER NO. 1
3405RE1P2	CONNECTOR, NAV RECEIVER NO. 1
3405RE1P3	CONNECTOR, NAV RECEIVER NO. 1
3405RE2	RECEIVER, NAV NO. 2
3405RE2P1	CONNECTOR, NAV RECEIVER NO. 2
3405RE2P2	CONNECTOR, NAV RECEIVER NO. 2
3405RE2P3	CONNECTOR, NAV RECEIVER NO. 2
3405S1	SWITCH, COURSE SET
3405S3	SWITCH, PILOT NAV BEARING POINTER
3405S4	SWITCH, COPILOT NAV BEARING POINTER
3405TB1	TERMINAL BLOCK, NAV, NOSE

Table 1. Equipment List (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
3405TB2	TERMINAL BLOCK, NAV, NOSE
3406C1	CAPACITOR, ADF SENSE ANTENNA
3406CP1	CONTROL PANEL, ADF
3406CP1P1	CONNECTOR, ADF CONTROL PANEL
3406E1	ANTENNA, ADF SENSE
3406E1J1	DISCONNECT, ADF SENSE ANTENNA
3406E1P1	DISCONNECT, ADF SENSE ANTENNA
3406E2	ANTENNA, ADF LOOP
3406E2P1	CONNECTOR, ADF LOOP ANTENNA
3406J1	DISCONNECT, NOSE
3406P1	DISCONNECT, NOSE
3406RE1	RECEIVER, ADF
3406RE1P1	CONNECTOR, ADF RECEIVER
3406RE1P2	CONNECTOR, ADF RECEIVER
3406RE1P3	CONNECTOR, ADF RECEIVER
3407CP1	CONTROL PANEL, PILOT COMPASS
3407CP1P1	CONNECTOR, PILOT COMPASS CONTROL PANEL
3407CP2	CONTROL PANEL, COPILOT COMPASS
3407CP2P1	CONNECTOR, COPILOT COMPASS CONTROL PANEL
3407J1	DISCONNECT, TAILBOOM
3407J2	DISCONNECT, TAILBOOM
3407L1	COMPENSATOR, DUAL REMOTE COMPASS
3407L1P1	CONNECTOR, DUAL REMOTE COMPASS COMPENSATOR
3407L1P2	CONNECTOR, DUAL REMOTE COMPASS COMPENSATOR
3407MP1	SENSOR, 3-AXIS REFERENCE NO. 1 (PILOT)
3407MP1P1	CONNECTOR, 3-AXIS REFERENCE SENSOR NO. 1
3407MP2	SENSOR, 3-AXIS REFERENCE NO. 2 (COPILOT)
3407MP2P1	CONNECTOR, 3-AXIS REFERENCE SENSOR NO. 2
3407P1	DISCONNECT, TAILBOOM
3407P2	DISCONNECT, TAILBOOM
3407TB1	TERMINAL BLOCK, GYRO COMPASS, NOSE
3407TR1	FLUX VALVE, NO. 1 TRANSMITTER
3407TR2	FLUX VALVE, NO. 2 TRANSMITTER
3408DS1	INDICATOR, RADAR ALTIMETER INDICATOR
3408DS1P1	CONNECTOR, RADAR ALTIMETER INDICATOR
3408E1	ANTENNA, RADAR ALTIMETER RECEIVE
3408E1P1	CONNECTOR, RADAR ALTIMETER RECEIVE ANTENNA
3408E2	ANTENNA, RADAR ALTIMETER TRANSMIT
3408E2P1	CONNECTOR, RADAR ALTIMETER TRANSMIT ANTENNA
3408J4	DISCONNECT, TAILBOOM
3408J5	DISCONNECT, TAILBOOM
3408P4	DISCONNECT, TAILBOOM
3408P5	DISCONNECT, TAILBOOM

Table 1. Equipment List (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
3408TR1	RECEIVER-TRANSMITTER (R/T), RADAR ALTIMETER
3408TR1P1	CONNECTOR, RADAR ALTIMETER R/T
3408TR1P2	CONNECTOR, RADAR ALTIMETER R/T
3408TR1P3	CONNECTOR, RADAR ALTIMETER R/T
3410CP1	CONTROL PANEL, TRANSPONDER
3410CP1P1	CONNECTOR, TRANSPONDER CONTROL PANEL
3410E1	ANTENNA, TRANSPONDER
3410E1P1	CONNECTOR, TRANSPONDER ANTENNA
3410J1	DISCONNECT, NOSE
3410P1	DISCONNECT, NOSE
3410TR1	TRANSPONDER
3410TR1P1	CONNECTOR, TRANSPONDER
3410TR1P2	CONNECTOR, TRANSPONDER
3410TR1P3	CONNECTOR, TRANSPONDER
3412CP1	INDICATOR, PILOT MARKER BEACON
3412CP1P1	CONNECTOR, PILOT MARKER BEACON INDICATOR
3412CP2	INDICATOR, COPILOT MARKER BEACON
3412CP2P1	CONNECTOR, COPILOT MARKER BEACON INDICATOR
3412E1	ANTENNA, MARKER BEACON
3412E1P1	CONNECTOR, MARKER BEACON ANTENNA
3412RE1	RECEIVER, MARKER BEACON
3412RE1P1	CONNECTOR, MARKER BEACON RECEIVER
3413CP1	COUPLER, GLIDESLOPE ANTENNA
3413CP1P1	CONNECTOR, GLIDESLOPE COUPLER
3413CP1P2	CONNECTOR, GLIDESLOPE COUPLER
3413CP1P3	CONNECTOR, GLIDESLOPE COUPLER
3413E1	ANTENNA, GLIDESLOPE
3413E1P1	CONNECTOR, GLIDESLOPE ANTENNA
3414DS1	INDICATOR, DME
3414DS1P1	CONNECTOR, DME INDICATOR
3414E1	ANTENNA, DME
3414E1P1	CONNECTOR, DME ANTENNA
3414TR1	RECEIVER-TRANSMITTER DME
3414TR1P1	CONNECTOR, DME RECEIVER-TRANSMITTER
3414TR1P2	CONNECTOR, DME RECEIVER-TRANSMITTER
3414TR1P3	CONNECTOR, DME RECEIVER-TRANSMITTER
3414TR1P4	CONNECTOR, DME RECEIVER-TRANSMITTER

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent)

REF DES	NOMENCLATURE
J308L	CONN, COPILOT HEADSET
J308R	CONN, PILOT HEADSET
TB22	

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

TERMINAL BLOCK, COPILOT HEADSET
TB23
TERMINAL BLOCK, PILOT HEADSET
4A1J1
CONN, PILOT CYCLIC STICK
4A1P1
CONN, PILOT CYCLIC STICK
5A1J1
CONN, COPILOT CYCLIC STICK
5A1P1
CONN, COPILOT CYCLIC STICK
*9J3
CONN, AFCS
*9P3
CONN, AFCS
*9TB33
RECEPTACLE, INTEGRATED WIRE
*9TB33P1
CONN, INTEGRATED WIRE RECEPTACLE
*9WT1
RECEPTACLE, GROUND
*9WT1P1
CONN, GROUND RECEPTACLE
22J14
CONN, LEFT WINDSHIELD POST DISCONNECT
22J17
CONN, LEFT INSTRUMENT PANEL DISCONNECT
22P14
CONN, LEFT WINDSHIELD POST DISCONNECT
22P17
CONN, LEFT INSTRUMENT PANEL DISCONNECT
2300J1
CONN, LEFT NOSE DISCONNECT
2300J2
CONN, LEFT DOORPOST DISCONNECT
2300P1
CONN, LEFT NOSE DISCONNECT
2301A1
PANEL, EMERGENCY KEY
2301A1P1
CONN, EMERGENCY KEY PANEL
2301A2
PANEL, INTERFACE
2301A2P1
CONN, INTERFACE PANEL
2301CP1
CONTROL UNIT, COPILOT INTERCOMM
2301CP1P1
CONN, COPILOT INTERCOMM CONTROL UNIT
2301CP2
CONTROL UNIT, PILOT INTERCOMM
2301CP2P1
CONN, PILOT INTERCOMM CONTROL UNIT
2301CP3
CONTROL UNIT, LEFT AFT CREW INTERCOMM
2301CP3P1
CONN, LEFT AFT CREW INTERCOMM CONTROL UNIT
2301CP4
CONTROL UNIT, RIGHT AFT CREW INTERCOMM
2301CP4P1
CONN, RIGHT AFT CREW INTERCOMM CONTROL UNIT
2301DS1
ANNUNCIATOR, COPILOT AFT INT/CALL/ON
2301DS2
ANNUNCIATOR, PILOT AFT INT/CALL/ON

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

2301J3
CONN, LEFT AFT CREW HEADSET
2301J4
CONN, RIGHT AFT CREW HEADSET
2301P3
CONN, LEFT AFT CREW HEADSET
2301P4
CONN, RIGHT AFT CREW HEADSET
2301S1
SWITCH, PILOT FOOT
2301S2
SWITCH, COPILOT FOOT
2301TB1
TERMINAL BLOCK, INTERCOMM
2301TB2
TERMINAL BLOCK, INTERCOMM
2301WT1
MODULE, ICS GROUND
*2301WT1
RECEPTACLE, GROUND
*2301WT1P1
CONN, GROUND RECEPTACLE
2301WT2
MODULE, ICS GROUND
*2301WT2
RECEPTACLE, GROUND
*2301WT2
CONN, GROUND RECEPTACLE
2301WT3
MODULE, ICS GROUND
2301WT4
MODULE, ICS GROUND
2301WT5
MODULE, ICS GROUND
2301WT6
MODULE, ICS GROUND
2301WT7
MODULE, ICS GROUND
2301WT8
MODULE, ICS GROUND
2301WT9
MODULE, ICS GROUND
2303CP1
CONTROL, VHF COMM NO. 1
2303CP1P1
CONN, VHF COMM NO. 1 CONTROL
2303CP2
CONTROL, VHF COMM NO. 2
2303CP2P1
CONN, VHF COMM NO. 2 CONTROL
2303E1
ANTENNA, VHF COMM NO. 1
2303E1P1
CONN, VHF COMM NO. 1 ANTENNA
2303E2
ANTENNA, VHF COMM NO. 2
2303E2P1
CONN, VHF COMM NO. 2 ANTENNA
2303J1
CONN, SERVICE DECK DISCONNECT
2303P1
CONN, SERVICE DECK DISCONNECT
2303TR1
TRANSCEIVER, VHF COMM NO. 1
2303TR1P1

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

CONN, VHF COMM NO. 1 TRANSCEIVER
2303TR1P2
CONN, VHF COMM NO. 2 TRANSCEIVER
2303TR2
TRANSCEIVER, VHF COMM NO. 2
2303TR2P1
CONN, VHF COMM NO. 2 TRANSCEIVER
2303TR2P2
CONN, VHF COMM NO. 2 TRANSCEIVER
3400CP1
CONTROL PANEL, NAV AUDIO
3400CP1P1
CONN, NAV AUDIO CONTROL PANEL
3400J2
CONN, WINDSHIELD POST DISCONNECT
3400P2
CONN, WINDSHIELD POST DISCONNECT
3400TB1
TERMINAL BLOCK, NAV, NOSE
*3400TB1
RECEPTACLE, INTEGRATED WIRE
*3400TB1P1
CONN, INTEGRATED WIRE RECEPTACLE
3400TB2
TERMINAL BLOCK, NAV, NOSE
*3400TB2
RECEPTACLE, INTEGRATED WIRE
*3400TB2P1
CONN, INTEGRATED WIRE RECEPTACLE
3400TB3
TERMINAL BLOCK, NAV, NOSE
*3400TB3
RECEPTACLE, INTEGRATED WIRE
*3400TB3P1
CONN, INTEGRATED WIRE RECEPTACLE
3400WT1
MODULE, NAV GROUND
*3400WT1
RECEPTACLE, GROUND
*3400WT1P1
CONN, GROUND RECEPTACLE
3400WT2
MODULE, NAV GROUND
*3400WT2
RECEPTACLE, GROUND
*3400WT2P1
CONN, GROUND RECEPTACLE
3400WT3
MODULE, NAV GROUND
3400WT4
MODULE, NAV GROUND
3400WT5
MODULE, NAV GROUND
3400WT6
MODULE, NAV GROUND
3400WT7
MODULE, NAV GROUND
3400WT8
MODULE, NAV GROUND
3405CP1
CONTROL PANEL, NAV NO. 1
3405CP1P1
CONN, NAV NO. 1 CONTROL PANEL
3405CP2
CONTROL PANEL, NAV NO. 2

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

3405CP2P1
CONN, NAV NO. 2 CONTROL PANEL
3405CP3
COUPLER, NAV ANTENNA
3405CP3P1
CONN, NAV ANTENNA COUPLER
3405CP3P2
CONN, NAV ANTENNA COUPLER
3405CP3P3
CONN, NAV ANTENNA COUPLER
3405CP4
CONTROL PANEL, PILOT NAV BEARING POINTER
3405CP4P1
CONN, PILOT NAV BEARING POINTER SWITCH
3405CP5
CONTROL PANEL, COPILOT NAV BEARING POINTER
3405CP5P1
CONN, COPILOT NAV BEARING POINTER SWITCH
3405CP6
NAV ANTENNA COUPLER RELAY, COURSE SET
3405DS1
INDICATOR, COPILOT HORIZONTAL SITUATION INDICATOR
3405DS1P1
CONN, COPILOT HORIZONTAL SITUATION INDICATOR
3405DS2
INDICATOR, PILOT HORIZONTAL SITUATION INDICATOR
3405DS2P1
CONN, PILOT HORIZONTAL SITUATION INDICATOR
3405DS3
INDICATOR, COURSE SELECT
3405DS3P1
CONN, COURSE SELECT INDICATOR
3405E1
ANTENNA, RIGHT OMNI
3405E1P1
CONN, RIGHT OMNI ANTENNA
3405E2
ANTENNA, LEFT OMNI
3405E2P1
CONN, LEFT OMNI ANTENNA
3405J1
CONN, TAILBOOM DISCONNECT
3405K1
RELAY, COURSE SET
3405K2
RELAY, COURSE SET
3405P1
CONN, TAILBOOM DISCONNECT
3405P2
CONN, OMNI ANTENNA
3405RE1
RECEIVER, NAV NO. 1
3405RE1P1
CONN, NAV RECEIVER NO. 1
3405RE1P2
CONN, NAV RECEIVER NO. 1
3405RE1P3
CONN, NAV RECEiVER NO. 1
3405RE1P4
CONN, NAV RECEIVER NO. 1
3405RE1P5
CONN, NAV RECEIVER NO. 1
3405RE2
RECEIVER, NAV NO. 2
3405RE2P1

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

CONN, NAV RECEIVER NO. 2
3405RE2P2
CONN, NAV RECEIVER NO. 2
3405RE2P3
CONN, NAV RECEIVER NO. 2
3405RE2P5
CONN, NAV RECEIVER NO. 2
3405S1
SWITCH, COPILOT NAV BEARING POINTER
3405S2
SWITCH, PILOT NAV BEARING POINTER
3405S3
SWITCH, COURSE SET
3406CP1
CONTROL PANEL, ADF
3406CP1P1
CONN, ADF CONTROL PANEL
3406E1
ANTENNA, ADF
3406E1P1
CONN, ADF ANTENNA
3406RE1
RECEIVER, ADF
3406RE1P1
CONN, ADF RECEIVER
3406RE1P2
CONN, ADF RECEIVER
3407B1 (3)
AHRS COOLING FAN
3407B1P1 (3)
CONNECTOR
3407CP1
CONTROL PANEL, COPILOT COMPASS
3407CP1P1
CONN, COPILOT COMPASS CONTROL PANEL
3407CP2
CONTROL PANEL, PILOT COMPASS
3407CP2P1
CONN, PILOT COMPASS CONTROL PANEL
3407F1 (2)
FUSE
3407F2 (2)
FUSE
3407J1
CONN, TAILBOOM DISCONNECT
3407J2
CONN, TAILBOOM DISCONNECT
3407J3
CONN, LEFT NOSE DISCONNECT
3407J4
CONN, RIGHT NOSE DISCONNECT
3407J6
CONN, LEFT BEAM DISCONNECT
3407L1 (1)
COMPENSATOR, DUAL REMOTE COMPASS
3407L1P1 (1)
CONN, DUAL REMOTE COMPASS COMPENSATOR
3407L1P2 (1)
CONN, DUAL REMOTE COMPASS COMPENSATOR
3407MP1 (1)
SENSOR 3-AXIS REFERENCE NO. 2 (PILOT)
3407MP1 (3)
ATTITUDE HEADING REFERENCE UNIT (AHRU) NO. 1 COPILOT
3407MP1P1 (1) (3)
CONNECTOR

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

3407MP1P2 (1) (3)
CONNECTOR
3407MP1P3 (3)
CONNECTOR
3407MP1P4 (3)
CONNECTOR
3407MP1P5 (3)
CONNECTOR (CAL PROM)
3407MP2 (1)
SENSOR, 3-AXIS REFERENCE NO. 2 (PILOT)
3407MP2 (3)
ATTITUDE HEADING REFERENCE UNIT (AHRU) NO. 2 (PILOT)
3407MP2P1 (1) (3)
CONNECTOR
3407MP2P2 (1) (3)
CONNECTOR
3407MP2P3 (3)
CONNECTOR
3407MP2P4 (3)
CONNECTOR
3407MP2P5 (3)
CONNECTOR (CAL PROM)
3407P1
CONN, TAILBOOM DISCONNECT
3407P2
CONN, TAILBOOM DISCONNECT
3407P3
CONN, LEFT NOSE DISCONNECT
3407P4
CONN, RIGHT NOSE DISCONNECT
3407P6
CONN, LEFT BEAM DISCONNECT
3407TR1
FLUX VALVE, NO. 1 TRANSMITTER
3407TR2
FLUX VALVE, NO. 2 TRANSMITTER
3407XF1 (2)
HOLDER, FUSE
3407XF2 (2)
HOLDER, FUSE
3408DS1
INDICATOR, RADAR ALTIMETER
3408DS1P1
CONN, RADAR ALTIMETER INDICATOR
3408E1
ANTENNA, RADAR ALTIMETER TRANSMIT
3408E1P1
CONN, RADAR ALTIMETER TRANSMIT ANTENNA
3408E2
ANTENNA, RADAR ALTIMETER RECEIVE
3408E2P1
CONN, RADAR ALTIMETER RECEIVE ANTENNA
3408J1
CONN, LEFT NOSE DISCONNECT
3408J2
CONN, TAILBOOM DISCONNECT
3408J3
CONN, TAILBOOM DISCONNECT
3408P1
CONN, LEFT NOSE DISCONNECT
3408P2
CONN, TAILBOOM DISCONNECT
3408P3
CONN, TAILBOOM DISCONNECT
3408TR1

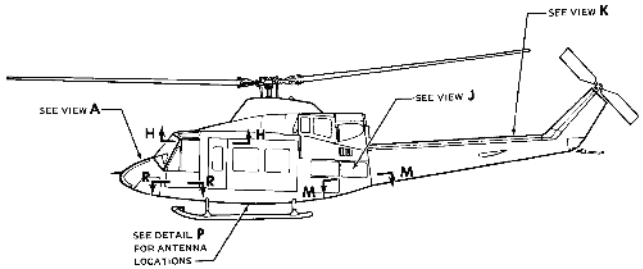
Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

RECEIVER-TRANSMITTER (R/T), RADAR ALTIMETER
3408TR1P1
CONN, RADAR ALTIMETER R/T
3408TR1P2
CONN, RADAR ALTIMETER R/T
3408TR1P3
CONN, RADAR ALTIMETER R/T
3410CP1
CONTROL PANEL, TRANSPONDER
3410CP1P1
CONN, TRANSPONDER CONTROL PANEL
*3410CR10
RECEPTACLE, INTEGRATED DIODE
*3410CR10P1
CONN, INTEGRATED DIODE RECEPTACLE
3410E1
ANTENNA, TRANSPONDER
3410E1P1
CONN, TRANSPONDER ANTENNA
3410TR1
TRANSPONDER
3410TR1P1
CONN, TRANSPONDER
3410TR1P2
CONN, TRANSPONDER
3412DS1
INDICATOR, COPILOT MARKER BEACON
3412DS1P1
CONN, COPILOT MARKER BEACON INDICATOR
3412DS2
INDICATOR, PILOT MARKER BEACON
3412DS2P1
CONN, PILOT MARKER BEACON INDICATOR
3412E1
ANTENNA, MARKER BEACON
3412E1P1
CONN, MARKER BEACON ANTENNA
3413CP1
COUPLER, GLIDESLOPE ANTENNA
3413CP1P1
CONN, GLIDESLOPE COUPLER
3413CP1P2
CONN, GLIDESLOPE COUPLER
3413CP1P3
CONN, GLIDESLOPE COUPLER
3413CP2
COUPLER, GLIDESLOPE ANTENNA
3413E1
ANTENNA, GLIDESLOPE
3413E1P1
CONN, GLIDESLOPE ANTENNA
3414DS1
INDICATOR, DME
3414DS1P1
CONN, DME INDICATOR
3414E1
ANTENNA, DME
3414E1P1
CONN, DME ANTENNA
3414TR1
RECEIVER-TRANSMITTER, DME
3414TR1P1
CONN, DME RECEIVER-TRANSMITTER
3414TR1P2
CONN, DME RECEIVER-TRANSMITTER

Table 2. Equipment List (S/N 33101 through 36086 and 36087 and Subsequent) (continued)

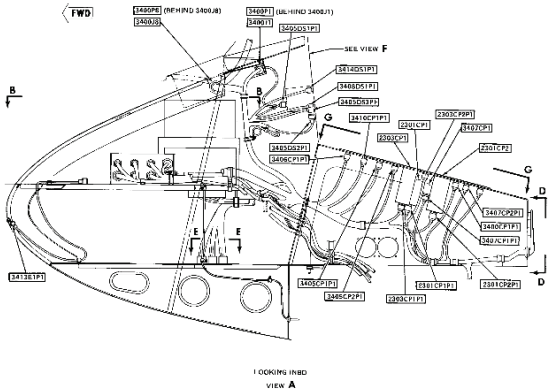
NOTES:
SN: 36087-Subsequent:
* Indicates reference designator peculiar to.
1
2
3

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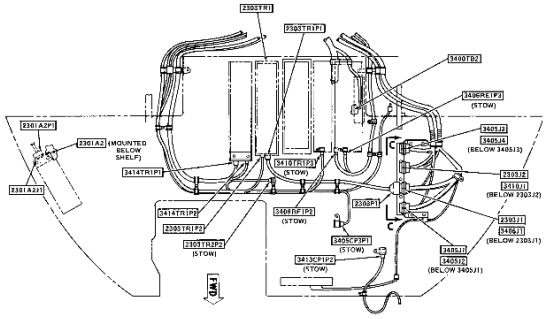
ICN-412-A-970000-B-97499-00001-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 1 of 13)



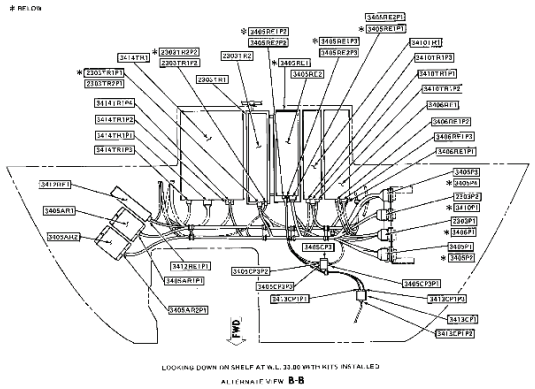
ICN-412-A-970000-B-97499-00002-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 2 of 13)



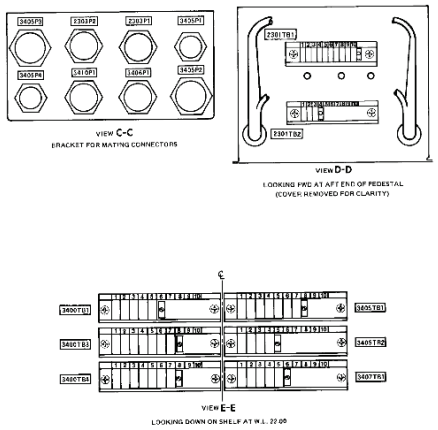
ICN-412-A-970000-B-97499-00003-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 3 of 13)



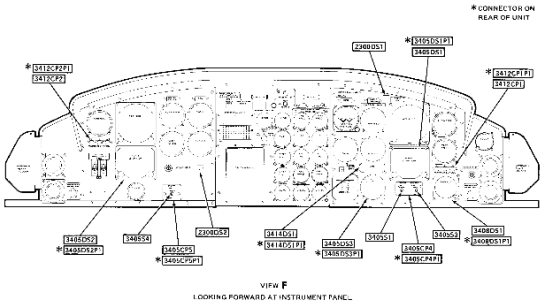
ICN-412-A-970000-B-97499-00004-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 4 of 13)



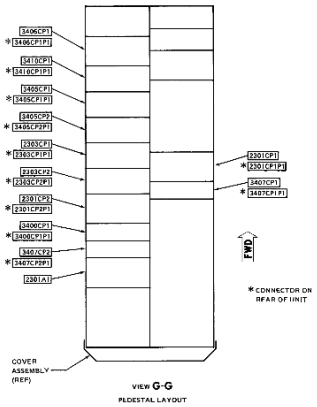
ICN-412-A-970000-B-97499-00005-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 5 of 13)



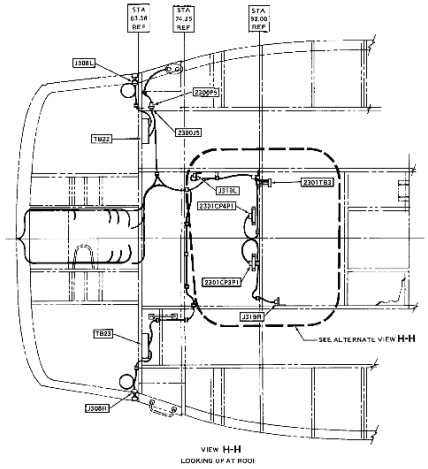
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Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 6 of 13)



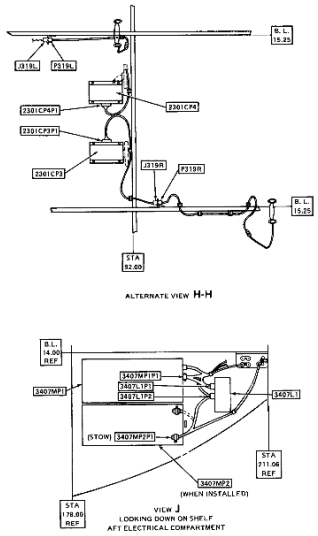
ICN-412-A-970000-B-97499-00007-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 7 of 13)



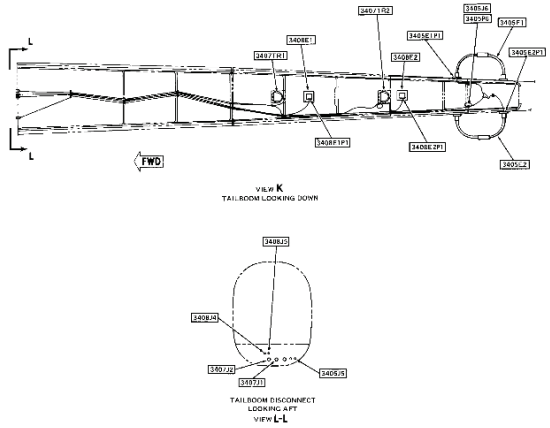
ICN-412-A-970000-B-97499-00008-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 8 of 13)



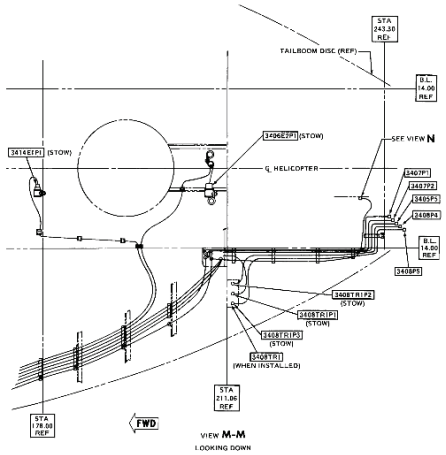
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Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 9 of 13)



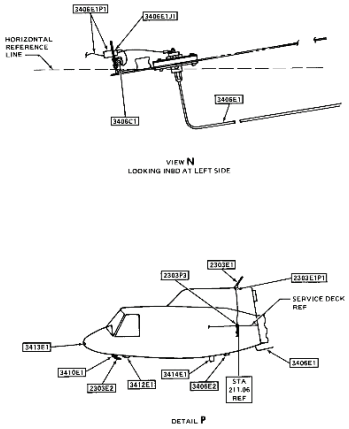
ICN-412-A-970000-B-97499-00010-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 10 of 13)



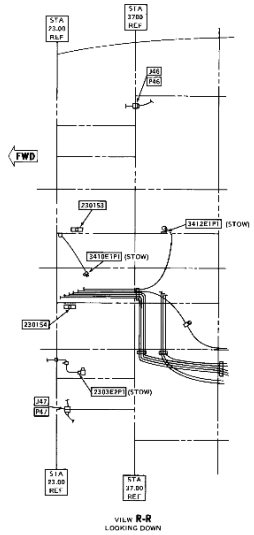
ICN-412-A-970000-B-97499-00011-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 11 of 13)



ICN-412-A-970000-B-97499-00012-A-001-01

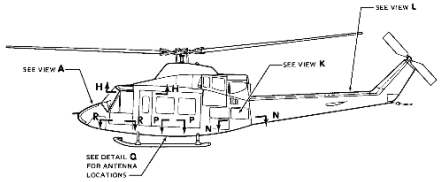
Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 12 of 13)



ICN-412-A-970000-B-97499-00013-A-001-01

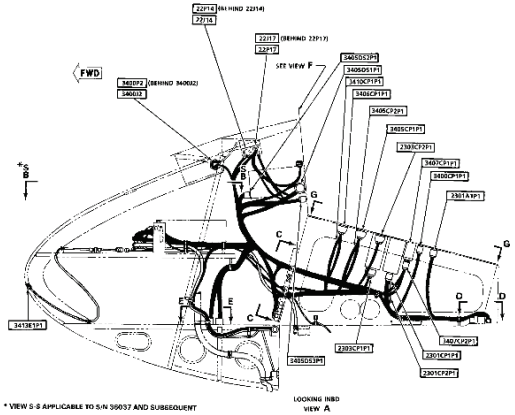
Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 13 of 13)

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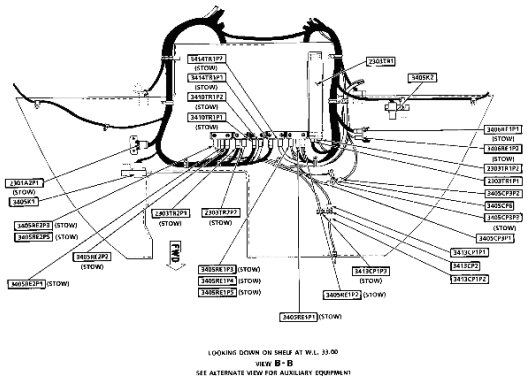
ICN-412-A-970000-B-97499-00014-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 20)



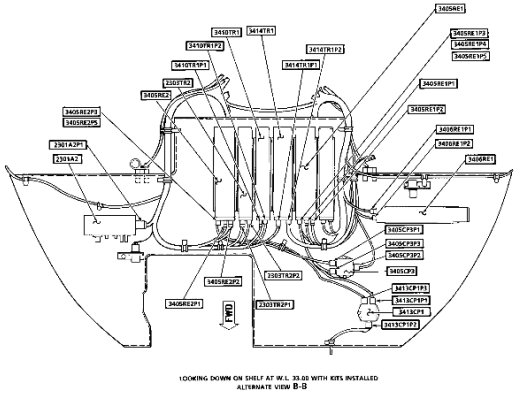
ICN-412-A-970000-B-97499-00015-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 2 of 20)



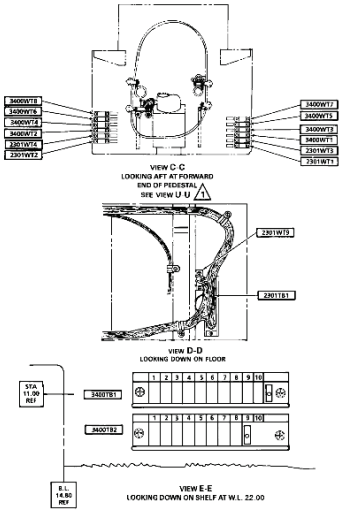
ICN-412-A-970000-B-97499-00016-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 3 of 20)



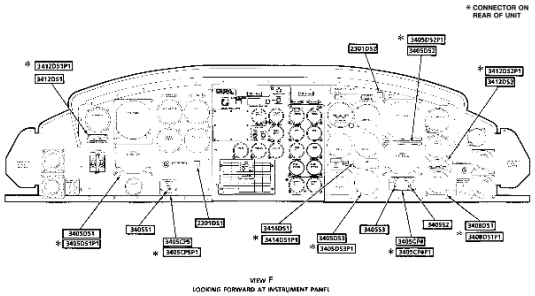
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Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 4 of 20)



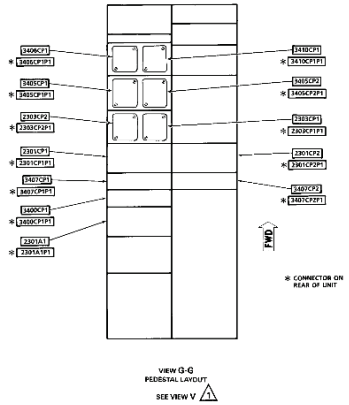
ICN-412-A-970000-B-97499-00018-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 5 of 20)



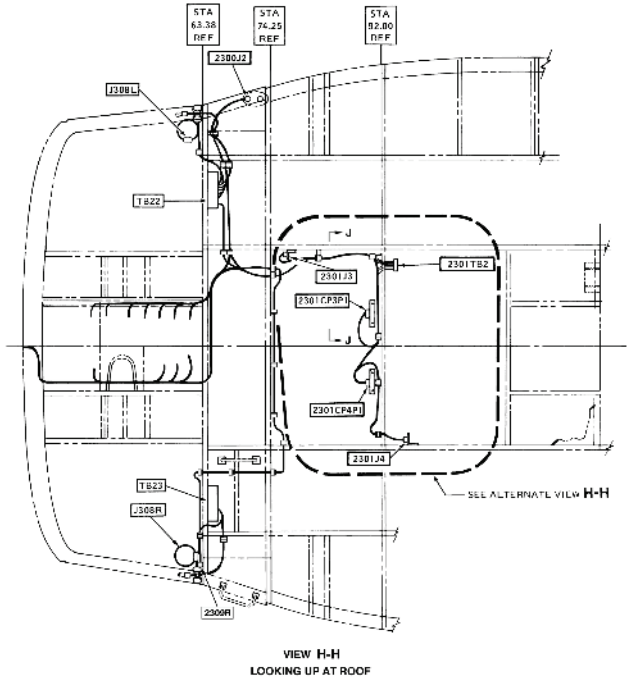
ICN-412-A-970000-B-97499-00019-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 6 of 20)



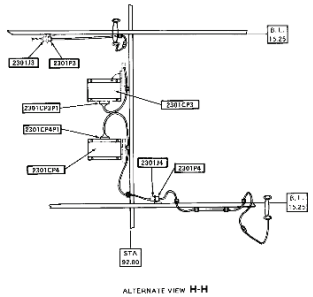
ICN-412-A-970000-B-97499-00020-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 7 of 20)



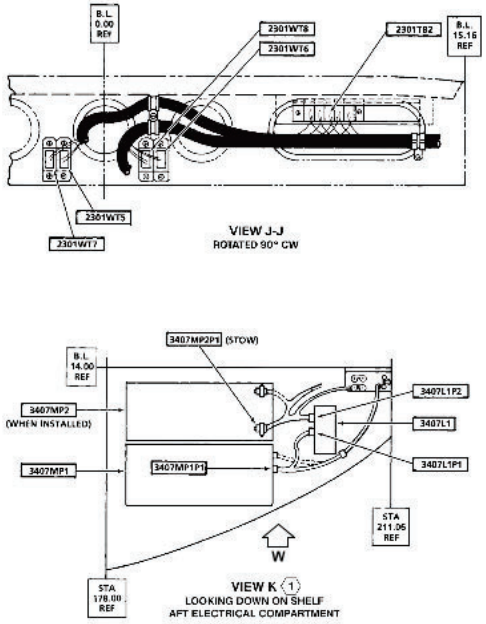
ICN-412-A-970000-B-97499-00021-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 8 of 20)



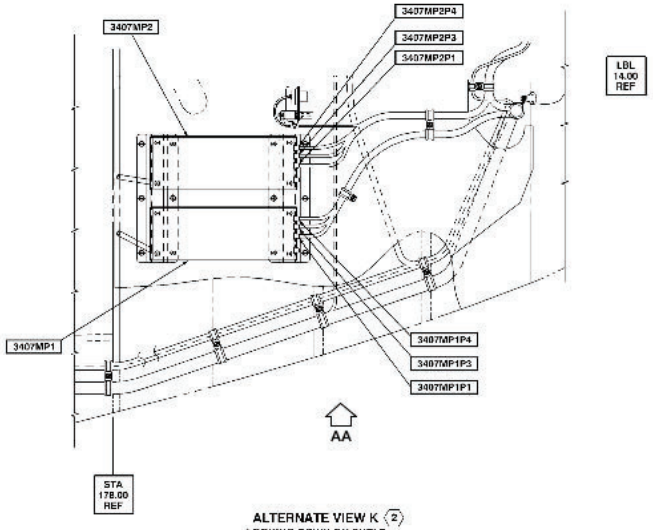
ICN-412-A-970000-B-97499-00022-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 9 of 20)



ICN-412-A-970000-B-97499-00023-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 10 of 20)



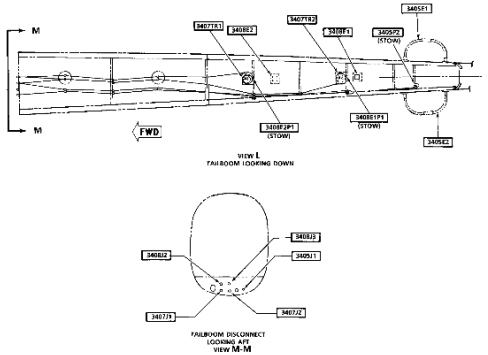
ALTERNATE VIEW K (2)
LOOKING DOWN ON SHELF
AFT ELECTRICAL COMPARTMENT

EFFECTIVITY

(1) S/N 33101 THROUGH 33213 and 36001 THROUGH 36247.
(2) S/N 36248 THROUGH 36999.

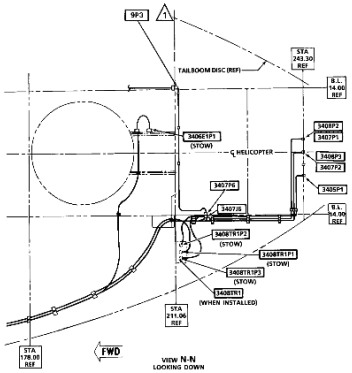
ICN-412-A-970000-B-97499-00024-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 11 of 20)



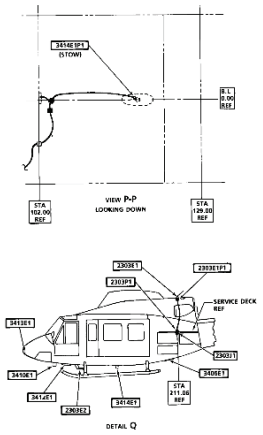
ICN-412-A-970000-B-97499-00025-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 12 of 20)



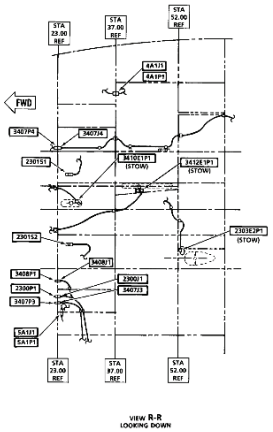
ICN-412-A-970000-B-97499-00026-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 13 of 20)



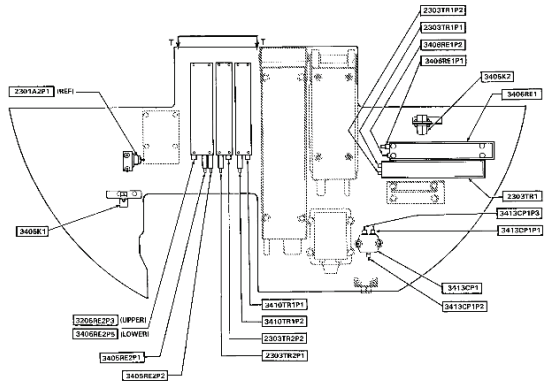
ICN-412-A-970000-B-97499-00027-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 14 of 20)



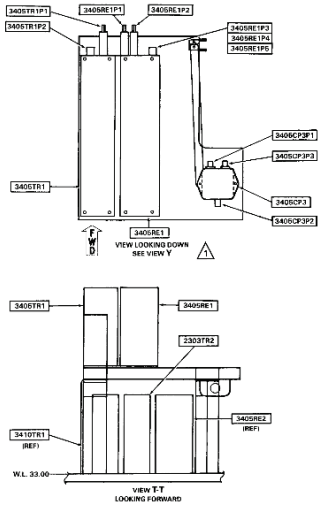
ICN-412-A-970000-B-97499-00028-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 15 of 20)



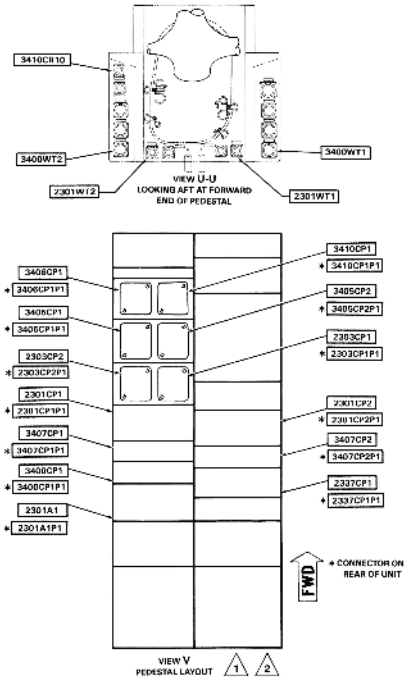
ICN-412-A-970000-B-97499-00029-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 16 of 20)



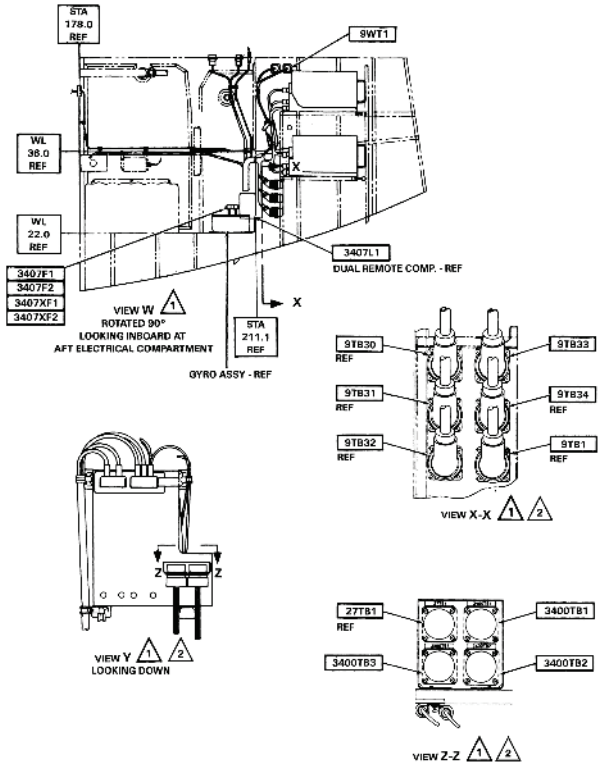
ICN-412-A-970000-B-97499-00030-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 17 of 20)



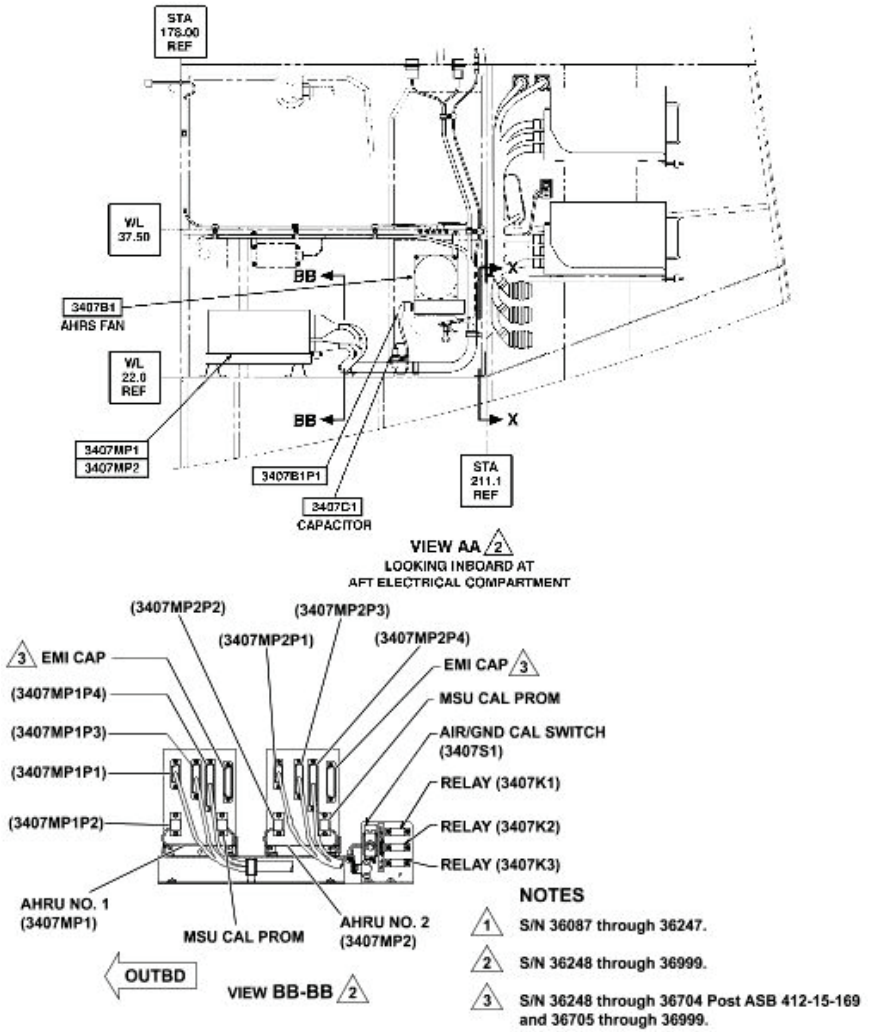
ICN-412-A-970000-B-97499-00031-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 18 of 20)



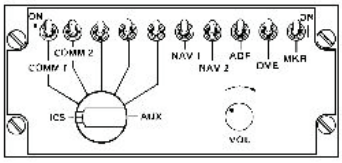
ICN-412-A-970000-B-97499-00032-A-001-01

Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 19 of 20)

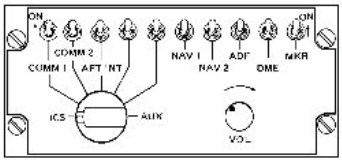


ICN-412-A-970000-B-97499-00033-A-001-01

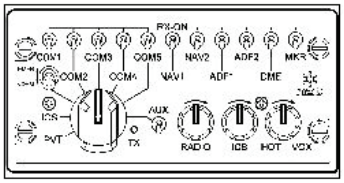
Figure 2. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 20 of 20)



1 2



3



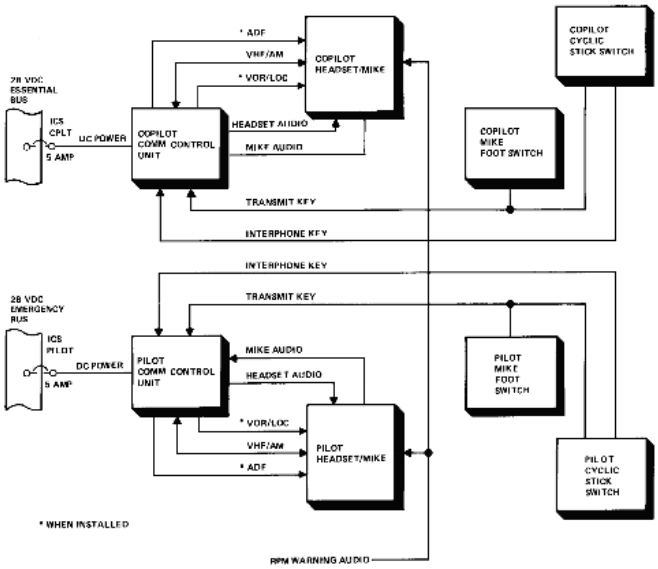
4

NOTES

1. Pilot and copilot intercom control unit on S/N 33001 through 33107 when aft cabin ICS is not installed.
2. Pilot and copilot intercom control unit on S/N 33108 through 36086 and 36087 through 36637. Aft left and right side cabin intercom control unit, if installed.
3. Pilot and copilot intercom control unit on S/N 33001 through 33107 when aft ICS is installed.
4. Pilot and copilot intercom control unit on S/N 36638 and subsequent. Aft left and right side cabin intercom control unit, if installed.

ICN-412-A-970000-B-97499-00034-A-001-01

Figure 3. Intercom Control Unit (Sheet 1 of 1)



ICN-412-A-970000-B-97499-00035-A-001-01

Figure 4. Intercommunications System (ICS) Block Diagram (S/N 33001 through 36086 and 36087 and Subsequent (Sheet 1 of 1)

Avionics System Wiring Diagrams - Description

This chapter contains wiring diagrams for each of the avionics systems that may be installed in the Bell Models 412/412EP helicopter (Figure 1 through Figure 27 and Figure 28). Individual circuit diagrams are provided to assist maintenance personnel in understanding the circuits and components installed in the helicopter and in troubleshooting and tracing of inoperative and malfunctioning circuits and components.

1.1. Wiring Diagrams

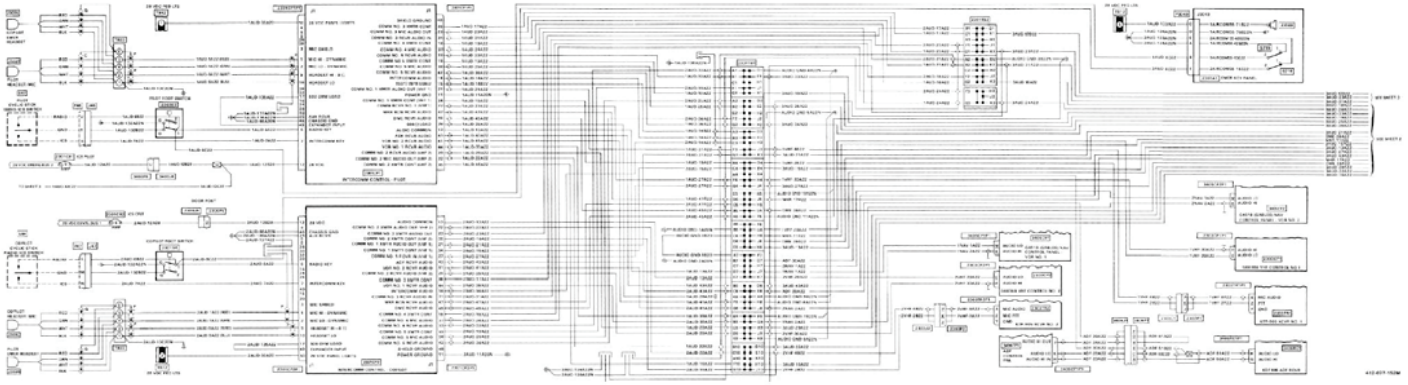
All wires on the wiring diagrams are identified by codes exactly as they are marked in the helicopter.

The code or wire number indicates the circuit function, wire number, wire segment letter and the wire size of the conductor. Identification of wires may be accomplished by using the wiring identification code figure in (412-A-98-00-00-00A-042A-A / 02456).

1.2. Use of Underlined Capital Letters

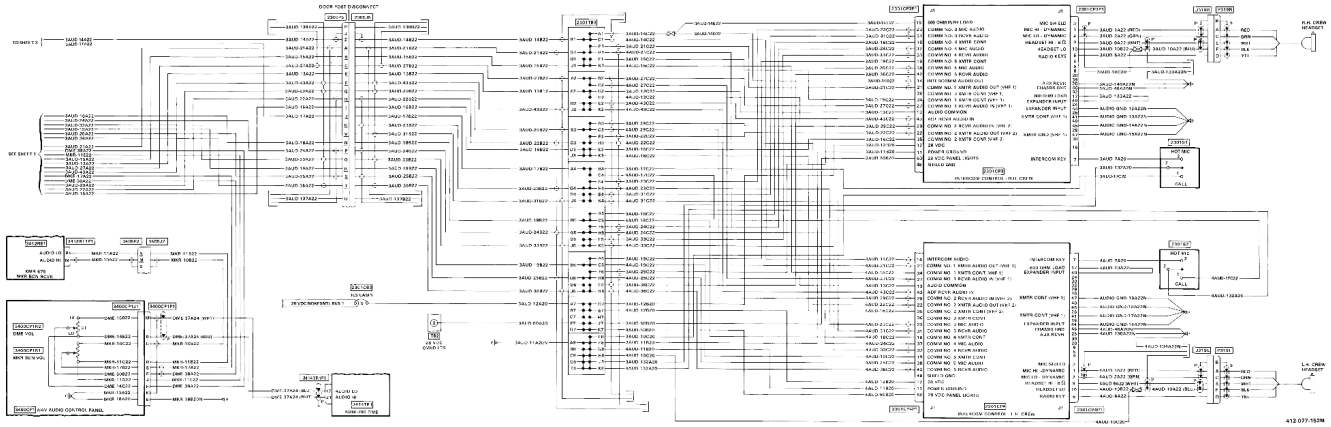
Many letters have similar upper case and lower case forms. To eliminate this possible source of error in identifying connector pin designations, underlined upper case (Capital) letters (A, B, C, etc.) are used to signify the lower case pin letters (a, b, c, etc).

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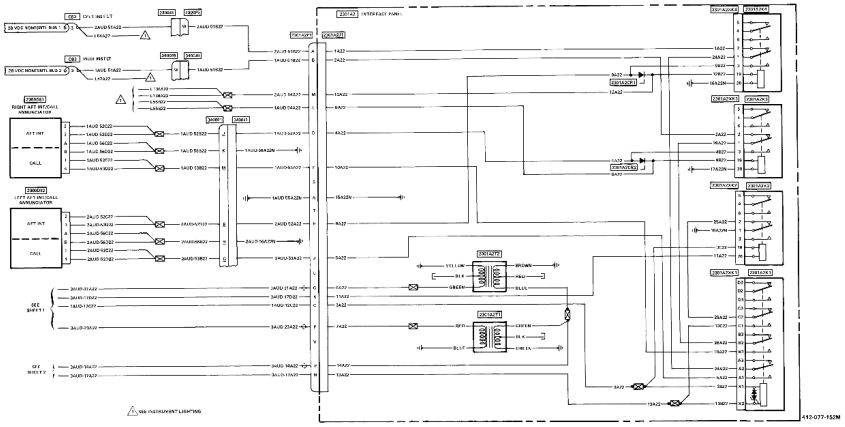
ICN-412-A-972900-B-97499-00001-A-001-01

Figure 1. Intercommunication System (ICS) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 3)



ICN-412-A-972900-B-97499-00002-A-001-01

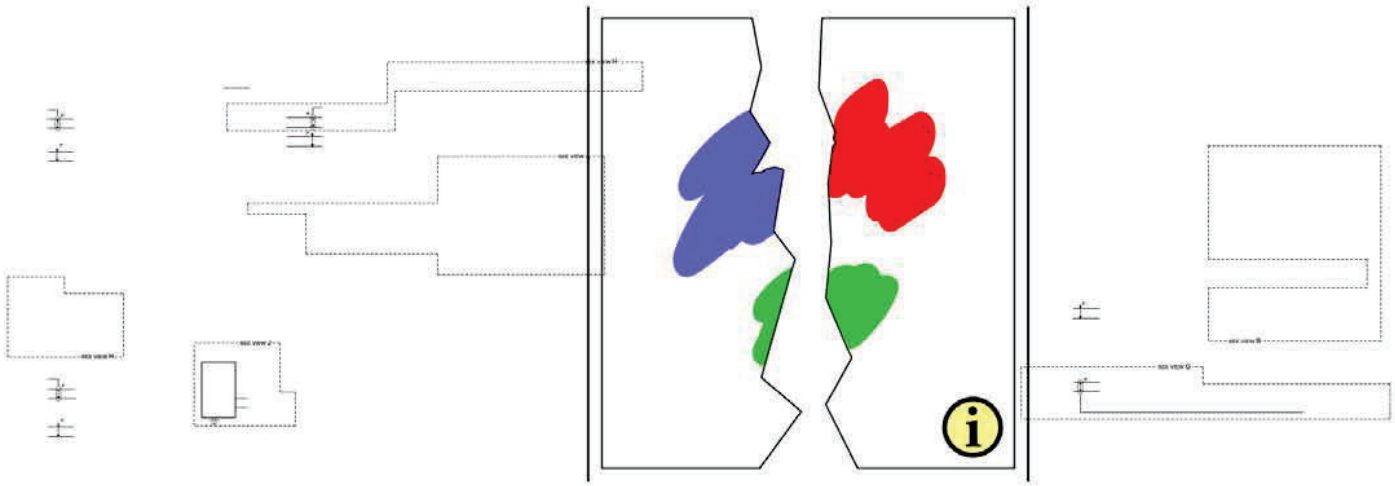
Figure 1. Intercommunication System (ICS) Wiring Diagram (S/N 33001 through 33100) (Sheet 2 of 3)



ICN-412-A-972900-B-97499-00003-A-001-01

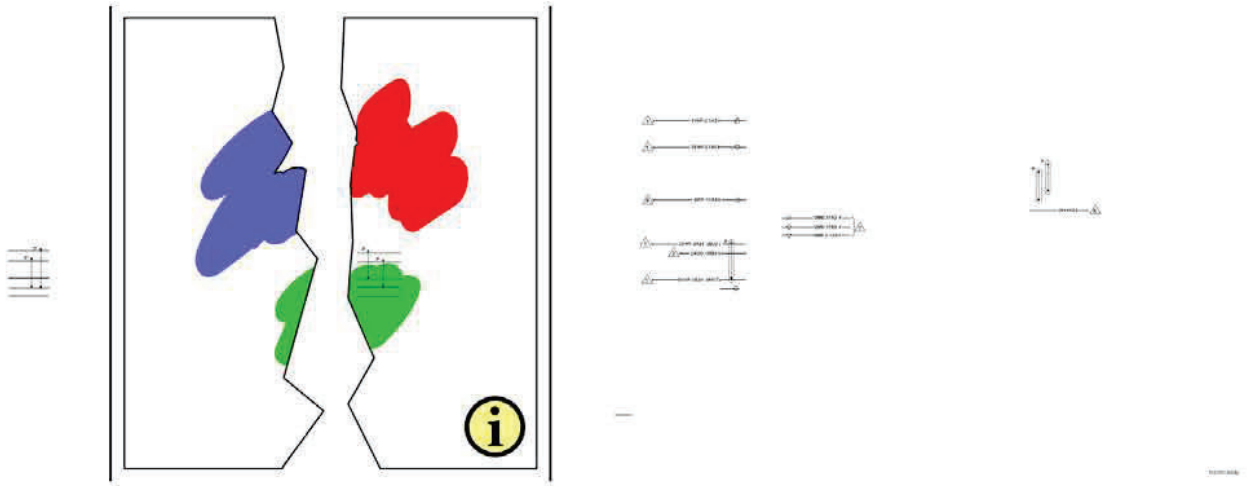
Figure 1. Intercommunication System (ICS) Wiring Diagram (S/N 33001 through 33100) (Sheet 3 of 3)

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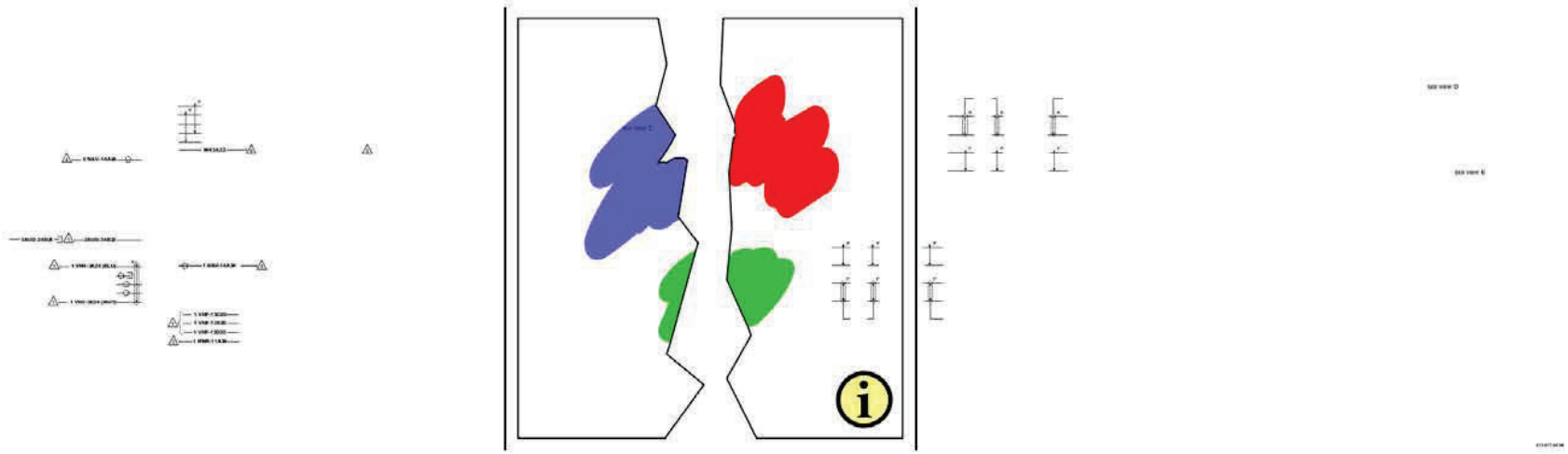
ICN-412-A-972900-B-97499-00004-A-001-01

Figure 2. Intercommunication System (ICS) Wiring Diagram (S/N 33101 through 36086 and 36087 through 36637) (Sheet 1 of 5)



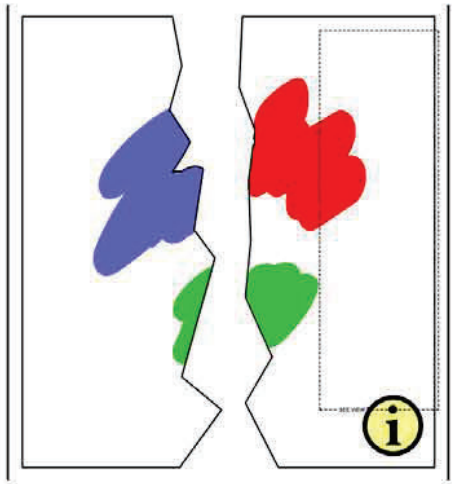
ICN-412-A-972900-B-97499-00005-A-001-01

Figure 2. Intercommunication System (ICS) Wiring Diagram (S/N 33101 through 36086 and 36087 through 36637) (Sheet 2 of 5)



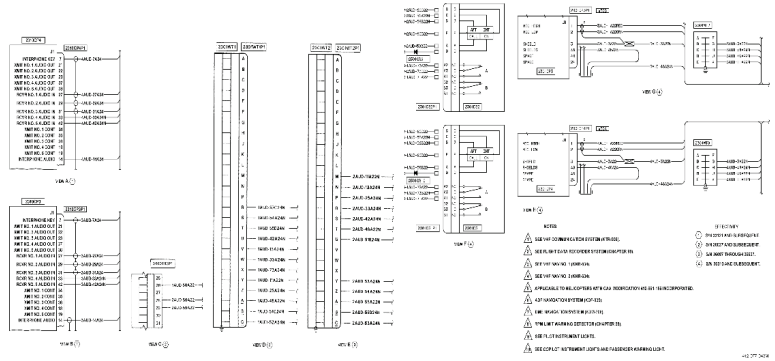
ICN-412-A-972900-B-97499-00006-A-001-01

Figure 2. Intercommunication System (ICS) Wiring Diagram (S/N 33101 through 36086 and 36087 through 36637) (Sheet 3 of 5)



ICN-412-A-972900-B-97499-00007-A-001-01

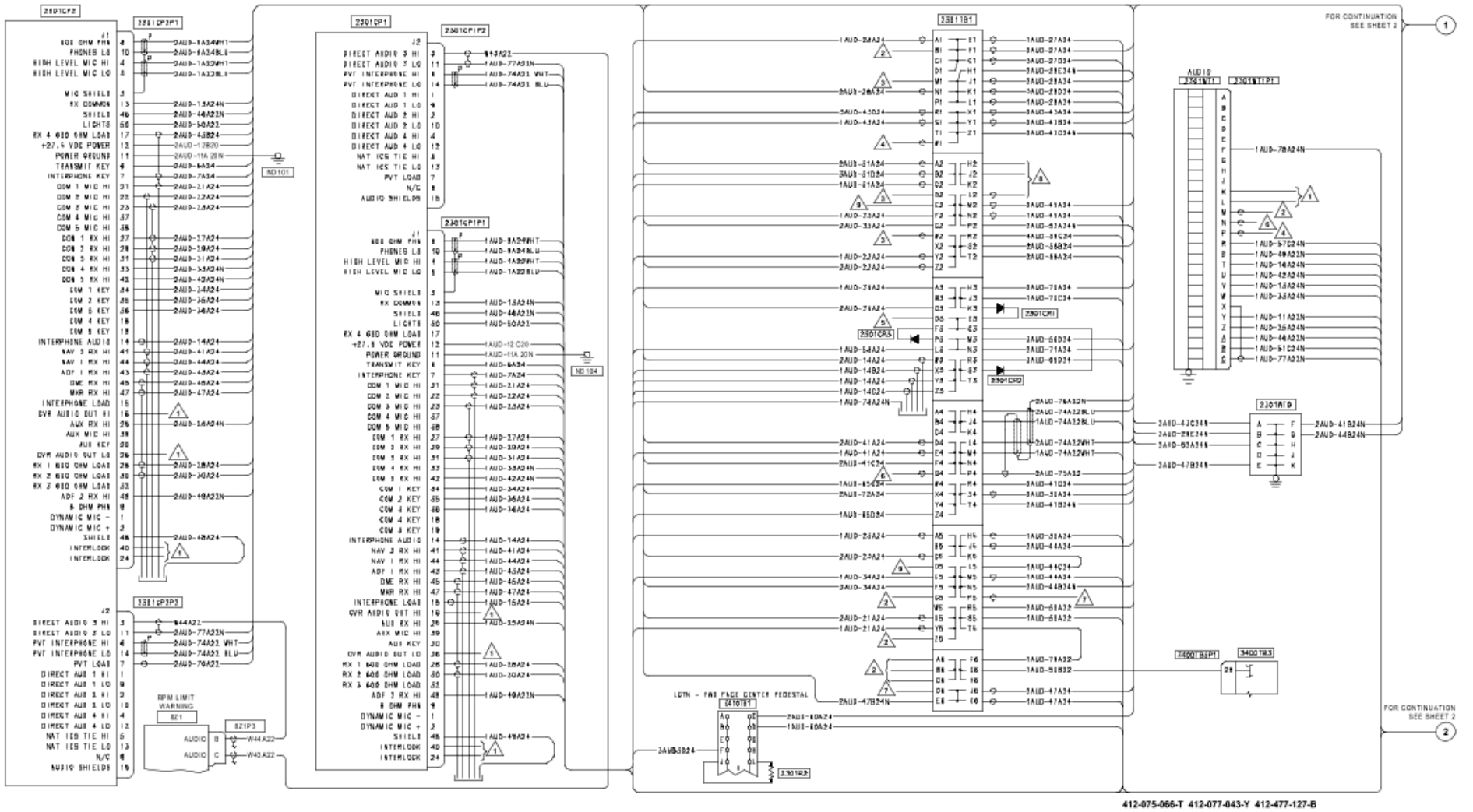
Figure 2. Intercommunication System (ICS) Wiring Diagram (S/N 33101 through 36086 and 36087 through 36637) (Sheet 4 of 5)



ICN-412-A-972900-B-97499-00008-A-001-01

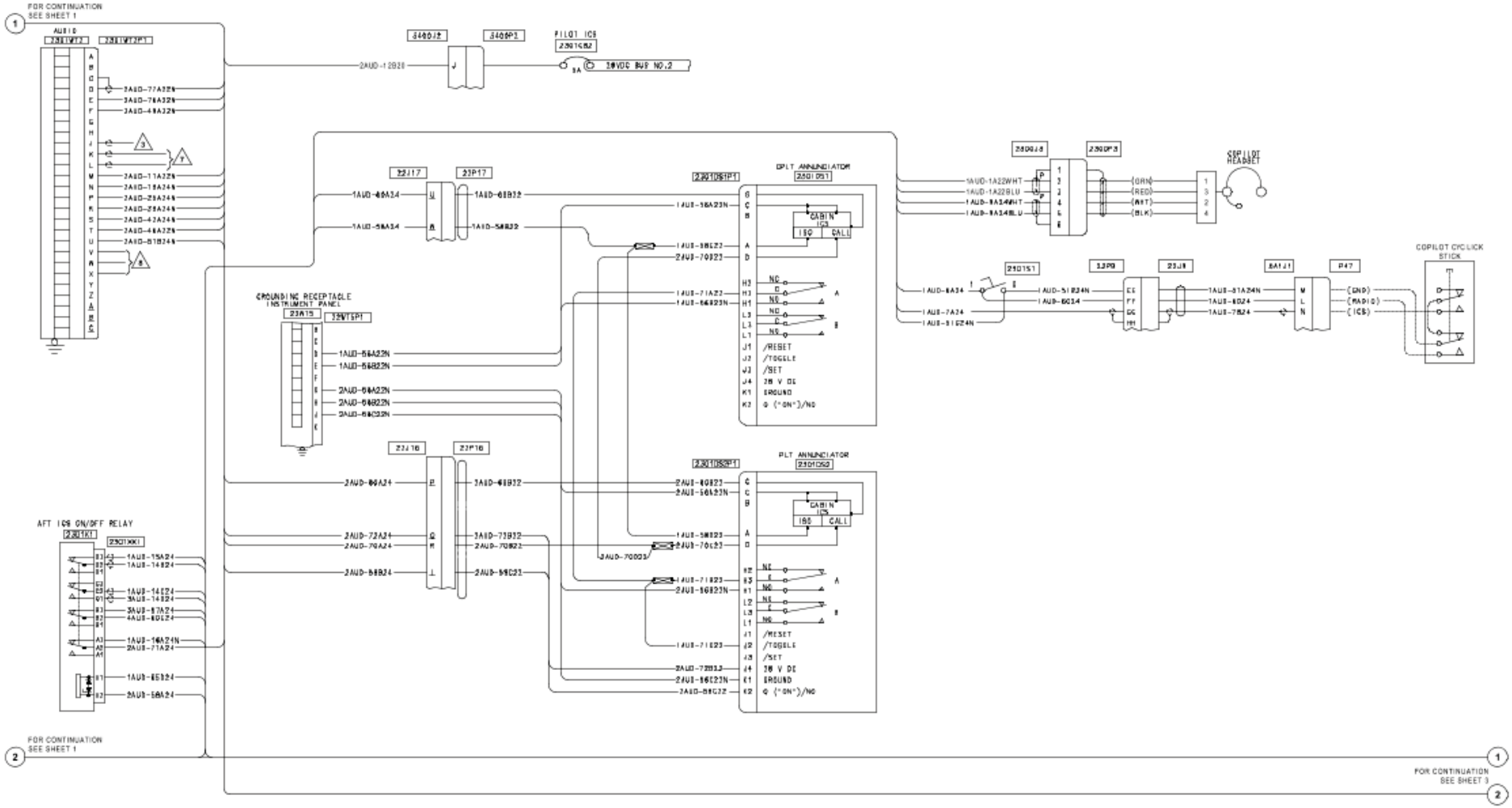
Figure 2. Intercommunication System (ICS) Wiring Diagram (S/N 33101 through 36086 and 36087 through 36637) (Sheet 5 of 5)

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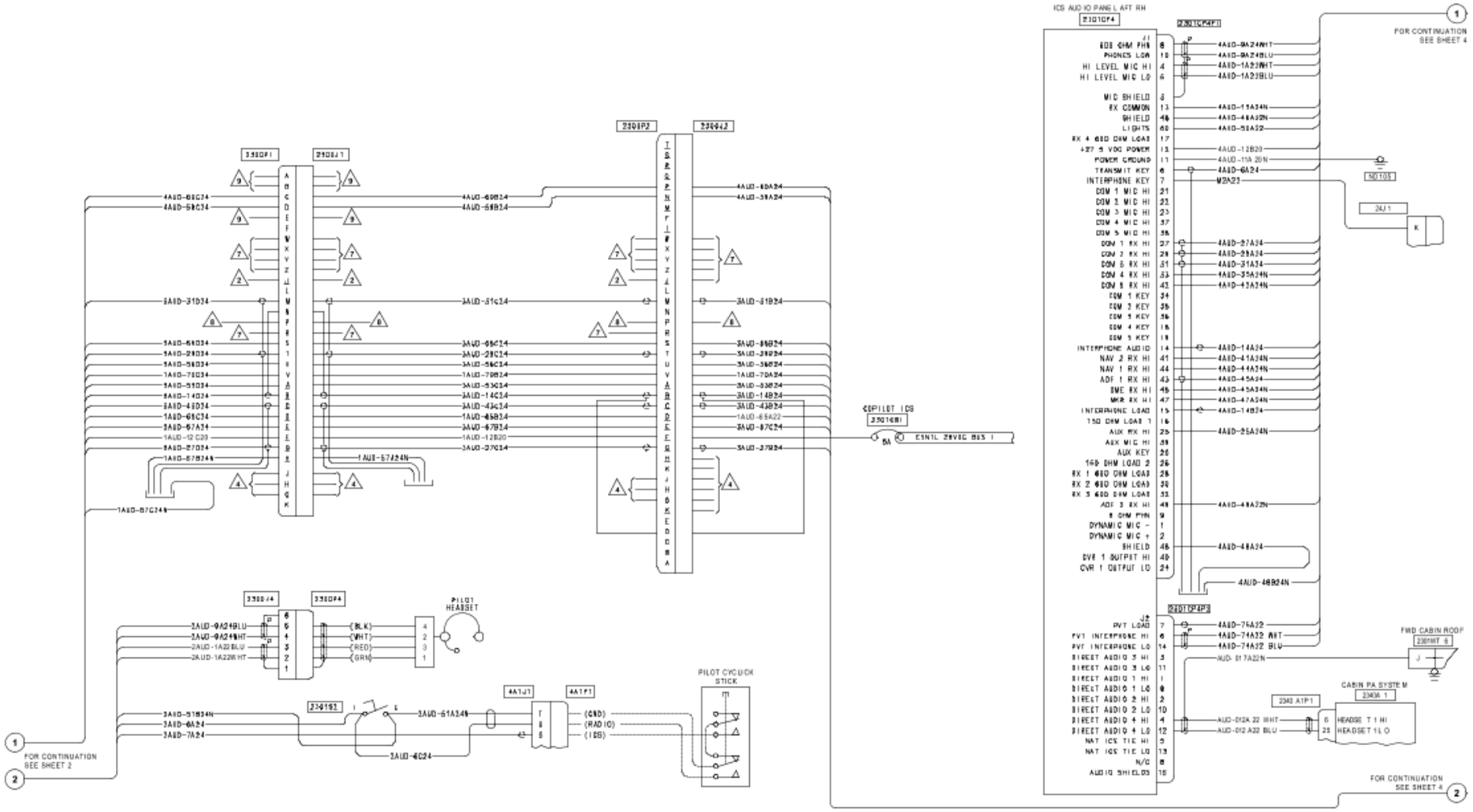
ICN-412-A-972900-B-97499-00009-A-001-01

Figure 3. Intercommunication System (ICS) Wiring Diagram (S/N 36638 through 36999) (Sheet 1 of 5)



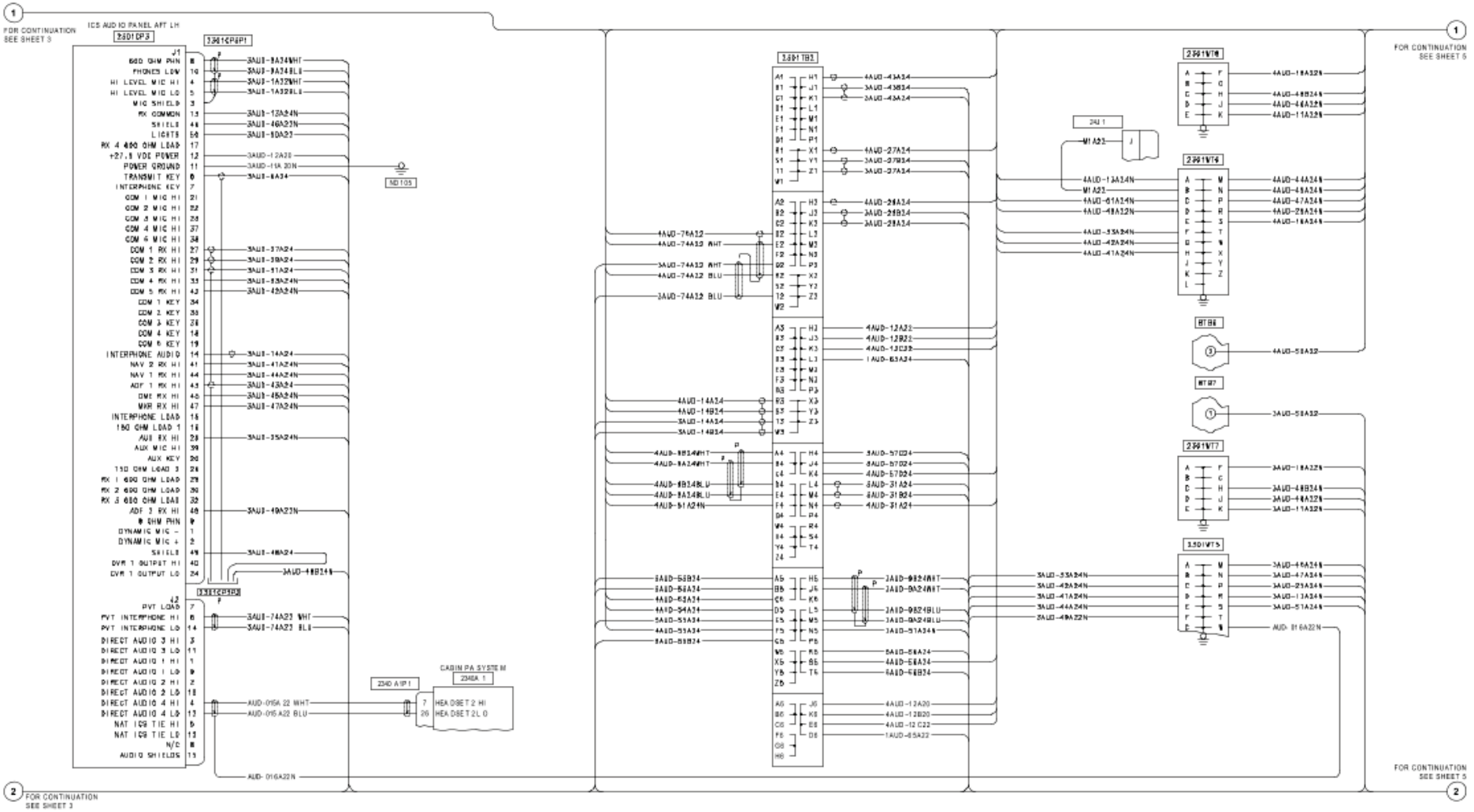
ICN-412-A-972900-B-97499-00010-A-001-01

Figure 3. Intercommunication System (ICS) Wiring Diagram (S/N 36638 through 36999) (Sheet 2 of 5)



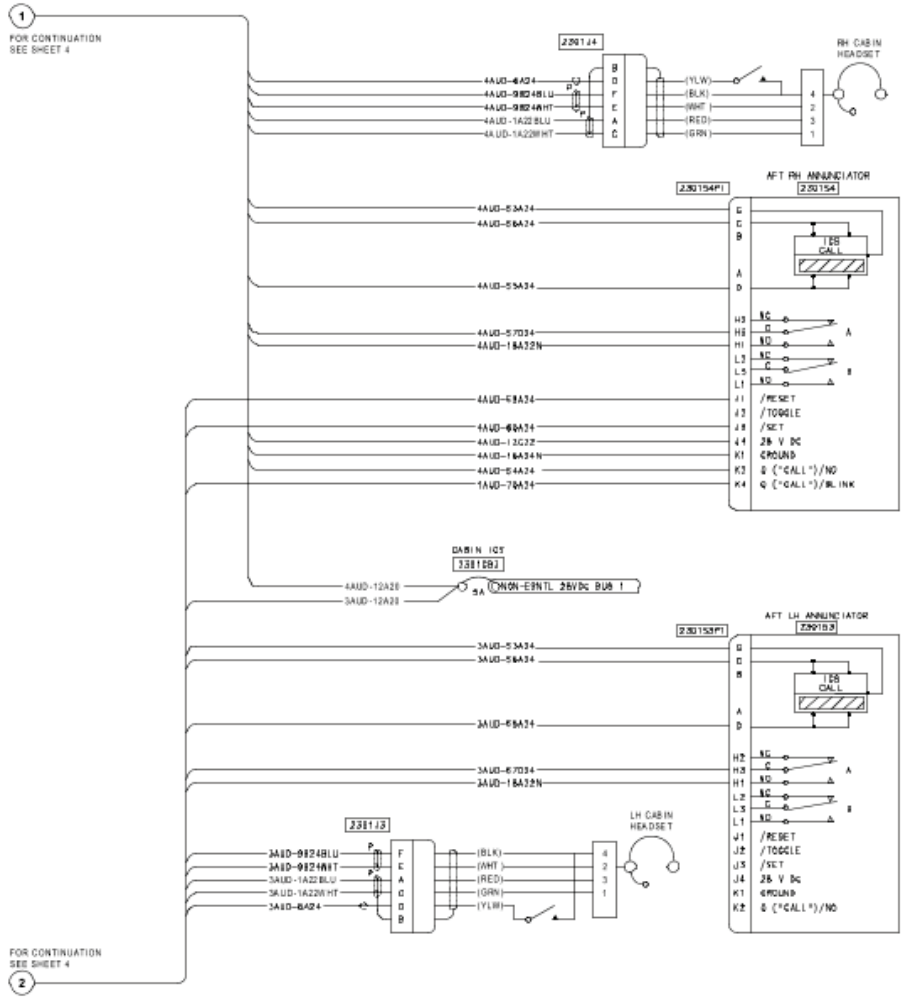
ICN-412-A-972900-B-97499-00011-A-001-01

Figure 3. Intercommunication System (ICS) Wiring Diagram (S/N 36638 through 36999) (Sheet 3 of 5)



ICN-412-A-972900-B-97499-00012-A-001-01

Figure 3. Intercommunication System (ICS) Wiring Diagram (S/N 36638 through 36999) (Sheet 4 of 5)



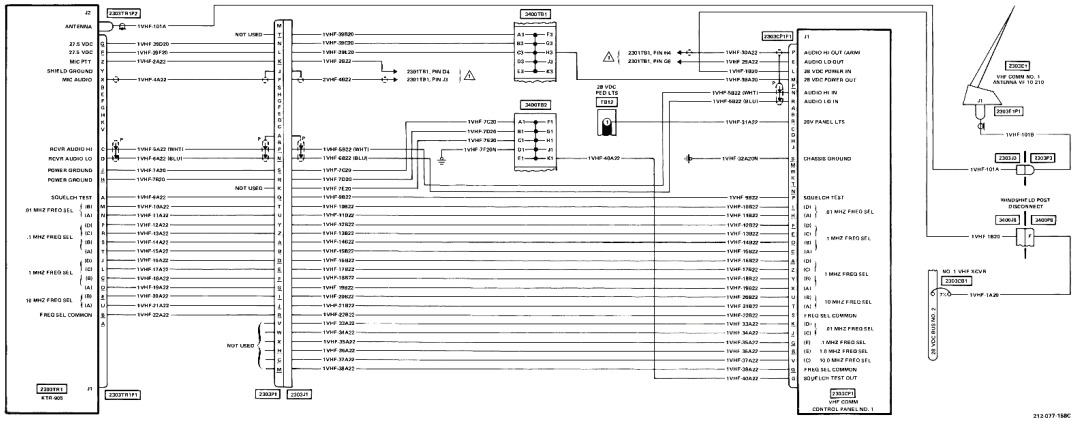
NOTES

1. REFER TO COCKPIT VOICE RECORDER WIRING DIAGRAM CHAPTER 98.
2. REFER TO VHF COMM NO. 1 SYSTEM WIRING DIAGRAM CHAPTER 97.
3. REFER TO VHF COMM NO. 2 SYSTEM WIRING DIAGRAM CHAPTER 97.
4. REFER TO ADF SYSTEM WIRING DIAGRAM CHAPTER 97.
5. REFER TO DIGITAL AFCS SYSTEM WIRING DIAGRAM CHAPTER 98.
6. REFER TO VHF NO. 2 SYSTEM WIRING DIAGRAM CHAPTER 97.
7. REFER TO VHF NAV NO. 1 SYSTEM WIRING DIAGRAM CHAPTER 97.
8. REFER TO DME WIRING DIAGRAM CHAPTER 97.
9. REFER TO FLIGHT DATA RECORDER WIRING DIAGRAM CHAPTER 98.

ICN-412-A-972900-B-97499-00013-A-001-01

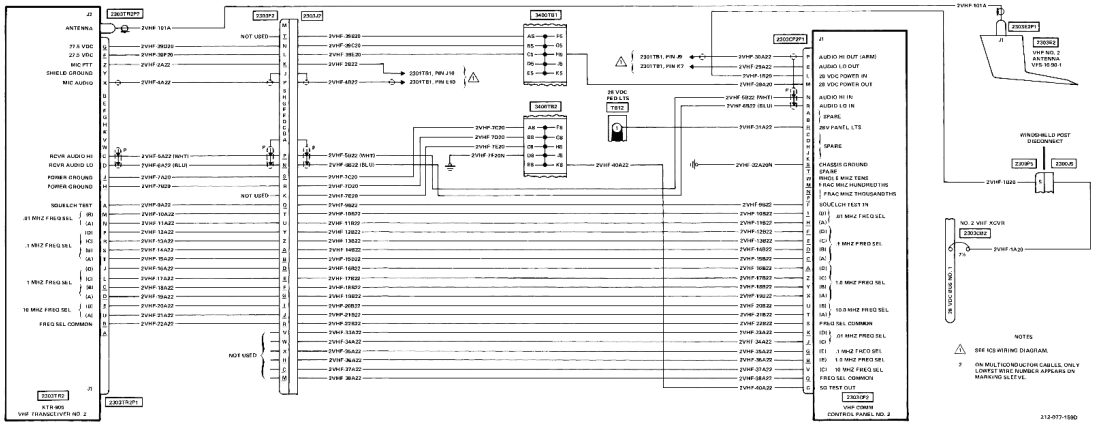
Figure 3. Intercommunication System (ICS) Wiring Diagram (S/N 36638 through 36999) (Sheet 5 of 5)

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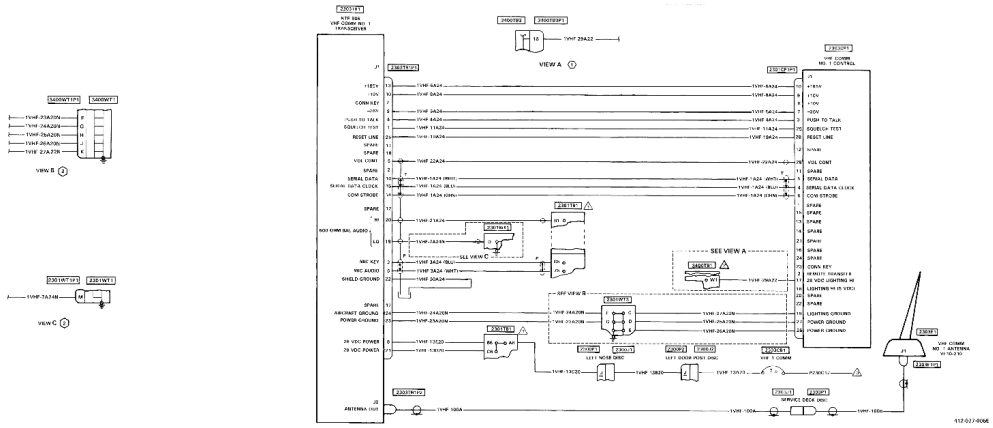
ICN-412-A-972900-B-97499-00014-A-001-01

Figure 4. VHF Communication System (KTR-905) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 2)



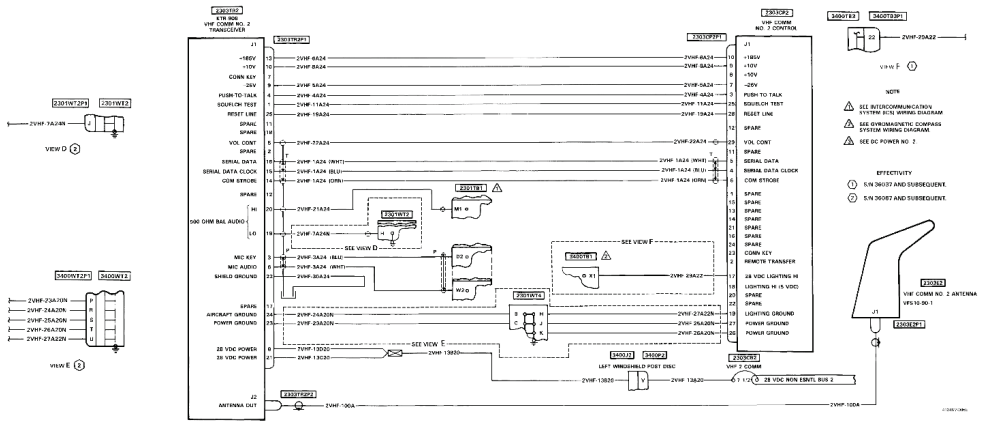
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Figure 4. VHF Communication System (KTR-905) Wiring Diagram (S/N 33001 through 33100) (Sheet 2 of 2)



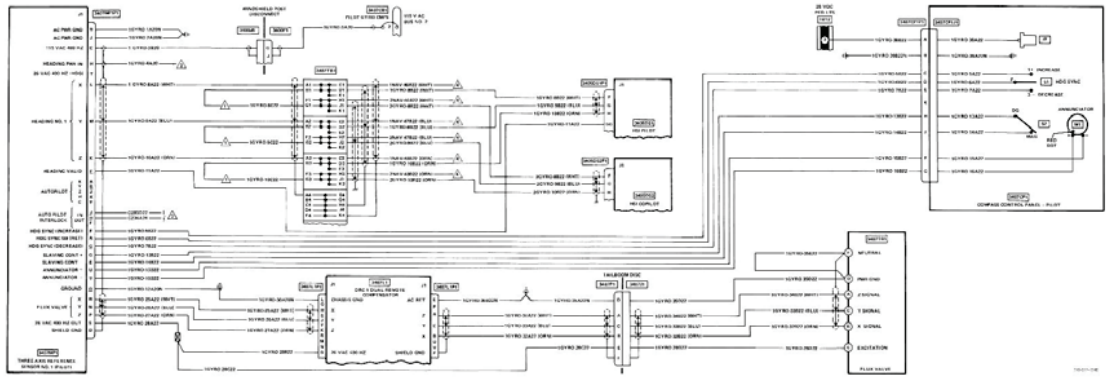
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Figure 5. VHF Communication System (KTR-908) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 2)



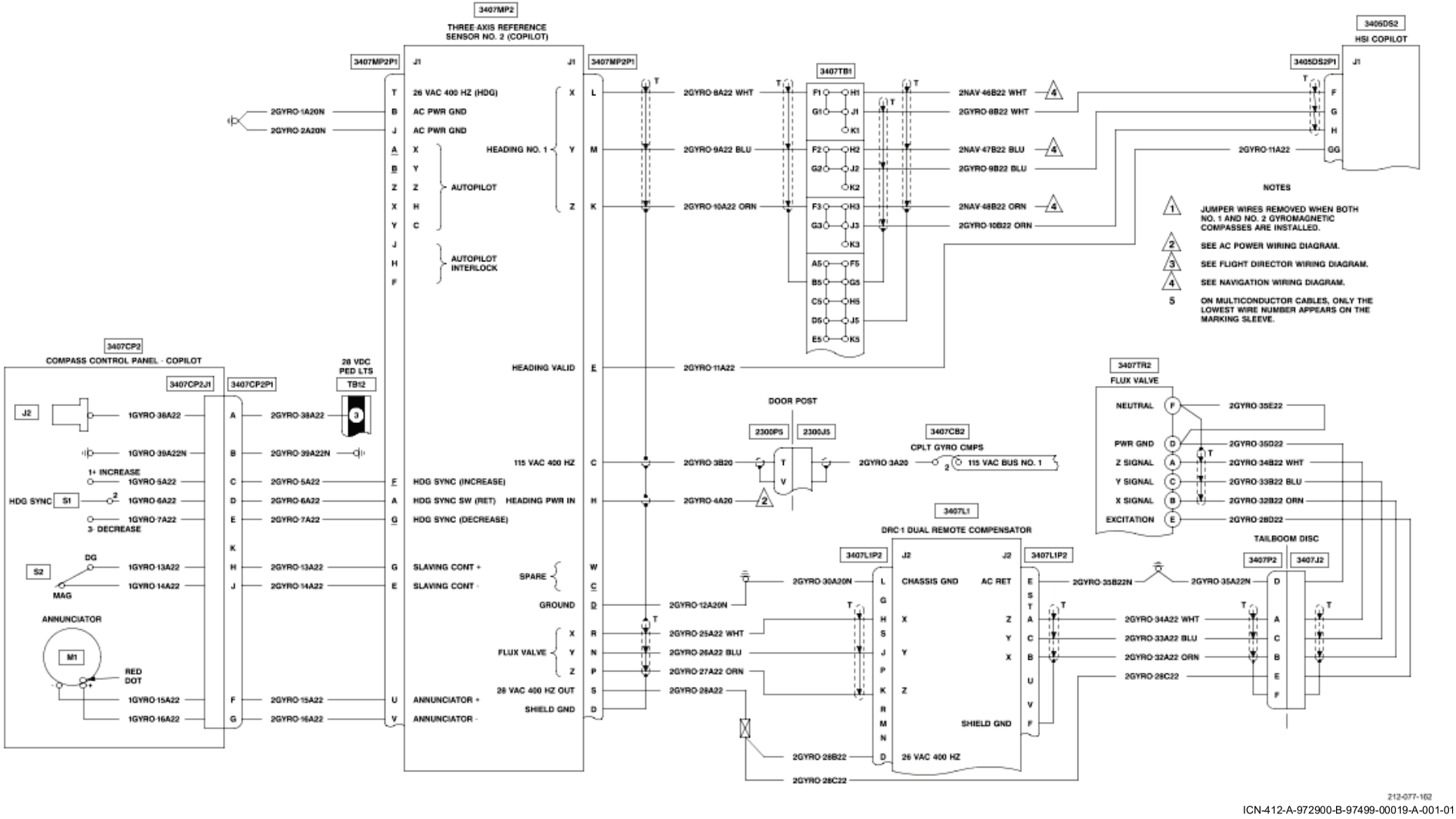
ICN-412-A-972900-B-97499-00017-A-001-01

Figure 5. VHF Communication System (KTR-908) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 2 of 2)



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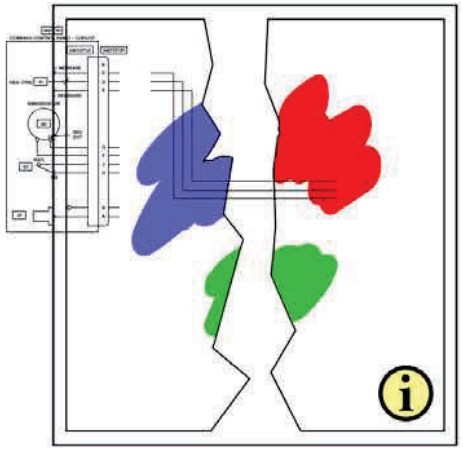
Figure 6. Gyromagnetic Compass System Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 2)



212-077-102

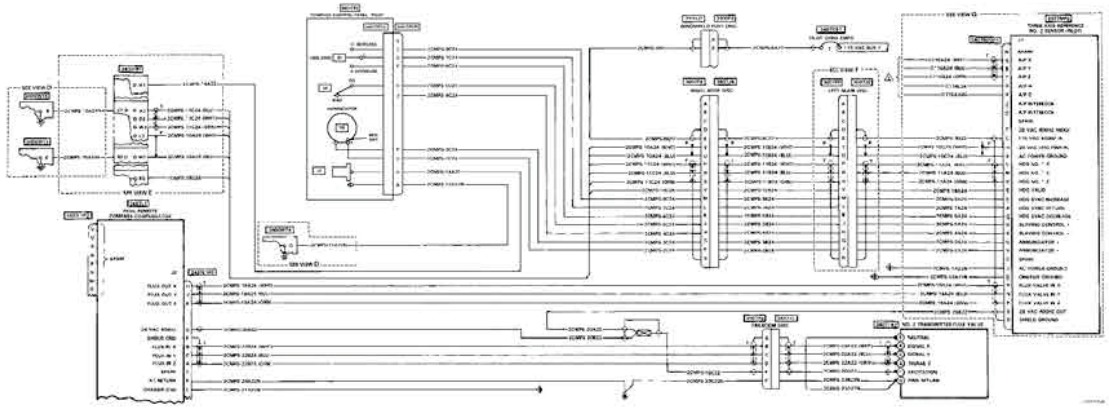
ICN-412-A-972900-B-97499-00019-A-001-01

Figure 6. Gyromagnetic Compass System Wiring Diagram (S/N 33001 through 33100) (Sheet 2 of 2)



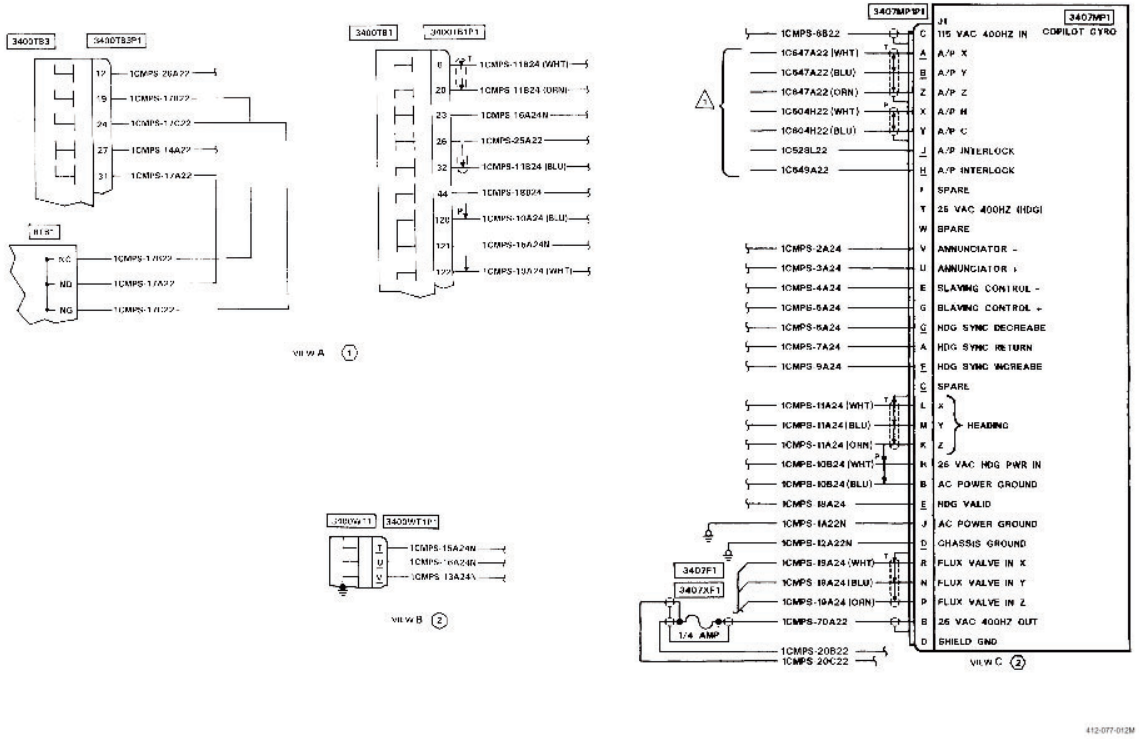
ICN-412-A-972900-B-97499-00020-A-001-01

Figure 7. Gyromagnetic Compass System Wiring Diagram (S/N 33101 through 36086 and 36087 through 36247) (Sheet 1 of 4)



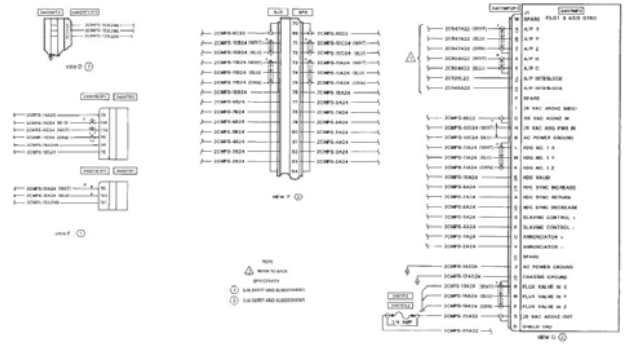
ICN-412-A-972900-B-97499-00021-A-001-01

Figure 7. Gyromagnetic Compass System Wiring Diagram (S/N 33101 through 36086 and 36087 through 36247) (Sheet 2 of 4)



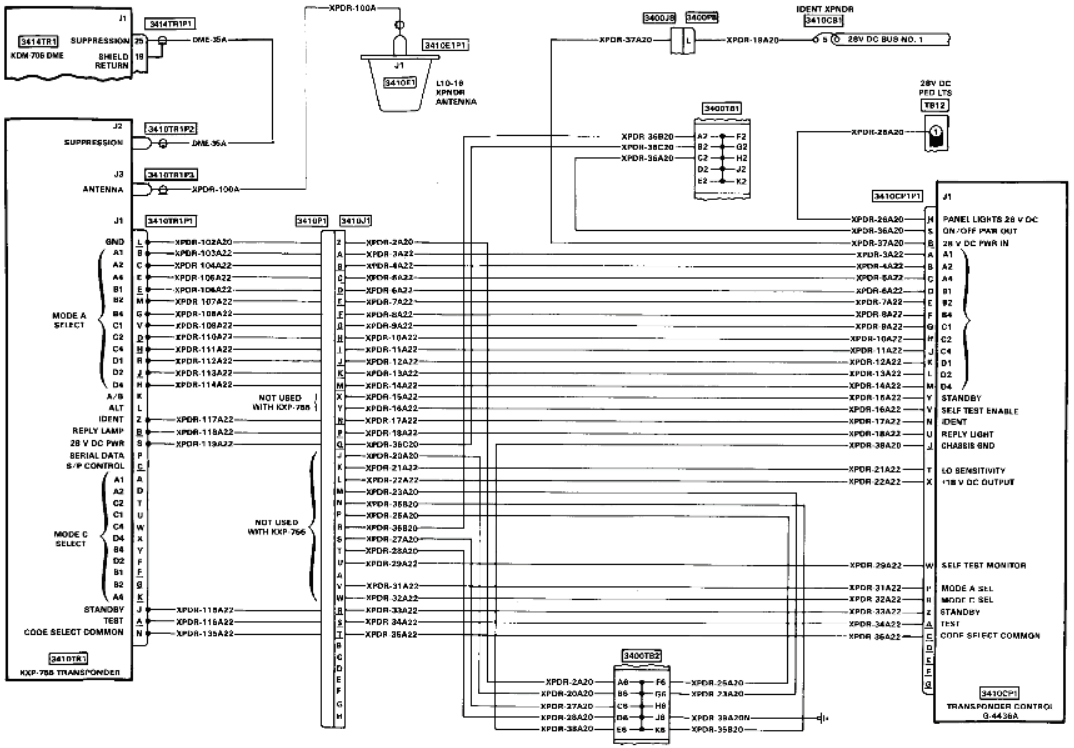
ICN-412-A-972900-B-97499-00022-A-001-01

Figure 7. Gyromagnetic Compass System Wiring Diagram (S/N 33101 through 36086 and 36087 through 36247) (Sheet 3 of 4)



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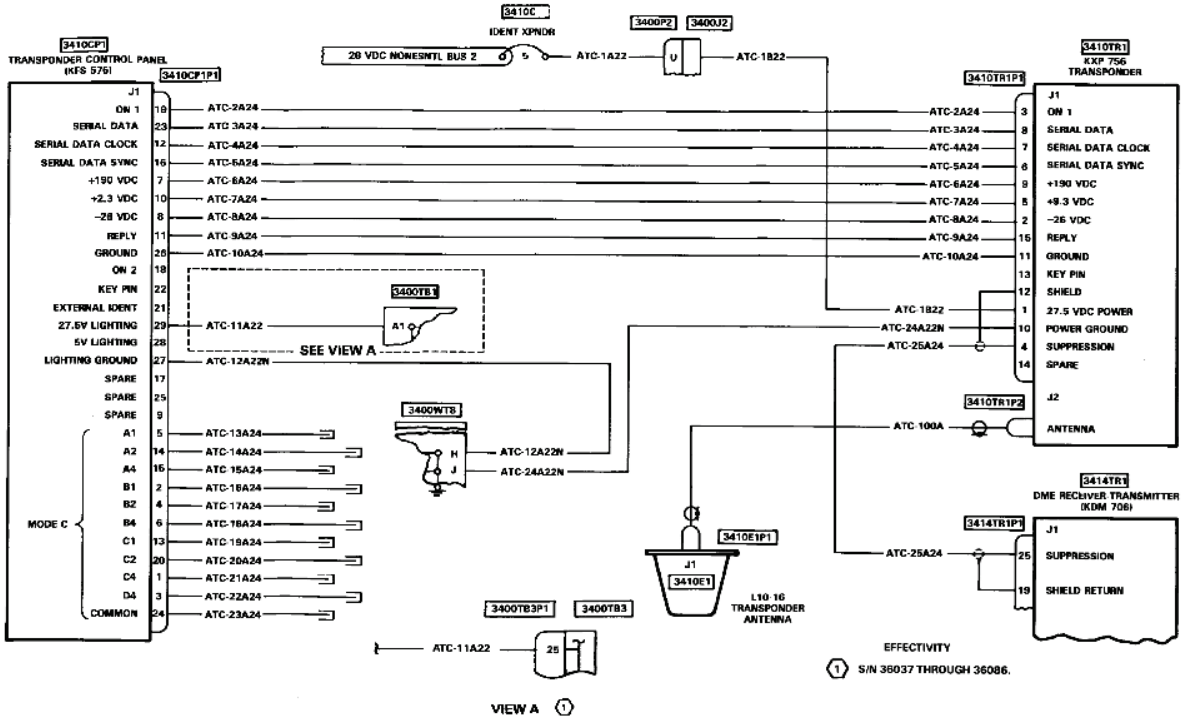
Figure 7. Gyromagnetic Compass System Wiring Diagram (S/N 33101 through 36086 and 36087 through 36247) (Sheet 4 of 4)



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Figure 8. ATC Transponder System (KXP-755) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 1)

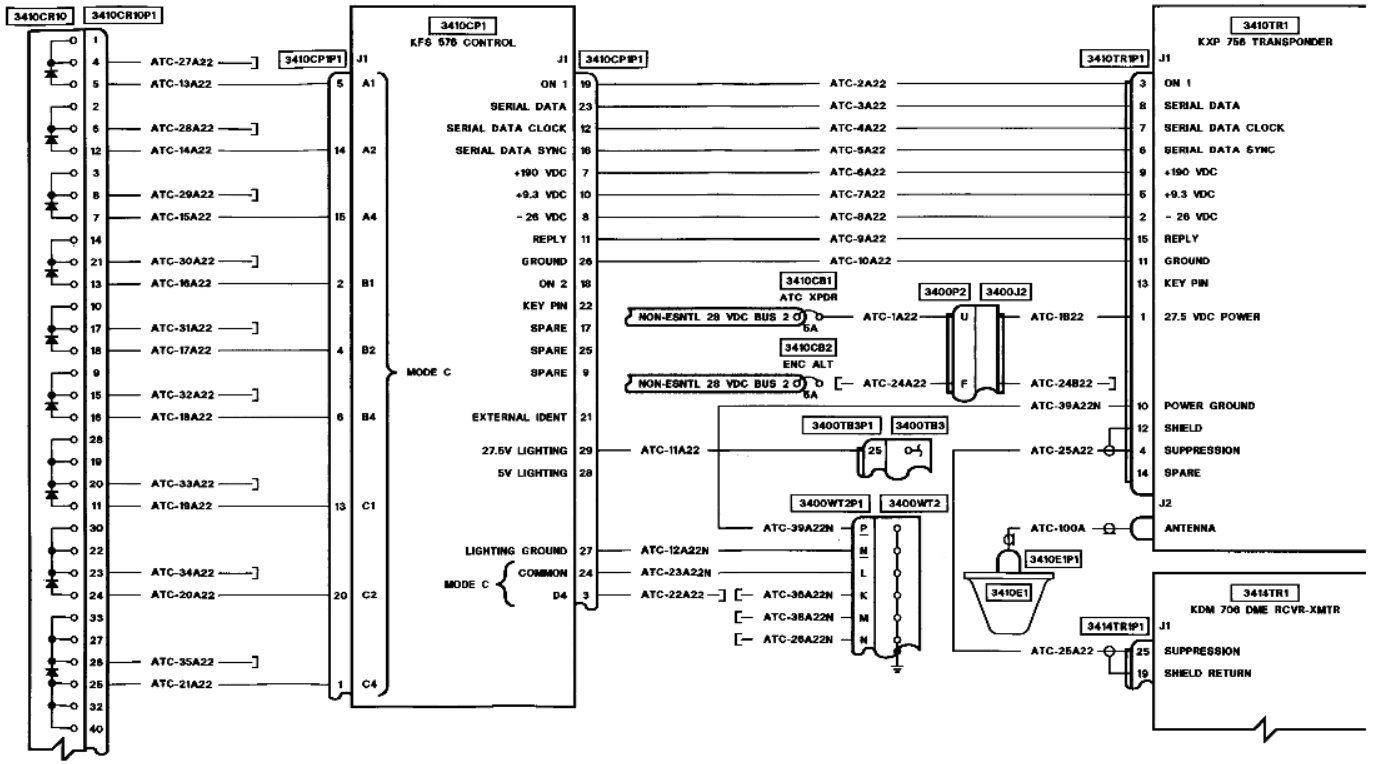
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Figure 9. ATC Transponder System (KXP-755) Wiring Diagram (S/N 33101 through 36086) (Sheet 1 of 1)

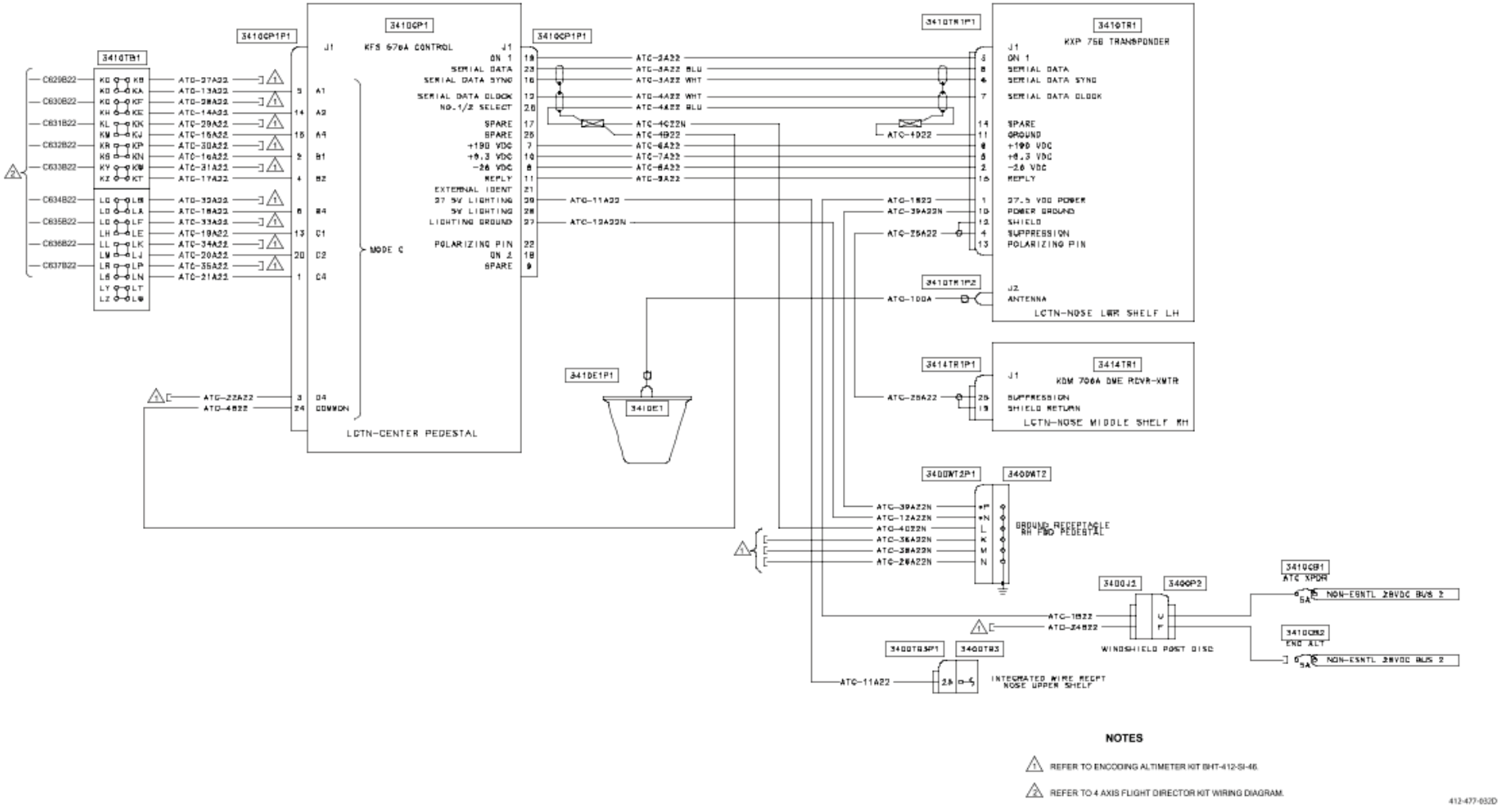
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Figure 10. ATC Transponder System (KXP-755) Wiring Diagram (S/N 36087 through 36124) (Sheet 1 of 1)

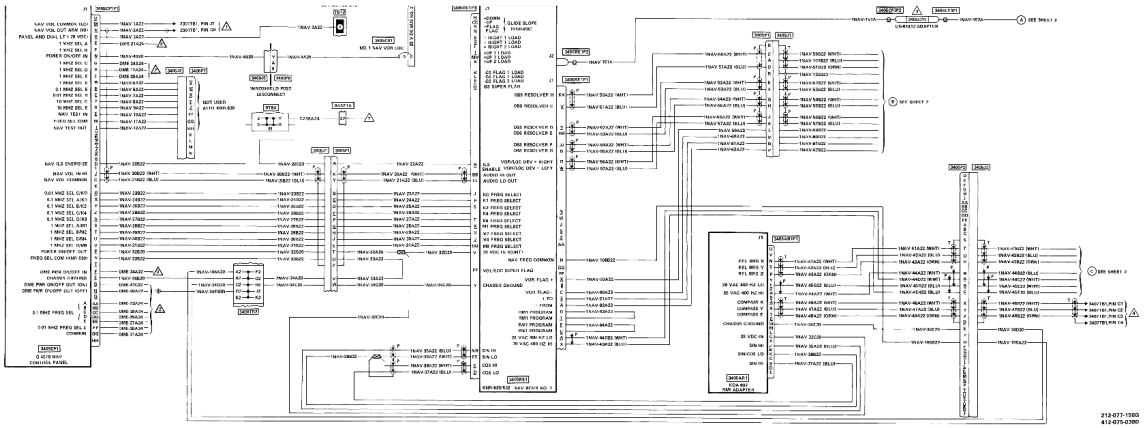
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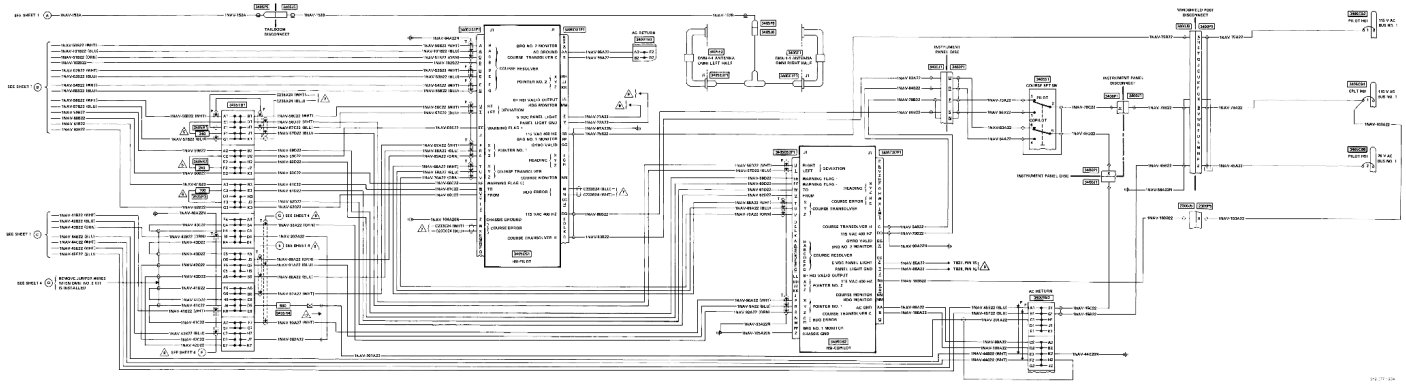
Figure 11. ATC Transponder System (KXP-756) Wiring Diagram (S/N 36125 through 36272 and 36274 and 36275) (Sheet 1 of 1)

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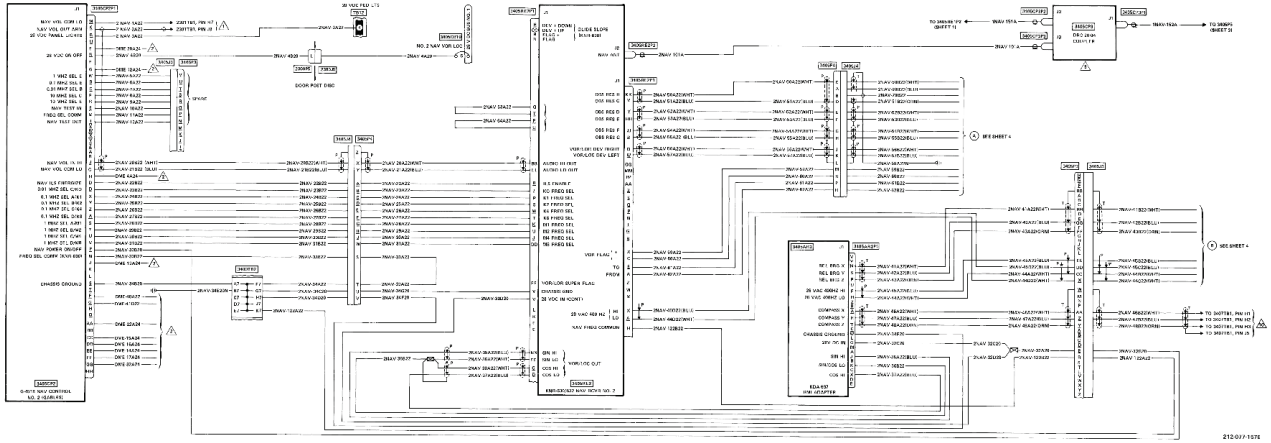
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Figure 12. VOR/LOC/GS Navigation System(s) (KNR-630/632) (S/N 33001 through 33100) (Sheet 1 of 5)



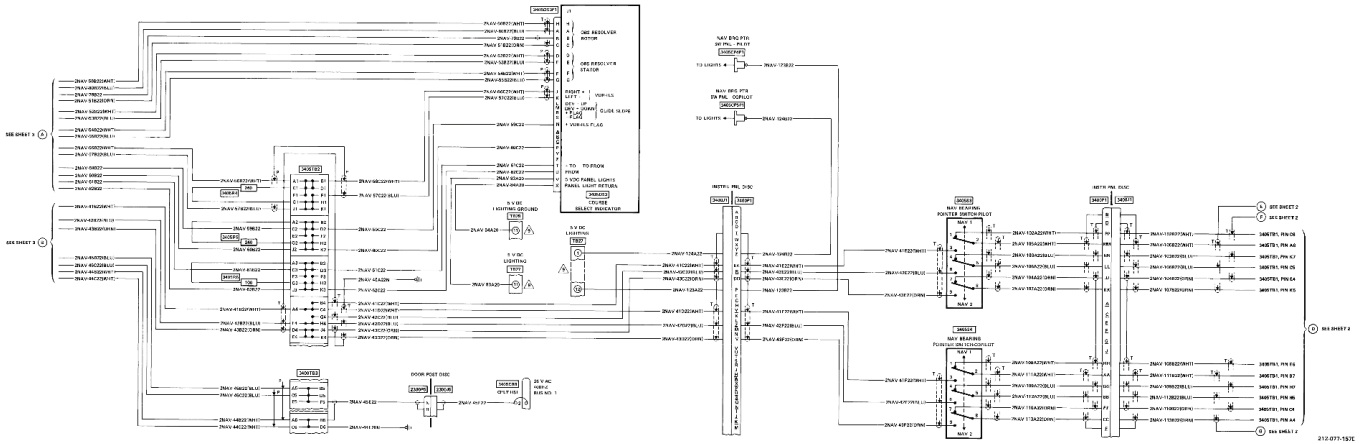
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Figure 12. VOR/LOC/GS Navigation System(s) (KNR-630/632) (S/N 33001 through 33100) (Sheet 2 of 5)



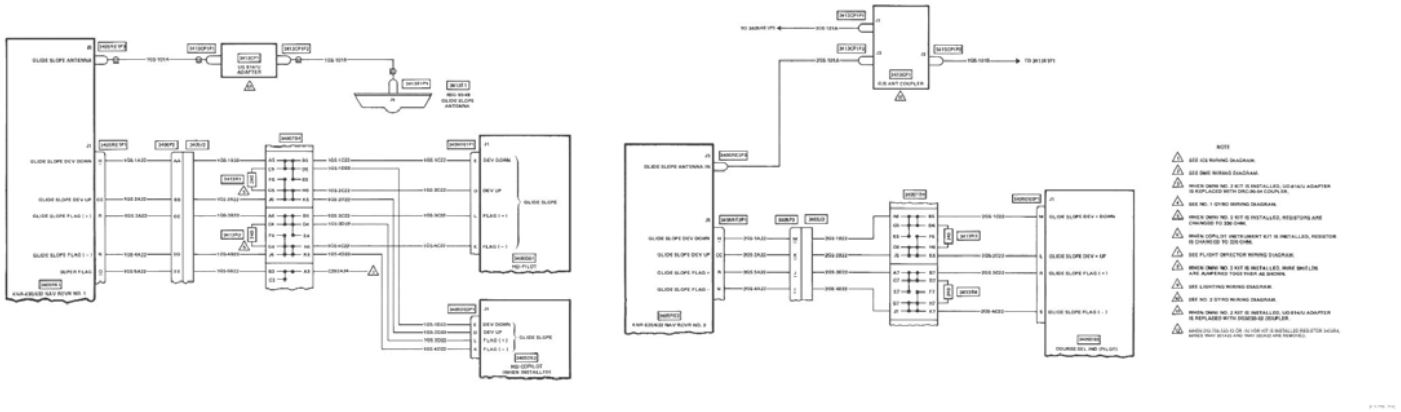
ICN-412-A-972900-B-97499-00030-A-001-01

Figure 12. VOR/LOC/GS Navigation System(s) (KNR-630/632) (S/N 33001 through 33100) (Sheet 3 of 5)



ICN-412-A-972900-B-97499-00031-A-001-01

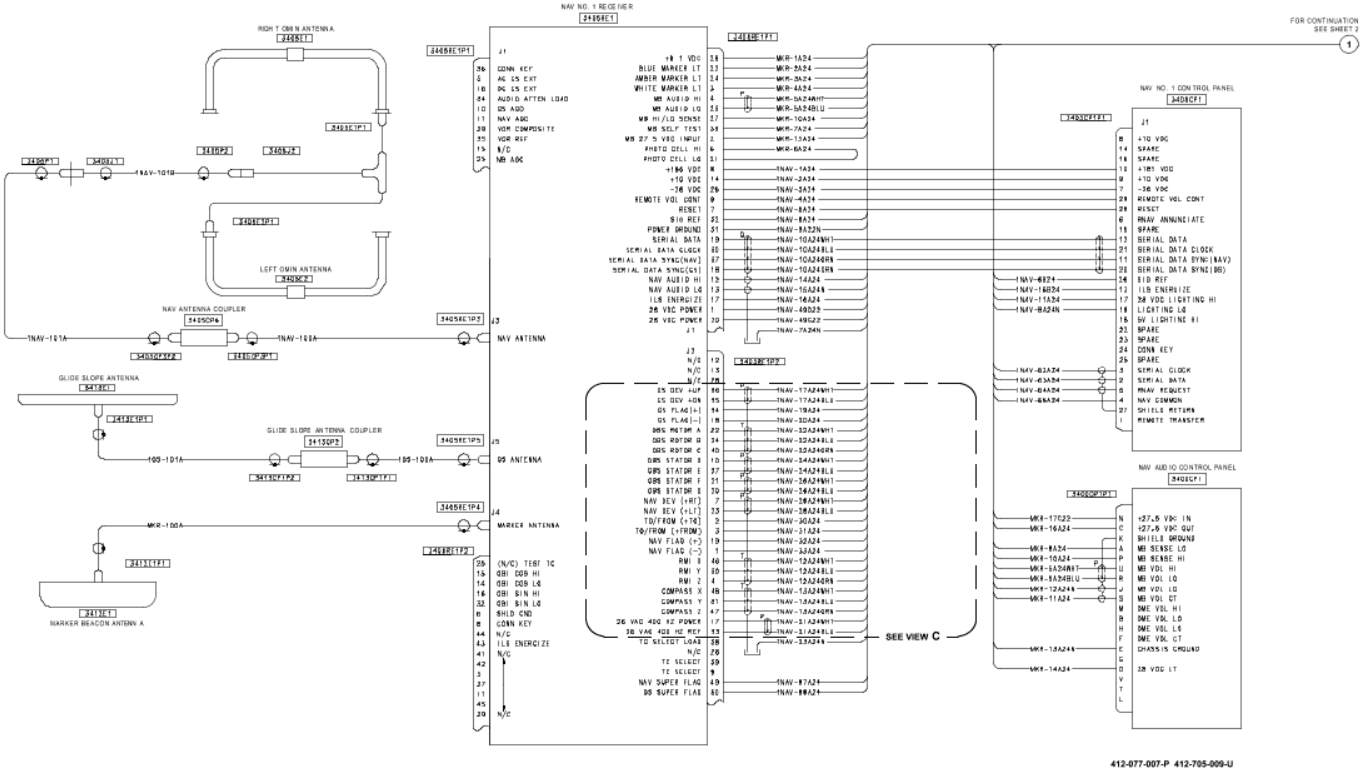
Figure 12. VOR/LOC/GS Navigation System(s) (KNR-630/632) (S/N 33001 through 33100) (Sheet 4 of 5)



ICN-412-A-972900-B-97499-00032-A-001-01

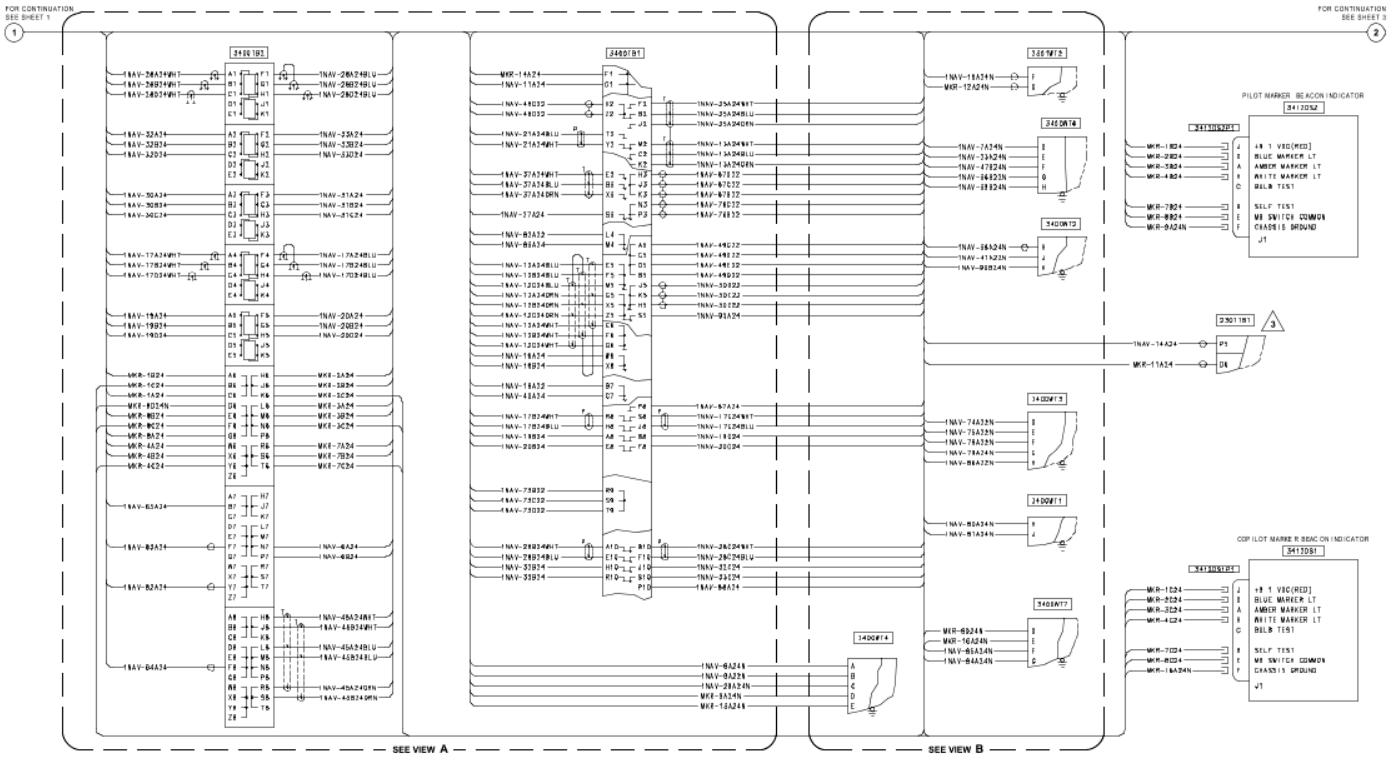
Figure 12. VOR/LOC/GS Navigation System(s) (KNR-630/632) (S/N 33001 through 33100) (Sheet 5 of 5)

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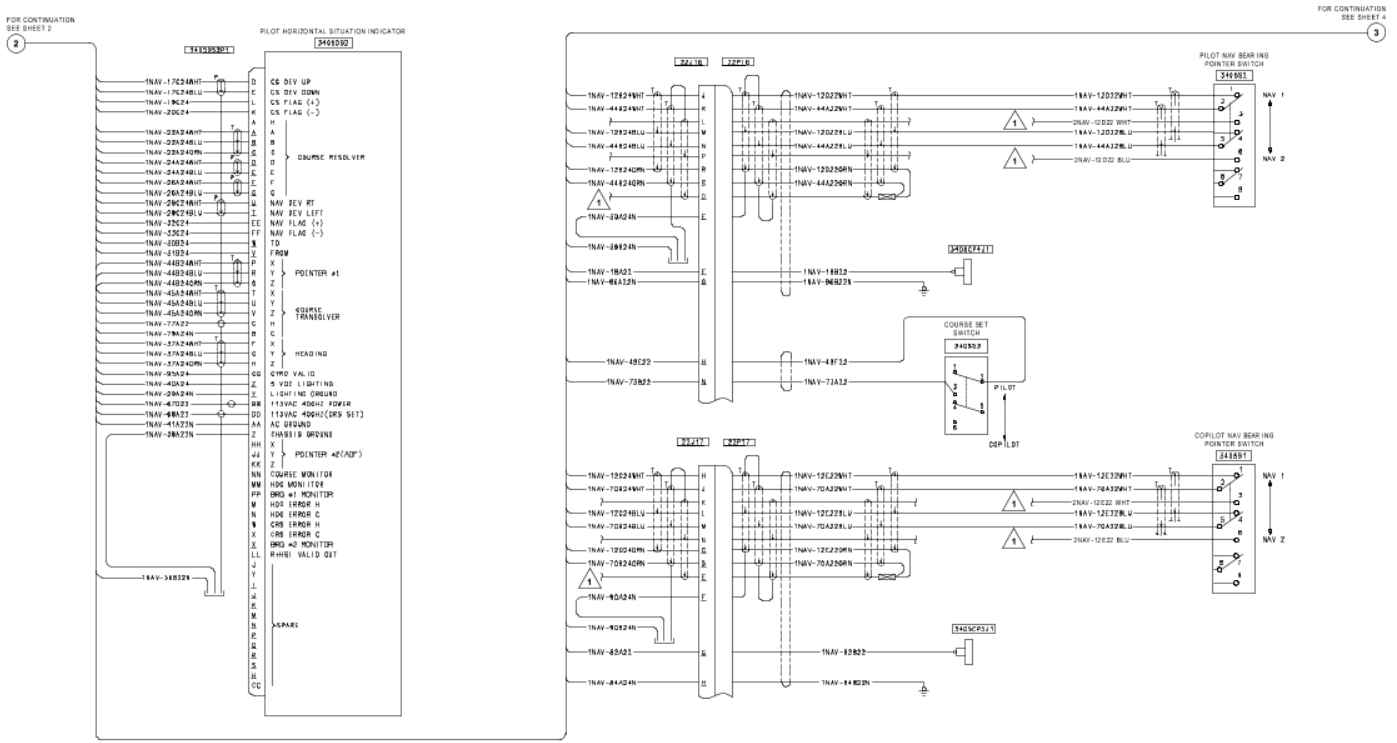
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Figure 13. VHF NAV System No. 1 (KNR-634/634A) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 6)



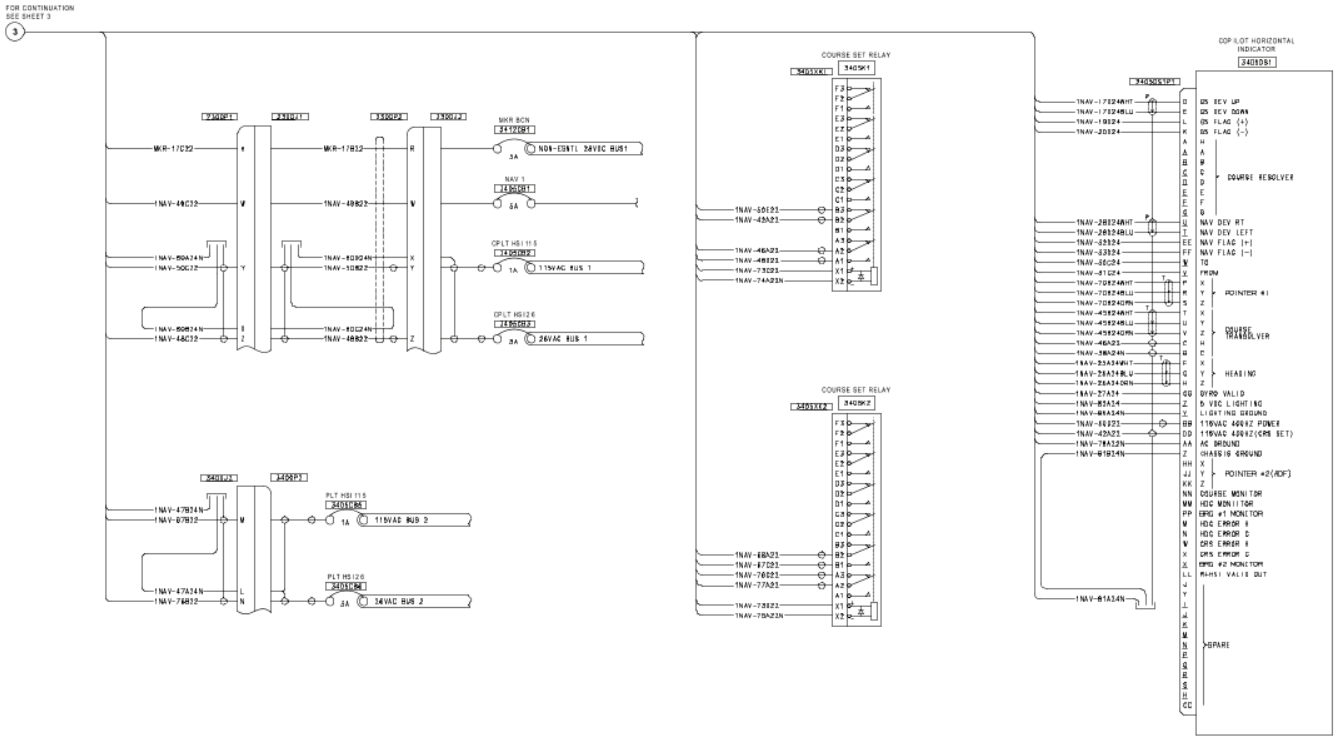
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Figure 13. VHF NAV System No. 1 (KNR-634/634A) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 2 of 6)



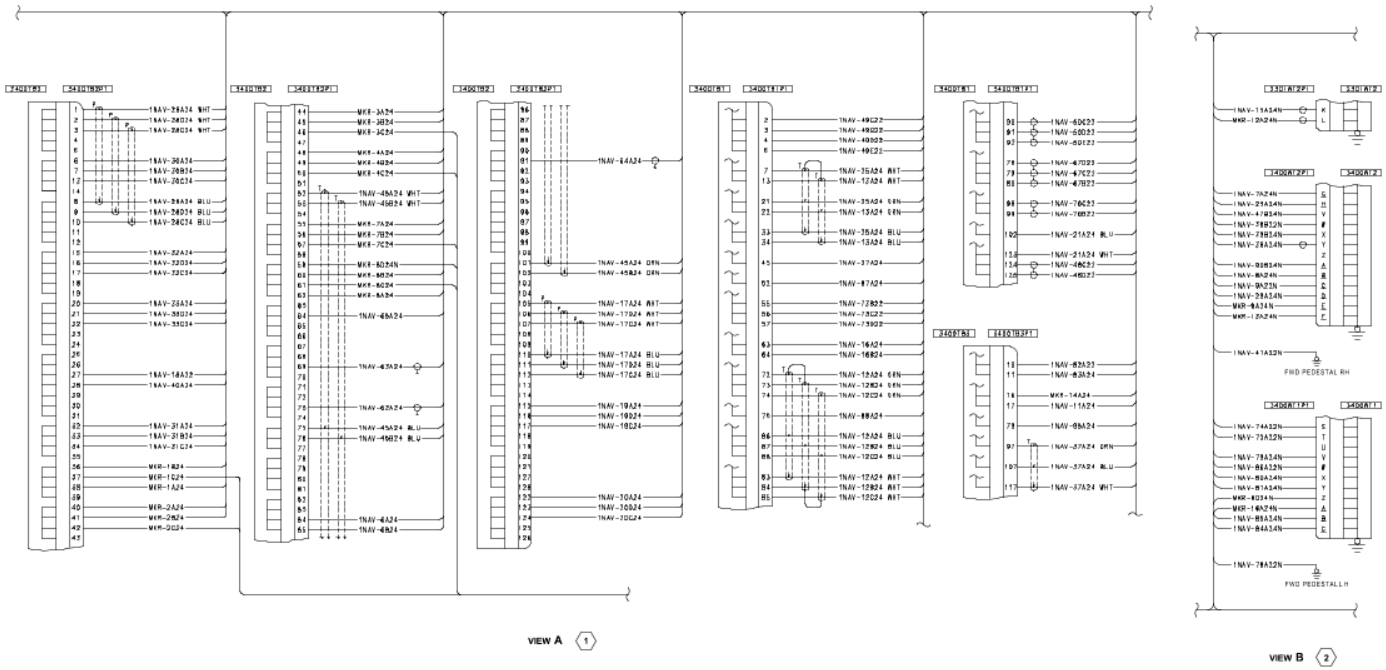
ICN-412-A-972900-B-97499-00035-A-001-01

Figure 13. VHF NAV System No. 1 (KNR-634/634A) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 3 of 6)



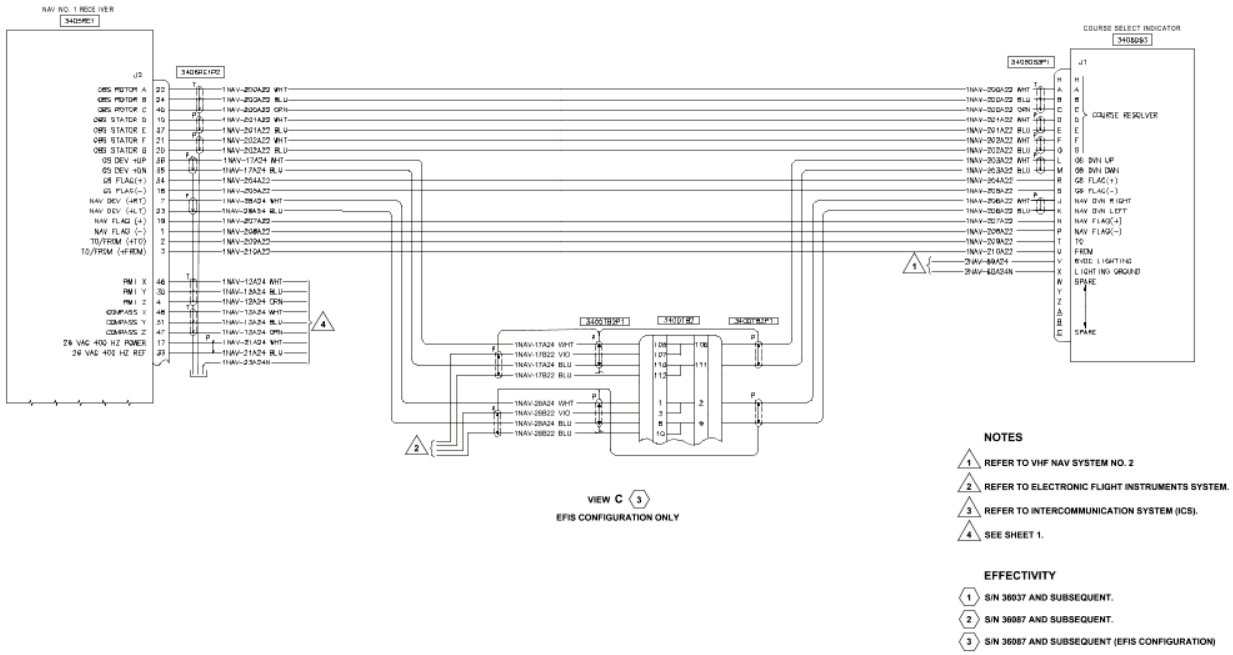
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Figure 13. VHF NAV System No. 1 (KNR-634/634A) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 4 of 6)



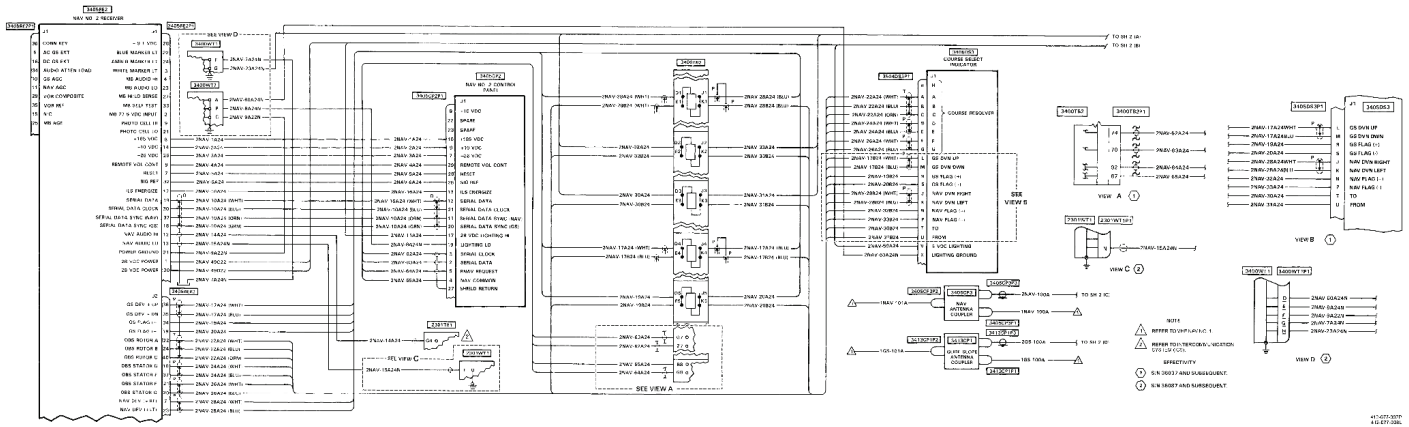
ICN-412-A-972900-B-97499-00037-A-001-01

Figure 13. VHF NAV System No. 1 (KNR-634/634A) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 5 of 6)



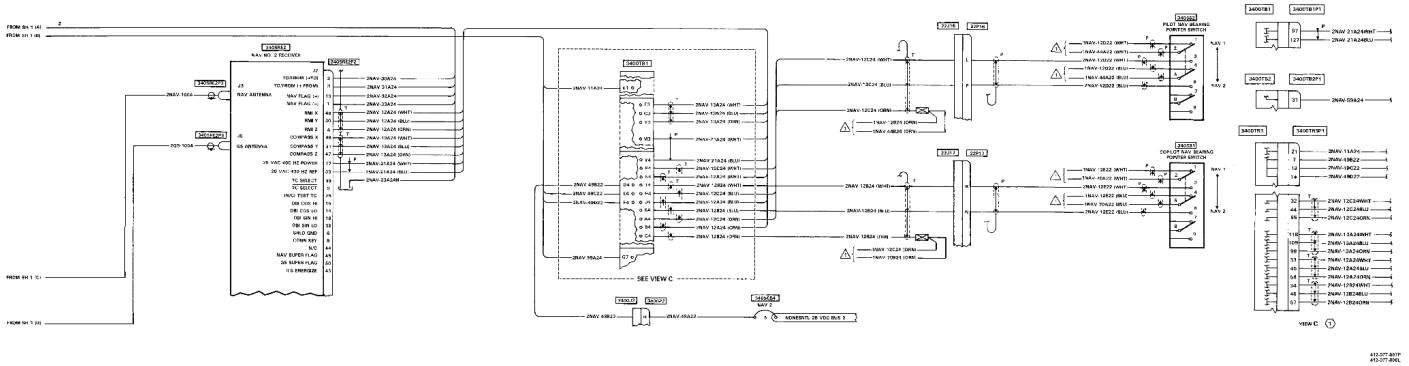
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Figure 13. VHF NAV System No. 1 (KNR-634/634A) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 6 of 6)



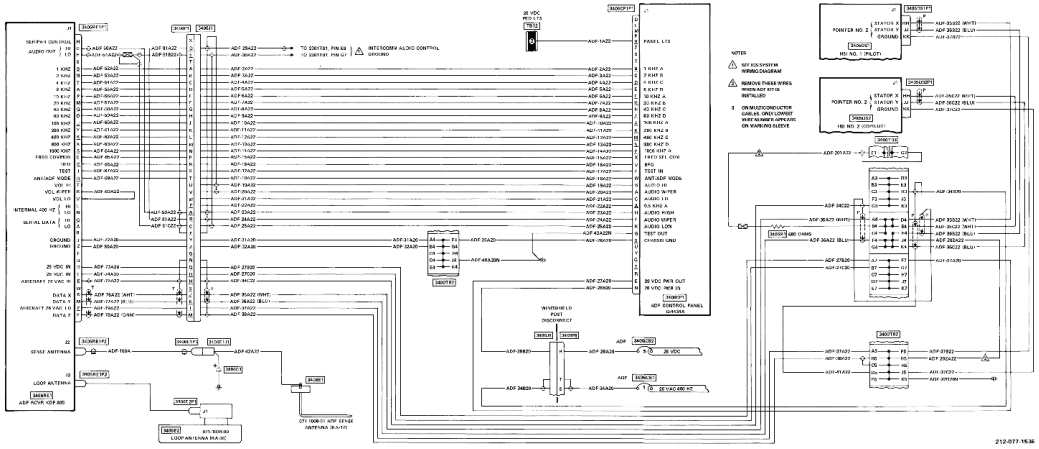
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Figure 14. VHF NAV System No. 2 (KNR-634) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 2)



ICN-412-A-972900-B-97499-00040-A-001-01

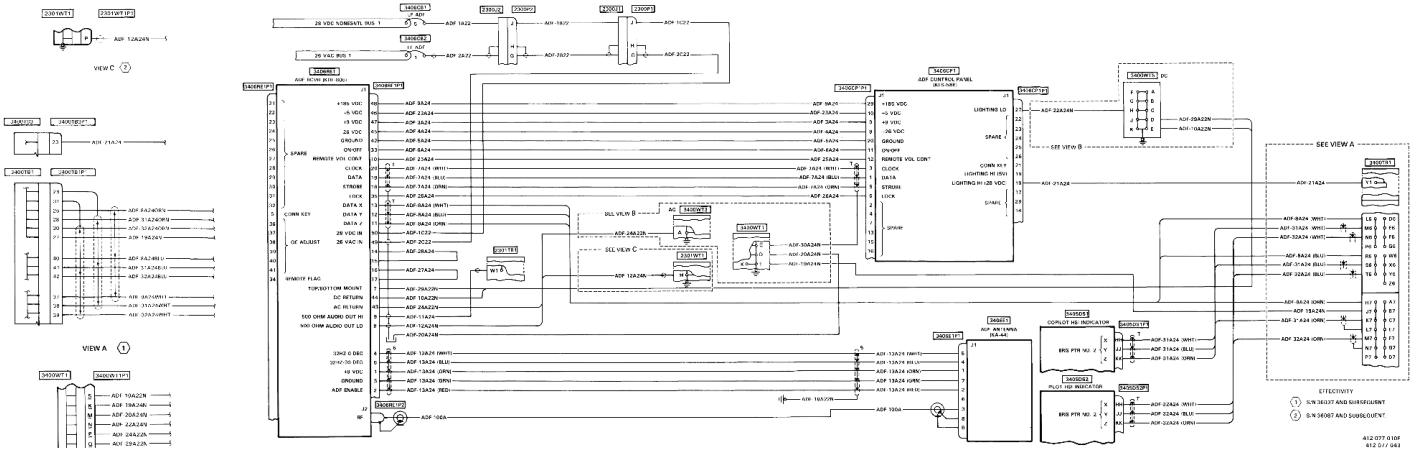
Figure 14. VHF NAV System No. 2 (KNR-634) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 2 of 2)



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Figure 15. VHF Navigation System (KDF-805) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 1)

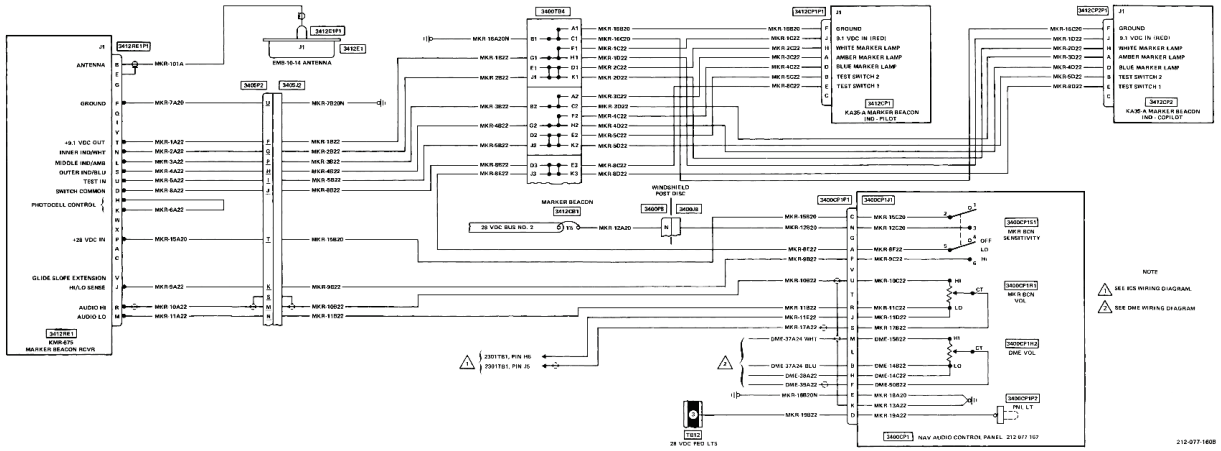
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Figure 16. ADF Navigation System (KDF-806) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 1)

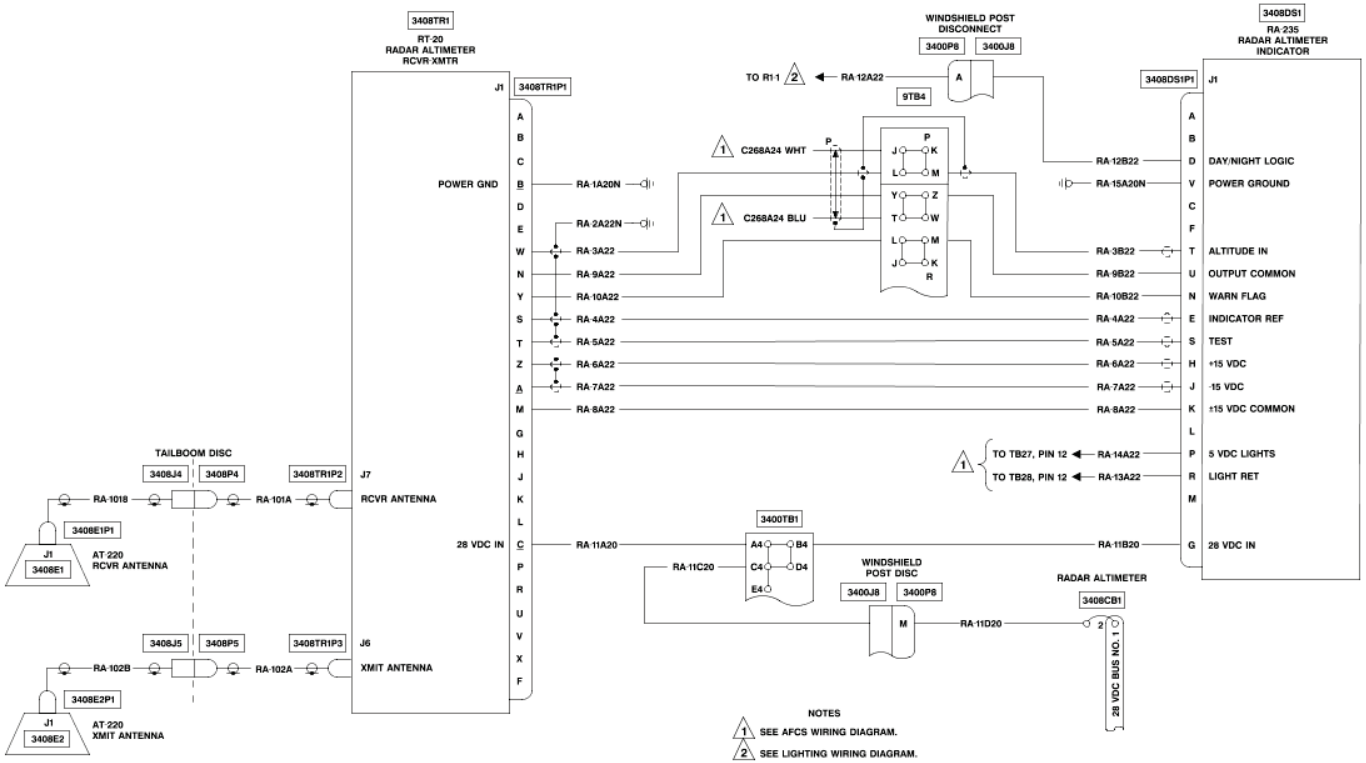
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Figure 17. Marker Beacon System (KMR-675) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 1)

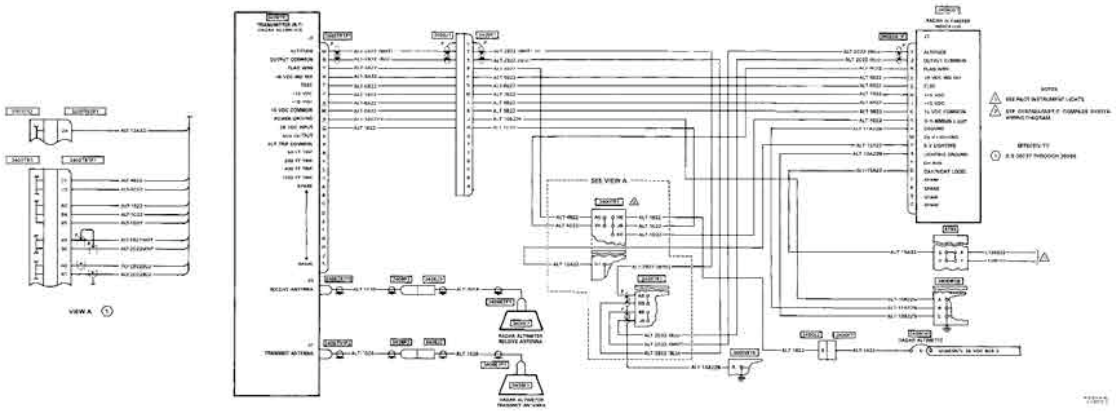
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Figure 18. Radar Altimeter Navigation System (RA-235) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 1)

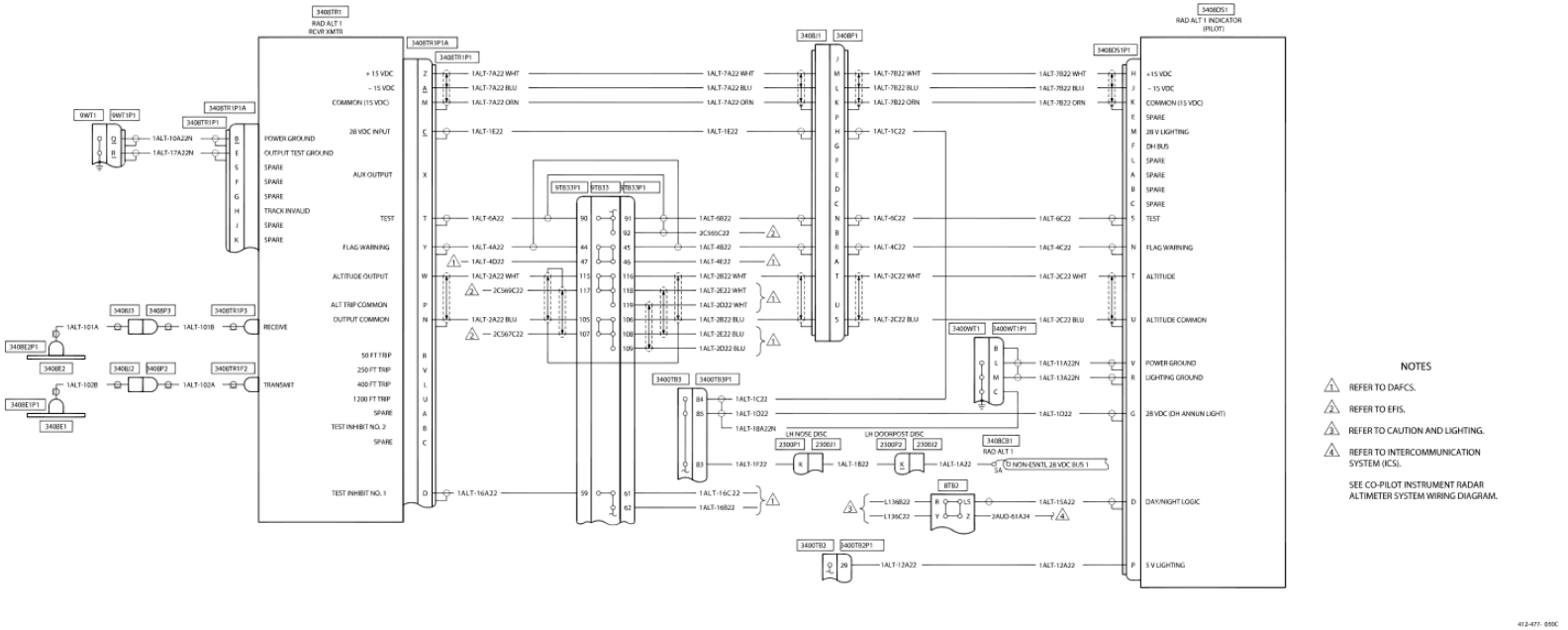
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Figure 19. Radar Altimeter Navigation System Wiring Diagram (S/N 33101 through 36086) (Sheet 1 of 1)

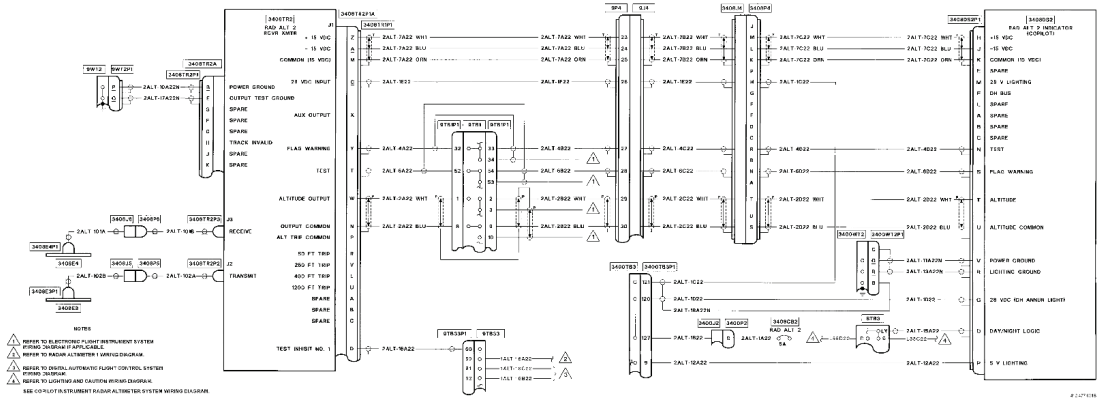
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Figure 20. Radar Altimeter 1 Wiring Diagram (S/N 36087 and Subsequent) (Sheet 1 of 1)

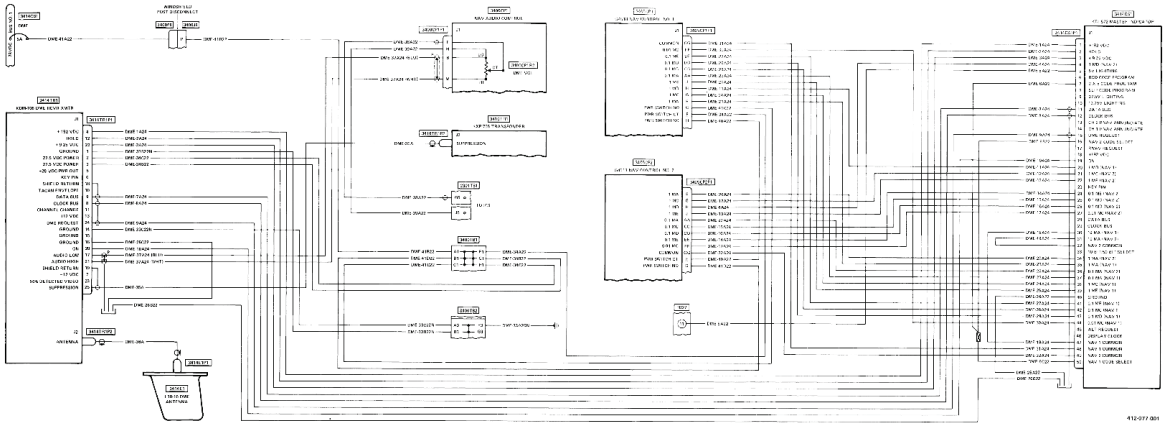
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Figure 21. Radar Altimeter 2 Wiring Diagram (S/N 36087 and Subsequent) (Sheet 1 of 1)

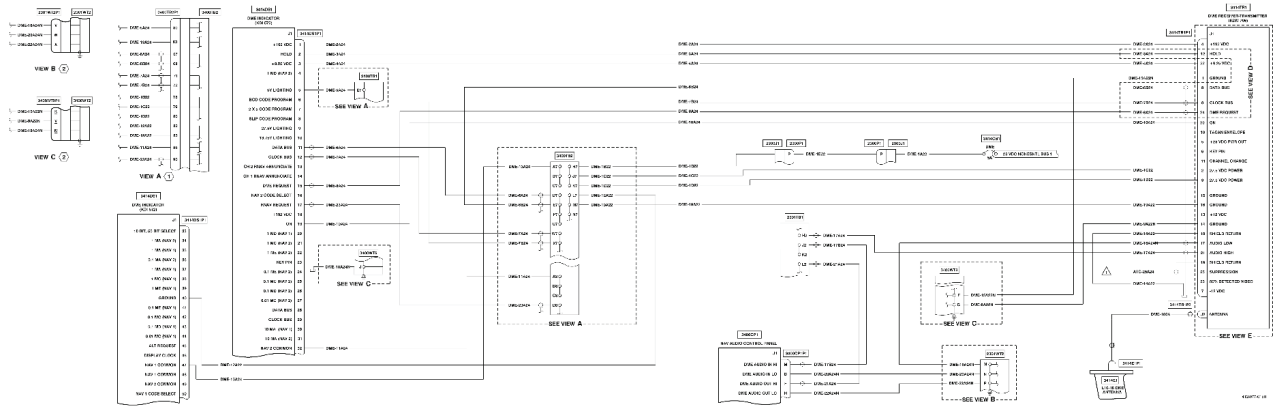
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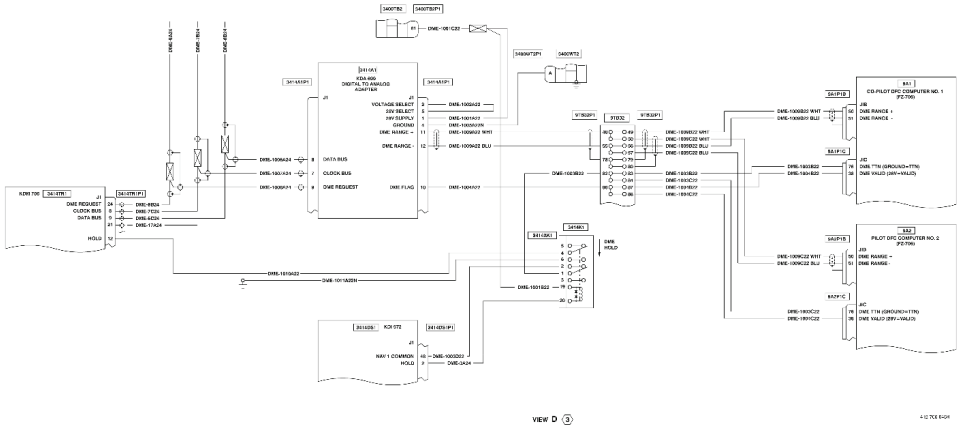
Figure 22. DME Navigation System (KDM-706) Wiring Diagram (S/N 33001 through 33100) (Sheet 1 of 1)

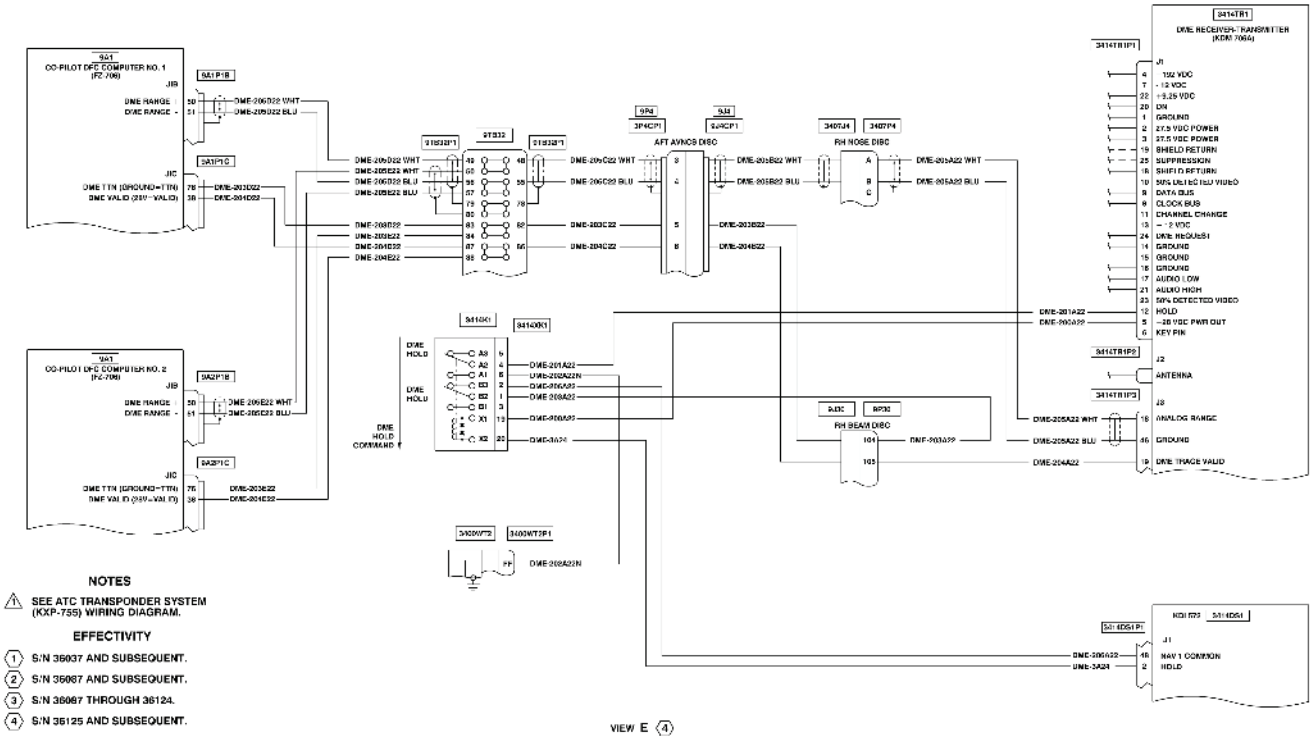
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Figure 23. DME Navigation System (KDM-706) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 3)

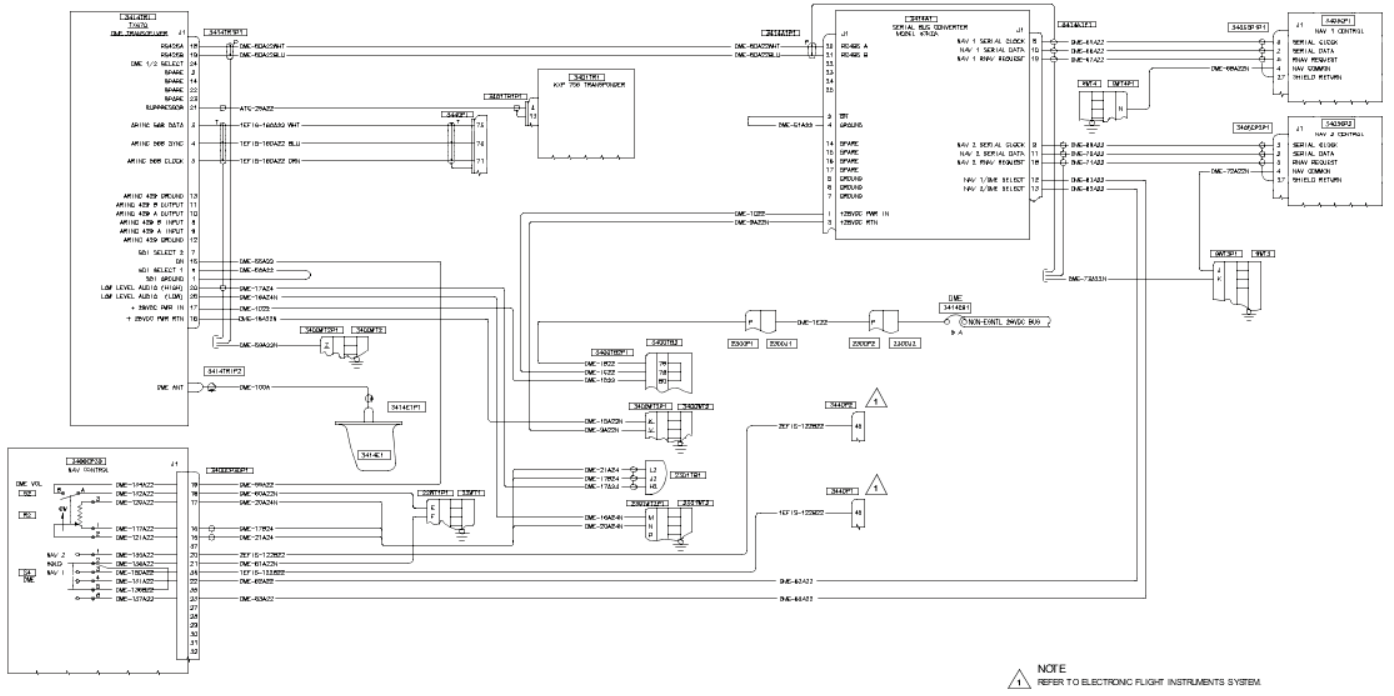

KDM 706
DME REQUEST
CLOCK BUS
DATA BUS
HOLD
DME-1601A22
DME-1601A22H
DME-1601E22
DME-1601C22
DME-1601D22
DME-1601B22
DME-1601A23
DME-1600A23
DME-1600A22 WHT
DME-1600A22 BLU
DME-1600A24
DME-1600A21
DME-1600A22
DME-1600B22
DME-1600C22
DME-1600D22
DME-1600E22
DME-1600F22
DME-1600G22 WHT
DME-1600G22 BLU
DME-1600H22
DME-1600J22
DME-1600K22
DME-1600L22
DME-1600M22
DME-1600N22
DME-1600P22
DME-1600R22
DME-1600S22
DME-1600T22
DME-1600U22
DME-1600V22
DME-1600W22
DME-1600X22
DME-1600Y22
DME-1600Z22
DME-1600AA22
DME-1600AB22
DME-1600AC22
DME-1600AD22
DME-1600AE22
DME-1600AF22
DME-1600AG22
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DME-1600AK22
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DME-1600AN22
DME-1600AP22
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ICN-412-A-972900-B-97499-00051-A-001-01

Figure 23. DME Navigation System (KDM-706) Wiring Diagram (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 3 of 3)

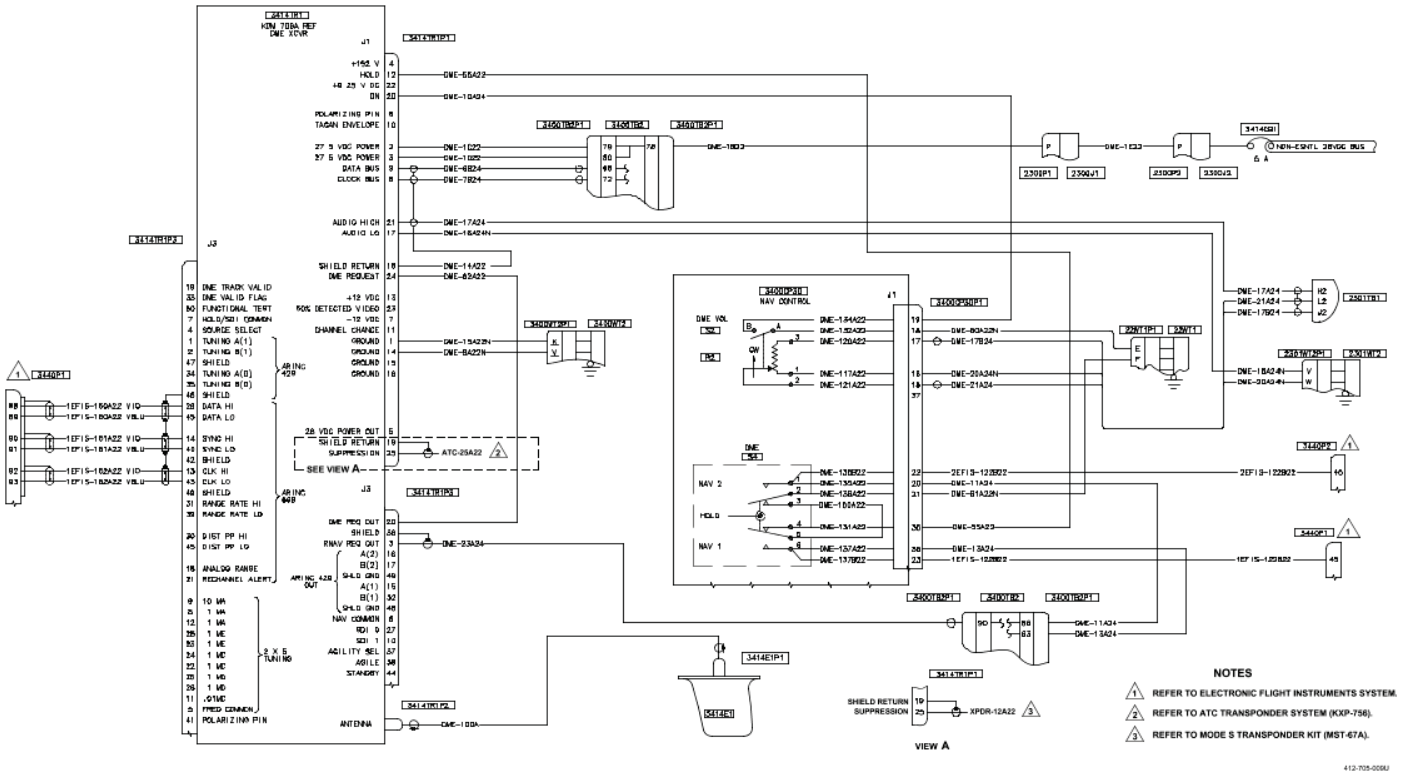
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Figure 24. DME Navigation System (KDM-706) Wiring Diagram (EFIS Configuration) (S/N 36087 through 36171) (Sheet 1 of 1)

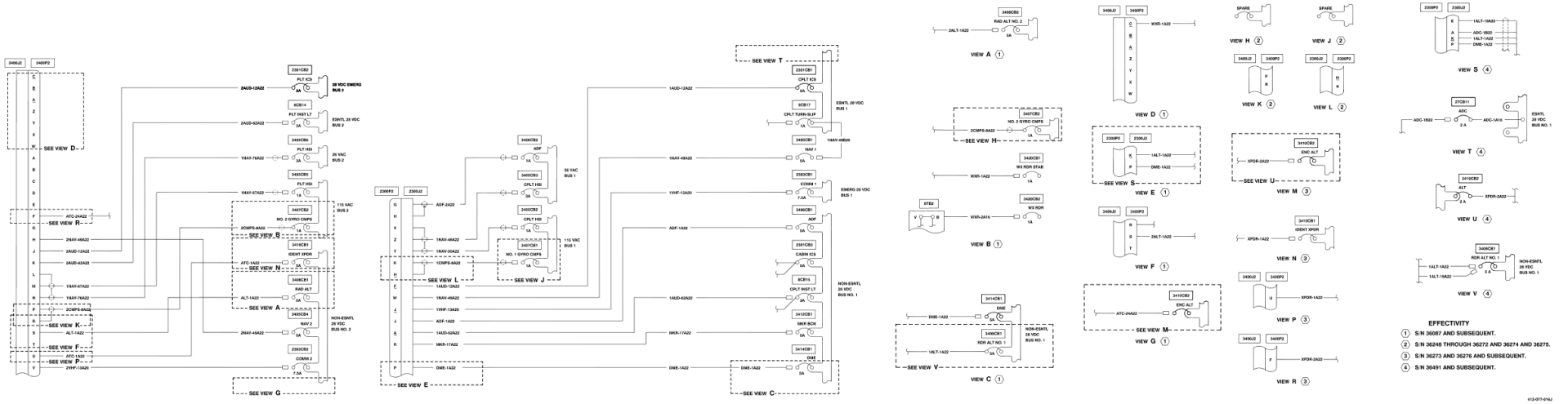
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ICN-412-A-972900-B-97499-00053-A-001-01

Figure 25. DME Navigation System (KDM-706) Wiring Diagram (EFIS Configuration) (S/N 36172 and Subsequent) (Sheet 1 of 1)

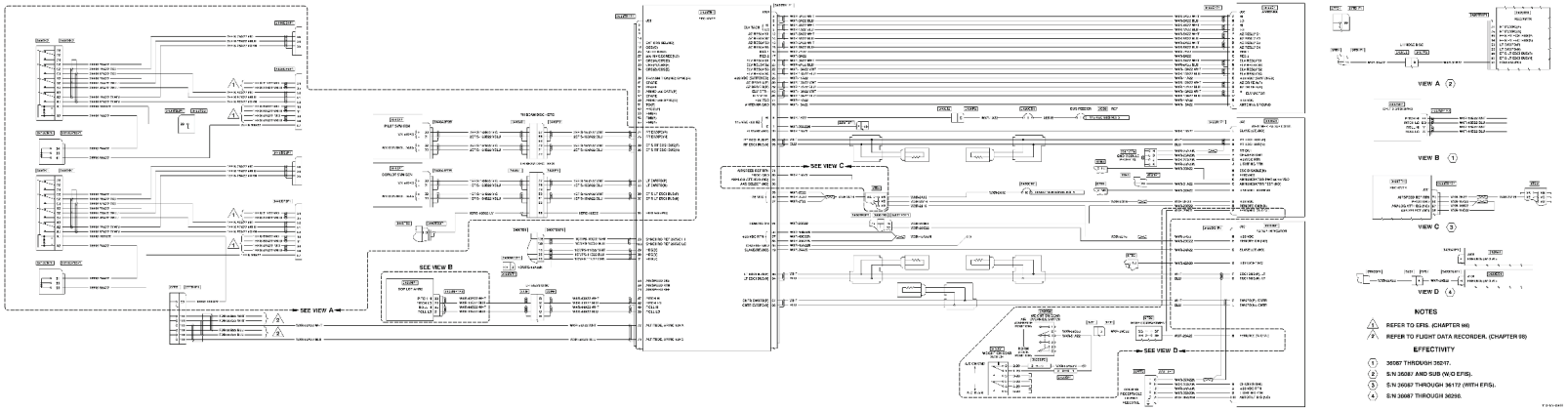
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Figure 26. Avionics Circuit Breakers (Sheet 1 of 1)

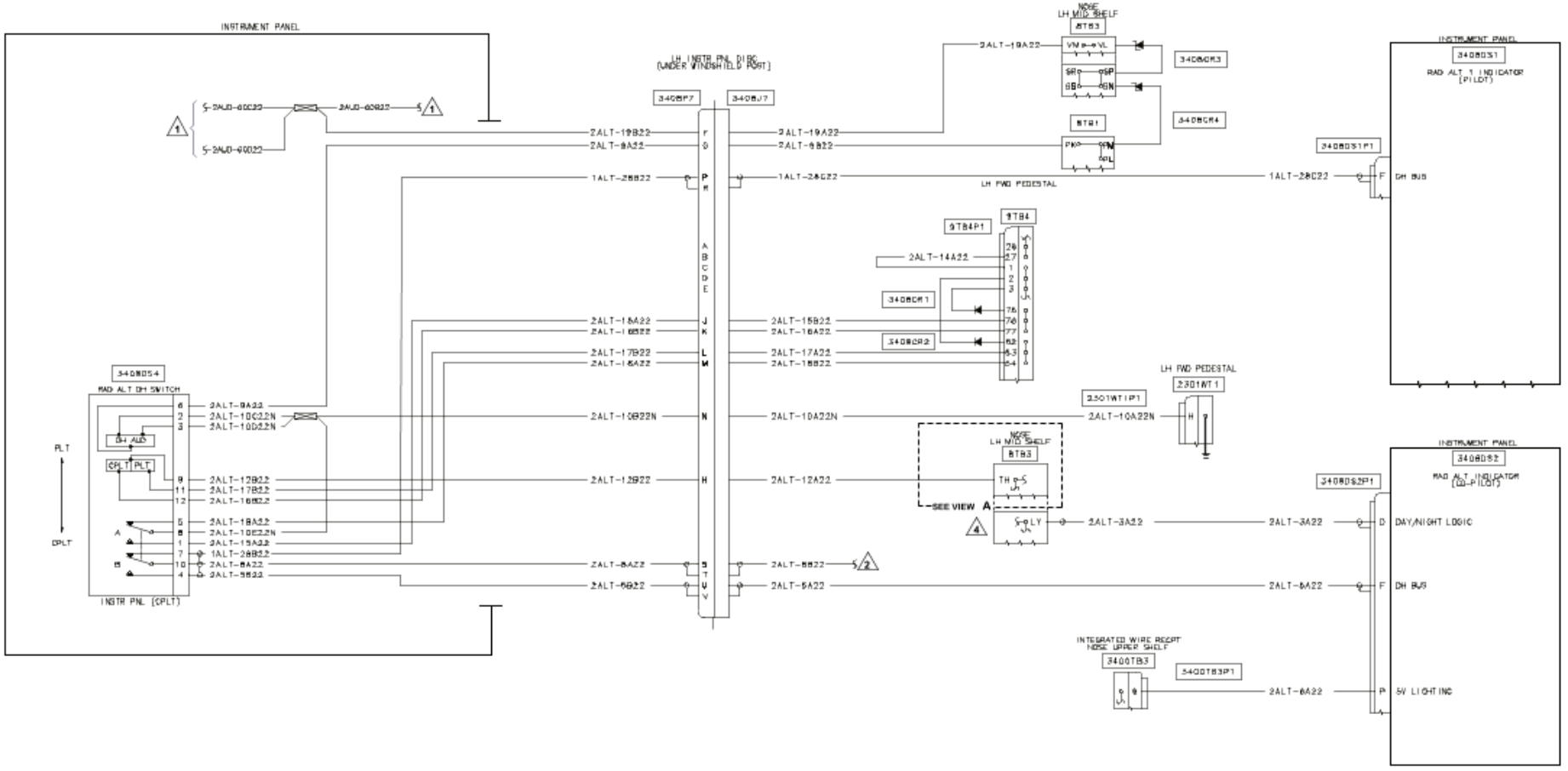
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Figure 27. Weather Radar Kit Wiring Diagram (S/N 36087 and Subsequent) (Sheet 1 of 1)

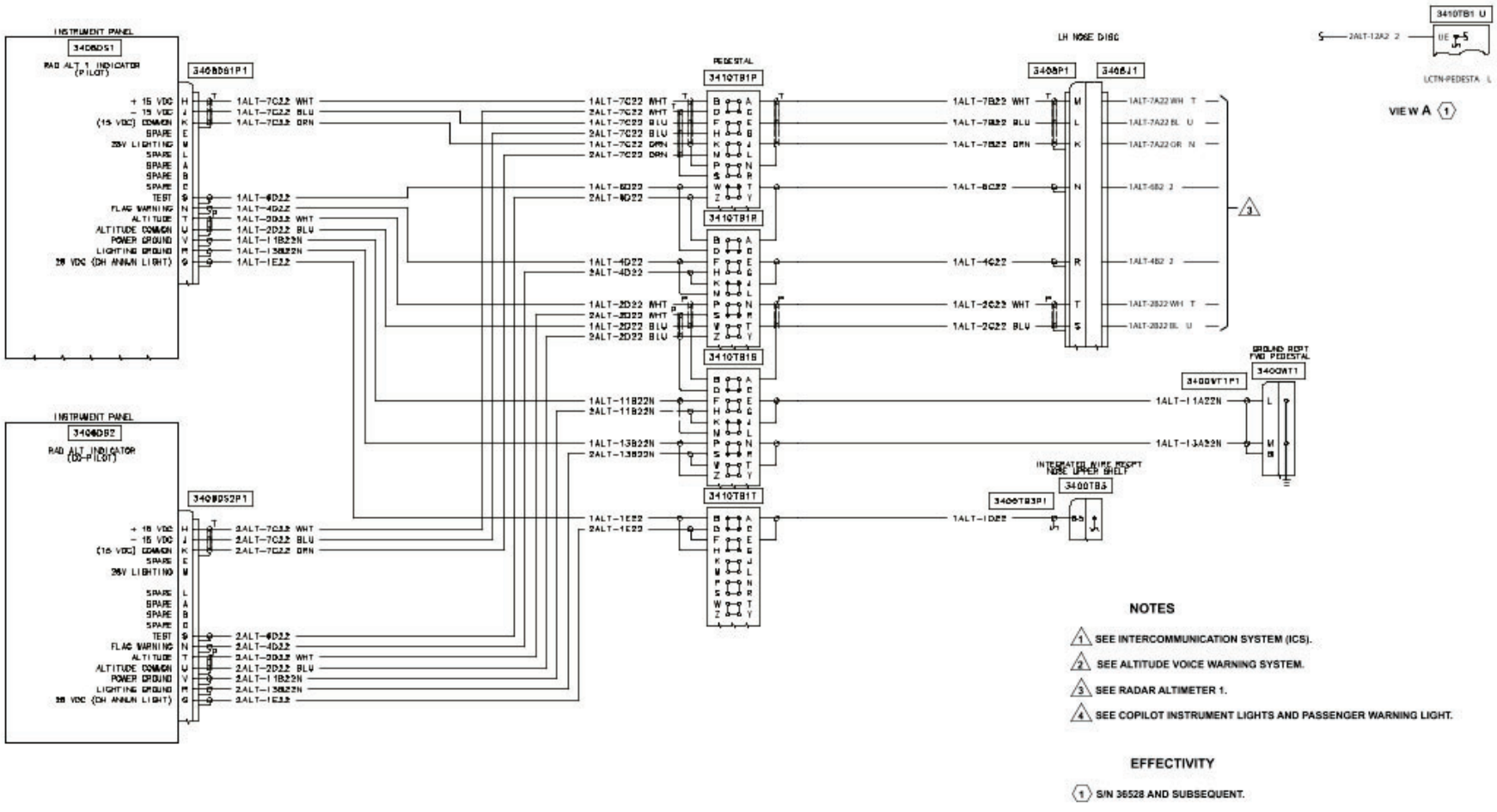
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ICN-412-A-972900-B-97499-00056-A-001-01

Figure 28. Copilot Instrument Radio Altimeter Wiring Diagram (SN 36087 and Subsequent) (Sheet 1 of 2)



ICN-412-A-972900-B-97499-00057-A-001-01

Figure 28. Copilot Instrument Radio Altimeter Wiring Diagram (SN 36087 and Subsequent) (Sheet 2 of 2)

Intercommunication System (ICS) - Description

The basic Intercommunication System (ICS) consists of the following units at both the pilot and the copilot stations:

- Communication (intercom) control unit
- Headset/mike
- Foot switch (mike)
- Cyclic stick radio/ICS switch
- Emergency key provision

The helicopter also has a provision for an Aft Cabin Intercommunications System (ICS) Kit (BHT-412-SI-15).

Applicable: SN: 33001-33107 or 33108-36086 or 36087-36637:

Installation includes two aft intercom control units, two hot microphone/call switches, and two instrument panel mounted AFT INT/CALL/ON annunciators. Aft stations can be isolated from pilot/copilot ICS.

Applicable: SN: 36638-Subsequent:

Installation includes two aft intercom control units, two ICS CALL annunciators, and two instrument panel mounted CABIN ICS/ISO/CALL annunciators. Aft stations can be isolated from pilot/copilot ICS audio.

Applicable: SN: 33001-33107 or 33108-36086 or 36087-36637:

The intercom control units (Figure 1) enable monitoring of up to 10 receivers with 10 on-off input (receiver) switches. Audio level is controlled by VOL control on the intercom control unit and receivers. A seven-position rotary selector (transmit) enables voice transmission from up to six transmitters. An emergency keying/comm panel allows the pilot emergency access to the copilot audio panel in the event of a pilot audio panel failure.

Applicable: SN: 36638-Subsequent:

The intercom control unit (Figure 1), enables monitoring of up to 11 receivers with 11 on-off input (receiver) switches. The audio level is controlled by the VOL control on the intercom control unit and the associated receivers. An eight-position rotary selector is provided for transmitting. In addition, this ICS unit has a built-in emergency selection switch.

The pilot ICS is powered by the 28 VDC emergency bus and protected by the ICS PILOT circuit breaker. The copilot ICS is powered by the 28 VDC essential bus and protected by the ICS CPLT circuit breaker. The aft crew ICS is powered by the 28 VDC non-essential bus and protected by the ICS CABIN circuit breaker. All receivers can be monitored by the pilot and copilot. ICS keying is by either first position (ICS) on the cyclic stick RADIO/ICS switch when rotary selector switch is in any position, or with the foot switch when ICS is selected. All transmitters can be keyed from the pilot or copilot station by the use of second position (RADIO) on the cyclic stick RADIO/ICS switch or the foot switch.

Applicable: SN: 33001-33107:

When the Aft Cabin Intercommunications System (ICS) Kit (BHT-412-SI-15) is installed, the pilot and copilot intercom control units have an AFT INT position. Selection of the call position of the HOT MIC/CALL switch at either aft ICS station illuminates the AFT INT/CALL annunciators on the instrument panel. The annunciators will remain illuminated until the pilot or copilot respond by selecting AFT INT on the comm control and keying the foot switch or second position of the cyclic stick switch.

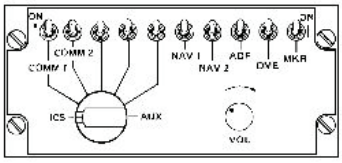
Applicable: SN: 33108-36086 or 36087-36637:

When the Aft Cabin Intercommunications System (ICS) Kit (BHT-412-SI-15) is installed, the pilot and copilot are provided with the AFT INT/CALL/ON annunciator located on the instrument panel. Selection of the CALL position of the HOT MIC/CALL switch at either aft ICS station illuminates the CALL segment of the AFT INT/CALL/ON annunciators. The CALL segment will remain illuminated until the pilot or copilot respond by depressing the AFT INT/CALL/ON annunciator causing both CALL segments to extinguish and both ON segments to illuminate. The aft ICS is now connected to the pilot and copilot ICS allowing all four stations to communicate with each other. The ICS can be returned to normal operation by the pilot or copilot depressing the AFT INT/CALL/ON annunciator causing both ON segments to extinguish, and disconnecting the aft ICS from the pilot and copilot ICS. This operation is independent of the ICS rotary switch positioning.

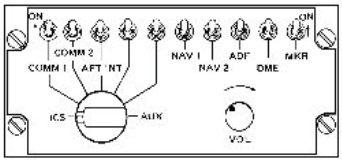
Applicable: SN: 36638-Subsequent:

When the Aft Cabin Intercommunications System (ICS) Kit (BHT-412-SI-15) is installed, the pilot and copilot are provided with the CABIN ICS/ISO/CALL annunciators located on the instrument panel. Momentarily pressing the ICS CALL annunciator (identified by illumination of blue Barber Pole) at either aft ICS station will cause the blue CALL segment of the pilot and copilot CABIN ICS/ISO/CALL annunciators to flash. The blue CALL segments will

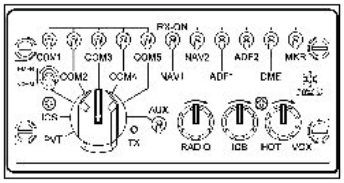
continue to flash until the pilot or copilot respond by depressing either of the CABIN ICS/ISO/CALL annunciators causing both blue CALL segments to stop flashing and remain illuminated. In addition, the green ISO segments of the pilot and copilot CABIN ICS/ISO/CALL annunciators will extinguish and the blue Barber Pole segments of the aft cabin ICS CALL annunciators will also extinguish. The aft ICS is now connected to the pilot and copilot ICS allowing all four stations to communicate with each other. The ICS can be returned to normal operation by the pilot or copilot depressing either of the CABIN ICS/ISO/CALL annunciators. This will cause the green ISO segments of the pilot and copilot CABIN ICS/ISO/CALL annunciators to illuminate indicating that ICS communication between the crew and aft passengers is isolated. This operation is independent of the ICS rotary switch positioning.



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NOTES

1. Pilot and copilot intercom control unit on S/N 33001 through 33107 when aft cabin ICS is not installed.
2. Pilot and copilot intercom control unit on S/N 33108 through 36086 and 36087 through 36637. Aft left and right side cabin intercom control unit, if installed.
3. Pilot and copilot intercom control unit on S/N 33001 through 33107 when aft ICS is installed.
4. Pilot and copilot intercom control unit on S/N 36638 and subsequent. Aft left and right side cabin intercom control unit, if installed.

ICN-412-A-970000-B-97499-00034-A-001-01

Figure 1. Intercom Control Unit (Sheet 1 of 1)

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Intercom Control Unit - Switch Functions - Description

The rotary selector switch selects communication sets for transmitting audio signals, and has seven positions that are connected at the pilot, copilot, and aft cabin intercom control units, if installed, as follows:

POSITION	FUNCTION
ICS	Intercommunications
COMM 1	VHF No. 1
COMM 2	VHF No. 2
(3) (2)	Not Used (1)
(4)	Not Used (1)
(5)	Not Used (1)
AUX	Not Used

NOTES:

Aft crew stations are not capable of transmitting on communication sets.

1 Not engraved on intercom control unit.

2 Reads AFT INT on pilot and copilot units when installed.

The 10 toggle (mixing) switches are used for monitoring receiver audio and are connected at each intercom control unit as follows:

POSITION	FUNCTION
COMM 1	VHF No. 1
COMM 2	VHF No. 2
(3)	Not Used (1)
(4)	Not Used (1)
(5)	Not Used (1)
NAV 1 (2)	VOR No. 1
NAV 2 (2)	VOR No. 2
ADF	ADF
DME (2)	DME
MKR (2)	Marker Beacon

NOTES:

1 Not engraved on intercom control unit.

2 Audio is not connected to aft crew stations.

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Intercom Control Unit - Switch Functions - Description

The rotary selector switch selects communication sets for transmitting audio signals, and has eight positions that are connected at the pilot, copilot, and aft cabin intercom control units, if installed, as follows:

POSITION	FUNCTION
PVT	Private
ICS	Intercommunications
COMM 1	VHF No. 1
COMM 2	VHF No. 2
(5)	Not Used (1)
(6)	Not Used (1)
(7)	Not Used (1)
AUX	Not Used

NOTES:

Aft crew stations are incapable of transmitting on communication sets.

1 Not engraved on intercom control unit.

The 11 toggle (mixing) switches are used for monitoring receiver audio and are connected at each intercom control unit as follows:

POSITION	FUNCTION
COMM 1	VHF No. 1
COMM 2	VHF No. 2
(3)	Not Used (1)
(4)	Not Used (1)
(5)	Not Used (1)
NAV 1 (2)	VOR/ILS No. 1
NAV 2 (2)	VOR/ILS No. 2
ADF	ADF
(9)	Not Used (1)
DME (2)	DME
MKR (2)	Marker Beacon

NOTES:

1 Not engraved on intercom control unit.

2 Audio is not connected to aft crew stations.

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Intercom System - Operational Check

Procedure

1. 412-A-97-01-00-01A-320A-A / 02245, 412-A-97-01-00-02A-320A-A / 02246, 412-A-97-01-00-03A-320A-A / 02247, 412-A-97-01-00-04A-320A-A / 02248, 412-A-97-01-00-05A-320A-A / 02249, 412-A-97-01-00-06A-320A-A / 02250, 412-A-97-01-00-07A-320A-A / 02251 and 412-A-97-01-00-08A-320A-A / 02252 are associated with the operational check.

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Intercom System - Preliminary Check Procedures - Operational Check

Procedure

1. Visually inspect each component to ensure that it is properly installed, and electrical connectors are properly mating and tight. Connect an external power source to the helicopter.

2. Energize the power buses (412-A-97-00-00-00A-340A-A / 02238 or 412-A-97-00-00-00A-340B-A / 02239) as applicable.

3. Close the circuit breakers ICS PILOT, ICS CPLT, and ICS CABIN (if aft ICS is installed), and the circuit breakers of any associated kits. Warm up the equipment for a period of 3 minutes with the initial setting of controls and switches at each intercom control unit as follows:

CONTROL	SETTING
Volume Control (VOL)	Mid Position
Mixing Switches	All Down
Rotary Selector Switch	ICS Position

NOTE

After warmup of the system, including all associated receivers and transmitters, the various operational checks may be performed in any sequence.

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Intercom System - Pilot and Copilot Stations - Operational Check

Procedure

Applicable: SN: 33001-33100:

1. Verify the pilot and copilot AFT INT/CALL annunciators are not illuminated, if installed.

Applicable: SN: 33101-36637:

2. Verify both ON segments of the pilot and copilot AFT INT/CALL/ON annunciators are not illuminated, if installed.

3. Actuate the pilot cyclic stick RADIO/ICS switch to first position (ICS), and speak into the microphone. Sidetone should be heard in the pilot headset, and received audio should be heard in the copilot headset. Rotation of the volume control should vary audio to the pilot headset. Verify received audio is not heard in the aft ICS stations, if installed.

4. Actuate the pilot cyclic stick switch to second position (RADIO), and speak into the microphone. Sidetone should be heard in the pilot headset, and received audio should be heard in the headset at the copilot station. Verify received audio is not heard in the aft ICS stations, if installed.

5. Actuate the copilot cyclic stick RADIO/ICS switch to first position (ICS), and speak into the microphone. Sidetone should be heard in the copilot headset, and received audio should be heard in the headset at the pilot station. Rotation of the volume control should vary audio to the copilot headset. Verify received audio is not heard in the aft ICS stations, if installed.

6. Actuate the copilot cyclic stick switch to second position (RADIO), and speak into the microphone. Sidetone should be heard in the copilot headset, and received audio should be heard in the headset at the pilot station. Verify received audio is not heard in the aft ICS stations, if installed.

7. Key the pilot foot switch and speak into the microphone. Sidetone should be heard in the pilot headset and received audio in the copilot headset. Verify received audio is not heard in the aft ICS stations, if installed.

8. Key the copilot foot switch and speak into the microphone. Sidetone should be heard in the copilot headset and received audio in the pilot headset. Verify received audio is not heard in the aft ICS stations, if installed.

9. Set the pilot and copilot rotary selector switches to COMM 1. Actuate the pilot cyclic stick switch to first position (ICS), and speak into the microphone. Sidetone should be heard in the pilot headset, and received audio should be heard in the headset at the copilot station. Verify received audio is not heard in the aft ICS stations, if installed.

10. Actuate the copilot cyclic stick switch to first position (ICS), and speak into the microphone. Sidetone should be heard in the copilot headset, and received audio should be heard in the headset at the pilot station. Verify received audio is not heard in the aft ICS stations, if installed. Return the pilot and copilot rotary selector switches to ICS position.

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Intercom System - Aft ICS Stations (When Installed) - Operational Check

Procedure

1. Set the HOT MIC/CALL switch to HOT MIC position at each crew station, one at a time, and speak into the microphone. Sidetone should be heard in the headset of the crew station being keyed, and received audio should be heard in the headset at the other crew station. Verify received audio is not heard in the pilot or copilot station.

2. Repeat Step 1. for hand-held trigger switches. Verify rotation of the VOL control on each unit adjusts the audio level in the respective headset.

3. Set the HOT MIC/CALL switch to CALL position at each crew station. Verify the AFT INT/CALL annunciators CALL lights illuminate and remain illuminated.

4. Set the pilot rotary selector switch to AFT INT position. Set each aft crew ICS HOT MIC/CALL switch to HOT MIC position one at a time, and speak into the respective microphone. Verify received audio is heard in the pilot headset but not in the copilot headset.

5. Key the pilot foot switch and speak into the microphone. Verify received audio is heard in the aft crew ICS stations. Verify the CALL lights extinguish. Return the pilot rotary selector switch to ICS position.

6. Set the crew station HOT MIC/CALL switches to CALL position to illuminate the CALL lights. Repeat Step 4. and Step 5. for the copilot station.

NOTE

If the internal hoist ICS is installed, refer to the applicable Service Instruction (BHT-412-SI-7).

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Intercom System - Aft ICS Stations (When Installed) - Operational Check

Procedure

1. Verify both ON segments, and both CALL segments of the AFT INT/CALL/ON annunciators are extinguished and both aft crew station rotary selector switches are set to the ICS position. Set the left and right HOT MIC/CALL switches to the HOT MIC position at each crew station, one at a time, and speak into the microphone. Sidetone should be heard in the headset of the crew station being keyed, and received audio should be heard in the headset at the other crew station. Verify received audio is not heard in the pilot or copilot station.

2. Repeat Step 1. for the hand-held trigger switches. Verify rotation of VOL control on each unit adjusts audio level in the respective headset.

3. Set the HOT MIC/CALL switch to the CALL position at each crew station. Verify CALL segments of the AFT INT/CALL/ON annunciators illuminate and remain illuminated.

4. Press the pilot AFT INT/CALL/ON annunciator and verify CALL segment extinguishes and ON segment illuminates. Set each aft crew ICS HOT MIC/CALL switch to the HOT MIC position, one at a time, and speak into the respective microphone. Verify received audio is heard in the pilot and the copilot headsets.

5. Key the pilot foot switch and speak into the microphone. Verify received audio is heard in the aft crew ICS stations. Depress the pilot AFT INT/CALL/ON annunciator and verify the ON segments extinguish.

6. Position the crew station HOT MIC/CALL switches to the CALL position to illuminate CALL segments. Repeat Step 4. and Step 5. for the copilot station.

NOTE

If the internal hoist ICS is installed, refer to the applicable Service Instruction (BHT-412-SI-7 or BHT-412-SI-32).

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Intercom System - Receiver Mixing Switch Selections - Operational Check

Procedure

NOTE

Radio reception or radio background noise heard from the selected receiver may be used to determine satisfactory reception during this test.

1. When testing navigation position, turn the installed navigational receiver volume controls fully counterclockwise, then turn any one of the receiver volume controls to listening level and check reception. Return the volume control to full counterclockwise position. Repeat this procedure on each of the navigational receivers. Finally, position all volume controls to the desired listening level.

2. Set the mixing switches on each intercom control unit to the down position, and set each rotary selector switch to the ICS position.

NOTE

Each time a mixing switch is positioned to ON, the previous mixing switch should be returned to the down (off) position.

3. Set the COMM 1 mixing switch on the pilot intercom control unit to the ON position. Depress the SQ TEST pushbutton on the VHF comm control unit. Receiver background noise should be heard in the headset. Return the COMM 1 mixing switch to the down (off) position.

4. Repeat Step 2. at the copilot and aft stations. Repeat for COMM 2, if installed.

5. Set the NAV 1 and ADF mixing switches on the pilot and copilot intercom control unit to the ON position, one at a time. Receiver background noise should be heard in the headsets. Repeat the ADF test for the aft stations. Return the NAV 1 and ADF mixing switches to the down (off) position.

6. Repeat Step 5. for NAV 2, if installed.

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Intercom System - Receiver/Transmitter Selections - Operational Check

Procedure

NOTE

Allow a 3 minute warmup period before performing the following test.

1. Set all mixing switches on the pilot intercom control unit to the down (off) position, and set the volume control to the mid position. Position the pilot rotary selector switch to COMM 1. Actuate the pilot cyclic stick switch to the RADIO position, and speak into the microphone. The selected transmitter shall have been placed into operation and sidetone should be audible in the headset. Release the pilot cyclic stick switch.

2. Repeat Step 1. for the copilot station.

3. Repeat Step 1. while actuating the pilot and copilot foot switch, one at a time.

4. Repeat Step 1. through Step 3. for COMM 2, if installed.

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Intercom System - Emergency Keying/Comm - Operational Check

Procedure

1. Insert the pilot headphone plug into the emergency jack.

2. Turn the rotary switch on the copilot audio panel to the ICS position.

3. Activate the emergency Comm switch on the Emergency/Keying/Comm panel and speak into the microphone. Sidetone should be heard in the pilot headset.

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Intercom System - Post Operational Check Procedures

Procedure

1. At completion of checks, if required, disconnect the headsets, disengage the circuit breakers, and remove the external power.

2. Return the helicopter to the standard configuration after electrical maintenance of repairs (412-A-96-00-01-02A-913A-A / 01875).

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Intercom System - Operational Check

Procedure

Applicable: SN: 36638-Subsequent:

1. 412-A-97-01-01-01A-320A-A / 02254, 412-A-97-01-01-02A-320A-A / 02255, 412-A-97-01-01-03A-320A-A / 02256, 412-A-97-01-01-04A-320A-A / 02257, 412-A-97-01-01-05A-320A-A / 02258, 412-A-97-01-01-06A-320A-A / 02259, 412-A-97-01-01-07A-320A-A / 02260, 412-A-97-01-01-08A-320A-A / 02261, 412-A-97-01-01-09A-320A-A / 02262, 412-A-97-01-01-10A-320A-A / 02263, 412-A-97-01-01-11A-320A-A / 02264 and 412-A-97-01-01-12A-320A-A / 02265 are associated with the operational check. The procedures also reflect the Aft Cabin Intercommunications System (ICS) Kit 412-706-012-115 installed in the helicopter. In addition, 412-A-97-01-01-13A-320A-A / 02266 and 412-A-97-01-01-14A-320A-A / 02267 provide associated ground run and in-flight checks of the intercom system.

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Intercom System - Preliminary Check Procedures - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
David Clarke Head Set	H10-13H	4	or Equivalent

Procedure

1. Inspect the equipment installation to detect faulty mechanical mountings or electrical connections.

2. On the left and right sides of each of the four installed audio control panels, verify the switch positions are set as follows:

LEFT SIDE	
CVR Out Level	Set to maximum (full CW)
TRANSMIT MIC LEVEL COM1	Do not adjust
TRANSMIT MIC LEVEL COM2	Do not adjust
TRANSMIT MIC LEVEL COM3	Do not adjust
TRANSMIT MIC LEVEL COM4	Do not adjust
TRANSMIT MIC LEVEL COM5	Do not adjust
TRANSMIT MIC LEVEL AUX	Do not adjust
MODEL (103/105)	Set to 105
RIGHT SIDE	
TIE LINE MODE SELECT (1)	
Switch 1	Down
Switch 2	Up
Switch 3	Up
Switch 4	Up

NOTE:

1 Configure for ANDREA Tie Line Ref N301A Installation and Operation Manual (SM45)

3. On each of the four audio control panels, set the switch positions as follows:

EMER/NORM	NORM
Transmit Selector (Rotary) Switch	ICS
AUX Switch	Down
All Mixer Switches	Down (off)
RADIO (Volume)	Mid position
ICS (Volume)	Mid position
HOT/VOX	Full CW (VOX off)

4. Connect the david clarke head set (H10-13H) to the drop cords of each ICS station.

5. Make sure the pilot, copilot, and aft ICS circuit breakers are engaged.

NOTE

Use an external power unit, if available, to prevent the battery from being drained of its stored electrical power.

6. Connect external power to the helicopter.

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Crew Intercommunication System - Operational Check

Procedure

1. Apply power to the 28V DC non-essential bus. Verify the default power-up status of the ICS annunciators is as follows:
- White CABIN ICS segments of pilot and copilot CABIN ICS annunciators are illuminated.
- Green ISO segments of pilot and copilot CABIN ICS annunciators are illuminated.
- Blue CALL segments of pilot and copilot CABIN ICS annunciators are extinguished.
- White ICS CALL segments of cabin left and right side ICS CALL annunciators are illuminated.
- Blue Barber Pole (CALL) segments of cabin left and right side ICS CALL annunciators are extinguished.

2. Verify minimal ambient background noise in the pilot and copilot headsets.

3. Without keying ICS, speak into the pilot headset microphone, verify no sidetone or ICS audio in either the pilot or copilot headsets. Repeat for the copilot headset.

4. Key the pilot ICS through first click of the Pilot Cyclic Key switch and speak into the pilot headset microphone. While modulating the headset microphone, verify the following:
- Verify acceptable quality ICS audio is heard in the pilot and copilot headsets with minimal background noise.
- Verify acceptable ICS volume range is available in the corresponding headsets while adjusting ICS volume control on the pilot and copilot audio selector panels.
- Verify no effect on ICS volume while adjusting RADIO volume control on the pilot and copilot audio selector panels.

5. Repeat Step 4. for the copilot ICS.

6. Repeat Step 4. and Step 5. using the corresponding PTT foot switch for the pilot and copilot ICS.

7. On the pilot audio selector panel, select each of the 11 mixer switches in turn to the ON position. Verify no mixer switch introduces unacceptable ambient background noise to the pilot ICS.

8. Repeat Step 7. for the copilot ICS.

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Crew VOX Sensitivity - Operational Check

Procedure

1. On the pilot audio selector panel select the VOX control switch to full ccw (HOT) position. Without keying the ICS, speak into the pilot headset microphone and verify acceptable ICS audio quality and volume in both headsets (pilot and copilot).

2. While continuing to speak into the pilot headset, adjust the VOX sensitivity control on the pilot audio selector panel. Verify that a VOX sensitivity level can be obtained that provides voice activated keying of the pilot ICS with acceptable audio quality.

3. Repeat Step 1. and Step 2. for the copilot ICS.

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COM1 - Operational Check

Procedure

1. Switch on the COM1 and tune to an approved transmit frequency. Select the volume control on the COM1 control panel to mid position.

2. On the pilot audio selector panel, select the rotary transmit selector switch to the COM1 position. Establish two-way radio communication with an independent VHF radio platform (i.e., another aircraft, local ATC) utilizing COM1 through the pilot ICS station. During radio communication, verify the following:
   - Verify acceptable received audio quality and volume at the independent radio platform when transmitted from the COM1 through the pilot ICS station.
   - Verify acceptable audio sidetone quality in the pilot headset when transmitting.
   - Verify acceptable received audio quality in the pilot headset when receiving.
   - Verify acceptable transmit sidetone and receive audio volume range is available while adjusting RADIO volume control on the pilot audio selector panel.
   - Verify green TX LED on the pilot audio selector panel illuminates while transmitting.
   - Verify radio PTT can be accomplished through the pilot cyclic PTT switch (second click) and through the pilot PTT foot switch.
   - On the copilot audio selector panel, select the COM1 mixer switch to the down (OFF) position. Verify the COM1 Tx/Rx audio is not audible in the copilot headset.
   - On the copilot audio selector panel, select the COM1 mixer switch to the up (ON) position. Verify the COM1 Tx/Rx audio is audible in the copilot headset. Verify acceptable COM1 audio quality. Verify acceptable COM1 audio volume range can be selected through the RADIO volume control on the copilot audio selector panel.

3. Repeat Step 2. for COM1 radio communication through the copilot ICS station.

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COM2 - Operational Check

Procedure

1. Switch on the COM2 and tune to an approved transmit frequency. Select volume control on the COM2 control panel to the mid position.

2. On the pilot audio selector panel, select the rotary transmit selector switch to the COM2 position. Establish two-way radio communication with an independent VHF radio platform utilizing COM2 through the pilot ICS station. During radio communication, verify the following:
   - Verify acceptable received audio quality and volume at the independent radio platform when transmitted from COM2 through the pilot ICS station.
   - Verify acceptable audio sidetone quality in the pilot headset when transmitting.
   - Verify acceptable received audio quality in the pilot headset when receiving.
   - Verify acceptable transmit sidetone and receive audio volume range is available while adjusting RADIO volume control on the pilot audio selector panel.
   - Verify green TX LED on the pilot audio selector panel illuminates while transmitting.
   - On the copilot audio selector panel, select the COM2 mixer switch to the down (OFF) position. Verify COM2 Tx/Rx audio is not audible in the copilot headset.
   - On the copilot audio selector panel, select the COM2 mixer switch to the up (ON) position. Verify COM2 Tx/Rx audio is audible in the copilot headset. Verify acceptable COM2 audio quality. Verify acceptable COM2 audio volume range can be selected through the RADIO volume control on the copilot audio selector panel.

3. Repeat Step 2. for the COM2 radio communication through the copilot ICS station.

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Emergency Communication Function - Operational Check

Procedure

1. Switch on the COM1 and tune to an approved transmit frequency. Select volume control on the COM1 control panel to the mid position.

2. Disengage the pilot ICS circuit breaker. On the pilot audio selector panel, select the NORM/EMER switch to the EMER position and select rotary Transmit selector switch to the COM1 position. Establish two-way radio communication with an independent VHF radio platform utilizing COM1 through the pilot ICS station. During radio communication, verify the following:
   - Verify acceptable received audio quality and volume at the independent radio platform when transmitted from COM1 through the pilot ICS station.
   - Verify acceptable audio sidetone quality in the pilot headset when transmitting.
   - Verify acceptable received audio quality in the pilot headset when receiving.
   - Verify acceptable transmit sidetone and receive audio volume range is available while adjusting RADIO volume control on the pilot audio selector panel.

3. On completion of the test, re-engage the pilot ICS circuit breaker and return the NORM/EMER switch to the NORM position.

4. Repeat Step 2. and Step 3. for emergency communication through the copilot ICS station.

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Navigation Systems Audio - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Navigation Communication Test Set	IFR 4000	1	or Equivalent

Procedure

1. Switch on NAV1 and tune to an available frequency. Set up the NAV test set and tune to the same test frequency.

2. Configure the NAV test set to transmit NAV audio. On the pilot audio control panel, select the NAV1 mixer switch to the up (ON) position. Verify the following:
- Verify acceptable audio quality of NAV1 audio in the pilot headset when the NAV1 mixer switch is in the ON position.
- Verify that NAV1 audio is not heard in the pilot headset when the NAV1 mixer switch is in the OFF position.
- Verify acceptable NAV1 audio volume range is available while adjusting RADIO volume control on the pilot audio selector panel.

3. Repeat Step 2. for the copilot audio control panel and copilot headset.

4. Configure the NAV test set to transmit outer marker audio. On the pilot audio control panel select the MKR mixer switch to the up (ON) position. Verify the following:
- Verify acceptable audio quality of outer marker audio in the pilot headset when the MKR mixer switch is in the ON position.
- Verify that outer marker audio is not heard in the pilot headset when the MKR mixer switch is in the OFF position.
- Verify acceptable outer marker audio volume range is available while adjusting RADIO volume control on the pilot audio selector panel.

5. Repeat Step 4. for middle marker and inner marker audio.

6. Repeat Step 4. and Step 5. for the copilot audio control panel and copilot headset.

7. Switch off NAV1, switch on NAV2, and tune to test set frequency. Configure the NAV test set to transmit NAV audio. On the pilot audio control panel, select the NAV2 mixer switch to the up (ON) position. Verify the following:
- Verify acceptable audio quality of NAV2 audio in the pilot headset when the NAV2 mixer switch is in the ON position.
- Verify that NAV2 audio is not heard in the pilot headset when the NAV2 mixer switch is in the OFF position.
- Verify acceptable NAV2 audio volume range is available while adjusting RADIO volume control on the pilot audio selector panel.

8. Repeat Step 7. for the copilot audio control panel and copilot headset.

9. Switch on the DME. Tune the DME and test set to an appropriate DME test frequency. Configure the NAV test set to transmit DME audio. On the pilot audio control panel, select the DME mixer switch to the up (ON) position. Verify the following:
- Verify acceptable audio quality of DME audio in the pilot headset when the DME mixer switch is in the ON position.
- Verify that DME audio is not heard in the pilot headset when the DME mixer switch is in the OFF position.
- Verify acceptable DME audio volume range is available while adjusting the RADIO volume control on the pilot audio selector panel.

10. Repeat Step 9. for the copilot audio control panel and copilot headset.

11. Switch on the ADF and tune to an appropriate test frequency (local NDB or radio station).

12. On the pilot audio control panel, select the ADF mixer switch to the up (ON) position. Verify the following:
- Verify acceptable audio quality of ADF audio in the pilot headset when the ADF mixer switch is in the ON position.
- Verify that ADF audio is not heard in the pilot headset when the ADF mixer switch is in the OFF position.
- Verify acceptable ADF audio volume range is available while adjusting the RADIO volume control on the pilot audio selector panel.

13. Repeat Step 12. for the copilot audio control panel and copilot headset.

Radar Altimeter Voice Warning - Operational Check

Procedure

1. Audio volume for the RAD ALT as well as the RPM direct audio inputs is set in the factory.

The RAD ALT voice warning system provides a “Check height” voice warning as helicopter altitude descends through the RAD ALT Decision Height (DH) , plus a “One fifty feet” voice warning as helicopter altitude descends through the 150 feet radar altitude.

As the “One fifty feet” voice warning can only be evaluated in flight, the scope of this test is to verify the audio quality of the ICS interface to the RAD ALT voice warning system. Refer to 412-A-97-01-01-14A-320A-A / 02267 for the operation flight check.

Also, the RPM audio warning function can be tested after engine start (412-A-97-01-01-13A-320A-A / 02266).

1.1. Select RAD ALT DH voice warning to be referenced to the pilot RAD ALT indicator. Make sure the RAD ALT voice warning audio is not inhibited. Reduce DH setting on the pilot RAD ALT indicator until the DH light is extinguished, then increase DH setting until the DH light is illuminated, as the DH light illuminates, verify the following:

- Verify an aural warning tone followed by a “Check height” voice audio is heard in the pilot and copilot headsets.
- Verify acceptable audio quality and volume of the RAD ALT voice warning audio.
- Verify no affect on the volume of the RAD ALT audio in the corresponding headset while adjusting RADIO and ICS volume controls on each audio selector panel.

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Crew/Cabin Intercommunication System - Operational Check

Procedure

1. Momentarily press the left side ICS CALL annunciator in the aft cabin. Verify the ICS annunciators status is as follows:
- White CABIN ICS segments of the pilot and copilot CABIN ICS annunciators remain illuminated.
- Green ISO segments of pilot and copilot CABIN ICS annunciators remain illuminated.
- Blue CALL segments of pilot and copilot CABIN ICS annunciators flash and continue to do so when the ICS CALL annunciator is released.
- White ICS CALL segments of left and right side ICS CALL annunciators remain illuminated.
- Blue Barber Pole (CALL) segments of left and right side ICS CALL annunciators illuminate and remain illuminated when the ICS CALL annunciator is released.

2. Key the ICS and talk into the headset at each ICS station in turn. Verify as follows:
- Verify ICS audio communication between the pilot and copilot.
- Verify ICS audio Communication between the left and right side cabin ICS stations.
- Verify isolation of pilot/copilot ICS audio from cabin ICS audio.

3. Momentarily press the copilot CABIN ICS annunciator. Verify ICS annunciators status is as follows:
- White CABIN ICS segments of pilot and copilot CABIN ICS annunciators remain illuminated.
- Green ISO segments of pilot and copilot CABIN ICS annunciators extinguish and remain extinguished when the CABIN ICS annunciator is released.
- Blue CALL segments of pilot and copilot CABIN ICS annunciators cease flashing and remain illuminated when the CABIN ICS annunciator is released.
- White ICS CALL segments of the left and right side ICS CALL annunciators extinguish and remain extinguished when the CABIN ICS annunciator is released.
- Blue Barber Pole (CALL) segments of left and right side ICS CALL annunciators extinguish and remain extinguished when the CABIN ICS annunciator is released.

4. Key the ICS and talk into the headset at each ICS station in turn. Verify as follows:
- Verify ICS audio communication between all four ICS stations.

5. Momentarily press the left side aft cabin ICS CALL annunciator. Verify no change in status of pilot/copilot CABIN ICS annunciators or aft cabin ICS CALL annunciators.

6. Momentarily press the copilot CABIN ICS annunciator. Verify ICS annunciators status is as follows:
- White CABIN ICS segments of pilot and copilot CABIN ICS annunciators remain illuminated.
- Green ISO segments of pilot and copilot CABIN ICS annunciators illuminate and remain illuminated when the CABIN ICS annunciator is released.
- Blue CALL segments of pilot and copilot CABIN ICS annunciators extinguish and remain extinguished.
- White ICS CALL segments of the left and right side ICS CALL annunciators illuminate and remain illuminated when the CABIN ICS annunciator is released.
- Blue Barber Pole (CALL) segments of the left and right side ICS CALL annunciators remain extinguished.

7. Key the ICS and talk into the headset at each ICS station in turn. Verify as follows:
- Verify ICS audio communication between pilot and copilot.
- Verify ICS audio communication between the left and right side aft cabin ICS stations.
- Verify isolation of pilot/copilot ICS audio from aft cabin ICS audio.

8. Repeat Step 1. through Step 7. utilizing the switch functions of the pilot CABIN ICS annunciator and the right side ICS CALL annunciator.

9. Through the pilot or copilot CABIN ICS switch, select cabin ICS ON (ISO legend extinguished).

10. Verify minimal ambient background noise in the pilot, copilot, aft left ICS, and aft right ICS headsets.

11. Without keying the ICS, speak into the pilot headset microphone, verify no sidetone or ICS audio in any of the four headsets. Repeat for copilot, aft left ICS, and aft right ICS headset.

12. Key the pilot ICS and speak into the pilot headset microphone. While modulating the headset microphone, verify the following:
- Verify acceptable quality ICS audio is heard in each of the four headsets with minimal background noise.

- Verify acceptable ICS volume range is available in the corresponding headset while adjusting ICS volume control on each audio selector panel.
- Verify no effect on ICS volume while adjusting RADIO volume control on each audio selector panel.

13. Repeat Step 12. for copilot ICS.

14. Repeat Step 12. for aft left ICS.

15. Repeat Step 12. for aft right ICS.

16. On each audio selector panel in turn, select each of the 11 mixer switches in turn to the ON position. Verify no mixer switch introduces unacceptable ambient background noise in the corresponding headset.

17. While maintaining ICS communication between the pilot and copilot, switch aft ICS off and on in turn. Verify no discernible change in ICS volume or quality in the pilot or copilot headset as aft ICS is turned on or off.

18. Repeat Step 17. while monitoring ICS audio volume and quality in the aft left and aft right ICS headsets.

Aft Cabin VOX Sensitivity - Operational Check

Procedure

1. Select the aft cabin ICS ON (ISO legend extinguished). On the pilot audio selector panel, select the VOX control switch to full ccw (HOT) position. Without keying the ICS, speak into the pilot headset microphone, verify acceptable ICS audio quality and volume in each of the four headsets.

2. While continuing to speak into the pilot headset, adjust VOX sensitivity control on the pilot audio selector panel. Verify that a VOX sensitivity level can be obtained that provides voice activated keying of the pilot ICS with acceptable audio quality.

3. Repeat Step 1. and Step 2. for the copilot ICS.

4. Repeat Step 1. and Step 2. for the aft cabin left ICS.

5. Repeat Step 1. and Step 2. for the aft cabin right ICS.

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PVT (Private) Net - Operational Check

Procedure

1. Select the aft cabin ICS ON (ISO legend extinguished). On the pilot and copilot audio selector panels, select the rotary transmit selector switch to the Private (PVT) position.

2. Activate the pilot radio key switch and talk into the pilot headset microphone. Verify the following:
   - Verify private intercom audio from the pilot station is heard at the pilot and copilot ICS stations and is not heard at the aft left and aft right cabin ICS stations.
   - *Verify acceptable private intercom volume range is available in the corresponding headset while adjusting ICS volume control on pilot and copilot audio selector panels.*
   - *Verify no effect on private intercom volume while adjusting radio volume control on pilot and copilot audio selector panels.*
   - Key and modulate ICS from either aft cabin ICS station. Verify ICS communication is heard at the pilot and copilot ICS stations and is controlled by the corresponding ICS volume control.

3. Repeat Step 2. for the copilot ICS.

4. On the aft left and right cabin audio selector panel, select the rotary the transmit selector switch to the Private (PVT) position. Key the aft left ICS and talk into the headset microphone. Verify the following:
   - Verify private intercom audio from the aft left cabin ICS station is heard only at the aft right ICS station and not at the pilot and copilot ICS stations.
   - *Verify acceptable private intercom volume range is available in the aft left cabin ICS headset while adjusting ICS volume control on the aft left audio selector panel.*
   - *Verify no effect on private intercom volume while adjusting radio volume control on the aft left cabin audio selector panel.*
   - Key and modulate ICS from pilot and copilot ICS station, verify ICS communication is heard at the aft left and right cabin ICS stations and is controlled by the corresponding ICS volume control.

5. Repeat Step 4. for the aft right cabin ICS.

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Intercom System - Completion of Ground Based - Operational Checks

Procedure

1. At the completion of the checks, if required, disconnect the headsets, disengage the circuit breakers, and remove the external power.

2. Return the helicopter to the standard configuration after electrical maintenance of repairs (412-A-96-00-01-02A-913A-A / 01875).

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Pre-flight - RPM Audio Warning - Operational Check

Procedure



A QUALIFIED PERSON MUST BE AT THE HELICOPTER CONTROLS DURING THE FOLLOWING PROCEDURE.

1. Start the helicopter (BHT-412-FM-4).

2. After engine start, adjust the rotor RPM to induce a “Low rotor RPM” audio warning. For the pilot and copilot ICS, verify the following:

- Verify RPM audio warning is audible in headsets and volume is adequate for ambient flight conditions.

3. Disengage the pilot ICS circuit breaker. On the pilot audio selector panel, select the NORM/EMER switch to the EMER position. Repeat Step 2. for the pilot ICS. On completion of the test, re-engage the pilot ICS circuit breaker and return the NORM/EMER switches to NORM.

4. Repeat Step 3. for the copilot ICS circuit breaker and the NORM/EMER switch on the copilot audio selector panel.

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RAD ALT Voice Warning - Operational Flight Check

Procedure

NOTE

The RAD ALT voice warning audio is interfaced to the pilot and copilot ICS as a non-volume controllable direct audio input.

1. The scope of this check is to verify the audio quality (of the ICS interface to the RAD ALT voice warning system) is acceptable for flight conditions.



A QUALIFIED PERSON MUST BE AT THE HELICOPTER CONTROLS DURING THE FOLLOWING PROCEDURE.

1.1. Start the helicopter (BHT-412-FM-4) and establish level flight.

1.2. Select the RAD ALT DH voice warning to be referenced to the pilot RAD ALT indicator. Verify the RAD ALT voice warning audio is not inhibited. Select the pilot RAD ALT DH to an appropriate altitude below current aircraft AGL altitude. Descend through DH altitude and as the DH light on the pilot RAD ALT indicator illuminates, verify the following:

- Verify an aural warning tone followed by a “Check height” voice audio is heard in the pilot and copilot headsets.
- Verify quality and volume of “Check height” voice warn audio is acceptable for flight conditions.

1.3. From an altitude above 150 feet AGL, descend through 150 feet AGL. As the helicopter descends through 150 feet AGL as indicated on the pilot RAD ALT indicator, verify the following:

- Verify an aural warning tone followed by a “One fifty feet” voice audio is heard in the pilot and copilot headsets.
- Verify quality and volume of the “One fifty feet” voice warning audio is acceptable for flight conditions.

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Intercommunication System - Troubleshooting

Procedure

NOTE

Troubleshooting is the same for both the cyclic sticks and foot switches.

1. Make sure that the appropriate circuit breakers are closed and interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate ICS wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. Intercommunication System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
No sidetone or interphone signals when cyclic stick or foot switch is depressed.	Defective headset.	Replace headset.
	Defective cyclic stick or foot switch.	Replace switch.
	Defective intercom control unit.	Replace intercom control unit.
No communication at one station.	Defective circuit breaker.	Replace circuit breaker.
	Defective intercom control unit.	Replace intercom control unit.
No communication or sidetone heard at any station.	Defective headset.	Try same headset at a known good location. If good, refer to next probable cause. Replace headset if result is the same.
	Defective intercom control unit.	Replace intercom control unit.
	Defective cyclic stick, foot or crew station (push-to-talk) switch.	Replace switch.
No reception or transmission at one station. Sidetone heard during transmission attempt. Rotary selector switch is in ICS position.	Open interphone communication line.	Continuity check interphone audio line between intercom control unit and common tie point. Repair or replace wiring as applicable.
With pilot or copilot intercom control unit rotary switch in COMM 1 position; interphone (ICS) sidetone not heard at transmitting station and ICS audio not heard at other station with pilot or copilot cyclic stick switch positioned to detent 1 (ICS position).	Defective cyclic stick switch.	Replace cyclic stick switch.
Received audio at other stations and sidetone at station is slurred, noisy, or otherwise unclear.	Defective intercom control unit.	Replace intercom control unit.
Received audio slurred, noisy, or otherwise unclear at one of the intercom control units.	Defective intercom control unit.	Replace intercom control unit.
No sidetone or interphone when emergency switch is depressed and when headset is connected to emergency jack.	Defective headset.	Replace headset.
	Defective emergency keying/comm switch.	Replace emergency keying/comm panel.
	Defective wiring.	Repair/replace wiring.

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Intercom Control Unit - Removal

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the intercom control unit to the pedestal or mount.
2. Lift the intercom control unit far enough to reach and disconnect the electrical connector.
3. Remove the intercom control unit.

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Foot Switches - Removal

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Disconnect and tag the wires from the foot switch.
2. Remove and retain the screws and other hardware that secures the foot switch.
3. Remove the foot switch from the mount.

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Cyclic Stick Switches - Removal

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Remove the plastic cover (left side threads) at the top of the cyclic stick grip.
2. Remove two screws that secure the switch in the grip.
3. Disconnect and tag all the wires and remove the RADIO/ICS switch.

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Emergency Keying/Comm Panel - Removal

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Loosen and retain the mounting hardware that secures the emergency keying/comm panel.
2. Lift the emergency keying/comm panel far enough to reach and disconnect the electrical connector.
3. Remove the emergency keying/comm panel.

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Intercom Control Unit - Installation

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the rear of the intercom control unit and fasten with captive screws.
2. Position and secure the intercom control unit to the pedestal with spring lock fasteners.
3. Do an operational check of the intercom system (412-A-97-01-00-00A-320A-A / 02244 or 412-A-97-01-01-00A-320A-A / 02253).

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Foot Switches - Installation

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Connect the tagged wires to the foot switch.
2. Secure the foot switch to the cabin floor with retained hardware.
3. Do an operational check of the foot switches.

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Cyclic Stick Switches - Installation

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Connect the tagged wires to the cyclic stick switch.
2. Slide the switch into the cyclic stick grip, and install two screws to secure the switch to the grip.
3. Replace the plastic cover.
4. Do an operational check of the cyclic stick switches.

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Emergency Keying/Comm Panel - Installation

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the rear of the emergency keying/comm panel.
2. Position the emergency keying panel in its mount and secure with retained hardware.
3. Do an operational check of the emergency keying/Comm feature (412-A-97-01-00-07A-320A-A / 02251).

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Intercom System - Inspection Procedures

Procedure

1. Inspect the intercom control units for completeness, proper installation, cleanliness, and proper mounting.
2. Inspect the switches and controls for proper mechanical action.
3. Inspect the intercom control units for proper illumination of panel lights, and the lamp housings for condition.
4. Inspect the electrical connectors for corroded or bent pins, interconnecting cables for frayed or broken insulation, and circuit breakers for proper operation.

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Intercom System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.

2. Remove dust and dirt from all front panel controls, and panel light housing with a soft, clean cloth. To remove fingerprints or difficult to remove dirt, use a cloth dampened with water. If necessary, mild soap may be used to make the cleaning more effective.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.

4. Remove dirt from the connector with a brush and remove moisture with a dry cloth.

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Intercom System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs.

2. Repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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VHF/AM Communication System - Description

The VHF/AM communication system provides two-way communications within frequency range 118.000 to 135.975 megahertz. Transmission and reception are on the same frequency using the VHF COMM No. 1 antenna.

The VHF communication system consists of an AM transceiver, a VHF COMM control unit, and a VHF antenna. These components are designated VHF No. 1, and when an optional second system kit consisting of a set of identical units (except for antenna) is installed, it is designated VHF No. 2.

VHF COMM No. 1 and VHF COMM No. 2 are independent systems of communication that may be used at the pilot or copilot stations.

Applicable: SN: 33101-33107:

The digital VHF COMM control unit features active and standby, gaseous discharge, frequency displays with automatic dimming. The control has non-volatile memory, which retains last selected frequencies when power is turned off. To tune desired operating frequency, the frequency is first entered into the standby (SBY) display, and then the transfer button must be pushed. This will trade contents of the active and standby displays. The transceiver is always tuned to the frequency appearing in the active display. In addition, a symbol T will appear on the display while transmitting.

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VHF/AM Communication System - Functional Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Thruline RF Wattmeter	Bird Model 43	1	or Equivalent

Procedure

1. VHF/AM Communication System - Preliminary Test Procedures

1.1. Visually inspect the VHF/AM communication system to detect and correct any faulty mechanical mountings and electrical connections. Connect an external power source, adjusted to 27.5 VDC, to the helicopter external power receptacle.

Applicable: SN: 33001-33100 or 33101-36086 or 36087-Subsequent:

1.2. Energize the power buses per 412-A-97-00-00-00A-340A-A / 02238 and 412-A-97-00-00-00A-340B-A / 02239, as applicable. Close the circuit breakers No. 1 VHF XCVR, No. 2 VHF XCVR (when installed), ICS PILOT, ICS CPLT, or VHF 1 COMM, VHF 2 COMM (when installed), ICS PILOT, ICS CPLT. Set the pilot and/or copilot intercom control unit mixing switches to ON.

2. VHF/AM Communication System - Receiver Test Procedure

2.1. After accomplishing Step 1., adjust VOL control on the VHF COMM control unit to the mid position.

2.2. Disable the receiver squelch circuit by depressing the SQ TEST switch on the VHF COMM control unit, and note receiver background noise is audible in the headset.(Appropriate mixing switch No. 1 or No. 2 on the intercom control unit must be ON.)

2.3. Make sure receiver background noise ceases when the SQ TEST switch is released.

3. VHF/AM Communication System - Transmitter Test VSWR Check

3.1. Connect the thruline rf wattmeter (Bird Model 43) in line with the antenna coax at the transceiver.

3.2. Select a frequency of 118.0 MHz on the VHF COMM control unit.

3.3. Turn the thruline rf wattmeter (Bird Model 43) so the arrow points toward the antenna load. Key the transmitter and read indicated transmitter output power. The rotary selector switch on the intercom control unit being used must be placed to COMM 1 to key No. 1 transmitter or COMM 2 to key No. 2 VHF transmitter.

3.4. Turn the thruline rf wattmeter (Bird Model 43) to read reflected power. Key transmitter and read indicated reflected power. The ratio of forward-to-reflected power should be not less than 4:1 (VSWR of 3:1). Forward power output should be 15 watts or more.

3.5. Repeat Step 3.3. and Step 3.4. for frequencies 123.0, 127.0, and 135.95 MHz.

3.6. Disconnect the thruline rf wattmeter (Bird Model 43) and connect the helicopter antenna coax to the transceiver.

4. VHF/AM Communication System - Transmitter Test Communication Check

4.1. After accomplishing the preliminary test procedure (Step 1.), establish communication with a cooperating station. Check transmitted and received signals are loud and clear, and that variation of VOL control on the VHF COMM control unit varies audio level of received signal. Also, note the level and adequacy of sidetone without excessive background noise.

4.2. Repeat Step 4.1. at each installed intercom control unit.

4.3. Turn off all equipment, open all circuit breakers, and disconnect external power source.

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VHF/AM Communication System - Troubleshooting

Procedure

NOTE

The isolation procedures in Table 1 are applicable to VHF COMM No. 1 or VHF COMM No. 2.

1. Make sure that the appropriate circuit breakers are closed and interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate VHF communication system wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. VHF/AM Communication System - Troubleshooting

TROUBLE INDICTION	PROBABLE CAUSE	CORRECTIVE ACTION
Equipment inoperative.	Defective VHF transceiver.	Replace VHV transceiver.
	Defective VHF COMM control unit.	Replace VHF COMM control unit.
	Defective wiring.	Repair or replace wiring.
Equipment inoperative; circuit breaker trips.	Defective VHF COMM transceiver.	Replace VHF COMM transceiver.
	Defective VHF COMM control unit.	Replace VHF COMM control unit.
	Shorted power wire.	Replace power wire.
No sidetone during transmission.	Defective VHF COMM transceiver.	Replace VHF transceiver.
	Loose or dirty cable connectors.	Check and clean connections.
	Defective wiring.	Repair or replace wiring.
No audio output at any ICS station	Defective VHF COMM transceiver.	Replace VHF COMM transceiver.
	Defective VHF COMM control unit.	Replace VHF COMM control unit.
	Defective wiring.	Repair or replace wiring.
Operating frequency is not correct.	Defective VHF COMM transceiver.	Replace VHF COMM transceiver
	Defective VHF COMM control unit.	Replace VHF COMM control unit.
No audio output at one ICS station.	Defective intercom control unit.	Replace intercom control unit.
	Defective wiring.	Repair or replace wiring.

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VHF COMM Transceiver (No. 1 or No. 2) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect electrical and antenna connectors from the front of the transceiver and protect open ends with electrical tape and caps.
2. Loosen the knurled retainer nut on the front of the mounting rack until the hold-down clamp can be slipped off the lug at the front of the transceiver.
3. Slide the transceiver out of its mount.

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VHF COMM Control Unit (No. 1 or No. 2) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel to the pedestal assembly.
2. Lift the control unit far enough out of the pedestal assembly to reach and disconnect the electrical connector at the back of the control unit.
3. Lift the control unit free of the pedestal assembly.

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VHF COMM Control Unit (No. 1 or No. 2) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel mounting plate assembly to the pedestal assembly.
2. Lift the control panel mounting plate assembly and control unit(s) far enough out of the pedestal assembly to reach and disconnect the electrical connector(s) at the back of the control unit(s).
3. Lift the control panel mounting plate assembly and control unit(s) free of the pedestal assembly.
4. Loosen the screws and nuts that secure the control unit(s) to the control panel mounting plate assembly.
5. Lift the control unit(s) free of the control panel mounting plate assembly.

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VHF Communication Antenna No. 1 - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the screws that secure the antenna to the mounting plate on the helicopter.
2. Lift the antenna and disconnect the cable, connected to the receptacle at base of the antenna, and remove the antenna.

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VHF Communication Antenna No. 2 - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the screws that secure the access panel upon which the antenna is mounted.
2. Lower the access panel and antenna from the helicopter skin until the antenna connector can be reached.
3. Disconnect the antenna cable connection.
4. Remove the antenna mounting screws and separate the antenna from the access panel.

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VHF COMM Transceiver (No. 1 and No. 2) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the transceiver on the mount and slide the transceiver into place.
2. Place the hold-down clamp of the mount over the lug on the front of the transceiver.
3. Tighten the knurled retainer nut until the transceiver is secure.
4. Remove the protective tape and caps from the electrical and antenna connectors and connect to the front of the transceiver.

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VHF COMM Control Unit (No. 1 and No. 2) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the back of the control unit.
2. Position the control panel in place on the pedestal.
3. Secure the panel to the pedestal assembly with the spring lock fasteners.

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VHF COMM Control Unit (No. 1 and No. 2) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the control unit(s) in the control panel mounting plate assembly and secure with the mounting screws and nuts.
2. Connect the electrical connectors to the back of the control unit(s).
3. Position the control panel mounting plate assembly and control unit(s) in place on the pedestal and secure with the spring lock fasteners.

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VHF Communication Antenna No. 1 - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the antenna cable connector to the receptacle at the base of the antenna.
2. Position the communication antenna on its mounting plate and secure with the mounting screws.

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VHF Communication Antenna No. 2 - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Secure the antenna on the access panel.
2. Connect the antenna cable connector to the receptacle at the base of the antenna.
3. Position the access panel and antenna in place on the helicopter skin.
4. Secure the access panel with the mounting screws.

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VHF/AM Communication System - Inspection Procedures

Procedure

1. Inspect the VHF COMM control unit for completeness, proper installation, cleanliness, and proper mounting.
2. Inspect the indicators and controls for proper mechanical action and readout.
3. Inspect the VHF COMM control unit lights for proper illumination when pedestal lighting is on.
4. Inspect the VHF COMM transceiver for proper mounting.
5. Inspect the VHF antennas and mountings for corrosion, dents, cracks, and breaks.
6. Inspect the electrical connections for corrosion, bent pins, and proper mating. Inspect the interconnecting cables for frayed or broken insulation and the circuit breakers for proper operation.

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VHF/AM Communication System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.
2. Remove dust and dirt from the panel, panel controls, panel light housings, antenna, and transceiver with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.
4. Remove dirt from the connectors with a brush. Remove moisture with a clean, dry cloth.

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VHF/AM Communication System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs.
2. Replace defective or inoperative units.
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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Gyromagnetic Compass System(s) - Description

The pilot gyromagnetic compass system (No. 1) is installed as basic equipment, and a copilot gyromagnetic compass system (No. 2) is available as optional equipment. Each system consists of miscellaneous airframe components and the following prime components:

BASIC EQUIPMENT
3-Axis Reference Sensor No. 1
Transmitter Flux Valve No. 1
Compass Control Panel No. 1
Dual Remote Compensator
(Part of) Horizontal Situation Indicator, Pilot (1)
Standby Magnetic Compass
OPTIONAL EQUIPMENT
RMI Adapter No. 1
3-Axis Reference Sensor No. 2
Transmitter Flux Valve No. 2
Compass Control Panel No. 2
(Part of) Horizontal Situation Indicator, Copilot (1)
RMI Adapter No. 2

NOTE:

1 For maintenance procedures, refer to (412-A-95-12-06-00A-042A-A / 01835).

Separate annunciators, MAG/DG switches, and synchronizing switches for each gyromagnetic compass system are provided on each pedestal mounted compass control panel. A single dual remote compass compensator is used for both systems. The No. 1 compass system furnishes compass information to the compass card of the pilot Horizontal Situation Indicator (HSI) and the No. 2 compass system (when installed) furnishes compass information to the compass card of the copilot HSI (optional).

In addition to supplying indicator heading data, the No. 1 compass system provides heading information to the No. 1 VOR/LOC/GS navigation system (when installed). The No. 2 compass system provides heading information to the No. 2 VOR/LOC/GS navigation system (if installed) and the Automatic Flight Control System (AFCS) (when installed).

NOTE

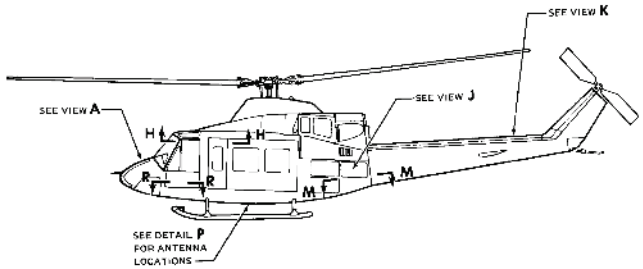
The vertical gyro portion of the No. 1 3-axis reference sensor provides pitch and roll information to the pilot Attitude Director Indicator (ADI) . The vertical gyro portion of the No. 2 3-axis reference sensor provides pitch and roll information for a copilot ADI and AFCS.

Heading synchronization for either gyromagnetic compass is accomplished by observing the appropriate annunciator on the compass control panel, and actuating the synchronization switch in the direction indicated to center annunciator needle (MAG/DG switch must be set to MAG).

The heading function of the 3-axis reference sensors for the pilot and copilot compass systems receives 115 VAC power from the No. 2 and No. 1 AC buses, respectively. The circuit breakers are labeled PILOT GYRO CMPS and CPLT GYRO CMPS. Synchro excitation (26 VAC) for both 3-axis reference sensors is normally supplied from the No. 1 26 VAC bus through the NORM position of the NAV AC switch on the overhead console. In the event of a No. 1 26 VAC bus failure, synchro excitation may be restored by manually placing the NAV AC switch to the BUS No. 2 position. 26 VAC synchro excitation circuits are protected by circuit breakers labeled PILOT HSI and CPLT HSI.

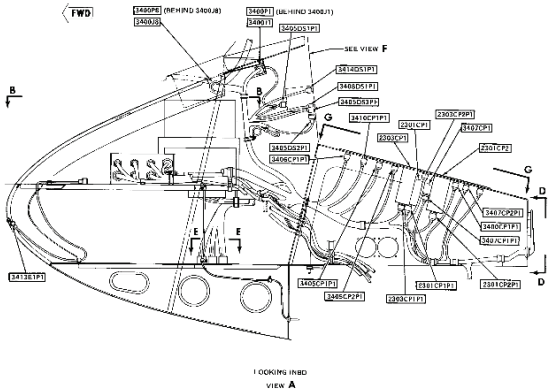
In addition to the gyromagnetic compass (or compasses), a pilot standby magnetic compass is installed and a copilots standby magnetic compass is available with the copilot instrument kit (optional). In each case, the standby magnetic compass is mounted on the upper windshield frame in front of the pilot and copilot, respectively. The standby magnetic compass (or wet compass) is a direct reading compass that indicates helicopter heading, with reference to magnetic north, and is independent of the helicopter electrical power. Refer to Figure 1 for avionics equipment location.

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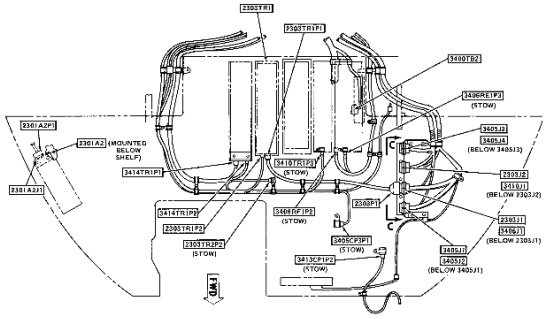
ICN-412-A-970000-B-97499-00001-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 1 of 13)



ICN-412-A-970000-B-97499-00002-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 2 of 13)



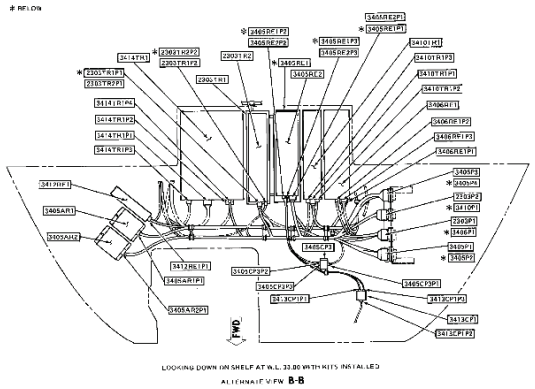
LOOKING DOWN ON SHELF AT W.L. 33.00

VIEW **B-B**

SEE ALTERNATE VIEW FOR AUXILIARY EQUIPMENT

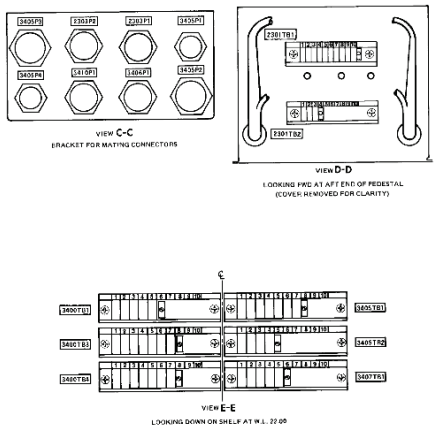
ICN-412-A-970000-B-97499-00003-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 3 of 13)



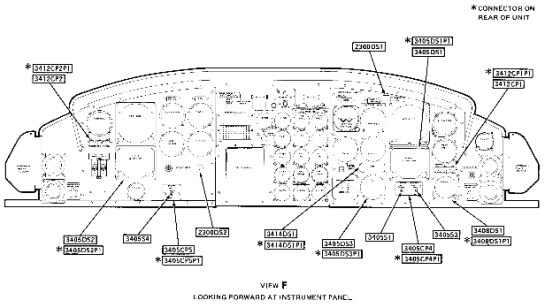
ICN-412-A-970000-B-97499-00004-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 4 of 13)



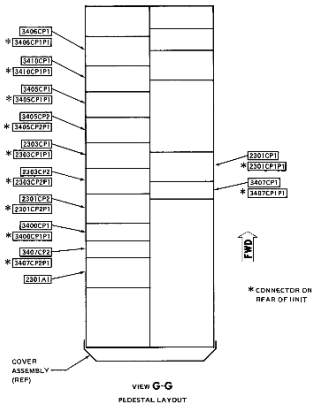
ICN-412-A-970000-B-97499-00005-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 5 of 13)



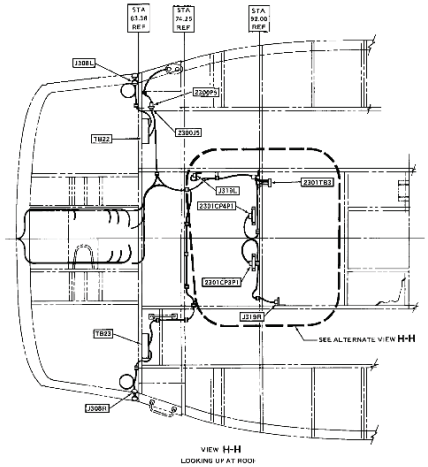
ICN-412-A-970000-B-97499-00006-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 6 of 13)



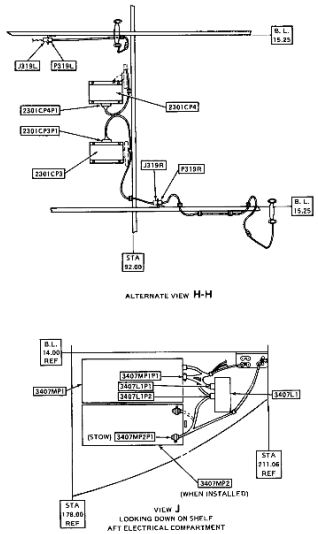
ICN-412-A-970000-B-97499-00007-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 7 of 13)



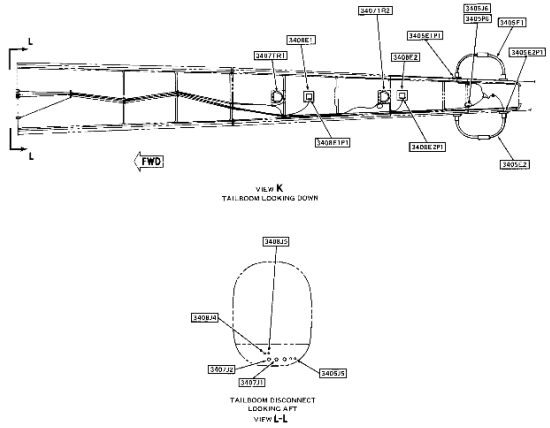
ICN-412-A-970000-B-97499-00008-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 8 of 13)



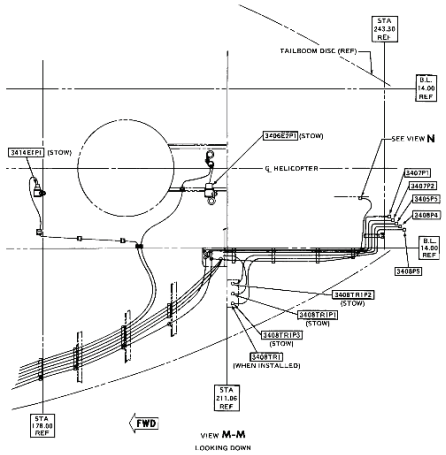
ICN-412-A-970000-B-97499-00009-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 9 of 13)



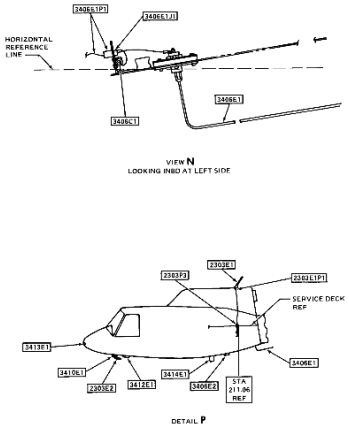
ICN-412-A-970000-B-97499-00010-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 10 of 13)



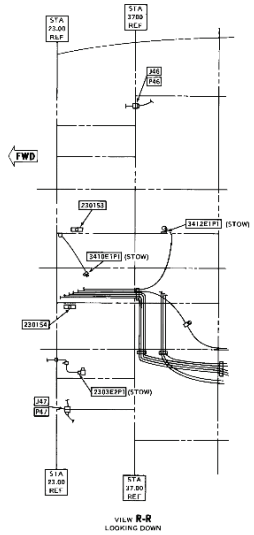
ICN-412-A-970000-B-97499-00011-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 11 of 13)



ICN-412-A-970000-B-97499-00012-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 12 of 13)



ICN-412-A-970000-B-97499-00013-A-001-01

Figure 1. Avionics Equipment Location (S/N 33001 through 33100) (Sheet 13 of 13)

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Gyromagnetic Compass System(s) - Description

The copilot gyromagnetic compass system (No. 1) and pilot gyromagnetic compass system (No. 2) are installed as basic equipment. Each system consists of miscellaneous airframe components and the following prime components:

BASIC EQUIPMENT
3-Axis Reference Sensor No. 1, Copilot
Transmitter Flux Valve No. 1
Compass Control Panel No. 1, Copilot
Dual Remote Compass Compensator
3-Axis Reference Sensor No. 2, Pilot
Transmitter Flux Valve No. 2
Compass Control Panel No. 2, Pilot
(Part of) Horizontal Situation Indicator, Pilot (1)
Standby Magnetic Compass, Pilot
OPTIONAL EQUIPMENT
(Part of) Horizontal Situation Indicator, Copilot (1)
Standby Magnetic Compass, Copilot

NOTE:

1 For maintenance procedures, refer to (412-A-95-12-06-00A-042A-A / 01835).

Separate annunciators, MAG/DG switches, and synchronizing switches for each gyromagnetic compass system are provided on each pedestal mounted compass control panel. A single dual remote compass compensator is used for both systems. No. 1 compass system furnishes compass information to the compass card of the copilot Horizontal Situation Indicator (HSI) (when installed) and the No. 2 compass system furnishes compass information to the compass card of the pilot HSI.

In addition to supplying indicator heading data, the No. 1 compass system provides heading information to the No. 1 VOR/ILS navigation system (when installed). The No. 2 compass system provides heading information to the No. 2 VOR/LOC/GS navigation system (if installed) and the Automatic Flight Control System (AFCS) (when installed).

NOTE

The vertical gyro portion of the No. 1 3-axis reference sensor provides pitch and roll information to the copilot Attitude Director Indicator (ADI) and AFCS. The vertical gyro portion of the No. 2 3-axis reference sensor provides pitch and roll information for pilot ADI and AFCS.

Heading synchronization for either gyromagnetic compass is accomplished by observing the appropriate annunciator on the compass control panel and actuating the synchronization switch in the direction indicated to center annunciator needle (MAG/DG switch must be set to MAG).

Applicable: SN: 33101-36086 or 36087-33247 or 33101-33107:

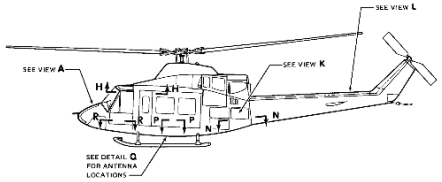
The heading function of the 3-axis reference sensors for pilot and copilot compass systems receive 115 VAC power from the No. 2 and No. 1 AC buses, respectively. Circuit breakers are labeled PILOT GYRO CMPS and CPLT GYRO CMPS or NO. 1 GYRO CMPS and NO. 2 GYRO CMPS.

Synchro excitation (26 VAC) for the No. 1 and No. 2 3-axis reference sensors is supplied from the No. 1 and No. 2 26 VAC buses respectively. The 26 VAC synchro excitation circuits are protected by circuit breakers labeled PILOT HSI and CPLT HSI.

In addition to gyromagnetic compasses, a pilot standby magnetic compass is installed and a copilot standby magnetic compass is available with the copilot instrument kit (optional). In each case, the standby magnetic compass is mounted on the upper windshield frame in front of the pilot and copilot, respectively. The standby magnetic compass (or wet compass) is a direct reading compass that indicates helicopter heading, with reference to magnetic north, and is independent of the helicopter electrical power system.

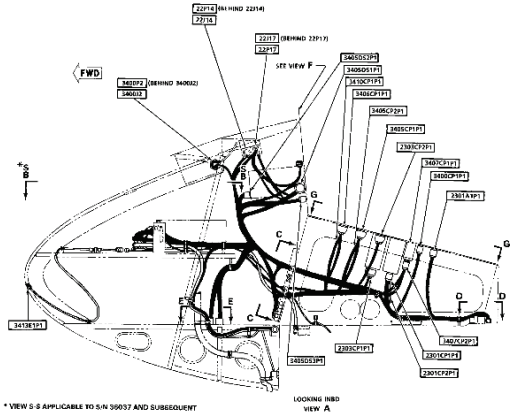
Refer to Figure 1 for avionics equipment location.

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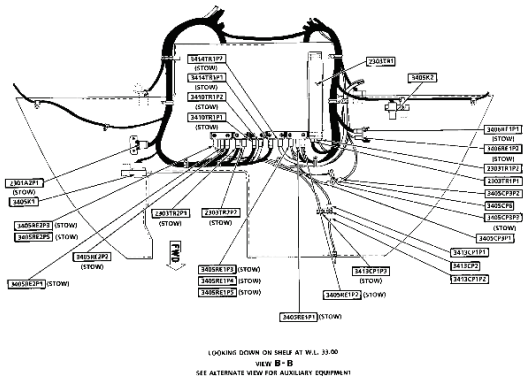
ICN-412-A-970000-B-97499-00014-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 20)



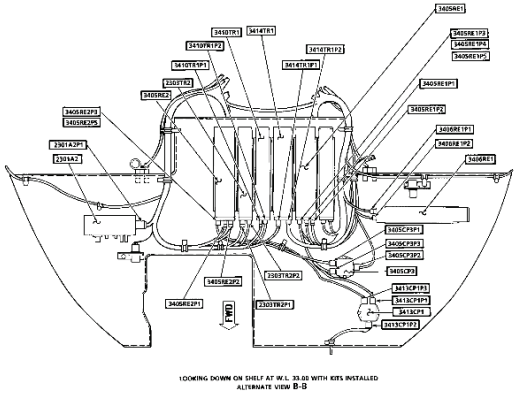
ICN-412-A-970000-B-97499-00015-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 2 of 20)



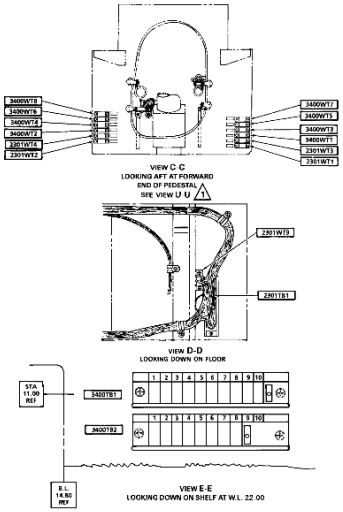
ICN-412-A-970000-B-97499-00016-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 3 of 20)



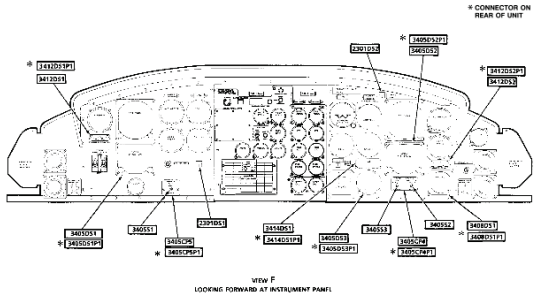
ICN-412-A-970000-B-97499-00017-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 4 of 20)



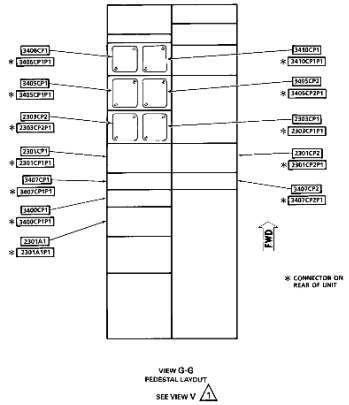
ICN-412-A-970000-B-97499-00018-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 5 of 20)



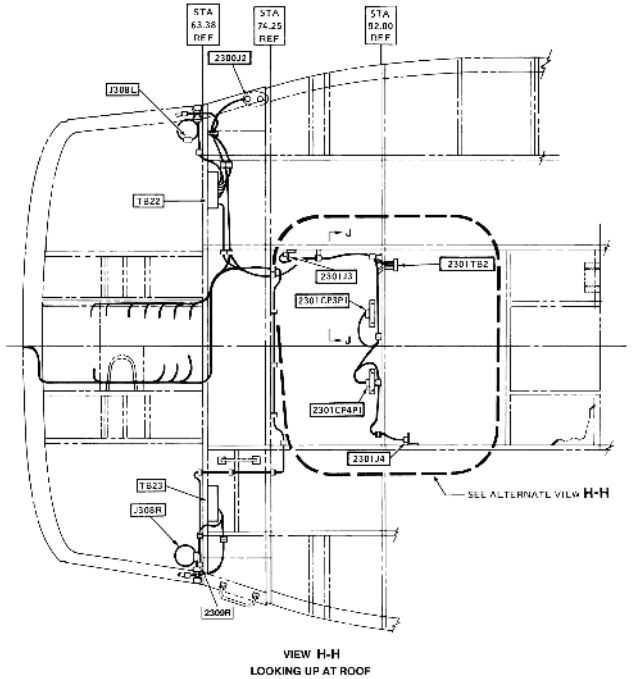
ICN-412-A-970000-B-97499-00019-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 6 of 20)



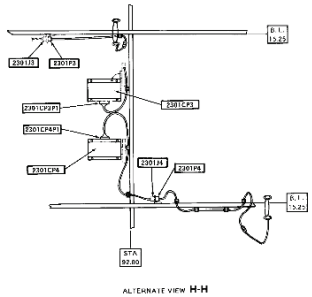
ICN-412-A-970000-B-97499-00020-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 7 of 20)



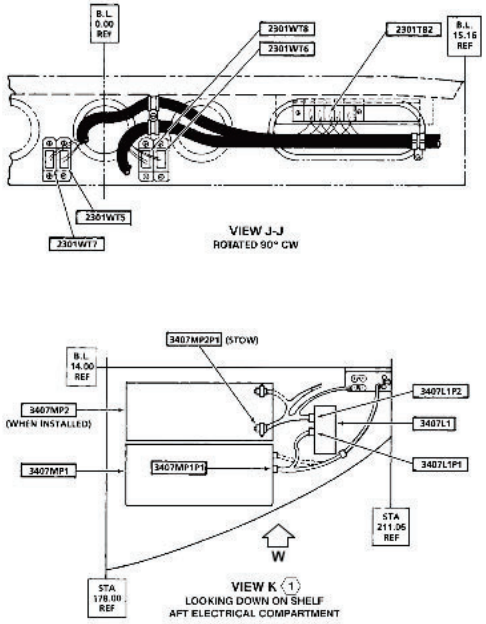
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 8 of 20)



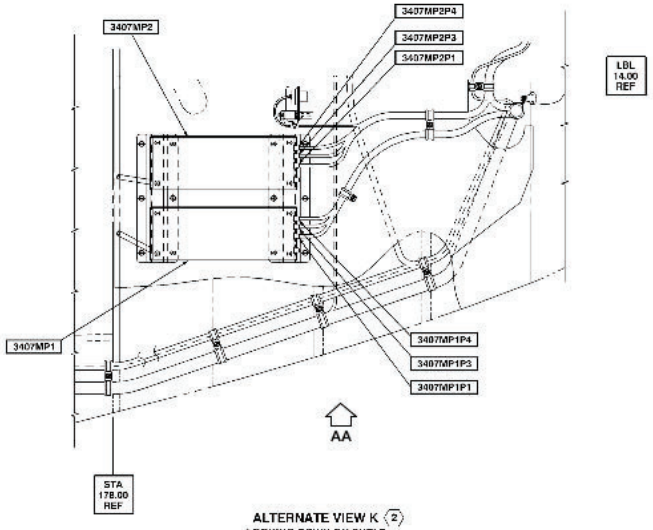
ICN-412-A-970000-B-97499-00022-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 9 of 20)



ICN-412-A-970000-B-97499-00023-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 10 of 20)



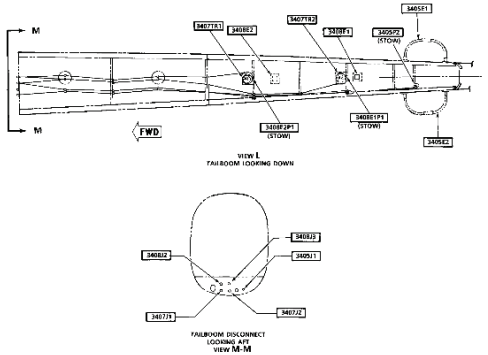
ALTERNATE VIEW K (2)
LOOKING DOWN ON SHELF
AFT ELECTRICAL COMPARTMENT

EFFECTIVITY

(1) S/N 33101 THROUGH 33213 and 36001 THROUGH 36247.
(2) S/N 36248 THROUGH 36999.

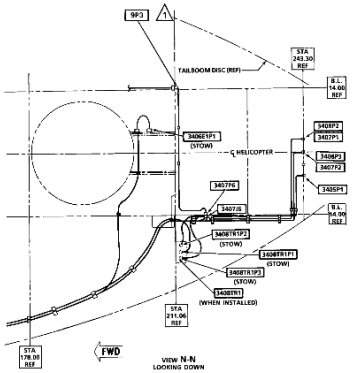
ICN-412-A-970000-B-97499-00024-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 11 of 20)



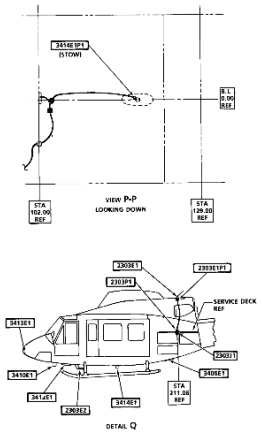
ICN-412-A-970000-B-97499-00025-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 12 of 20)



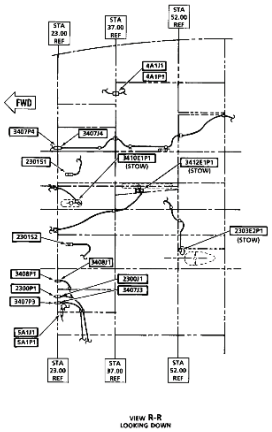
ICN-412-A-970000-B-97499-00026-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 13 of 20)



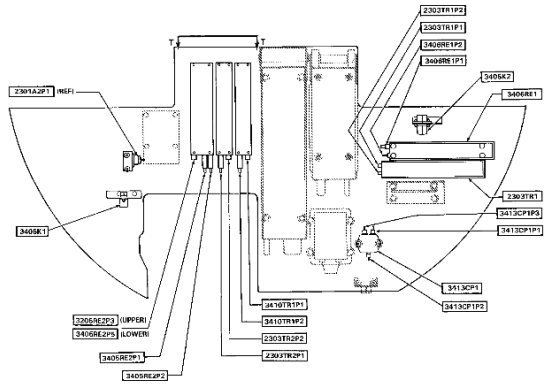
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 14 of 20)



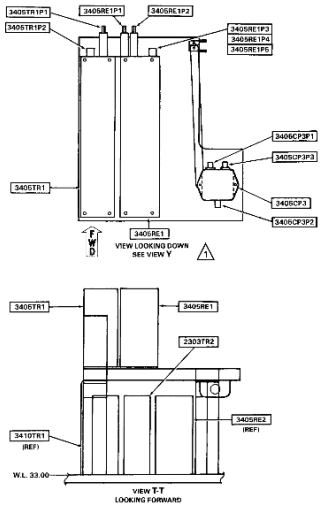
ICN-412-A-970000-B-97499-00028-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 15 of 20)



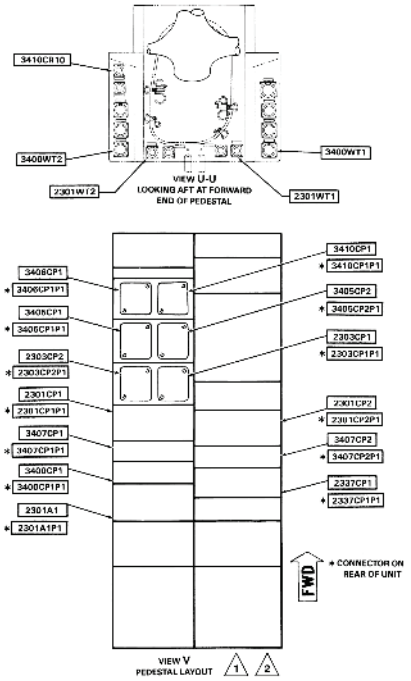
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 16 of 20)



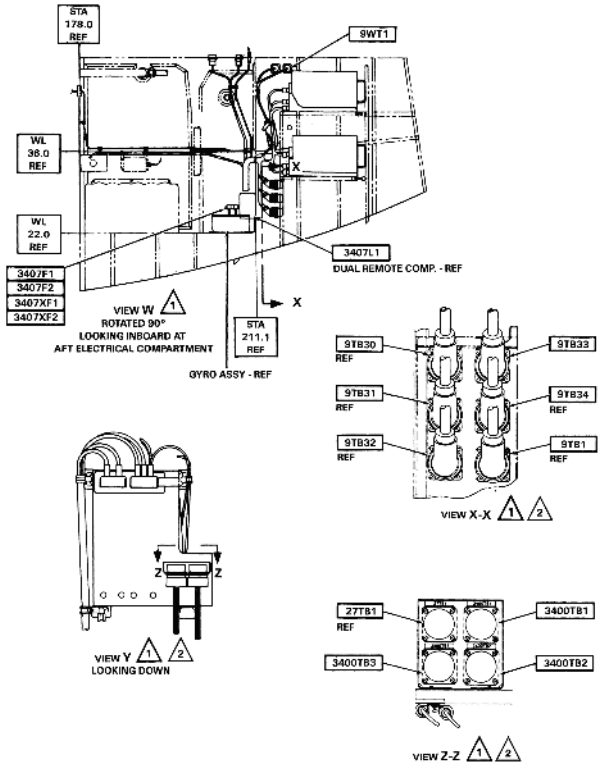
ICN-412-A-970000-B-97499-00030-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 17 of 20)



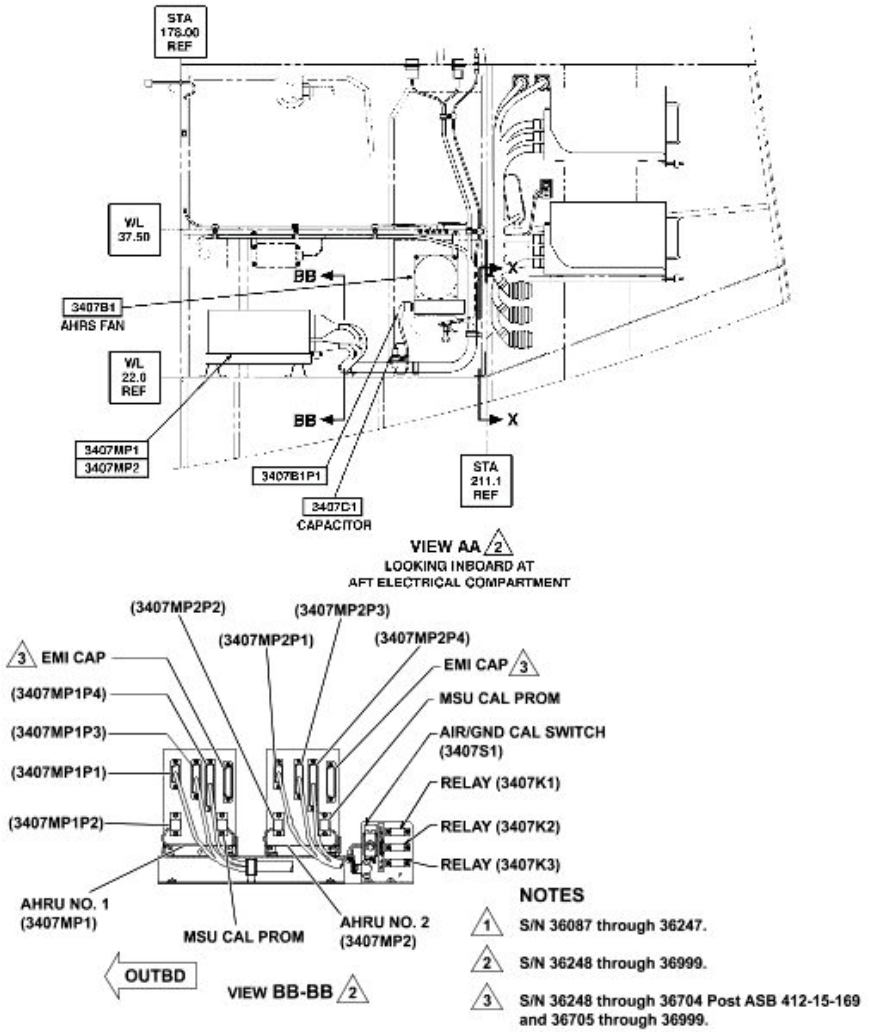
ICN-412-A-970000-B-97499-00031-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 18 of 20)



ICN-412-A-970000-B-97499-00032-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 19 of 20)



ICN-412-A-970000-B-97499-00033-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 20 of 20)

Gyromagnetic Compass System(s) - Functional Test

Procedure

1.

NOTE

Unless otherwise specified, the maintenance procedures are applicable to both No. 1 and No. 2 gyromagnetic compass systems.

NOTE

To conduct functional checks and calibration of the No. 1 compass system when a copilot HSI is not installed, it is necessary to connect an HSI to the copilot stowed HSI connector. The pilot HSI may be used for this purpose by disconnecting the pilot HSI connector and using a test harness to temporarily connect the copilot HSI wiring harness to the pilot HSI.

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Gyromagnetic Compass System(s) - Preliminary Operational Test

Procedure

NOTE

Unless otherwise specified, the following operational test procedures are applicable to both No. 1 and No. 2 gyromagnetic compass systems.

NOTE

This is a general test, conducted prior to compass rose alignment and calibration, to ensure the system is operational.

1. Set the compass MAG-DG switch to the MAG position.

NOTE

When the MAG-DG switch is in the DG position, directional gyro (unslaved) mode is selected and the annunciator feature is disengaged.

2. Energize the system by connecting external power source adjusted to 27.5 VDC to the helicopter external power receptacle.

3. Energize the power buses per 412-A-97-00-00-00A-340A-A / 02238 or 412-A-97-00-00-00A-340B-A / 02239 as applicable.

Applicable: SN: 33001-33100 or 33108-36086 or 36087-Subsequent:

4. Close the circuit breakers PILOT GYRO CMPS, CPLT GYRO CMPS, PILOT HSI (2), and CPLT HSI (2), as applicable.

Applicable: SN: 33101-33107:

5. Close the circuit breakers NO. 2 GYRO CMPS, NO. 1 GYRO CMPS, PILOT HSI (2), and CPLT HSI (2) as applicable.

6. The compass OFF flag on each installed HSI should be masked from view.

7. To synchronize the compass system under test, depress the appropriate synchronization switch (pilot or copilot) in the direction indicated by the annunciator (+ or •) until the needle is centered.

NOTE

It is normal for the HSI compass OFF flag to be in view while the synchronization switch is actuated.

8. Observe the appropriate annunciator for a few minutes to see that synchronization is maintained (no drift).

NOTE

Annunciators are very sensitive to transmitter flux valve movement and will appear to be unstable if slight tailboom movement occurs.

9. Observe the compass indication on the appropriate HSI, and depress the synchronization switch to plus (+) until the heading indication has increased 2°.

10. Observe the annunciator to verify that the needle is deflected off-center towards the dot (•), and observe the indicator for a sufficient period of time to verify that the system is slaving down scale.

11. To resynchronize the compass system under test, observe the compass indication on the HSI, then depress the synchronization switch to the dot (•) until the heading indication has decreased 2°.

12. Observe the annunciator to verify that the needle is deflected off-center towards the plus (+), and observe the indicator for a sufficient period of time to verify that the system is slaving up scale.

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Compass Swing - Initial Setup - Operational Test

Procedure

NOTE

Unless otherwise specified, the following initial setup procedures are applicable to both No. 1 and No. 2 gyromagnetic compass systems.

NOTE

The complete compass swing procedure outlined in this manual requires several hours to complete, and should be performed only when absolutely necessary.

1. Position the helicopter on a compass rose. The helicopter must be in its normal flight attitude.

2. Remove the cover from the dual remote compensator (located in the left aft electrical compartment) and loosen the locknuts on the N-S and E-W adjustment screws. Adjust the appropriate N-S and E-W compensator screws (No. 1 system or No. 2 system) to the mid-position.

NOTE

An erroneous heading indication may result if the area around the helicopter is not magnetically clean and free from equipment such as trucks, hangars, maintenance stands, etc.

3. Close all the circuit breakers, set the INV 1 and INV 2 switches to ON, set the MAG-DG switch to MAG, and turn on all the electrical/electronic equipment to simulate normal flight conditions. Allow at least 10 minutes for the directional gyro to attain full speed and settle to a null.

NOTE

Due to the influence of the helicopter generators, it is recommended that compass calibration be accomplished with the engines running. For safety purposes, the engines should be shut down when making adjustments to correct compass error.

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Compass Swing - Index Error Check - Operational Test

Procedure

NOTE

Unless otherwise specified, the following index error check procedures are applicable to both No. 1 and No. 2 gyromagnetic compass systems.

NOTE

Instead of a compass rose, a magnetic sighting compass may be used. To take a reading, locate the compass a considerable distance fore or aft of the helicopter and move it back and forth from line of sight coinciding with the fore and aft (longitudinal) axis of the helicopter. When a sight is taken facing aft, 180° must be added or subtracted from the sighting compass reading. When facing fore, the compass reads directly.

1. Rotate the helicopter to the four cardinal headings North, East, South, and West. At each heading, verify that the annunciator needle is centered, and record the difference in readings between the HSI scale dial and the compass rose. The difference will be plus if the scale dial reading is greater than that of the compass rose, and minus if the scale dial reading is less than that of the compass rose.

2. Add the errors algebraically and divide by four. The result is the index error.

3. Gain access to the appropriate transmitter flux valve.

NOTE

SN: 33001-33100:

The transmitter flux valve No. 1 for the pilot system is located forward in the tailboom. The transmitter flux valve No. 2 for the copilot system (when installed), is located aft in the tailboom.

NOTE

SN: 33101-36086 or 36087-Subsequent:

The transmitter flux valve No. 1 for the copilot system is located forward in the tailboom. The transmitter flux valve No. 2 for the pilot system is located aft in the tailboom.

4. Loosen the screws that hold the appropriate transmitter flux valve flange to its mounting surface, and rotate the flange of the unit to cancel out the index error. If the error is positive, the flange should be rotated counterclockwise (giving a minus reading on the flange), as observed from above the unit. If the error is negative, rotate the flange clockwise, giving a plus reading on the flange. The amount of rotation shall equal the index error.

5. Tighten the mounting screws and recheck the readings at the four cardinal headings. The procedure should be repeated, if necessary, to obtain the least index error possible.

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Compass Swing - One-Cycle Error Check - Operational Test

Procedure

NOTE

Unless otherwise specified, the following one-cycle error check procedures are applicable to both No. 1 and No. 2 gyromagnetic compass systems.

1. Place the helicopter on a north heading. Allow sufficient time for the scale dial to settle. Correct for all error by turning the appropriate North-South screw on the dual remote compensator.

NOTE

SN: 33001-33213 or 36001-36247:

The compensator screws labeled No. 1 SYSTEM are for the pilot compass system and the screws labeled No. 2 SYSTEM are for the copilot system.

2. Turn the helicopter to an east heading. To correct for all error, turn the appropriate East-West screw.

3. Turn the helicopter to a south heading. Take out half the error, if any, with the appropriate North-South screw.

4. Turn the helicopter to a west heading. Take out half the error, if any, with the appropriate East-West screw.

5. Repeat Step 3. and Step 4. as necessary.

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Compass Swing - Final Heading Check - Operational Test

Procedure

NOTE

Unless otherwise specified, the following final heading check procedures are applicable to both No. 1 and No. 2 gyromagnetic compass systems.

1. Swing the helicopter on 30° increments and note each indication. All readings should be within ±1° of the actual magnetic heading. At no heading should the pilot and copilot readings differ by more than 1°. If errors noted are not within the tolerance, repeat the index error check and one-cycle error check until the errors are all within ±1°.

2. Tighten the locknuts on the N-S and E-W compensator screws after final adjustment.

3. Replace the cover on the dual remote compass compensator.

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Standby Magnetic Compass - Swing Procedure

Procedure

NOTE

The standby magnetic compass (or compasses) can be aligned simultaneously, with the gyromagnetic compass system compensation swing.

1. Record the difference between indications on the standby magnetic compass and magnetic heading at all four cardinal headings during the index error check of the gyromagnetic compass system (412-A-97-21-00-01A-320B-A / 02301). Add errors algebraically and divide by four. Align the standby magnetic compass by adjusting mounting to satisfy the resultant error.

2. During the one-cycle error check of the applicable gyromagnetic compass system (412-A-97-21-00-01A-320C-A / 02302), correct errors on the standby magnetic compass until the error is 2° or less at the cardinal headings (0°, 90°, 180°, 270°) and 5° or less at other 30° points.

3. Proceed with the final swing (412-A-97-21-00-01A-320D-A / 02303).

4. Record the deviation errors on the compass correction card.

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Gyromagnetic Compass System - Troubleshooting

Procedure

1. Make sure the appropriate circuit breakers are closed and interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate gyromagnetic compass system wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. Gyromagnetic Compass System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
Compass card on HSI does not move, but annunciator does move.	Defective HSI.	Replace HSI.
	Defective 3-axis reference sensor.	Replace 3-axis reference sensor.
	Defective wiring to HSI.	Repair or replace wiring.
Annunciator will not center (null) with MAG-DG switch in MAG position.	Defective MAG-DG switch.	Replace compass control panel.
	Defective dual remote compensator.	Replace dual remote compensator.
	Defective wiring.	Repair or replace wiring.
Annunciator will not remain centered with switch in DG position.	Defective MAG-DG switch.	Replace compass control panel.
	Defective 3-axis reference sensor.	Replace 3-axis reference sensor.
	Defective wiring.	Repair or replace wiring.

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3-Axis Reference Sensor - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Disconnect the electrical connectors.
2. Loosen the wingnuts until the 3-axis reference sensor is free from the mounting tray.
3. Lift the 3-axis reference sensor from the mounting tray.
4.

NOTE

The isolation procedures in Table 1 are applicable to both the pilot and copilot gyromagnetic compass systems.

Table 1. Gyromagnetic Compass System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
Compass card on HSI does not move, but annunciator does move.	Defective HSI.	Replace HSI.
	Defective 3-axis reference sensor.	Replace 3-axis reference sensor.
	Defective wiring to HSI.	Repair or replace wiring.
Annunciator will not center (null) with MAG-DG switch in MAG position.	Defective MAG-DG switch.	Replace compass control panel.
	Defective dual remote compensator.	Replace dual remote compensator.
	Defective wiring.	Repair or replace wiring.
Annunciator will not remain centered with switch in DG position.	Defective MAG-DG switch.	Replace compass control panel.
	Defective 3-axis reference sensor.	Replace 3-axis reference sensor.
	Defective wiring.	Repair or replace wiring.

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Transmitter Flux Valve - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Tag and remove the electrical wires.
2. Remove and retain the mounting screws.
3. Remove the flux valve from the mounting bracket.

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Dual Remote Compass Compensator - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Disconnect the electrical connectors.
2. Remove and retain the mounting screws.
3. Remove the compensator from the shelf.

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Compass Control Panel - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Loosen the spring lock fasteners that secure the control panel to the pedestal assembly.
2. Lift the control panel far enough out of the pedestal assembly to reach and disconnect the electrical connectors.

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RMI Adapter - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Disconnect the electrical connector.
2. Loosen the mounting screws.
3. Lift the RMI adapter from the bracket.

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3-Axis Reference Sensor - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the 3-axis reference sensor in the mounting tray. Tighten the wingnuts until snug.
2. Connect the electrical connectors.

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Applicable to: ALL
ECCN EAR99

Transmitter Flux Valve - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the flux valve in the mounting bracket.
2. Connect the tagged electrical wires.
3. Install the mounting screws.

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Dual Remote Compass Compensator - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the compensator on the mounting shelf and reinstall the mounting screws.
2. Connect the electrical connectors.

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Applicable to: ALL
ECCN EAR99

Compass Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connectors.
2. Position the control panel in the pedestal and fasten the spring lock fasteners.

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RMI Adapter - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the RMI adapter on the bracket and tighten the screws.
2. Connect the electrical connector.

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Gyromagnetic Compass System(s) - Inspection

Procedure

1. Inspect the gyromagnetic and standby magnetic compasses for proper installation, cleanliness, loose or cracked indicator glass, and leakage.
2. Inspect the electrical connections and interconnecting cables for frayed or broken insulation.
3. Inspect the indicators and controls for proper mechanical action.
4. Inspect the standby magnetic compass for leaking float chamber. The fluid must be clear.

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Gyromagnetic Compass System(s) - Cleaning

Procedure

1. Remove moisture and loose dirt from the gyromagnetic and standby compasses with a clean, soft cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC. USE NECESSARY PRECAUTIONS TO PREVENT INJURY TO PERSONNEL AND DAMAGE TO HELICOPTER.

2. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with approved cleaning compound.



DO NOT USE POINTED OBJECTS OR TOOLS OTHER THAN THOSE PRESCRIBED TO CLEAN CONNECTORS. PERMANENT DAMAGE TO HELICOPTER COMPONENTS MAY OCCUR.

3. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a dry cloth.

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Gyromagnetic Compass System(s) - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or indicators and control panels.
2. Replace defective controls and indicators.
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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Applicable to: ALL
ECCN EAR99

Attitude and Heading Reference System(s) - Description

The Attitude and Heading Reference System (AHRS) is an all attitude inertial sensor system, which provides heading and flight dynamics information to the various helicopter navigation systems. Each AHRS uses fiber optic rate gyros tied to the principal helicopter axes. An internal computer mathematically integrates the rate data to provide information on heading, pitch, and roll. A flux valve and two internal level sensors provide other reference input data for each AHRS.

The No. 1 AHRS provides attitude and turn rate information to the copilot Attitude Director Indicator (ADI) and heading information to the copilot Horizontal Situation Indicator (HSI) . The No. 1 AHRS also provides yaw rate data to the No. 1 autopilot.

The No. 2 AHRS provides attitude and turn rate information to the pilot ADI and heading information to the pilot HSI. The No. 2 AHRS also provides yaw rate data to the No. 2 autopilot.

The AHRS also provides pitch, roll, and heading information to the digital Automatic Flight Control System (AFCS) . The following standard kits also receive AHRS data when installed:

KIT	DATA TYPE
Nav Rx	Heading
Wx Radar	Pitch, Roll, Heading
FDR	Pitch, Roll, Heading

Applicable: SN: 36248-36527:

The Model LCR-92 AHRS is installed in helicopters. For additional information on the LCR-92, refer to the LITEF GmbH Installation/Maintenance Instruction (Document No. 141450-0000-840).

Applicable: SN: 36528-Subsequent:

The Model LCR-100 AHRS is installed in helicopters and helicopters modified by LCR-92 to LCR-100 retrofit kit (412-704-047-101, -103, -119, -121, -123 and -125).

For additional information on the LCR-100, refer to the Northrop Grumman LITEF GmbH Installation/Maintenance Instruction (Document No. 145130-0000-840).

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AHRS Components - Description

The AHRS consists of the following main components:

- Attitude and Heading Reference Unit (AHRU) (Qty 2)
- Flux valve (MSU) (Qty 2)
- MSU Cal Prom (Qty 2)
- AHRS control panel (Qty 2)
- AHRS fan

AHRS test switches, MAG/DG mode switches, synchronization (•/+) switches, and slaving error (•/+) annunciators are provided on the pedestal-mounted pilot and copilot AHRS control panels.

The AHRS has an automatic synchronization feature that synchronizes to magnetic heading when power is applied. Automatic synchronization can only occur when the MAG/DG switch is set to MAG. When the MAG/DG switch is selected to the MAG position, the output data from the AHRS modules are slaved to magnetic heading. This is the normal operating switch position. When the MAG/DG switch is set to the DG position, the output data from the AHRS modules are no longer slaved to magnetic heading.

The synchronization switch has two positions. When held to the (•) position, the compass heading on the ADI is decreased. When held to the (+) position, the compass heading on the ADI is increased. The synchronization switch only functions when the MAG/DG switch is set to DG mode.

The synchronization annunciator, when centered, indicates that the compass heading is slaved to magnetic heading. When the annunciator is in the (•) position, it indicates that the heading is greater than detected magnetic heading. When in the (+) position, it indicates that the heading is less than magnetic heading.

The AHRS test switch engages the test mode when the switch is held in position. The AHRS test is inhibited when AFCS logic AOG condition is not set.

Primary power is supplied to the pilot AHRS from the 28 VDC emergency bus No. 2 through the circuit breaker labeled PLT AHRS. The copilot AHRS primary power is supplied from the 28 VDC essential bus No. 1 through the circuit breaker labeled CPLT AHRS. 26 VAC is supplied to the pilot AHRS for attitude reference through the circuit breaker ATTD 26V NO 2 on the emergency 26 VAC bus.

26 VAC is supplied to the copilot AHRS for attitude reference through the circuit breaker ATTD 26V NO 1 on 26 VAC bus 3. 26 VAC for heading reference is supplied to the pilot AHRS through the circuit breaker AFCS 26V NO 2 on the emergency 26 VAC bus. 26 VAC for heading reference is supplied to the copilot AHRS through the circuit breaker AFCS 26V NO 1.

Pilot and copilot Horizontal Situation Indicators (HSI) are supplied by the 115 VAC bus No. 2 and No. 1 respectively. The circuit breakers are labeled HSI PILOT and HSI CPLT.

Pilot and copilot Attitude Direction Indicators (ADI) are supplied by the 115 VAC emergency bus and 115 VAC bus No. 3 respectively. The circuit breakers are labeled ATTD SYS PILOT and ATTD SYS CPLT.

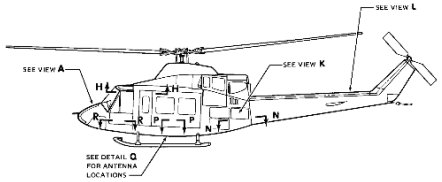
An integral fan on each AHRU mounting tray provides cooling. To ensure adequate cooling, an additional fan (AHRS fan) is installed in the same compartment as the AHRU No. 1 and No. 2. This fan operates on 115 VAC supplied from the 115 VAC emergency bus No. 2 through the AHRS FAN circuit breaker. Its activation is controlled by the AHRS relay powered by the 28 VDC bus No. 2 through the AHRS FAN RELAY circuit breaker. An annunciator on the master caution panel will indicate any malfunction of the AHRS fan.

A guarded two-position (AIR, GND) compass swing enable switch is mounted on a support assembly installed beside the pilot AHRU.

In addition to the compass functions available through the pilot or copilot HSI, two standby magnetic compasses are standard equipment on the helicopters. Each standby magnetic compass is mounted on the upper windshield frame in front of the pilot and copilot, respectively. The standby magnetic compass (or wet compass) is a direct reading compass that indicates helicopter heading, with reference to magnetic north, and is independent of helicopter electrical power.

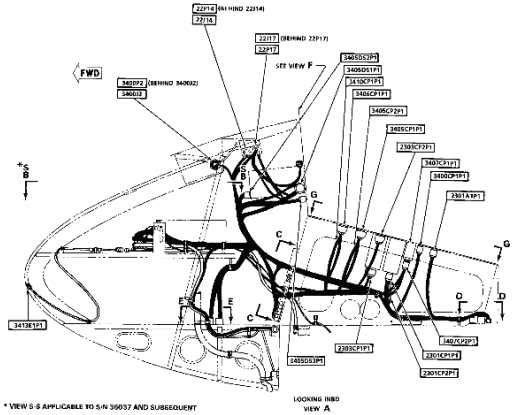
For avionics equipment location, refer to Figure 1.

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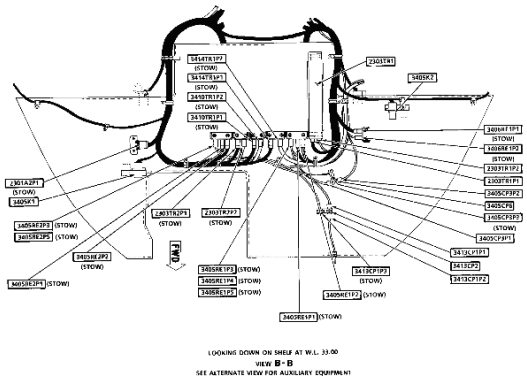
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 1 of 20)



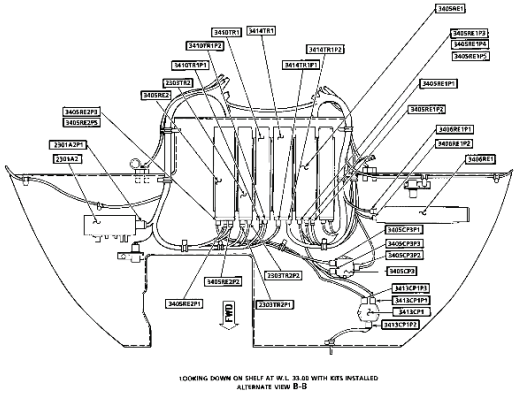
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 2 of 20)



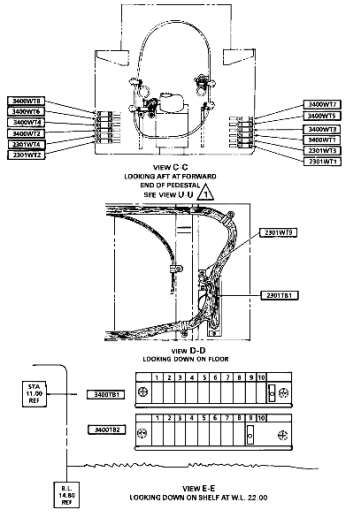
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 3 of 20)



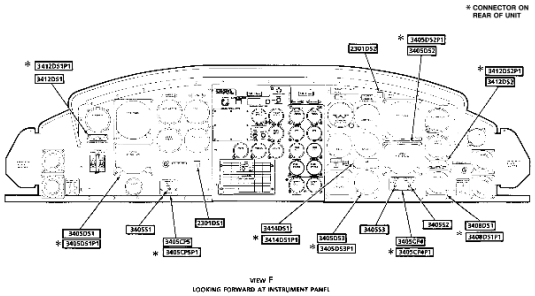
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 4 of 20)



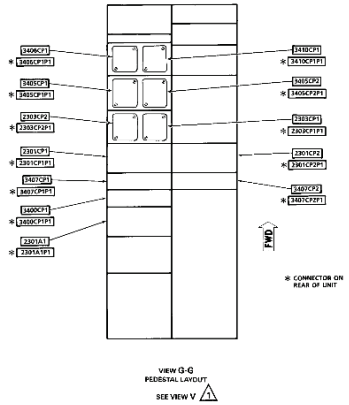
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 5 of 20)



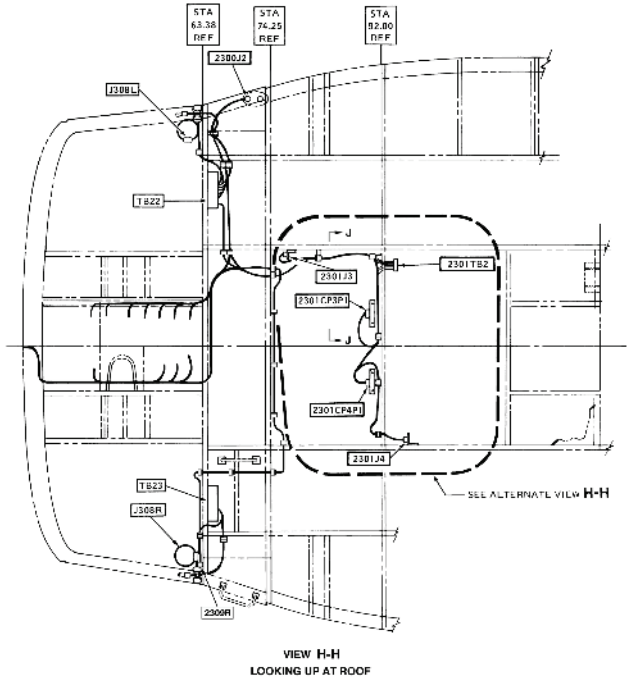
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 6 of 20)



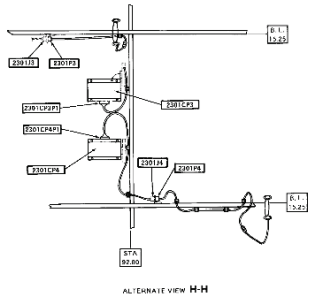
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 7 of 20)



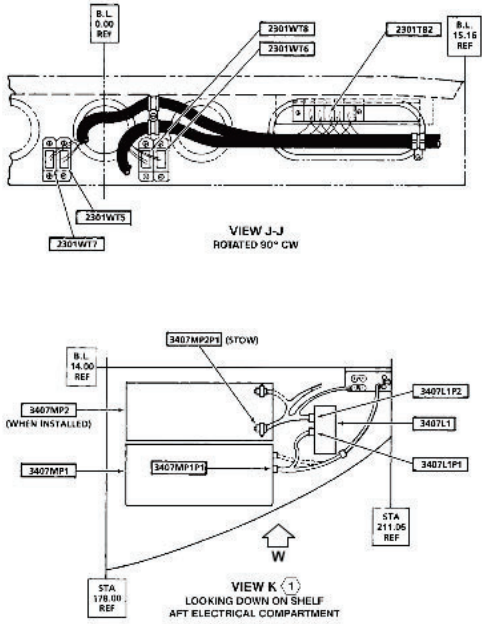
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 8 of 20)



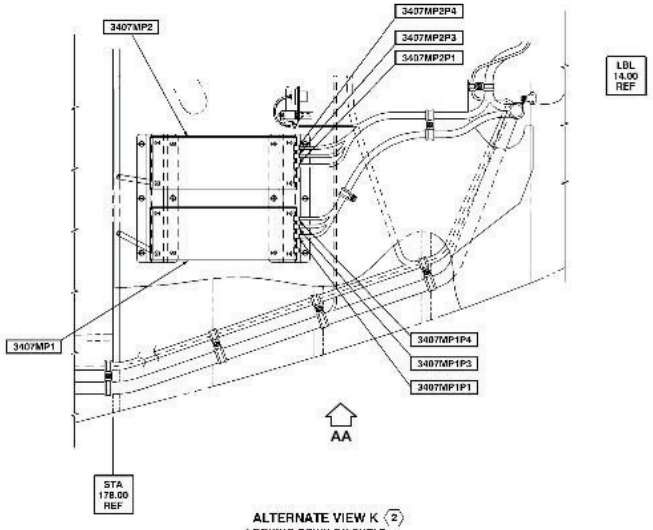
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 9 of 20)



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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 10 of 20)



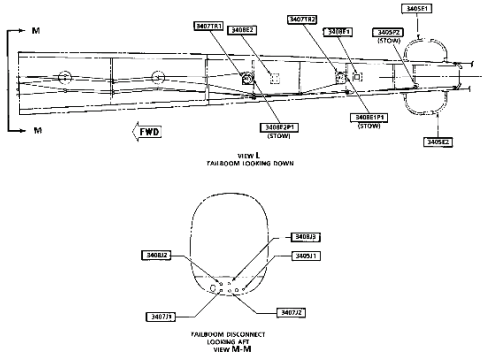
ALTERNATE VIEW K (2)
LOOKING DOWN ON SHELF
AFT ELECTRICAL COMPARTMENT

EFFECTIVITY

(1) S/N 33101 THROUGH 33213 and 36001 THROUGH 36247.
(2) S/N 36248 THROUGH 36999.

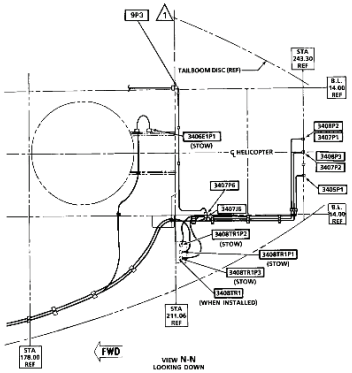
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 11 of 20)



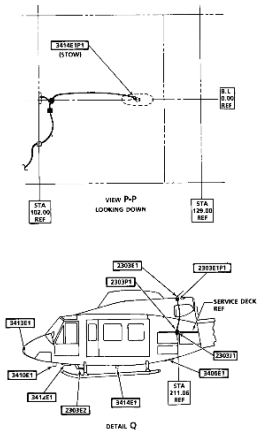
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 12 of 20)



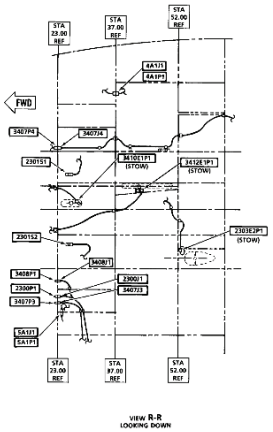
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 13 of 20)



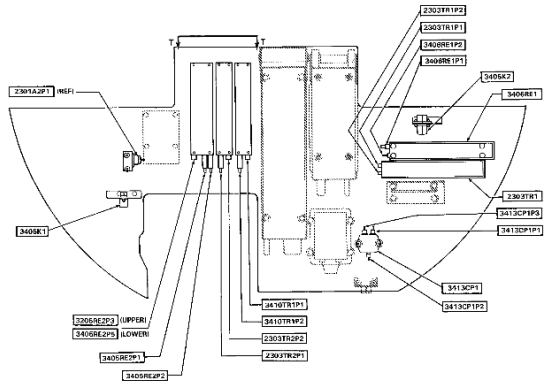
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Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 14 of 20)



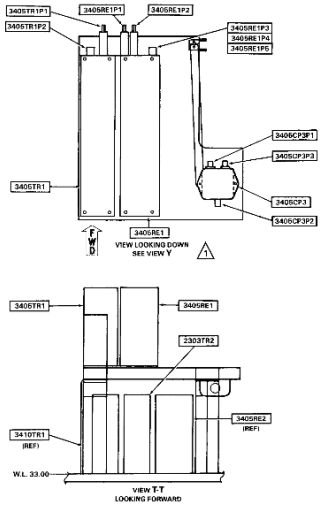
ICN-412-A-970000-B-97499-00028-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 15 of 20)



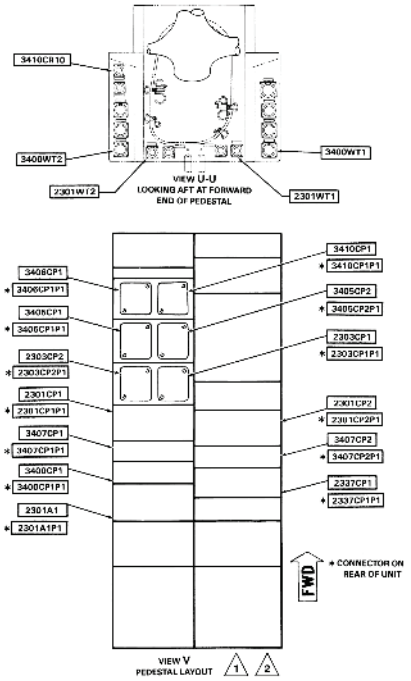
ICN-412-A-970000-B-97499-00029-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 16 of 20)



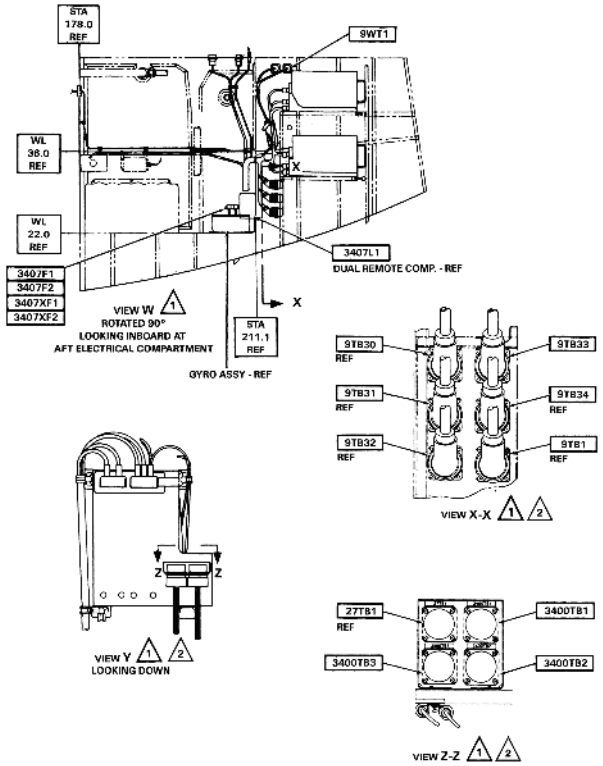
ICN-412-A-970000-B-97499-00030-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 17 of 20)



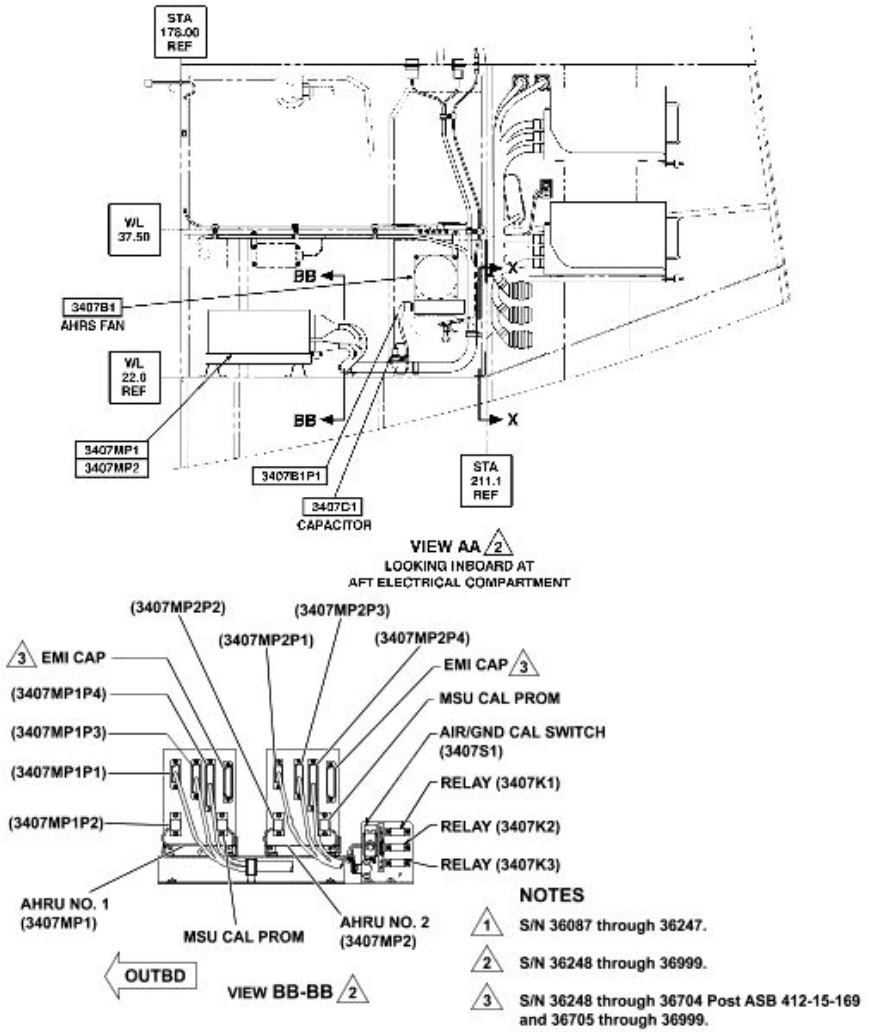
ICN-412-A-970000-B-97499-00031-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 18 of 20)



ICN-412-A-970000-B-97499-00032-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 19 of 20)



ICN-412-A-970000-B-97499-00033-A-001-01

Figure 1. Avionics Equipment Location (S/N 33101 through 36086 and 36087 and Subsequent) (Sheet 20 of 20)

AHRS - Functional Test

Procedure

1.

NOTE

This is a general functional test to ascertain that the system is operational prior to performing a compass swing.

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Applicable to: ALL
ECCN EAR99

Heading System - Functional Test

Procedure

1. Set the MAG/DG mode switches on the pilot and copilot AHRS control panel to the MAG position.
2. Energize the system by connecting the external power source, adjusted to 27.5 VDC, to the helicopter external power receptacle.
3. Energize the power buses (412-A-97-00-00-00A-340B-A / 02239).
4. Close the following circuit breakers:
   - AHRS CPLT - (28 VDC essential bus No. 1)
   - AHRS PILOT - (28 VDC emergency bus No. 2)
   - ATTD 26V NO 1 - (26 VAC bus No. 3)
   - ATTD 26V NO 2 - (26 VAC emergency bus)
   - ATTD SYS CPLT - (115 VAC bus No. 3)
   - ATTD SYS PILOT - (115 VAC emergency bus)
   - AFCS 26V NO 1 - (26 VAC bus No. 3)
   - AFCS 26V NO 2 - (26 VAC emergency bus)
   - HSI CPLT - (115 VAC bus No. 1)
   - HSI PILOT - (115 VAC bus No. 2)
   - HSI CPLT - (26 VAC bus No. 1)
   - HSI PILOT - (26 VAC bus No. 2)
   - AHRS FAN - (115 VAC emergency bus)
   - AHRS FAN RELAY - (28 VDC emergency bus)
5. Monitor the AHRS fan. Verify that it is operating.
6. After a short delay, verify that the compass OFF flag on each pilot and copilot HSI is removed from view.
7. Verify that the ATT and RT flag on the pilot and copilot ADI are removed from view.

NOTE

The compass system cannot be slewed when in MAG mode.

8. Verify that the pilot and copilot compass systems synchronize automatically by verifying that the slaving error annunciator on each AHRS panel is centered.
9. On the pilot AHRS control panel, operate the synchronization switch in both directions. Verify that there is no response on the pilot HSI or on the slaving error annunciator on the pilot AHRS control panel.
10. On the copilot AHRS control panel, operate the synchronization switch in both directions. Verify that there is no response on the copilot HSI or on the copilot slaving error annunciator on the copilot AHRS control panel.
11. Position the MAG/DG switch on the pilot AHRS control panel to DG. While observing the compass indication on the pilot HSI, select and hold the synchronization switch to the (+) position. Verify that the pilot HSI compass card slowly rotates counterclockwise to reflect the increase in heading direction. Hold the synchronization switch until the heading has increased 10°. Verify that the slaving error annunciator indicates a dot (•).
12. Return the pilot MAG/DG switch to MAG. Verify that compass heading on the pilot HSI returns to original value. Verify that the pilot slaving error annunciator returns to center.
13. Position the MAG/DG switch to DG. While observing compass indication on the pilot HSI, select and hold the synchronization switch on the pilot AHRS control panel to the dot position (•). Verify that the compass indication on the pilot HSI compass card slowly rotates clockwise to reflect the decrease in heading direction. Hold the synchronization switch until the heading has decreased 10°. Verify that the slaving error indicator indicates a cross (+).
14. Return the pilot MAG/DG switch to MAG. Verify that the compass heading on the pilot HSI returns to original value. Verify that the slaving error annunciator returns to center.
15. Repeat Step 11. through Step 14. for the copilot compass system using the copilot HSI and the copilot AHRS control panel.

NOTE

With the AHRS 28 VDC circuit breaker disengaged, the GS flag, GS pointer, FD flag, and FD command bars may be displayed on the applicable ADI.

16. Disengage the PILOT AHRS circuit breaker. Verify that the pilot HSI OFF flag and pilot ADI, ATT, and RT flags come into view.

17. Re-engage the PILOT AHRS circuit breaker and verify that the OFF flag on the pilot HSI and ATT and RT flags on the pilot ADI go out of view after approximately 30 seconds.

18. Disengage the AHRS FAN circuit breaker and verify that the AHRS FAN Caution is activated after several seconds.

19. Engage the AHRS FAN circuit breaker and verify that the AHRS FAN Caution is deactivated after several seconds.

20. Disengage AHRS FAN RELAY circuit breaker and verify that AHRS FAN Caution is activated after several seconds.

21. Engage the AHRS FAN RELAY circuit breaker and verify that the AHRS FAN Caution is deactivated.

22. Disengage the CPLT AHRS circuit breaker. Verify that the OFF flag on the copilot HSI comes into view and that the ATT and RT flags on the copilot ADI come into view.

23. Engage the CPLT AHRS circuit breaker. Verify that the OFF flag on the copilot HSI and the ATT and RT flags on the copilot ADI go out of view after approximately 30 seconds.

Attitude System - Functional Test

Procedure

NOTE

For the removal procedures of the pilot and copilot AHRS modules, refer to 412-A-97-22-00-01A-520A-A / 02329, 412-A-97-22-00-02A-520A-A / 02330, 412-A-97-22-00-03A-520A-A / 02331, 412-A-97-22-00-04A-520A-A / 02332, 412-A-97-22-00-05A-520A-A / 02333 and 412-A-97-22-00-06A-520A-A / 02334.

Space for tilting the AHRS modules is limited and wiring and cables may limit the actual tilt given. Tilting the AHRS 1 inch on its horizontal axis (pitch) will give an indication of approximately 8°. Tilting the AHRS 0.5 inch on its longitudinal axis (roll) will give an indication of approximately 11.5°.

1. Unfasten and tilt the No. 2 (pilot) AHRU to simulate a pitch up attitude between 5 and 10°. Verify that the pilot ADI indicates a pitch altitude of approximately 5 and 10° nose up.
2. Tilt the No. 2 AHRU to simulate a pitch down attitude of approximately 5 to 10°. Verify that the pilot ADI indicates a pitch down attitude of approximately 5 to 10°.
3. Tilt the No. 2 AHRU to simulate a 10° left roll. Verify that the pilot ADI indicates a 10° left roll. Repeat for a right roll.
4. Rotate the No. 2 AHRU in a clockwise direction. While the AHRU is in motion, verify that the pilot ADI turn rate indicator moves right to indicate a right turn.
5. Rotate the No. 2 AHRU in a counterclockwise direction. While the AHRU is in motion, verify that the pilot ADI turn rate indicator moves left to indicate a left turn.
6. Repeat Step 1. through Step 5. for the No. 1 (copilot) AHRU.
7. Secure the pilot and copilot AHRU modules to their mounting trays.

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Helicopters Equipped With Flight Director - Functional Test

Procedure

1. For helicopters equipped with the flight director, the following conditions must be met:

1.1. Collective must be below 25%.

1.2. Airspeed must be less than 20 knots.

1.3. Radar altimeter must indicate below 20 feet.

NOTE

Prior to performing the following test, set the applicable ADI pitch trim and roll trim knobs to center.

2. With the collective set below 25%, press and hold the AHRS test switch on the pilot AHRS control panel. Verify the following:

2.1. Heading OFF flag is in view on the pilot HSI.

2.2. Pilot ADI indicates 5° nose up.

2.3. Pilot ADI indicates a roll of 45°, right wing down.

2.4. ATT and RT flags are in view on the pilot ADI.

2.5. On the pilot AHRS control panel, the slaving error indicator indicates that heading is greater than magnetic heading (•) position.

2.6. On the autopilot control panel, AP switches indicate OFF.

3. Release the pilot AHRS test switch and verify that indications return to previous values.

4. Set the collective above 25%. Press and hold the pilot AHRS test switch and verify that test is inhibited.

5. Perform Step 1. through Step 4. for the copilot AHRS test switch.

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Helicopter Not Equipped With Flight Director - Functional Test

Procedure

1. For helicopters not equipped with the flight director, the following condition must be met:

1.1. Radar altimeter must indicate below 20 feet (if installed).

NOTE

Prior to performing the following test, set the applicable ADI pitch trim and roll trim knobs to center.

2. On the autopilot control panel, press and hold the AP CPL button. Press and hold the AHRS test switch on the pilot AHRS control panel. Verify the following:

2.1. Heading OFF flag is in view on the pilot HSI.

2.2. Pilot ADI indicates 5° nose up.

2.3. Pilot ADI indicates a roll of 45°, right wing down.

2.4. ATT and RT flags are in view on the pilot ADI.

2.5. On the pilot AHRS control panel, the slaving error indicator indicates that heading is greater than magnetic heading (•) position.

2.6. On the autopilot control panel, AP switches indicate OFF.

3. Release the pilot AHRS test switch and verify that indications return to previous values.

4. Release the AP CPL button. Press and hold the pilot AHRS test switch and verify that test is inhibited.

5. Perform Step 1. through Step 4. for the copilot AHRS test switch.

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Applicable to: ALL
ECCN EAR99

AHRS - Calibration

Procedure

1.

NOTE

The standby magnetic compass (or compasses) can be aligned simultaneously with the AHRS compass swing procedures.

1.1. Preparation for Calibration

1.1.1. Position the helicopter on a compass rose. The helicopter must be in its normal flight attitude.

1.1.2. All magnetic objects, such as vehicles, power carts, maintenance stands, or other aircraft, etc. shall be at a sufficient distance so as not to have any magnetic effect on the compass system.

1.1.3. Make sure that both the MAG/DG switches on their respective AHRS control panels are set to the MAG position.



FOR SAFETY PURPOSES, ENGINES SHALL BE SHUT DOWN PRIOR TO MAKING ANY ADJUSTMENTS TO CORRECT COMPASS ERROR.

NOTE

Due to magnetic interference created by helicopter generators, compass swing must be done with engines running.

1.1.4. Close all circuit breakers and energize all electrical equipment normally used during flight.

1.1.5. Start the helicopter engines and operate at flight RPM (100% N_R).

1.1.6. Wait 2 minutes and make sure that the center needle on slaving error annunciators on the pilot and copilot AHRS control panels are at the center position.

Applicable: SN: 36248-36527:

1.2. LCR-92 Index Error Check

NOTE

For LCR-100 index error correction, refer to Step 1.3..

NOTE

Instead of a compass rose, a magnetic sighting compass may be used. To take readings, the compass must be located a considerable distance either forward or aft of the helicopter and moved back or forth from line of sight coinciding with fore and aft (longitudinal) axis of the helicopter. When a sighting is taken facing aft, 180° must be added or subtracted from the sighting compass reading. When facing forward, the sighting compass reads directly.

NOTE

Prior to recording compass indications in the following steps, verify that slaving error annunciator needles are centered. When helicopter engines are running, the annunciator needles may fluctuate. If so, confirm that needle deviation is the same on each side of center.

1.2.1. Align the helicopter to North and wait 20 seconds.

1.2.2. Subtract the value indicated on the HSI from the Heading Nominal value and note resulting error value.

1.2.3. Repeat Step 1.2.1. and Step 1.2.2. for the other three cardinal headings.

1.2.4. Add the four resulting error values algebraically and divide by four. The result is the index error. For example:

NOMINAL	HSI	ERROR
0°	1.4°	1.4°
90°	87.6°	-1.4°

NOMINAL	HSI	ERROR
180°	182°	2°
270°	270.8°	0.8°
Resulting Index Error		2.8°

NOTE

If the resulting index error is less than 0.5°, no flux valve index compensation is required. If greater than 0.5°, do Step 1.2.5. through Step 1.2.7..

1.2.5. Gain access to the appropriate flux valve.

NOTE

The flux valve No. 1 for the copilot system is located forward in the tailboom. The flux valve No. 2 for the pilot system is located aft in the tailboom.

1.2.6. Loosen the screws that hold the appropriate flux valve flange to its mounting surface, and rotate the flange of the unit to cancel out index error. If index error is positive, the flange should be rotated counterclockwise (giving a minus reading on the flange), as observed from above the unit. If index error is negative, rotate the flange counterclockwise, giving a plus reading on the flange. Amount of rotation shall equal index error.

1.2.7. Tighten the mounting screws and verify readings at four cardinal headings. The procedure should be repeated if necessary to obtain the least index error possible.

Applicable: SN: 36528-Subsequent:

1.3. LCR-100 Index Error Correction

NOTE

For LCR-92 index error check and correction, refer to Step 1.2..

NOTE

Instead of compass rose, a magnetic sighting compass may be used. To take readings, the compass must be located a considerable distance either forward or aft of the helicopter and moved back or forth from line of sight coinciding with fore and aft (longitudinal) axis of the helicopter. When a sighting is taken facing aft, 180° must be added or subtracted from the sighting compass reading. When facing forward, the sighting compass reads directly.

1.3.1. Align the helicopter to a known heading within ±0.5°.

1.3.2. Enable the MSU calibration mode by pressing the MODE button on the front of each AHRU for more than 2 seconds. The MSU calibration mode is indicated by a continuous illumintation of the LED on the front of each AHRU. If the LED flashes, this indicates a fault and it is not possible to enter the MSU calibration mode.

1.3.3. On both AHRS control panels, set the MAG/DG switch to the DG position. If already set to DG, select MAG then back to DG. Each HSI will now indicate the uncorrected magnetic heading.

NOTE

Index error compensation will be incremented in 0.1° steps at a rate of 0.3° per second.

1.3.4. Use the AHRS control panel synchronization (slew) switch to set the desired heading.

NOTE

The maximum correction is limited to ±5°. If required, the flux valve can also be adjusted to compensate for larger index errors. Refer to Step 1.2. for the adjustment procedure.

1.3.5. When the desired heading is indicated, set the MAG/DG switches to the MAG position. Each HSI will now indicate the corrected magnetic heading.

1.4. Compass Swing - Magnetic Sensor Unit (MSU) Calibration

NOTE

The MSU calibration mode will be automatically disabled 10 seconds after the MAG/DG switches are set to the MAG position.

NOTE

This procedure is applicable to both models LCR-92 and LCR-100 AHRS except where indicated.

NOTE

A tolerance of ±5° exists for the positioning of the helicopter. If the stated tolerance is exceeded, incorrect data will be collected and the MSU calibration procedures must be started again.

1.4.1. Position the helicopter on an East magnetic heading.

1.4.2. Select the Compass Switch Enable switch to GND.

1.4.3. Enable the MSU calibration mode as follows:

Applicable: SN: 36249-36527:

1.4.3.1. For the LCR-92 AHRS, set the MSU CAL MODE switch on the front of each AHRU to ON.

Applicable: SN: 36528-Subsequent:

1.4.3.2. For the LCR-100 AHRS, press and hold the MODE button on the front of each AHRU for more than 2 seconds. The MSU calibration mode is indicated by a continuous illumination of the LED on the front of each AHRU. If the LED flashes, this indicates a fault and it is not possible to enter the MSU calibration mode.

1.4.4. Monitor the heading OFF flag on each HSI until both disappear from view.

NOTE

At each position, the helicopter must remain stationary until the heading OFF flag on each HSI disappears from view (approximately 10 seconds). The heading OFF flags reappear each time the helicopter is turned to a new heading. The ADI attitude warning flag (ATT) will be displayed continuously during this procedure.

1.4.5. Rotate the helicopter 7 times in 45° (±5°) increments. Once the helicopter starts moving, the HSI compass card moves to North for the start position. The 45° steps can then be read from North on the compass card.

1.4.6. After turning to 315° heading (7 times at 45°), the calibration coefficients are automatically calculated and stored in the MSU Cal Prom. Verify the heading OFF flags are out of view.

NOTE

If either MSU calibration was not successful, the corresponding heading OFF flag will remain in view in Step 1.4.6.. In this case, set the LCR-92 MSU CAL MODE switch to OFF or press the LCR-100 MODE button for more than 2 seconds (as applicable) to reset the MSU calibration mode. The old coefficients are now available and the compass swing can be repeated.

1.4.7. Observe the residual heading error (multiplied by 100) indicated on the compass card. 100 (corresponding to 1° of error) or less is acceptable. For example, a residual heading error of 0.8° will give an indication of 80°.

1.4.8. Select the Compass Swing Enable switch back to AIR and disable the MSU calibration mode using the MSU calibration switch on the front of each AHRU. The AHRS is now in normal mode and the new MSU calibration data has been saved.

1.4.9. Verify that the correct heading is displayed on both HSIs and the heading OFF flags are out of view.

1.5. Compass Swing - Final Heading Check

1.5.1. Proceed with final swinging by recording residual deviations at 30° increments. All readings should be within ±1° of actual magnetic heading. At no heading should the pilot and copilot readings differ by more than 2°.

1.5.2. If residual deviations are not within tolerance, repeat the index error check and the MSU calibration procedure.

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AHRS - Troubleshooting

Procedure

1. The AHRS system is designed to perform a series of self-tests to verify all functions. If test parameters are not met, warning signals for yaw, heading, attitude, and roll will cause applicable warning flags to be displayed on the helicopter flight instruments or complete AHRS shutdown. AHRS cooling fan failure will be indicated on the master caution panel.

2. Refer to Table 1 and the appropriate AHRS wiring diagram when troubleshooting. Make sure that all appropriate circuit breakers and switches are selected. Perform checks as necessary to isolate trouble.

Table 1. Attitude Heading Reference System (AHRS) - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
Annunciator on AHRS control panel will not center (null) with MAG/DG switch in MAG position.	Defective magnetic flux sensor.	Remove and replace applicable magnetic flux sensor.
	Defective AHRS control panel.	Remove and replace AHRS control panel.
	Defective annunciator.	Remove and replace AHRS control panel.
	Fault in wiring or shielding.	Repair or replace wiring as necessary.
Annunciator on control panel will not remain centered with switch in DG position.	Defective MAG/DG switch.	
MSU Cal Prom will not calibrate or invalid data on flight instruments during calibration.	MSU Cal Mode switch in OFF position.	Set switch to ON position during calibration.
	Defective MSU Cal Prom.	Remove and replace MSU Cal Prom.
	Defective AHRS module.	Remove and replace AHRS module (1) .
	Defective magnetic flux sensor.	Remove and replace flux sensor.
	Fault in wiring or connectors.	Repair or replace wiring.
Setting of synchronization switch in DG mode causes no change on annunciators.	Defective synchronization switch or MAG/DG switch on AHRS control panel.	Remove and replace AHRS control panel.
	Defective AHRS.	Remove and replace AHRS (1) .
AHRS fan not functioning.	Defective AHRS fan relay.	Remove and replace AHRS fan relay.
	Defective AHRS fan.	Remove and replace AHRS fan.
	Fault in wiring.	Repair or replace wiring.
Excessive fluctuation on annunciators.	Defective wiring.	Repair wiring.
	Defective magnetic flux sensors.	Remove and replace magnetic flux sensors.
AHRS shuts down.	Defective AHRS module.	Remove and replace AHRS module (1) .

NOTE:

1 Replacement of AHRS module does not require compass swing.

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AHRS Level 1 Test Set - Description

The optional level 1 test set (309444-0000) interfaces with the LCR-92 AHRU to measure and quantify binary output data for maintenance purposes including fault isolation and calibration. Refer to the LITEF GmbH Installation/Maintenance Instruction (Document No. 141450-0000-840) for information on the use of this equipment.

The optional level 1 test set (309946-XXXX) interfaces with the LCR-100 AHRU to measure and quantify binary output data for maintenance purposes, including fault isolation and calibration. Refer to the Northrop Grumman LITEF GmbH Installation/Maintenance Instruction (Document No. 145130-0000-840) for information on the use of this equipment.

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AHRU Module - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

NOTE

For LCR-92 and LCR-100 (145130-7000/-7001), the MSU Cal Prom is attached to the AHRU mounting tray by a nylon cord. It is not necessary to remove the MSU Cal Prom from the mounting tray.

For LCR-100 (145130-7010), the MSU Cal Prom and EMI cap are attached to the AHRU mounting tray with lanyards. It is not necessary to remove the MSU Cal Prom and EMI cap from the mounting tray.

1. Disconnect the MSU Cal Prom from the AHRU module.
2. Disconnect all electrical connectors.
3. For LCR-100 (145130-7010), remove the EMI cap from J6 on the AHRU module (3407MP1 or 3407MP2).
4. Loosen the self-torquing fastener by turning counterclockwise.
5. Slide the AHRU module slightly forward to disengage and carefully lift clear from the mounting tray.

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MSU Cal Prom - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Remove the AHRU (412-A-97-22-00-01A-520A-A / 02329).

2. Remove and retain the two screws that retain the MSU Cal Prom lanyard to the AHRU mounting tray. Lift the MSU Cal Prom clear from the mounting tray.

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Applicable to: ALL
ECCN EAR99

AHRU Mounting Tray - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

NOTE

The mounting tray includes an integral fan.

1. Remove the AHRU (412-A-97-22-00-01A-520A-A / 02329).
2. Remove the MSU Cal Prom (412-A-97-22-00-02A-520A-A / 02330).
3. For LCR-100 (145130-7010), remove and retain the hardware that secures the EMI cap lanyard to the mounting tray. Remove and retain the EMI cap.
4. Remove and retain the four screws and four washers that retain the AHRU mounting tray to the support assembly.
5. Remove the bonding strip.
6. Lift the mounting tray clear from the support assembly.

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AHRS Control Module - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Loosen the spring lock fasteners that secure the control panel to the pedestal assembly.
2. Lift the control panel far enough out of the pedestal assembly to reach and disconnect the electrical connectors.

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Flux Valve - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Tag and remove the electrical wires.
2. Remove and retain the mounting screws.
3. Remove the flux valve from the mounting bracket.

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AHRS Compartment Cooling Fan - Removal

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Disconnect the electrical connector.
2. Remove the four screws, four washers, and two nuts. Retain the hardware for reuse.
3. Lift the AHRS compartment cooling fan clear.

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AHRU Mounting Tray - Installation

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

NOTE

The mounting tray includes an integral fan.

1. Carefully position the mounting tray onto the support assembly.
2. Secure the AHRU mounting tray to the support assembly with four screws and four washers.
3. Attach the bonding strips from the AHRU mounting tray to the shelf.
4. Install the MSU Cal Prom (412-A-97-22-00-02A-720A-A / 02336).
5. For LCR-100 (145130-7010), secure the EMI cap lanyard to the mounting tray with the retained hardware.
6. Torque the screw 18 to 21 inch-pounds (2.05 to 2.35 Nm).
7. Install the AHRU (412-A-97-22-00-03A-720A-A / 02337).

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MSU Cal Prom - Installation

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Remove the AHRU (412-A-97-22-00-01A-520A-A / 02329).
2. Secure the MSU Cal Prom lanyard to the AHRU mounting tray using two screws.
3. Torque the two screws 7.5 to 8.5 inch-pounds (0.85 to 0.95 Nm).
4. Reinstall the AHRU module (412-A-97-22-00-03A-720A-A / 02337).

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AHRU Module - Installation

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.



VERIFY THAT AHRU MOUNTING TRAYS ARE FREE OF ALL OBJECTS PRIOR TO MOUNTING THE AHRU MODULES. DAMAGE TO AHRU MOUNTING TRAY FAN MAY OCCUR OTHERWISE.

NOTE

The AHRU model LCR-100 (145130-7000/-7001) must be identified as mod status 20 or higher.

The AHRU model LCR-100 (145130-7010) must be identified as mod 25.

1. Position the AHRU unit on the AHRU mounting tray engaging the rear indexing tab.
2. Secure the AHRU module to the mounting tray by engaging the self-torquing fastener and turning the knob clockwise. Verify that no looseness exists between the AHRU module and mounting tray. If the AHRU module is loose, the self-torquing fastener may be adjusted using the integral adjustment screw on the fastener.



EXERCISE CARE IN HANDLING OF AHRU MODULE. ELECTRICAL CONTACTS ARE EXPOSED IN PINS RENDERING THEM SUSCEPTIBLE TO DAMAGE.

3. Connect all electrical connectors.
4. For LCR-100 (145130-7010), install the EMI cap on the J6 of the AHRU.
5. Connect the MSU Cal Prom to the AHRU module.
6. Make sure that the MSU calibration switch is in the OFF position.

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AHRU Control Module - Installation

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Position the AHRS control panel above the pedestal and connect the electrical connectors.

2. Secure the AHRS control panel by tightening the two spring loaded fasteners.

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Flux Valve - Installation

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Position the flux valve in the mounting bracket.
2. Connect the tagged electrical wires.
3. Install the mounting screws.

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AHRS Compartment Cooling Fan - Installation

Procedure



MAKE SURE THAT ALL ELECTRICAL POWER IS REMOVED PRIOR TO REMOVAL AND INSTALLATION PROCEDURES.

1. Position the AHRS compartment cooling fan on the bulkhead and secure with the four screws, two washers, and two nuts retained at removal.

2. Connect the electrical connector.

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AHRS - Inspection

Procedure

1. Inspect the AHRS components and standby magnetic compasses for proper installation, cleanliness, and loose or cracked indicator glass.
2. Inspect the electrical connections and interconnecting cables for frayed or broken insulation.
3. Inspect the indicators and controls for proper mechanical action.
4. Inspect the standby magnetic compass for leaking float chamber. The fluid must be clear.

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AHRS - Cleaning

Procedure

1. Remove moisture and loose dirt from the AHRS components and standby compasses with a clean, soft cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

2. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.
3. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a dry cloth.

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AHRS - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or indicators and control panels.
2. Replace defective controls and indicator(s).
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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ATC Transponder System - (When Installed) - Description

The Air Traffic Control (ATC) transponder is an optional system that can be used with Air Traffic Control Radar Beacon Site (ATCRBS) . The system consists of a transponder, a transponder control, and a transponder antenna. The transponder system is designed for compliance with ATC requirements. When an ATCRBS interrogation signal is detected, the system transmits a coded replay enabling ground station to identify the helicopter and determine its range and azimuth. Also, the system contains a Special Pulse Identification (SPI) and operator test capability. The system requires 27.5 VDC at 1.5A maximum for proper operation. The transponder transmits at 1090 MHz with a peak pulse output of 500 watts and receives at 1030 MHz (crystal controlled). A receiver controlled time delay (60 seconds) is incorporated to allow adequate time for the transmitting components to warm-up.

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ATC Transponder System - (When Installed) - Description

The transponder control features a non-volatile memory that retains the last selected frequencies when power is turned off. No helicopter power or batteries are required for the non-volatile memory. Selected function and code are displayed by a gaseous discharge display with automatic dimming. For helicopters with a MST 67A (Mode S) transponder system that use a PFS 578A control unit, the display is LCD. Dimming for the display is preset in the unit (412-A-97-23-00-01A-042B-A / 02348). Edge lit panel lighting is controlled by the PED LT control on the overhead console. The function and ATC transponder code are selected by a serial data entry from the control panel to the receiver/transmitter.

The helicopter may have a KXP 756 (Mode C) or a MST 67A (Mode S) ATC transponder system. Each system has a transponder, control unit, and antenna. A helicopter with a transponder is identified and located in range and azimuth by an Air Traffic Control Radar Beacon System (ATCRBS) radar site. The helicopter transponder transmits a coded return after it receives a Mode A, C, or S interrogation. There are three modes of operation: Mode A, Mode C and Mode S.

Mode A

Mode A transmits an identity code that is set by the aircrew. It permits helicopter identification by ground controllers. You can also use it for emergency code transmissions if you can not use the radio communication transceivers.

Mode C

Mode C permits the transponder to transmit the identity code and helicopter altitude. The transponder Code Select Switch must be set to the ALT position to transmit the altitude. An encoding altimeter or Air Data Computer must also be installed to use the ALT function. The Mode C transponder will operate in Mode A if it is used without an encoding altimeter or Air Data Computer.

Mode S

Mode S gives identification and flight information to ground controllers and other aircraft with Mode S transponders. Mode S is the mode used for the Traffic Advisory and Collision Avoidance System (TCAS) . A code, supplied by the regulating authority, is implemented into the helicopter system.

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KSP 756 Mode C Transponder System - Description

The KSP 756 transponder system has the components that follow:

- KXP 756 transceiver
- KFS 576 or KFS 576A control unit
- Antenna

KXP 756 Transceiver

The KXP 756 transceiver is located in the nose on the avionics shelf.

The KXP 756 is used to transmit a coded response to a coded interrogation transmitted by an Air Traffic Control ground station. The KXP 756 operates in Mode A and Mode C modes of operation.

KFS 576 or KFS 576A Control Unit

The KFS576 and KFS 576A control unit functions are as follows:

IDT - When the identification (IDT) button is pushed momentarily and then released, it holds the IDT reply for approximately 25 seconds to make sure of a proper reply within at least one radar sweep. During this time, the IDT is displayed.

ON - When the function selector knob is in the ON position, the transponder replies to both Mode A and Mode C interrogations but without flight level information.

ALT - When the selector knob is in the ALT position, the transponder automatically selects the proper reply to Mode A or Mode C interrogation that transmits flight level information.

TST - When the function selector knob is on the TST position, it provides for a preflight or airborne check of transponder operation. This check will be independent of any ground operation.

R - (reply indicator) will be visual as an indication of a transmitted reply. The R (reply indicator) will flash or stay on when you switch the function selector knob to the test position.

FL - If an altitude encoder is a part of the installation, the flight level input to the transponder and the letters FL are shown.

The flight level readout is shown in 100-foot increments. A readout of 005 indicates 500 feet or a readout of 042 indicates a flight level of 4200 feet. If an encoder is not included in the installation, dashes are seen.

An automatic dimming function adjusts the brightness of the shown data and nomenclature to compensate for changes in the ambient light level.

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MST 67A Mode S Transponder System - Description

The MST 67A transponder system components are as follows:

- MST 67A transceiver
- KFS 578A control unit
- Antenna

MST 67A Transceiver

The MST 67A transceiver is located on the nose avionics shelf.

The MST 67A is used to transmit a coded response to a coded interrogation transmitted by an Air Traffic Control ground station. The MST 67A operates in Mode A, Mode C, and Mode S modes of operation.

The Mode S address code is implemented into the helicopter system by strapping certain pins on the terminal board 3410TB1 located on the forward face of the pedestal. Refer to wiring diagram (Figure 1). The Mode S address code is supplied by the control authority. When used, the helicopter details are automatically transmitted to the Air Traffic Control ground station and to other aircraft in the area that are equipped with Mode S transponder.

KFS 578A Control Unit

The KFS 578A control unit functions are as follows:

TST - Starts the transponder self test.

SBY - The transponder power is turned on, but the transponder does not transmit or reply to interrogations.

ON - The transponder transmits and replies to Mode S interrogations.

ALT - The transponder transmits and replies to Mode S interrogations and reports altitude in Mode C replies.

TA - Used only when TCAS is installed.

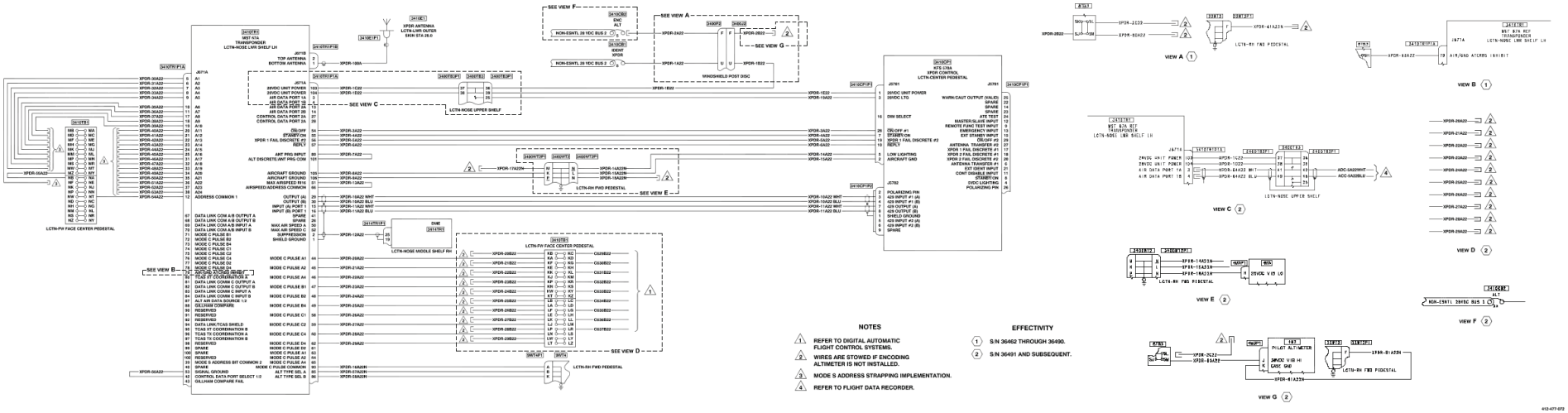
TA/RA - Used only when TCAS is installed.

FL - A pushbutton switch that permits the flight level altitude of the helicopter or other aircraft flight level altitude display to be selected.

IDT - A pushbutton switch, when pushed, will send a Mode S Special Position Identifier (SPI) to the ground stations.

R - The reply indicator (R) will be visible as an indication of a transmitted reply.

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ICN-412-A-972300-B-97499-00002-A-001-01

Figure 1. Mode S Transponder Kit (MST 67A) Wiring Diagram (Sheet 1 of 1)

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MST 67A Mode S (ELS) Transponder System - Description

The MST 67A transponder system components are as follows:

- MST 67A transceiver
- PS578A control unit
- Antenna

MST 67A Transceiver

The MST 67A transceiver is located on the avionics shelf in the nose.

The MST 67A is used to transmit a coded response to a coded interrogation transmitted by an Air Traffic Control ground station. The MST 67A operates in Mode A, Mode C, and Mode S modes of operation.

The Mode S address code is implemented into the helicopter system by strapping certain pins on the terminal board 3410TB1 located on the forward face of the pedestal. Refer to wiring diagram (Figure 1). The Mode S address code is supplied by the control authority. When used, the helicopter details are automatically transmitted to the Air Traffic Control ground station and to other aircraft in the area that are equipped with Mode S transponder.

PFS 578A Control Unit

The PFS 578A control unit functions are as follows:

TST - Starts the transponder self test.

SBY - The transponder power is turned on, but the transponder does not transmit or reply to interrogations.

ON - The transponder transmits and replies to Mode S interrogations.

ALT - The transponder transmits and replies to Mode S interrogations and reports altitude in Mode C replies.

FL - A pushbutton switch that permits the flight level altitude of the helicopter or other aircraft flight level altitude display to be selected.

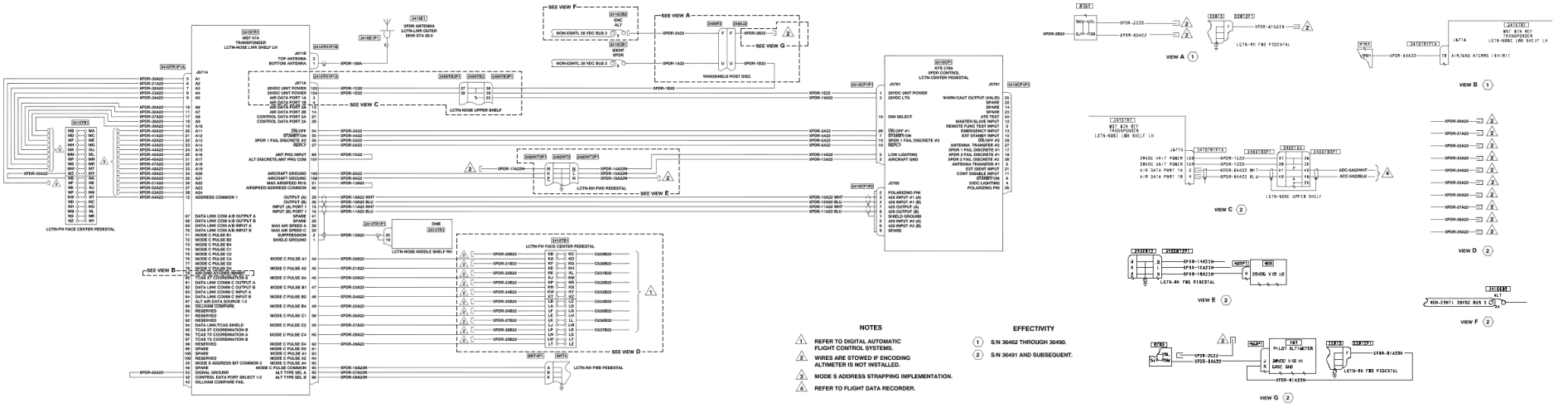
IDT - A pushbutton switch, when pushed, will send a Mode S Special Position Identifier (SPI) to the ground stations.

R - The reply indicator (R) will be visible as an indication of a transmitted reply.

FID - A pushbutton that allows entry of alpha numeric flight identification and announces FID on the display.

The PS 578A control unit has an LCD display and the adjustment of the lighting is done with an internal adjustment of the control unit and not with the PED LT control. Refer to (412-A-96-07-00-03A-271A-A / 02156) for adjustment of other Gold Crown control units. For the PS 578A refer to the manufacturer’s instructions.

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Figure 1. Mode S Transponder Kit (MST 67A) Wiring Diagram (Sheet 1 of 1)

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Mode C Transponder System - Functional Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Transponder Test Set	IFR Type ATC-600A	1	or Equivalent

Procedure

1. The following maintenance procedures provide complete functional check.

1.1. Preliminary Test Procedures

1.1.1. Visually inspect the equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 VDC, to the helicopter power receptacle.

1.1.2. Energize the power buses per 412-A-97-00-00-00A-340A-A / 02238 or 412-A-97-00-00-00A-340B-A / 02239, as applicable.

Applicable: SN: 33001-33100:

1.2. Self-Test



DO NOT SET THE CODE SELECTORS TO 0000, 7500, 7600, OR 7700 AS THEY ARE USED FOR MILITARY AIRCRAFT, HIJACK, COMMUNICATIONS FAILURE AND AIRCRAFT IN DISTRESS.

NOTE

When the prescribed test equipment, or suitable substitute, is not available, the following self-test check may be used for confidence testing at this level of maintenance.

1.2.1. Set controls on the transponder control as follows:

CONTROL	POSITION
Code Selectors	Any code (except 0000, 7500, 7600, 7700)
A/C/AC Switch	A
Function Selector Switch	ON

NOTE

Momentary flashes at the time the TEST button is depressed or released should be ignored.

Irregular flashes (other than at 10 to 15 second normal intervals) may be caused by noise or response to more than one interrogator. Transponder has no LO sensitivity, so the LO position of the FUNCTION switch is inoperative.

1.2.2. Allow 2 minutes for equipment warm-up. Depress the TEST switch and check that the REPLY lamp flashes.

Applicable: SN: 33101-33213 or 36001-36272 or 36274-36275:

1.3. Self-Test



DO NOT SET THE CODE SELECTORS TO 0000, 7500, 7600, OR 7700 AS THEY ARE USED FOR MILITARY AIRCRAFT, HIJACK, COMMUNICATIONS FAILURE AND AIRCRAFT IN DISTRESS.

1.3.1. Set the code selector to any code except 0000, 7500, 7600, 7700.

NOTE

If an encoding altimeter is not installed or if this function is inoperable, a readout of -002 will be displayed.

1.3.2. Set the function selector to test (TST) and check that the letter R (reply indication) starts to flash or illuminates continuously, indicating proper receiver operation, and that letters FL (flight level), along with a numerical altitude readout, illuminate indicating encoded altitude in 100-foot increments.

1.4. Setup of Transponder Test Set

1.4.1. Set up the tripod and remote test antenna in front of the transponder antenna, as shown in Figure 1.

1.4.2. Adjust the test antenna height (tripod height) such that its center approximates the same level as the transponder antenna. Bearing from transponder antenna is not critical and the test set antenna may be positioned for convenience.

1.4.3. Route the loose end of the coaxial cable into the helicopter cockpit through a vent window, door, etc. All testing, using transponder test set (IFR Type ATC-600A) is accomplished from the cockpit once the remote antenna is positioned.



DAMAGE TO TEST SET WILL RESULT IF REMOTE ANTENNA IS PLACED CLOSER THAN 1 FOOT (30.5 CM) TO THE TRANSPONDER ANTENNA WITH transponder test set (IFR Type ATC-600A) ON.

1.4.4. Energize transponder test set (IFR Type ATC-600A) by depressing the PWR rocker switch. Connect the remote antenna coaxial to the antenna connector. Position the INTERR and FRAMING controls to 0.

1.4.5. Position the test set mode switch to A/C CODE. Position the function selector switch to standby and allow approximately 2 minutes warm-up.

1.5. Pilot Code Checks

1.5.1. With the test set and transponder operating normally, and F2 light extinguished, select a code of 7777. Check that transponder test set (IFR Type ATC-600A) numerical display reads out the selected pilot code (7777). Binary lamps indicate pulses causing code display.

NOTE

These codes check interconnecting wiring between transponder and control unit.

1.5.2. Repeat Step 1.5.1. using codes 6563, 5226, 2315, and 0100.

1.6. IDENT Check

1.6.1. Set the transponder controls as follows:

CONTROL	POSITION
Code Selectors	1500
Function Selector Switch	ON

1.6.2. Momentarily press the IDENT switch on the transponder control.

1.6.3. Observe that the IDENT light on the test set illuminates for a 15 or 30 second period, then extinguishes.

1.7. Transmitter Frequency Test

1.7.1. Set the FREQ/POWER meter switch to FREQ.

1.7.2. Set the transponder code to 0100 and ensure that Mode C is disabled.

1.7.3. Adjust the GAIN control for a midscale meter indication.

1.7.4. Rotate the XMTR FREQ control for a peak indication of meter. At peak, read deviation in MHz from 1090 (±3.0) MHz directly from the XMTR FREQ control dial.

1.8. Receiver Sensitivity Test

NOTE

The transponder does not have a low sensitivity function. Sensitivity will not be reduced if the function switch is placed in the LO position.

1.8.1. Place the ATC-600A mode switch to A/C CODE (3/A). With XPDR SIG level, verify that the XPDR RPLY meter indicates 100%.

NOTE

The MTL for all transponders is -73 (±7) dBm.

1.8.2. Rotate the XPDR SIG control CW until the XPDR RPLY meter indicates 90% reply. Record the transponder Minimum Trigger Level (MTL) reading on the XPDR SIG level scale in - dBm.

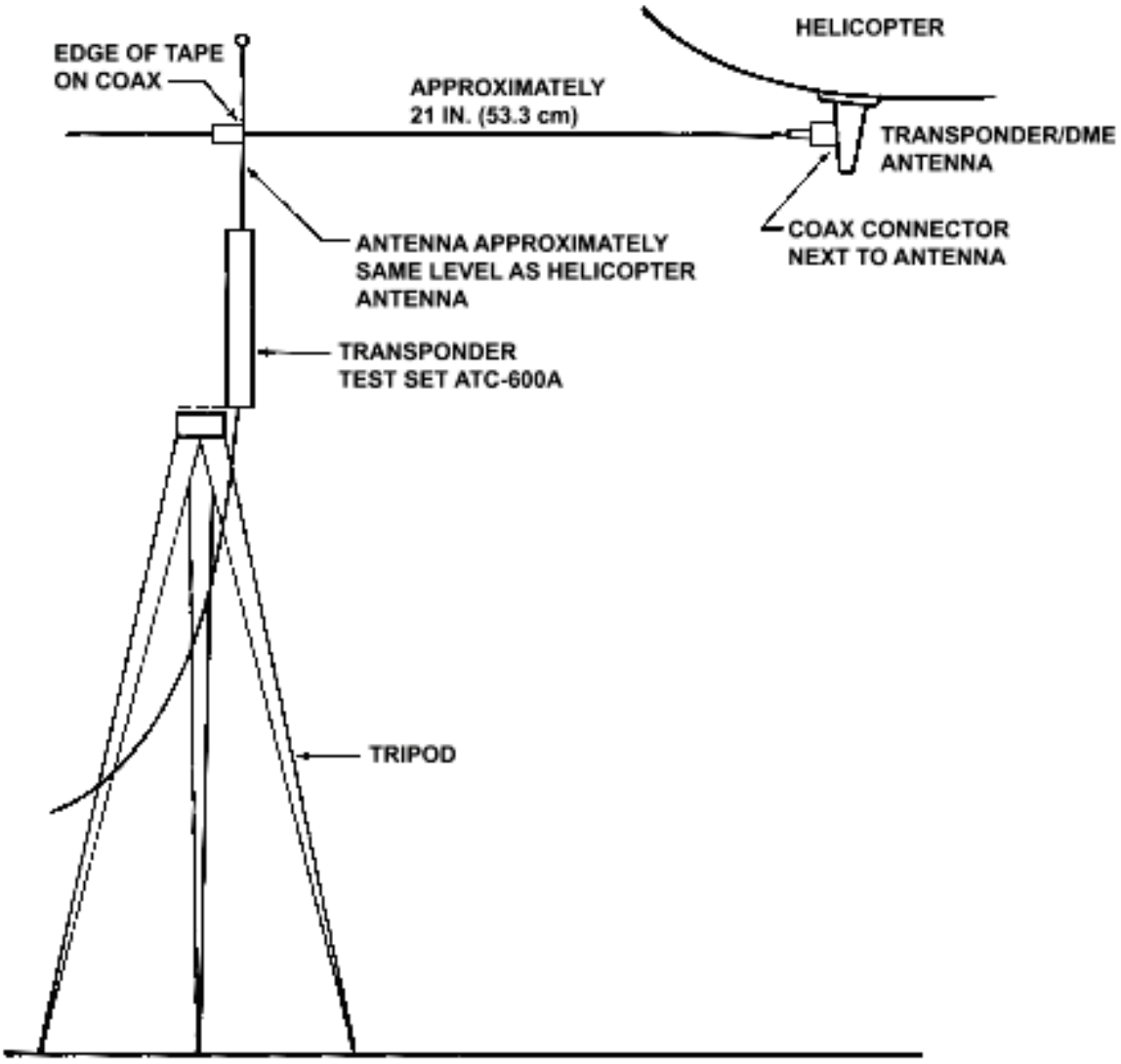
1.8.3. Return the XPDR SIG level control to full CCW and check that the XPDR meter reads 100%.

1.9. Side Lobe Suppression Tests

1.9.1. Using the XPDR SIG control, set the XPDR SIG level to 3 dB above MTL previously recorded.

1.9.2. Set the SLS switch to 0 dB and observe that the transponder replies do not exceed three replies per second (1.3%). Check the numerical and lamp displays are out and both meters indicate zero.

1.9.3. Set the SLS switch to -9 dB and observe that the transponder replies at a rate of at least 211 replies per second (90%).



ICN-412-A-972300-B-97499-00001-A-001-01

Figure 1. Transponder Test Setup (Sheet 1 of 1)

MST 67A Mode S Transponder - Functional Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Transponder Test Set	IFR Type ATC-601	1	
Static Test Set for Mode C Tests	Barfield Model 1811G264 Static Tester	1	or Equivalent

Procedure

1. The following maintenance procedures provide complete functional check.



DO NOT SET THE CODE SELECTORS TO 0000, 7500, 7600, OR 7700 AS THEY ARE USED FOR MILITARY AIRCRAFT, HIJACK, COMMUNICATIONS FAILURE, AND EMERGENCY-AIRCRAFT IN DISTRESS.

NOTE

Use an external power unit for ground tests to avoid battery depletion. If you use battery power, keep electrical loads and test time to a minimum.

1.1. On the transponder control unit (3410CP1), set the ON/OFF switch to ON. Make sure the control unit display illuminates.

1.2. Set the code selector switches to any permitted codes.

1.3. Put a cover or your finger on the photocell of the transponder control head. Verify that the display dims.

NOTE

SN: 36273 or 36276-36461 BHT-412-SI-84: installed:

The Weight-On-Gear (WOG) switch may be installed. For helicopters with this kit installed, make sure the WOG OVERRIDE switch 3420S2 is set to AIR.

Applicable: SN: 36462-Subsequent:

1.4. On the hourmeter panel on the right side of the pedestal, set the WOG OVERRIDE switch 3420S2 to AIR.

1.5. On the transponder control unit (3410CP1), set the mode select knob to the test (TST) position. Make sure of the following:

- All segments on the display come on for at least 2 seconds
- The altitude indication shows for 4 seconds

1.6. Set the mode select knob to the ON position.

2. **MST 67A Mode S Transponder - Ground Evaluation Test**

2.1. Set up the transponder test set (IFR Type ATC-601) as described in 412-A-97-23-00-02A-340A-A / 02349.

2.2. Push the POWER key of the transponder test set (IFR Type ATC-601).

2.3. Push the SETUP key to enter the SETUP menu.

NOTE

Push the SLEW key to replace a value. Push the SELECT key to replace an item.

Push and release the slew key to change the value by a 1 foot increment.

Push and hold the slew key to change the RANGE value by 10 feet increments and height value by 5 feet increments.

2.4. Push the SELECT key until the display shows BOTTOM RANGE. Set the horizontal distance between the test set and the Mode S antenna (located at the bottom of the helicopter).

2.5. Push the SELECT key until the display shows BOTTOM HEIGHT. Set the vertical height from the ground to the bottom of the Mode S antenna.

2.6. Push the SELECT key until the display shows GAIN (gain of the test set antenna). Set the gain of the test set antenna (written on the antenna).

2.7. Push the SELECT key until the display shows LOSS (loss of the test set coaxial cable). Set the loss of the coaxial cable (written on the cable).

2.8. Push the SELECT key until the display shows BOTTOM ANTENNA.

2.9. Push the SETUP Key to exit the SETUP menu.



DO NOT SET THE CODE SELECTORS TO 0000, 7500, 7600, OR 7700 AS THEY ARE USED FOR MILITARY AIRCRAFT, HIJACK, COMMUNICATIONS FAILURE AND AIRCRAFT IN DISTRESS.

NOTE

The ATC-601 shows Test Passed if the radio frequency reply is within 1090 ±3 MHz. For Mode S, the necessary range is 1090 ±1 MHz.

It is not necessary to do a test for Mode S diversity isolation because there is only one antenna in this configuration.

2.10. On the transponder control unit (3410CP1), make sure the mode select knob is still set to ON.

2.11. Push the AUTO TEST switch on the test set. The auto test screen shows. The results of the last tests are seen on the screen.

2.12. Push the RUN/STOP switch.

- The DISPLAY shows the modes under test, the modes that passed and the modes that failed. The radio frequency used, power measurements and diversity channel isolation (not necessary) are also shown.
- The reply frequency is 1090 ±3 MHz.

2.13. Push the RUN/STOP switch.

2.14. Push the AUTO TEST switch.

3. Mode A Test

3.1. Select ATCRBS Reply Test on the transponder test equipment. Make sure the Reply Light on the test set is illuminated and R is shown on the transponder control unit display.

3.2. Verify that the code displayed on the transponder test set matches with the code selected on the transponder control unit display.

3.3. On the transponder control unit, press the IDT button. Make sure the Ident information is displayed on both the transponder control unit display and the transponder test set.

4. Mode C Test

NOTE

Perform this test if an encoder altimeter is installed.

4.1. Set the transponder control unit function switch to ALT (Mode C altitude reporting).

4.2. Set the pilot altimeter to 29.92 inches of mercury (1013.2 millibars or hectopascal).

4.3. Connect a pitot/static test set for mode c tests (Barfield Model 1811G264 Static Tester) to the pitot/static system of the helicopter (412-A-95-11-05-00A-320A-A / 01808).

4.4. On the transponder test set, push the RUN/STOP switch.

NOTE

The rate of change of altitude must be less than 750 feet (228 m) per minute.

4.5. Increase the test altitude until the transponder output changes and indicates the first test point in Table 1 or Table 2. Make a note of the altitude indications of the pitot/static test points. Make sure the pilot altimeter indication is within ±125 feet (±38 m) of the altitude indication on the test set display.

4.6. Change the altitude to the next test point at a rate of change that is less than 750 feet (228 m) per minute.

4.7. Do Step 4.5. through Step 4.6. for each of the test points in Table 1 or Table 2.

4.8. Decrease the test altitude slowly and do Step 4.5. and Step 4.6. for each test points in Table 1 or Table 2.

5. FID Mode (PS 578 only)

5.1. Press the FID button on the transponder control unit. Verify that FID is shown on the display.

5.2. Momentarily press the encoder knob to select the most significant digit (left side) and rotate the encoder knob to change the alpha-numeric value of the selected digit. The change will be accepted after a few seconds or if the FID or IDT pushbutton is pressed.

6. Post-test Procedures

6.1. Push the RUN/STOP switch on the transponder test set (IFR Type ATC-601).

6.2. Turn the transponder function switch to OFF and remove the test equipment.

6.3. If applicable, on the hourmeter panel on the right side of the pedestal, set the WOG OVERRIDE switch 3420S2 to NORM.

Table 1. Test Points (Feet)

-1000	2000	8000
0	2600	10000
500	2700	12000
1000	2800	14000
1100	3000	14800
1300	4000	16000
1500	6000	18000
1800	6800	20000

Table 2. Test Points (Meters)

-300	610	2438
0	792	3048
152	823	3958
305	853	4267
335	914	4511
396	1219	4877
457	1829	5486
548	2073	6096

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Mode C Transponder System - Troubleshooting

Procedure

1. Make sure the appropriate circuit breakers are closed and interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate ATC transponder wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. ATC Transponder System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
(R) REPLY lamp does not flash with TEST switch depressed.	Transponder circuit breaker on overhead console inoperative.	Replace circuit breaker.
	Defective REPLY lamp.	Replace lamp.
	Defective wiring.	Repair or replace wiring.
	Defective transponder control.	Replace transponder control.
	Defective transponder.	Replace transponder.
	Defective transponder antenna.	Replace transponder antenna.

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Transponder - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and antenna coax connectors at the front of the transponder.
2. Cap the open ends of the connectors for protection, and secure the caps with electrical tape.
3. Loosen the knurled nut, at the front of the transponder mounting rack, until the hold-down clamp can be slipped clear of the lug at the front of transponder.
4. Slide the transponder out of its mounting rack.

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Transponder Control - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the four spring lock fasteners that secure the transponder control to the pedestal assembly.
2. Lift the transponder control sufficiently to gain access to the electrical connector.
3. Disconnect the electrical connector and remove the transponder control.

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Transponder Control - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel mounting plate assembly to the pedestal assembly.
2. Lift the control panel mounting plate assembly and control unit(s) far enough out of the pedestal assembly to reach and disconnect the electrical connector(s) at the back of the unit(s).
3. Lift the control panel mounting plate assembly and control unit(s) free of the pedestal assembly.
4. Loosen the screws and nuts that secure the control unit to the control panel mounting plate assembly.
5. Lift the control unit free of the control panel mounting plate assembly.

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Transponder Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the attaching screws from the antenna mount located on the lower surface of the nose section.
2. Cut and remove sealant between the airfoil and the antenna mount.
3. Disconnect the antenna coaxial cable connector from the antenna base and protect the open end with a cap and electrical tape.

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Transponder - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the transponder in the mounting rack and slide into place.
2. Lift the holddown clamp, at the front of the mount, over the lug at the front of the transponder.
3. Tighten the knurled retainer nut to secure the transponder.
4. Remove the protective caps from the coaxial and electrical connectors and connect each to its respective receptacle.
5. Perform the transponder self-test per 412-A-97-23-00-02A-340A-A / 02349, step 1.2 or step 1.3 as applicable.

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Transponder Control - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Place the transponder control near the mounting hole on the pedestal.
2. Connect the electrical connector to the transponder control.
3. Place the transponder control in its mounting and secure the four spring lock fasteners.
4. Perform the transponder self-test (412-A-97-23-00-02A-340A-A / 02349).

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Transponder Control - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the control unit in the control panel mounting plate assembly and secure with the mounting screws and nuts.
2. Connect the electrical connector(s) to the back of the control unit(s).
3. Position the control panel mounting plate assembly and control unit(s) in place on the pedestal and secure with the spring lock fasteners.
4. Perform the transponder self-test (412-A-97-23-00-02A-340A-A / 02349).

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Transponder Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the protective cap from the antenna coaxial cable connector.
2. Connect the antenna cable to the connector receptacle at the antenna base.
3. Position the antenna on its mounting surface and secure with the antenna base attaching screws.
4. Apply a sealant bead to the mounting surface at the antenna base periphery.
5. Perform the transponder self-test per 412-A-97-23-00-02A-340A-A / 02349, step 1.2 or step 1.3 as applicable.

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Transponder System - Inspection

Procedure

1. Inspect the transponder system for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect the switches and controls on the control panel for proper mechanical action.
3. Inspect the transponder control to ensure the panel lights illuminate with power on.
4. Inspect the antenna and mounting for corrosion, dents, cracks, and breaks.
5. Inspect the electrical connections for corroded or bent pins and proper mating, the interconnecting cable for frayed or broken insulation, and the circuit breakers for proper operation.

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Transponder System - Cleaning

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.

2. Remove dust and dirt from the panel, controls, indicators, panel light housings, antenna, and transponder with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.

4. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a clean, dry cloth.

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Transponder System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs on the transponder control.
2. Replace defective panel light bulbs. Replace defective or inoperative units.
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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VHF Navigation (VOR/LOC/GS) System - Description

The VHF navigation (VOR/LOC/GS) system provides VHF omnidirectional radio range (VOR), localizer (LOC), and glideslope(GS) information. The helicopter may be equipped with either single or dual NAV (VOR/LOC) receivers or single or dual NAV (VOR/LOC/GS) receivers, coupled to either single or dual pilot instruments. In addition to NAV receivers, components include: NAV control panels (one for each receiver), RMI adapters (one for each receiver), VOR coupler (with dual receivers), Course Select Indicator (CSI) (with dual receivers), GS antenna, and GS antenna coupler (with dual GS installed).

On the helicopter with the single navigation (VOR/LOC or VOR/LOC/GS) system installed, VOR bearing information is presented on the No. 1 pointer of each installed Horizontal Situation Indicator (HSI) . (Copilot HSI is part of optional copilot instrument kit.) If the copilot HSI is not installed, the COURSE SET knob on the pilot HSI controls the manual VOR. If the copilot HSI is installed, the COURSE SET switch is provided to select either the pilot or copilot COURSE SET knob to control manual VOR. Course deviation, vertical steering, and flag information for the glideslope (when installed) is presented on both HSIs (when copilot HSI is installed). The glideslope (when installed) is channeled with the No. 1 VHF NAV control panel to the glideslope pointer of each installed HSI.

With the dual navigation (VOR/LOC or VOR/LOC/GS) systems installed, the HSIs present course deviation for either receiver selected with the associated BRG PTR No. 1 switch. If the copilot HSI is not installed, the COURSE SET knob on the pilot HSI controls manual VOR for the selected receiver. If the copilot HSI is installed, the COURSE SET switch is provided to select either the pilot or copilot COURSE SET knob to control manual VOR for the selected receiver. The No. 1 and No. 2 glideslope (when installed) are channeled with the No. 1 and No. 2 VHF NAV control panels respectively. The No. 1 glideslope (when installed) furnishes vertical steering and flag information to both HSIs and the No. 2 glideslope (when installed) furnishes vertical steering and flag information to the pilot CSI.

Heading information is supplied to the No. 1 VOR/LOC or VOR/LOC/GS receiver from the pilot 3-axis reference sensor and to the No. 2 VOR/LOC or copilot 3-axis reference sensor. To maintain proper phase relationship for the synchro circuits, 26 VAC excitation for both VOR/LOC or VOR/LOC/GS receivers, both 3-axis reference sensors, and both HSIs is normally supplied from the No. 1 26 VAC bus. Also, 115 VAC for both HSIs is normally supplied from the No. 1 115 VAC bus.

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VHF Navigation (VOR/ILS and VOR/LOC/GS) System - Description

The VHF navigation (VOR/ILS) system No. 1 provides VHF omnidirectional radio range (VOR), and Instrument Landing System (ILS) information. The VHF navigation (VOR/LOC/GS) system No. 2 provides VHF omnidirectional radio range (VOR), localizer (LOC), and glideslope (GS) information. The helicopter may be equipped with either single or dual NAV receivers coupled to either single or dual pilot instruments. In addition to NAV receivers, components include NAV control panels (one for each receiver), OMNI antenna, NAV antenna coupler (with dual receivers), Course Select Indicator (CSI) (with dual receivers), GS antenna, and GS antenna coupler (with dual GS installed).

On the helicopters with the single navigation (VOR/ILS) system installed, VOR bearing information is presented on the No. 1 pointer of each installed Horizontal Situation Indicator (HSI) . (The copilot HSI is part of optional copilot instrument kit.) If the copilot HSI is not installed, the COURSE SET knob on the pilot HSI controls manual VOR. If the copilot HSI is installed, the COURSE SET switch is provided to select either the pilot or copilot COURSE SET knob to control manual VOR. Course deviation, glideslope deviation (if installed), and signal strength flag information is presented on both HSIs (when the copilot HSI is installed). The glideslope (when installed) is channeled with the No. 1 VHF NAV control panel to the glideslope pointer of each installed HSI.

With the dual navigation (VOR/ILS and VOR/LOC/GS) systems installed, HSIs present course deviation for the NAV 1 receiver selected with the associated BRG PTR No. 1 switch, while the selection of the BRG PTR No. 1 NAV 2 presents on the CSI. If the copilot HSI is not installed, the COURSE SET knob on the pilot HSI controls the manual VOR for the selected receiver. If the copilot HSI is installed, a COURSE SET switch is provided to select either the pilot or copilot COURSE SET knob to control the manual VOR for the selected receiver. No. 1 and No. 2 glideslope receivers (when installed) are channeled with the No. 1 and No. 2 VHF NAV control panels respectively. The No. 1 glideslope receiver (when installed) furnishes vertical steering and flag information to the pilot HSI and copilot HSI (when installed). The No. 2 glideslope receiver (when installed) furnishes vertical steering and flag information to the CSI.

Heading information is supplied to the No. 1 VOR/ILS receiver from the No. 1 3-axis reference sensor and to the No. 2 VOR/LOC/GS receiver from No. 2 3-axis reference sensor.

26 VAC excitation of the No. 1 and No. 2 NAV receivers is supplied from the No. 1 and No. 2 26 VAC buses respectively and the 115 VAC is supplied from the No. 1 and No. 2 115 VAC buses respectively. The No. 1 NAV receiver and No. 1 NAV control panel are powered by the 28 VDC ESNTL BUS 1, No. 2 NAV receiver and No. 2 NAV control panel are powered by the 28 VDC NONESNTL BUS 2. The No. 1 NAV receiver and control panel, and No. 2 NAV receiver and control panel are protected by the circuit breakers NAV 1 and NAV 2 respectively.

The marker beacon receiver is incorporated into the No. 1 VHF navigation (VOR/ILS) system (412-A-97-26-00-00A-028B-A / 02411).

The digital VHF navigation control unit features active and standby gaseous discharge frequency displays with automatic dimming. The control has a nonvolatile memory that retains last selected frequencies when the power is turned off. Desired operating frequency may be entered directly by pulling out the inner frequency selector knob. If the inner frequency selector knob is pushed in, frequency is first entered into the standby (SBY) display, and then the transfer button must be pushed. This will trade the contents of the active and standby displays. The receiver is always tuned to the frequency appearing in the active display.

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Preliminary Test Procedures - Functional Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Signal Generator	Collins Type 479T-2	1	or Equivalent
MKR/NAV/COMM Test Set	IFR Type NAV-401	1	

Procedure

NOTE

SN: 33001-33100:

Separate navigation test procedures are presented for single VOR/LOC or VOR/LOC/GS system, dual VOR/LOC or VOR/LOC/GS system, and the glideslope portion of the system.

NOTE

SN: 33101-36086 or 36087-Subsequent:

Separate navigation test procedures are presented for single VOR/ILS system, dual VOR/ILS and VOR/LOC/GS system, and the glideslope portion of the system.

1. Visually inspect the navigation system to detect and correct any faulty mechanical mountings and electrical connections.

2. Position the signal generator (Collins Type 479T-2) near the helicopter and turn the SET LINE TO 21V control to OFF.

3. Connect the power cord to 21 VDC to 29 VDC connector and to the 21 to 29 VDC power source.

4. Set the METER switch to LINE and adjust the SET LINE TO 21V control until the meter indicates 21 (±0.5) volts.

5. Set the METER switch to B+. The meter should read 230 volts. Allow 3 minutes warmup.

6. Set the METER switch to CAR and turn the CARRIER SET control fully counterclockwise. Adjust the METER ZERO control until the meter reads zero. This adjustment is not critical.

7. Set the NAV-GS switch to NAV (for 108.0 to 135.9 MHz operation).

8. If a specific signal strength is required, rotate the MICROVOLTS dial to the desired output level and, with the meter switch in the CAR position, set the CARRIER SET control until the meter red lines.

NOTE

The meter must be reset to the red line each time frequency is changed.

9. Connect the signal generator (Collins Type 479T-2) antenna, fully extended, to the NAV OUTPUT connector.

10. Position the LOC-GS control in center position.

11. Connect an external power source adjusted to 27.5 VDC to the helicopter external power receptacle.

12. Energize the power buses per 412-A-97-00-00-00A-340A-A / 02238 or 412-A-97-00-00-00A-340B-A / 02239, as applicable. Place all communications and navigation switches and controls in the OFF or normal position. Close all communications and navigation circuit breakers.

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Single Receiver System - Functional Test

Procedure

1. The following VHF navigation receiver checks are applicable for single or dual pilot configurations using a single VHF navigation receiver.

1.1. Tune the VHF NAV control and test set to 108.0 MHz. Set the test set to zero (00.0) degrees signal. Check the red VOR warning flag on each installed HSI is out of view. If a copilot HSI is installed, position the COURSE SET switch to PILOT.

1.2. Adjust the COURSE SET knob (when installed) until the course deviation bar is centered. Check the course window and the No. 1 bearing pointer indicates within 3.0° of the VOR bearing selected on the signal generator and proper ambiguity is indicated (triangular flag pointing toward arrow end of course select pointer). The bearing pointer No. 1 should indicate within 3.0° of course window indication.

NOTE

The course deviation bar will also center on the reciprocal course. However, the ambiguity meter will indicate from the station.

1.3. Check the copilot HSI (when installed) course window indicates within 2.0° of the pilot course window. Check the copilot course deviation bar centers within 2.0° of the pilot. The bearing pointer No. 1 on the copilot HSI should indicate within 3.0° of the pilot No. 1 pointer whenever the two compass cards agree.

1.4. Repeat Step 1.2. and Step 1.3. for the test set VOR bearings of 45.0, 90.0, 135.0, 180.0, 225.0, 270.0, and 315.0°.

1.5. Set the copilot HSI (when installed) COURSE SET switch to COPILOT. On the copilot HSI, adjust the COURSE SET knob until the course deviation bar is centered. Check the course window indicates within 3.0° of the VOR bearing selected on the signal generator and proper ambiguity is indicated (triangular flag pointing toward arrow end of course select pointer). The No. 1 bearing pointer should indicate a bearing within 3.0° of that indicated in the course window, and the pilot course deviation bar should center within 2.0° of copilot. Return the COURSE SET switch to PILOT.

1.6. Rotate the COURSE SET knob on the pilot HSI through 360.0° and check for proper to-from indications on each installed HSI.

1.7. Set the COURSE SET switch to COPILOT and check that the rotation of the copilot COURSE SET knob through 360.0° results in proper to-from indication on both HSIs. Return the COURSE SET switch to PILOT.

1.8. Adjust the test set for an audio signal. Check aural output of the VHF navigation receiver by monitoring headset audio for a tone (appropriate mixer switches must be ON). Determine that the rotating volume control on the NAV control panel will vary the audio level. Leave the volume control at a comfortable listening level.

1.9. Select the test set frequency of 117.9 MHz and a bearing of 0°. Select a frequency of 117.0 MHz on the NAV control.

1.10. Adjust the COURSE SET knob on the pilot HSI until the course deviation bar is centered. With the course deviation bar centered and the ambiguity flag indicating to the station, the COURSE digits and course deviation pointer should be indicating approximately 0°. The No. 1 bearing pointer should also indicate a bearing of approximately 0°.

1.11. Rotate the COURSE SET knob on the pilot HSI to full-scale (second dot) left and right course deviations. In each case, the COURSE window indication in both the pilot and copilot (when installed) HSIs should be 10.0 ± 2.0° from the center course (0°).

1.12. Set the COURSE SET switch, when copilot HSI is installed, to COPILOT and repeat Step 1.10. and Step 1.11. for the copilot HSI.

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Dual Receiver (Single Pilot) System - Functional Test

Procedure

1. The following VOR checks are applicable for dual receiver installations when the copilot instruments are not installed.

1.1. Tune the VHF NAV controls and test set to 108.0 MHz. Set the test set for 0° (00.0) signal. Check the red VOR warning flags on the HSI and CSI are out of view.

1.2. Set the pilot BRG PTR NO. 1 switch to NAV 1.

1.3. Adjust the COURSE SET knob, on the pilot HSI, until the course window and No. 1 bearing pointer indicate within 3.0° of VOR bearing selected on the signal generator and that proper ambiguity is indicated (triangular flag pointing toward arrow end of course select pointer). The No. 1 pointer should indicate a bearing within 3.0° of that indicated in the course window.

1.4. Adjust the CSI OBS knob until the vertical needle is centered. Check the course card indicates within 3.0° of VOR bearing selected on the signal generator and TO flag is in view.

NOTE

The course deviation pointers will also center on the reciprocal course. However, the ambiguity meter will indicate from the the station.

1.5. Set the pilot BRG PTR NO. 1 switch to NAV 2. Observe that the No. 1 pointer on the HSI indicates within 3.0° of bearing indicated on the CSI, and within 3.0° of bearing selected on the signal generator. Return the BRG PTR NO. 1 switch to NAV 1.

NOTE

Several degrees difference may exist between NAV 1 and NAV 2 indications if the two gyromagnetic compass systems have not been swung or synchronized.

1.6. Repeat Step 1.3. through Step 1.5. for test set VOR headings of 45.0, 90.0, 135.0, 180.0, 225.0, 270.0, and 315.0°.

1.7. Rotate the COURSE SET knob on pilot HSI through 360.0° and check for proper to-from indication.

1.8. Rotate the OBS knob on the CSI through 360.0° and check for proper to-from indication.

1.9. Adjust the test set for an audio signal. Check the aural output to both navigation receivers by monitoring headset audio for a tone (appropriate mixer switches must be ON). Determine that the rotating volume control on each NAV control panel will vary the audio level.

1.10. Select a test set frequency of 117.9 MHz and a heading of 0°. Select a frequency of 117.0 MHz on both NAV control panels.

1.11. Adjust the COURSE SET knob on the pilot HSI until the course deviation bar is centered. With the course deviation bar centered and the ambiguity flag indicating to the station, the COURSE digits should be indicating approximately 0° and the course deviation bar should be approximately centered. The No. 1 pointer should also indicate a bearing of approximately 0°.

NOTE

There may be several degrees difference between NAV 1 and NAV 2 indications if the two gyromagnetic compass systems have not been swung or are not synchronized.

1.12. Set the pilot BRG PTR NO. 1 switch to NAV 2 and determine that the No. 1 pointer on the pilot HSI indicates VOR bearing for No. 2 VHF navigation receiver.

1.13. Adjust the CSI OBS knob until the vertical pointer is centered. Check that the TO flag is in view and the course card indicates approximately 0°.

1.14. Rotate the COURSE SET knob on the pilot HSI for full scale (second dot) left and right deviations. In each case, the COURSE window indication should be 10.0 ±2.0° from center course.

1.15. Rotate the CSI OBS knob for full scale (inside edge of dash marks) left and right course deviations. In each case, the course card indication should be 10.0 ±2.0 ° from the center course.

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Dual Receiver (Dual Pilot) System - Functional Test

Procedure

1. The following tests are applicable to dual receiver installations when both the pilot and copilot instruments are installed.

1.1. Tune the VHF NAV controls and test set to 108.0 MHz. Set the test set for a 0° (00.0) signal. Check that the red VOR warning flags on both HSIs and CSI are out of view.

1.2. Set the pilot BRG PTR NO. 1 switch to NAV 1, copilot BRG PTR NO. 1 switch to NAV 2, and COURSE SET switch to PILOT.

NOTE

With the preceding setup, the pilot HSI will present VOR bearing for No. 1 VHF navigation receiver, and copilot HSI will present VOR bearing for No. 2 VHF navigation receiver. Course deviation for the No. 1 VHF navigation receiver is displayed on both HSIs, and course deviation for No. 2 VHF navigation receiver is displayed on CSI.

1.3. Adjust the COURSE SET knob on the pilot HSI until the course deviation bar is centered. Check the course window and No. 1 bearing pointer indicate within 3.0° of VOR bearing selected on signal the generator and proper ambiguity is indicated (triangular flag pointing toward arrow end of course select pointer). The pilot No. 1 pointer should indicate a bearing within 3.0° of that indicated in the COURSE window and within 4.0° of the copilot No. 1 pointer (NAV 2) whenever the two compass cards agree.

NOTE

There may be several degrees difference if the gyromagnetic compasses have not been swung or are not synchronized.

1.4. Set the COURSE SET switch to COPILOT and adjust the copilot COURSE SET knob until the course deviation bar is centered. The copilot COURSE window indication should be within 3.0° of bearing indicated by the pilot HSI pointer No. 1. Observe the pilot HSI COURSE window indication and pilot course deviation bar centers within 2.0° of the copilot.

1.5. Return the COURSE SET switch to PILOT and check that the copilot HSI COURSE window is within 2.0° of the pilot COURSE window. Check the copilot course deviation bar centers within 2.0° of the pilot.

1.6. Adjust the CSI OBS knob until the vertical pointer on the CSI is centered. Check the indicated course on the CSI (NAV 2) and No. 1 bearing pointer on the copilot HSI are within 3.0° of VOR bearing selected on the signal generator and proper ambiguity is indicated (TO-FROM flag indicates TO).

1.7. Repeat Step 1.3. through Step 1.6. for test set VOR bearings of 45.0, 90.0, 135.0, 180.0, 225.0, 270.0, and 315.0°.

1.8. Rotate the COURSE SET knob on the pilot HSI through 360.0° and check for proper to-from indications on both HSIs.

1.9. Set the COURSE SET switch to COPILOT and check that rotation of the copilot COURSE SET knob through 360.0° results in proper to-from indication on both HSIs. Return the COURSE SET switch to PILOT.

1.10. Rotate the OBS knob on the CSI through 360.0°and check for proper TO-FROM indications on the CSI.

1.11. Adjust the test set for an audio signal. Check aural output of both navigation receivers by monitoring the headset audio for a tone (appropriate mixer switch must be ON). Determine that the rotating volume control on each NAV control panel will vary the audio level.

1.12. Select the test set frequency of 117.9 MHz and a bearing of 0°. Select a frequency of 117.0 MHz on both NAV controls.

1.13. Set the pilot and copilot BRG PTR NO. 1 switches to NAV 1. On the pilot HSI, adjust the COURSE SET knob until the course deviation bar is centered. With the course deviation bar centered and the ambiguity flag indicating to the station, the COURSE digits and course pointers on both HSIs should be indicating 0.0 ±3.0°. The No. 1 pointer on both HSIs should also indicate a bearing of 0.0 ±3.0°.

1.14. Set the pilot and copilot BRG PTR NO. 1 switches to NAV 2. On the CSI, adjust the OBS knob until the vertical pointer on the CSI is centered. With the vertical pointer centered and the ambiguity flag indicating TO, the indicated course should be 0.0 ±3.0°. The No. 1 pointer of both HSIs should also indicate a bearing of 0.0 ±3.0°. Return the pilot BRG PTR NO. 1 switch to NAV 1.

1.15. Rotate the COURSE SET knob on the pilot HSI for full scale (second dot) left and right course deviations. In each case, the COURSE window indication should be 10.0 ±2.0° from the center (zero) course.

1.16. Rotate the CSI OBS knob for full scale (inside edge of dash marks) left and right course deviations. In each case, the course indication should be 10.0 ±2.0° from the center (zero) course.

NOTE

NAV 1 and NAV 2 No. 1 pointer indications may differ by several degrees if the two gyromagnetic compass systems have not been swung or synchronized.

1.17. Position the pilot BRG PTR NO. 1 switch to NAV 2 and determine that the No. 1 pointer on the pilot HSI indicates VOR bearing for the No. 2 VHF navigation receiver.

1.18. Position the copilot BRG PTR NO. 1 switch to NAV 1 and determine that the No. 1 pointer on the copilot HSI indicates VOR bearing for the No. 1 VHF navigation receiver.

Localizer - Functional Test

Procedure

1. Set up the test set for localizer operation.
2. Select the test set and each installed VHF navigation receiver to 110.1 MHz.
3. Set the COURSE SET switch on the copilot HSI (when installed) to PILOT. On the pilot HSI, set the course pointer to aircraft heading with the COURSE SET knob (pointer vertical with arrow up).
4. Increase the test set output level until the red warning flags on the installed HSI and CSI (when installed) are out of view.
5. Adjust the test set for right deflection. Observe that the course deviation bars on each installed HSI and the vertical needle on the CSI (when installed) deflect to the right.
6. Adjust the test set for left deflection. Observe the course deviation bars on each installed HSI and the vertical needle on the CSI (when installed) deflect to the left.
7. Adjust the test set for centered indications. Observe that the course deviation bar on each installed HSI and the vertical needle on the CSI (when installed) are centered.

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Glideslope (GS) - Functional Test

Procedure

1. The following tests are applicable for single or dual pilot installations.

1.1. Set up the test set for glideslope operation.

1.2. Set the test set and each installed VHF navigation receiver for a glideslope frequency of 334.7 MHz (108.1 MHz-localizer).

1.3. Increase the test set output level until the GS warning flags on the pilot HSI, pilot CSI (when installed), and copilot HSI (when installed) are out of view.

NOTE

If two glideslope receivers are installed, they may be checked simultaneously.

1.4. Adjust the test set for up-deflection (150.0 Hz modulation greater than 90.0 Hz). Check the glideslope pointer on each installed indicator is at maximum upward deflection.

1.5. Adjust the test set for down-deflection (90.0 Hz modulation greater than 150.0 Hz). Check each glideslope pointer has moved through zero and is at full down position (bottom of display) on each indicator. The GS flags should remain out of view.

1.6. Adjust the test set for on-course (center) output. Check that the glideslope pointer on each installed indicator is centered and the glideslope flags remain out of view.

1.7. Select a frequency (when two VHF navigation receivers are installed) other than 108.1 MHz on the No. 1 NAV control panel and check the pilot and copilot HSI GS warning flags come into view and the GS warning flag on the CSI remains out of view.

1.8. Observe the GS warning flags on the pilot HSI (No. 1 glideslope) and CSI (No. 2 glideslope) are out of view for each of the glideslope/localizer frequency pairings listed as follows:

GLIDESLOPE FREQ (MHZ) (TEST SET)	LOCALIZER FREQ (MHZ) (NAV CONTROLS)
329.6	110.5
330.2	110.7
330.8	110.9
331.4	109.1
331.7	111.1
332.6	109.5
333.2	109.7
333.8	109.9
334.1	108.3
334.4	110.1
335.0	110.3

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VHF Navigation System - Troubleshooting

Procedure

1. Make sure the appropriate circuit breakers are closed and interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. VHF Navigation System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
No. 1 or No. 2 NAV control panel dial lights inoperative when pedestal lighting is turned on (other pedestal lights operate properly with PEDESTAL lights control).	Defective NAV control panel.	Replace NAV control panel.
	Defective wiring to lighting bus.	Repair or replace wiring.
GS warning flag is masked from view but no fly-up or fly-down indication from pointers.	Defective HSI or CSI.	Replace HSI or CSI.
	Defective wiring to GS pointers.	Repair or replace wiring.
Glideslope deviation indicators and/or GS warning flags inoperative on all indicator presentations.	Defective NAV receiver.	Replace NAV receiver.
	Defective NAV control panel	Replace NAV control panel.
	Defective glideslope antenna.	Replace glideslope antenna.
	Defective wiring.	Repair or replace wiring.

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VHF Navigation Receiver (No. 1 or No. 2) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and antenna connectors from the front of the receiver and protect open ends with electrical tape and caps.
2. Loosen the knurled retainer nut on the front of the mounting rack until the holddown clamp can be slipped off the lug on the front of the receiver.
3. Slide the receiver out of its mount.

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NAV Control Panel (No. 1 or No. 2) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the NAV control panel to the pedestal assembly.
2. Lift the NAV control panel far enough out of the pedestal assembly to reach and disconnect the electrical connector at the back of the NAV control panel.
3. Lift the NAV control panel from the pedestal assembly.

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NAV Control Panel (No. 1 or No. 2) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel mounting plate assembly to the pedestal assembly.
2. Lift the control panel mounting plate assembly and control unit(s) far enough out of the pedestal assembly to reach and disconnect the electrical connector(s) at the back of the unit(s).
3. Lift the control panel mounting plate assembly and control unit(s) free of the pedestal assembly.
4. Loosen the screws and nuts that secure the control unit to the control panel mounting plate assembly.
5. Lift the control unit free of the control panel mounting plate assembly.

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CSI (Dual VHF Navigation Systems) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the three screws and nuts that secure the indicator.
2. Disconnect the electrical connector and cap and tape.

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NAV Antenna Coupler (Installed with No. 2 VHF Navigation System) - Removal Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the three antenna cable connectors from the antenna coupler and tag the cable connectors for identification.

2. Remove the mounting screws and remove the NAV antenna coupler from the mounting shelf.

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Omni Antenna (Left or Right Half Element) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the 12 mounting screws that secure the antenna element base plates (left or right half, as applicable) to the skin of the helicopter.
2. Pull the antenna element a short distance away from the helicopter skin.
3. Disconnect the coaxial connector from the antenna element base.

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Glideslope Antenna Coupler - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the three antenna connectors from the GS antenna coupler and tag the cable connectors for identification.
2. Remove the mounting screws and remove the GS antenna coupler from the mounting shelf.

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Glideslope Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Gain access to the antenna by lowering the hinged lower nose door.
2. Disconnect the antenna cable connector from the antenna.
3. Remove the two mounting screws and washers from the outside skin, while supporting the antenna inside the access door, and remove the antenna.

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VHF Navigation Receiver (No. 1 or No. 2) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the receiver on the mount and carefully slide the receiver into place.
2. Place the holddown clamp of the mount over the lug on the front of the receiver.
3. Tighten the knurled retainer nut until the receiver is secure.
4. Remove the protective tape and caps from the electrical and antenna connectors and connect to the front of the receiver.

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NAV/DME Control Unit (No. 1 or No. 2) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the back of the control unit.
2. Position the control unit in place on the pedestal assembly and secure with the spring lock fasteners.

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NAV/DME Control Unit (No. 1 or No. 2) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the control unit in the control panel mounting plate assembly and secure with the mounting screws and nuts.
2. Connect the electrical connector(s) to the back of the control unit(s).
3. Position the control panel mounting plate assembly and control unit(s) in place on the pedestal and secure with the spring lock fasteners.

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CSI (Dual VOR/LOC/GS or VOR/LOC Systems) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the back of the CSI.
2. Position the CSI in place on the instrument panel.
3. Secure the CSI to the instrument panel with the three screws and nuts.

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NAV Antenna Coupler (Installed with No. 2 VHF Navigation System) - Installation Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Secure the antenna coupler in place on the mounting shelf with the mounting screws.
2. Connect the three tagged antenna cable connectors to the respective connectors on the VOR/LOC antenna coupler.

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Omni Antenna (Left or Right Half Element) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Place the antenna element (left or right half, as applicable) a short distance from its mounting and connect the coaxial cable to the antenna element connector.

2. Secure the antenna element base plates to the helicopter skin with 12 mounting screws.

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Glideslope Antenna Coupler - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Secure the GS antenna coupler in place on the mounting shelf with the mounting screws.
2. Connect the three tagged antenna cable connectors to their respective connectors on the GS antenna coupler.

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Glideslope Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the antenna in place in the housing inside access door and secure with two mounting screws and washers from the outside skin.
2. Connect the antenna cable connector to the antenna.
3. Close and secure the access door.

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VHF Navigation System - Inspection Procedures

Procedure

1. Inspect the VHF navigation system components for completeness. Inspect for proper installation, cleanliness, and proper mounting.
2. Inspect the switches and controls on the NAV control panels and the COURSE SET and BRG PTR switches (when installed) on the instrument panel for proper mechanical action.
3. Inspect the NAV control panels to ensure that the panel lights illuminate when the power is on.
4. Inspect the VHF navigation receivers for proper mounting.
5. Inspect the antennas, antenna couplers (when installed), and mounting for corrosion, dents, cracks, and breaks.
6. Inspect the electrical connectors for corroded or bent pins and proper mating, interconnecting cables for frayed or broken insulation, and circuit breakers for proper operation.

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VHF Navigation System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.
2. Remove dust and dirt from the panels, panel controls, panel light housings, antennas, antenna couplers, and NAV receivers with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE TOXIC AND FLAMMABLE.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.
4. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a dry cloth.

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VHF Navigation System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs on the NAV control panels COURSE SET or BRG PTR switches (dual navigation systems).
2. Replace defective or inoperative units.
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair station.

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ADF Navigation System (When Installed) - Description

The Automatic Direction Finder (ADF) navigation system consists of a receiver, control panel, sense antenna, and a loop antenna.

ADF bearing information is presented on the No. 2 pointer on each installed Horizontal Situation Indicator (HSI) .

The ADF navigation system provides both visual and aural facilities of ADF homing, radio range navigation, and position fixing. The operating frequency range is 200 to 1790 kHz. Magnetic bearing of a selected radio station is displayed on the HSI. Aural reception of audio transmissions from the station is heard in the headset when the ADF mixing switch, on the intercom control unit, is ON.

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ADF Navigation System (When Installed) - Description

The Automatic Direction Finder (ADF) navigation system consists of a receiver, control panel, and a loop/sense antenna.

ADF bearing information is presented on the No. 2 pointer on each installed Horizontal Situation Indicator (HSI) .

The ADF navigation system provides both visual and aural facilities of ADF homing, radio range navigation, and position fixing. The operating frequency range is 200 to 1790 kHz. The magnetic bearing of a selected radio station is displayed on the HSI. Aural reception of the audio transmissions from the station is heard in the headset when the ADF mixing switch on the intercom control unit is ON.

The digital ADF control unit features active and standby gaseous discharge frequency displays with automatic dimming. The control has a nonvolatile memory that retains last selected frequencies when the power is turned off. To tune a desired operating frequency, the frequency is first entered into the standby (SDY) display, and then the transfer button must be pushed. This will trade the contents of the active and standby displays. If the transfer button is held in for approximately 2 seconds, the standby display will become blank and the active display will be controlled directly by the frequency select knobs. The receiver is always tuned to the frequency appearing in the active display. An X is displayed when the ADF is not pointing to a station. If an invalid ADF frequency is selected, the display will flash. In the ANT mode, the loop portion of the antenna will be disabled allowing clearer audio reception and the No. 2 pointer on each installed HSI will rotate to a relative bearing of 90°.

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ADF Navigation System - Functional Test

Procedure

1. Preliminary Test Procedures

1.1. Visually inspect the ADF navigation system to detect and correct any faulty mechanical mountings and electrical connections. Connect an external power source adjusted to 27.5 VDC to the helicopter external power receptacle.

1.2. Energize the power buses per 412-A-97-00-00-00A-340A-A / 02238 or 412-A-97-00-00-00A-340B-A / 02239, as applicable. Place all communications and navigation switches and control in the OFF or normal position. Close all communications and navigation circuit breakers.

1.3. Place the ADF mixing switch, on the helicopter intercom control unit, to the ON position.

2. Test Procedures

2.1. Accomplish Step 1.. Adjust the VOL control on the ADF control panel to midposition, and set the ADF/ANT switch to ANT.

2.2. Tune several local stations across the band. For each station selected, check for proper frequency indication, satisfactory headset audio, and adequate volume control. Make sure the ADF mixer switch at each intercom control unit is in the ON position.

Applicable: SN: 33001-33100:

2.3. While in ANT mode and tuned to a station, turn on the BFO switch and determine that a 1020 Hz tone is audible; check the No. 2 pointer on each installed HSI rotates to a relative bearing of 270°.

Applicable: SN: 33101-36086 or 36087-Subsequent:

2.4. While in ANT mode and tuned to a station, turn on the BFO switch and determine that a 1020 Hz tone is audible; check that No. 2 pointer on each installed HSI rotates to a relative bearing of 90°.

2.5. On the ADF control panel, position the ADF/ANT switch in ADF and select several local station frequencies across the band. For each station selected, make sure that the proper frequency is indicated, that the No. 2 pointer on each installed HSI indicates the approximate direction to the station, and that the headset audio is satisfactory.

Applicable: SN: 33001-33100:

2.6. While in ADF mode, and locked on to a station, depress the TEST switch. The No. 2 pointer on each HSI should drive counterclockwise, and return to the approximate station bearing when the switch is released.

2.7. Turn off all equipment, open all circuit breakers, and disconnect the external power source.

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ADF Navigation System - Troubleshooting

Procedure

1. Make sure the appropriate circuit breakers are closed and interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. ADF Navigation System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
No audio output.	Defective ADF control panel.	Replace ADF control panel.
	Defective ADF receiver.	Replace ADF receiver.
	Defective wiring.	Repair or replace wiring.
Operating frequency is not correct.	Defective ADF control panel.	Replace ADF control panel.
	Defective ADF receiver.	Replace ADF receiver.
	Defective wiring.	Repair or replace wiring.
ADF indication 180° in error on HSI(s).	Defective ADF receiver.	Replace ADF receiver.
	SN: 33001-33100: Defective ADF sense antenna (whip).	SN: 33001-33100: Replace ADF sense antenna (whip).
	SN: 33001-33100: Defective ADF sense antenna coax cable.	SN: 33001-33100: Check and repair ADF sense antenna coax cable.
	SN: 33101-36086 or 36087-Subsequent: Defective ADF loop/sense antenna.	SN: 33101-36086 or 36087-Subsequent: Replace ADF loop/sense antenna.
	SN: 33101-36086 or 36087-Subsequent: Defective ADF loop/sense antenna coax cable.	SN: 33101-36086 or 36087-Subsequent: Check and repair ADF loop/sense antenna coax cable.
	If only one HSI, defective HSI.	Replace HSI.
	Defective wiring.	Repair or replace wiring.
All functions inoperative.	Defective ADF circuit breaker.	Replace ADF circuit breaker.
	Defective ADF receiver.	Replace ADF receiver.
	Defective ADF control panel.	Replace ADF control panel.
	Defective wiring.	Repair or replace wiring.
ADF bearing pointers on HSI(s) do not point to station.	Defective ADF control panel.	Replace ADF control panel.
	Defective ADF receiver.	Replace ADF receiver.
	SN: 33001-33100: Defective ADF sense antenna or coax cable.	SN: 33001-33100: Repair or replace sense antenna or coax cable.
	SN: 33001-33100: Defective ADF loop antenna.	SN: 33001-33100: Replace ADF loop antenna.
	SN: 33101-36086 or 36087-Subsequent: Defective ADF loop/sense antenna or coax cable.	SN: 33101-36086 or 36087-Subsequent: Repair or replace loop/sense antenna or coax cable.
	Defective wiring.	Repair or replace wiring.
ADF bearing pointer on one HSI does not point to station with two HSI(s) installed.	Defective HSI.	Replace defective HSI.
	Defective wiring.	Repair or replace wiring.

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ADF Receiver - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and antenna connectors from the front of the receiver and protect open ends with electrical tape and caps.

2. Loosen the knurled retainer nut on the front of the mounting rack until the holddown clamp can be slipped off the lug on the front of the receiver.

3. Slide the receiver out of its mount.

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ADF Control Panel - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the mounting panel to the pedestal assembly.
2. Lift the ADF control panel, far enough out of the pedestal assembly to reach and disconnect the electrical connector from the back of the control unit.
3. Lift the ADF control panel free of the pedestal assembly.

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ADF Control Panel - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel mounting plate assembly to the pedestal assembly.
2. Lift the control panel mounting plate assembly and control unit(s) far enough out of the pedestal assembly to reach and disconnect the electrical connector(s) at the back of the unit(s).
3. Lift the control panel mounting plate assembly and control unit(s) free of the pedestal assembly.
4. Loosen the screws and nuts that secure the control unit to the control panel mounting plate assembly.
5. Lift the control unit free of the control panel mounting plate assembly.

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ADF Loop Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the screws that secure the loop antenna access panel.
2. Lower the access panel and antenna from the helicopter skin until the antenna cable connector can be reached.
3. Disconnect the antenna cable connector and remove the access panel and antenna.
4. Remove the screws that attach the loop antenna to the access panel.

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ADF Loop/Sense Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the screws that secure the loop/sense antenna access panel.
2. Lower the access panel and antenna from the helicopter skin until the antenna cable connector can be reached.
3. Disconnect the antenna cable connector and remove the access panel and antenna.
4. Remove the screws that attach the loop/sense antenna to the access panel.

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ADF Sense Antenna (Whip) - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Open the heater compartment access door on the right side of the helicopter.
2. Disconnect the cable assembly from the whip sense antenna by removing the screw from the mount end of the whip antenna.
3. From outside the helicopter, loosen the clamping screw on each antenna standoff.
4. Remove the whip antenna by sliding the antenna out of each standoff.

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ADF Receiver - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the receiver on the mount and slide the receiver into place.
2. Place the holddown clamp of the mount over the lug on the front of the receiver.
3. Tighten the knurled retainer nut until the receiver is secure.
4. Remove the protective tape and caps from the electrical and antenna connectors and connect to the front of the receiver.

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ADF Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the back of the ADF control panel.
2. Position the ADF control panel in place on the pedestal assembly.
3. Secure the ADF control panel to the pedestal assembly with the spring lock fasteners.

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ADF Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the control unit in the control panel mounting plate assembly and secure with the mounting screws and nuts.
2. Connect the electrical connector(s) to the back of the control unit(s).
3. Position the control panel mounting plate assembly and control unit(s) in place on the pedestal and secure with the spring lock fasteners.

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ADF Loop Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the ADF loop antenna in place on the helicopter access panel. Make sure the forward side of the antenna is oriented with the forward edge of the panel.
2. Install the washers between the backup plate and panel to obtain proper bonding and secure the antenna to the access panel and backup plate with the mounting screws.
3. Apply a suitable weatherproofing sealant to the outside perimeter of the ADF loop antenna base.
4. Connect the antenna cable to the antenna receptacle.
5. Position the access panel in place on the helicopter skin.
6. Secure the access panel with the mounting screws.

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ADF Loop/Sense Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the ADF loop/sense antenna in place on the helicopter access panel. Make sure the forward side of the antenna is oriented with the forward edge of the panel.
2. Install the washers between the backup plate and panel to obtain proper bonding and secure the antenna to the access panel and backup plate with the mounting screws.
3. Apply a suitable weatherproofing sealant to the outside perimeter of the ADF loop/sense antenna base.
4. Connect the antenna cable to the antenna receptacle.
5. Position the access panel in place on the helicopter skin.
6. Secure the access panel with the mounting screws.

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ADF Sense Antenna (Whip) - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Place the ADF whip sense antenna into each stand-off outside the helicopter, and position the mount end of the antenna into the mounting base.

2. Gain access through the heater compartment access door inside the helicopter, secure the ADF sense antenna, and the connect cable assembly with the mounting screw.

3. Tighten the clamp screw on each stand-off.

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ADF Navigation System - Inspection Procedures

Procedure

1. Inspect the ADF system components for completeness, proper installation, cleanliness, and proper mounting.
2. Inspect the switches and controls for proper mechanical action.
3. Inspect the ADF control panel to ensure that the panel lights illuminate when the power is on.
4. Inspect the ADF antennas and mountings for corrosion, dents, cracks, and breaks.
5. Inspect the electrical connections for corroded or bent pins and proper mating, interconnecting cables for frayed or broken insulation, and circuit breakers for proper operation.

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ADF Navigation System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.
2. Remove dust and dirt from the panel, panel controls, panel light housings, antennas, and receiver with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.
4. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a dry cloth.

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ADF Navigation System - Repair or Replacement

Procedure

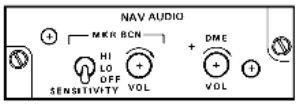
1. Tighten or replace loose or cracked control knobs on the ADF control panel.
2. Replace defective or inoperative units.
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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Marker Beacon System (When Installed) - Description

The marker beacon system consists of a receiver, antenna, NAV AUDIO control panel, and either one or two three-lamp indicators.

The marker beacon system may be installed dependent upon avionics equipment kits installed. The marker beacon system has a remote mounted receiver, with a power requirement of 27.5 VDC, operating on 75 MHz, which provides both visual (three-lamp indicators) and audible signals (to the ICS) when the helicopter is within marker beacon areas of transmission along airways and ILS courses. A test function is also provided, which checks the operation of the tone amplifiers, lamps and lamp circuitry, and audio circuit. The receiver provides outputs for three lamps. The reception of a 400 Hz tone activates the blue lamp, a 1.3 kHz tone activates the amber lamp, and a 3.0 kHz tone activates the white lamp. Audio output is continuously variable, independent of lamp outputs. An optional automatic dimming photocell is used for night-time operation. Marker beacon audio is routed through the MKR BCN VOL control on the NAV AUDIO control panel (Figure 1) to the MKR mixing switch inputs of the pilot and copilot intercom control units. A marker beacon MKR BCN SENSITIVITY - OFF/LO/HI switch is also located on the NAV AUDIO control panel. Indicator light units for both the pilot and copilot are provided on the instrument panel.



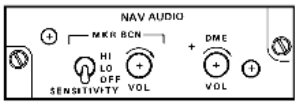
ICN-412-A-972600-B-97499-00001-A-001-01

Figure 1. NAV AUDIO Control Panel (Sheet 1 of 1)

Marker Beacon System (When installed) - Description

The marker beacon system consists of a marker beacon portion of VHF NAV receiver No. 1, marker beacon antenna, NAV AUDIO control panel, and either one or two three-lamp indicators.

The marker beacon system may be installed dependent upon avionics equipment kits installed. The marker beacon system is incorporated into the VHF NAV receiver No. 1, with a power requirement of 27.5 VDC, operating on 75 MHz, which provides both visual (three-lamp indicators) and audible signals (to ICS) when the helicopter is within marker beacon areas of transmission along airways and ILS courses. A test function is also provided, which checks the operation of tone amplifiers, lamps and lamp circuitry, and audio circuit. A receiver provides outputs for three lamps. The reception of a 400 Hz tone activates the blue lamp, a 1.3 kHz tone activates the amber lamp, and a 3.0 kHz tone activates the white lamp. Audio output is continuously variable, independent of lamp outputs. An optional automatic dimming photocell is used for night-time operation. Marker beacon audio is routed through the MKR BCN VOL control on the NAV AUDIO control panel (Figure 1) to the MKR mixing switch inputs of the pilot and copilot intercom control units. A marker beacon MKR BCN SENSITIVITY - OFF/LO/HI switch is also located on the NAV AUDIO control panel. Indicator light units for both the pilot and copilot are provided on the instrument panel.



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Figure 1. NAV AUDIO Control Panel (Sheet 1 of 1)

Marker Beacon System - Functional Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Marker Beacon Tester	TIC Type T-18A	1	or Equivalent
MKR/NAV/COMM Test Set	IFR Type NAV-401	1	

Procedure

1. Preliminary Test Procedures

1.1. Visually inspect the system installation to discover and correct damaged mechanical and electrical hardware, such as component mounts, connectors, cables, etc.

1.2. Connect an external power source adjusted to 27.5 VDC to the helicopter external power receptacle and energize power buses.

Applicable: SN: 33001-33100:

1.2.1. 412-A-97-00-00-00A-340A-A / 02238.

Applicable: SN: 33100-36086 or 36087-Subsequent:

1.2.2. 412-A-97-00-00-00A-340B-A / 02239.

1.3. Place all communications and navigation switches and controls in the OFF position.

1.4. Close all required communication and navigation circuit breakers.

2. Marker Beacon Test Procedures

2.1. Perform preliminary test procedures prior to accomplishing following tests (Step 1.).

2.2. Set the MKR BCN SENSITIVITY - OFF/LO/HI switch, on the NAV AUDIO control panel, to HI and rotate the MKR BCN VOL control fully clockwise.

2.3. Depress the test switch on each installed MARKER BEACON indicator and observe the 800 Hz tone is audible in the pilot and copilot headsets and the indicator(s) lamps illuminate. Place a finger over the photocell and observe the lamps dim.

2.4. Set the MKR mixing switch, on the pilot and copilot intercom control unit (when installed), to ON.

2.5. Position the marker beacon tester (TIC Type T-18A) near the helicopter and energize the tester. Depress the O switch on the marker beacon tester (TIC Type T-18A). Check the blue lamp on the indicator(s) illuminates, and a 400 Hz tone is audible in the pilot and copilot headsets. Rotate the MKR VOL control to determine proper operation.

2.6. Depress the M switch on the marker beacon tester (TIC Type T-18A). Check the yellow lamp on the indicator(s) illuminates and a 1300 Hz tone is audible in the pilot and copilot headsets.

2.7. Depress the A switch on the marker beacon tester (TIC Type T-18A). Check the white lamp on the indicator(s) illuminates and a 3000 Hz tone is audible in the pilot and copilot headsets.

2.8. Depress the O switch on the marker beacon tester (TIC Type T-18A) and move the tester as far away from the helicopter as possible while retaining the blue marker light(s) and audio. Position the MKR BCN SENSITIVITY - OFF/HI/LO switch to LO and note that the marker light and audio indications cease.

2.9. With the O switch depressed, move the tester toward the helicopter until the blue marker light(s) illuminate and a 400 Hz audio is heard in the headsets.

2.10. If the crew ICS kit is installed, check the marker beacon audio is NOT audible at the crew ICS stations.

2.11. De-energize and remove the marker beacon tester (TIC Type T-18A).

2.12. Turn off all equipment, open all circuit breakers, and remove the external power source.

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Marker Beacon System - Troubleshooting

Procedure

1. Make sure that the appropriate circuit breakers are closed and the interconnecting cables are in normal condition. Refer to Table 1 and the appropriate wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. Marker Beacon System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
No indication on either marker beacon indicator when test switch is depressed.	SN: 33001-33100: Defective marker beacon receiver.	SN: 33001-33100: Replace marker beacon receiver.
	SN: 33101-36086 or 36087-Subsequent: Defective marker beacon circuitry in VHF NAV receiver No. 1.	SN: 33101-36086 or 36087-Subsequent: Replace VHF NAV receiver No. 1.
	Defective wiring.	Repair or replace wiring.
No marker beacon audio signals heard in one headset (pilot or copilot); indications on both marker beacon indicators normal.	Defective NAV AUDIO control panel.	Replace NAV AUDIO control panel.
	Defective wiring.	Repair or replace wiring.
No marker beacon audio signals heard in either pilot or copilot headset; indications on both marker beacon indicators normal.	SN: 33001-33100: Defective audio circuits in marker beacon receiver.	SN: 33001-33100: Replace marker beacon receiver.
	SN: 33101-36086 or 36087-Subsequent: Defective marker beacon circuitry in VHF NAV receiver No. 1.	SN: 33101-36086 or 36087-Subsequent: Replace VHF NAV receiver No. 1.
	Defective NAV AUDIO control panel.	Replace NAV AUDIO control panel.
	Defective audio wiring.	Repair or replace wiring.
Marker beacon audio signals heard in both headsets; no indication on marker beacon indicator(s).	SN: 33001-33100: Defective marker beacon receiver.	SN: 33001-33100: Replace marker beacon receiver.
	SN: 33101-36086 or 36087-Subsequent: Defective marker beacon circuitry in VHF NAV receiver No. 1.	SN: 33101-36086 or 36087-Subsequent: Replace VHF NAV receiver No. 1.
	Defective marker beacon indicator.	Replace marker beacon indicator.
	Defective wiring.	Repair or replace wiring.
Marker beacon audio signals normal; no indication on any one light (white, blue or amber) on marker beacon indicator(s).	Defective lamp in indicator.	Replace applicable lamp.
	Defective wiring.	Repair or replace wiring.

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Marker Beacon Receiver - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and antenna connectors from the front of the receiver and protect open ends with electrical tape and caps.
2. Loosen the knurled retainer nut on the front of the mounting rack until the holddown clamp can be slipped off the lug on the front of the receiver.
3. Slide the receiver out of its mount.

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VHF NAV Receiver No. 1 - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and antenna connectors from the front of the receiver and protect open ends with electrical tape and caps.
2. Loosen the knurled retainer nut on the front of the mounting rack until the holddown clamp can be slipped off the lug on the front of the receiver.
3. Slide the receiver out of its mount.

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NAV AUDIO Control Panel - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the NAV AUDIO control panel to the pedestal assembly.
2. Lift the NAV AUDIO control panel far enough out of the pedestal assembly to reach and disconnect the electrical connector from the back of the NAV AUDIO control panel.
3. Lift the NAV AUDIO control panel free of the pedestal assembly.

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Marker Beacon Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the antenna mounting screws from the base of the antenna.
2. Lower the antenna from the helicopter skin until the antenna connector can be reached.
3. Disconnect the antenna cable connector and remove the antenna.

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Marker Beacon Receiver - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the receiver on the mount and slide the receiver into place.
2. Place the holddown clamp of the mount over the lug on the front of the receiver.
3. Tighten the knurled retainer nut until the receiver is secure.
4. Remove the protective tape and caps from the antenna and electrical connectors and connect to the front of the receiver.

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VHF NAV Receiver No. 1 - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the receiver on the mount and carefully slide the receiver into place.
2. Position the holddown clamp of the mount over the lug on the front of the receiver.
3. Tighten the knurled retainer nut until the receiver is secure.
4. Remove the protective tape and caps from the electrical and antenna connectors and connect to the front of the receiver.

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NAV AUDIO Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connectors to the back of the NAV AUDIO control panel.
2. Position the NAV AUDIO control panel in place on the pedestal.
3. Secure the NAV AUDIO control panel to the pedestal assembly with the spring lock fasteners.

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Marker Beacon Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the antenna cable connector to the antenna receptacle.
2. Position the antenna in place on the helicopter.
3. Secure the antenna with the mounting screws.

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Marker Beacon System - Inspection Procedures

Procedure

1. Inspect the system components for completeness, proper installation, cleanliness, and proper mounting.
2. Inspect the controls and indicator on the NAV AUDIO control panel for proper operation and mounting.
3. Inspect the receiver for proper mounting.
4. Inspect the antennas and mountings for corrosion, dents, cracks, and breaks.
5. Inspect the cable connections for corroded or bent pins and proper mating and the interconnecting cables for frayed or broken insulation.
6. Inspect the circuit breakers for proper operation.

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Marker Beacon System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.
2. Remove dust and dirt from the panel, panel controls, antennas, and receiver with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE TOXIC AND FLAMMABLE.

3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with an approved cleaning compound.
4. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a dry cloth.

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Marker Beacon System - Repair or Replacement

Procedure

1. Tighten or replace loose or broken knobs.
2. Replace defective or inoperative units.
3. Further repairs, other than removal and replacement, must be conducted by an authorized repair facility.

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Radar Altimeter Navigation System - Description

The radar altimeter system is optional equipment that consists of a receiver-transmitter, indicator (altimeter), and two antennas, one transmit and one receive.

The radar altimeter is a high resolution absolute altitude indicating system with a range of 0 to 1500 feet. The indicator has, in addition to the altitude display, a red power OFF warning flag, a TEST switch (button) for confidence testing, a Decision Height (DH) knob for setting a DH index, and a DH warning annunciator. Two antennas are used to allow simultaneous reception and transmission.

The radar altimeter is powered by the No. 1 bus with 28 VDC through the RDR ALTM circuit breaker.

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Radar Altimeter Navigation System - Functional Test

Procedure

1. Preliminary Test Procedures

1.1. Visually inspect the equipment to detect and correct any faulty mechanical mountings and electrical connections.

1.2. Connect an external power source adjusted to 27.5 VDC to the helicopter external power receptacle.



DO NOT ENERGIZE THE SYSTEM WITH TRANSMIT ANTENNA DISCONNECTED.

1.3. Energize the power buses per 412-A-97-00-00-00A-340A-A / 02238 or 412-A-97-00-00-00A-340B-A / 02239, as applicable.

2. Test Procedures

NOTE

Positioning the DH SET knob to X deactivates the DH function and DH warning annunciator.

2.1. Rotate the DH SET knob on the radar altimeter to set the DH index to 2 (200 feet). Close the RDR ALTM circuit breaker. The indicator pointer should immediately mask. After approximately 35 seconds warmup, the red power OFF flag on the indicator should mask, DH annunciator should illuminate, and indicator pointer should indicate 0 (±50) feet. If this accuracy is not indicated, allow a 2 minute stabilization period and adjust zero height control located in the upper right corner on the aft side of the receiver-transmitter as mounted in the helicopter, by removing the hole-plug and adjusting the control for a pointer indication of 0 (±5) feet.

2.2. Depress and hold the TEST button on the indicator. The pointer should indicate 100 (±20) feet, DH annunciator remain lighted, and power OFF flag appear.

2.3. Release the TEST button. The pointer should return to 0 (±5) feet, DH annunciator remain lighted, and power OFF flag mask.

NOTE

On the indicator, the DH index may be set to below 0 feet to deactivate the DH function and annunciator.

2.4. Set the DH index below zero and observe the DH annunciator extinguishes.

2.5. Turn off all equipment, open all circuit breakers, and disconnect the external power source.

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Radar Altimeter Navigation System - Troubleshooting

Procedure

1. Make sure the appropriate circuit breakers are closed and the interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. Radar Altimeter Navigation System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
Power OFF flag remains in view with pointer remaining in zero position after warmup.	Defective radar altimeter receiver-transmitter.	Replace radar altimeter receiver-transmitter.
	Defective radar altimeter indicator.	Replace radar altimeter indicator.
	Defective power wiring.	Repair or replace wiring.
Power OFF flag in view with pointer behind the mask after warmup.	Defective radar altimeter receiver-transmitter.	Replace radar altimeter receiver-transmitter.
	Defective radar altimeter antennas.	Replace radar altimeter antennas (one or both).
	Defective coaxial cables or connectors.	Repair or replace coaxial cables or connectors.
Pointer remains behind mask but responds correctly to depressed TEST button condition.	Defective radar altimeter receiver-transmitter.	Replace radar altimeter receiver-transmitter.
	Defective coaxial cable or antenna.	Replace radar altimeter antennas (one or both) or cable(s).
	NOTE TEST button tolerance is 100 (±20) feet.	
Pointer fails to respond, or responds incorrectly, to the TEST condition, after power OFF flag is masked from view.	Defective radar altimeter receiver-transmitter.	Replace radar altimeter receiver-transmitter.
	Defective radar altimeter indicator.	Replace radar altimeter indicator.
	Defective wiring.	Replace or repair wiring.
Indicator shows TEST altitude at all times.	Defective radar altimeter indicator.	Replace radar altimeter indicator.
	Defective radar altimeter receiver-transmitter.	Replace radar altimeter receiver-transmitter.
	NOTE Ground test tolerance is ±5 feet. (This may be adjusted with zero height control located on top of receiver-transmitter.)	
Altimeter pointer reading out of tolerance. All other indication normal.	Defective radar altimeter receiver-transmitter.	Replace radar altimeter receiver-transmitter.
	Defective radar altimeter indicator.	Replace radar altimeter indicator.
Low warning lamp inoperative. (DH SET knob not positioned X.)	Defective radar altimeter indicator or lamp.	Replace radar altimeter indicator.
	Defective wiring.	Repair or replace wiring.

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Radar Altimeter Receiver-Transmitter - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and two antenna connectors from the receiver-transmitter and protect with caps and electrical tape.
2. Remove the screws that secure the receiver-transmitter to its mounting inserts in the aft compartment bulkhead.
3. Remove the receiver-transmitter.

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Radar Altimeter Indicator - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

NOTE

Access may be gained through the upper nose access door.

1. Disconnect the cable connector from the receptacle on the rear of the indicator.
2. Loosen the mounting clamp and remove the indicator from the instrument panel.

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Radar Altimeter Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

NOTE

The following procedure is for the forward or the aft radar altimeter antenna.

1. Remove the screws attaching the tailboom access panel.
2. Lower the access panel and disconnect the antenna connector.
3. Remove the screws that attach the antenna to its panel and remove antenna.

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Radar Altimeter Receiver-Transmitter - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the receiver-transmitter in place (electrical connectors down) on the mounting inserts in the aft compartment bulkhead and install the mounting screws.

2. Connect the electrical and two antenna connectors to the receiver-transmitter.

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Radar Altimeter Indicator - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Carefully position the indicator in the instrument panel and tighten the screw of the mounting clamp.



EXCESSIVE CLAMP TENSION MAY DEFORM INDICATOR AND CAUSE ERRONEOUS READING.

2. Connect the cable connector to the receptacle on the rear of the indicator.

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Radar Altimeter Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

NOTE

The following procedure is for the forward or the aft radar altimeter antenna.

1. Position the antenna in place on the access panel and install the mounting screws.
2. Lift the panel into the approximate position and connect the antenna connector.
3. Position the access panel in place and install the mounting screws.

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Radar Altimeter Navigation System - Inspection Procedures

Procedure

1. Inspect the electrical connectors for corrosion, breakage, and proper mating.
2. Inspect the interconnecting cables for frayed or broken insulation and the circuit breakers for proper operation.
3. Inspect the receiver-transmitter for security of mounting.
4. Inspect the indicator for damage and proper mounting.
5. Inspect the antennas for corrosion, dents, cracks, and breaks.

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Radar Altimeter Navigation System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, dry, soft cloth.
2. Remove dust and dirt from the indicator, receiver-transmitter, and antennas with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.
4. Remove dirt from the connectors with a soft bristle brush.

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Radar Altimeter Navigation System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked DH SET knob on indicator.
2. Replace defective or damaged indicator.
3. Tighten or reinstall loose or improperly installed receiver-transmitter or antenna.
4. Further repairs, other than removal and replacement, must be conducted at an authorized repair shop.

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DME Navigation System (When Installed) - Description

The DME navigation system is optional equipment that consists of a receiver-transmitter, a DME (master) indicator, and an antenna.

The DME navigation system is powered from 28 VDC bus No. 1 through the DME circuit breaker. The system is energized with the NAV No. 1 or NAV No. 2 control panels.

The DME navigation system displays digital readouts of NAUTICAL MILES to or from selected VORTAC station, ground speed in knots, and time-to-station in minutes. The DME indicator has a four-position switch to provide OFF, N1, HLD, and N2 functions. The OFF position provides power control. The N1 (NAV 1) and N2 (NAV 2) provides a dual channeling source and causes indicator to display 1 or 2 to indicate which NAV system channels the DME. An HLD (hold) function allows last channel selected to be held, even when the frequency of both NAV channels has been changed. Also, an H1 or H2 is displayed to indicate channel being held.

The DME indicator displays dashes while in search or, if input power has been momentarily interrupted, while in frequency hold mode.

The DME audio is connected through the DME volume control on the NAV AUDIO control panel to the DME receiver audio control switch of the pilot and copilot intercom controls.

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DME Navigation System - Functional Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
DME Ramp Tester	TIC Type T-11A	1	or Equivalent
DME and Transponder Test Set	IFR Type ATC-600A	1	

Procedure

1. Preliminary Test Procedures

1.1. Visually inspect the equipment to detect and connect any faulty mechanical mountings and electrical connections.

1.2. Connect an external power source, adjusted to 27.5 VDC, to the helicopter external power receptacle.

1.3. Energize the power buses.

Applicable: SN: 33001-33100:

1.3.1. 412-A-97-00-00-00A-340A-A / 02238.

Applicable: SN: 33101-36086 or 36087-Subsequent:

1.3.2. 412-A-97-00-00-00A-340B-A / 02239.

2. Test Procedures

2.1. Energize the system by setting the OFF/N1/HLD/N2 selector to N1. Check the indicator displays dashes, and a 1 is displayed to indicate NAV 1 has been selected as the channeling source.

2.2. Select a frequency of 108.0 MHz (dme and transponder test set (IFR Type ATC-600A)) or 109.4 MHz (dme ramp tester (TIC Type T-11A)) on the NAV 1 control.

2.3. On the NAV AUDIO control panel, rotate the DME/VOL control fully clockwise. On the pilot and copilot intercom control, set the DME receiver audio switch to ON.

2.4. Turn on the test set and position the tester 2 to 3 feet from the helicopter DME antenna on dme and transponder test set (IFR Type ATC-600A), set the SQTR/OFF switch to SQTR.

2.5. Set the test set to 97-N/M. When using dme and transponder test set (IFR Type ATC-600A), position the RANGE/VELOCITY switch to RANGE and actuate the FAST and SLOW slewing switches until 097.0 NMi is displayed in the dme and transponder test set (IFR Type ATC-600A) display window. Observe the helicopter DME indicator displays 97 (±5) NM and a tone is audible in the pilot and copilot headsets. Also, check the tone can be shut off with the DME switch on the pilot and copilot ICS controls.

NOTE

The dme ramp tester (TIC Type T-11A) is broadbanded and operates within 8 to 10 DME channels of nominal. Consequently, when using the dme ramp tester (TIC Type T-11A), if proper indications are not observed, it may be necessary to select adjacent DME channels on the NAV control.

2.6. Set the test set to 30-N/M. Observe the DME indicator displays 30 (±2) NM.

NOTE

When using the dme and transponder test set (IFR Type ATC-600A), position the DME RANGE/VELOCITY switch to RANGE and actuate the FAST and SLOW slewing switches to obtain desired distance.

2.7. Set the test set to 5-N/M. Observe the DME indicator displays 5 (±1) NM.

2.8. On the DME indicator, set the selector switch to HLD, and change the frequency on the NAV 1 control to 108.0 MHz. Observe the DME indicator continues to display 5 (±1) NM, and H1 is displayed to indicate the channeling source being held.

2.9. Select a frequency of 115.7 MHz (dme ramp tester (TIC Type T-11A)) or 108.0 MHz (dme and transponder test set (IFR Type ATC-600A)) on the NAV 2 control, (when installed).

2.10. On the DME indicator, set selector switch to N2. Check the indicator displays 5 (±1) NM, and H2 is displayed to indicate that NAV 2 has been selected as the channeling source.

2.11. Set the test set to 97-N/M. Observe the DME indicator displays 97 (±5) NM, and a tone is audible in the pilot headset.

2.12. Cover the photocell on the DME indicator, and observe display dims.

DME Navigation System - Troubleshooting

Procedure

1. Make sure that the appropriate circuit breakers are closed and the interconnecting cabling is in normal condition. Refer to Table 1 and the appropriate DME navigation system wiring diagram when troubleshooting. Perform checks as necessary to isolate trouble.

Table 1. DME Navigation System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
No DME audio heard in either pilot or copilot headset, DME indicator inoperative.	Defective circuit breaker.	Replace circuit breaker.
	Defective DME receiver-transmitter.	Replace DME receiver-transmitter.
	Defective wiring.	Repair or replace wiring.
No DME audio heard in one headset (DME mixing switch in ON position) DME indicator operation normal.	Defective respective NAV AUDIO control panel.	Replace NAV AUDIO control panel.
	Defective wiring.	Repair or replace wiring.
No DME audio heard in either pilot or copilot headset. DME indicator operation normal.	Defective NAV AUDIO control panel.	Replace NAV AUDIO control panel.
	Defective DME receiver-transmitter.	Replace DME receiver-transmitter.
	Defective audio wiring.	Repair or replace wiring.
DME indicator does not lock on to station in either OMNI 1 or 2 switch position.	Defective DME indicator.	Replace DME indicator.
	Defective DME receiver-transmitter.	Replace DME receiver-transmitter.
	Defective DME antenna or coaxial cable.	Replace DME antenna; repair or replace coaxial cable.
	Defective wiring.	Repair or replace wiring.
DME indicator does not lock on in OMNI 1 or 2 position, but operation normal (searches, locks on, and tracks) in other OMNI position.	Defective selector switch on DME indicator.	Replace DME indicator.
	Defective respective NAV control panel.	Replace NAV control panel.
	Defective wiring.	Repair or replace wiring.
DME indicator locks on and tracks normally at certain distances, then goes into search in both OMNI positions.	Defective DME receiver-transmitter.	Replace DME receiver-transmitter.
	Defective DME antenna or coax cable.	Replace DME antenna; repair or replace coax cable.
DME indicator does not display correct ground speed (KTS position selected) and/or time-to-station (TTS position selected).	Defective DME indicator.	Replace DME indicator.
	Defective DME receiver-transmitter.	Replace DME receiver-transmitter.

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DME Receiver-Transmitter - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Disconnect the electrical and antenna connectors from the front of the receiver-transmitter and protect open ends with electrical tape and caps.

2. Loosen the knurled retainer nuts on the front of the mounting rack until the holddown clamps can be slipped off the lugs on the front of the receiver-transmitter.

3. Slide the receiver-transmitter out of its mount.

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NAV Control Panel - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel to the pedestal assembly.
2. Lift the control panel far enough out of the pedestal assembly to reach and disconnect the electrical connector.
3. Lift the control panel free of the pedestal assembly.

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NAV Control Panel - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel mounting plate assembly to the pedestal assembly.
2. Lift the control panel mounting plate assembly and control unit(s) far enough out of the pedestal assembly to reach and disconnect the electrical connector(s) at the back of the unit(s).
3. Lift the control panel mounting plate assembly and control unit(s) free of the pedestal assembly.
4. Loosen the screws and nuts that secure the control unit to the control panel mounting plate assembly.
5. Lift the control unit free of the control panel mounting plate assembly.

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NAV AUDIO Control Panel - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the control panel to the pedestal assembly.
2. Lift the control panel far enough out of the pedestal assembly to reach and disconnect the electrical connector from the back of the control panel.
3. Lift the control panel free of the pedestal assembly.

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DME Antenna - Removal

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Remove the screws that secure the antenna.
2. Lower the antenna from the helicopter skin until the antenna connector can be reached.
3. Disconnect the antenna cable connector and remove the antenna.

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DME Receiver-Transmitter - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the receiver-transmitter on the mount and slide the receiver into place.
2. Place the holddown clamps of the mount over the lugs on the front of the receiver-transmitter.
3. Tighten the knurled retainer nuts until the receiver-transmitter is secure.
4. Remove the protective tape and caps from the electrical connectors and connect to the front of the receiver-transmitter.

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Applicable to: ALL
ECCN EAR99

NAV Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the back of the control panel.
2. Position the control panel in place on the pedestal.
3. Secure the control panel to the pedestal assembly with the spring lock fasteners.

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NAV Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Position the control unit in the control panel mounting plate assembly and secure with the mounting screws and nuts.
2. Connect the electrical connector(s) to the back of the control unit(s).
3. Position the control panel mounting plate assembly and control unit(s) in place on the pedestal and secure with the spring lock fasteners.

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NAV AUDIO Control Panel - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the back of the control panel.
2. Position the control panel in place in the pedestal assembly.
3. Secure the control panel to the pedestal with the spring lock fasteners.

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DME Antenna - Installation

Procedure



MAKE SURE ELECTRICAL POWER IS OFF.

1. Connect the antenna cable to the antenna receptacle.
2. Position the antenna in place on the helicopter skin.
3. Secure the antenna with the mounting screws.

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DME Navigation System - Inspection Procedures

Procedure

1. Inspect the DME system components for completeness, proper installation, cleanliness, and security of mounting.
2. Inspect the switches and controls on the DME indicator and controls on the NAV control panels for proper mechanical action.
3. Inspect the DME indicator to ensure good condition and that the glass is clean and not cracked or damaged.
4. Inspect the control panels to ensure the panel lights illuminate when the power is on.
5. Inspect the receiver-transmitter for proper mounting.
6. Inspect the antenna and the mounting for corrosion, dents, cracks, and breaks.
7. Inspect the electrical connections for corroded or bent pins and proper mating.
8. Inspect the interconnecting cables for frayed or broken insulation.
9. Inspect the circuit breakers for proper operation.

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DME Navigation System - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.
2. Remove dust and dirt from the panel, panel controls, antenna, and receiver-transmitter with a soft, clean cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

3. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with an approved cleaning compound.
4. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a dry cloth.

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DME Navigation System - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs on the DME indicator, NAV control, or NAV AUDIO control panels.
2. Replace defective or inoperative units.
3. Further repair, other than removal and replacement, must be conducted at an authorized repair facility.

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CHAPTER 98
WIRING DIAGRAMS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Wiring Diagrams - General	DMC-412-A-98-00-00-00A-028A-A / 02455	2023-05-31	ALL
Wiring Identification - Description	DMC-412-A-98-00-00-00A-042A-A / 02456	2023-05-31	ALL
Airline Wiring Diagrams - Description	DMC-412-A-98-00-00-01A-042A-A / 02457	2023-05-31	ALL
Use of Underlined Capital Letters - Description	DMC-412-A-98-00-00-02A-042A-A / 02458	2023-05-31	ALL
Component Replacement Diagrams - Description	DMC-412-A-98-00-00-03A-042A-A / 02459	2023-05-31	ALL
Electrical and Electronic Symbols Chart - Wiring Diagrams	DMC-412-A-98-00-00-00A-051A-A / 02460	2023-05-31	ALL
Simplified Electrical Power and Busing Schematic - Wiring Diagrams	DMC-412-A-98-00-00-01A-051A-A / 02461	2023-05-31	SN: 33001-33107
Simplified Electrical Power and Busing Schematic - Wiring Diagrams	DMC-412-A-98-00-00-02A-051A-A / 02462	2023-05-31	SN: 33108-33213 or 36001-36999
Electrical Power Distribution Schematic - Wiring Diagrams	DMC-412-A-98-00-00-03A-051A-A / 02463	2023-05-31	SN: 33001-33107
Electrical Power Distribution Schematic - Wiring Diagrams	DMC-412-A-98-00-00-04A-051A-A / 02464	2023-05-31	SN: 33108-33213 or 36001-36999
AC Load Analysis Chart - Wiring Diagrams	DMC-412-A-98-00-00-05A-051A-A / 02465	2023-05-31	ALL
DC Load Analysis Chart - Wiring Diagrams	DMC-412-A-98-00-00-06A-051A-A / 02466	2023-05-31	ALL
DC Power No. 1 - Wiring Diagrams	DMC-412-A-98-00-00-07A-051A-A / 02467	2023-05-31	SN: 33001-33100
DC Power No. 1 - Wiring Diagrams	DMC-412-A-98-00-00-08A-051A-A / 02468	2023-05-31	SN: 33101-33107
DC Power No. 1 - Wiring Diagrams	DMC-412-A-98-00-00-09A-051A-A / 02469	2023-05-31	SN: 33108-Subsequent
DC Power No. 1 (EFIS Configuration) - Wiring Diagrams	DMC-412-A-98-00-00-10A-051A-A / 02470	2023-05-31	SN: 36087-36999
DC Power No. 2 - Wiring Diagrams	DMC-412-A-98-00-00-11A-051A-A / 02471	2023-05-31	SN: 33001-33100
DC Power No. 2 - Wiring Diagrams	DMC-412-A-98-00-00-12A-051A-A / 02472	2023-05-31	SN: 33101-33107
DC Power No. 2 - Wiring Diagrams	DMC-412-A-98-00-00-13A-051A-A / 02473	2023-05-31	SN: 33108-33213 or 36001-36086
DC Power No. 2 (EFIS Configuration) - Wiring Diagrams	DMC-412-A-98-00-00-14A-051A-A / 02474	2023-05-31	SN: 36087-36999
Battery and External Power - Wiring Diagrams	DMC-412-A-98-00-00-15A-051A-A / 02475	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
Battery and External Power - Wiring Diagrams	DMC-412-A-98-00-00-16A-051A-A / 02476	2023-05-31	SN: 33101-33107
Battery and External Power - Wiring Diagrams	DMC-412-A-98-00-00-17A-051A-A / 02477	2023-05-31	SN: 33108-33213 or 36001-36999
AC Power System - Wiring Diagrams	DMC-412-A-98-00-00-18A-051A-A / 02478	2023-05-31	SN: 33001-33100
AC Power System - Wiring Diagrams	DMC-412-A-98-00-00-19A-051A-A / 02479	2023-05-31	SN: 33101-33107
AC Power System - Wiring Diagrams	DMC-412-A-98-00-00-20A-051A-A / 02480	2023-05-31	SN: 33108-33213 or 36001-36999
Engine Start - Wiring Diagrams	DMC-412-A-98-00-00-21A-051A-A / 02481	2023-05-31	SN: 33001-33100
Engine Start - Wiring Diagrams	DMC-412-A-98-00-00-22A-051A-A / 02482	2023-05-31	SN: 33101-33213 or 36001-36999
Idle Stop - Wiring Diagrams	DMC-412-A-98-00-00-23A-051A-A / 02483	2023-05-31	SN: 33001-33100
Idle Stop - Wiring Diagrams	DMC-412-A-98-00-00-24A-051A-A / 02484	2023-05-31	SN: 33101-Subsequent
Particle Separator - Wiring Diagrams	DMC-412-A-98-00-00-25A-051A-A / 02485	2023-05-31	SN: 33001-33100
Particle Separator - Wiring Diagrams	DMC-412-A-98-00-00-26A-051A-A / 02486	2023-05-31	SN: 33101-Subsequent
RPM Limit Warning Detector - Wiring Diagrams	DMC-412-A-98-00-00-27A-051A-A / 02487	2023-05-31	SN: 33001-33100
RPM Limit Warning Detector - Wiring Diagrams	DMC-412-A-98-00-00-28A-051A-A / 02488	2023-05-31	SN: 33101-Subsequent
Governor RPM - Wiring Diagrams	DMC-412-A-98-00-00-29A-051A-A / 02489	2025-05-13	SN: 33001-33100 BHT-412-SI-97: Not installed
Governor RPM - Wiring Diagrams	DMC-412-A-98-00-00-30A-051A-A / 02490	2023-05-31	SN: 33101-33107
Governor RPM - Wiring Diagrams	DMC-412-A-98-00-00-31A-051A-A / 02491	2023-05-31	SN: 33108-33213 or 36001-36019
Governor RPM and Engine 2 RPM Actuator - Wiring Diagrams	DMC-412-A-98-00-00-32A-051A-A / 02492	2023-05-31	SN: 36020-36999 or 412-570-001-103: Installed
Governor RPM and Engine 2 RPM Actuator - Wiring diagram	DMC-412-A-98-00-00-32C-051A-A / 02492.1	2025-05-13	BHT-412-SI-97: Installed
Fire Extinguisher - Wiring Diagrams	DMC-412-A-98-00-00-33A-051A-A / 02493	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
Fire Extinguisher - Wiring Diagrams	DMC-412-A-98-00-00-34A-051A-A / 02494	2023-05-31	SN: 33101-33213 or 36001-36999
Engine Fire Detection System - Wiring Diagrams	DMC-412-A-98-00-00-35A-051A-A / 02495	2023-05-31	SN: 33001-33100
Engine Fire Detector System - Wiring Diagrams	DMC-412-A-98-00-00-36A-051A-A / 02496	2023-05-31	SN: 33101-33107
Engine Fire Detection System - Wiring Diagrams	DMC-412-A-98-00-00-37A-051A-A / 02497	2023-05-31	SN: 33108-33213 or 36001-36999
Oil Pressure and Temperature Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-38A-051A-A / 02498	2023-05-31	SN: 33001-33100
Oil Pressure and Temperature Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-39A-051A-A / 02499	2023-05-31	SN: 33101-Subsequent
Triple Tachometer Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-40A-051A-A / 02500	2023-05-31	SN: 33001-33100
Triple Tachometer Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-41A-051A-A / 02501	2023-05-31	SN: 33101-Subsequent
Interturbine Temperature Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-42A-051A-A / 02502	2023-05-31	SN: 33001-33100
Interturbine Temperature Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-43A-051A-A / 02503	2023-05-31	SN: 33101-33213 or 36001-36999
Torque Indicator Systems - Wiring Diagrams	DMC-412-A-98-00-00-44A-051A-A / 02504	2025-05-13	SN: 33001-33100 BHT-412-SI-97: Not installed
Torque Indicator Systems - Wiring Diagrams	DMC-412-A-98-00-00-45A-051A-A / 02505	2023-05-31	SN: 33101-33213 or 36001-36019
Engine Torque Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-46A-051A-A / 02506	2023-05-31	SN: 36020-36999 or 412-570-001-103: Installed
Engine Torque Indicating Systems - Wiring diagram	DMC-412-A-98-00-00-46C-051A-A / 02506.1	2025-05-13	BHT-412-SI-97: Installed
Mast Torque Indicating Systems - Wiring Diagrams	DMC-412-A-98-00-00-47A-051A-A / 02507	2023-05-31	SN: 36020-36999 or 412-570-001-103: Installed
Mast Torque Indicating Systems - Wiring diagrams	DMC-412-A-98-00-00-47C-051A-A / 02507.1	2025-05-13	BHT-412-SI-97: Installed
Hydraulic Pressure and Temperature Indicting System - Wiring Diagrams	DMC-412-A-98-00-00-48A-051A-A / 02508	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
Hydraulic Pressure and Temperature Indicating System - Wiring Diagrams	DMC-412-A-98-00-00-49A-051A-A / 02509	2023-05-31	SN: 33101-33213 or 36001-36999
Hourmeter - Wiring Diagrams	DMC-412-A-98-00-00-50A-051A-A / 02510	2023-05-31	SN: 33001-33100
Hourmeter - Wiring Diagrams	DMC-412-A-98-00-00-51A-051A-A / 02511	2023-05-31	SN: 33101-33107
Hourmeter - Wiring Diagrams	DMC-412-A-98-00-00-52A-051A-A / 02512	2023-05-31	SN: 33108-33213 or 36001-36999
Pilot Attitude Director System - Wiring Diagrams	DMC-412-A-98-00-00-53A-051A-A / 02513	2023-05-31	SN: 33001-33100
Pilot Attitude Director System - Wiring Diagrams	DMC-412-A-98-00-00-54A-051A-A / 02514	2023-05-31	SN: 33101-33213 or 36001-36999
Copilot Attitude Director System - Wiring Diagrams	DMC-412-A-98-00-00-55A-051A-A / 02515	2023-05-31	SN: 33001-33100
Copilot Attitude Director System - Wiring Diagrams	DMC-412-A-98-00-00-56A-051A-A / 02516	2023-05-31	SN: 33101-33213 or 36001-36999
Fuel Quantity Indicating System - Wiring Diagrams	DMC-412-A-98-00-00-57A-051A-A / 02517	2023-05-31	SN: 33001-33100
Fuel Quantity Indicating System - Wiring Diagrams	DMC-412-A-98-00-00-58A-051A-A / 02518	2023-05-31	SN: 33101-33107
Fuel Quantity Indicating System - Wiring Diagrams	DMC-412-A-98-00-00-59A-051A-A / 02519	2023-05-31	SN: 33108-33213 or 36001-36999
Fuel Control No. 1 - Wiring Diagrams	DMC-412-A-98-00-00-60A-051A-A / 02520	2023-05-31	SN: 33001-33100
Fuel Control No. 1 - Wiring Diagrams	DMC-412-A-98-00-00-61A-051A-A / 02521	2023-05-31	SN: 33101-33107
Fuel Control No. 1 - Wiring Diagrams	DMC-412-A-98-00-00-62A-051A-A / 02522	2023-05-31	SN: 33108-Subsequent
Fuel Control No. 2 - Wiring Diagrams	DMC-412-A-98-00-00-63A-051A-A / 02523	2023-05-31	SN: 33001-33100
Fuel control No. 2 - Wiring Diagrams	DMC-412-A-98-00-00-64A-051A-A / 02524	2023-05-31	SN: 33101-33107
Fuel Control No. 2 - Wiring Diagrams	DMC-412-A-98-00-00-65A-051A-A / 02525	2024-05-09	SN: 33108-Subsequent
Fuel Crossfeed - Wiring Diagrams	DMC-412-A-98-00-00-66A-051A-A / 02526	2023-05-31	SN: 33001-33100
Fuel Crossfeed - Wiring Diagrams	DMC-412-A-98-00-00-67A-051A-A / 02527	2023-05-31	SN: 33101-33107
Fuel Crossfeed - Wiring Diagrams	DMC-412-A-98-00-00-68A-051A-A / 02528	2023-05-31	SN: 33108-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Fuel Interconnect Valve - Wiring Diagrams	DMC-412-A-98-00-00-69A-051A-A / 02529	2023-05-31	SN: 33108-33213 or 36001-36999
Pilot Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-70A-051A-A / 02530	2023-05-31	SN: 33001-33100
Pilot Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-71A-051A-A / 02531	2023-05-31	SN: 33101-33213 or 36001-36999
Copilot Instrument Lights and Passenger Warning Light - Wiring Diagrams	DMC-412-A-98-00-00-72A-051A-A / 02532	2023-05-31	SN: 33001-33100
Copilot Instrument Lights and Passenger Warning Light - Wiring Diagrams	DMC-412-A-98-00-00-73A-051A-A / 02533	2023-05-31	SN: 33101-33213 or 36001-36999
Engine Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-74A-051A-A / 02534	2023-05-31	SN: 33001-33100
Engine Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-75A-051A-A / 02535	2023-05-31	SN: 33101-33213 or 36001-36999
Master Caution and RPM Caution Light - Wiring Diagrams	DMC-412-A-98-00-00-76A-051A-A / 02536	2023-05-31	SN: 33001-33100
Master Caution and RPM Caution Lights - Wiring Diagrams	DMC-412-A-98-00-00-77A-051A-A / 02537	2023-05-31	SN: 33101-33107
Master Caution, RPM Caution, and Cyclic Stick Centering Indicator Lights - Wiring Diagrams	DMC-412-A-98-00-00-78A-051A-A / 02538	2023-05-31	SN: 33108-33213 or 36001-36999
Master Caution Panel - Wiring Diagrams	DMC-412-A-98-00-00-79A-051A-A / 02539	2023-05-31	SN: 33001-33100
Master Caution Panel - Wiring Diagrams	DMC-412-A-98-00-00-80A-051A-A / 02540	2023-05-31	SN: 33101-33107
Master Caution Panel - Wiring Diagrams	DMC-412-A-98-00-00-81A-051A-A / 02541	2023-05-31	SN: 33108-33213 or 36001-36999
Pilot/Copilot Cyclic Centering Indicator Light - Wiring Diagrams	DMC-412-A-98-00-00-82A-051A-A / 02542	2023-05-31	SN: 33001-33100
Pilot/Copilot Cyclic Centering Indicator Light - Wiring Diagrams	DMC-412-A-98-00-00-83A-051A-A / 02543	2023-05-31	SN: 33101-33213 or 36001-36999
Pedestal Lights - Wiring Diagrams	DMC-412-A-98-00-00-84A-051A-A / 02544	2023-05-31	SN: 33001-33100
Pedestal Lights - Wiring Diagrams	DMC-412-A-98-00-00-85A-051A-A / 02545	2023-05-31	SN: 33101-33213 or 36001-36999
Secondary Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-86A-051A-A / 02546	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
Secondary Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-87A-051A-A / 02547	2023-05-31	SN: 33101-33213 or 36001-36086
Secondary Instrument Lights - Wiring Diagrams	DMC-412-A-98-00-00-88A-051A-A / 02548	2023-05-31	SN: 36087-36999
Overhead Console Lights - Wiring Diagrams	DMC-412-A-98-00-00-89A-051A-A / 02549	2023-05-31	SN: 33001-33100
Overhead Console Lights - Wiring Diagrams	DMC-412-A-98-00-00-90A-051A-A / 02550	2023-05-31	SN: 33101-33213 or 36001-36999
Cockpit, Map, and Utility Light - Wiring Diagrams	DMC-412-A-98-00-00-91A-051A-A / 02551	2023-05-31	SN: 33001-33100
Cockpit, Map, and Utility Lights - Wiring Diagrams	DMC-412-A-98-00-00-92A-051A-A / 02552	2023-05-31	SN: 33101-33213 or 36001-36999
Cabin Dome Lights - Wiring Diagrams	DMC-412-A-98-00-00-93A-051A-A / 02553	2023-05-31	SN: 33001-33100
Cabin Dome Lights - Wiring Diagrams	DMC-412-A-98-00-00-94A-051A-A / 02554	2023-05-31	SN: 33101-33213 or 36001-36999
Searchlight - Wiring Diagrams	DMC-412-A-98-00-00-95A-051A-A / 02555	2023-05-31	SN: 33001-33100
Searchlight - Wiring Diagrams	DMC-412-A-98-00-00-96A-051A-A / 02556	2023-05-31	SN: 33101-36086 or 36087-Subsequent
Anticollision Lights - Wiring Diagrams	DMC-412-A-98-00-00-97A-051A-A / 02557	2023-05-31	SN: 33001-33100
Anti-collision Lights - Wiring Diagrams	DMC-412-A-98-00-00-98A-051A-A / 02558	2023-05-31	SN: 33101-Subsequent
Position Lights - Wiring Diagrams	DMC-412-A-98-00-00-99A-051A-A / 02559	2023-05-31	SN: 33001-33100
Position Lights - Wiring Diagrams	DMC-412-A-98-00-00-00B-051A-A / 02560	2023-05-31	SN: 33101-33213 or 36001-36999
Landing Lights - Wiring Diagrams	DMC-412-A-98-00-00-01B-051A-A / 02561	2023-05-31	SN: 33001-33100
Landing Lights - Wiring Diagrams	DMC-412-A-98-00-00-02B-051A-A / 02562	2023-05-31	SN: 33101-33213 or 36001-36999
Baggage Compartment Lights - Wiring Diagrams	DMC-412-A-98-00-00-03B-051A-A / 02563	2023-05-31	SN: 33001-33100
Baggage Compartment Lights - Wiring Diagrams	DMC-412-A-98-00-00-04B-051A-A / 02564	2023-05-31	SN: 33101-33213 or 36001-36999
Hydraulic Systems - Wiring Diagrams	DMC-412-A-98-00-00-05B-051A-A / 02565	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
Hydraulic Systems - Wiring Diagrams	DMC-412-A-98-00-00-06B-051A-A / 02566	2023-05-31	SN: 33101-33107
Hydraulic Systems - Wiring Diagrams	DMC-412-A-98-00-00-07B-051A-A / 02567	2023-05-31	SN: 33108-33213 or 36001-36247
Hydraulic Systems - Wiring Diagrams	DMC-412-A-98-00-00-08B-051A-A / 02568	2023-05-31	SN: 36248-36999
AFCS - Wiring Diagrams	DMC-412-A-98-00-00-09B-051A-A / 02569	2023-05-31	SN: 33001-33100
AFCS - Wiring Diagrams	DMC-412-A-98-00-00-10B-051A-A / 02570	2023-05-31	SN: 33101-33213 or 36001-36086
Flight Director System - Wiring Diagrams	DMC-412-A-98-00-00-11B-051A-A / 02571	2023-05-31	SN: 33001-33100
Flight Director System - Wiring Diagrams	DMC-412-A-98-00-00-12B-051A-A / 02572	2023-05-31	SN: 33101-33213 or 36001-36086
Digital Automatic Flight Control Systems - Wiring Diagrams	DMC-412-A-98-00-00-13B-051A-A / 02573	2023-05-31	SN: 36087-36999
4-Axis Flight Director Kit - Wiring Diagrams	DMC-412-A-98-00-00-14B-051A-A / 02574	2023-05-31	SN: 36087-36999
Flight Data Recorder - Wiring Diagrams	DMC-412-A-98-00-00-15B-051A-A / 02575	2023-05-31	SN: 36037-36490
Flight Data Recorder - Wiring Diagrams	DMC-412-A-98-00-00-16B-051A-A / 02576	2023-05-31	SN: 36491-36999
Cockpit Voice Recorder - Wiring Diagrams	DMC-412-A-98-00-00-17B-051A-A / 02577	2023-05-31	SN: 36037-36507
Baggage Compartment Fire Detection - Wiring Diagrams	DMC-412-A-98-00-00-18B-051A-A / 02578	2023-05-31	SN: 33001-33100
Baggage Compartment Fire Detection - Wiring Diagrams	DMC-412-A-98-00-00-19B-051A-A / 02579	2023-05-31	SN: 33101-Subsequent
Chip Detectors - Wiring Diagrams	DMC-412-A-98-00-00-20B-051A-A / 02580	2023-05-31	SN: 33001-33100
Chip Detectors - Wiring Diagrams	DMC-412-A-98-00-00-21B-051A-A / 02581	2023-05-31	SN: 33101-Subsequent
Pitot-Static Heaters - Wiring Diagrams	DMC-412-A-98-00-00-22B-051A-A / 02582	2023-05-31	SN: 33001-33100
Pitot-Static Heaters - Wiring Diagrams	DMC-412-A-98-00-00-23B-051A-A / 02583	2023-05-31	SN: 33101-33213 or 36001-36999
Vent Blowers - Wiring Diagrams	DMC-412-A-98-00-00-24B-051A-A / 02584	2023-05-31	SN: 33001-33100
Vent Blowers - Wiring Diagrams	DMC-412-A-98-00-00-25B-051A-A / 02585	2023-05-31	SN: 33101-33213 or 36001-36999
Step Actuators - Wiring Diagrams	DMC-412-A-98-00-00-26B-051A-A / 02586	2023-05-31	SN: 33001-33100

Document title	Data module code / Seq number	Issue date	Applicable to
Step Actuators - Wiring Diagrams	DMC-412-A-98-00-00-27B-051A-A / 02587	2023-05-31	SN: 33101-33213 or 36001-36086
Step Actuators - Wiring Diagrams	DMC-412-A-98-00-00-28B-051A-A / 02588	2023-05-31	SN: 36087-36124
Step Actuators - Wiring Diagrams	DMC-412-A-98-00-00-29B-051A-A / 02589	2023-05-31	SN: 36125-36999
Cargo Hook Release - Wiring Diagrams	DMC-412-A-98-00-00-30B-051A-A / 02590	2023-05-31	SN: 33001-33100
Cargo Hook Release - Wiring Diagrams	DMC-412-A-98-00-00-31B-051A-A / 02591	2023-05-31	SN: 33101-36086
Cargo Hook Release - Wiring Diagrams	DMC-412-A-98-00-00-32B-051A-A / 02592	2023-05-31	SN: 36087-36999
Cabin Heating - Wiring Diagrams	DMC-412-A-98-00-00-33B-051A-A / 02593	2023-05-31	SN: 33001-33100
Cabin Heating - Wiring Diagrams	DMC-412-A-98-00-00-34B-051A-A / 02594	2023-05-31	SN: 33101-33107
Cabin Heating - Wiring Diagrams	DMC-412-A-98-00-00-35B-051A-A / 02595	2023-05-31	SN: 33108-33213 or 36001-36999
Cabin Heating - Wiring Diagrams	DMC-412-A-98-00-00-36B-051A-A / 02596	2023-05-31	SN: 37002-37999
Windshield Heating - Wiring Diagrams	DMC-412-A-98-00-00-37B-051A-A / 02597	2023-05-31	SN: 36087-36999
Door Switches - Wiring Diagrams	DMC-412-A-98-00-00-38B-051A-A / 02598	2023-05-31	SN: 33001-33100
Door Switches - Wiring Diagrams	DMC-412-A-98-00-00-39B-051A-A / 02599	2023-05-31	SN: 33101-33213 or 36001-36999
Windshield Wipers - Wiring Diagrams	DMC-412-A-98-00-00-40B-051A-A / 02600	2023-05-31	SN: 33001-33100
Windshield Wipers - Wiring Diagrams	DMC-412-A-98-00-00-41B-051A-A / 02601	2023-05-31	SN: 33101-33213 or 36001-36999
Emergency Floats - Wiring Diagrams	DMC-412-A-98-00-00-42B-051A-A / 02602	2023-05-31	SN: 33001-33100
Emergency Floats - Wiring Diagrams	DMC-412-A-98-00-00-43B-051A-A / 02603	2023-05-31	SN: 33101-36086
Emergency Floats - Wiring Diagrams	DMC-412-A-98-00-00-44B-051A-A / 02604	2023-05-31	SN: 36087-36999
Hoist Control - Wiring Diagrams	DMC-412-A-98-00-00-45B-051A-A / 02605	2023-05-31	SN: 36087-36999
Rotor Brake - Wiring Diagrams	DMC-412-A-98-00-00-46B-051A-A / 02606	2023-05-31	SN: 36087-36999
Digital Clocks - Wiring Diagrams	DMC-412-A-98-00-00-47B-051A-A / 02607	2023-05-31	SN: 36087-36999
Standby Attitude Indicator Provisions - Wiring Diagrams	DMC-412-A-98-00-00-48B-051A-A / 02608	2023-05-31	SN: 36087-36999

Document title	Data module code / Seq number	Issue date	Applicable to
Component Replacement - Wiring Diagrams	DMC-412-A-98-00-00-49B-051A-A / 02609	2023-05-31	SN: 33108-36086
Component Replacement - Wiring Diagrams	DMC-412-A-98-00-00-50B-051A-A / 02610	2023-05-31	SN: 36087-36999
Pilot AHRS 2 (LCR-92) - Wiring Diagrams	DMC-412-A-98-00-00-51B-051A-A / 02611	2023-05-31	SN: 36248-36527
Pilot AHRS 2 (LCR-100) - Wiring Diagrams	DMC-412-A-98-00-00-52B-051A-A / 02612	2023-05-31	SN: 36248-36999
Copilot AHRS 1 (LCR-92) - Wiring Diagrams	DMC-412-A-98-00-00-53B-051A-A / 02613	2023-05-31	SN: 36248-36527
Copilot AHRS 1 (LCR-100) - Wiring Diagrams	DMC-412-A-98-00-00-54B-051A-A / 02614	2023-05-31	SN: 36248-36999
Electronic Flight Instruments System - Wiring Diagrams	DMC-412-A-98-00-00-55B-051A-A / 02615	2023-05-31	SN: 36087-36247
Electronic Flight Instruments System - Wiring Diagrams	DMC-412-A-98-00-00-56B-051A-A / 02616	2023-05-31	SN: 36248-36999
Component Replacement, EFIS - Wiring Diagrams	DMC-412-A-98-00-00-57B-051A-A / 02617	2023-05-31	SN: 36087-36999
SAR Interface Kit - Wiring Diagrams	DMC-412-A-98-00-00-58B-051A-A / 02618	2023-05-31	ALL
Weight-on-Gear Annunciator - Wiring Diagrams	DMC-412-A-98-00-00-59B-051A-A / 02619	2023-05-31	SN: 36528-36999
Rotor Track and Balance Provisions - Wiring Diagrams	DMC-412-A-98-00-00-60B-051A-A / 02620	2023-05-31	SN: 33108-33213 or 36001-36999
Emergency Locator Transmitter (ELT ARTEX) Provision - Wiring Diagrams	DMC-412-A-98-00-00-61B-051A-A / 02621	2023-05-31	SN: 36537-36651

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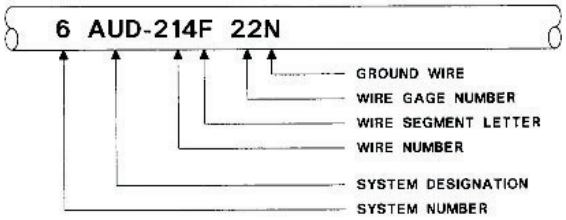
Wiring Diagrams - General

This chapter contains electrical schematic diagrams, ac and dc load analysis charts, electrical wiring diagrams, and component replacement charts. This information is to assist maintenance personnel in performing maintenance and troubleshooting of electrical and avionic systems installed on the Model 412/412EP helicopters.

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Wiring Identification - Description

All wires on the wiring diagrams are identified exactly as they are marked in the helicopter. The wire number indicates the circuit function, wire number, wire segment, and wire size. Identification of wires may be accomplished by using the wiring data in Figure 1.



CIRCUIT FUNCTIONS

PREFIX CODE	NOMENCLATURE
APX	RECOGNITION (IFF)
ARC	COMMUNICATIONS RADIO
ARN	NAVIGATION RADIO
AUD	AUDIO (INTERCOMMUNICATIONS)
C	CONTROL SYSTEM
CMPS	GYRO COMPASS
CVFDR	COCKPIT VOICE FLIGHT DATA RECORDER
D	INSTRUMENTS (OTHER THAN FLIGHT OR ENGINE)
DF	DIRECTION FINDER
DME	DISTANCE MEASURING EQUIPMENT
DNS	DOPPLER NAVIGATION
E	ENGINE INSTRUMENTS
F	FLIGHT INSTRUMENTS
H	HEATING, VENTILATION AND DE-ICING
J	IGNITION
K	ENGINE CONTROL
L	LIGHTING

CIRCUIT FUNCTIONS (Cont)

PREFIX CODE	NOMENCLATURE
M	MISCELLANEOUS ELECTRICAL
P	DC POWER
Q	FUEL AND OIL
V	DC CONTROL FOR AC POWER
W	WARNING AND EMERGENCY
X	AC POWER
1	SYSTEM NO. 1
2	SYSTEM NO. 2
3	SYSTEM NO. 3
4	SYSTEM NO. 4

WIRE FUNCTIONS

SUFFIX CODE	NOMENCLATURE
N	GROUND
CHROM	CHROMEL (HEAT DETECTION)
ALML	ALUMEL (HEAT DETECTION)
WHT	WHITE WIRE
BLU	BLUE WIRE
ORN	ORANGE WIRE

ICN-412-A-980000-B-97499-00001-A-001-01

Figure 1. Wiring Identification Code (Sheet 1 of 1)

Airline Wiring Diagrams - Description

Some of the larger wiring diagrams are presented in a format commonly called "airline" style. The airline style minimizes difficulty in tracing signal paths and simplifies component layout. To use airline wiring diagrams, locate the desired wire at the component connection. Trace the wire to the termination point. The termination point (pin letter, terminal number, etc.).

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Use of Underlined Capital Letters - Description

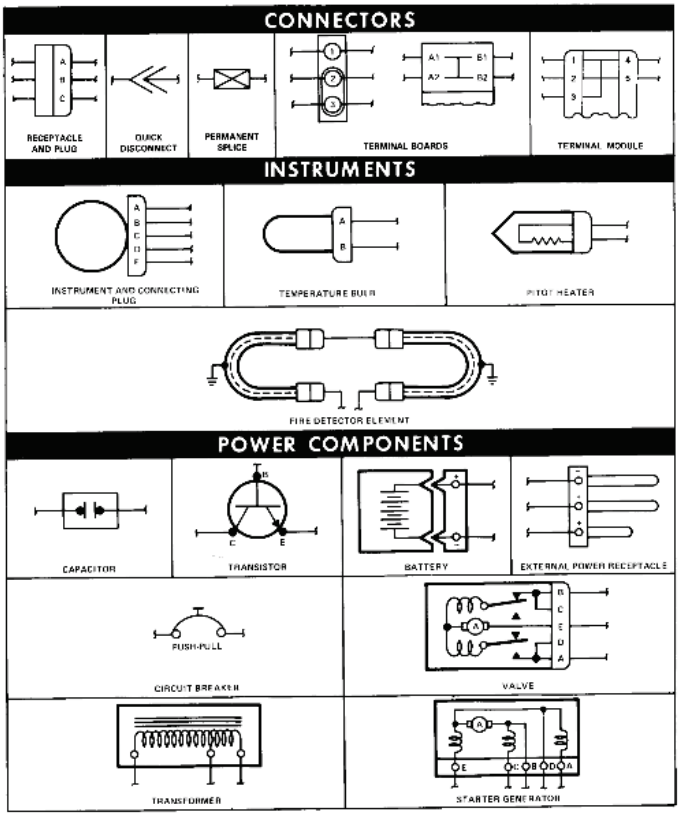
Many letters have similar upper and lower case forms. To eliminate this as a possible source of error in identifying connector pin designations, underlined upper case letters (A, B, C, etc.) are used to show lower case pin letters (a, b, c, etc.).

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Component Replacement Diagrams - Description

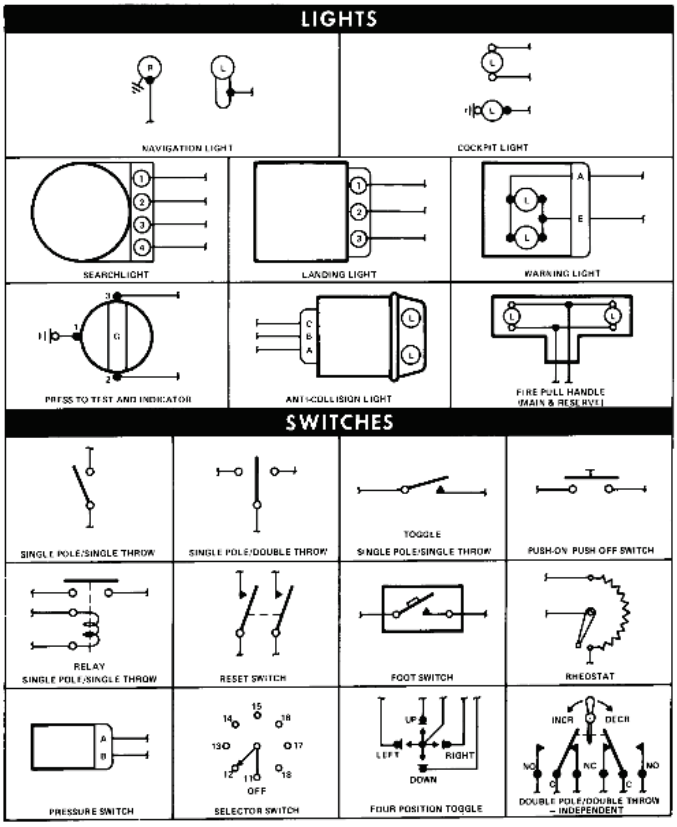
Some components are not complete on some wiring diagrams. If a component is to be replaced, all wire numbers and connections required to complete the task are shown in the component replacement diagrams.

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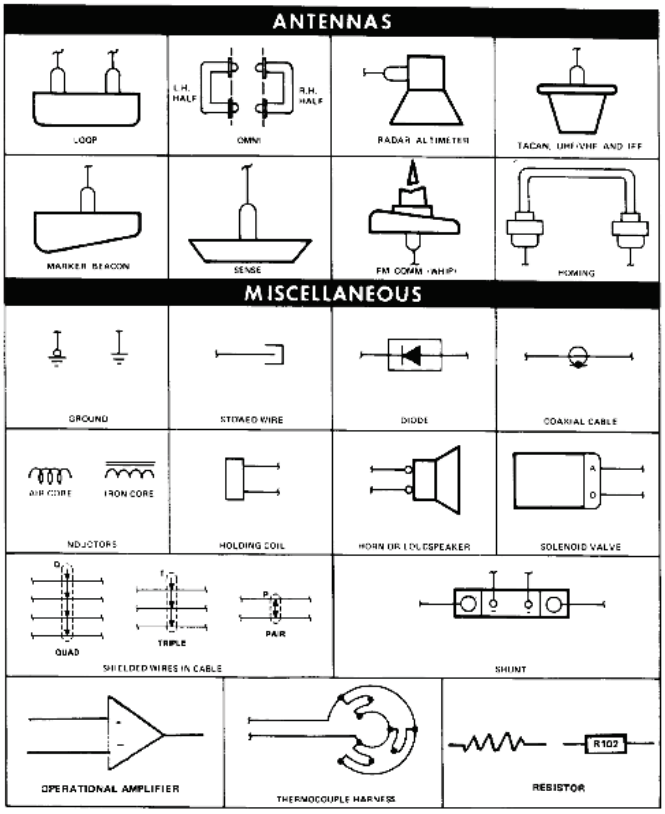
ICN-412-A-980000-B-97499-00002-A-001-01

Figure 1. Electrical and Electronic Symbols Chart (Sheet 1 of 3)



ICN-412-A-980000-B-97499-00003-A-001-01

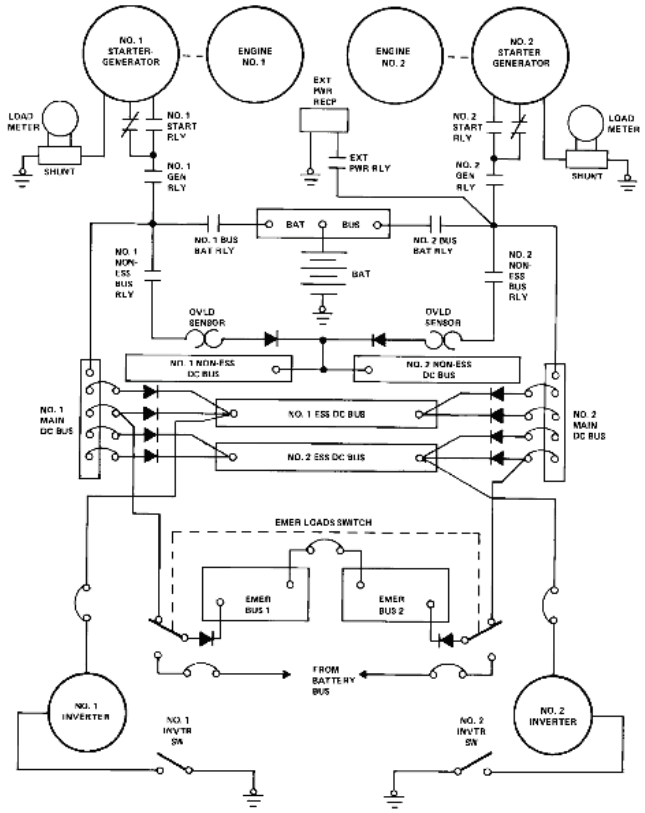
Figure 1. Electrical and Electronic Symbols Chart (Sheet 2 of 3)



ICN-412-A-980000-B-97499-00004-A-001-01

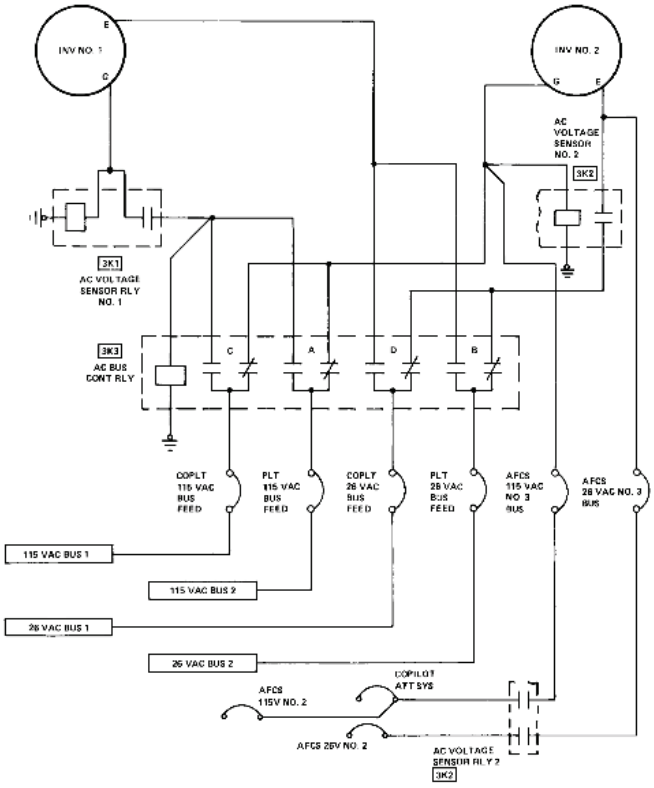
Figure 1. Electrical and Electronic Symbols Chart (Sheet 3 of 3)

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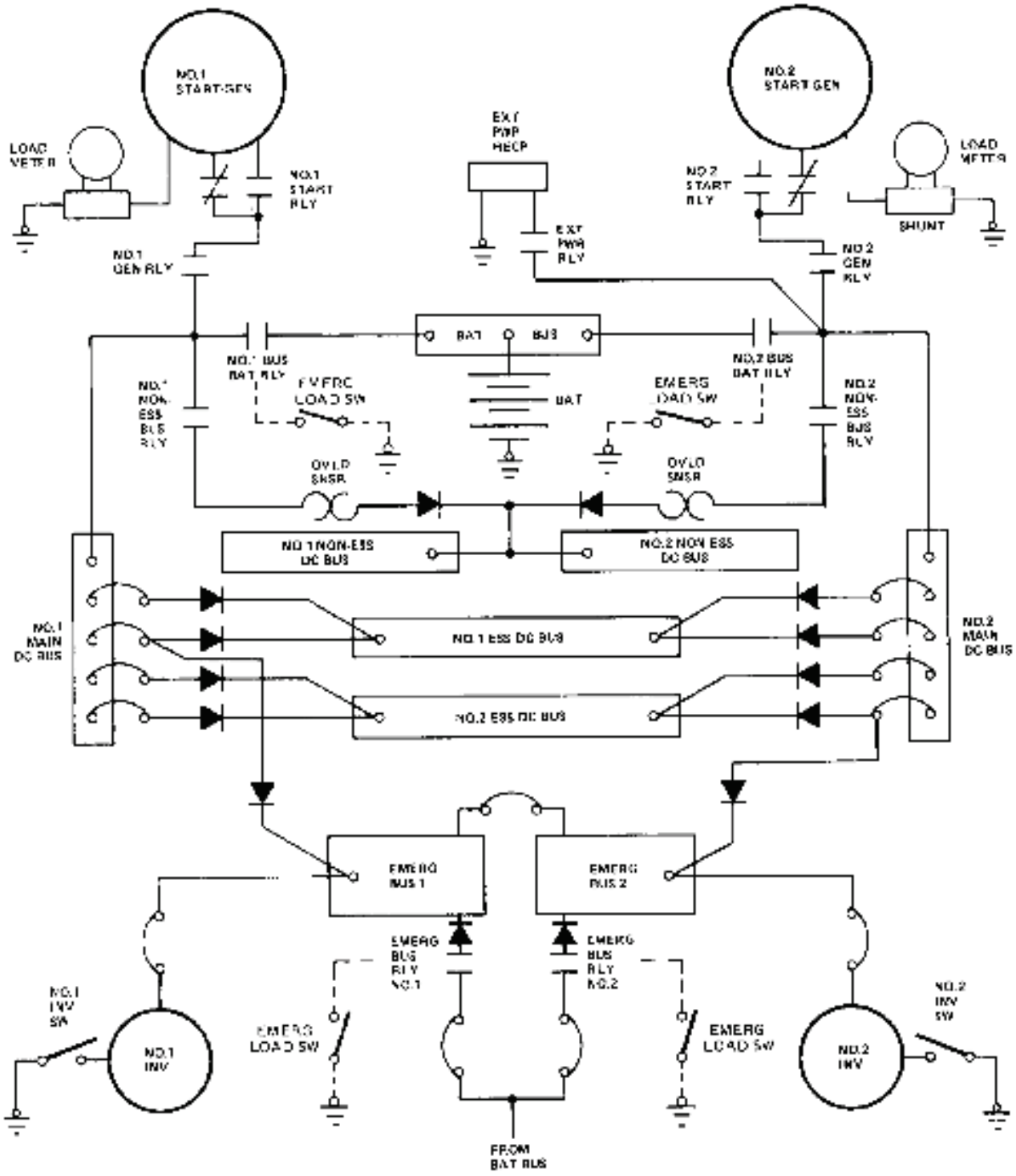
ICN-412-A-980000-B-97499-00005-A-001-01

Figure 1. Simplified Electrical Power and Busing Schematic (Sheet 1 of 2)



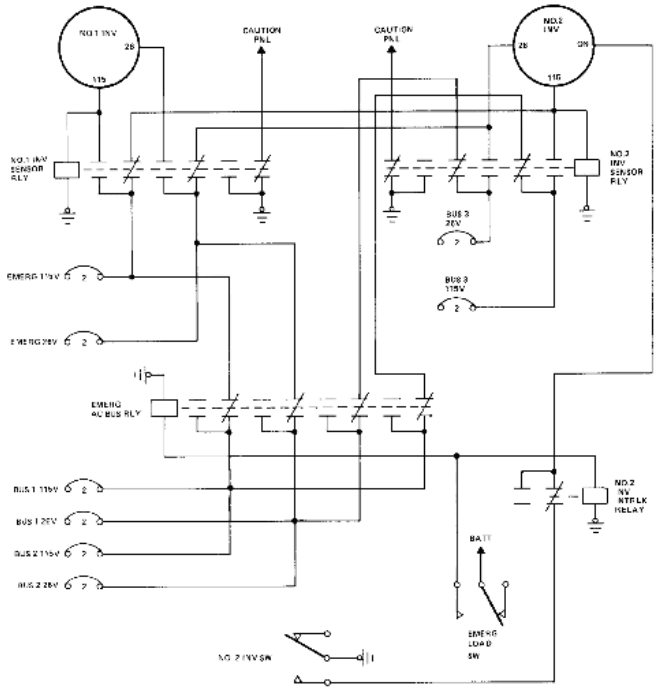
ICN-412-A-980000-B-97499-00006-A-001-01

Figure 1. Simplified Electrical Power and Busing Schematic (Sheet 2 of 2)



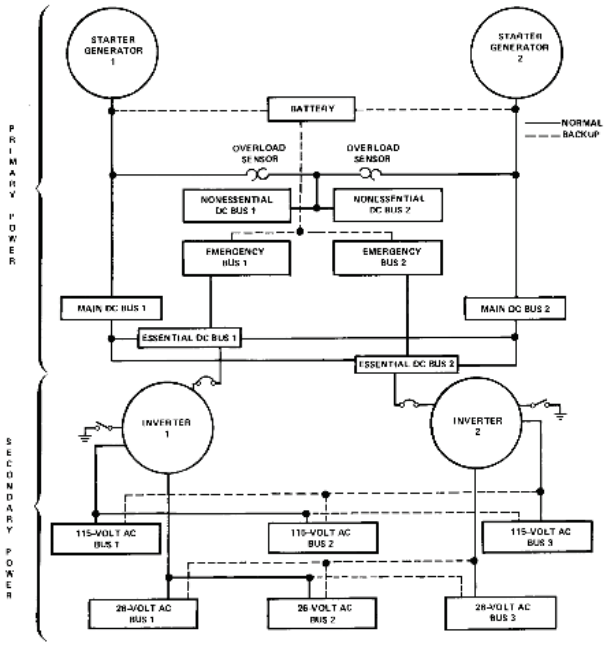
ICN-412-A-980000-B-97499-00007-A-001-01

Figure 1. Simplified Electrical Power and Busing Schematic (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00008-A-001-01

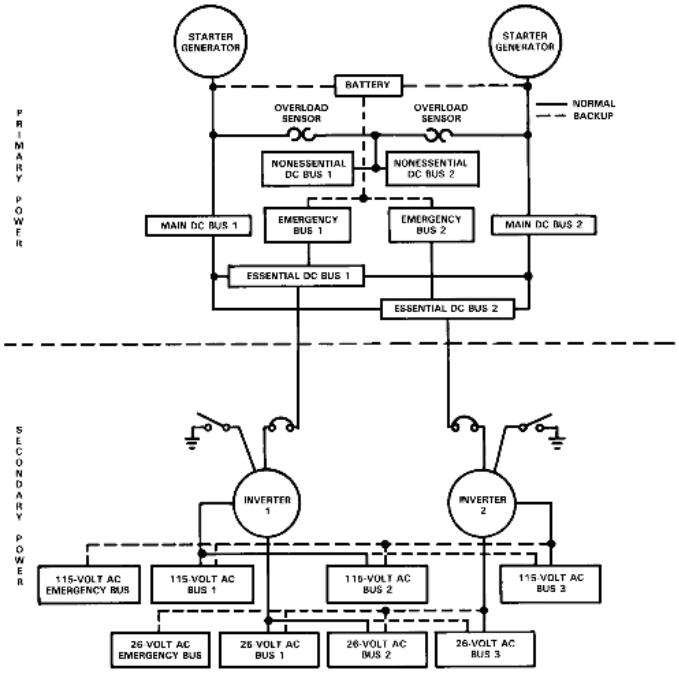
Figure 1. Simplified Electrical Power and Busing Schematic (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00009-A-001-01

Figure 1. Electrical Power Distribution Schematic (Sheet 1 of 1)

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Figure 1. Electrical Power Distribution Schematic (Sheet 1 of 1)

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AC LOAD ANALYSIS

AC EMERGENCY BUS-BASIC

EFFECTIVITY	LOAD (VA)
33001 through 33100	89.37
33101 through 33107	
33108 through 33120	89.37
33121 through 36019	89.37
36020 through 36036	89.37
36037 and subsequent	89.37

AC BUS NO. 1 – BASIC

EFFECTIVITY	LOAD (VA)
33001 through 33100	46.74
33101 through 33107	48.74
33108 through 33120	49.74
33121 through 36019	49.74
36020 through 36036	39.93
36037 and subsequent	39.93

AC BUS NO. 2 – BASIC

EFFECTIVITY	LOAD (VA)
33001 through 33100	43.02
33101 through 33107	127.60
33108 through 33120	40.16
33121 through 36019	40.16
36020 through 36036	30.16
36037 and subsequent	30.16

AC BUS NO. 3 – BASIC

EFFECTIVITY	LOAD (VA)
33001 through 33100	27.05
33101 through 33107	27.05
33108 through 33120	27.05
33121 through 36019	27.05
36020 through 36047	27.05
36037 and subsequent	43.69

TOTAL AC LOAD – BASIC

EFFECTIVITY	LOAD (VA)
33001 through 33100	
Inverter No. 1	173.05
Inverter No. 2	27.05
Single inverter	198.35
33101 through 33107	
Inverter No. 1	174.97
Inverter No. 2	27.05
Single inverter	200.24
33108 through 33120	
Inverter No. 1	172.00
Inverter No. 2	27.00
Single inverter	198.00
33121 through 36019	
Inverter No. 1	172.00
Inverter No. 2	27.00
Single inverter	198.00
36020 through 36036	
Inverter No. 1	154.00
Inverter No. 2	27.00
Single inverter	180.00
36037 and subsequent	
Inverter No. 1	154.00
Inverter No. 2	44.00
Single inverter	180.00

AC EMERGENCY BUS – IFR

EFFECTIVITY	LOAD (VA)
33001 through 33100	7.97
33101 through 33107	
33108 through 33120	
33121 through 36019	
36020 through 36036	
36037 and subsequent	

ICN-412-A-980000-B-97499-00011-A-001-01

Figure 1. AC Load Analysis Chart (Sheet 1 of 3)

AC LOAD ANALYSIS

AC BUS NO. 1 — IFR

EFFECTIVITY	LOAD (VA)
33001 through 33100	29.12
33101 through 33107	48.41
33108 through 33120	32.08
33121 through 36019	32.08
36020 through 36036	23.00
36037 and subsequent	23.00

AC BUS NO. 2 — IFR

EFFECTIVITY	LOAD (VA)
33001 through 33100	10.00
33101 through 33107	11.99
33108 through 33120	12.99
33121 through 36019	12.00
36020 through 36036	3.00
36037 and subsequent	3.00

AC BUS NO. 3 (INV #2) — IFR

EFFECTIVITY	LOAD (VA)
33001 through 33100	8.00
33101 through 33107	47.00
33108 through 33120	
33121 through 36019	
36020 through 36036	
36037 and subsequent	

TOTAL AC LOAD — BASIC + IFR

EFFECTIVITY	LOAD (VA)
33001 through 33100	
Inverter No. 1	219.28
Inverter No. 2	34.64
Single inverter	252.35

TOTAL AC LOAD — BASIC + IFR

EFFECTIVITY	LOAD (VA)
33101 through 33107	
Inverter No. 1	234.36
Inverter No. 2	74.84
Single inverter	306.18
33108 through 33120	
Inverter No. 1	217.00
Inverter No. 2	27.00
Single inverter	242.00
33121 through 36019	
Inverter No. 1	217.00
Inverter No. 2	27.00
Single inverter	242.00
36020 through 36036	
Inverter No. 1	180.00
Inverter No. 2	27.00
Single inverter	205.00
36037 and subsequent	
Inverter No. 1	180.00
Inverter No. 2	44.00
Single inverter	205.00

AC BUS NO. 2 — AUXILIARY KIT

EFFECTIVITY	LOAD (VA)
33001 through 33100	17.00
33101 through 33107	17.00
33108 through 33120	17.00
33121 through 36019	17.00
36020 through 36036	17.00
36037 and subsequent	17.00

ICN-412-A-980000-B-97499-00012-A-001-01

Figure 1. AC Load Analysis Chart (Sheet 2 of 3)

AC LOAD ANALYSIS

TOTAL AC LOAD – BASIC + IFR + AUXILIARY

EFFECTIVITY	LOAD (VA)
33001 through 33100	
Inverter No. 1	235.38
Inverter No. 2	34.84
Single inverter	268.67
33101 through 33107	
Inverter No. 1	250.53
Inverter No. 2	74.84
Single inverter	322.70
33108 through 33120	
Inverter No. 1	233.00
Inverter No. 2	27.00
Single inverter	258.00

TOTAL AC LOAD – BASIC + IFR + AUXILIARY

EFFECTIVITY	LOAD (VA)
33121 through 36019	
Inverter No. 1	233.00
Inverter No. 2	27.00
Single inverter	258.00
36020 through 36036	
Inverter No. 1	196.00
Inverter No. 2	27.00
Single inverter	222.00
36037 and subsequent	
Inverter No. 1	196.00
Inverter No. 2	44.00
Single inverter	222.00

ICN-412-A-980000-B-97499-00013-A-001-01

Figure 1. AC Load Analysis Chart (Sheet 3 of 3)

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DC LOAD ANALYSIS

MAIN DC BUS NO. 1 BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
36037 and subsequent	0.85	0.85	0.85	0.85

MAIN DC BUS NO. 2 — BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
36037 and subsequent	0.85	0.85	0.85	0.85

EMERGENCY DC BUS NO. 1 — BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	12.20	12.13	12.13	12.20
33101 through 33107	1.39	1.32	1.32	1.39
33108 through 33120	14.35	14.28	14.28	14.35
33121 through 36019	14.35	14.28	14.28	14.35
36020 through 36036	14.62	14.55	14.55	14.62
36037 and subsequent	14.35	14.28	14.28	14.35

EMERGENCY DC BUS NO. 2 — BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	13.68	10.28	15.83	13.69
33101 through 33107	8.06	4.66	10.21	8.06
33108 through 33120	15.48	12.08	17.63	15.49
33121 through 36019	15.48	12.08	17.63	15.49
36020 through 36036	15.66	12.26	17.81	15.67
36037 and subsequent	16.25	12.85	18.40	16.25

ICN-412-A-980000-B-97499-00014-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 1 of 8)

DC LOAD ANALYSIS

ESSENTIAL DC BUS NO. 1 — BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	9.69	9.50	9.52	9.67
33101 through 33107	19.17	18.99	19.01	19.15
33108 through 33120	10.89	10.70	10.72	10.87
33121 through 36019	10.89	10.70	10.72	10.87
36020 through 36036	10.89	10.70	10.72	10.87
36037 and subsequent	10.89	10.70	10.72	10.87

ESSENTIAL DC BUS NO. 2 — BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	14.87	14.70	14.71	14.87
33101 through 33107	18.48	18.31	18.32	18.48
33108 through 33120	17.03	16.85	16.86	17.02
33121 through 36019	17.03	16.85	16.86	17.02
36020 through 36036	17.03	16.85	16.86	17.02
36037 and subsequent	17.03	16.85	16.86	17.02

NON-ESSENTIAL DC BUS NO. 1 — BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	9.82	9.82	9.82	17.01
33101 through 33107	11.32	11.32	11.32	18.51
33108 through 33120	9.81	9.81	9.81	16.99
33121 through 36019	9.81	9.81	9.81	16.99
36020 through 36036	9.81	9.81	9.81	16.99
36037 and subsequent	9.81	9.81	9.81	16.99

ICN-412-A-980000-B-97499-00015-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 2 of 8)

DC LOAD ANALYSIS

NON-ESSENTIAL DC BUS NO. 2 – BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	14.26	14.05	14.05	14.25
33101 through 33107	14.26	14.05	14.05	14.25
33108 through 33120	14.26	14.05	14.05	14.25
33121 through 36019	14.26	14.05	14.05	14.25
36020 through 36036	14.26	14.05	14.05	14.25
36037 and subsequent	14.26	14.05	14.05	14.25

BATTERY BUS – BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	54.37	29.19	14.47	8.96
33101 through 33107	54.37	29.19	14.47	8.96
33108 through 33120	54.36	29.18	14.46	8.95
33121 through 36019	57.18	30.68	15.18	9.38
36020 through 36036	57.18	30.68	15.18	9.38
36037 and subsequent	57.18	30.68	15.18	9.38

TOTAL ALL DC BUSSES – BASIC

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	128.90	99.69	90.54	90.64
33101 through 33107	127.05	97.84	88.69	88.80
33108 through 33120	136.18	106.97	97.82	97.93
33121 through 36019	139.01	108.47	98.55	98.36
36020 through 36036	139.46	108.92	99.00	98.81
36037 and subsequent	141.48	110.94	101.01	100.83

ICN-412-A-980000-B-97499-00016-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 3 of 8)

DC LOAD ANALYSIS

EMERGENCY DC BUS NO. 1 – IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	4.89	4.89	4.89	4.89
33101 through 33107				
33108 through 33120	1.42	1.42	1.42	1.42
33121 through 36019	1.42	1.42	1.42	1.42
36020 through 36036	1.16	1.16	1.16	1.16
36037 and subsequent	1.42	1.42	1.42	1.42

EMERGENCY DC BUS NO. 2 – IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	4.69	4.69	4.69	4.69
33101 through 33107	4.07	3.98	3.94	3.92
33108 through 33120	3.90	3.90	3.90	3.90
33121 through 36019	3.90	3.90	3.90	3.90
36020 through 36036	4.04	4.04	4.04	4.04
36037 and subsequent	3.90	3.90	3.90	3.90

ESSENTIAL DC BUS NO. 1 – IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	7.60	7.60	7.60	7.60
33101 through 33107	9.78	9.78	9.78	9.78
33108 through 33120	7.60	7.60	7.60	7.60
33121 through 36019	7.60	7.60	7.60	7.60
36020 through 36036	7.60	7.60	7.60	7.60
36037 and subsequent	7.60	7.60	7.60	7.60

ICN-412-A-980000-B-97499-00017-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 4 of 8)

DC LOAD ANALYSIS

ESSENTIAL DC BUS NO. 2 — IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	1.60	1.60	1.60	1.60
33101 through 33107	2.16	2.16	2.16	2.16
33108 through 33120				
33121 through 36019				
36020 through 36036				
36037 and subsequent				

NON-ESSENTIAL DC BUS NO. 1 — IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	3.45	3.45	3.45	3.45
33101 through 33107	2.48	2.48	2.48	2.48
33108 through 33120	3.88	3.88	3.88	3.88
33121 through 36019	3.88	3.88	3.88	3.88
36020 through 36036	3.88	3.88	3.88	3.88
36037 and subsequent	3.88	3.88	3.88	3.88

NON-ESSENTIAL DC BUS NO. 2 — IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	0.50	0.50	0.50	0.50
33101 through 33107	2.59	2.59	2.59	2.59
33108 through 33120	1.49	1.49	1.49	1.49
33121 through 36019	1.49	1.49	1.49	1.49
36020 through 36036	1.49	1.49	1.49	1.49
36037 and subsequent	1.49	1.49	1.49	1.49

ICN-412-A-980000-B-97499-00018-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 5 of 8)

DC LOAD ANALYSIS

TOTAL ALL DC BUSSES BASIC + IFR

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	151.63	122.41	113.26	113.37
33101 through 33107	148.14	118.83	109.64	109.74
33108 through 33120	154.48	125.27	116.12	116.22
33121 through 36019	157.31	126.77	116.84	116.66
36020 through 36036	157.63	127.09	117.17	116.99
36037 and subsequent	159.77	129.23	119.31	119.12

EMERGENCY DC BUS NO. 1 — AUXILIARY KIT

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100	0.78	0.78	0.85	0.78
33101 through 33107			0.07	
33108 through 33120	0.78	0.78	0.85	0.78
33121 through 36019	0.78	0.78	0.85	0.78
36020 through 36036	0.78	0.78	0.85	0.78
36037 and subsequent	0.78	0.78	0.85	0.78

EMERGENCY DC BUS NO. 2 — AUXILIARY KIT

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.
33001 through 33100				
33101 through 33107				
33108 through 33120				
33121 through 36019				
36020 through 36036				
36037 and subsequent				

ICN-412-A-980000-B-97499-00019-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 6 of 8)

DC LOAD ANALYSIS

EMERGENCY DC BUS 1 — AUXILIARY KIT

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.	AMPS PER UNIT
33001 and subsequent					
CARGO HOOK KIT			0.07		10.25
INVERTER SYSTEM	0.78	0.78	0.78	0.78	
ROTOR BRAKE KIT					0.040

ESSENTIAL DC BUS NO. 1 — AUXILIARY KIT

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.	AMPS PER UNIT
33001 and subsequent					
HEATING	0.80	0.08	0.08	0.08	0.040
RADIO NAV/COMM	1.28	1.28	1.28	1.28	0.779
RADAR	1.60	1.60	1.60	1.60	1.60
33101 through 33107					
INVERTER SYSTEM	0.78	0.78	0.78	0.78	

NON-ESSENTIAL DC BUS NO. 1 — AUXILIARY KIT

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.	AMPS PER UNIT
33001 and subsequent					
HEATING	71.40	71.40	71.40	71.40	35.70

ICN-412-A-980000-B-97499-00020-A-001-01

Figure 1. DC Load Analysis Chart (Sheet 7 of 8)

DC LOAD ANALYSIS

NON-ESSENTIAL DC BUS NO. 2 – AUXILIARY KIT

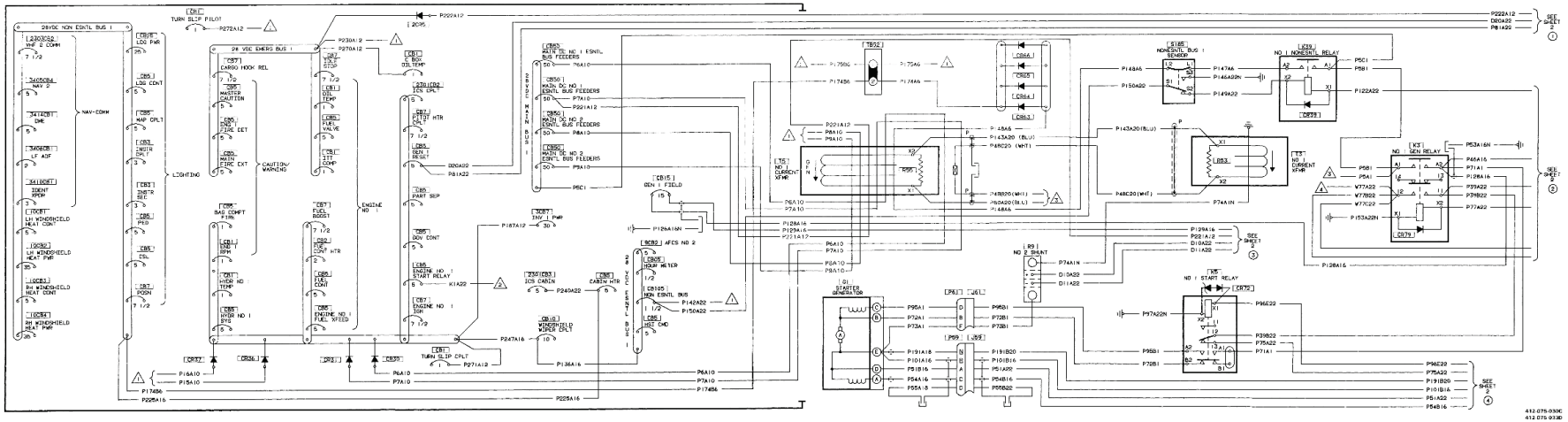
EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.	AMPS PER UNIT
33001 and subsequent					
FLIGHT CONTROLS	0.31	0.30	0.30	0.30	1.084
HEATING	2.01	2.00	2.00	2.00	6.600
RESCUE HOIST KIT			50.81		63.410
33108 and subsequent					
RESCUE HOIST KIT			62.10		143.70

TOTAL ALL DC BUSSES – BASIC + IFR + AUXILIARY

EFFECTIVITY	START AND WARMUP 15 MIN.	TAKEOFF 15 MIN.	CRUISE 15 MIN.	LAND 15 MIN.	AMPS PER UNIT
33001 through 33100	229.08	199.86	241.58	190.81	
33101 through 33107	225.60	196.28	237.96	187.18	
33108 through 33120	231.94	202.71	255.73	193.67	
33121 through 36019	234.76	204.21	256.46	194.10	
36020 through 36036	235.09	204.54	256.78	194.43	
36037 and subsequent	237.23	206.68	258.92	196.57	

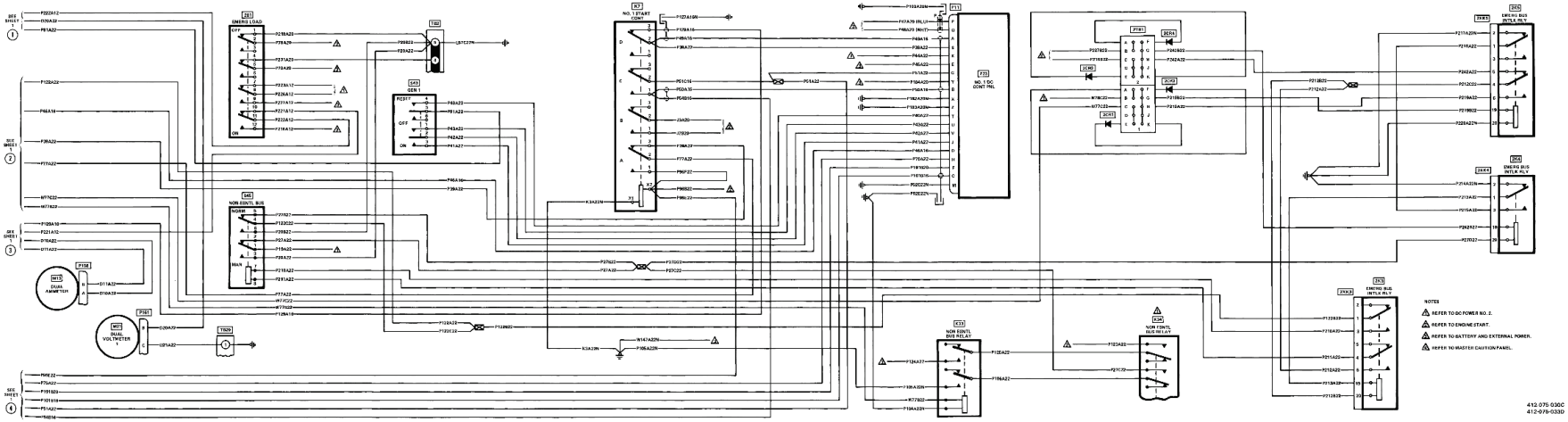
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Figure 1. DC Load Analysis Chart (Sheet 8 of 8)



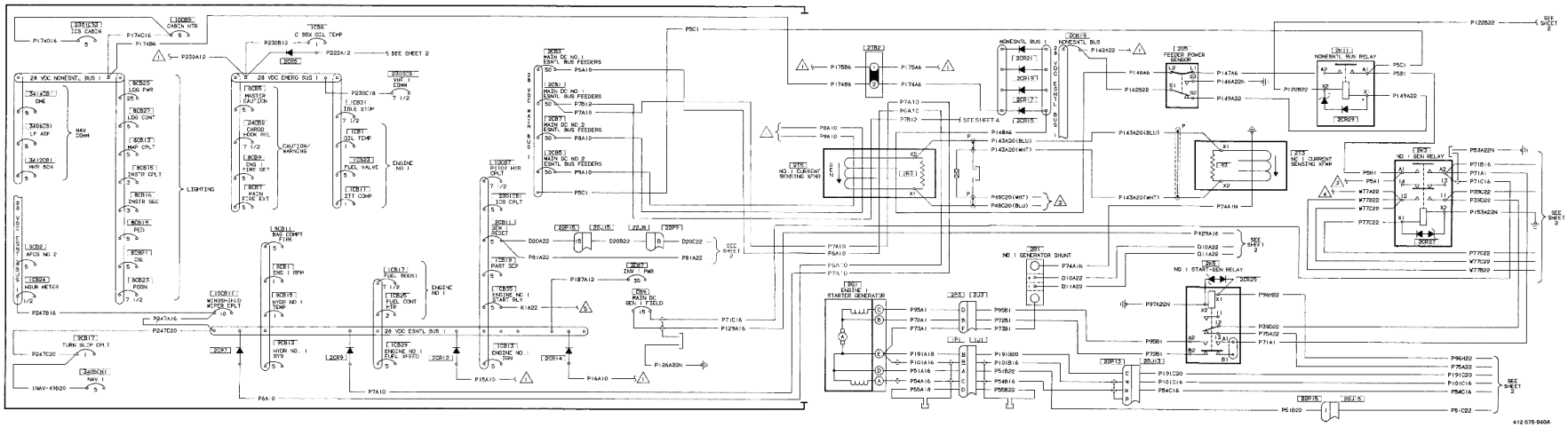
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Figure 1. DC Power No. 1 (Sheet 1 of 2)



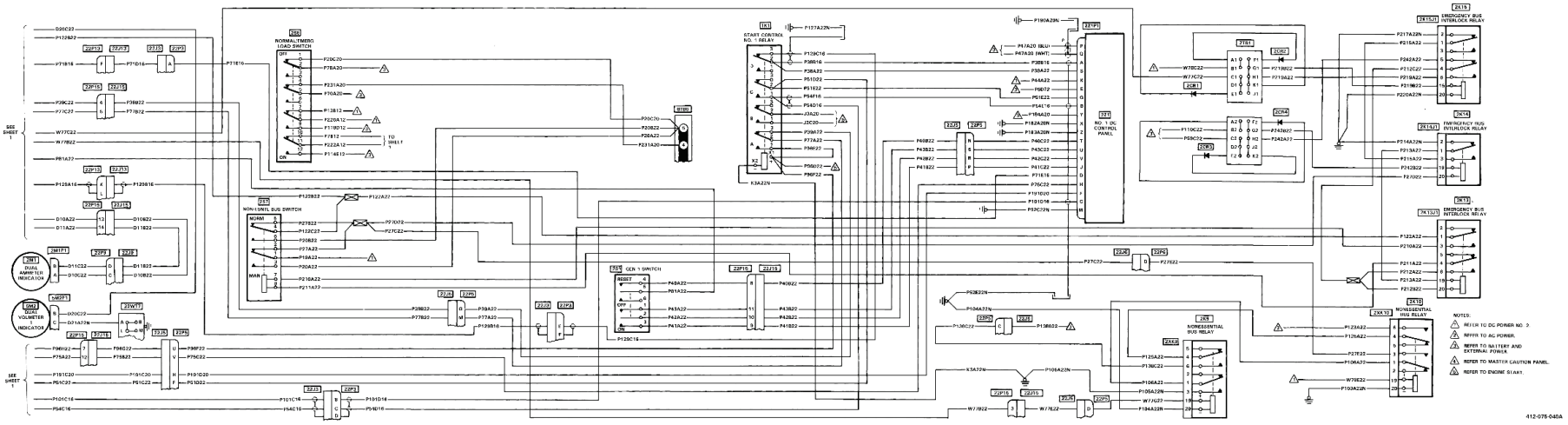
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Figure 1. DC Power No. 1 (Sheet 2 of 2)



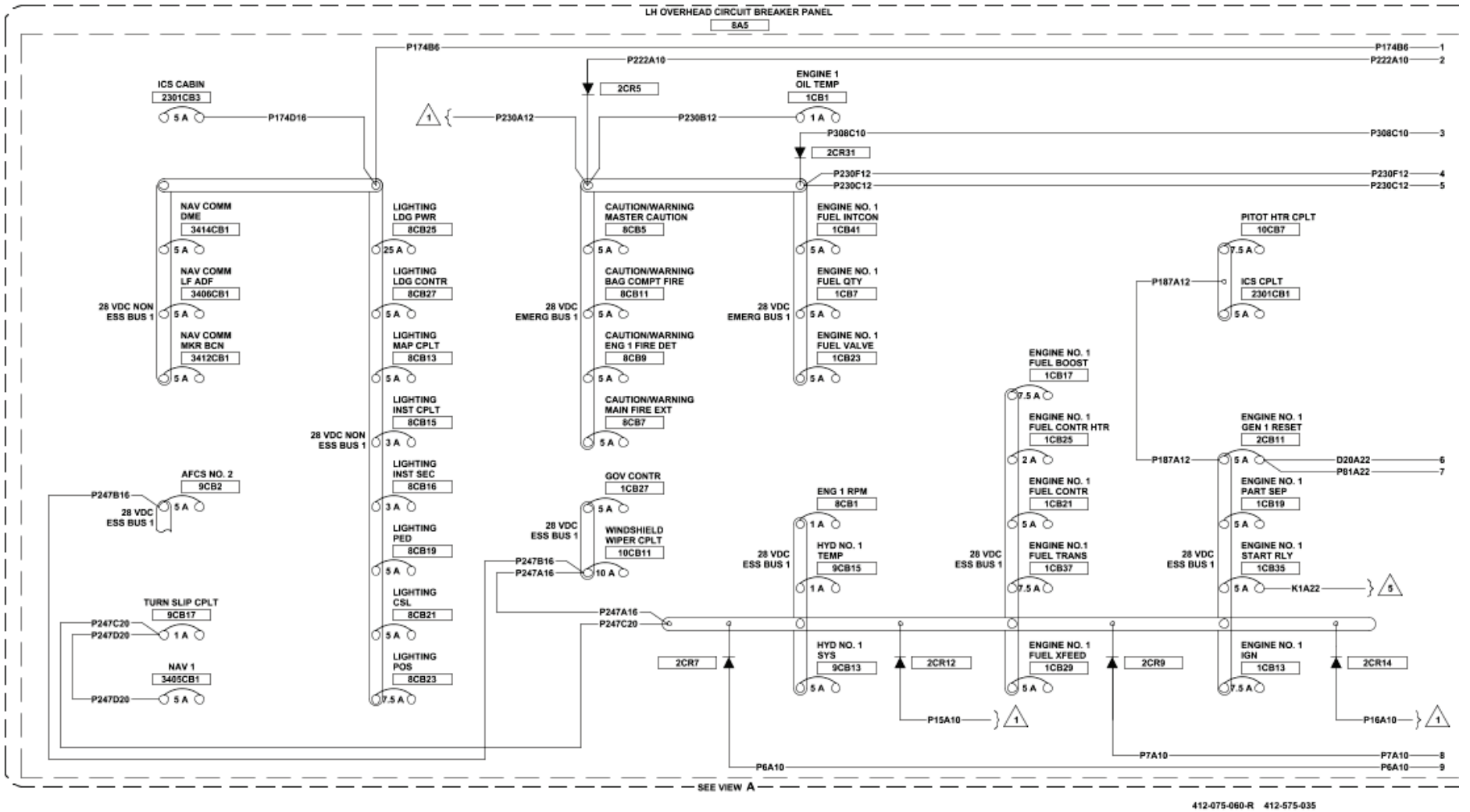
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Figure 1. DC Power No. 1 (Sheet 1 of 2)



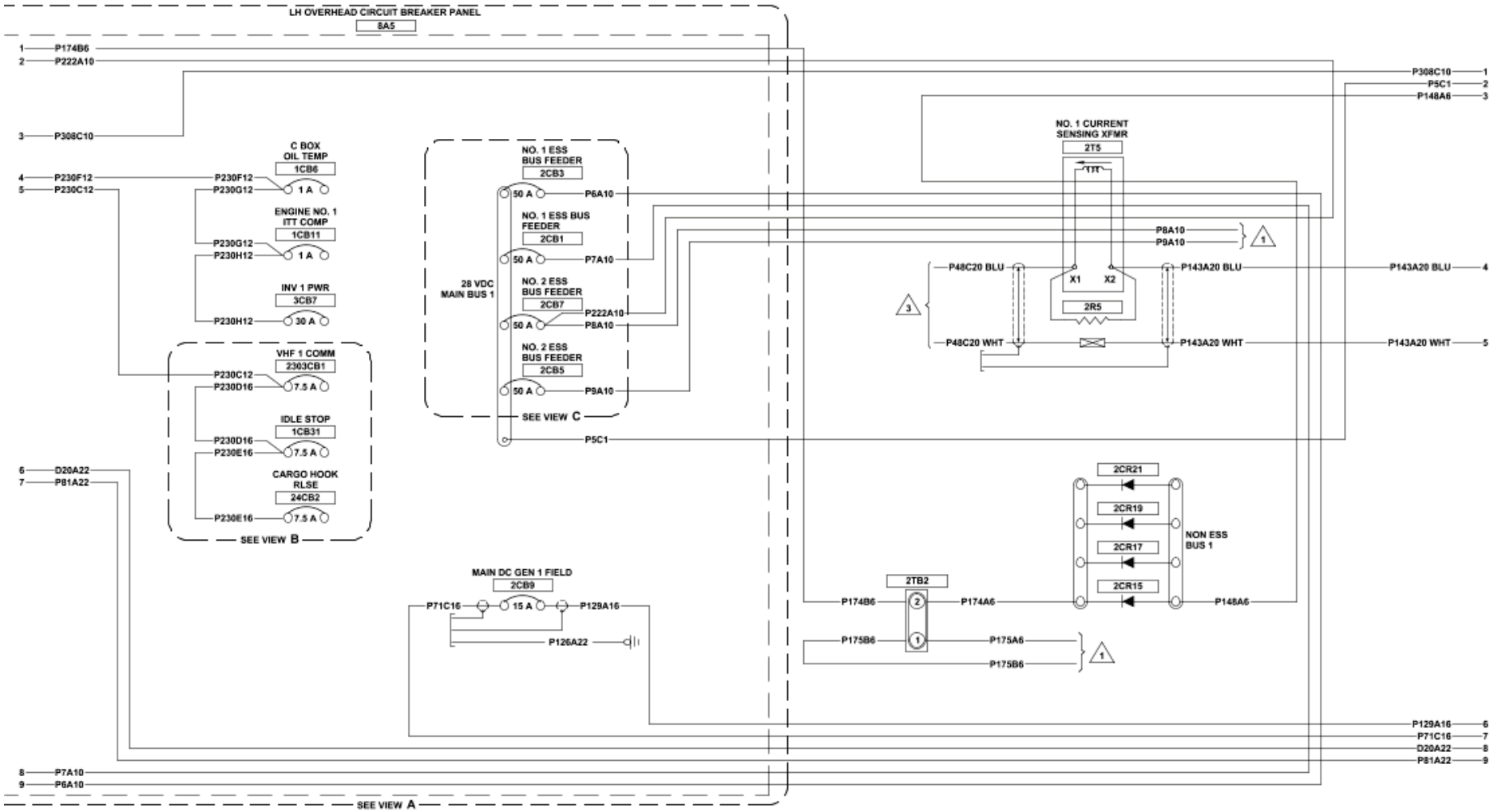
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Figure 1. DC Power No. 1 (Sheet 2 of 2)



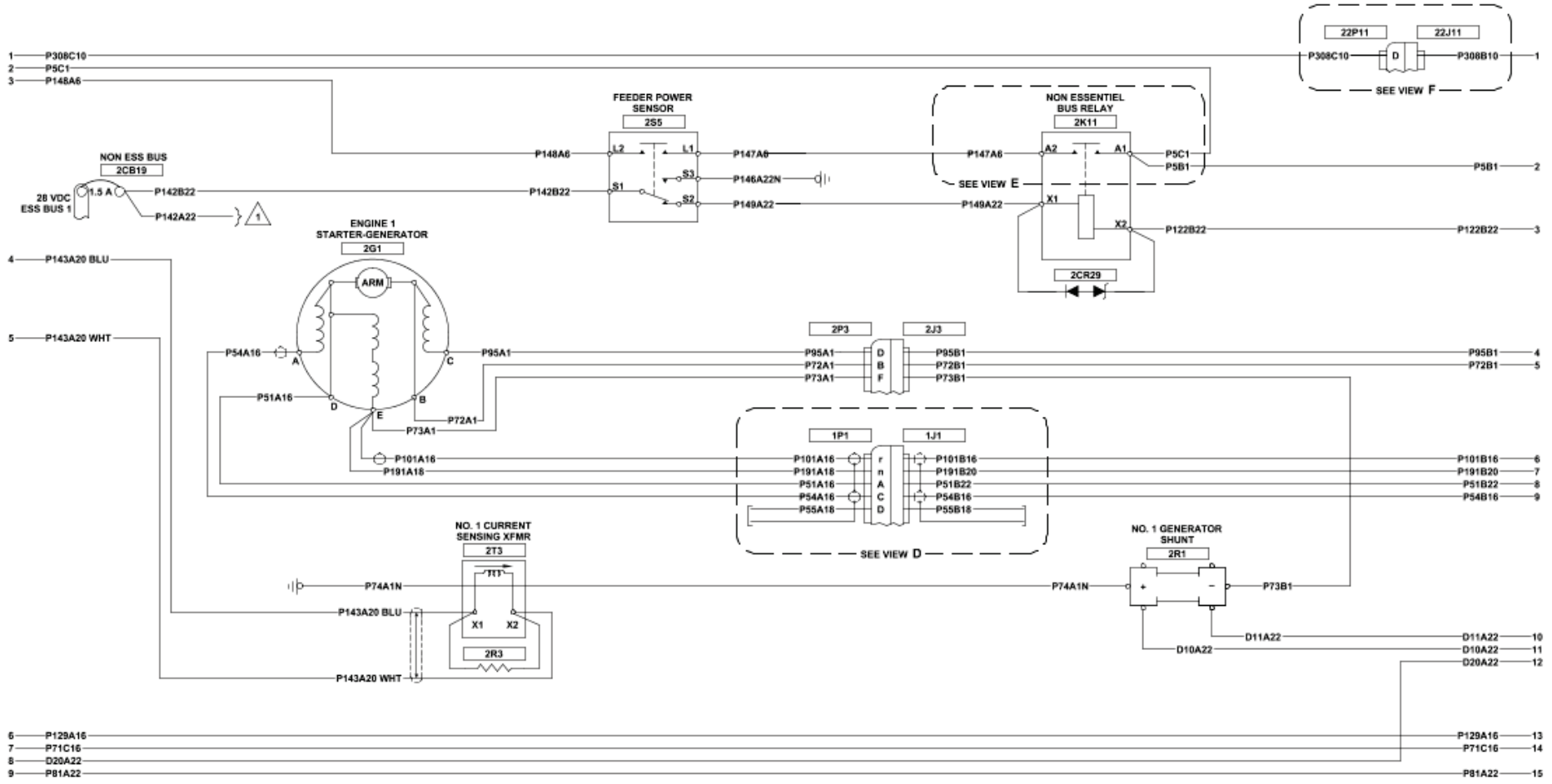
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Figure 1. DC Power No. 1 (Sheet 1 of 9)



ICN-412-A-980000-B-97499-00027-A-001-01

Figure 1. DC Power No. 1 (Sheet 2 of 9)



ICN-412-A-980000-B-97499-00028-A-001-01

Figure 1. DC Power No. 1 (Sheet 3 of 9)

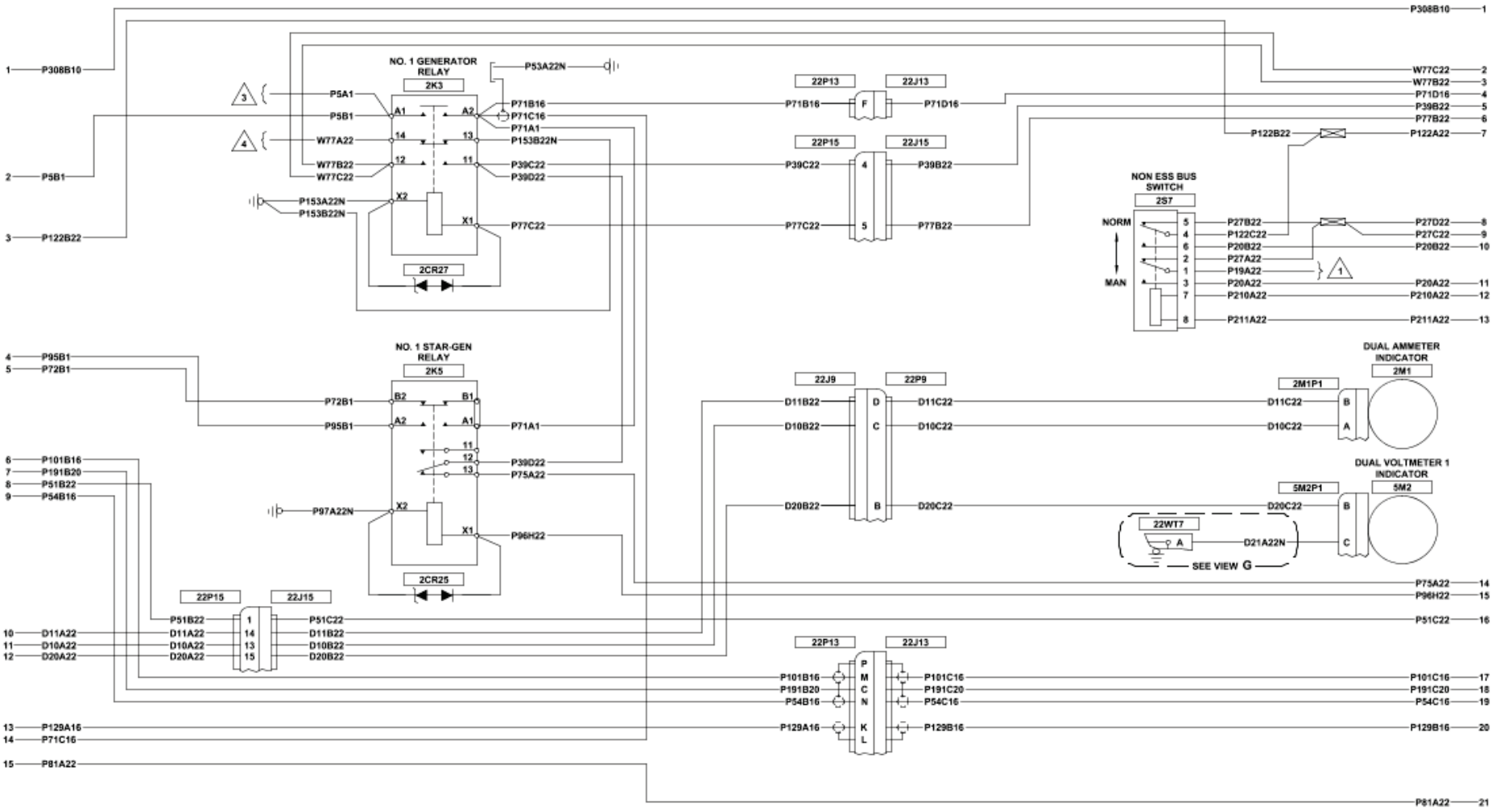
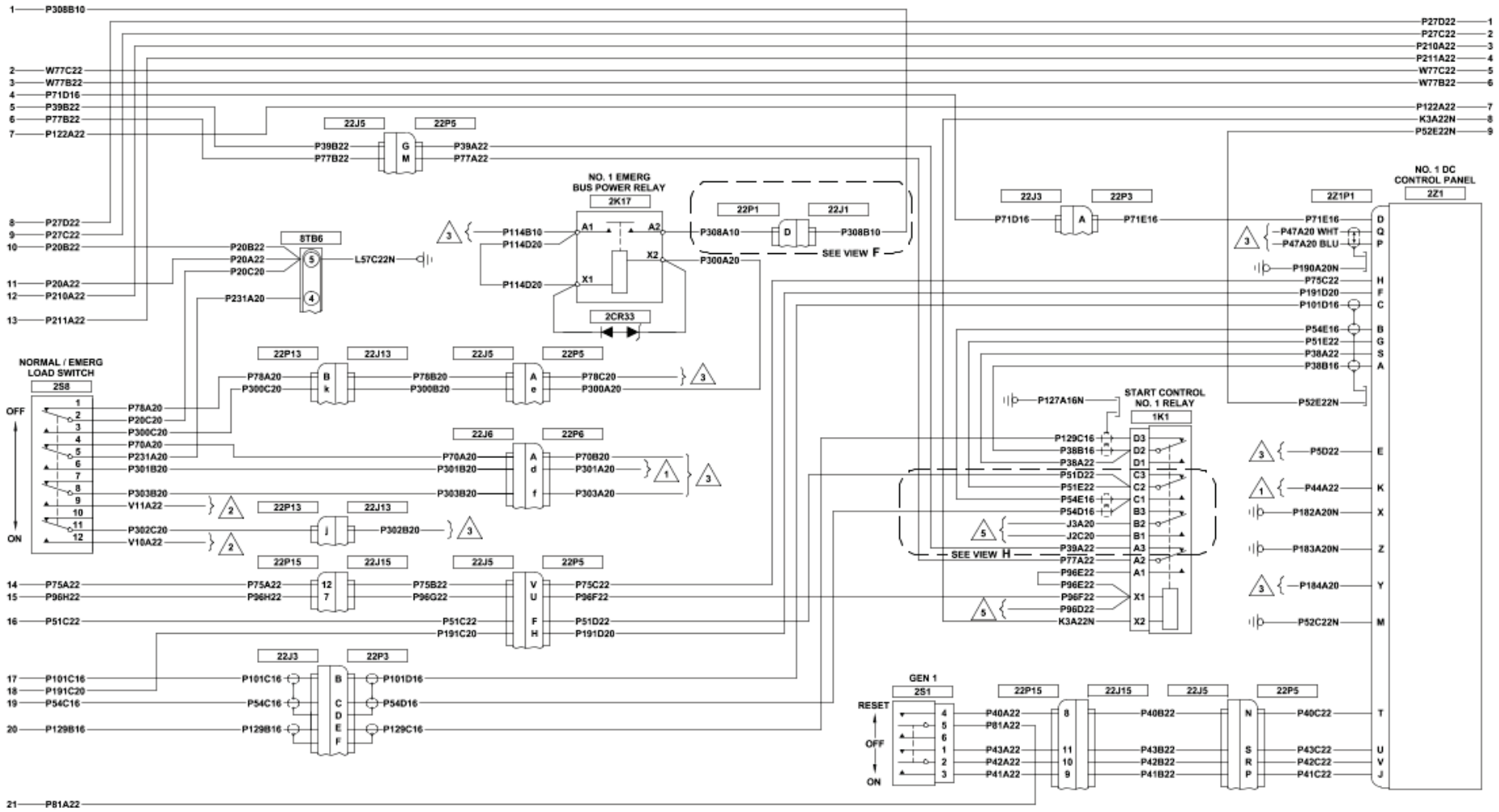
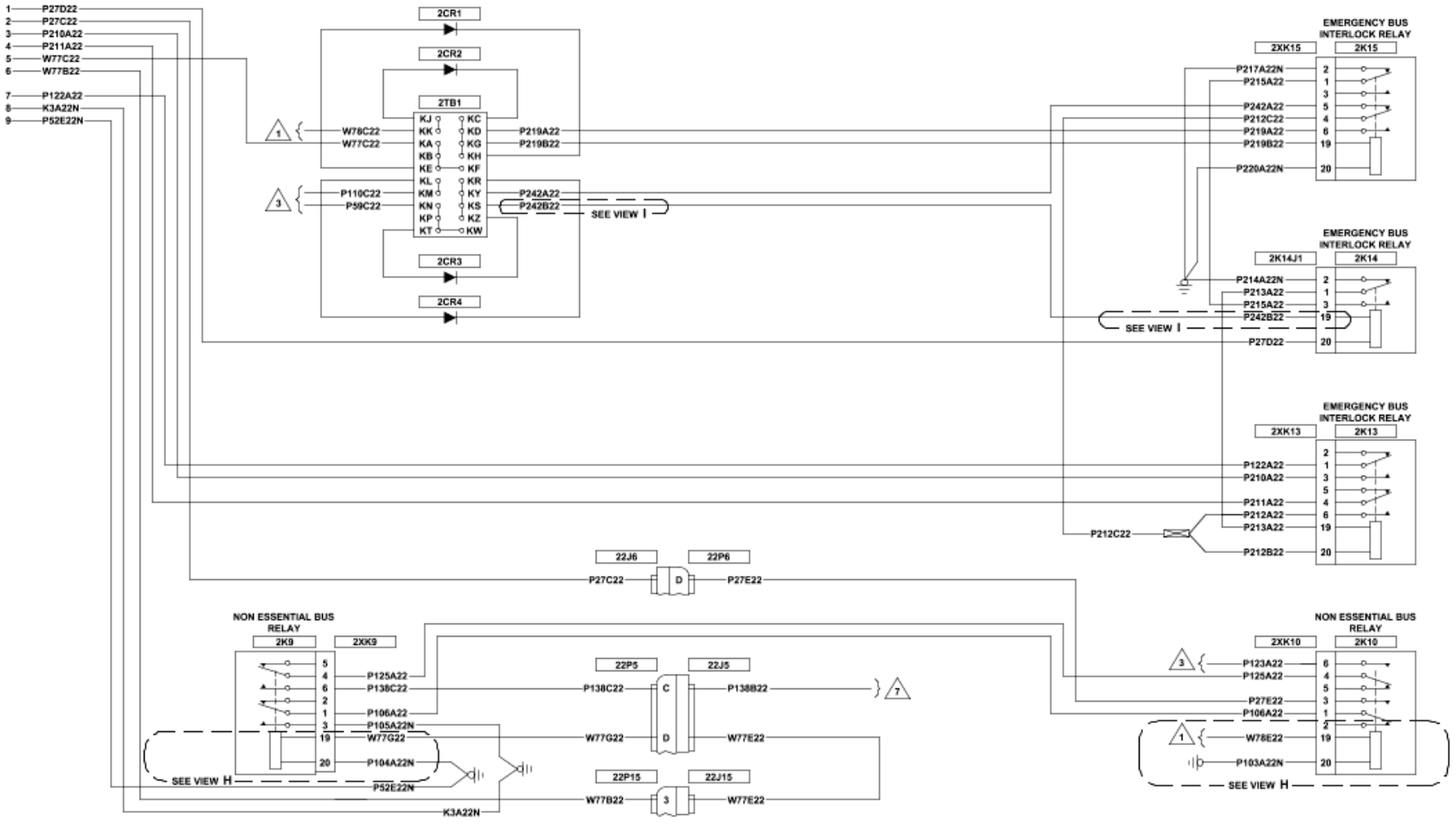


Figure 1. DC Power No. 1 (Sheet 4 of 9)



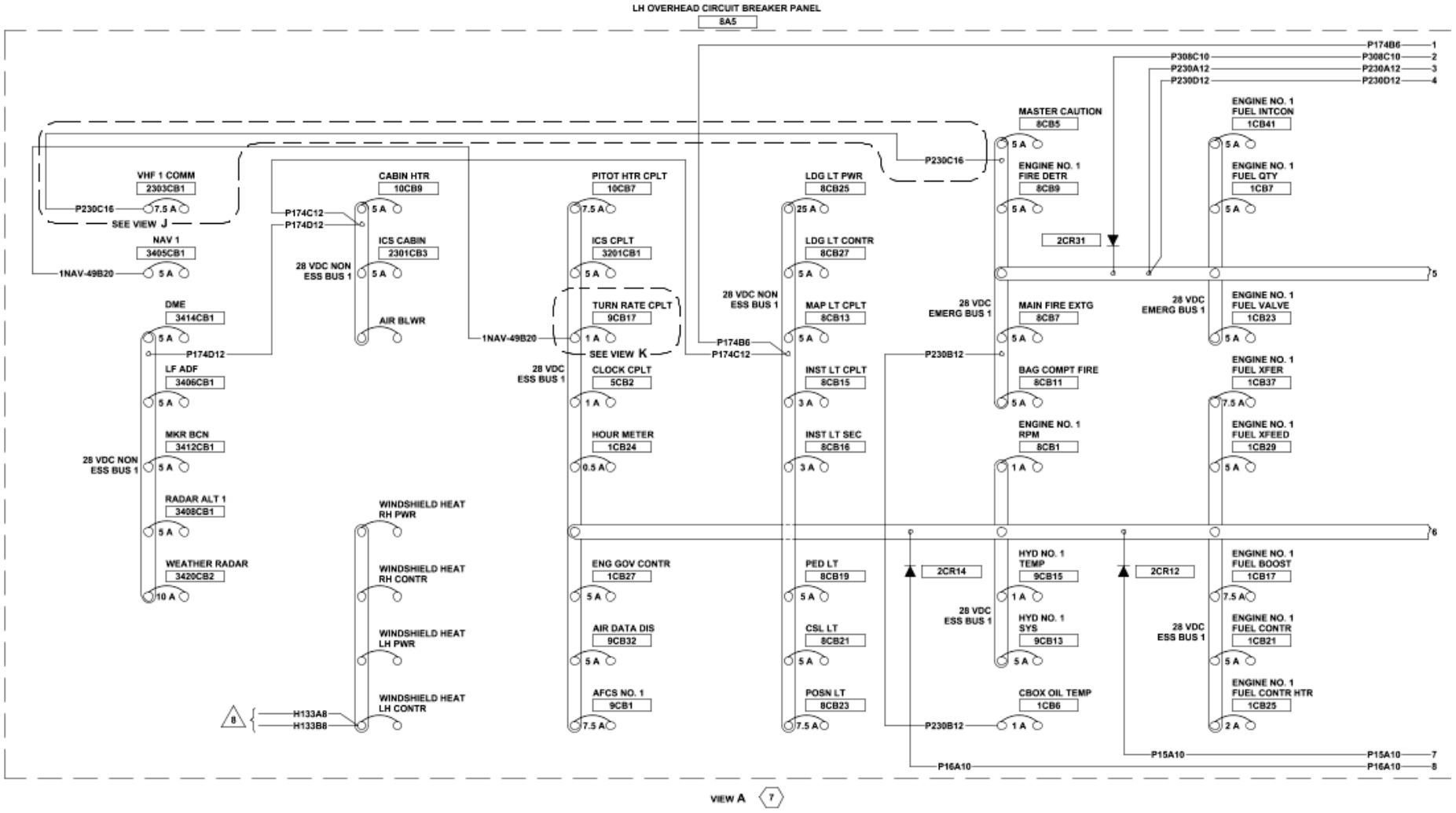
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Figure 1. DC Power No. 1 (Sheet 5 of 9)



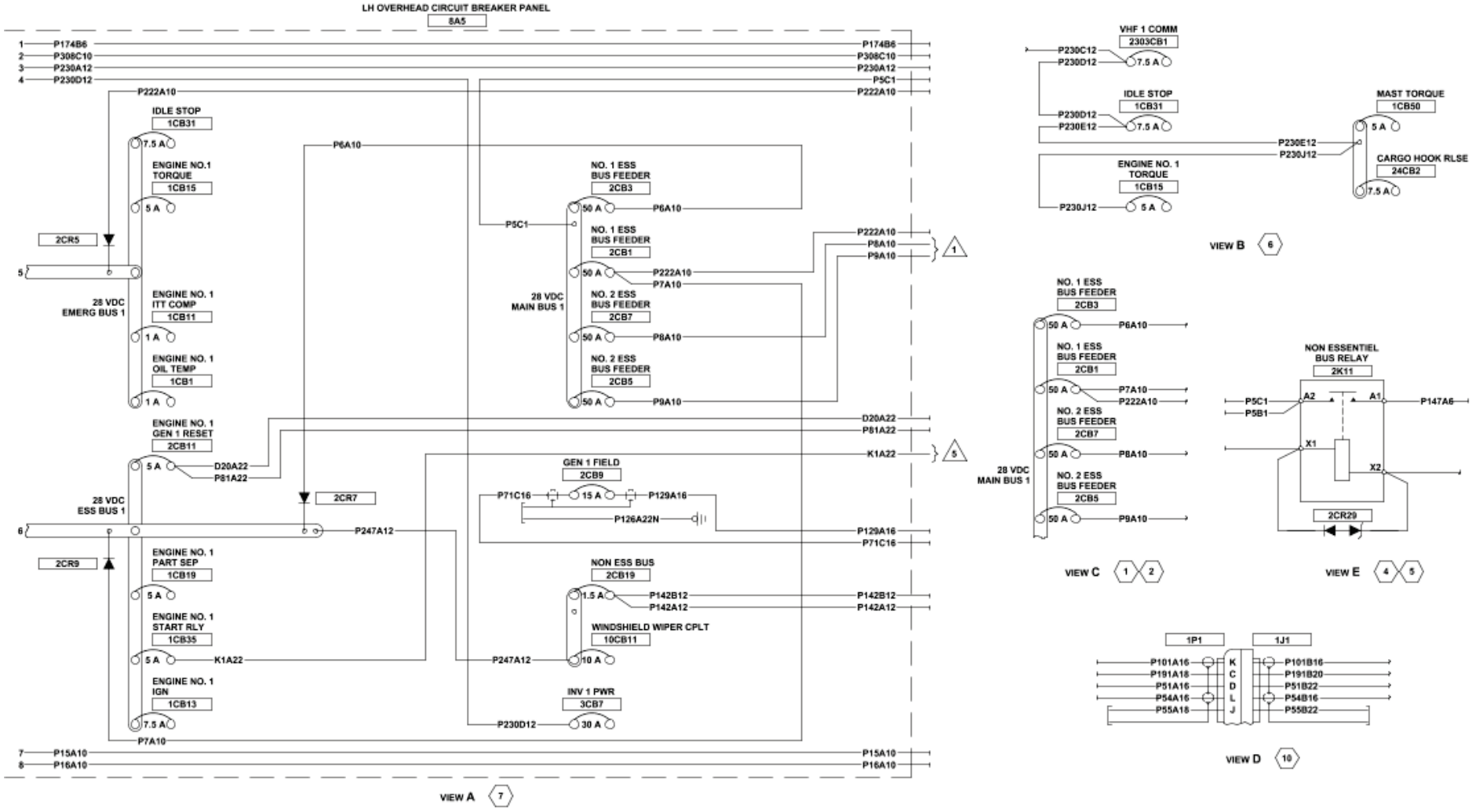
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Figure 1. DC Power No. 1 (Sheet 6 of 9)



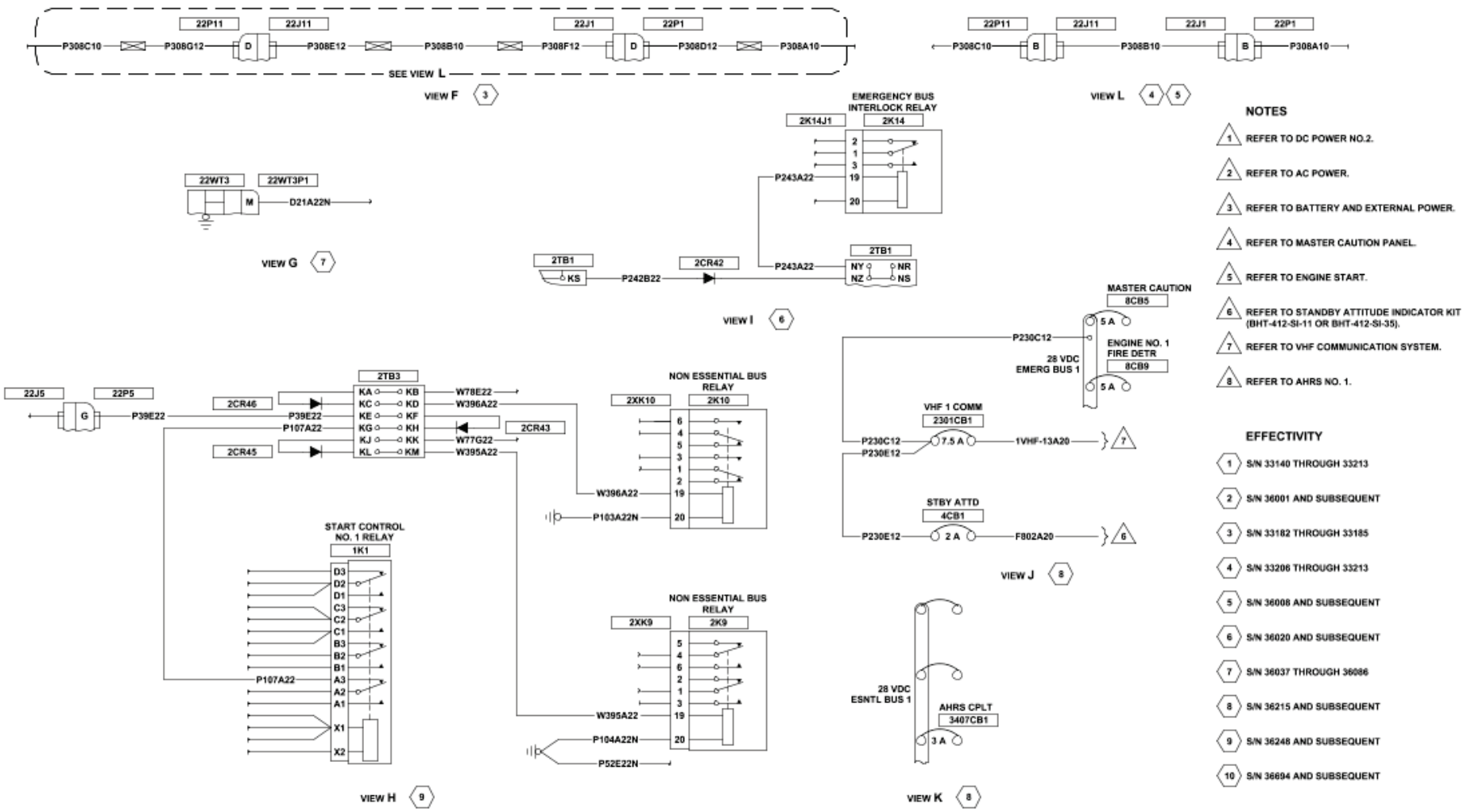
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Figure 1. DC Power No. 1 (Sheet 7 of 9)



ICN-412-A-980000-B-97499-00033-A-001-01

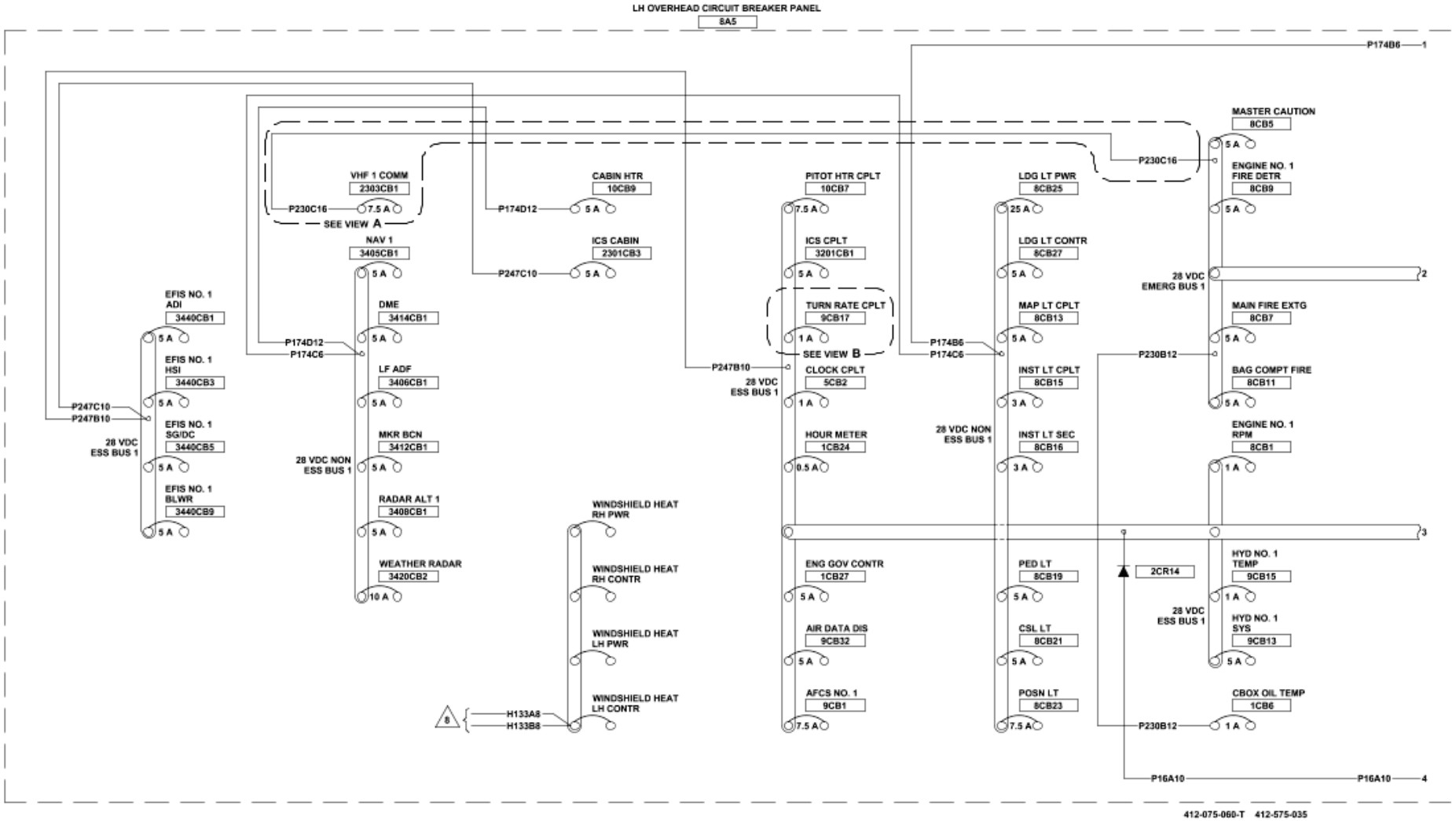
Figure 1. DC Power No. 1 (Sheet 8 of 9)



ICN-412-A-980000-B-97499-00034-A-001-01

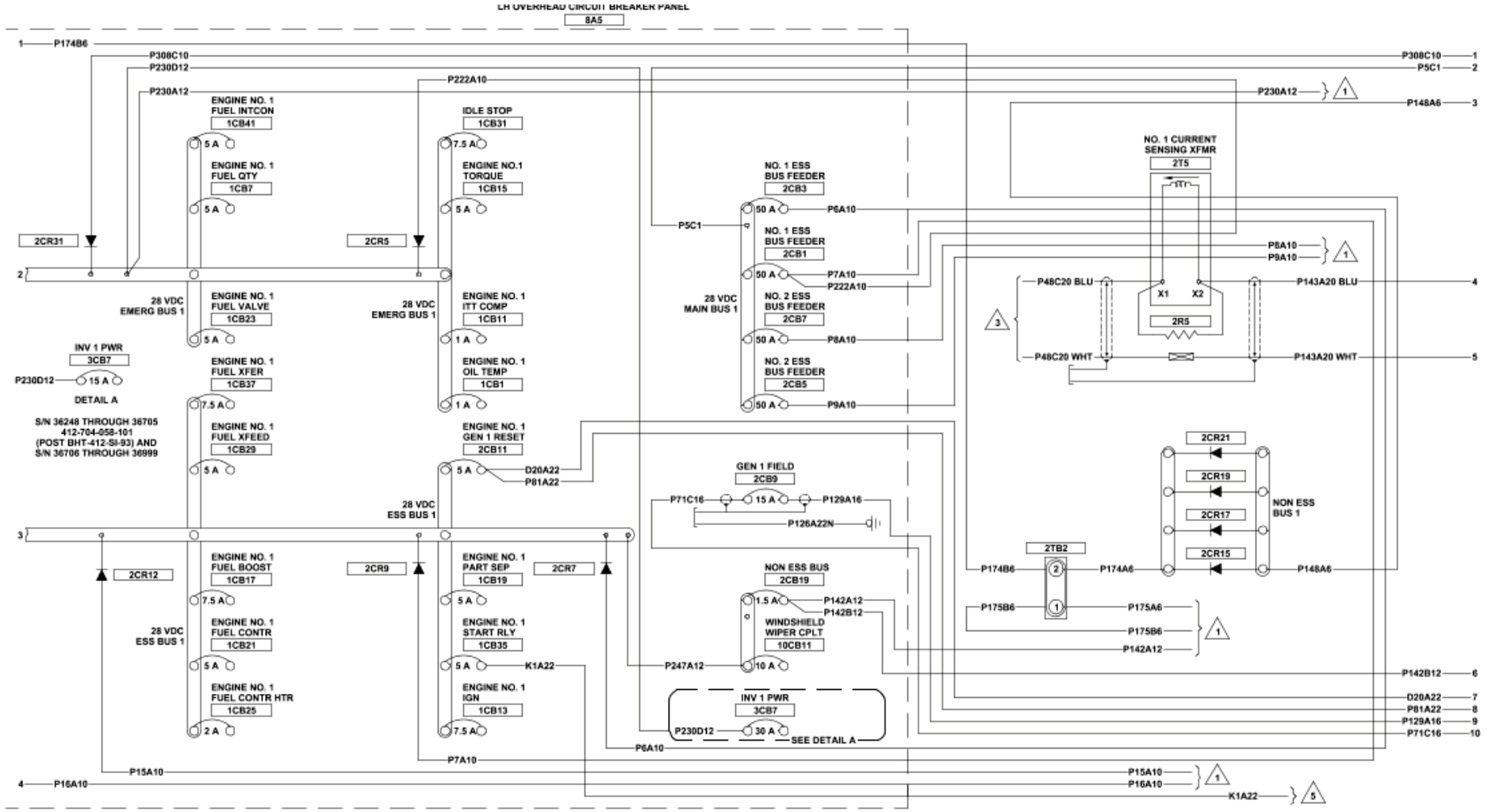
Figure 1. DC Power No. 1 (Sheet 9 of 9)

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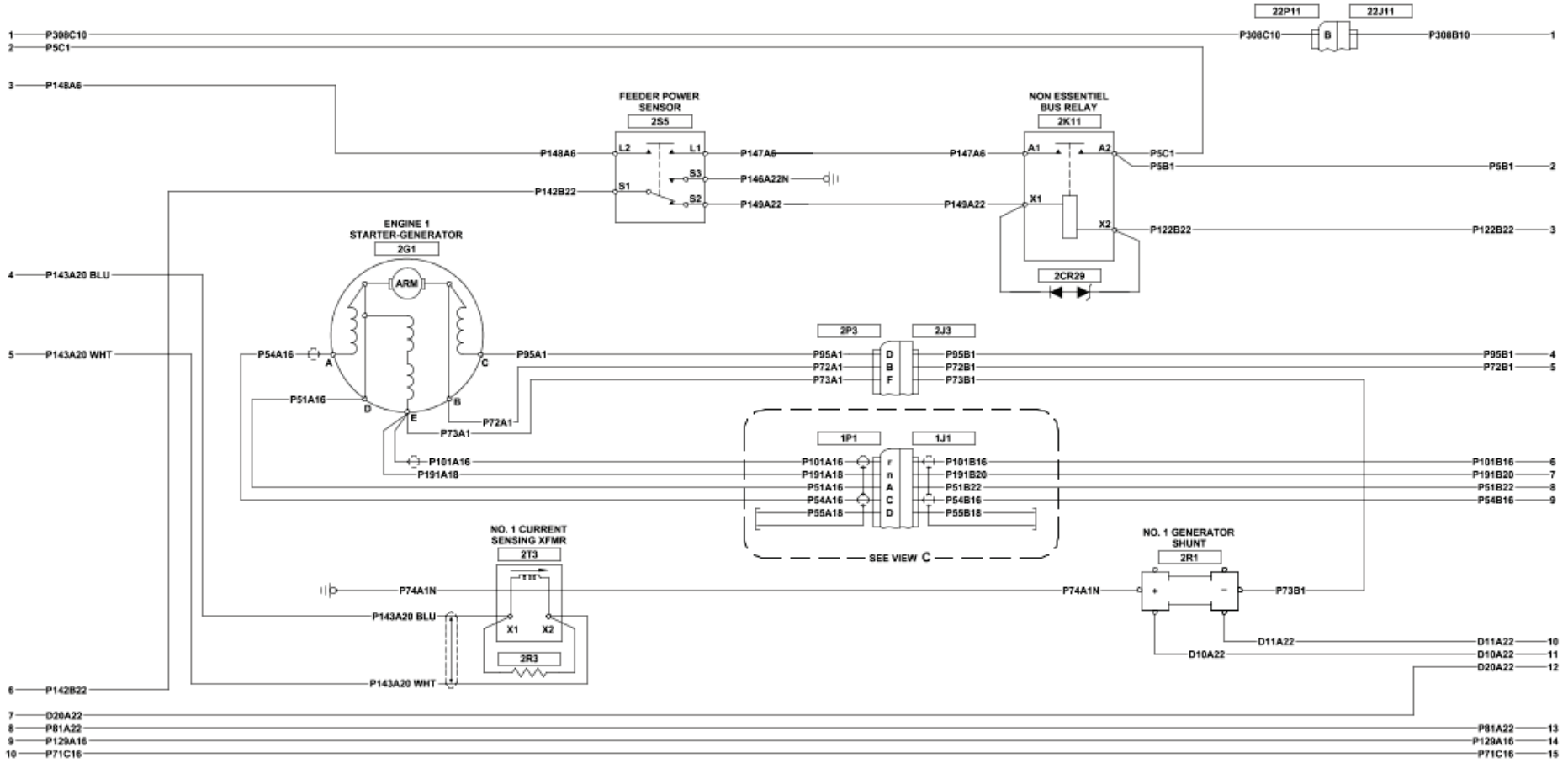
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Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 1 of 7)



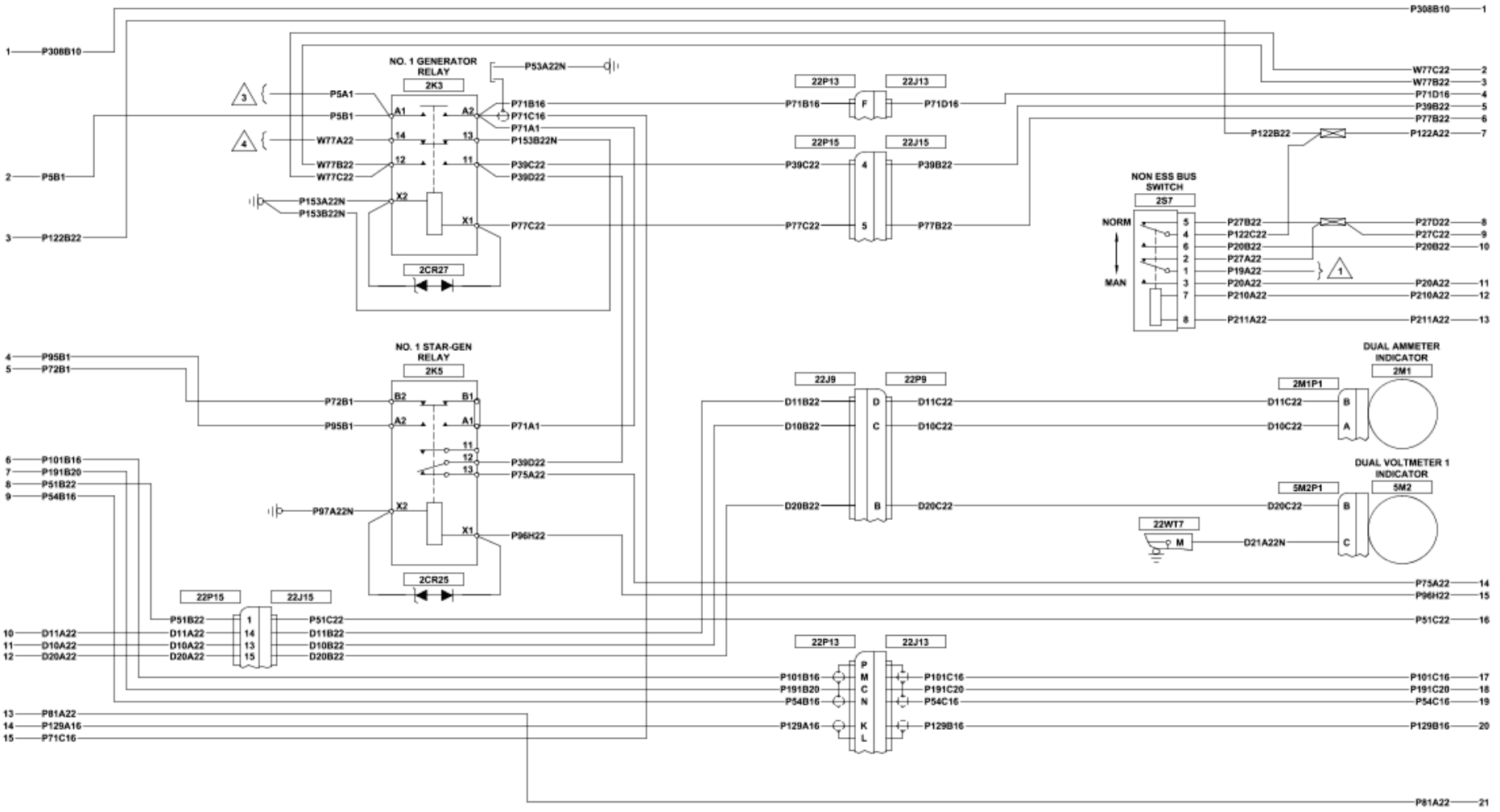
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Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 2 of 7)



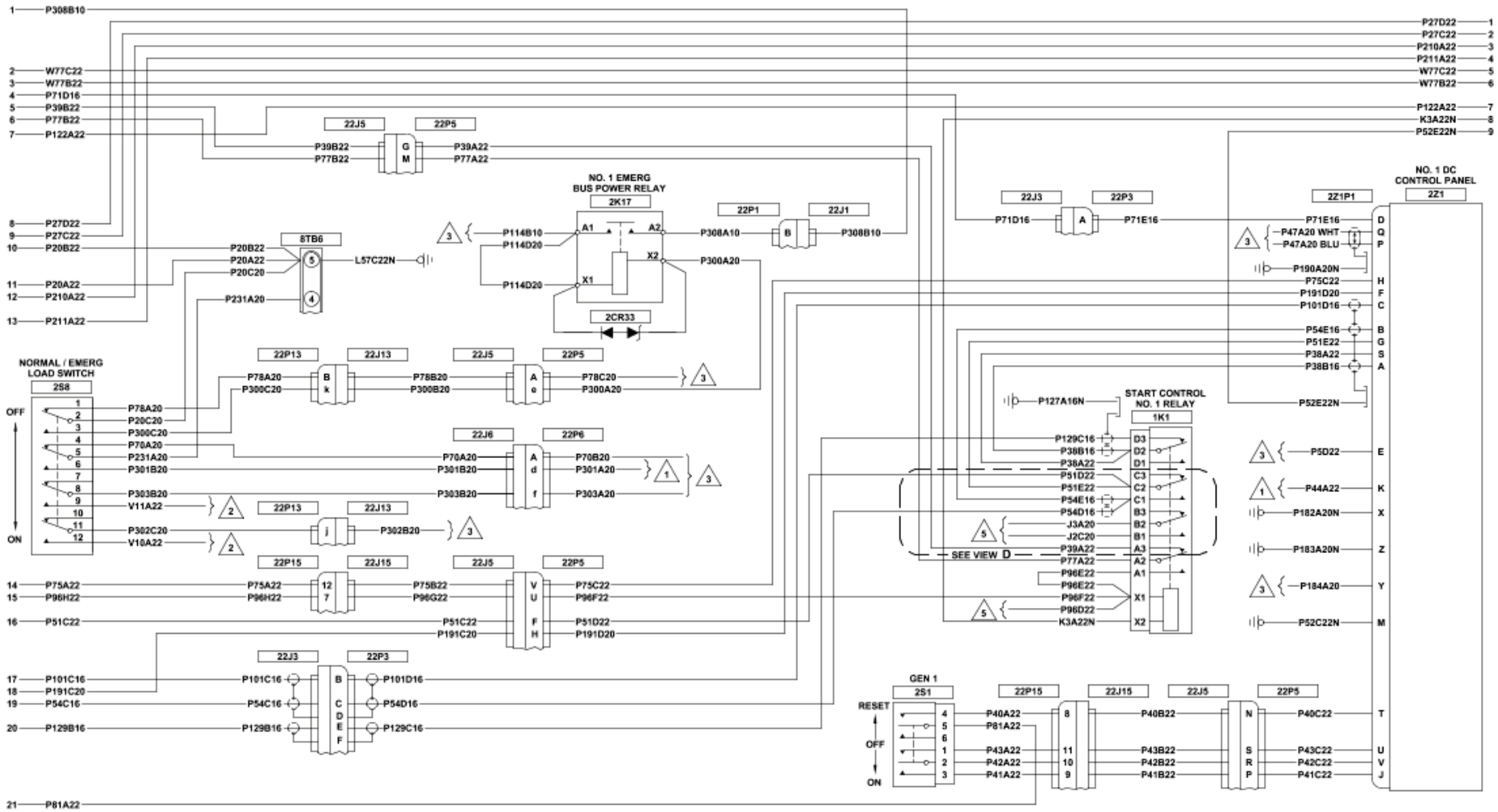
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Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 3 of 7)



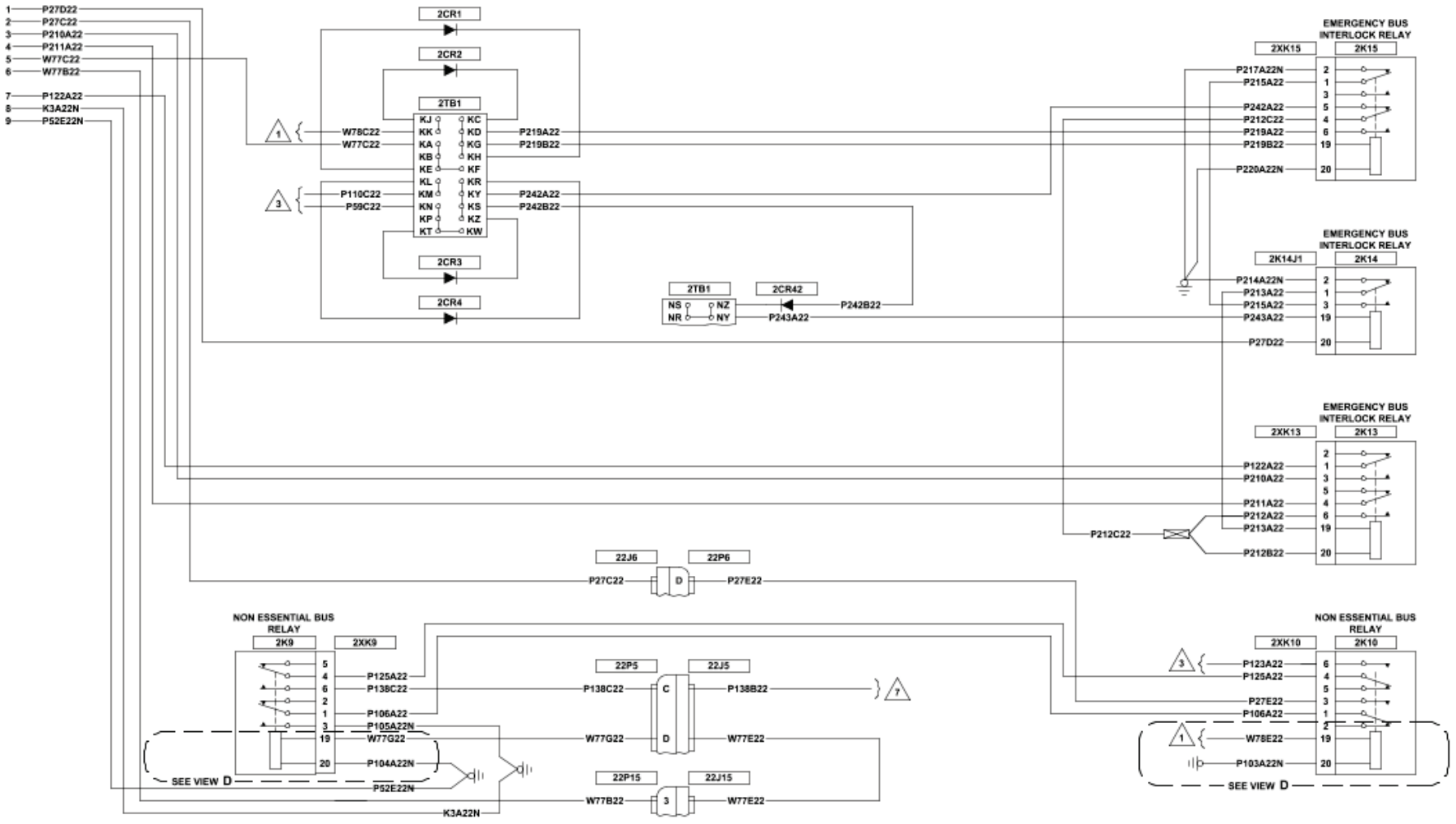
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Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 4 of 7)



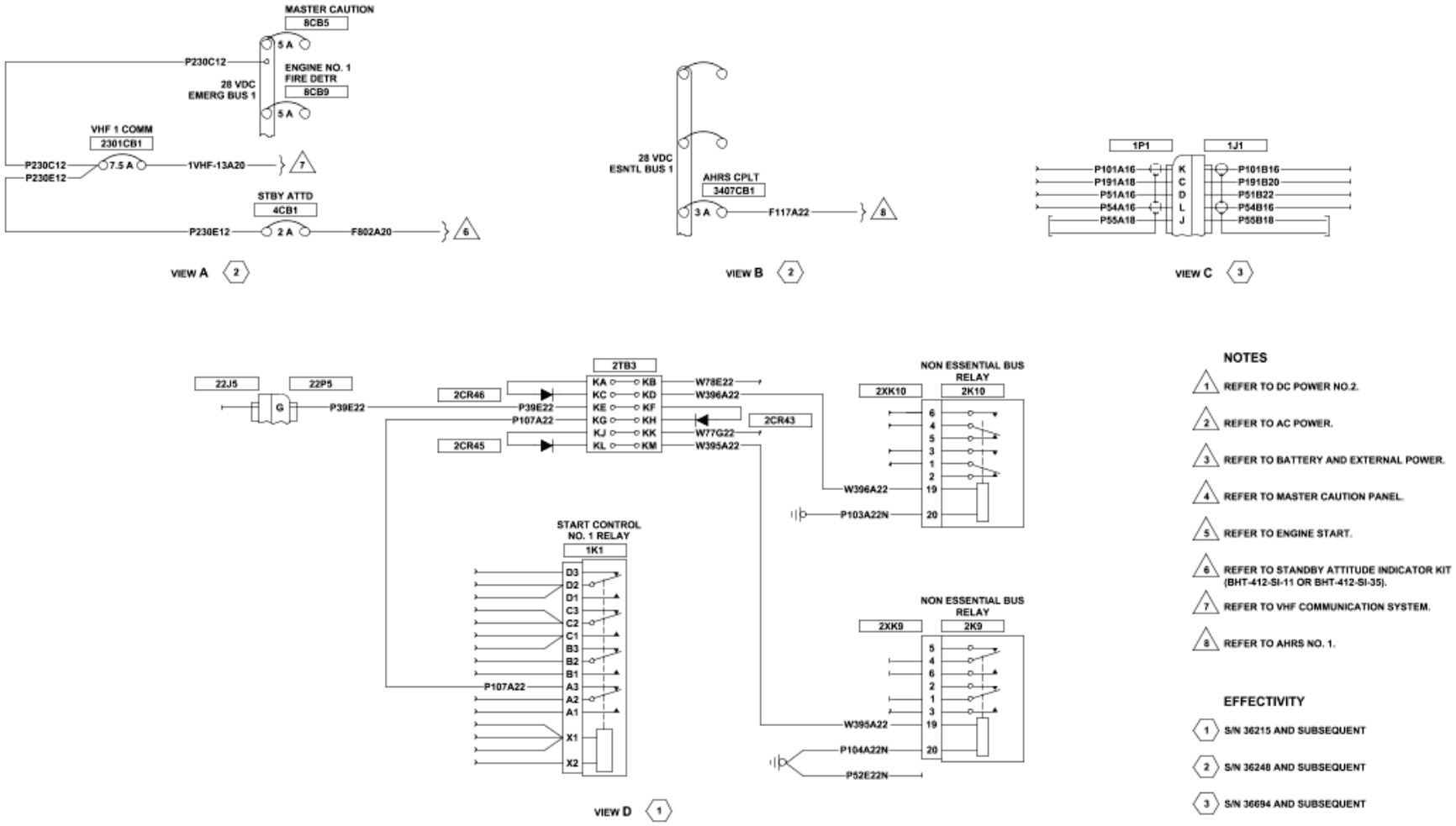
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Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 5 of 7)



ICN-412-A-980000-B-97499-00040-A-001-01

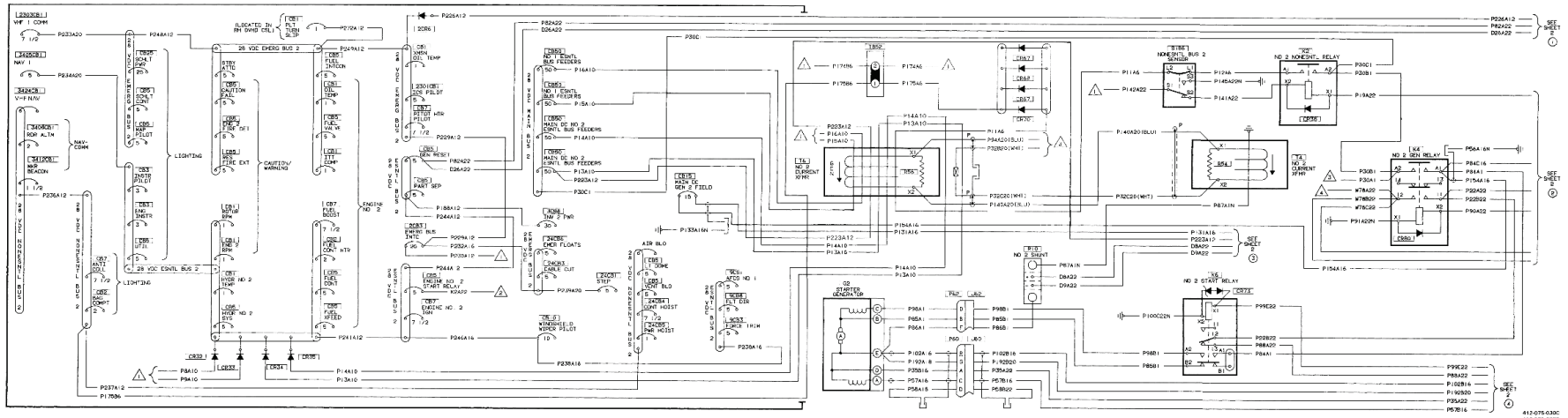
Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 6 of 7)



ICN-412-A-980000-B-97499-00041-A-001-01

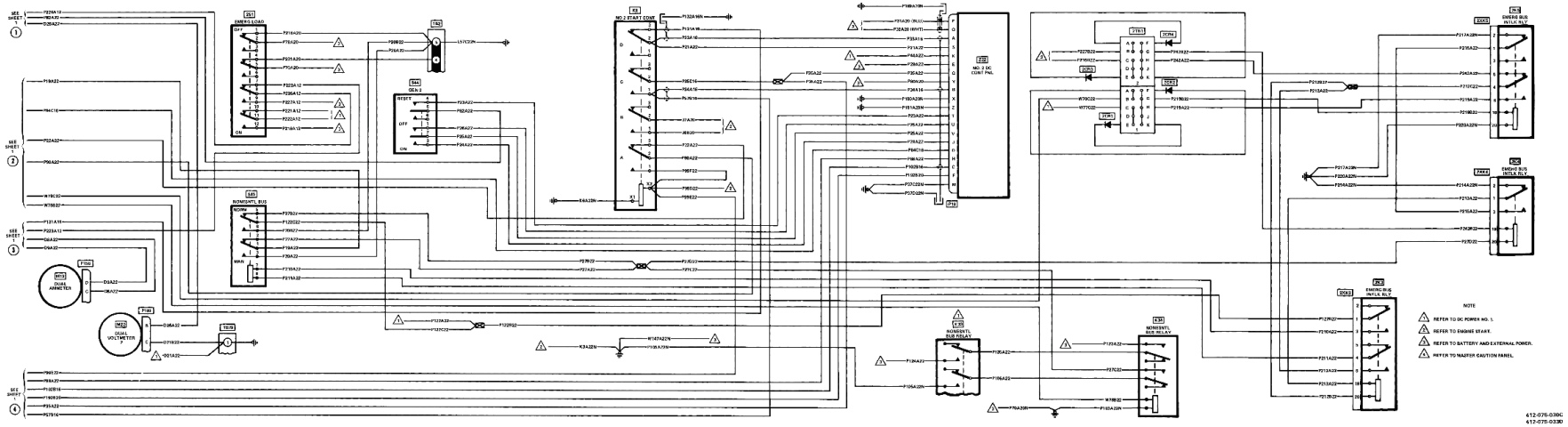
Figure 1. DC Power No. 1 (EFIS Configuration) (Sheet 7 of 7)

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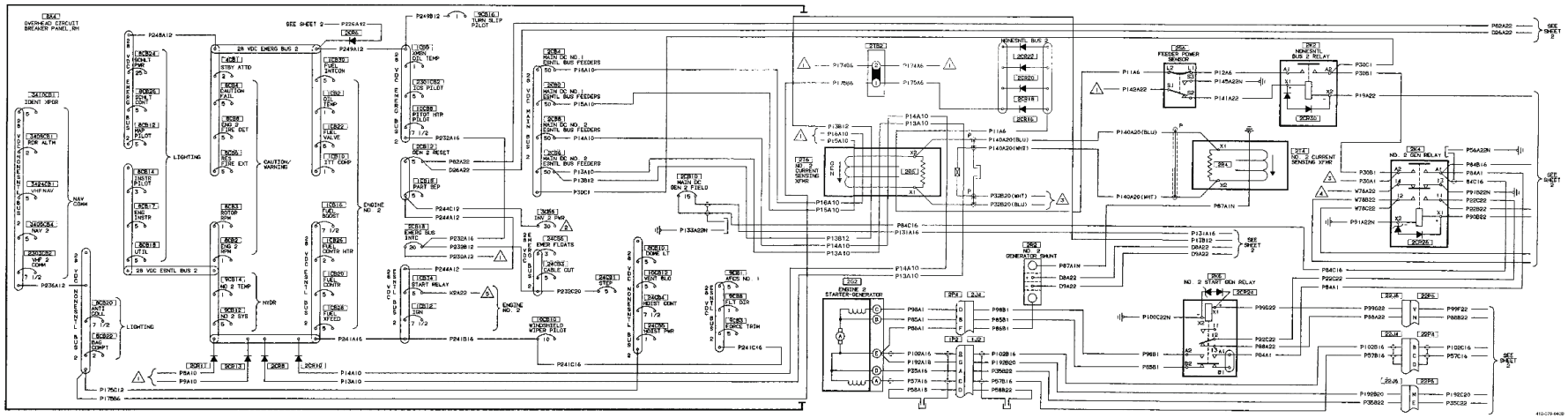
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Figure 1. DC Power No. 2 (Sheet 1 of 2)



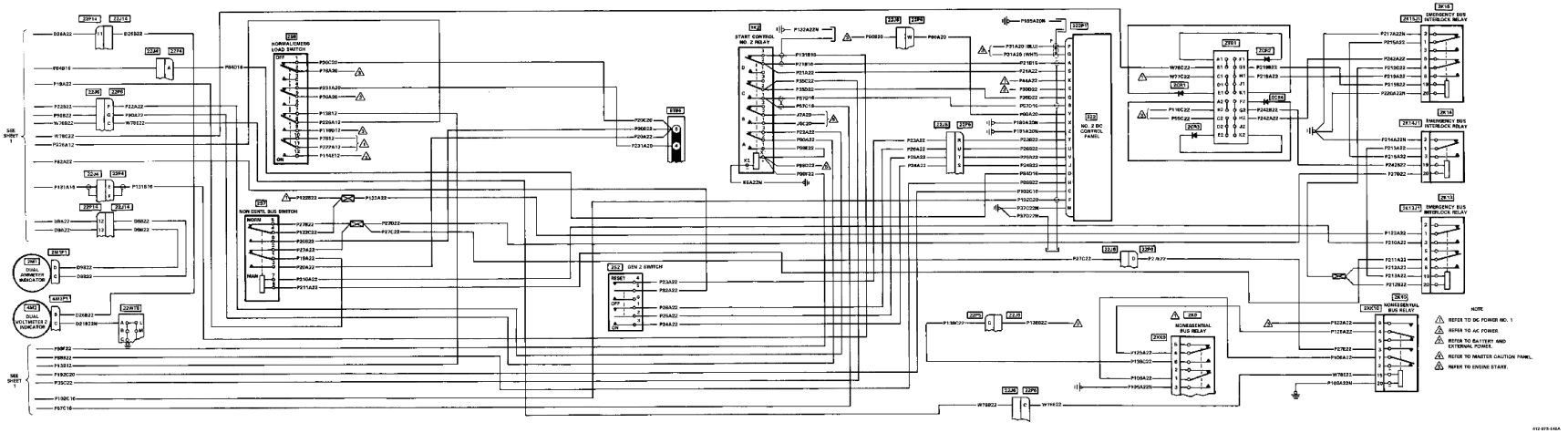
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Figure 1. DC Power No. 2 (Sheet 2 of 2)



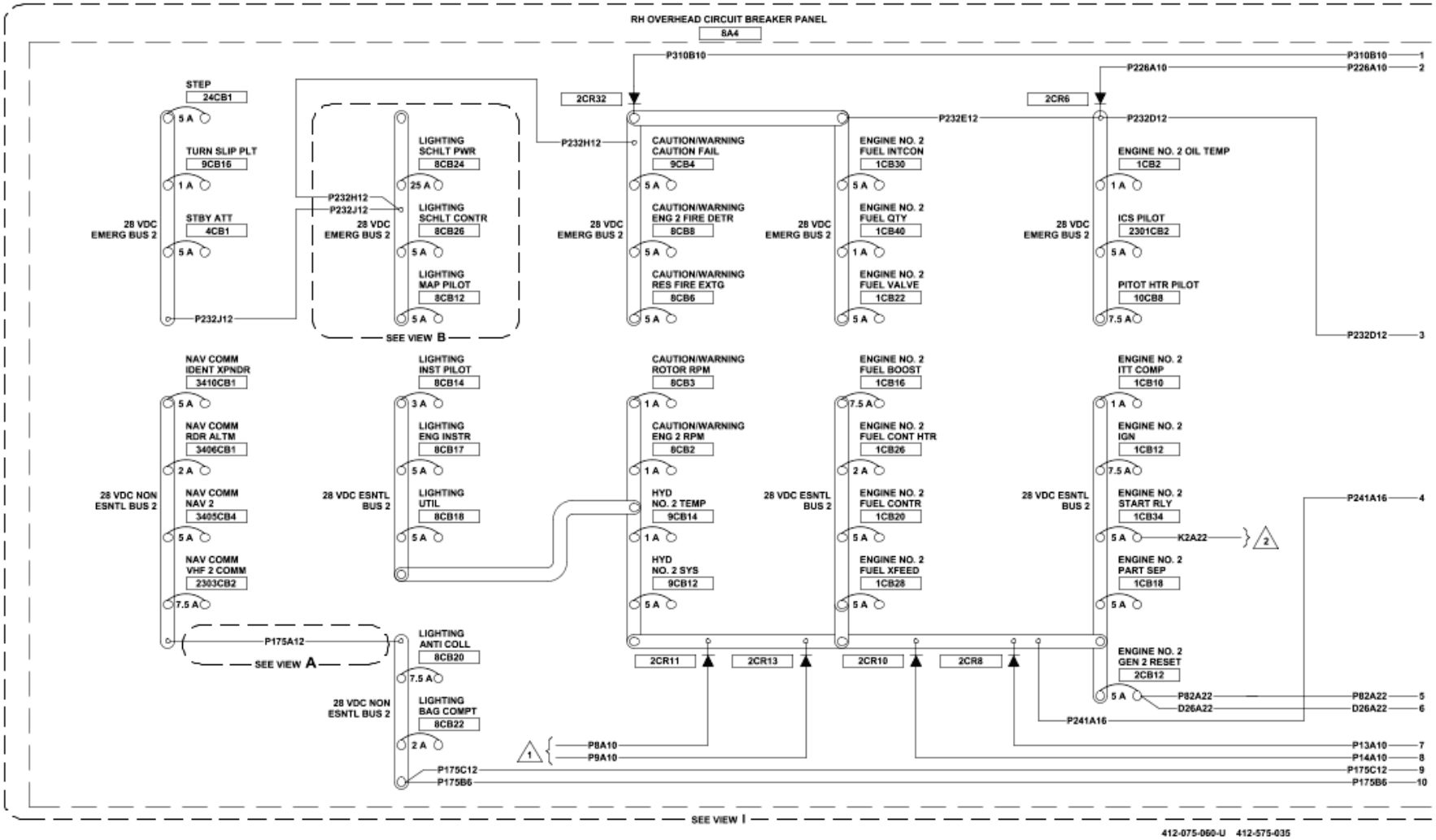
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Figure 1. DC Power No. 2 (Sheet 1 of 2)



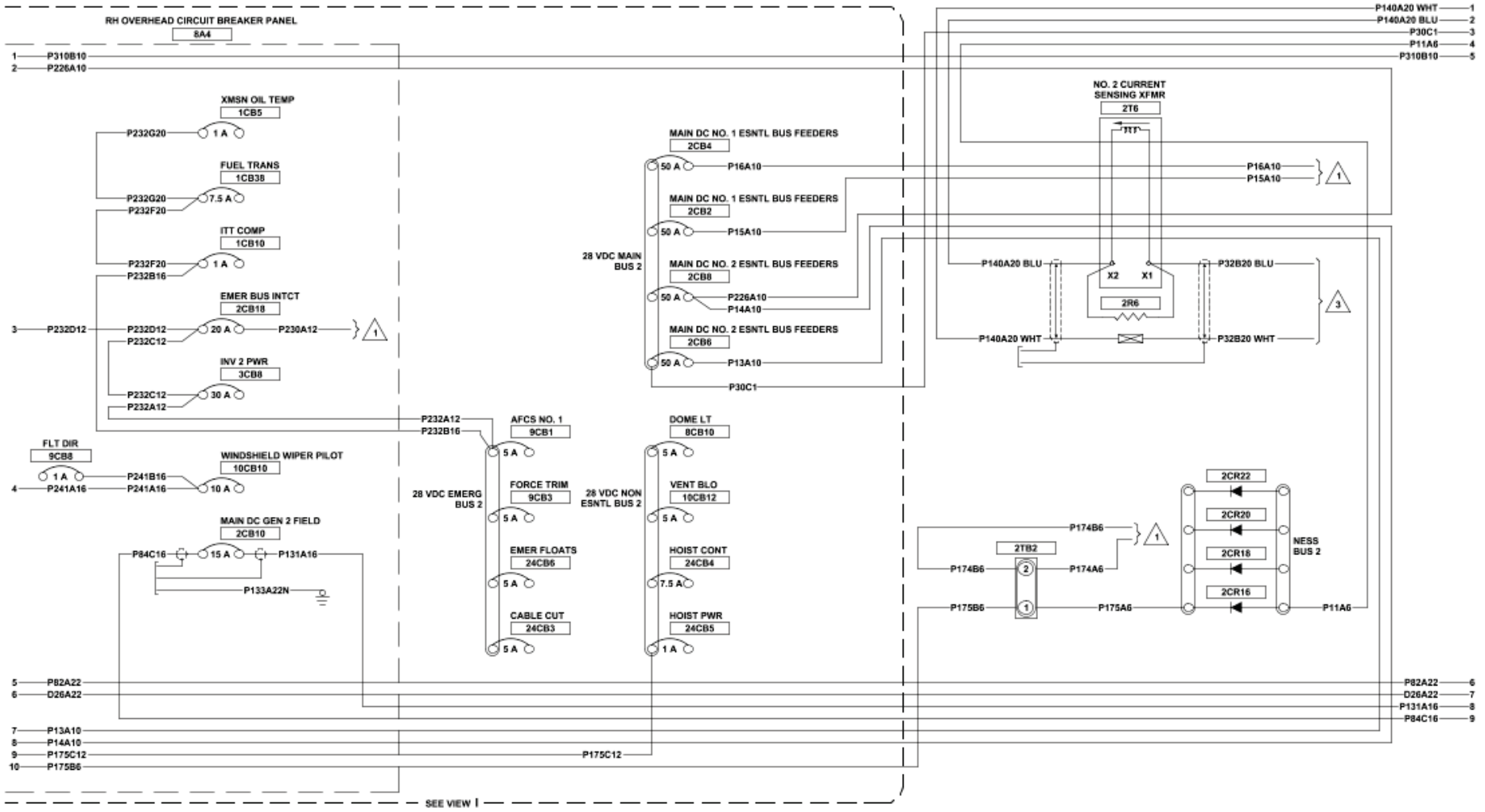
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Figure 1. DC Power No. 2 (Sheet 2 of 2)



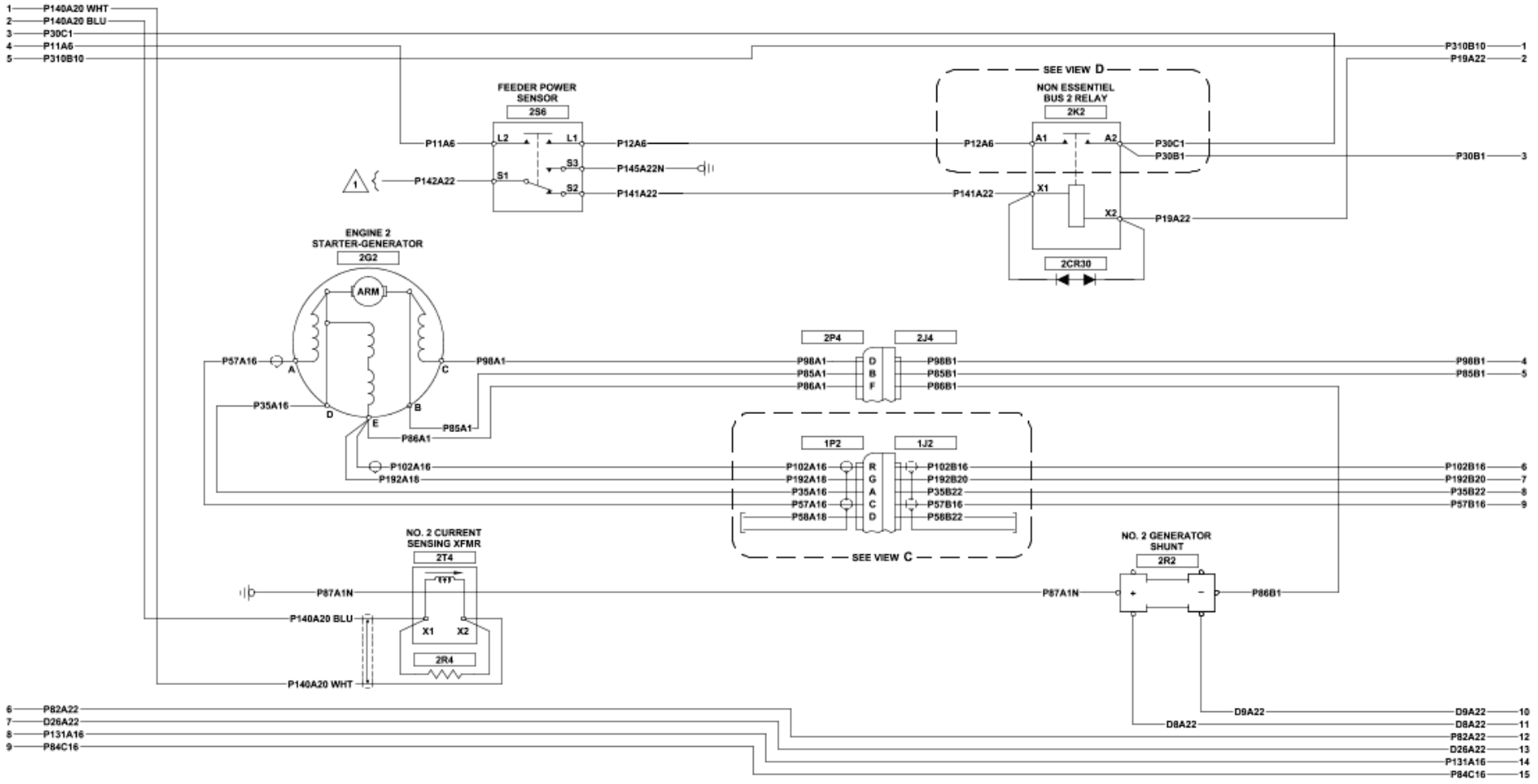
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Figure 1. DC Power No. 2 (Sheet 1 of 9)



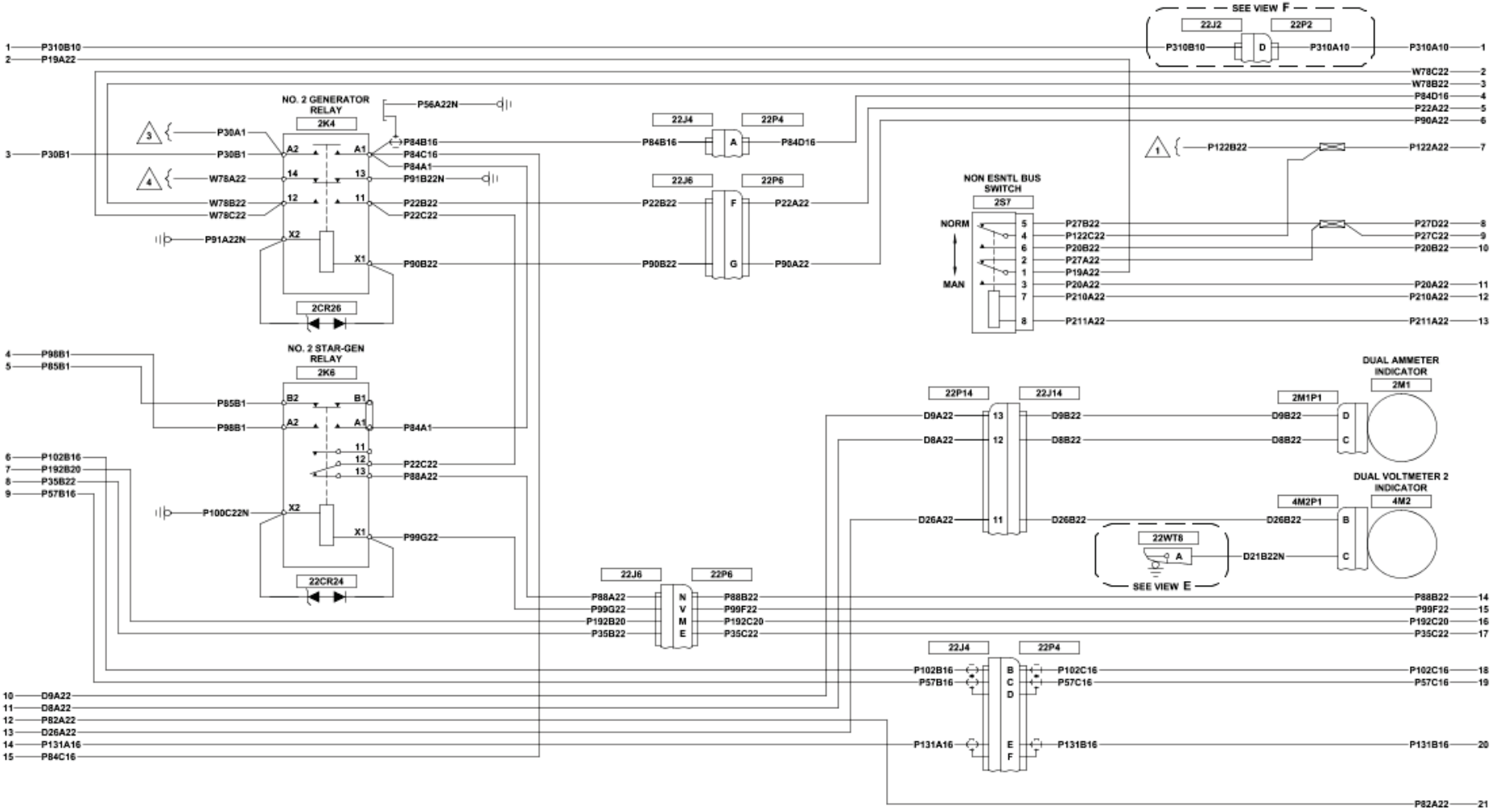
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Figure 1. DC Power No. 2 (Sheet 2 of 9)



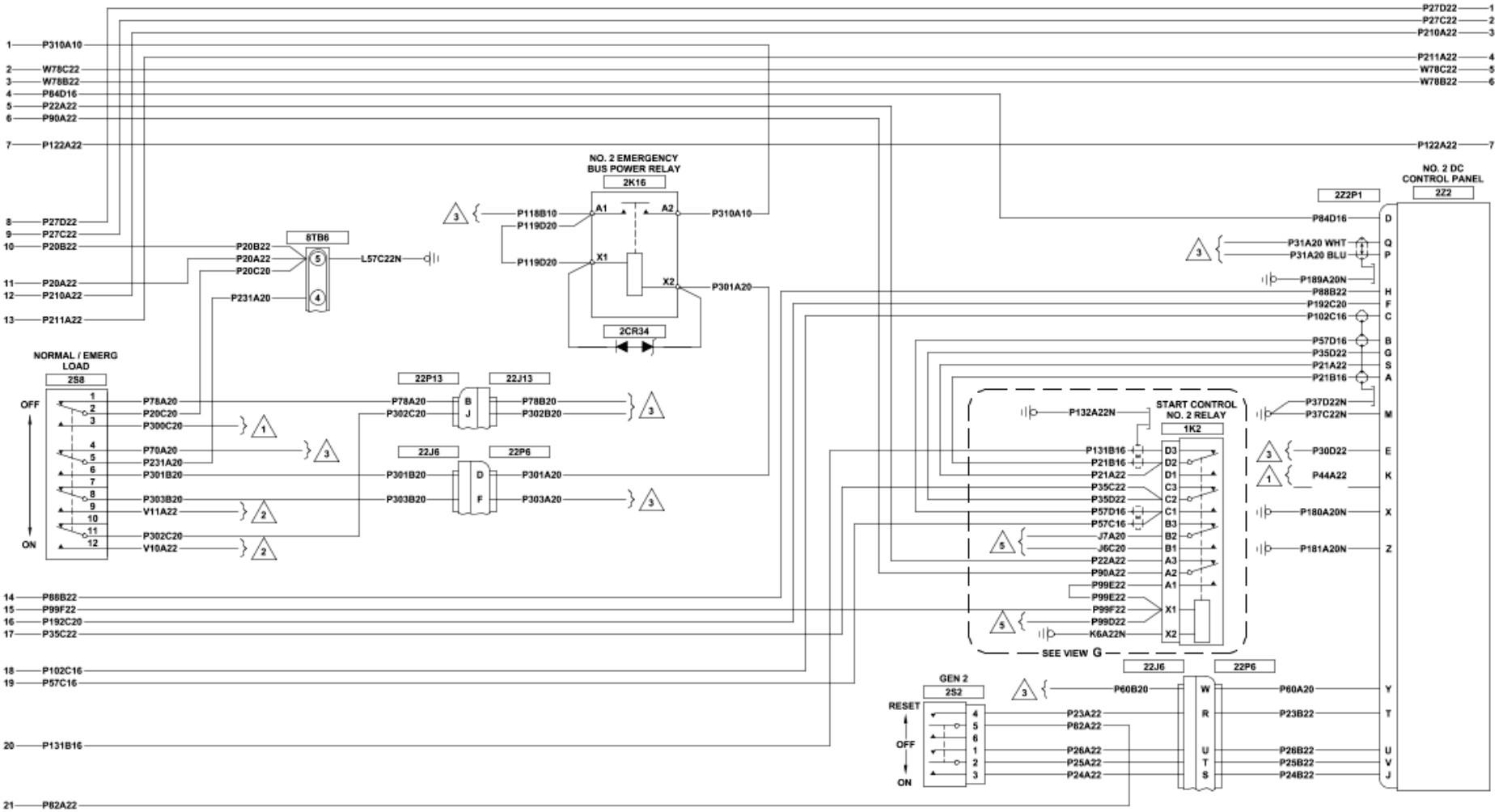
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Figure 1. DC Power No. 2 (Sheet 3 of 9)



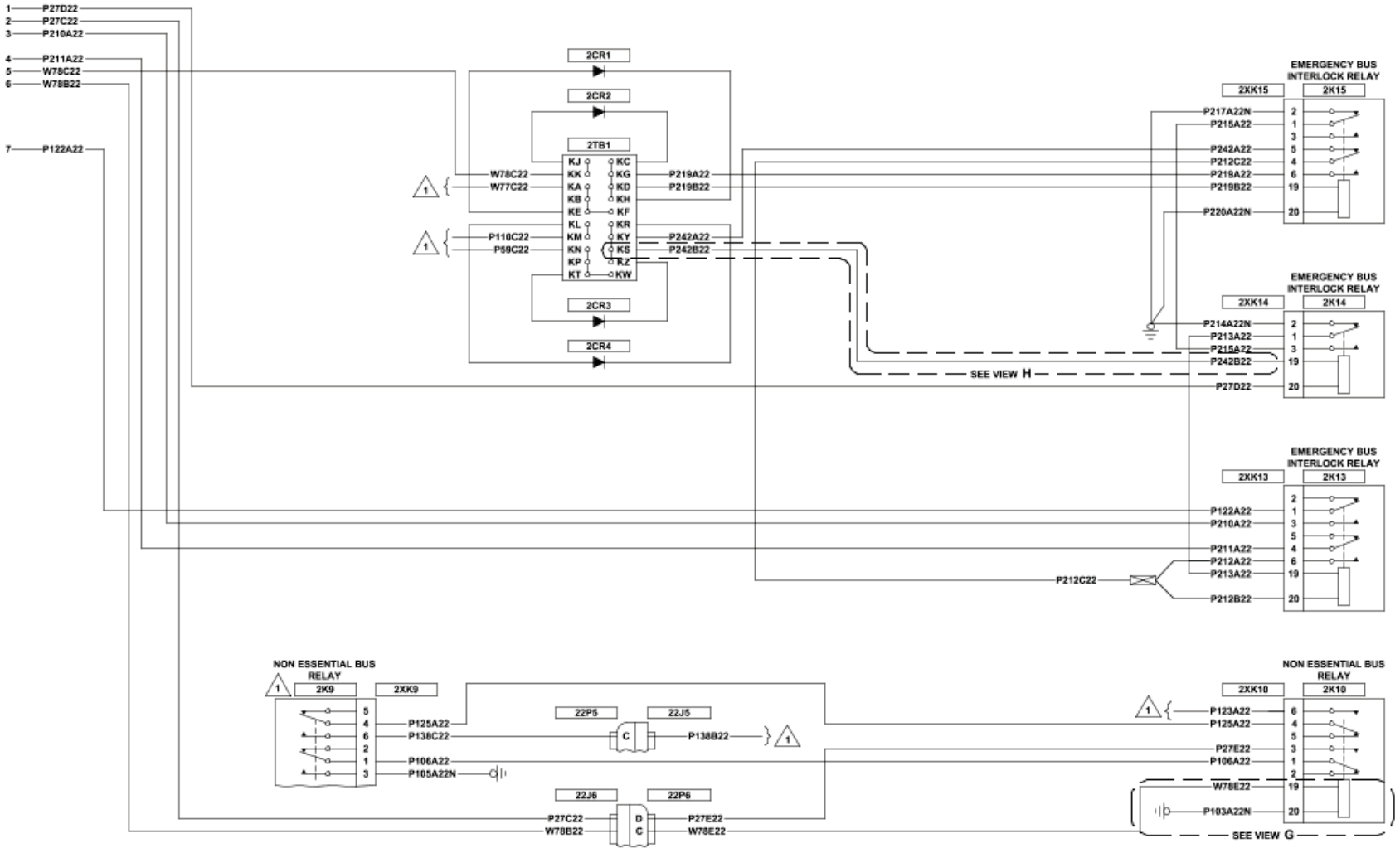
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Figure 1. DC Power No. 2 (Sheet 4 of 9)



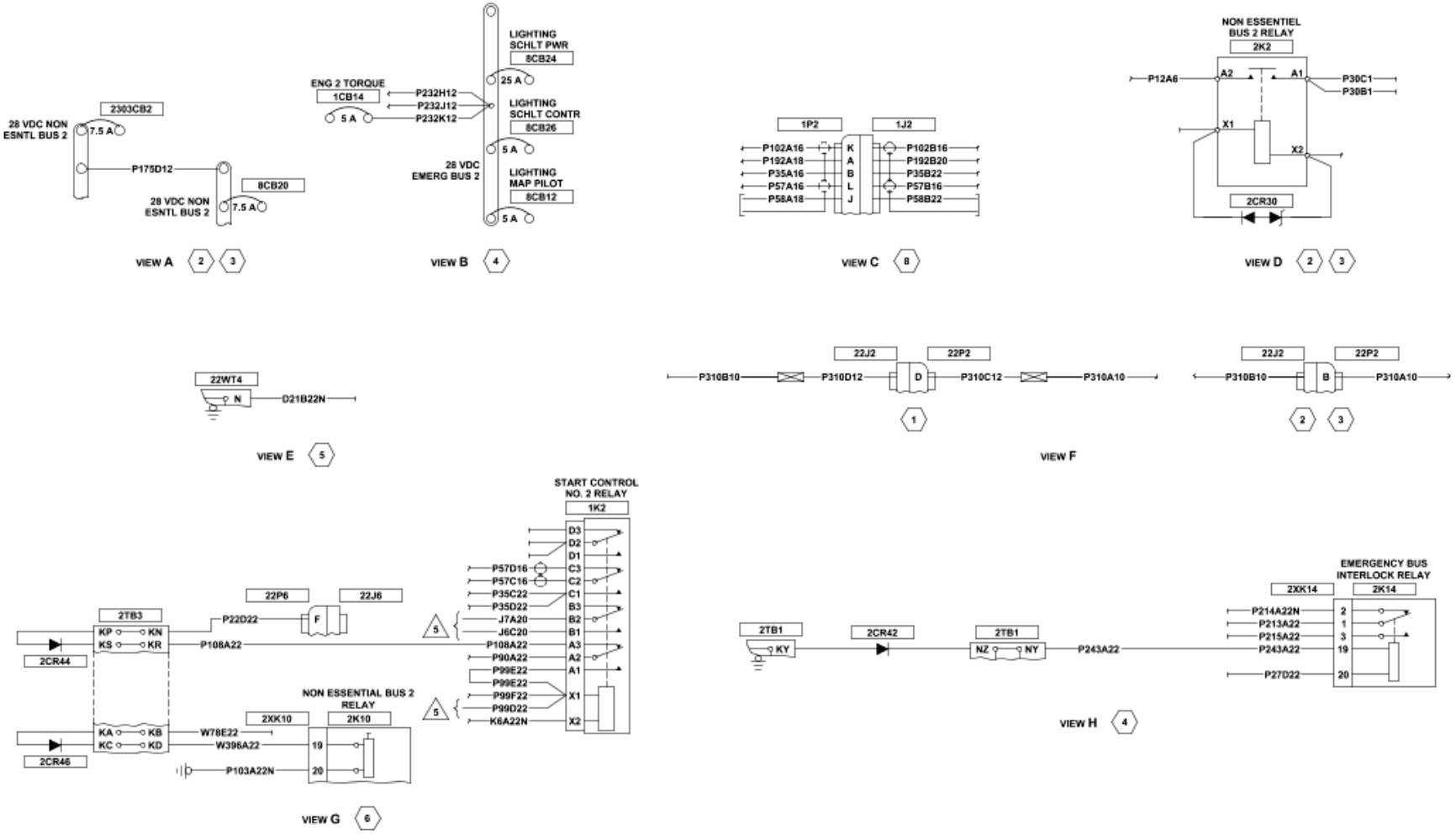
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Figure 1. DC Power No. 2 (Sheet 5 of 9)



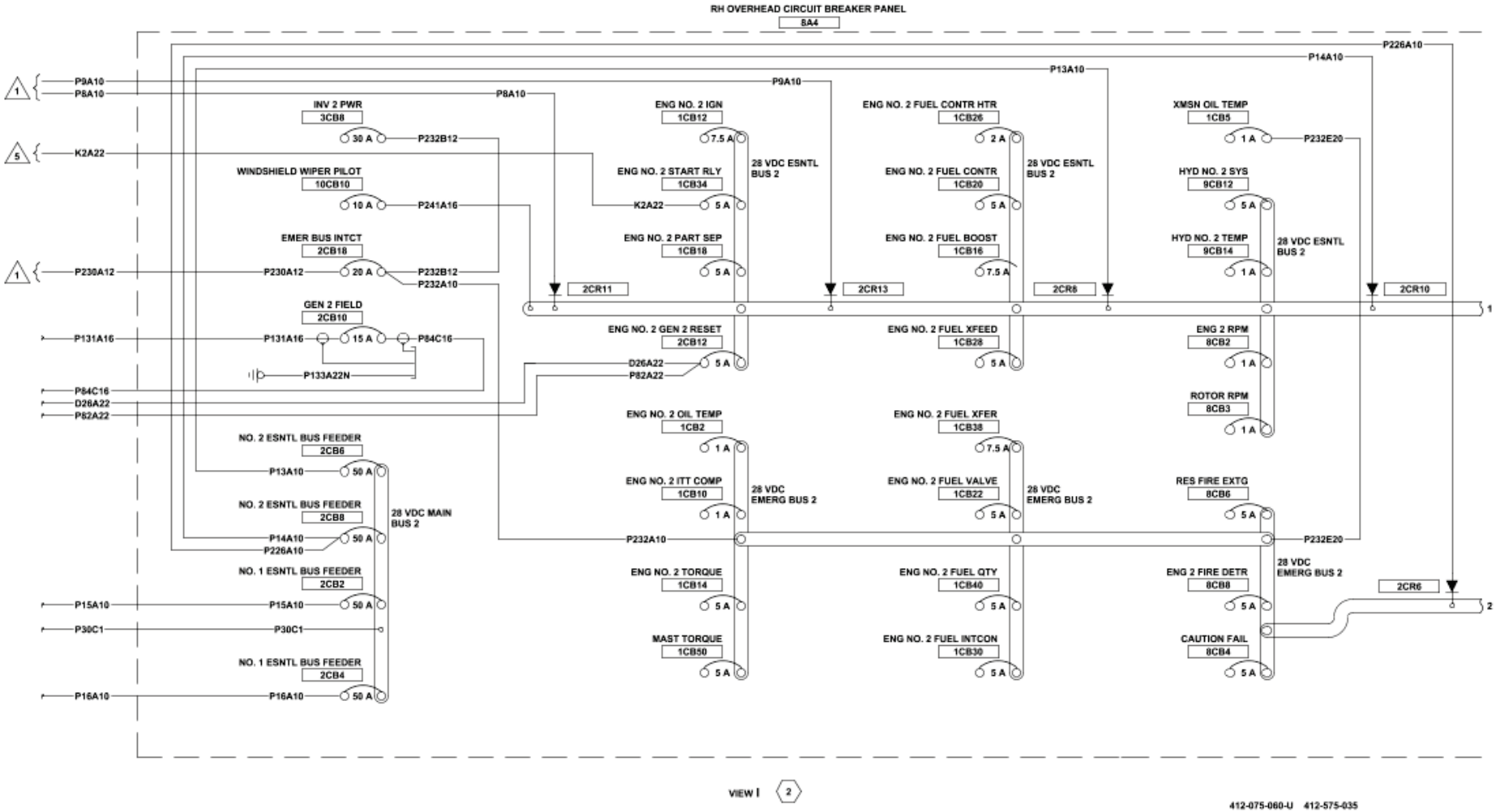
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Figure 1. DC Power No. 2 (Sheet 6 of 9)



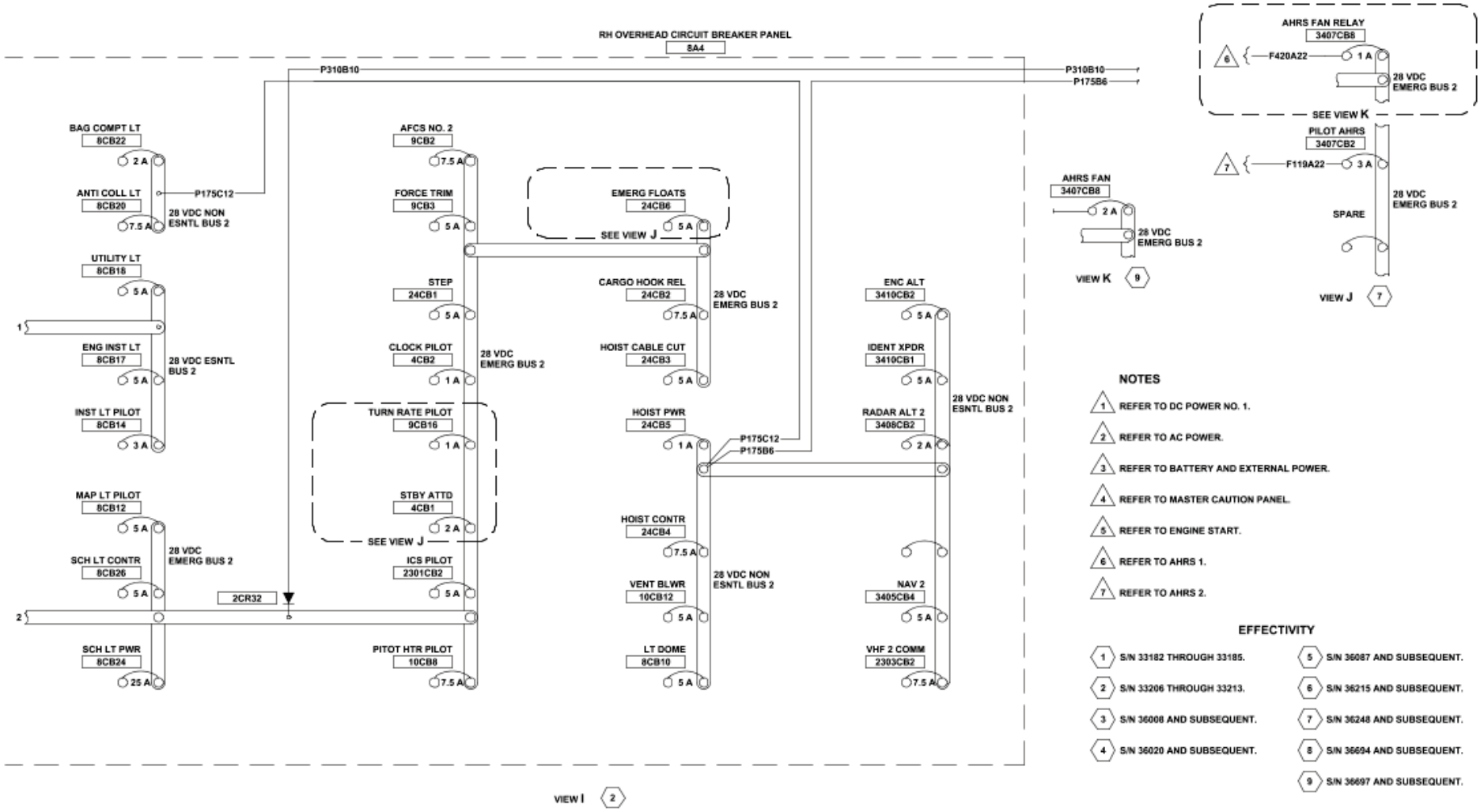
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Figure 1. DC Power No. 2 (Sheet 7 of 9)



ICN-412-A-980000-B-97499-00053-A-001-01

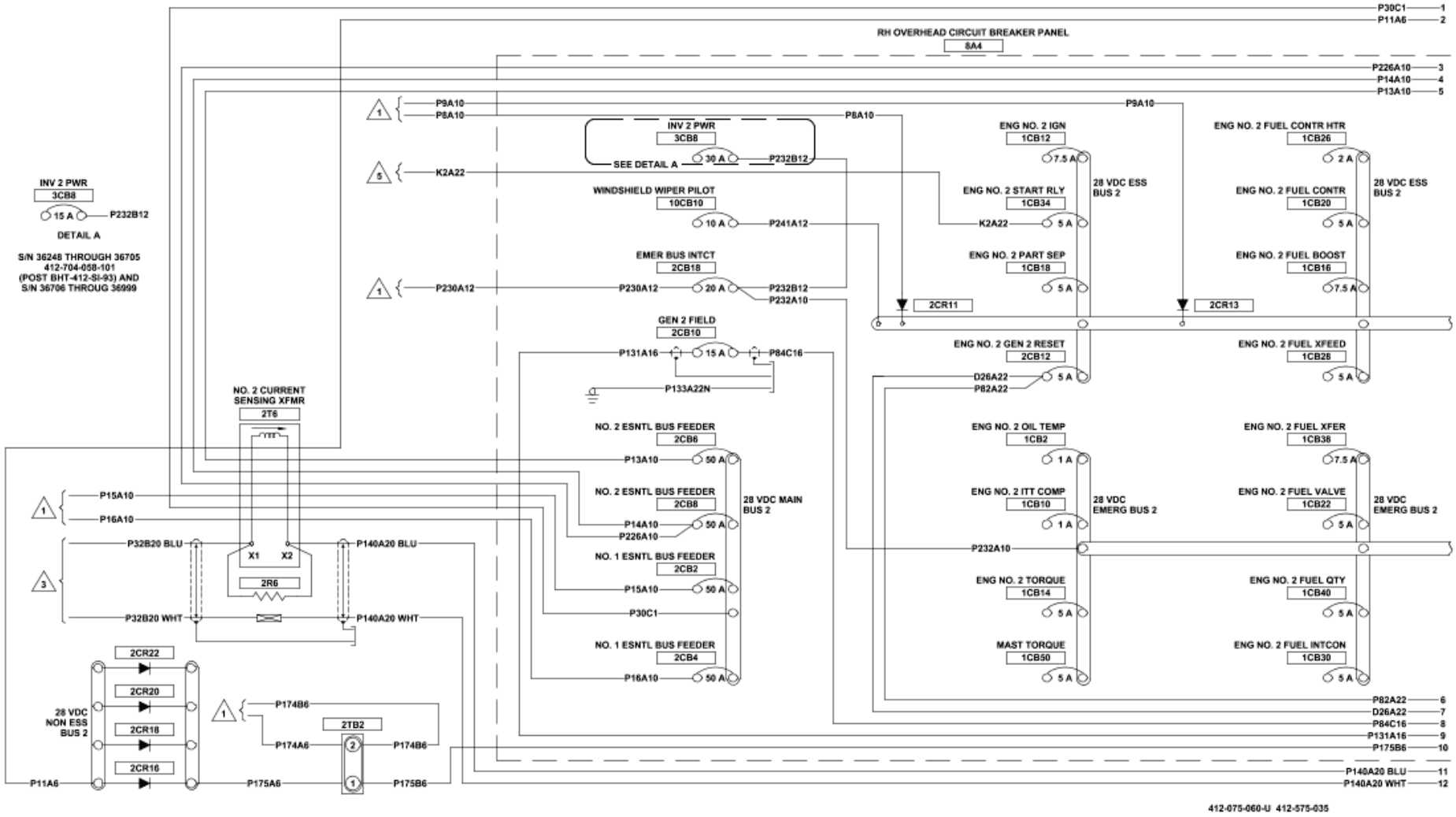
Figure 1. DC Power No. 2 (Sheet 8 of 9)



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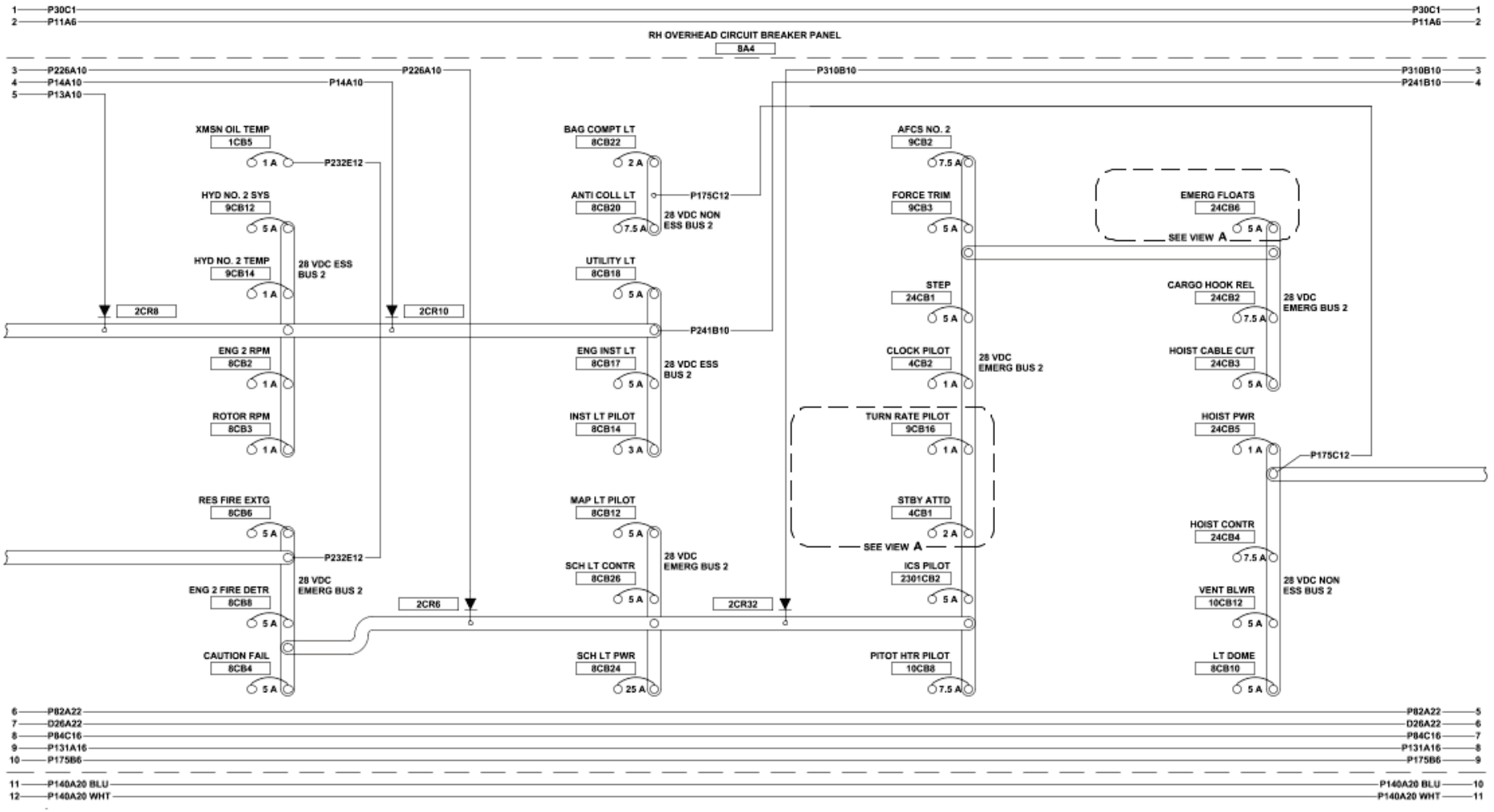
Figure 1. DC Power No. 2 (Sheet 9 of 9)

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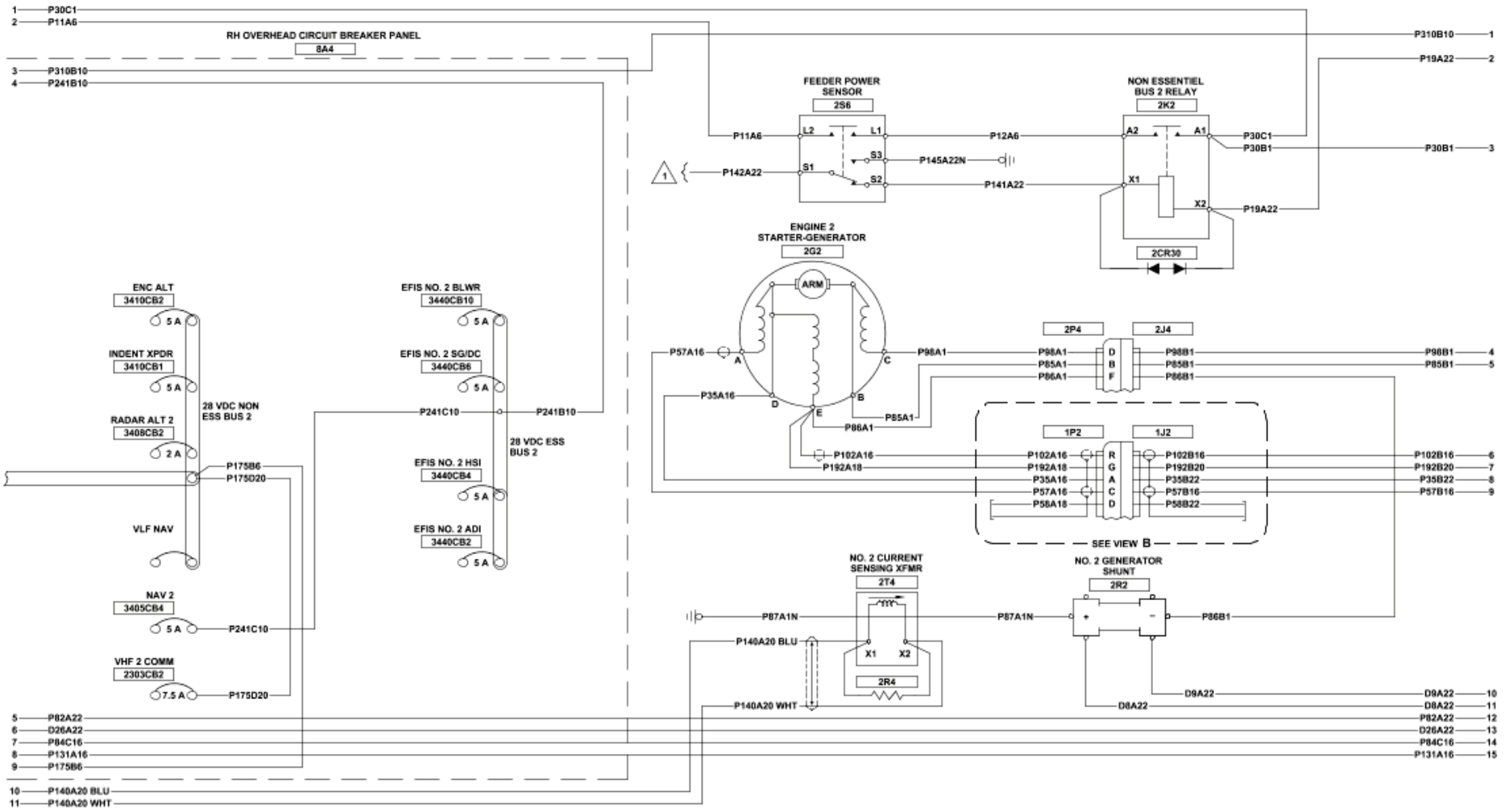
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Figure 1. DC Power No. 2 (EFIS Configuration) (Sheet 1 of 6)



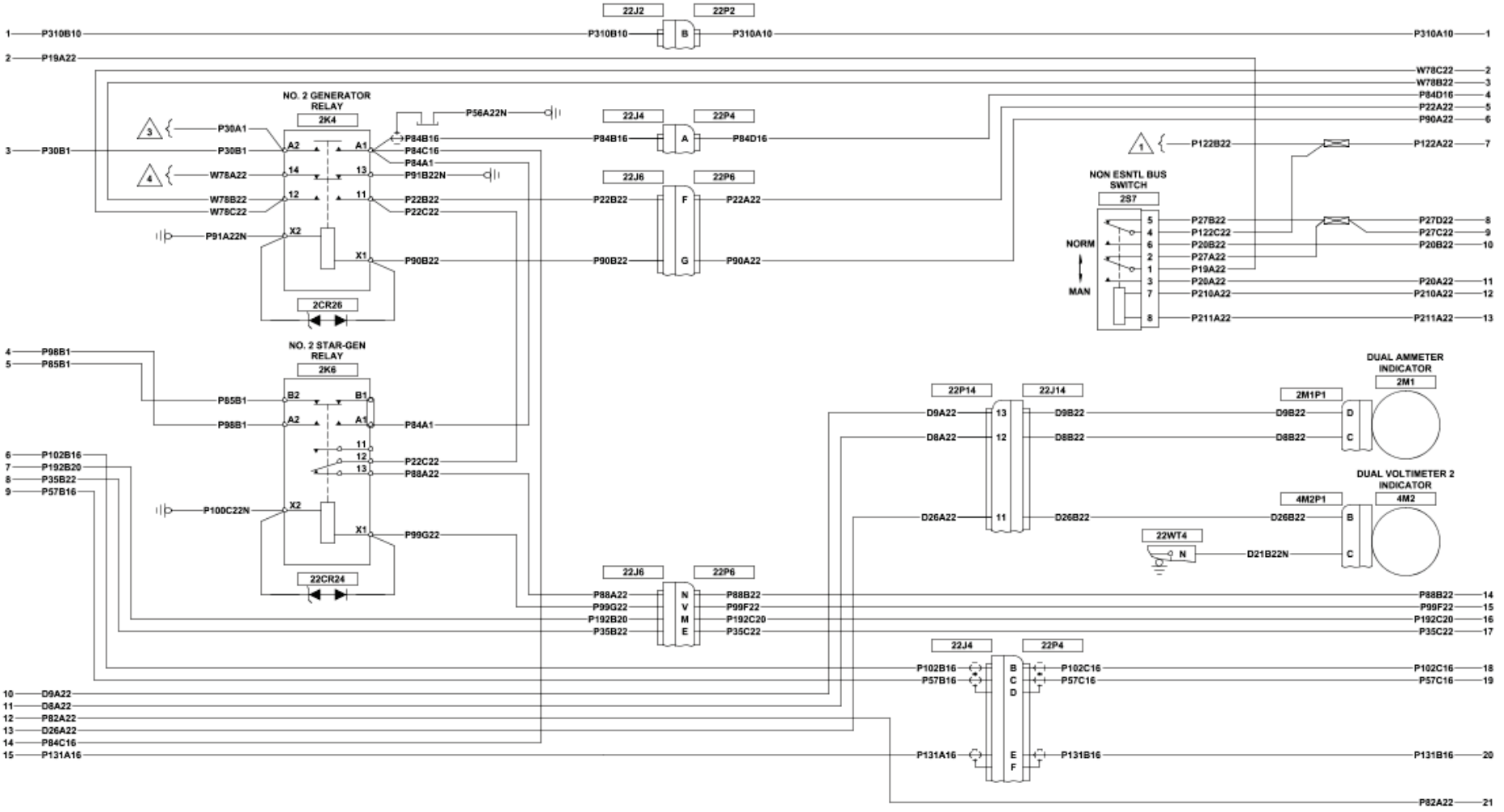
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Figure 1. DC Power No. 2 (EFIS Configuration) (Sheet 2 of 6)



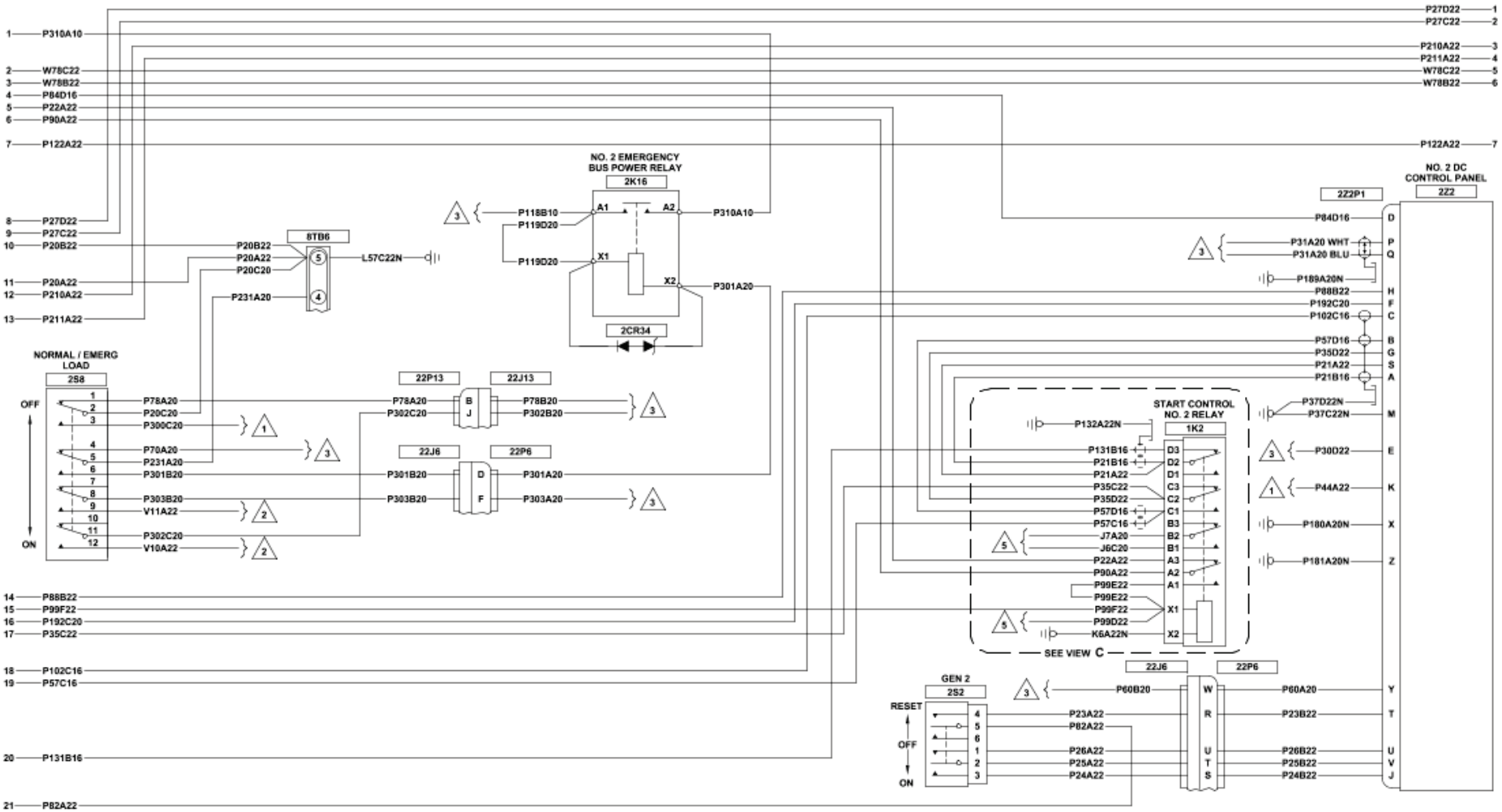
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Figure 1. DC Power No. 2 (EFIS Configuration) (Sheet 3 of 6)



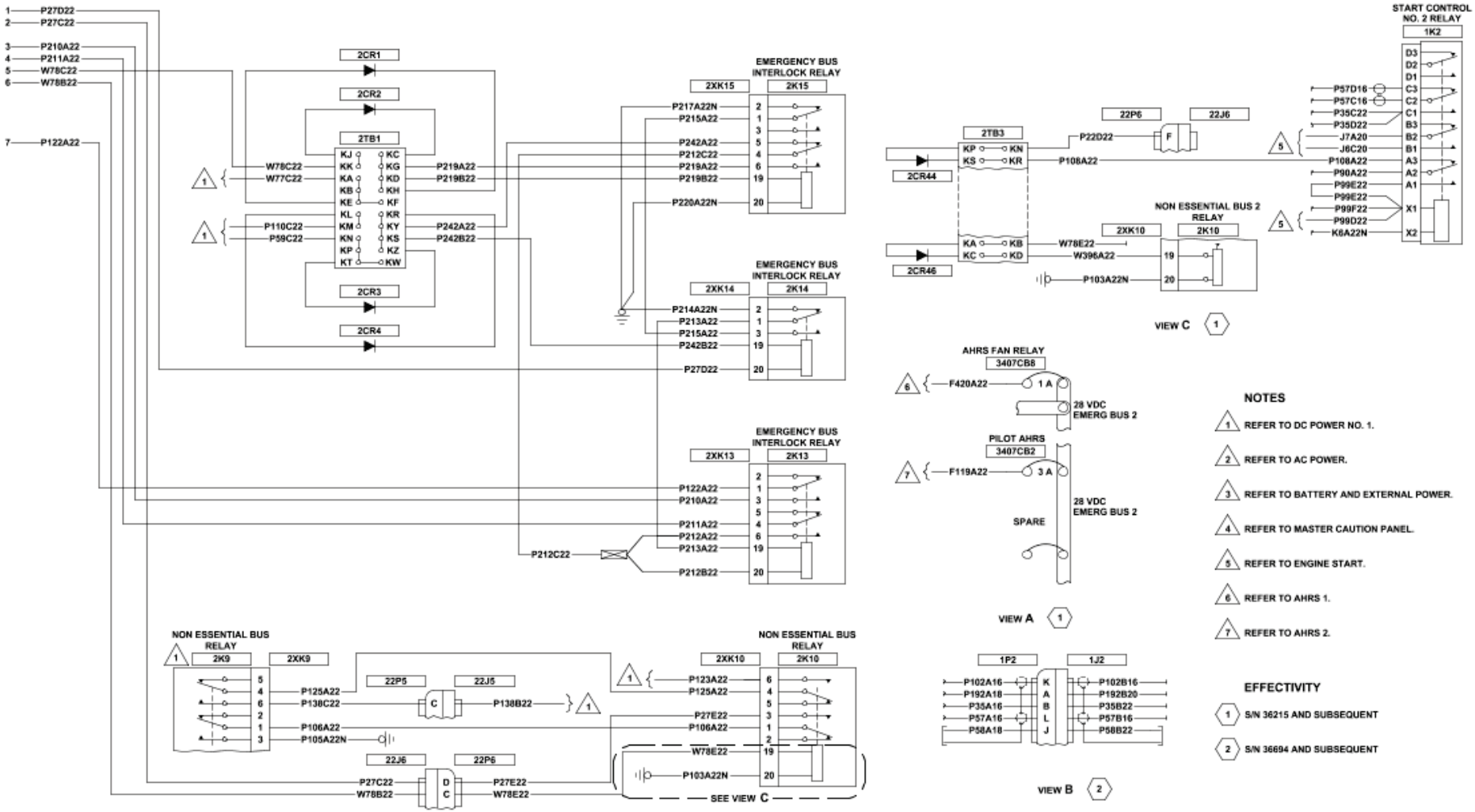
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Figure 1. DC Power No. 2 (EFIS Configuration) (Sheet 4 of 6)



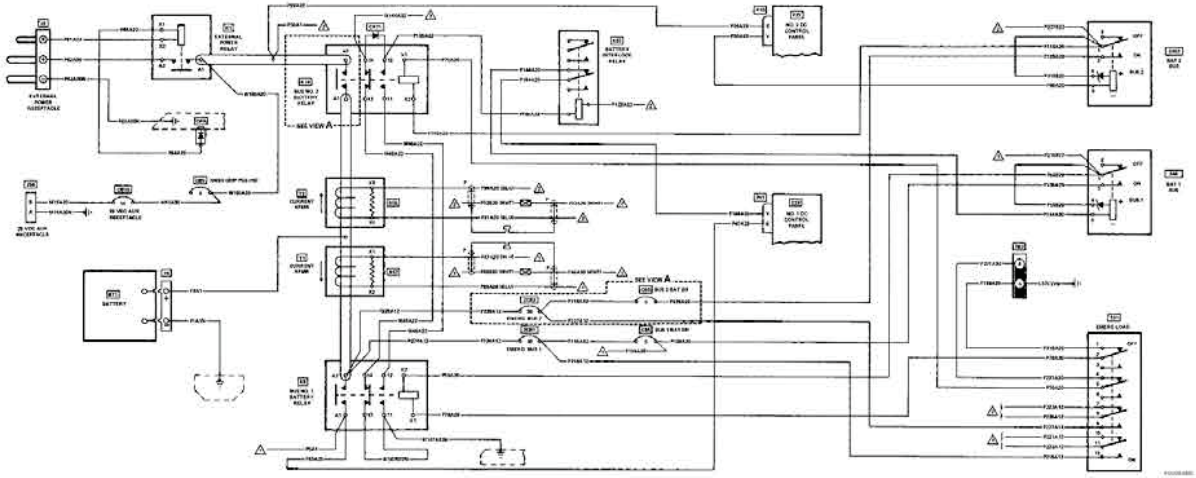
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Figure 1. DC Power No. 2 (EFIS Configuration) (Sheet 5 of 6)



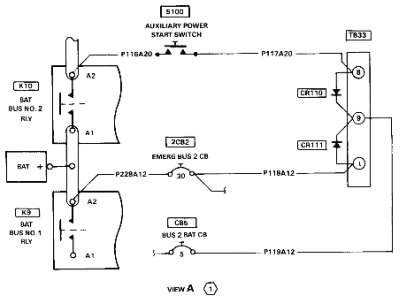
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Figure 1. DC Power No. 2 (EFIS Configuration) (Sheet 6 of 6)



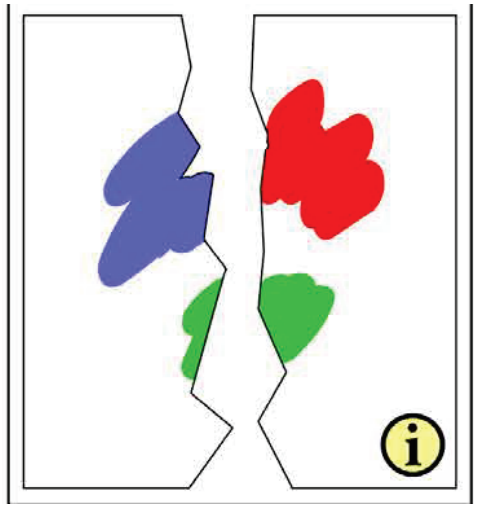
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Figure 1. Battery and External Power (Sheet 1 of 2)



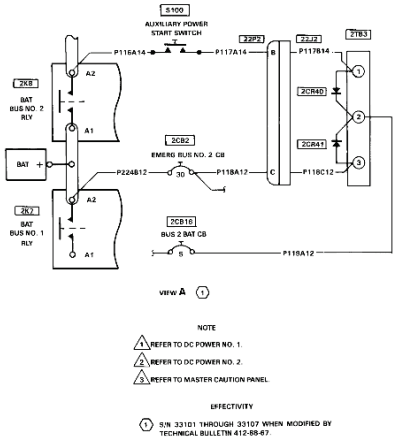
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Figure 1. Battery and External Power (Sheet 2 of 2)



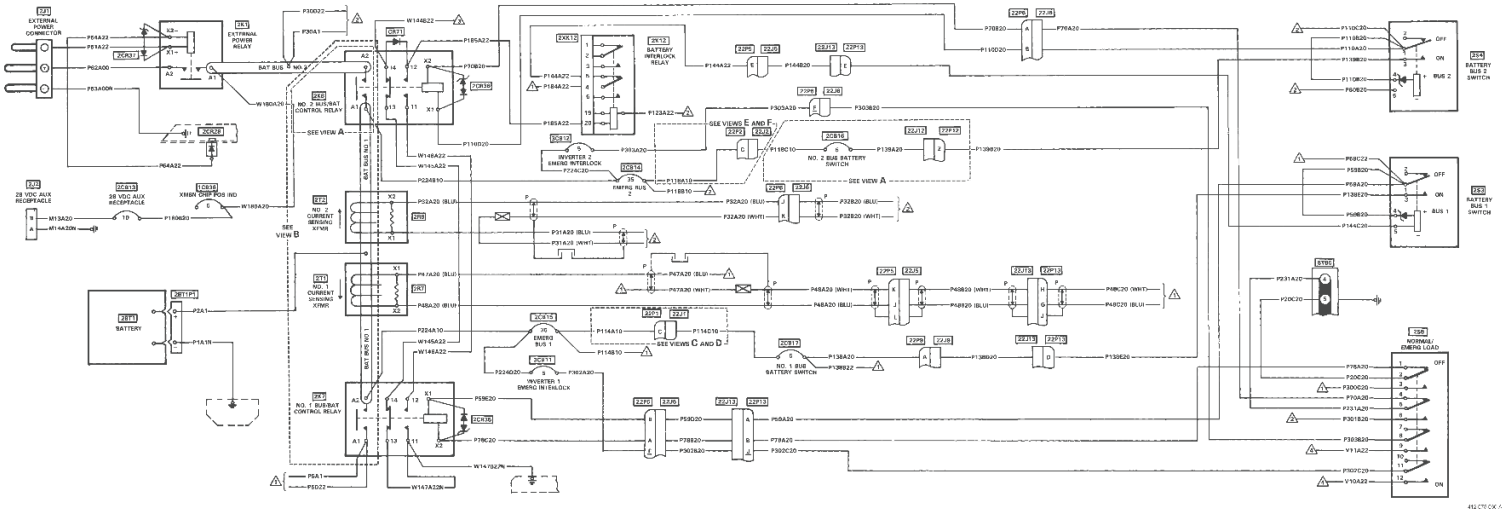
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Figure 1. Battery and External Power (Sheet 1 of 2)



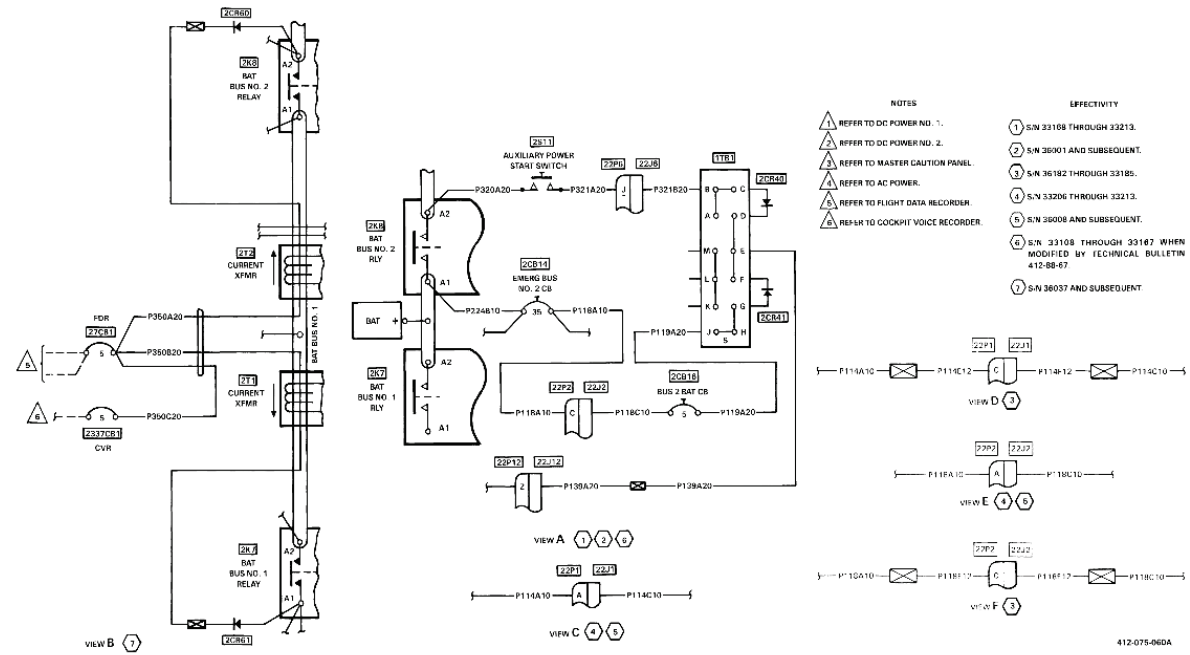
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Figure 1. Battery and External Power (Sheet 2 of 2)



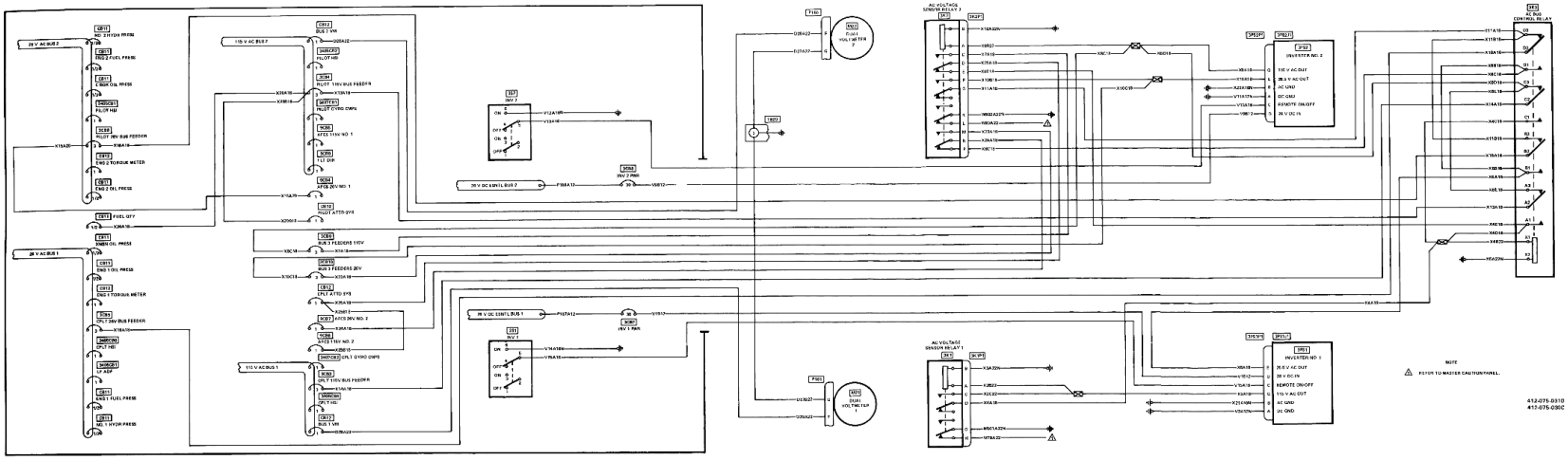
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Figure 1. Battery and External Power (Sheet 1 of 2)



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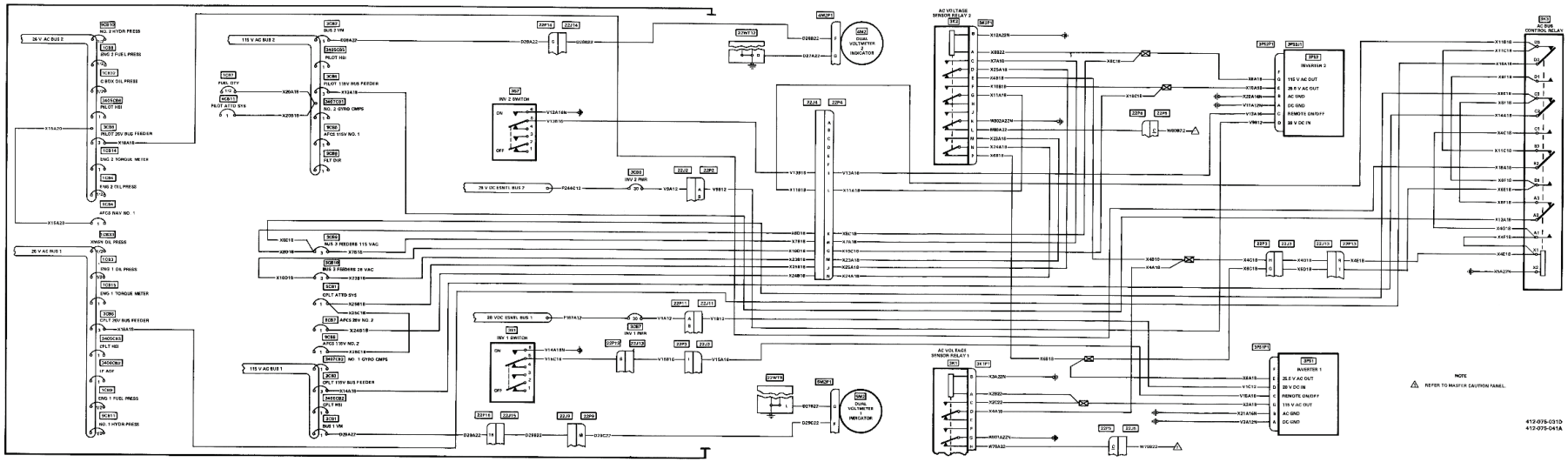
Figure 1. Battery and External Power (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00067-A-001-01

Figure 1. AC Power System (Sheet 1 of 1)

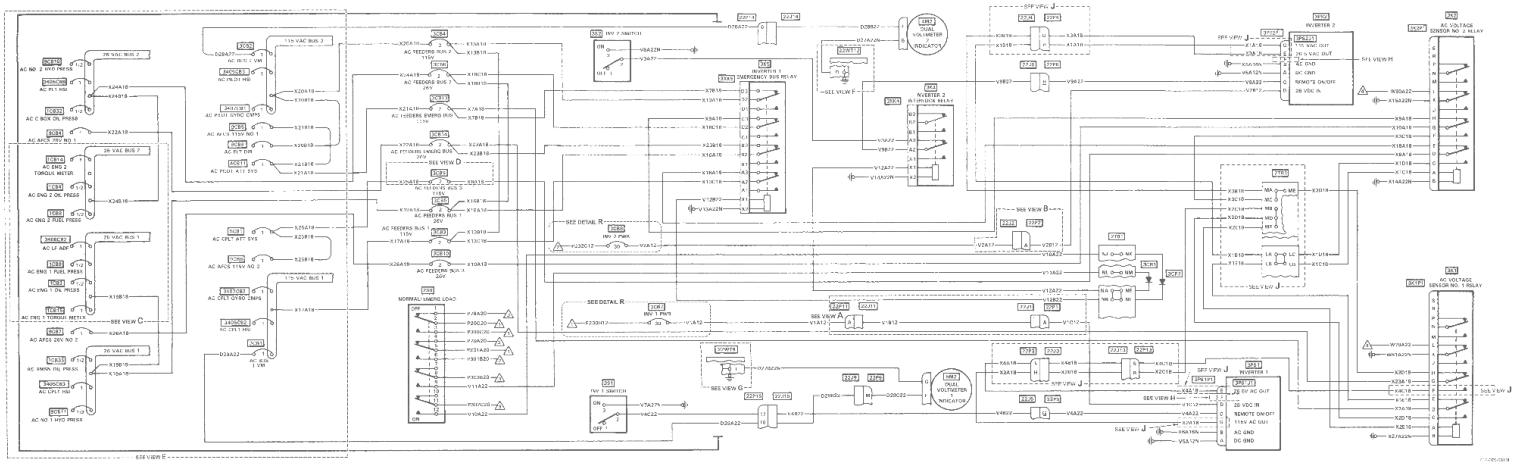
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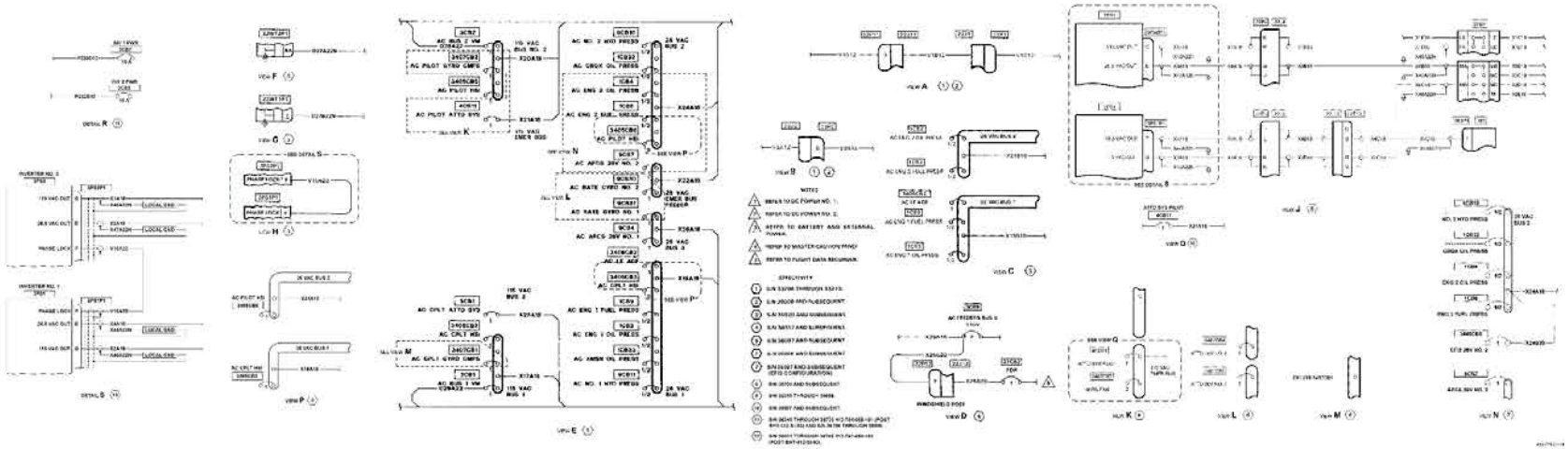
Figure 1. AC Power System (Sheet 1 of 1)

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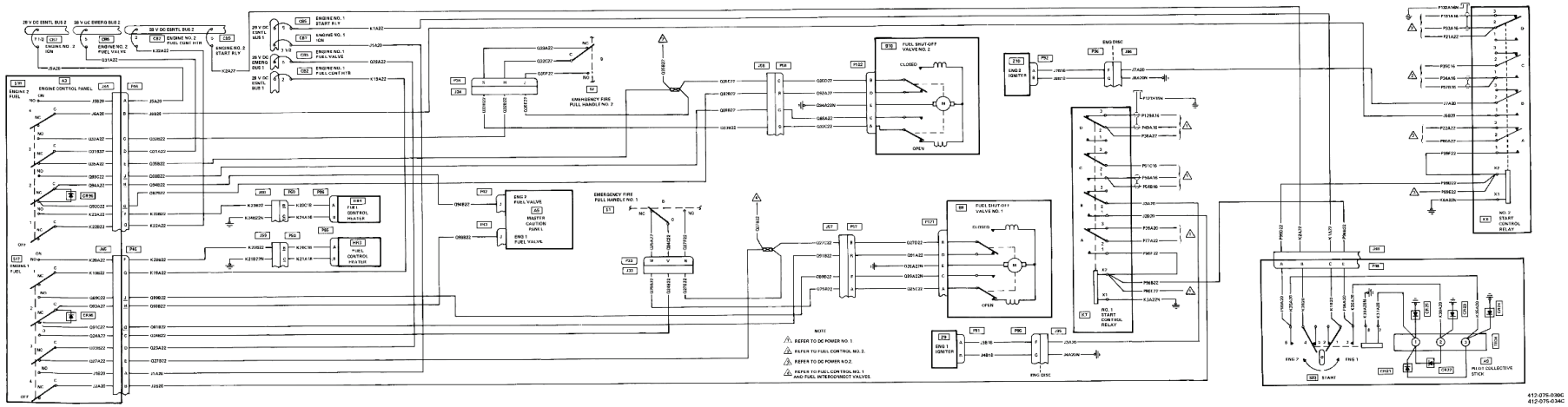
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Figure 1. AC Power System (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00070-A-001-01

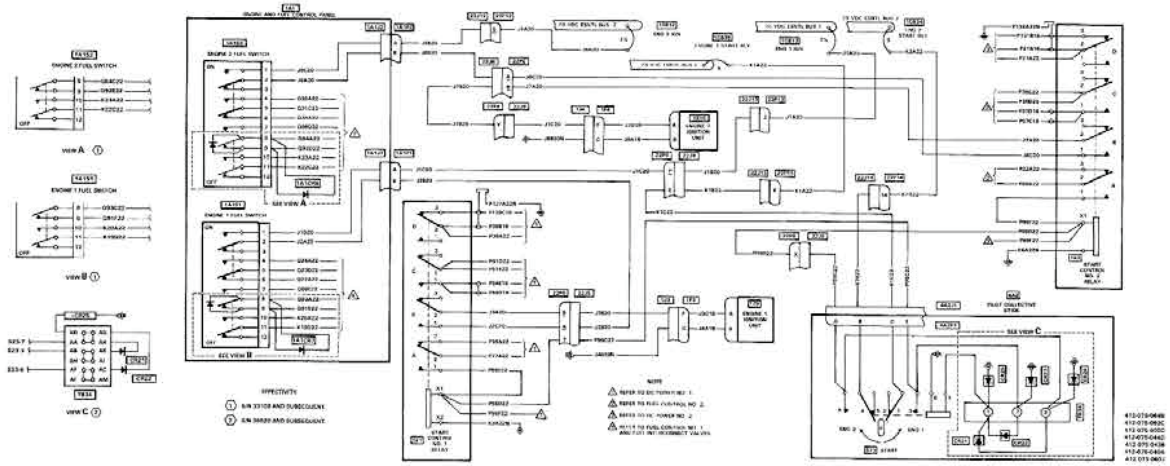
Figure 1. AC Power System (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00071-A-001-01

Figure 1. Engine Start (Sheet 1 of 1)

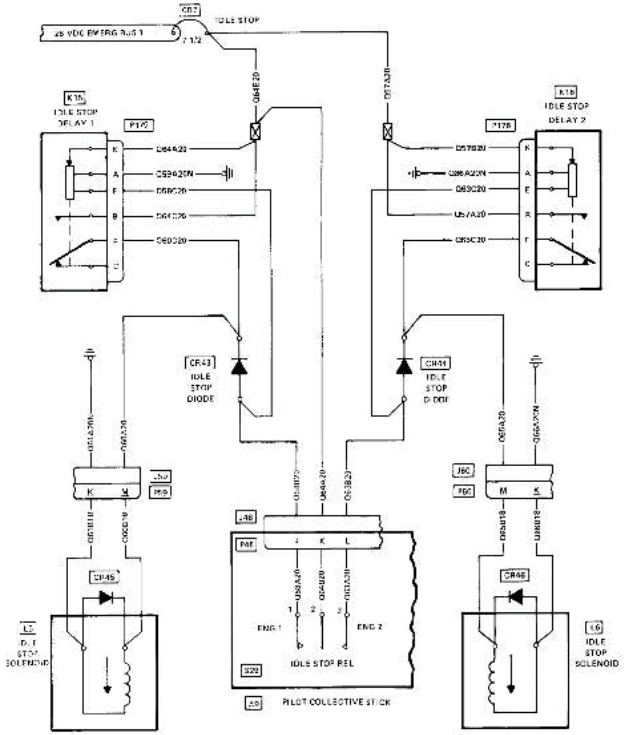
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Figure 1. Engine Start (Sheet 1 of 1)

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Figure 1. Idle Stop (Sheet 1 of 1)

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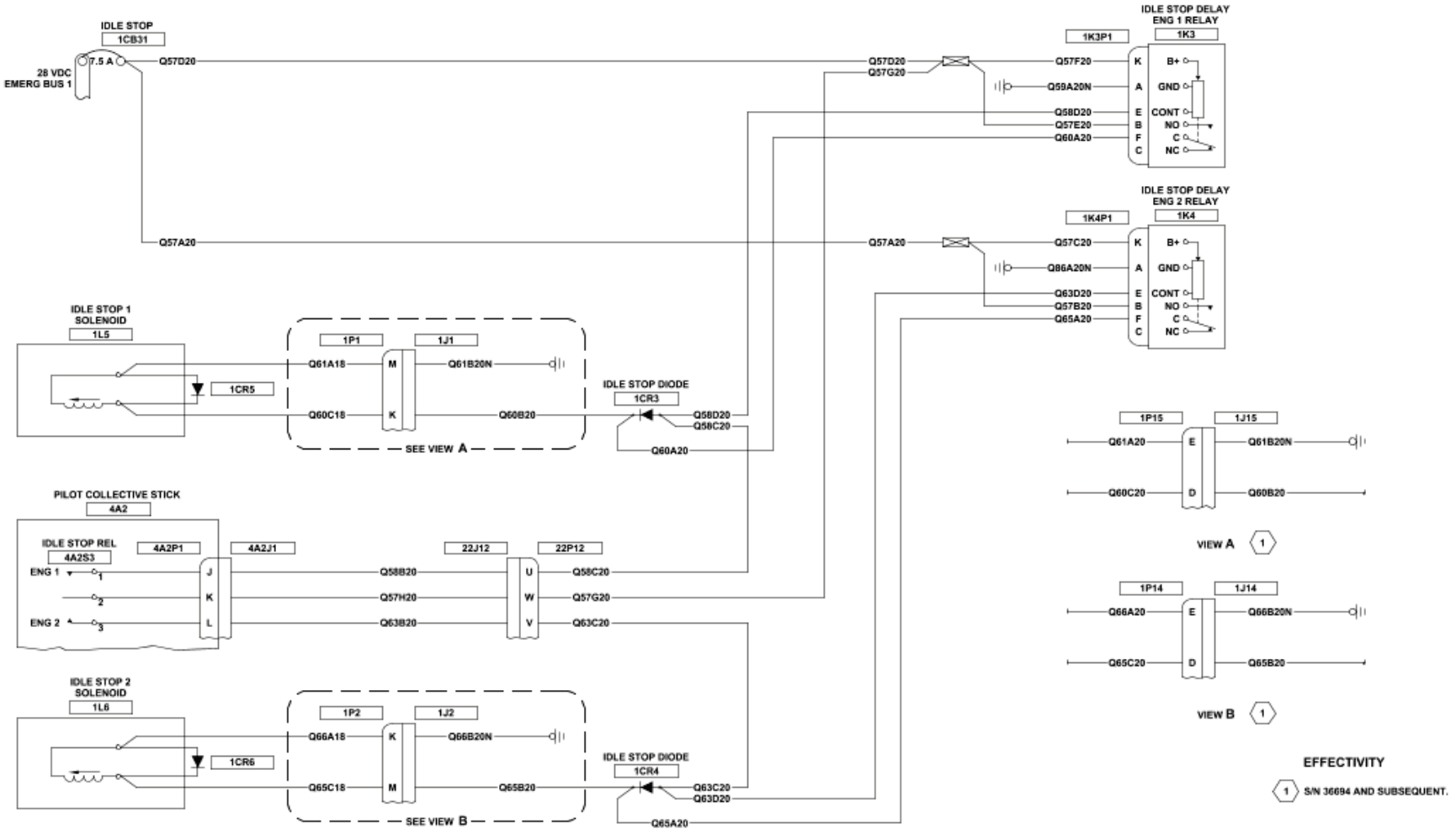
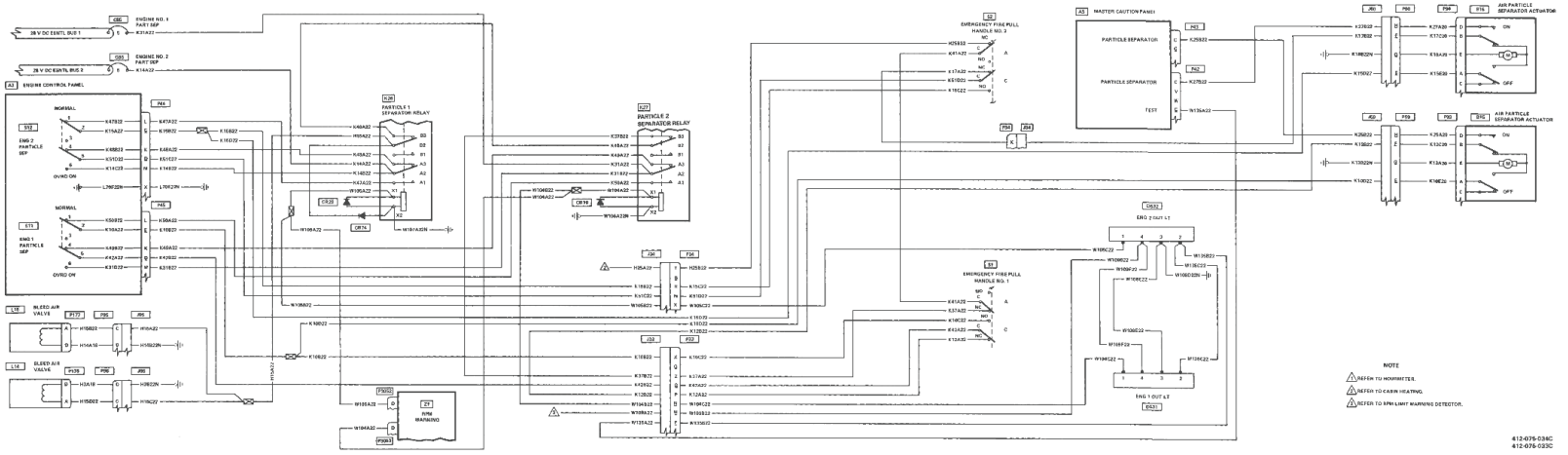


Figure 1. Idle Stop (Sheet 1 of 1)

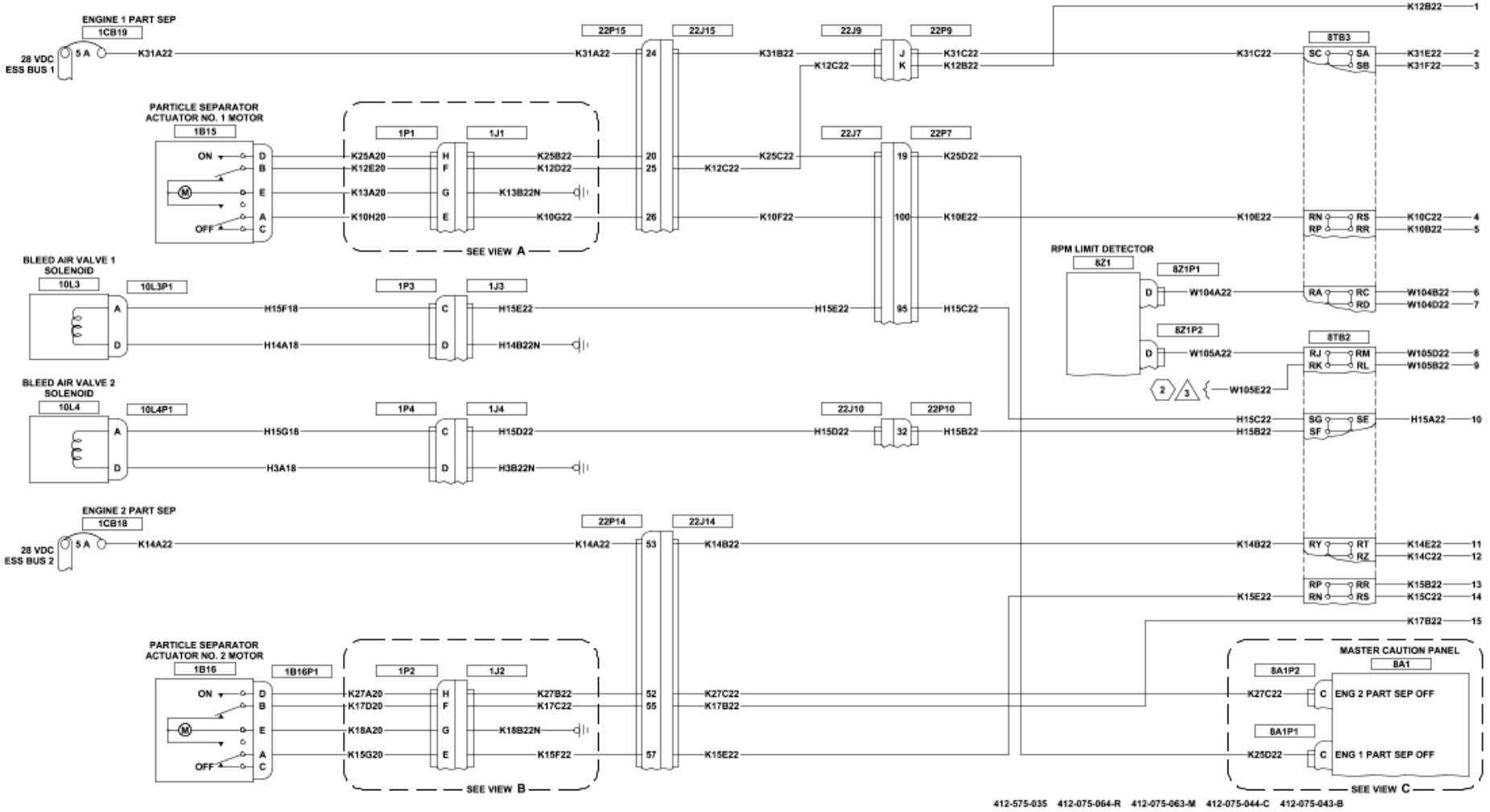
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Figure 1. Particle Separator (Sheet 1 of 1)

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Figure 1. Particle Separator (Sheet 1 of 3)

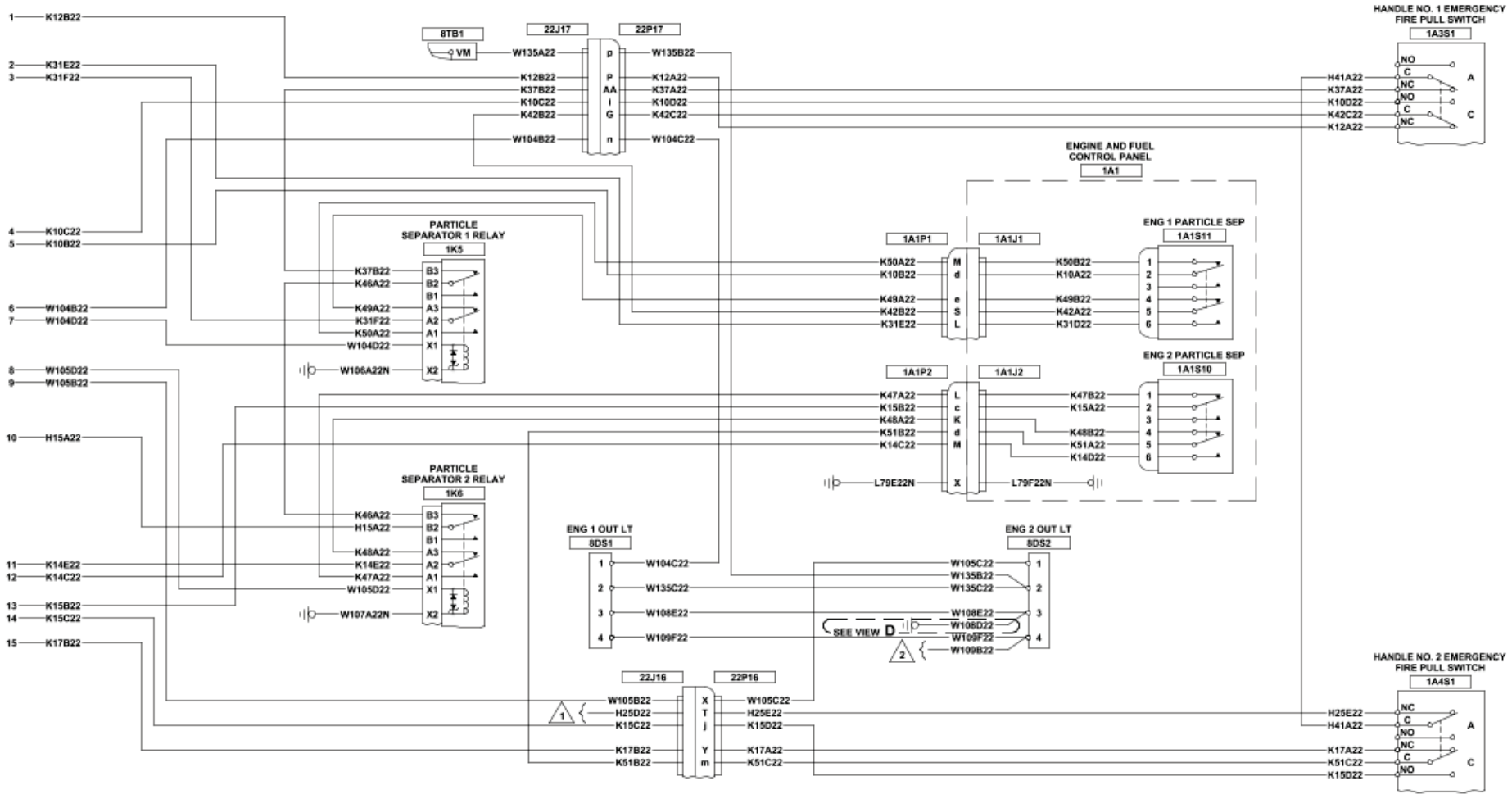
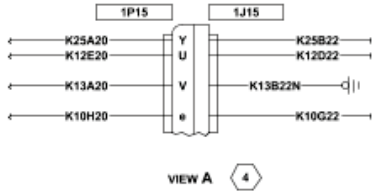
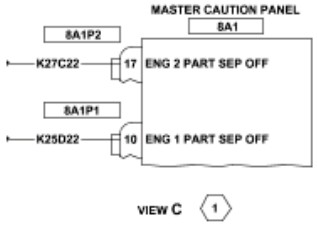
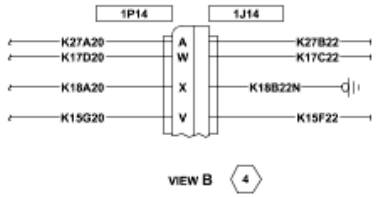
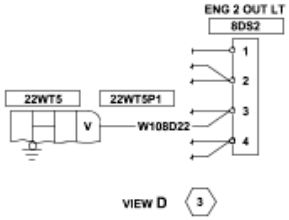


Figure 1. Particle Separator (Sheet 2 of 3)









NOTES

1. REFER TO CABIN HEATING.
2. REFER TO MASTER CAUTION, RPM CAUTION AND PILOT/COPILOT CYCLIC STICK CENTERING INDICATION LIGHT.
3. REFER TO FLIGHT DATA RECORDER.

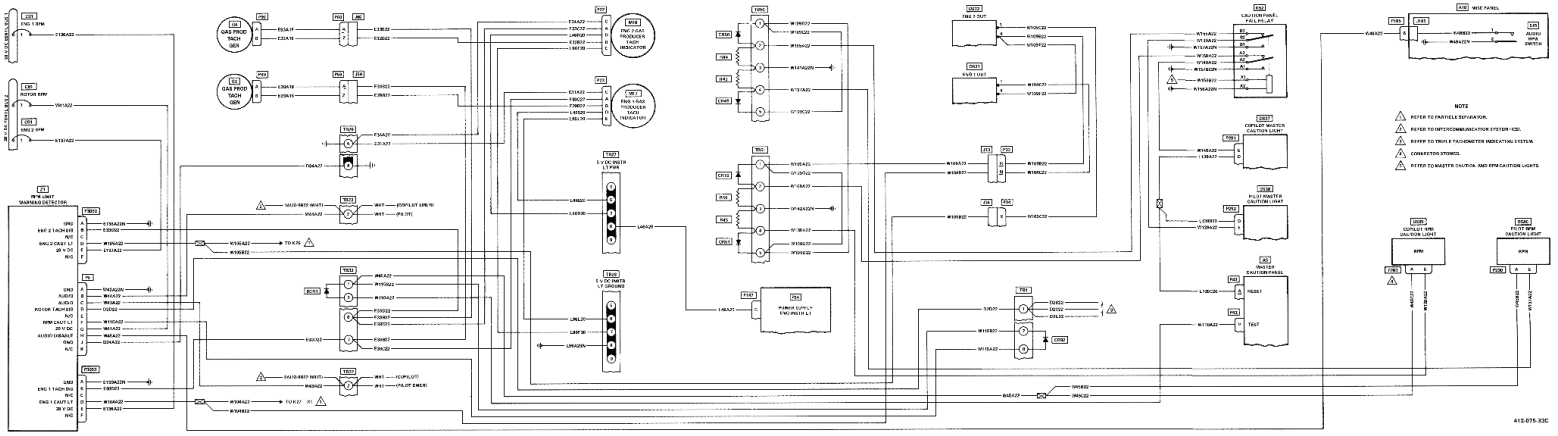
EFFECTIVITY

1. S/N 33108 AND SUBSEQUENT
2. S/N 36037 THROUGH 36086
3. S/N 36087 AND SUBSEQUENT
4. S/N 36694 AND SUBSEQUENT

ICN-412-A-980000-B-97499-00078-A-001-01

Figure 1. Particle Separator (Sheet 3 of 3)

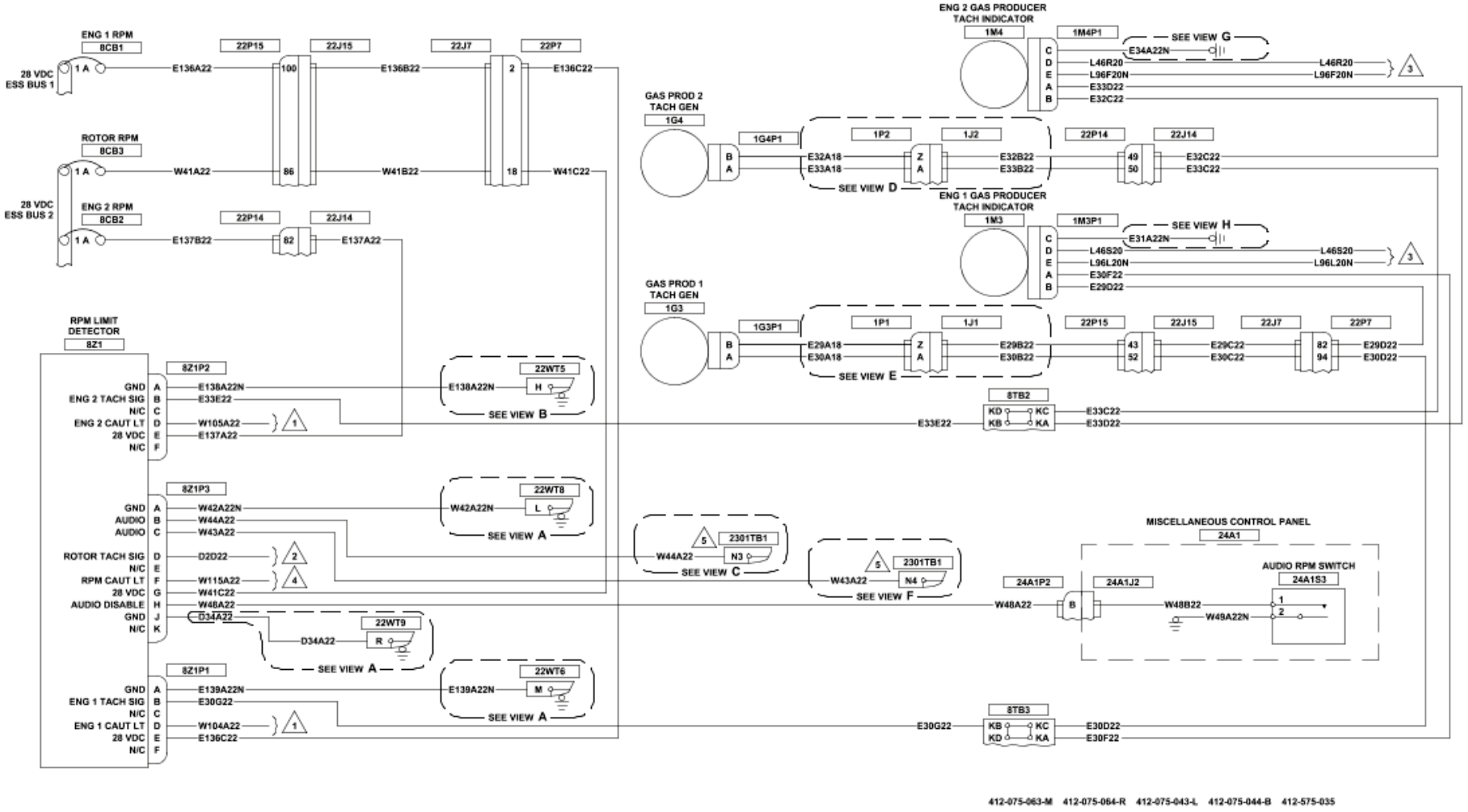
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Figure 1. RPM Limit Warning Detector (Sheet 1 of 1)

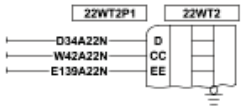
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412-075-063-M 412-075-064-R 412-075-043-L 412-075-044-B 412-575-035

ICN-412-A-980000-B-97499-00080-A-001-01

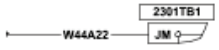
Figure 1. RPM Limit Warning Detector (Sheet 1 of 2)



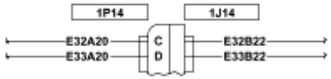
VIEW A (2)



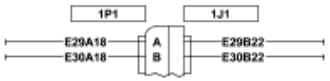
VIEW B (2)



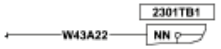
VIEW C (1)



VIEW D (3)



VIEW E (3)



VIEW F (1)



VIEW G (2)



VIEW H (2)

NOTES

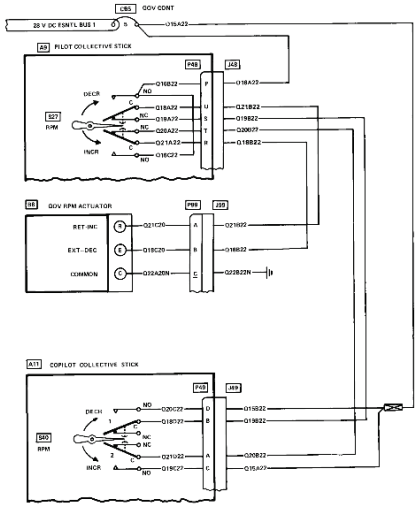
1. REFER TO PARTICLE SEPARATOR.
2. REFER TO TRIPLE TACHOMETER INDICATING SYSTEM.
3. REFER TO ENGINE INSTRUMENT LIGHTS.
4. REFER TO MASTER CAUTION, RPM CAUTION, AND CYCLIC STICK CENTERING INDICATOR LIGHT.
5. REFER TO INTERCOMMUNICATION SYSTEM (ICS) (CHAPTER 97).

EFFECTIVITY

1. S/N 33121 THRU 33138
2. S/N 36087 AND SUBSEQUENT
3. S/N 36694 AND SUBSEQUENT

ICN-412-A-980000-B-97499-00081-A-001-01

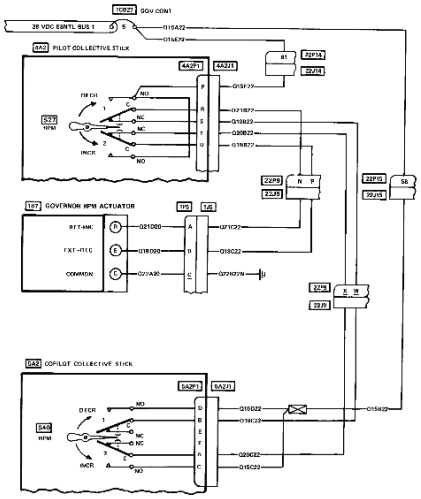
Figure 1. RPM Limit Warning Detector (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00082-A-001-01

Figure 1. Governor RPM (Sheet 1 of 1)

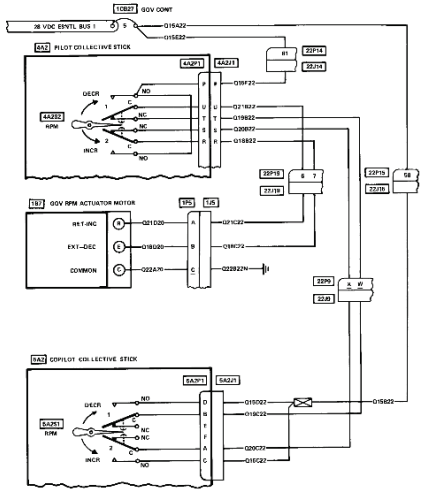
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Figure 1. Governor RPM (Sheet 1 of 1)

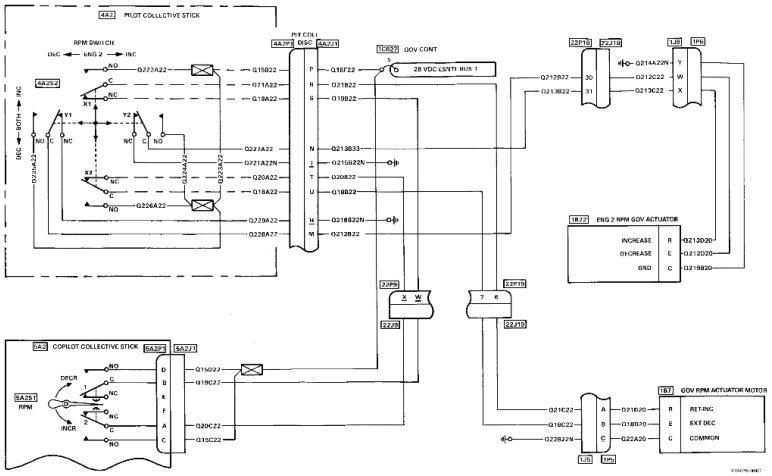
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Figure 1. Governor RPM (Sheet 1 of 1)

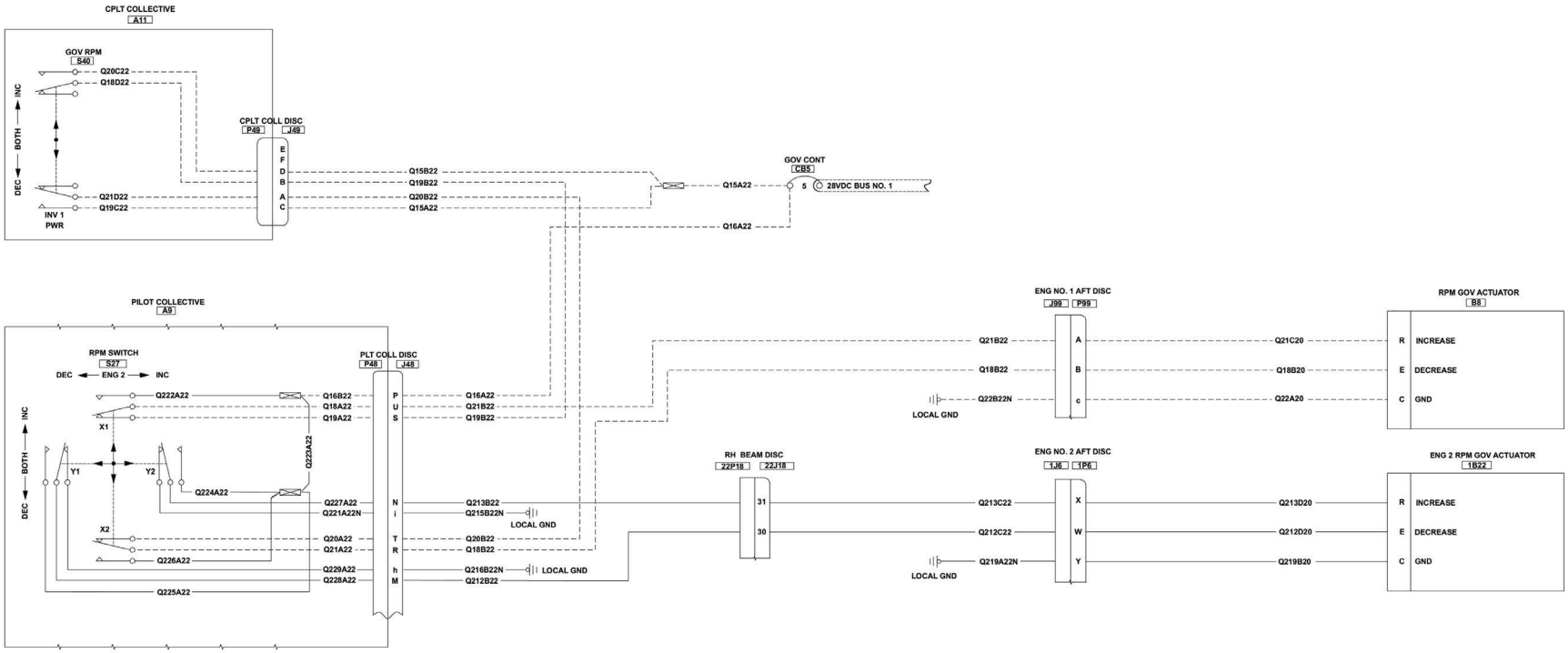
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Figure 1. Governor RPM and Engine 2 RPM Actuator (Sheet 1 of 1)

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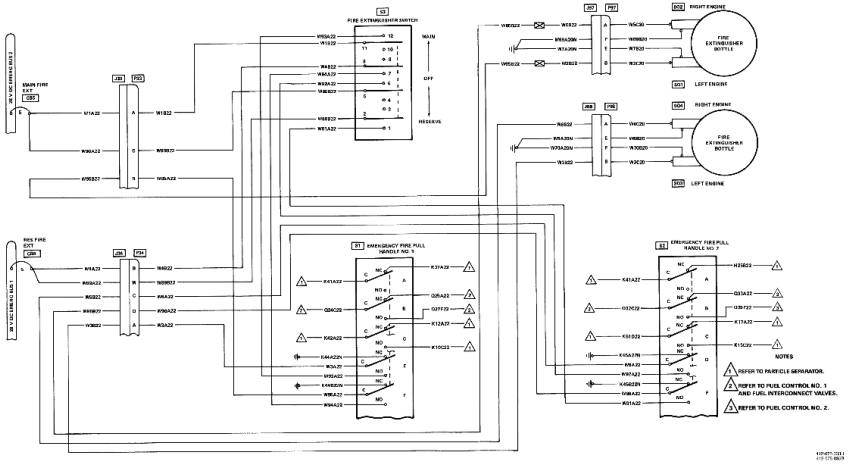


drawing number

ICN-412-A-980000-B-97499-02955-A-001-01

Figure 1. Governor RPM and Engine 2 RPM Actuator - Wiring Diagrams (Sheet 1 of 1)

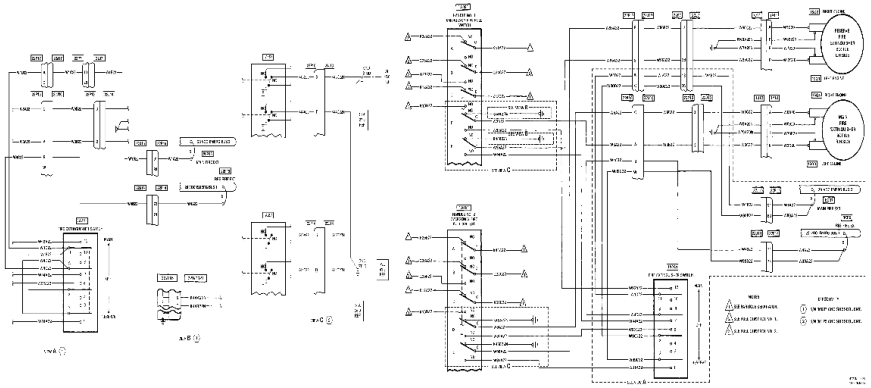
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Figure 1. Fire Extinguisher (Sheet 1 of 1)

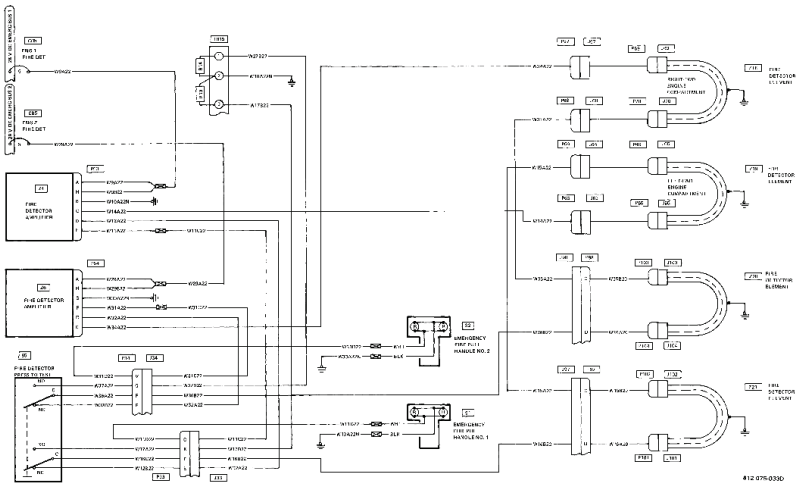
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Figure 1. Fire Extinguisher (Sheet 1 of 1)

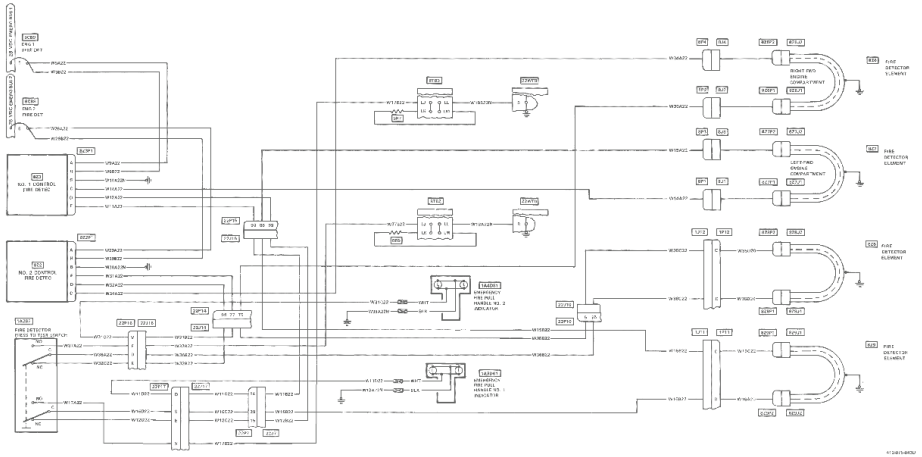
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ICN-412-A-980000-B-97499-00088-A-001-01

Figure 1. Engine Fire Detection System (Sheet 1 of 1)

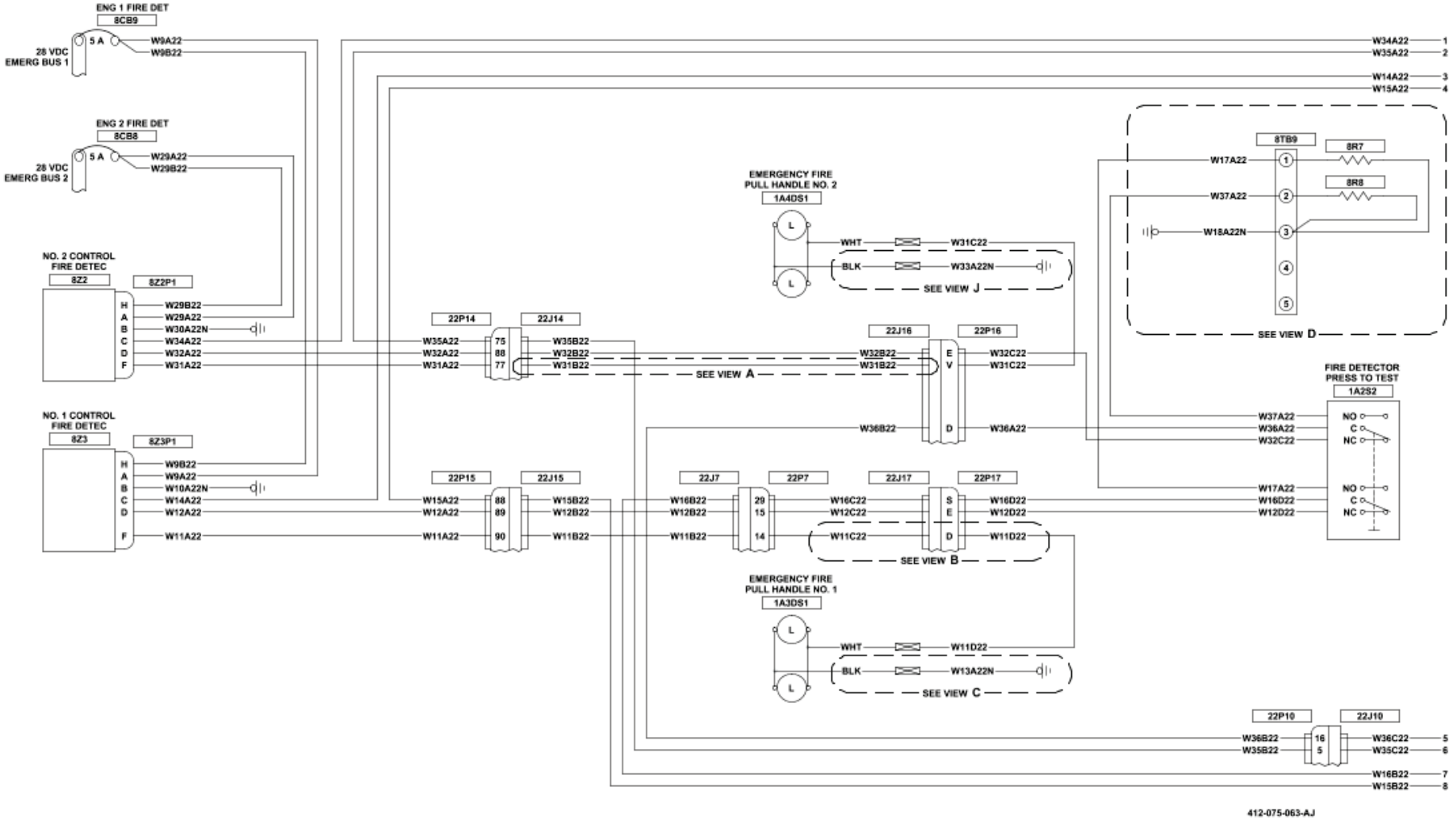
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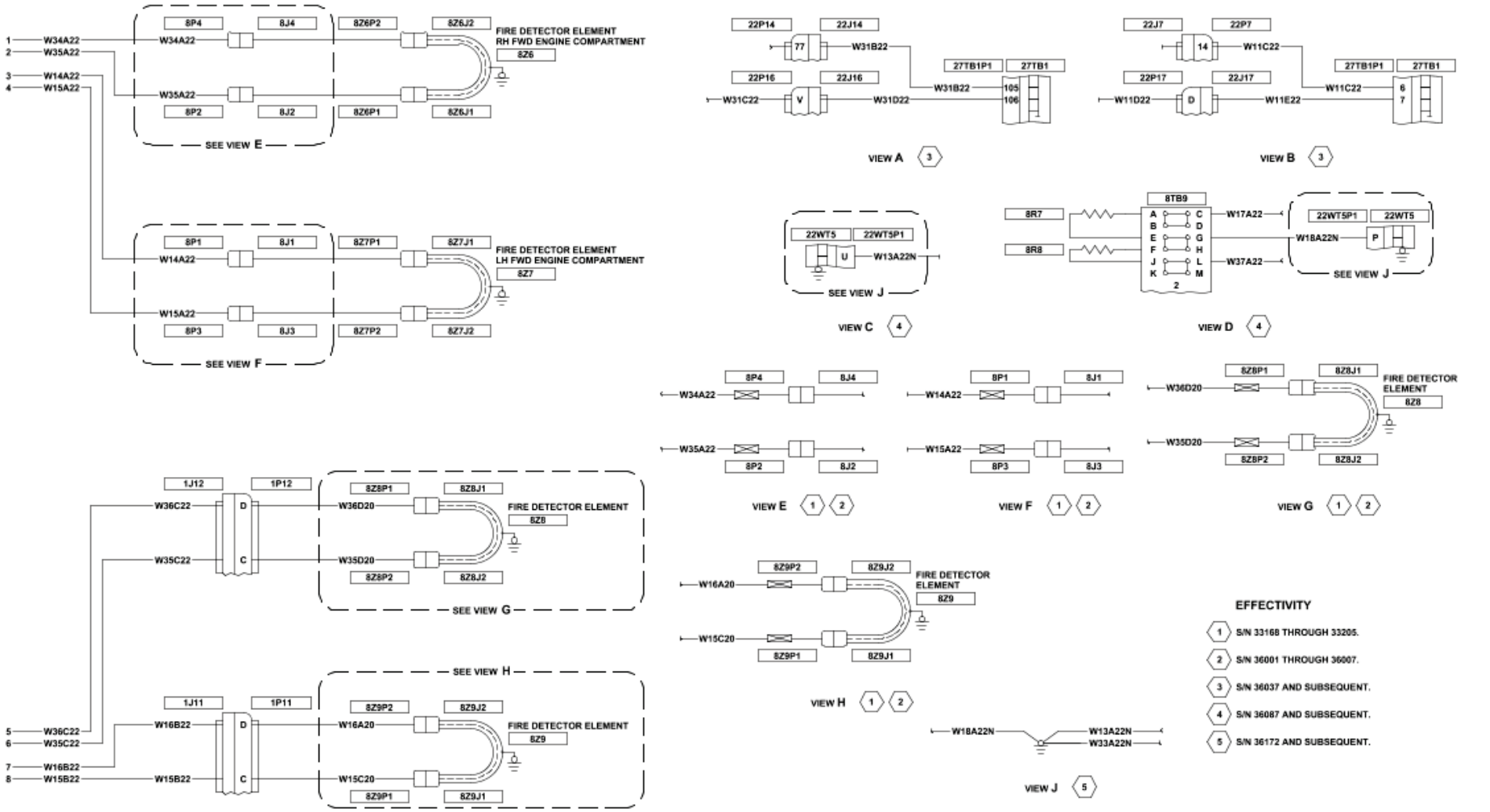
Figure 1. Engine Fire Detector System (Sheet 1 of 1)

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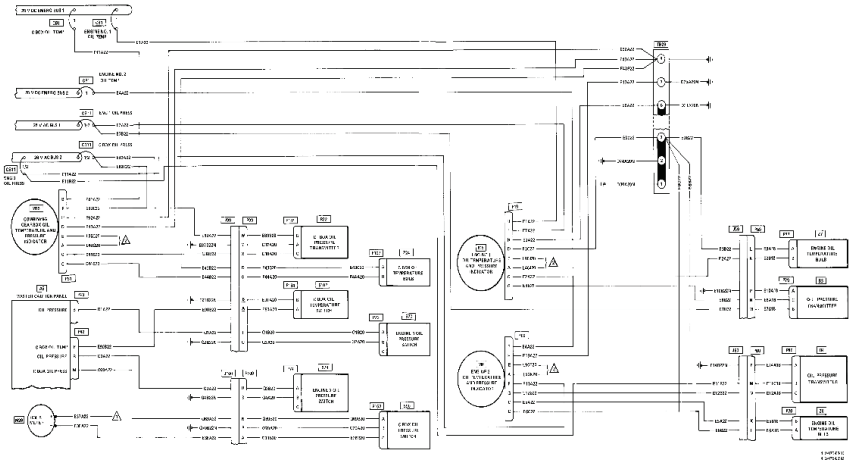
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Figure 1. Engine Fire Detection System (Sheet 1 of 2)



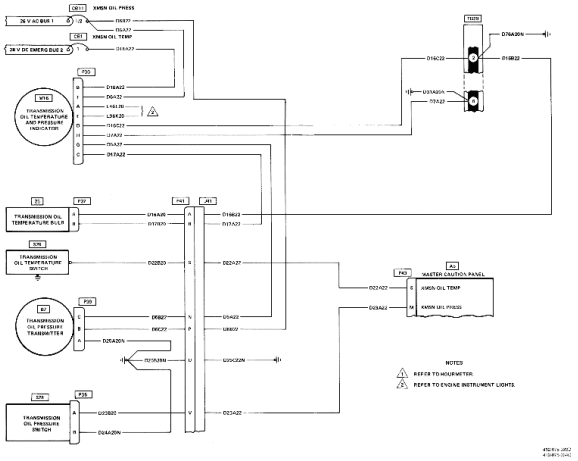
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Figure 1. Engine Fire Detection System (Sheet 2 of 2)



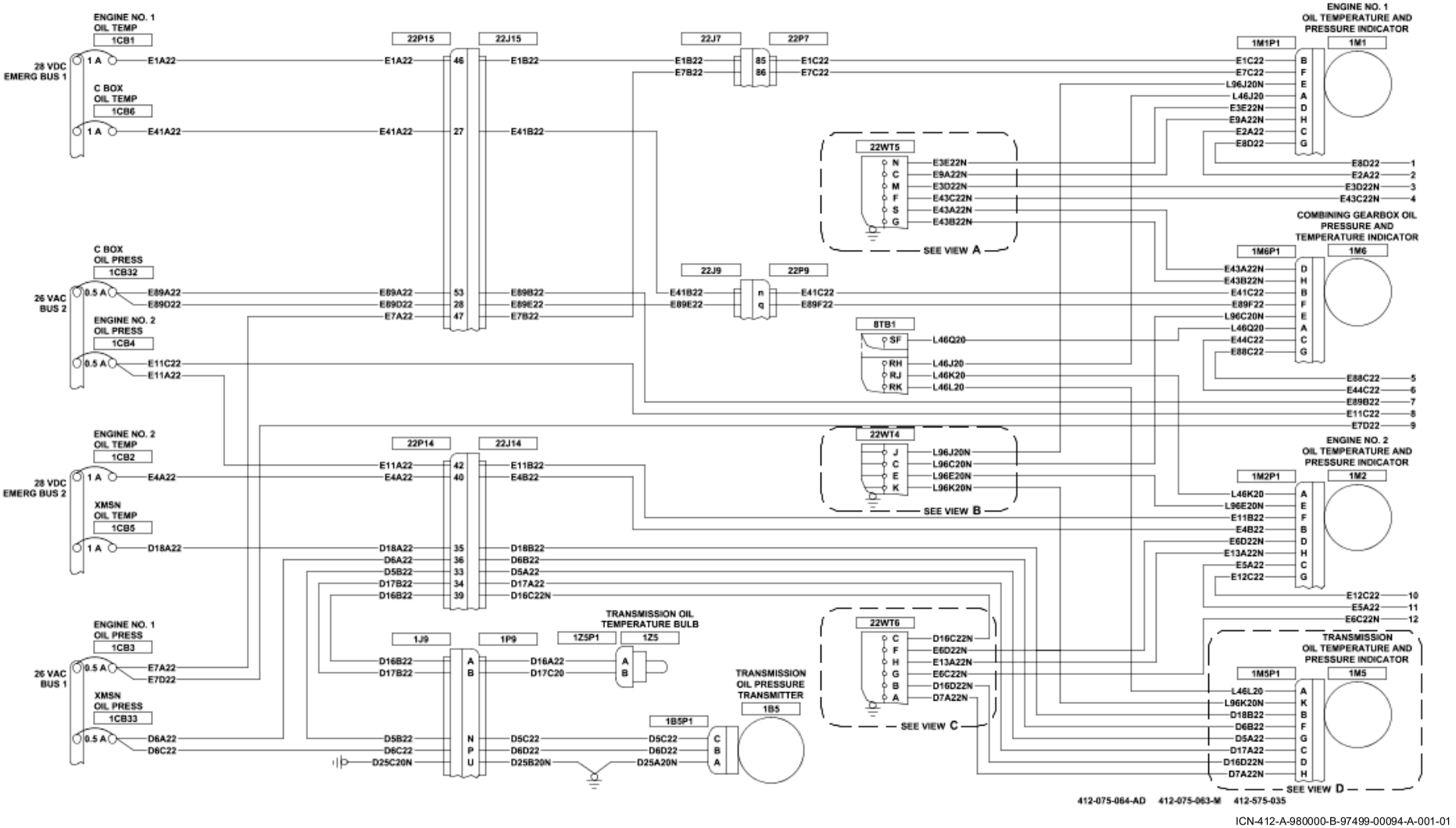
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Figure 1. Oil Pressure and Temperature Indicating Systems (Sheet 1 of 2)



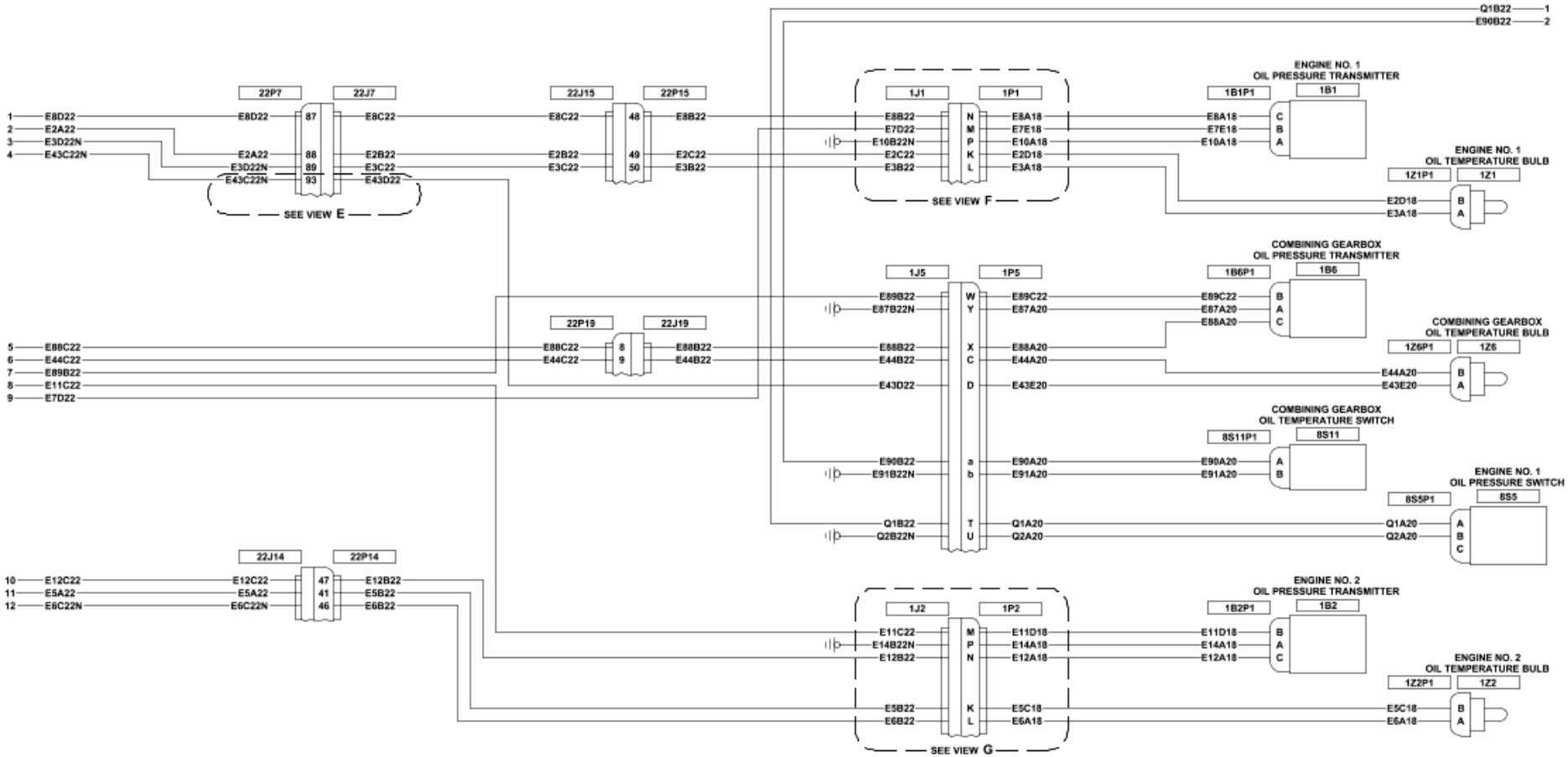
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Figure 1. Oil Pressure and Temperature Indicating Systems (Sheet 2 of 2)



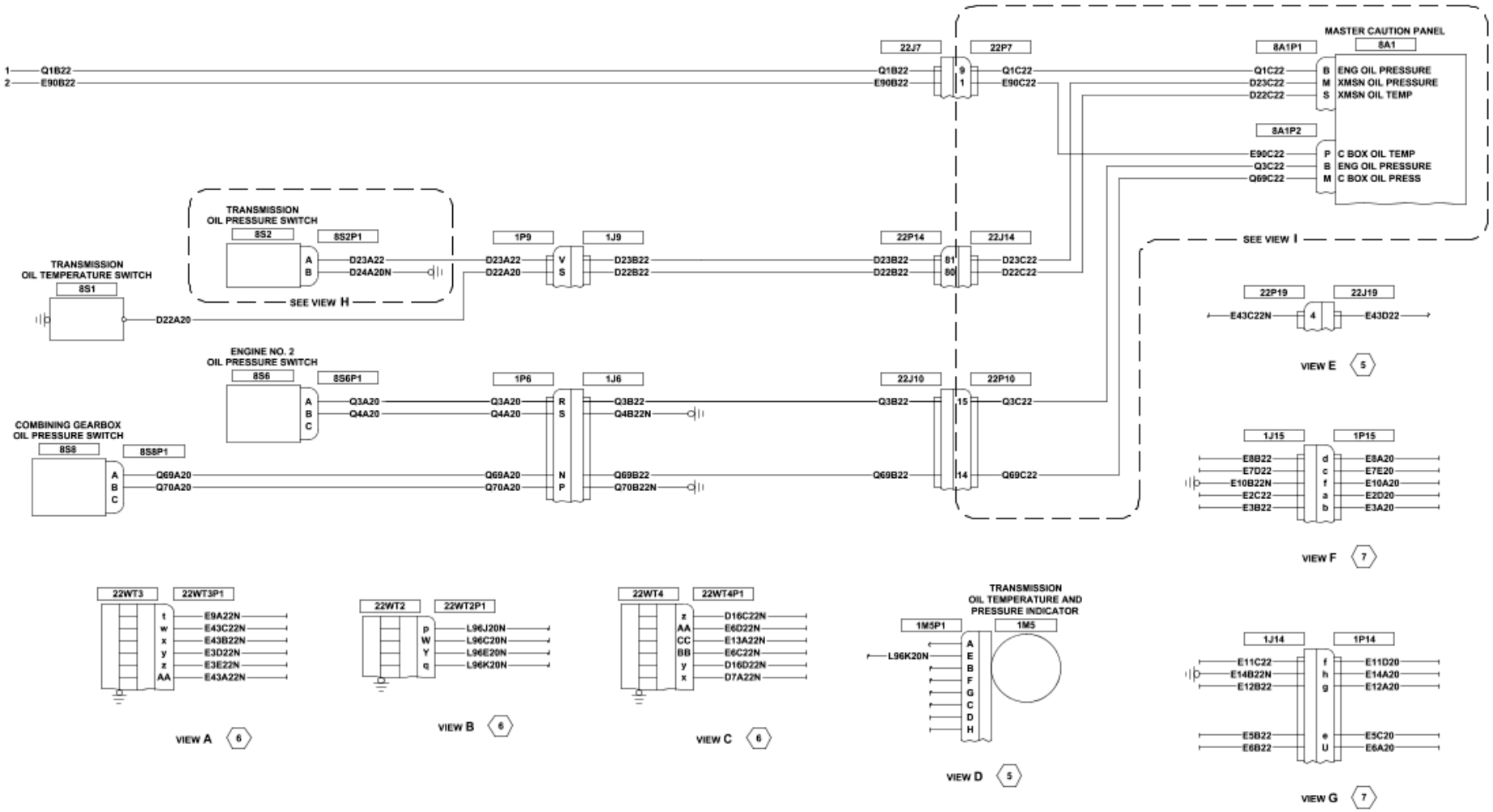
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Figure 1. Oil Pressure and Temperature Indicating Systems (Sheet 1 of 4)



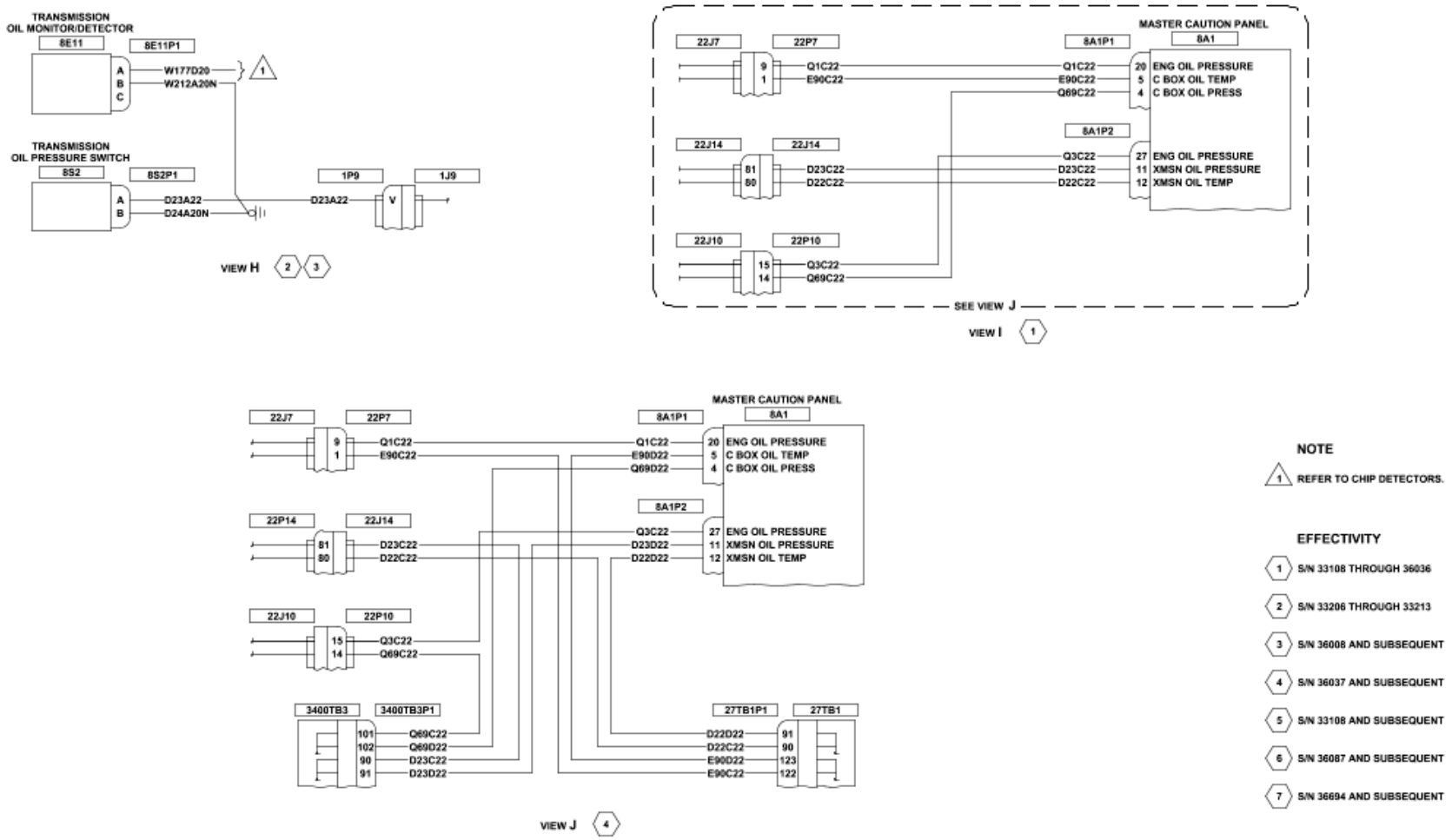
ICN-412-A-980000-B-97499-00095-A-001-01

Figure 1. Oil Pressure and Temperature Indicating Systems (Sheet 2 of 4)



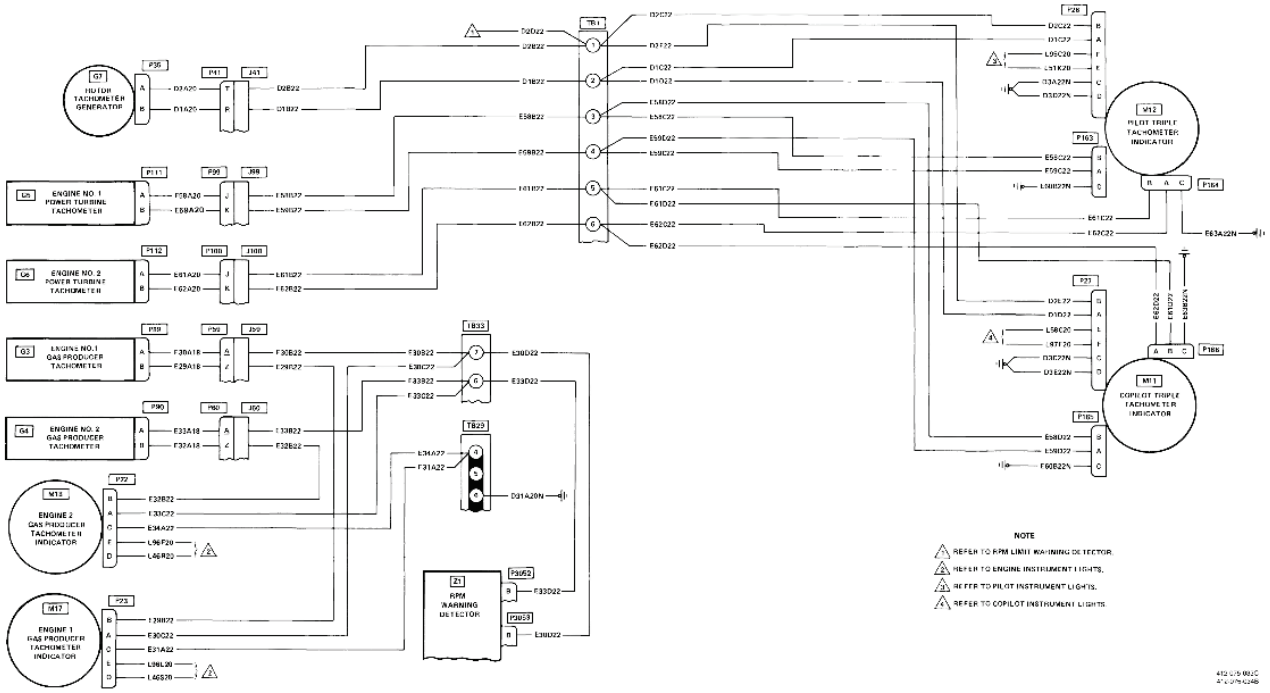
ICN-412-A-980000-B-97499-00096-A-001-01

Figure 1. Oil Pressure and Temperature Indicating Systems (Sheet 3 of 4)



ICN-412-A-980000-B-97499-00097-A-001-01

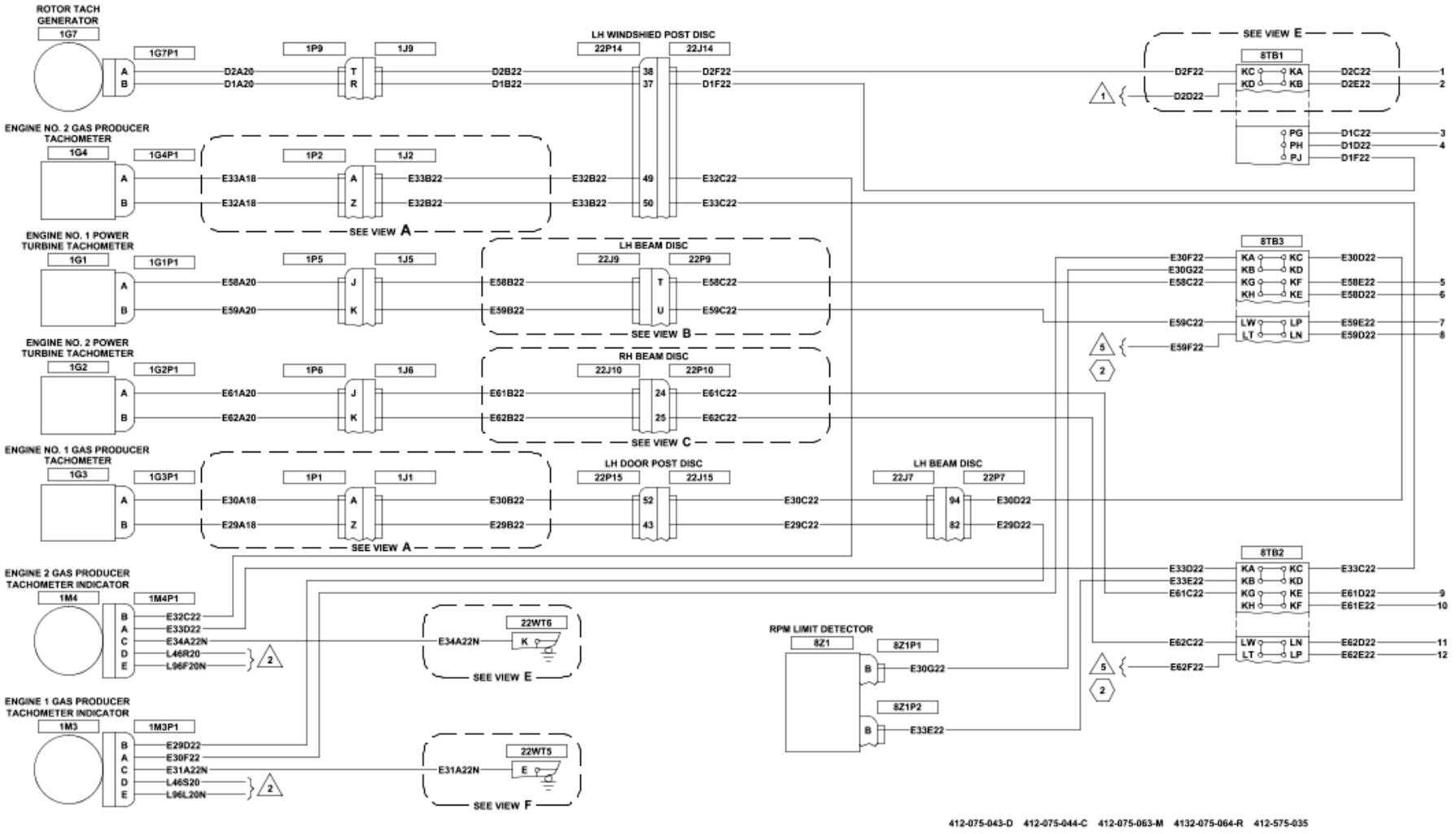
Figure 1. Oil Pressure and Temperature Indicating Systems (Sheet 4 of 4)



ICN-412-A-980000-B-97499-00098-A-001-01

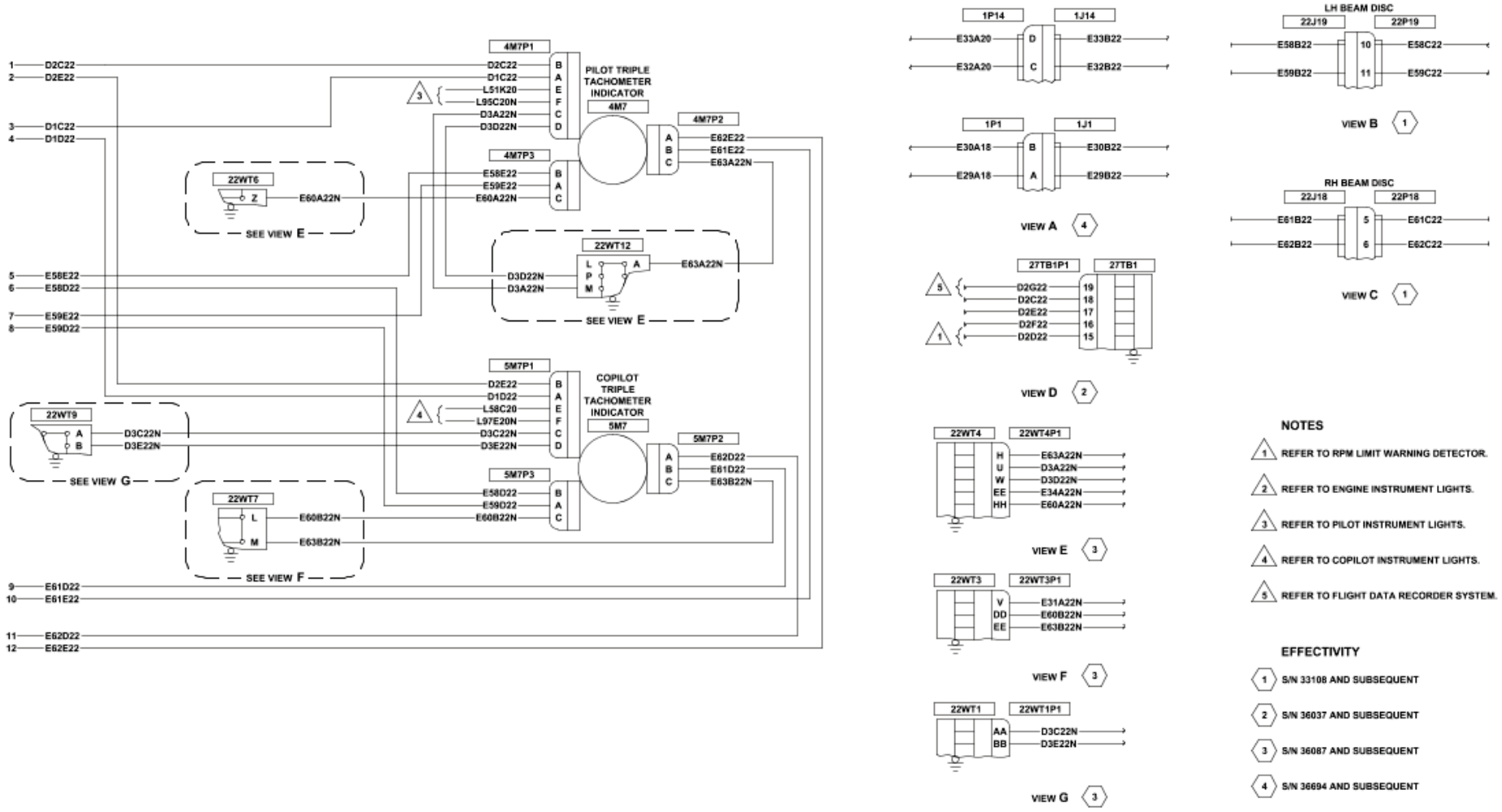
Figure 1. Triple Tachometer Indicating Systems (Sheet 1 of 1)

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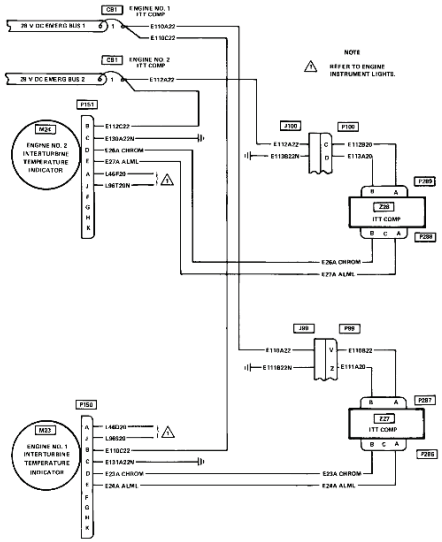
ICN-412-A-980000-B-97499-00099-A-001-01

Figure 1. Triple Tachometer Indicating Systems (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00100-A-001-01

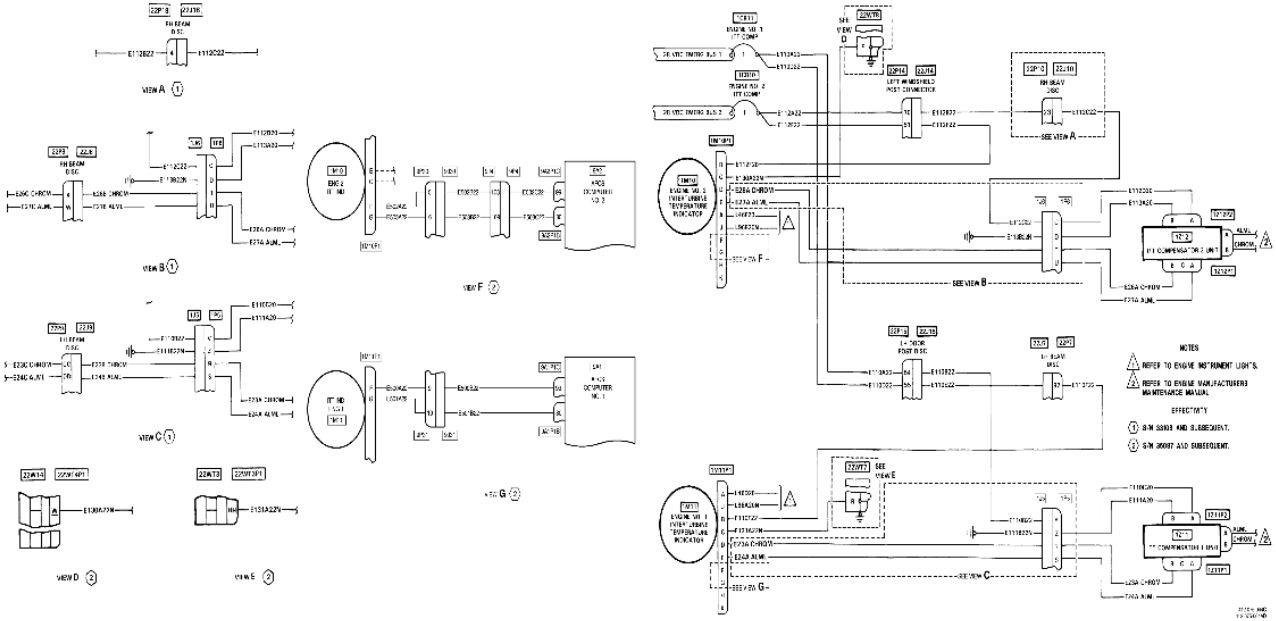
Figure 1. Triple Tachometer Indicating Systems (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00101-A-001-01

Figure 1. Interturbine Temperature Indicating Systems (Sheet 1 of 1)

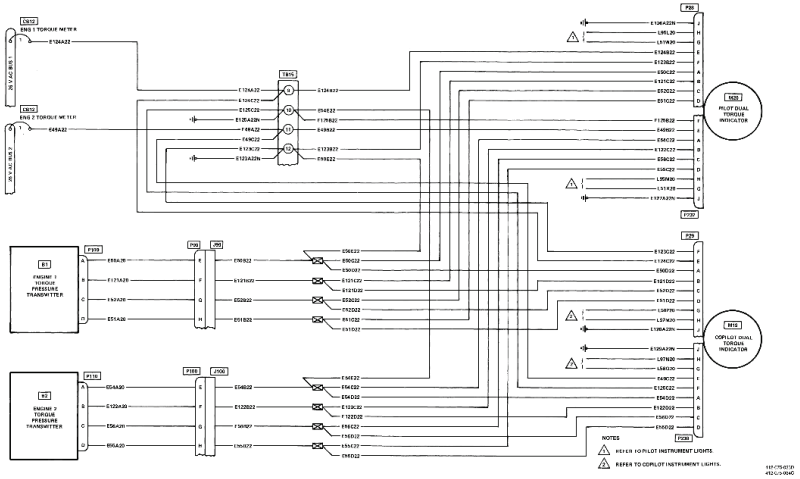
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Figure 1. Interturbine Temperature Indicating Systems (Sheet 1 of 1)

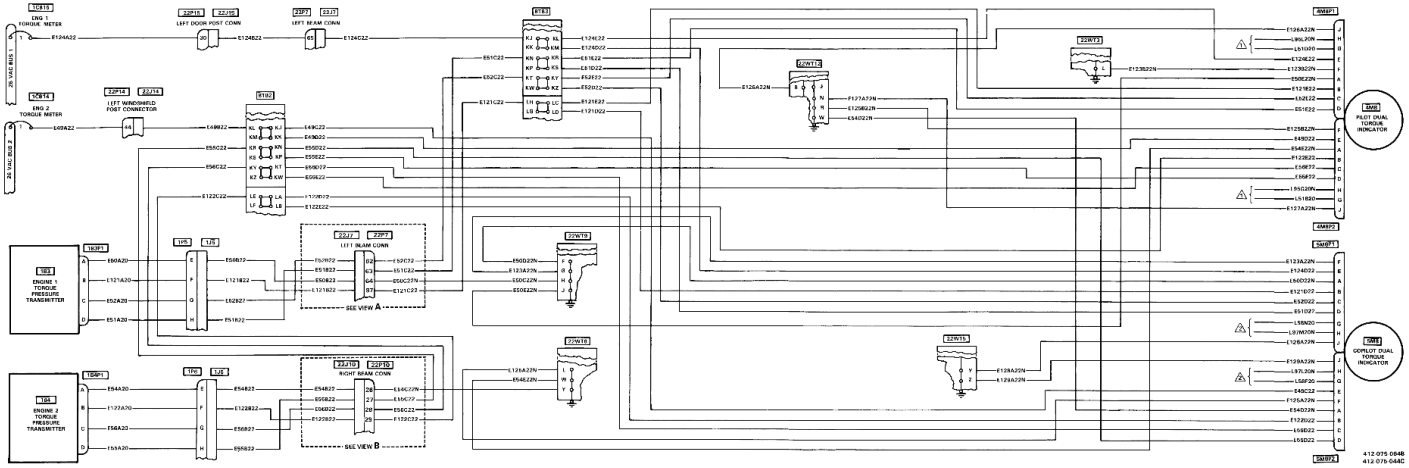
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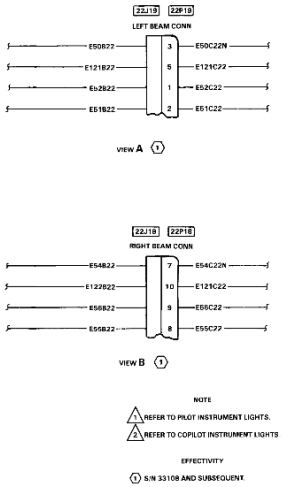
Figure 1. Torque Indicator Systems (Sheet 1 of 1)

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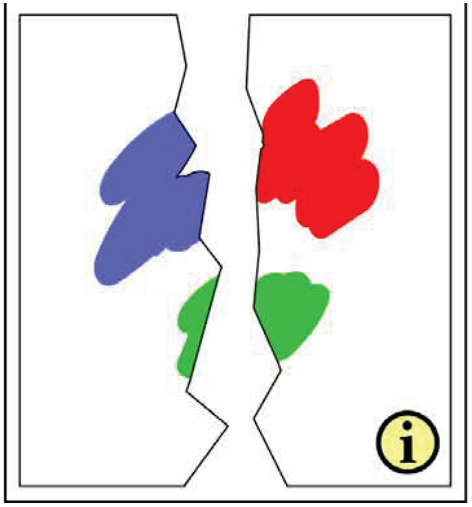
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Figure 1. Torque Indicator Systems (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00105-A-001-01

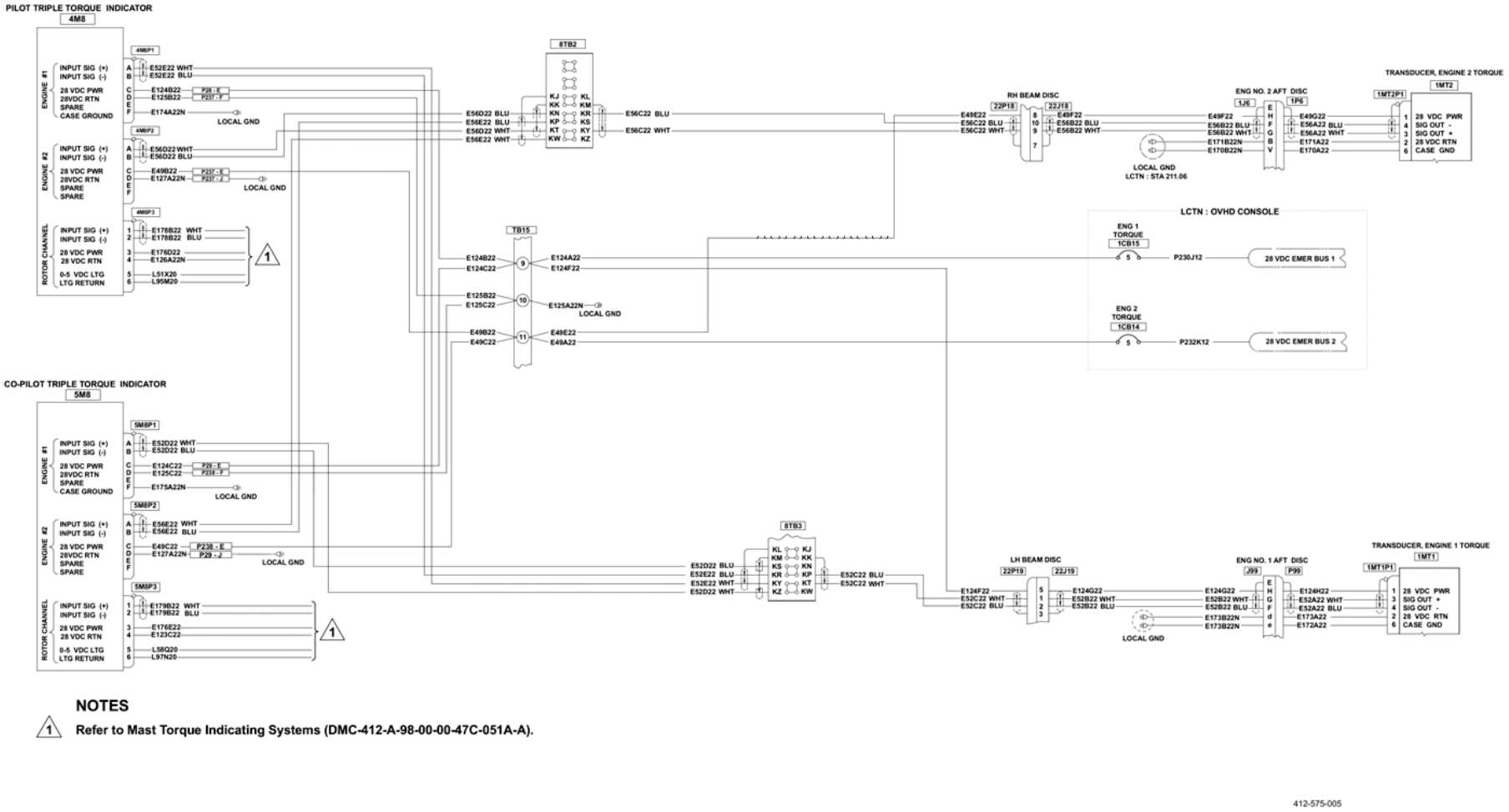
Figure 1. Torque Indicator Systems (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00106-A-001-01

Figure 1. Engine Torque Indicating Systems (Sheet 1 of 1)

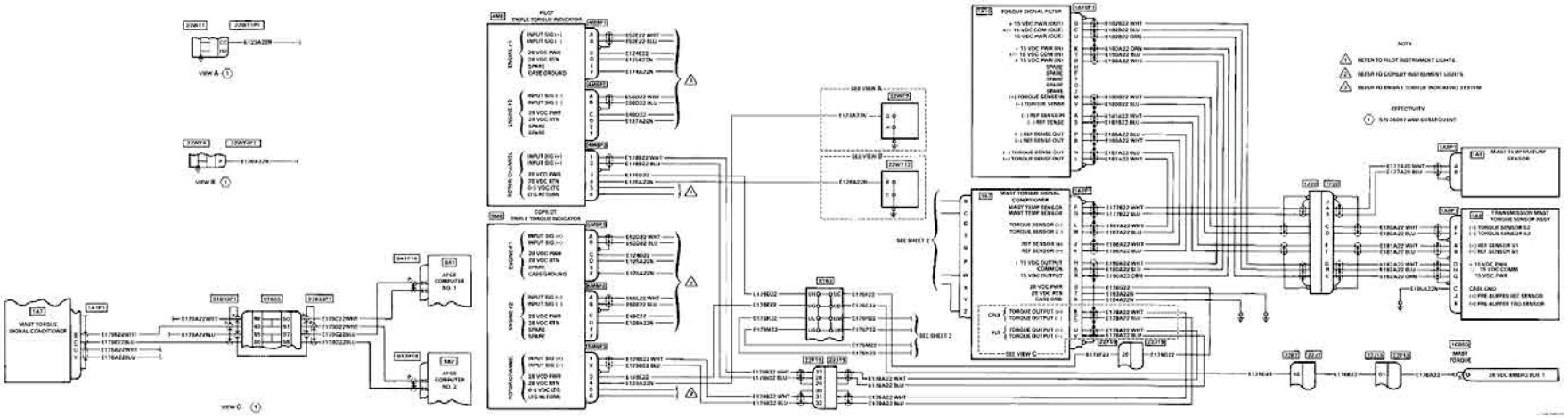
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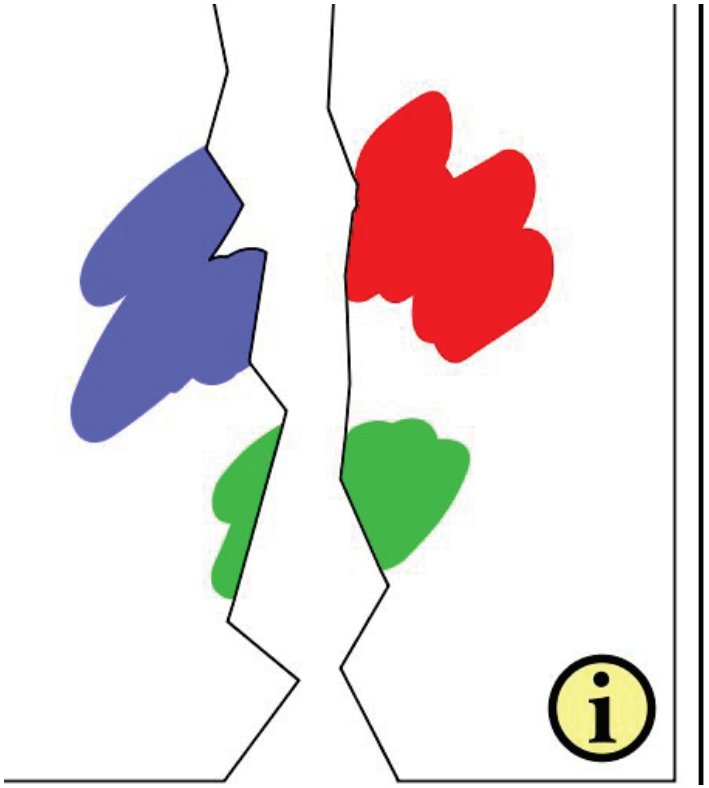
Figure 1. Engine Torque Indicating Systems - Wiring Diagrams (Sheet 1 of 1)

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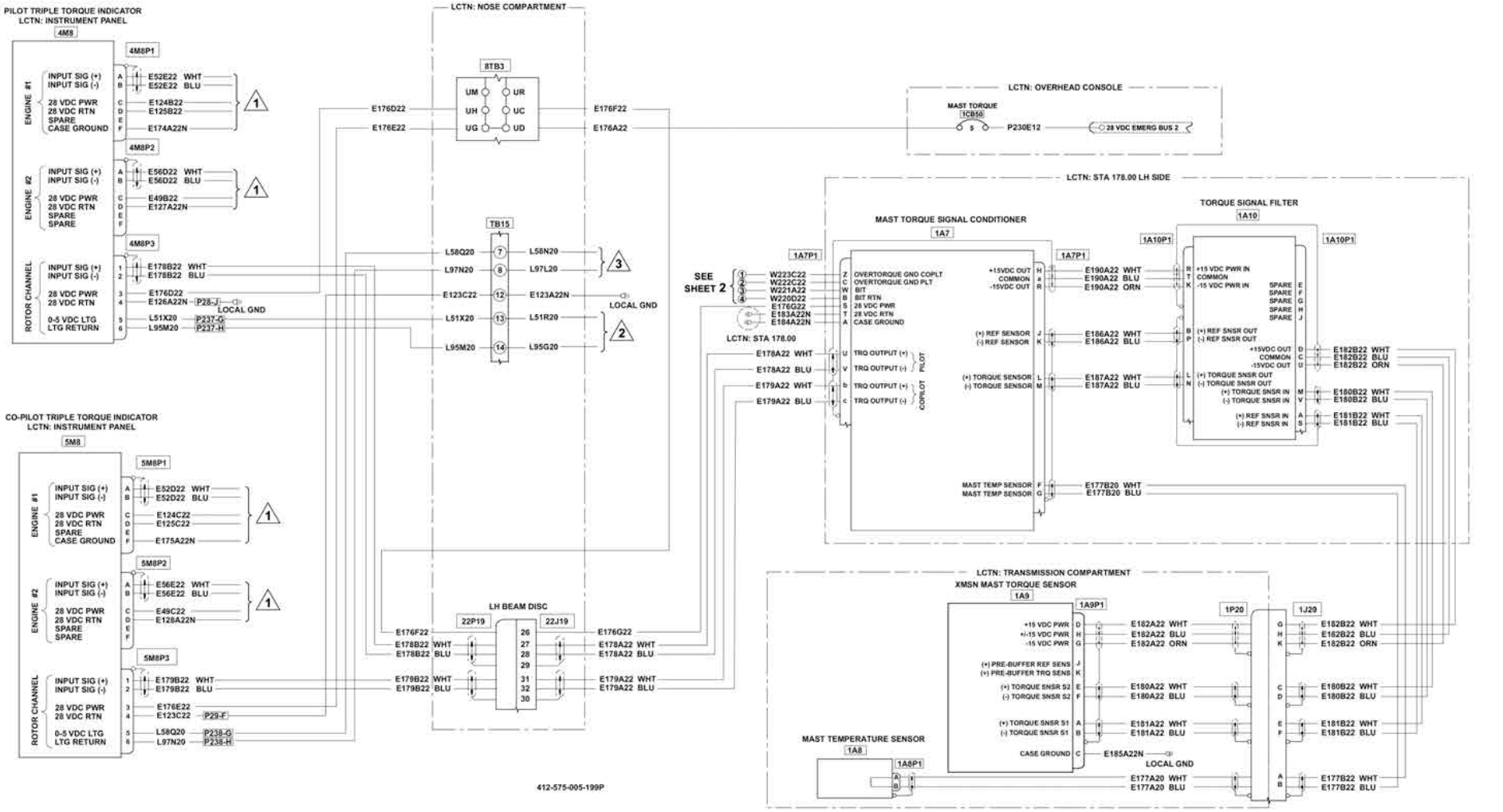
Figure 1. Mast Torque Indicating Systems (Sheet 1 of 2)



412-075-063M
412-075-064R

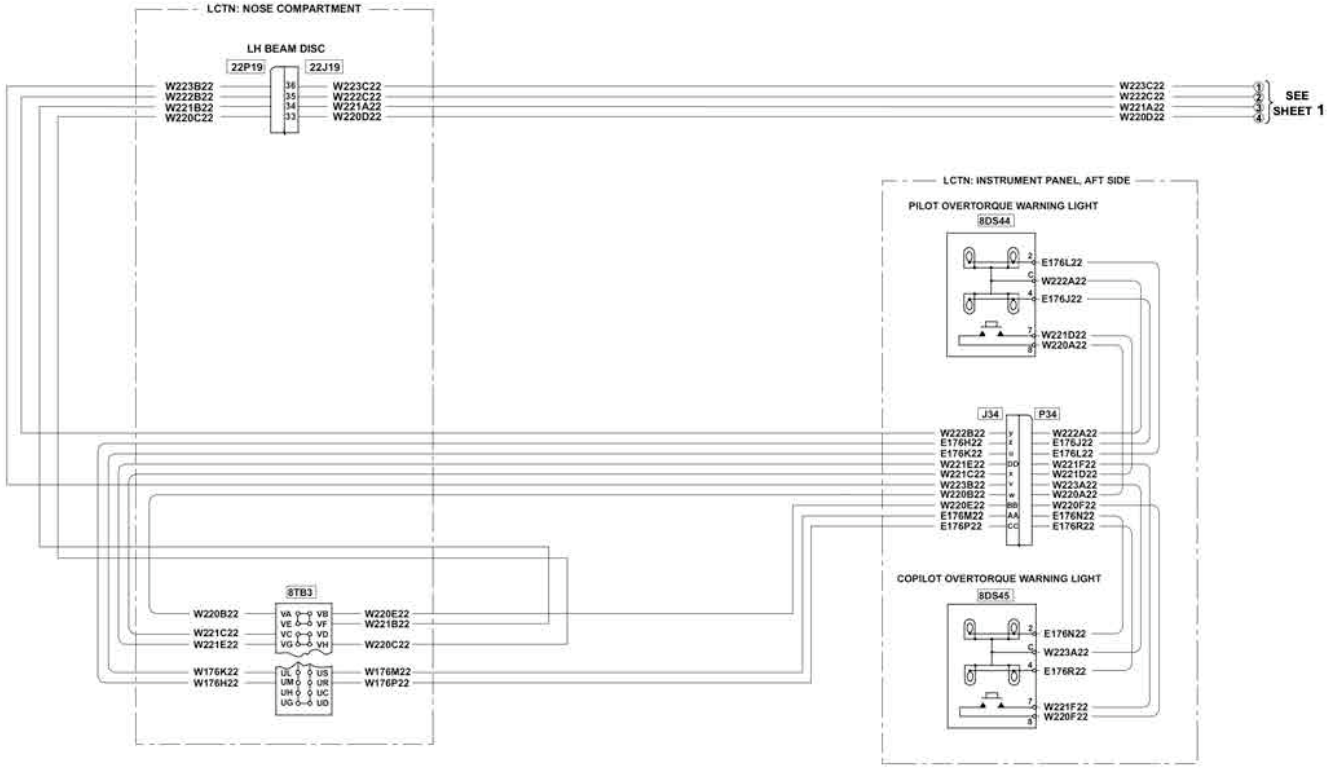
ICN-412-A-980000-B-97499-00108-A-001-01

Figure 1. Mast Torque Indicating Systems (Sheet 2 of 2)



ICN-412-A-980000-B-97499-02952-A-001-01

Figure 1. Engine Torque Indicating Systems - Wiring Diagrams (Sheet 1 of 2)

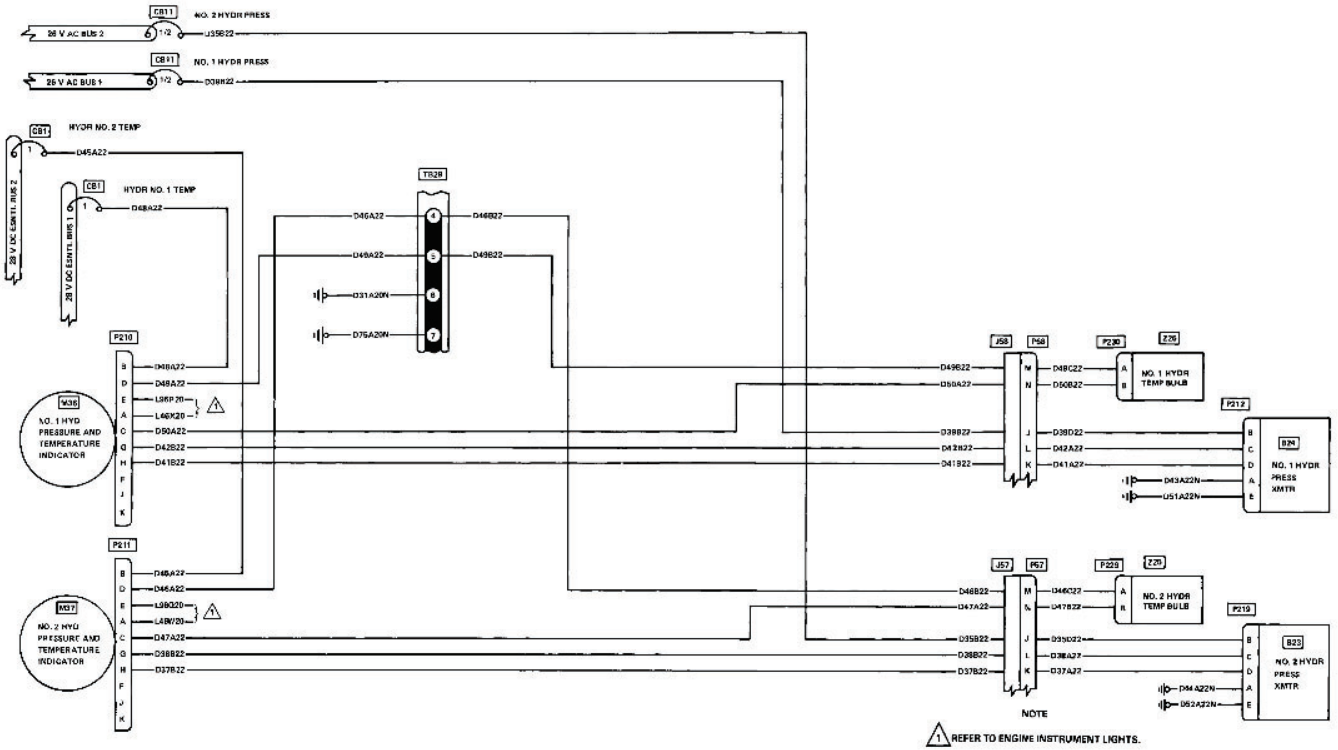


NOTES

1. Refer to engine torque indicating systems (DMC-412-A-98-00-00-46C-051A-A).
2. Refer to pilot instrument (DMC-412-A-98-00-00-70A-051A-A).
3. Refer to copilot instrument lights and passenger warning light (DMC-412-A-98-00-00-72A-051A-A).

ICN-412-A-980000-B-97499-02953-A-001-01

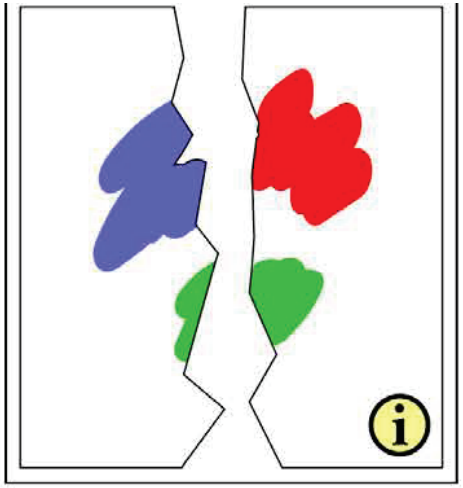
Figure 1. Engine Torque Indicating Systems - Wiring Diagrams (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00109-A-001-01

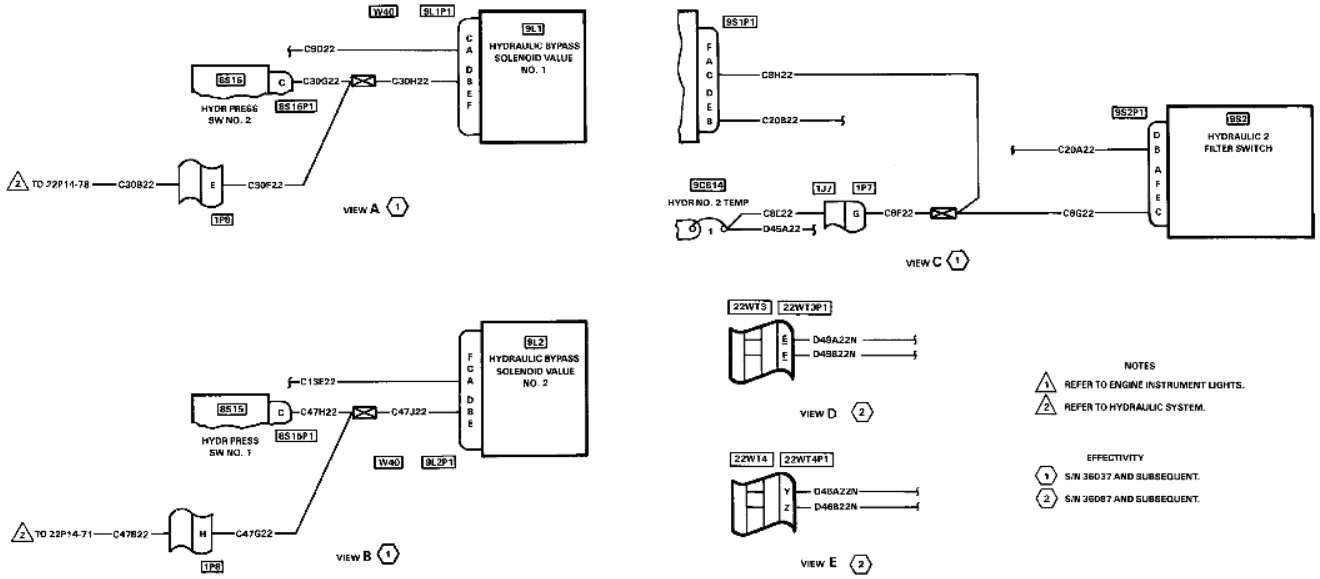
Figure 1. Hydraulic Pressure and Temperature Indicting System (Sheet 1 of 1)

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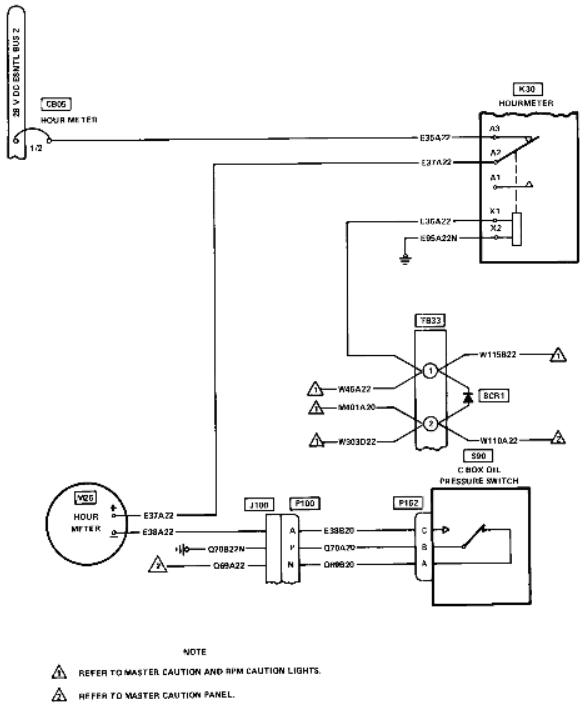
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Figure 1. Hydraulic Pressure and Temperature Indicating System (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00111-A-001-01

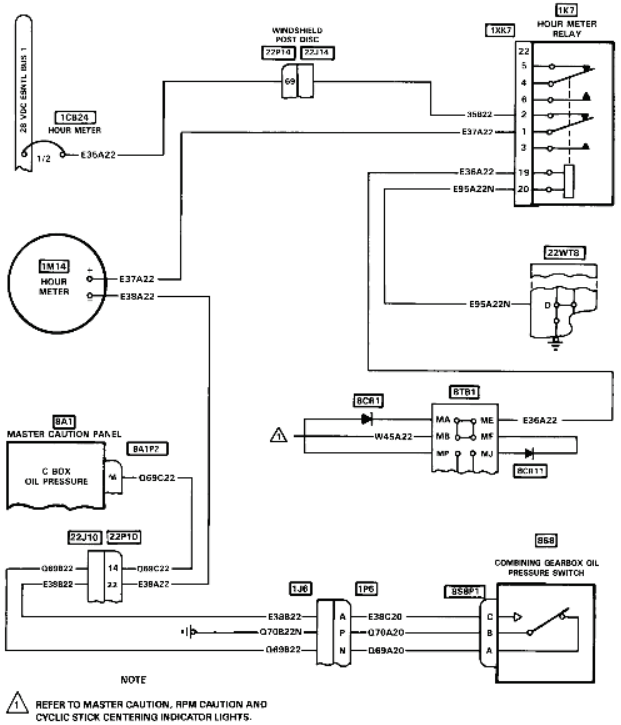
Figure 1. Hydraulic Pressure and Temperature Indicating System (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00112-A-001-01

Figure 1. Hourmeter (Sheet 1 of 1)

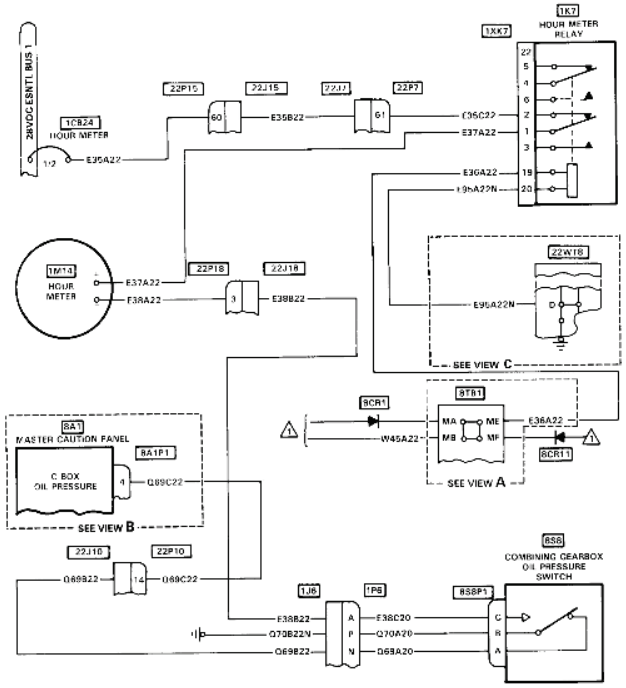
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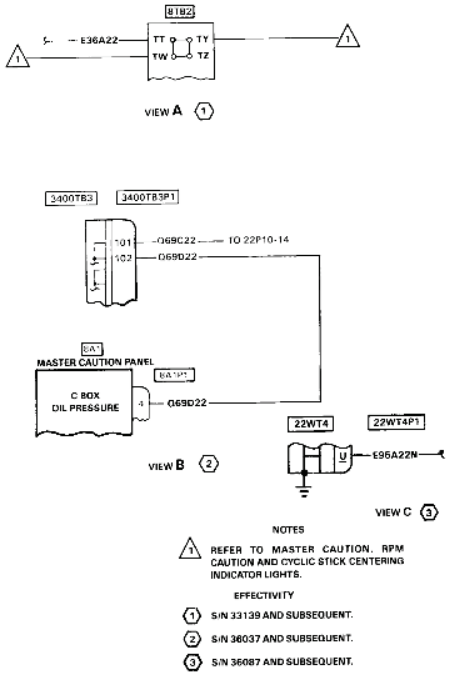
Figure 1. Hourmeter (Sheet 1 of 1)

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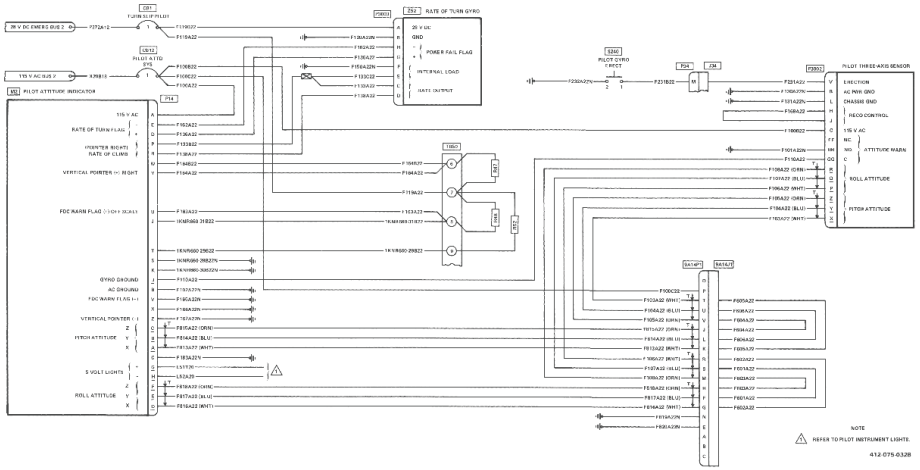
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Figure 1. Hourmeter (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00115-A-001-01

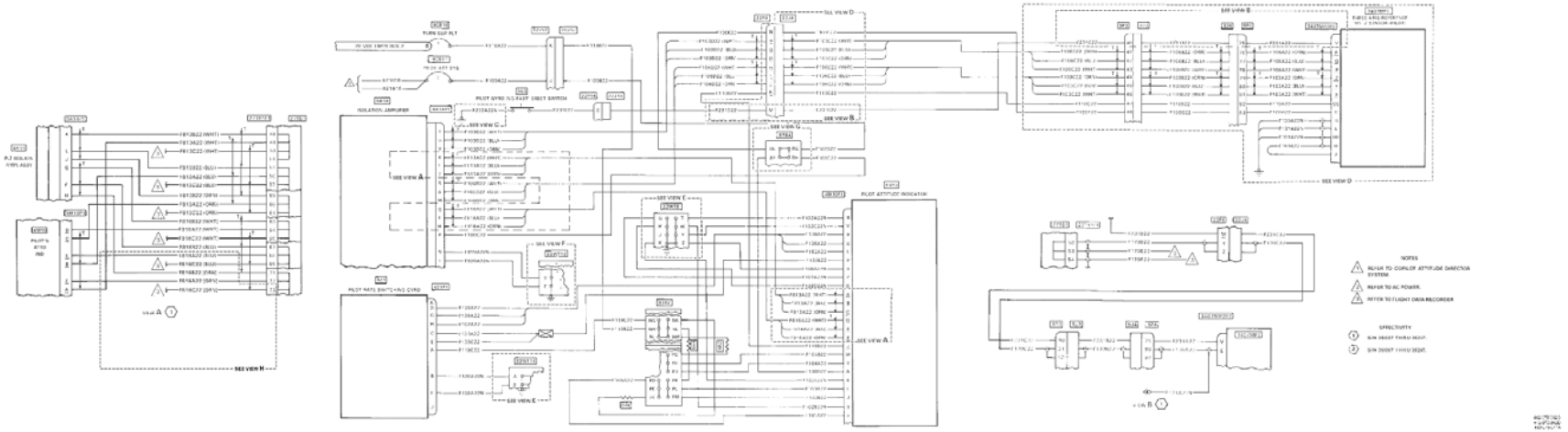
Figure 1. Hourmeter (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00116-A-001-01

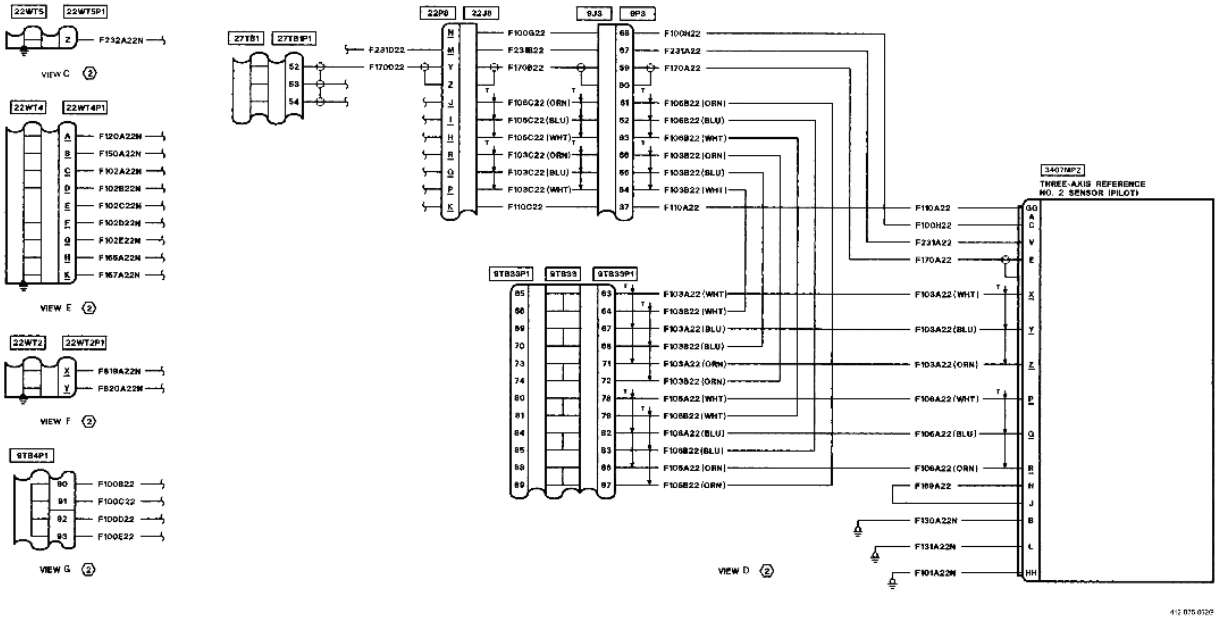
Figure 1. Pilot Attitude Director System (Sheet 1 of 1)

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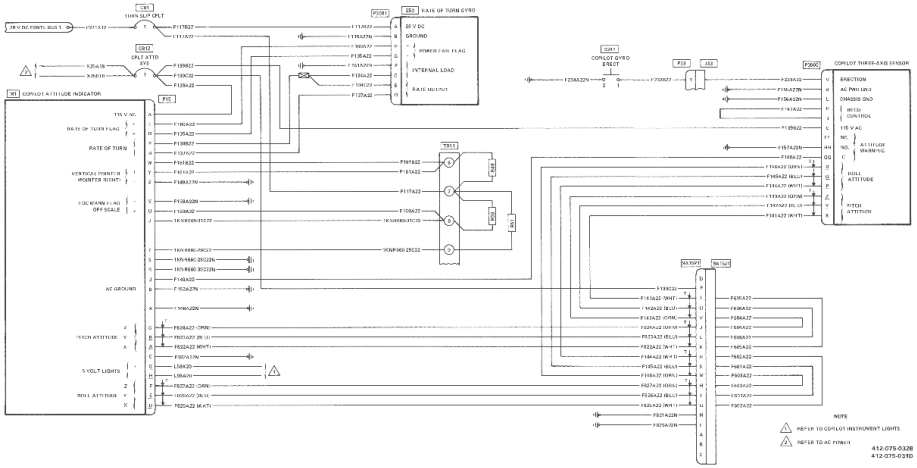
ICN-412-A-980000-B-97499-00117-A-001-01

Figure 1. Pilot Attitude Director System (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00118-A-001-01

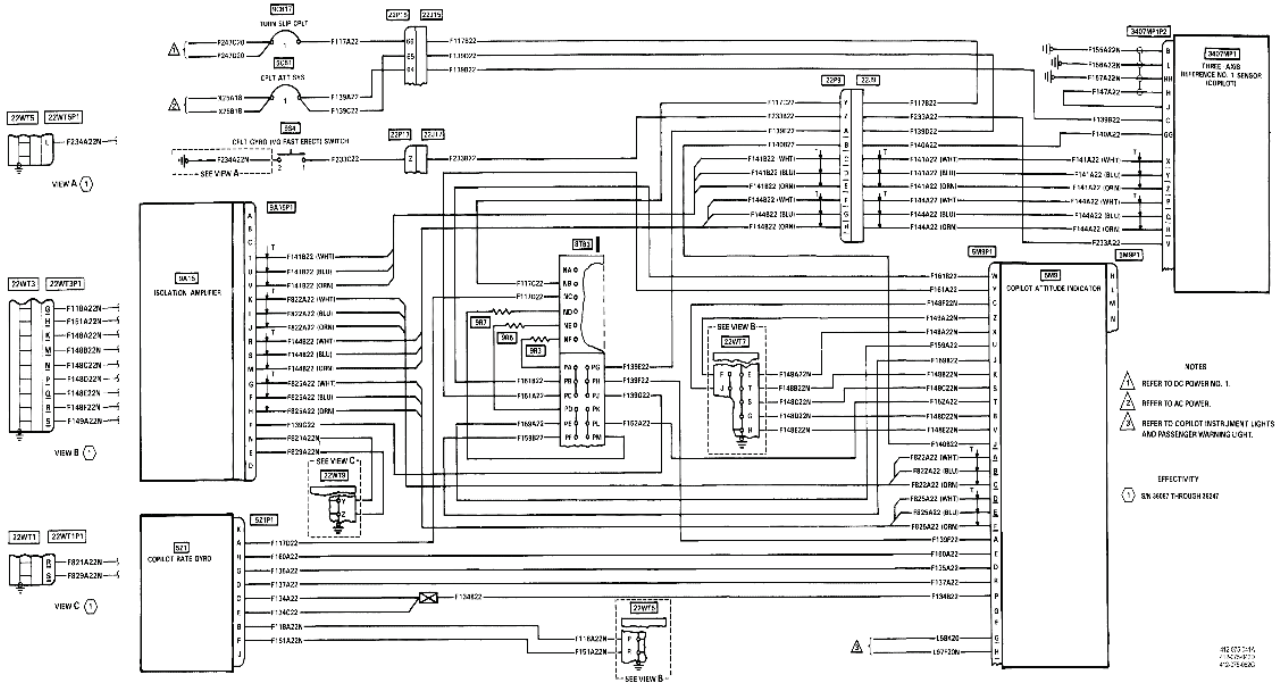
Figure 1. Pilot Attitude Director System (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00119-A-001-01

Figure 1. Copilot Attitude Director System (Sheet 1 of 1)

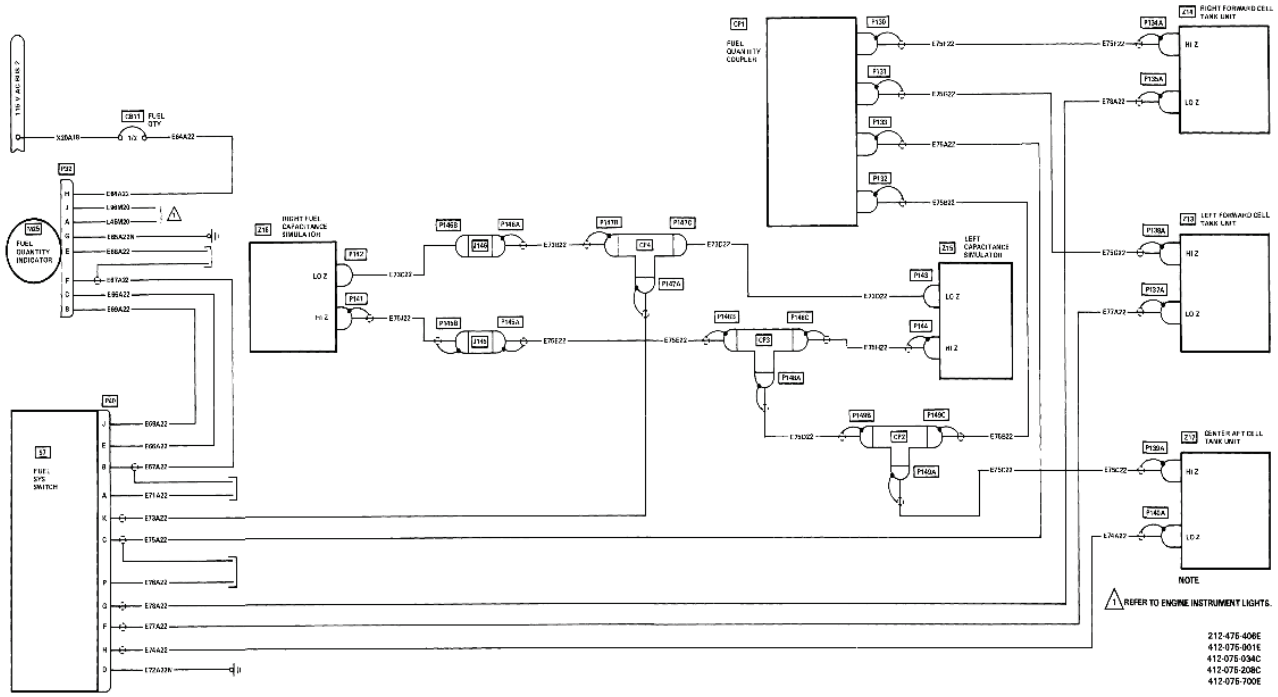
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ICN-412-A-980000-B-97499-00120-A-001-01

Figure 1. Copilot Attitude Director System (Sheet 1 of 1)

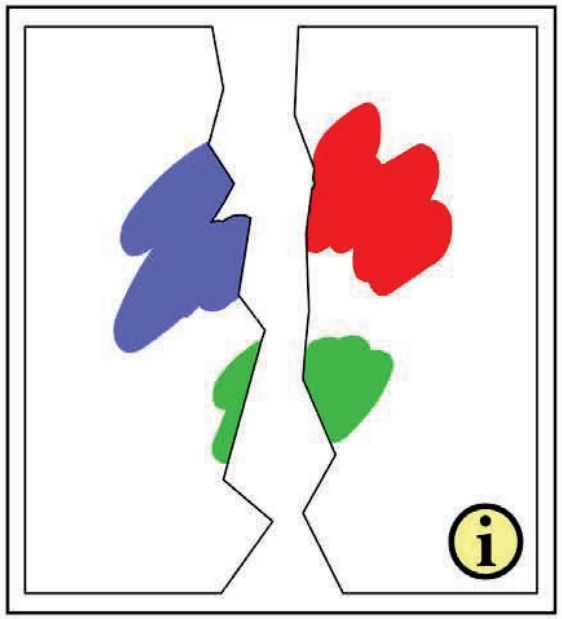
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ICN-412-A-980000-B-97499-00121-A-001-01

Figure 1. Fuel Quantity Indicating System (Sheet 1 of 1)

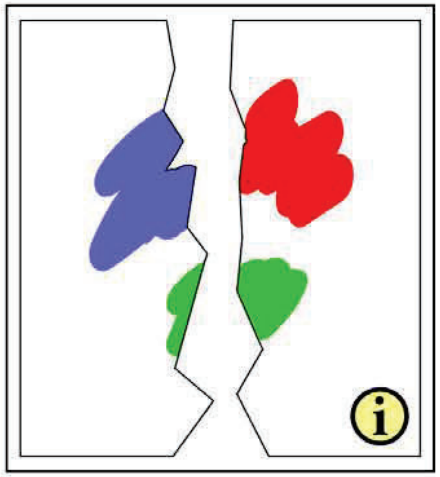
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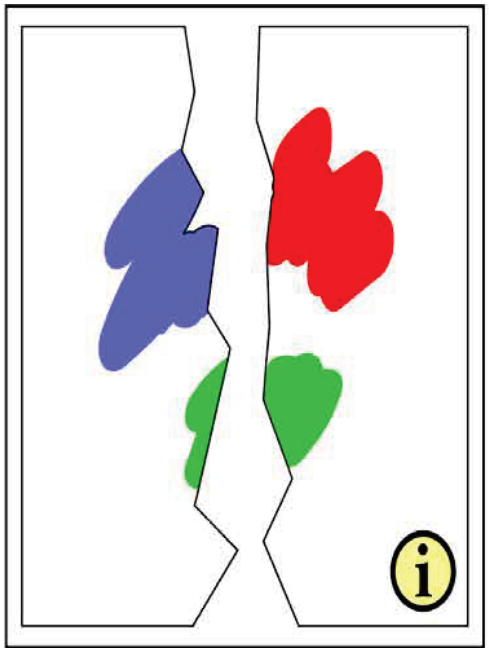
Figure 1. Fuel Quantity Indicating System (Sheet 1 of 1)

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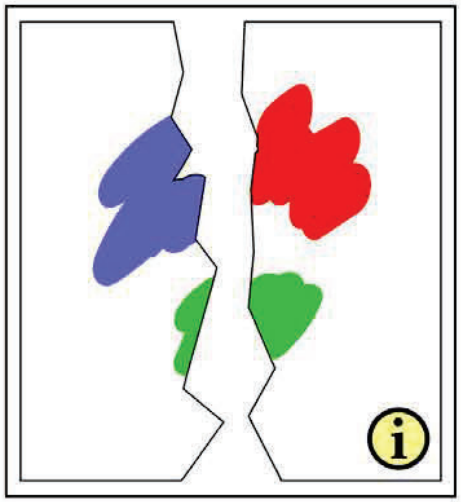
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Figure 1. Fuel Quantity Indicating System (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00124-A-001-01

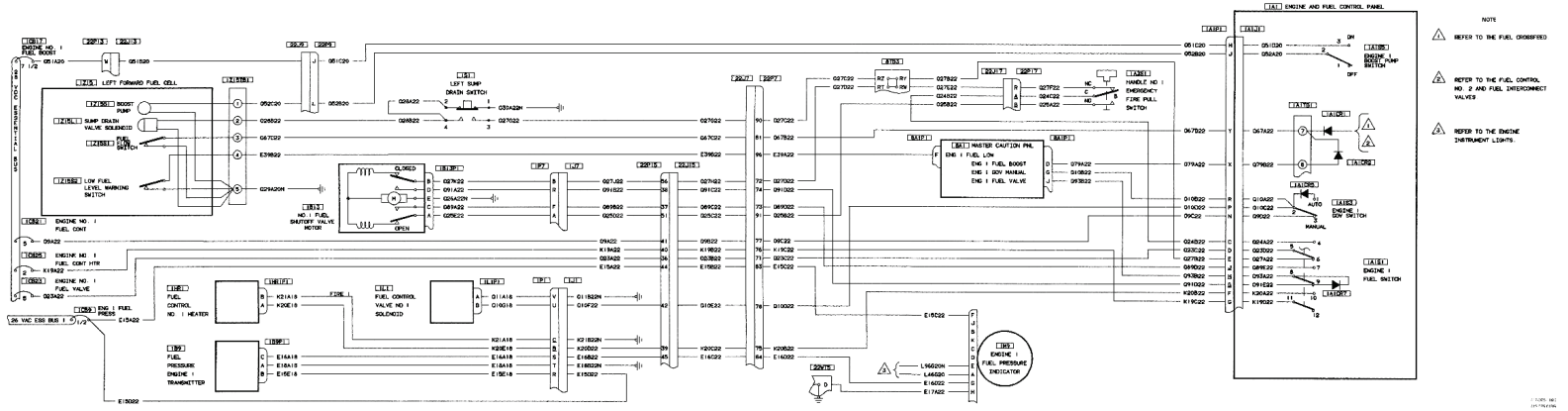
Figure 1. Fuel Quantity Indicating System (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00125-A-001-01

Figure 1. Fuel Control No. 1 (Sheet 1 of 1)

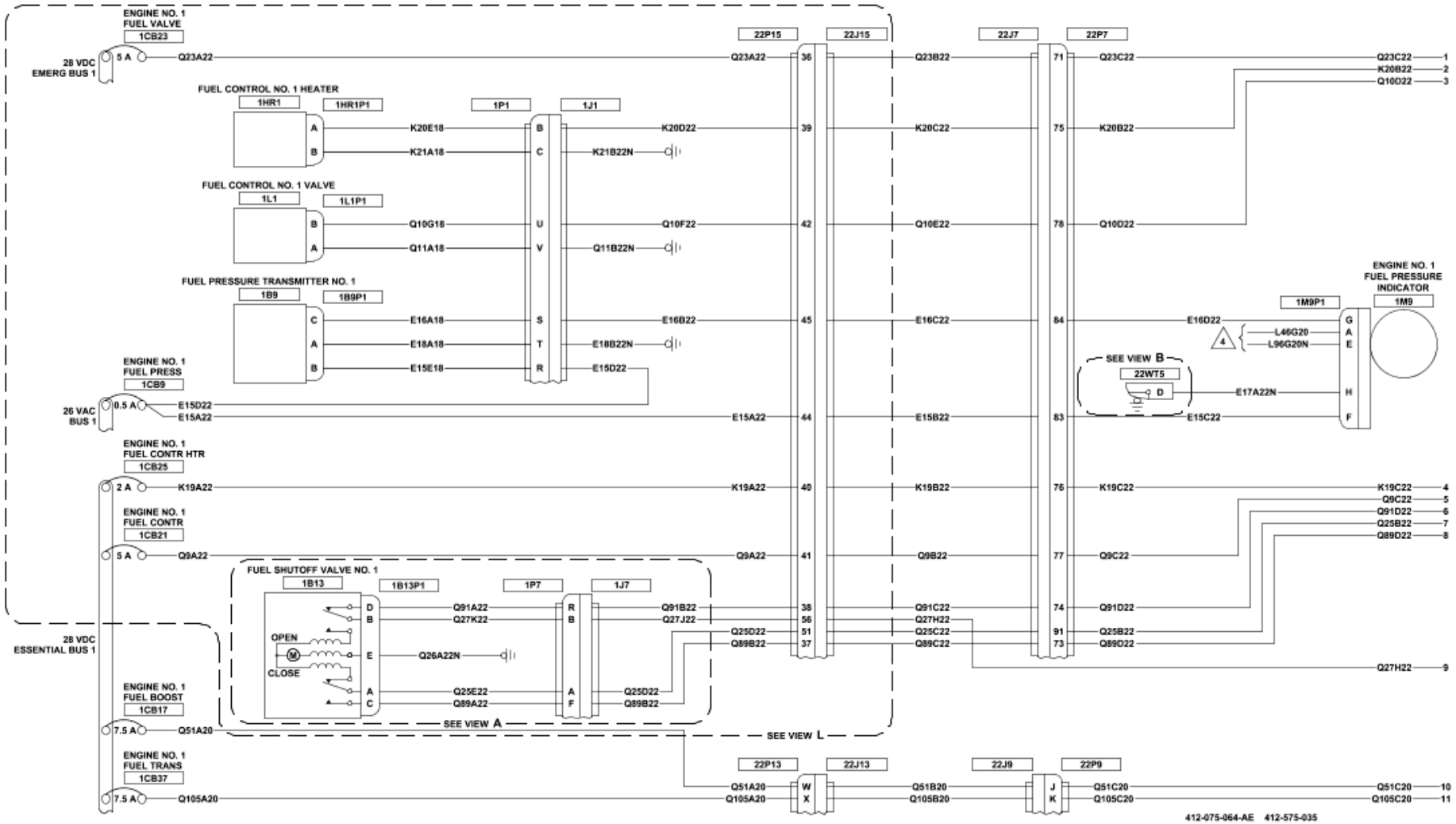
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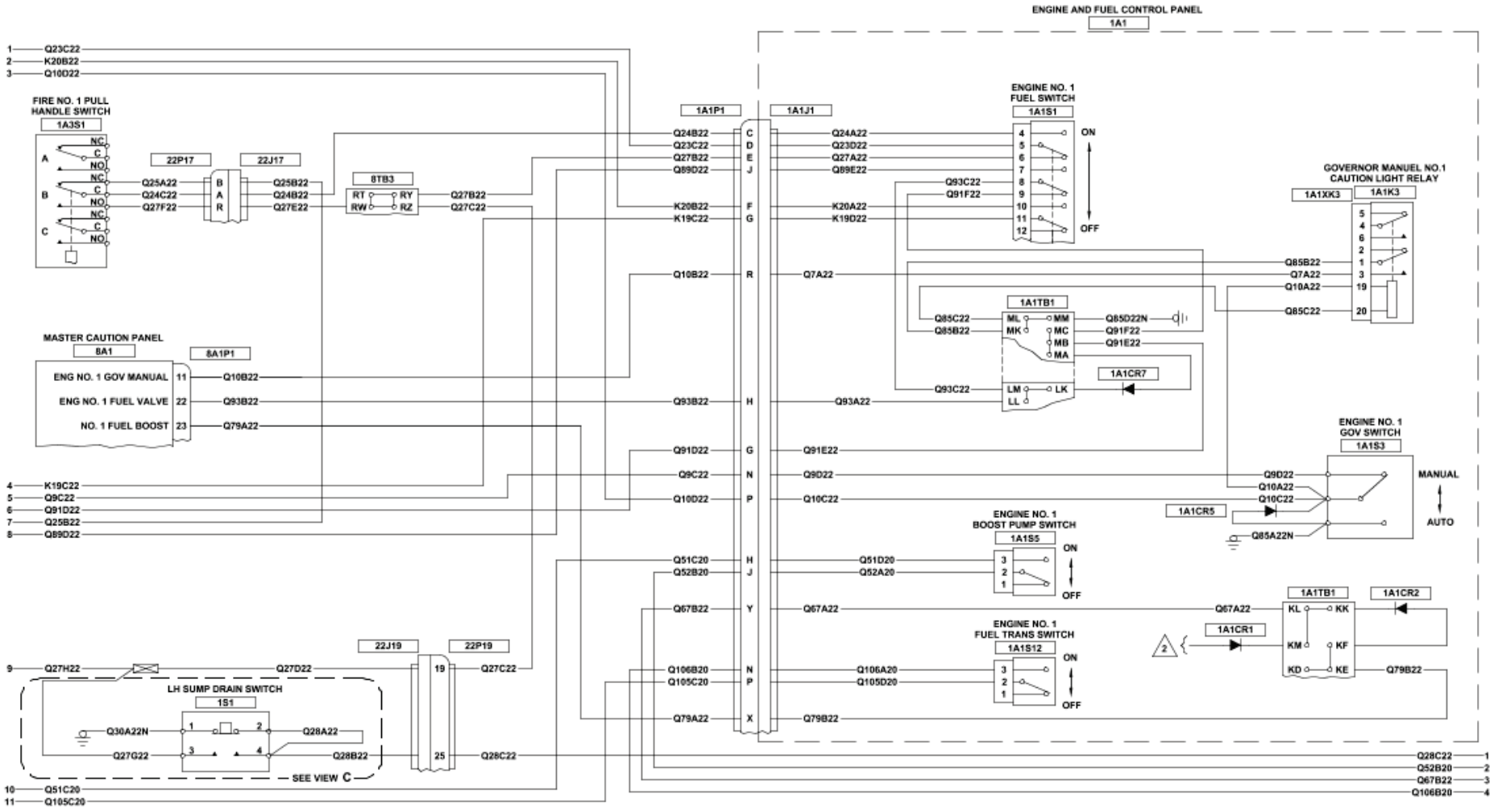
Figure 1. Fuel Control No. 1 (Sheet 1 of 1)

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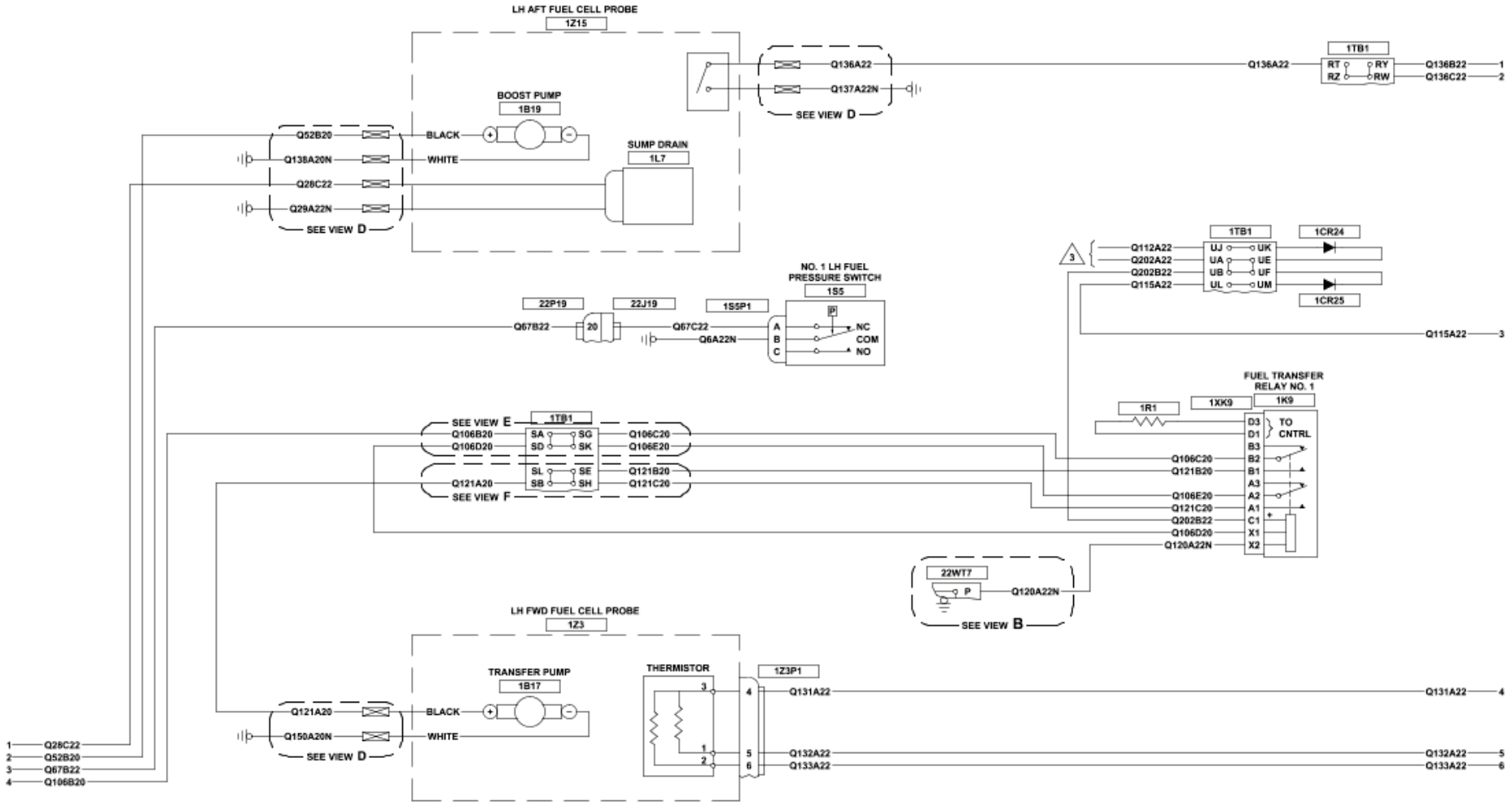
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Figure 1. Fuel Control No. 1 (Sheet 1 of 6)



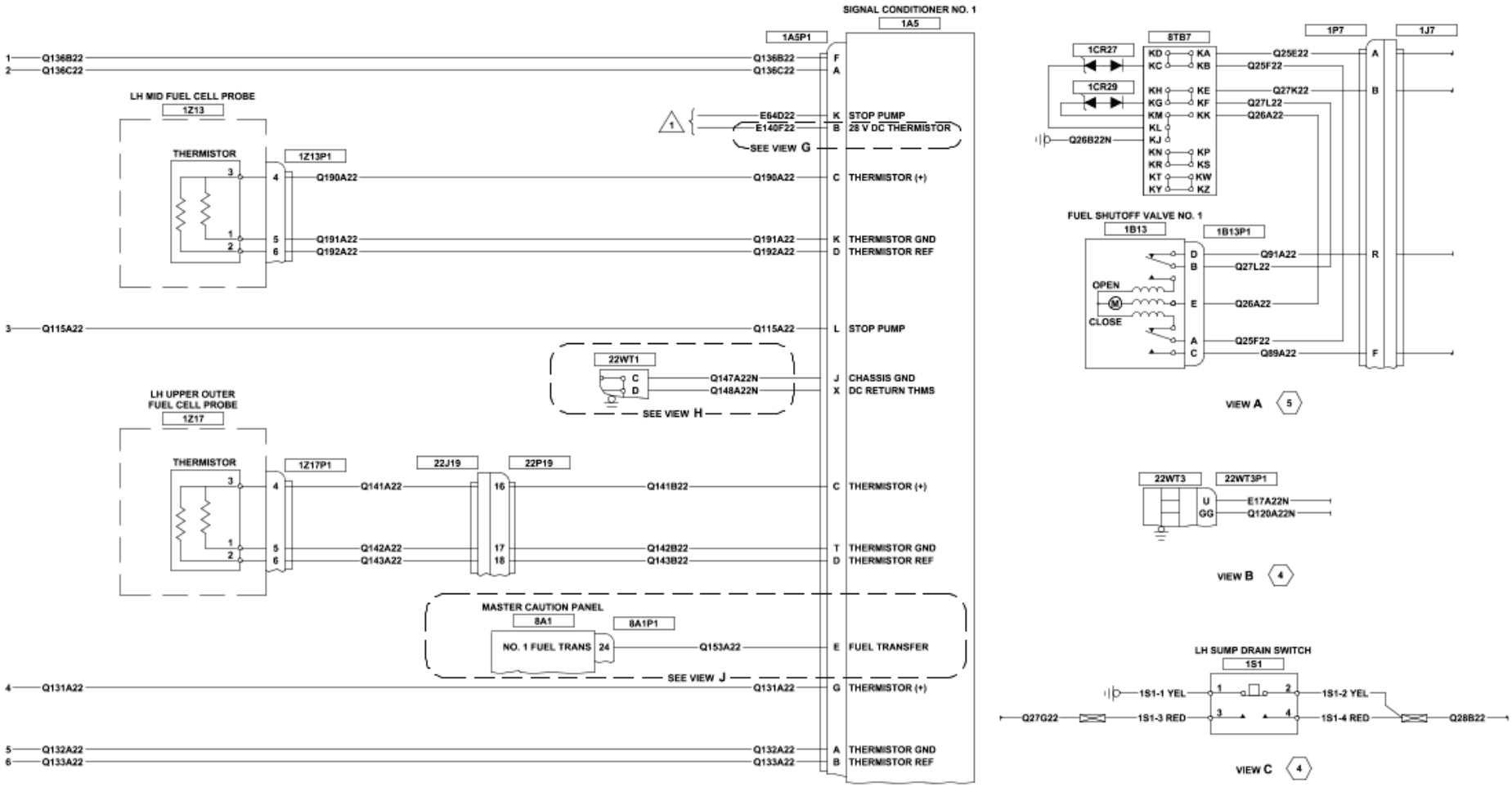
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Figure 1. Fuel Control No. 1 (Sheet 2 of 6)



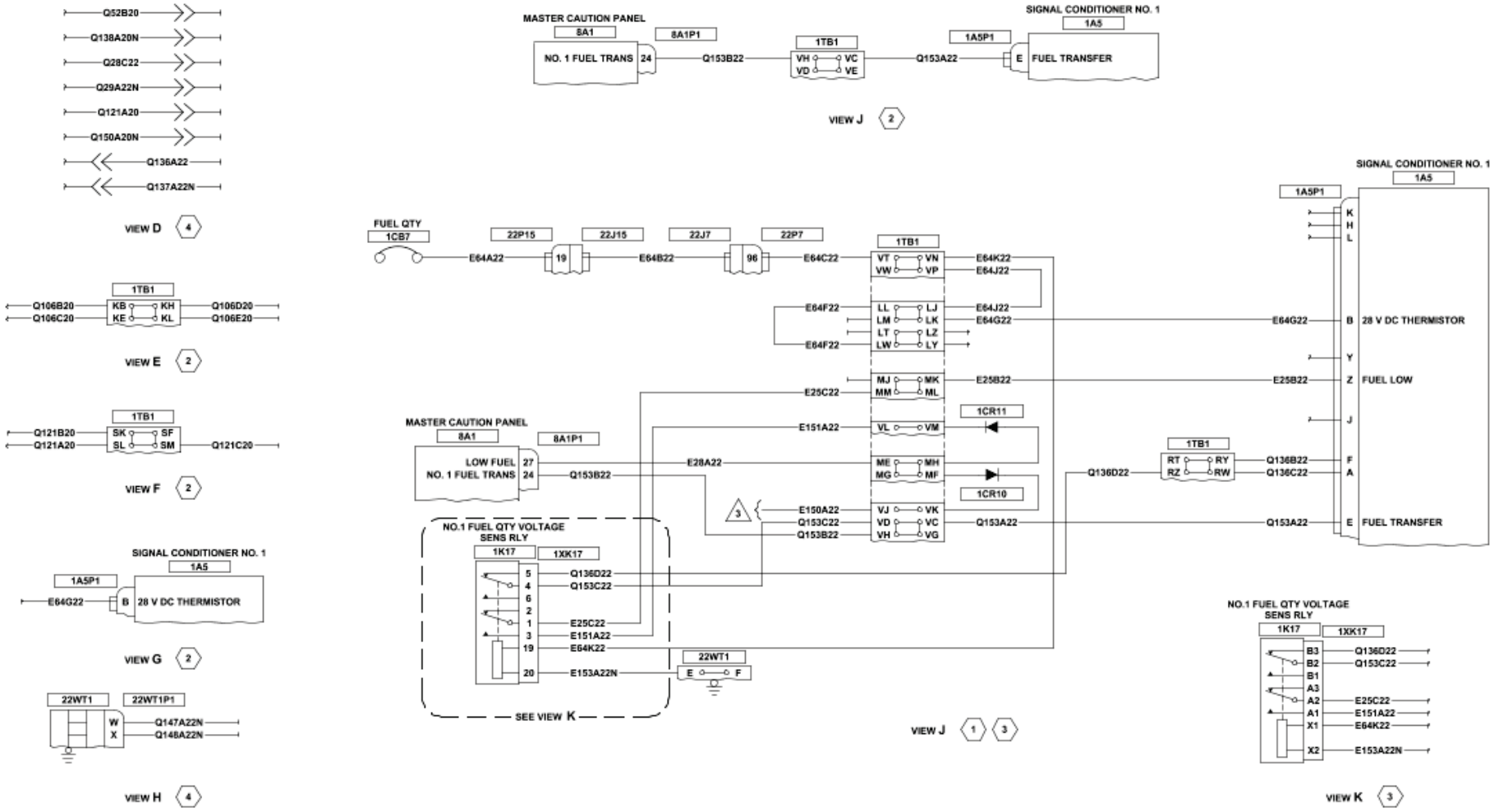
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Figure 1. Fuel Control No. 1 (Sheet 3 of 6)



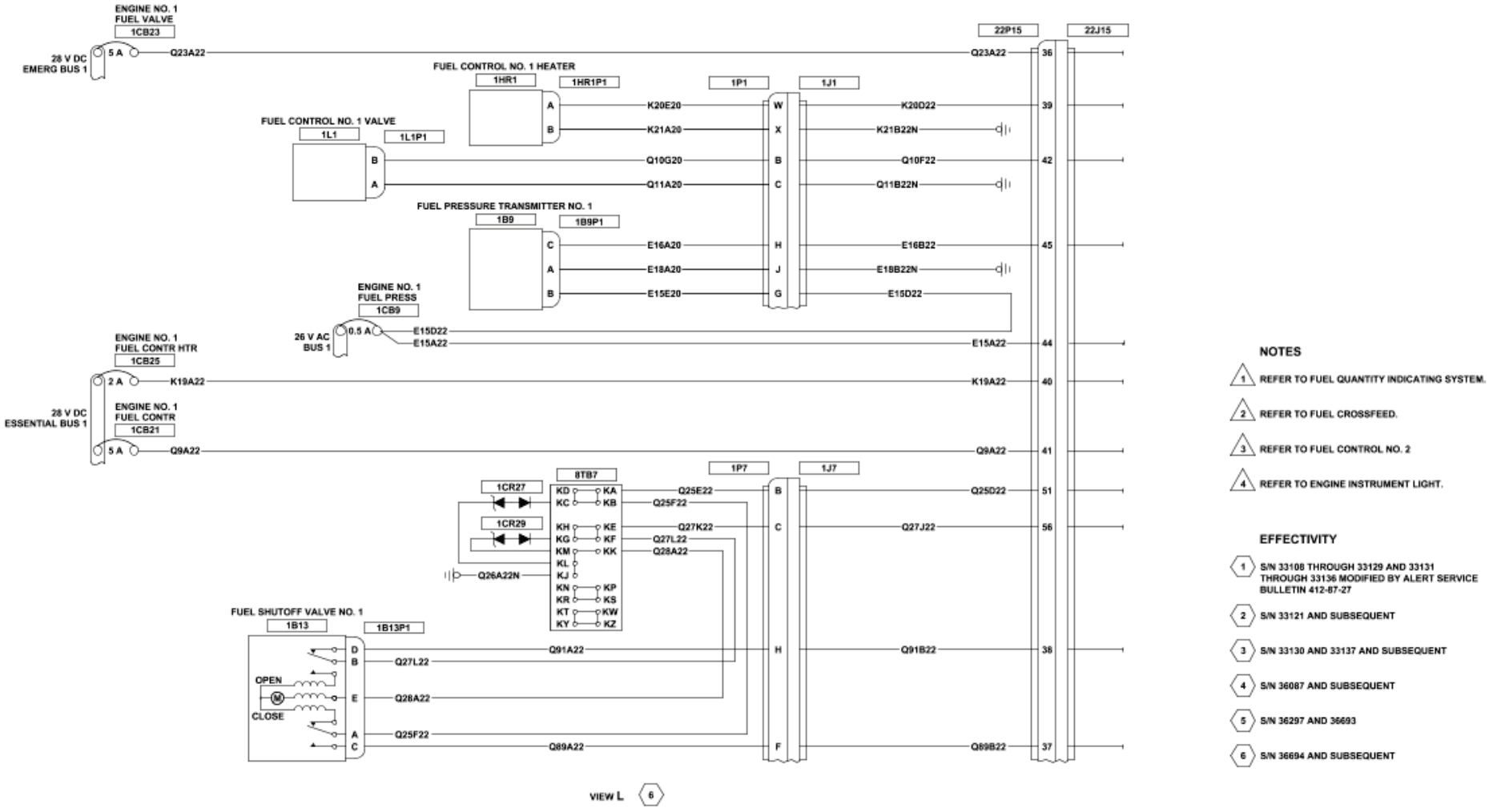
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Figure 1. Fuel Control No. 1 (Sheet 4 of 6)



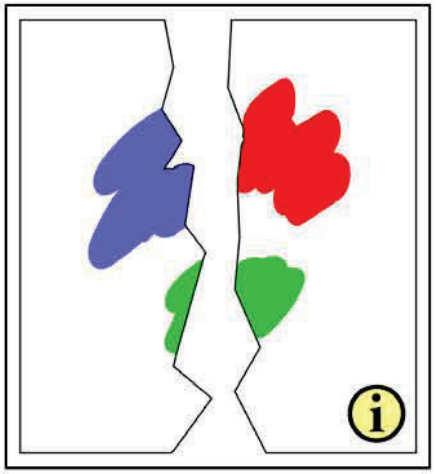
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Figure 1. Fuel Control No. 1 (Sheet 5 of 6)



ICN-412-A-980000-B-97499-00132-A-001-01

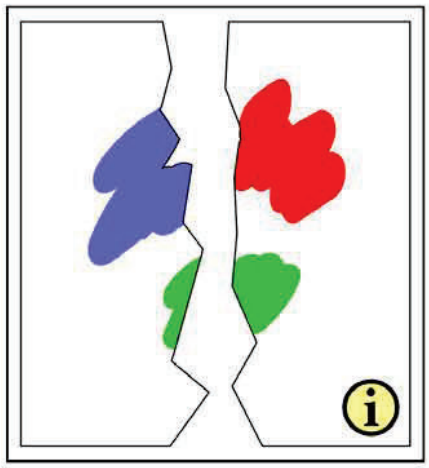
Figure 1. Fuel Control No. 1 (Sheet 6 of 6)



ICN-412-A-980000-B-97499-00133-A-001-01

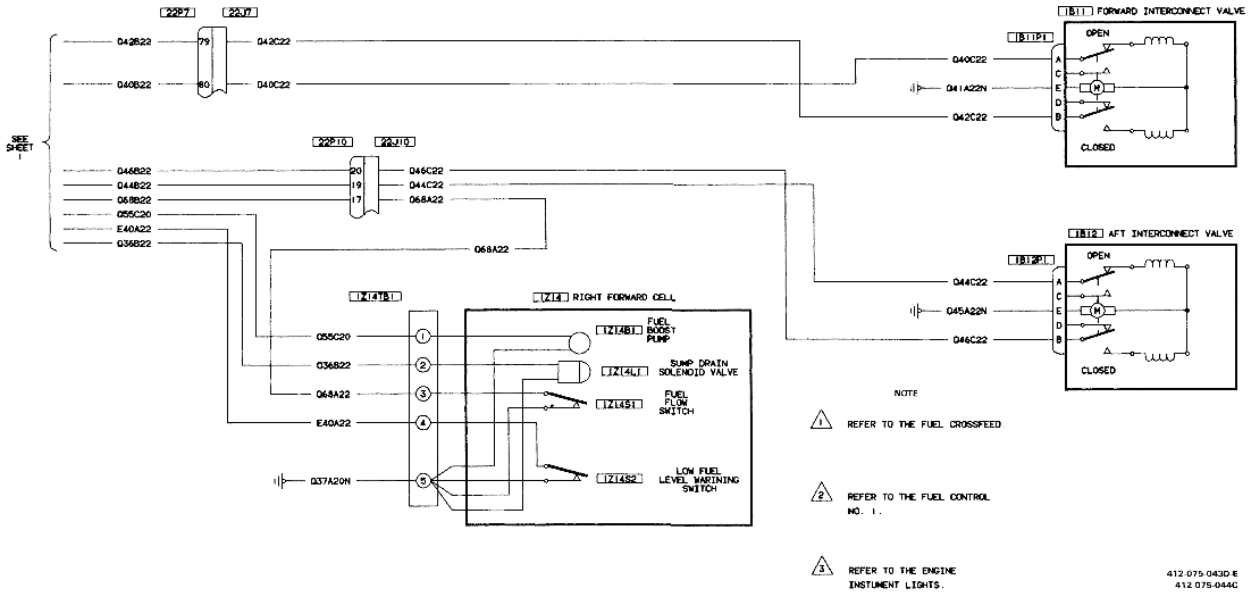
Figure 1. Fuel Control No. 2 (Sheet 1 of 1)

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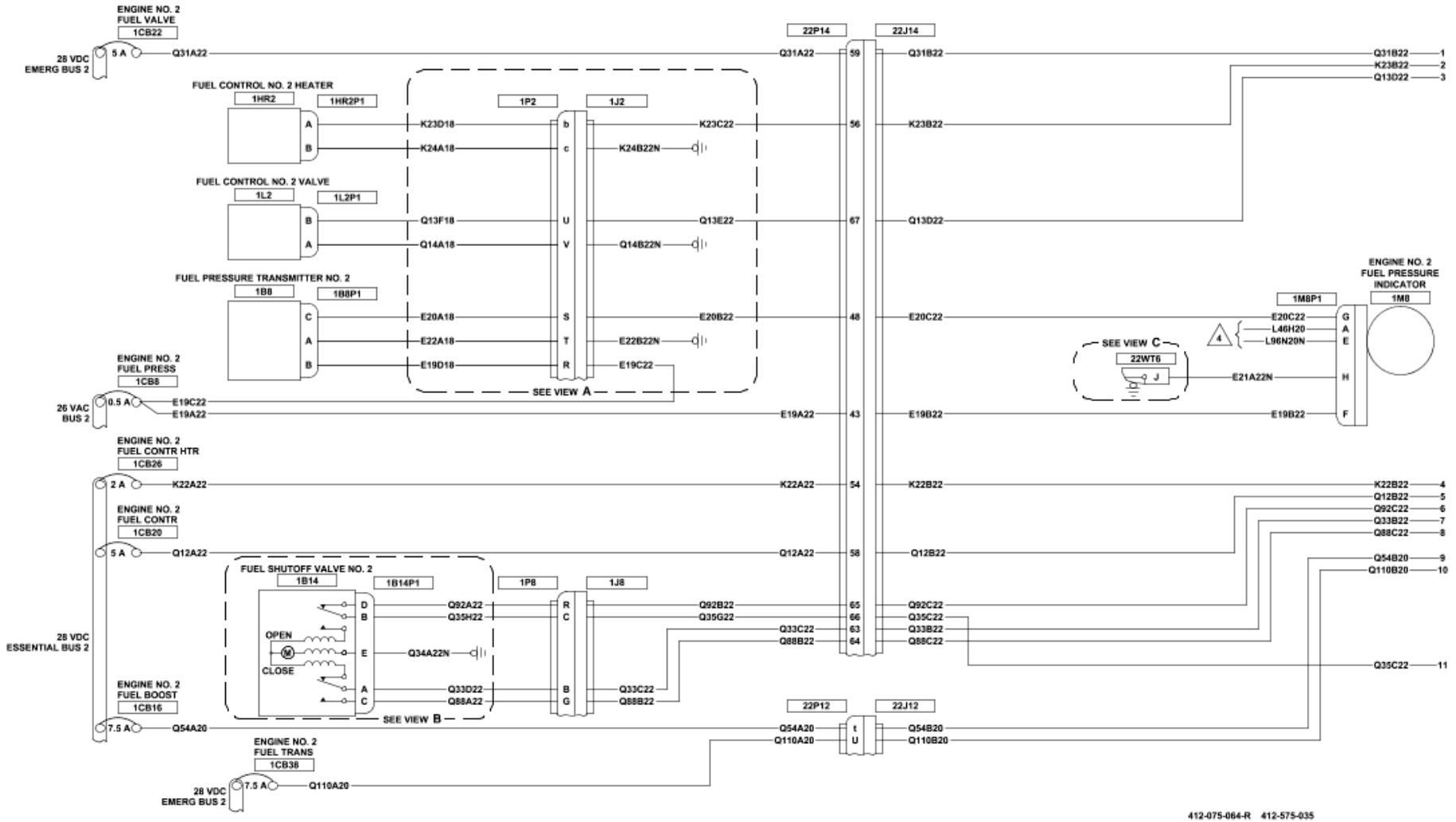
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Figure 1. Fuel Control No. 2 (Sheet 1 of 2)



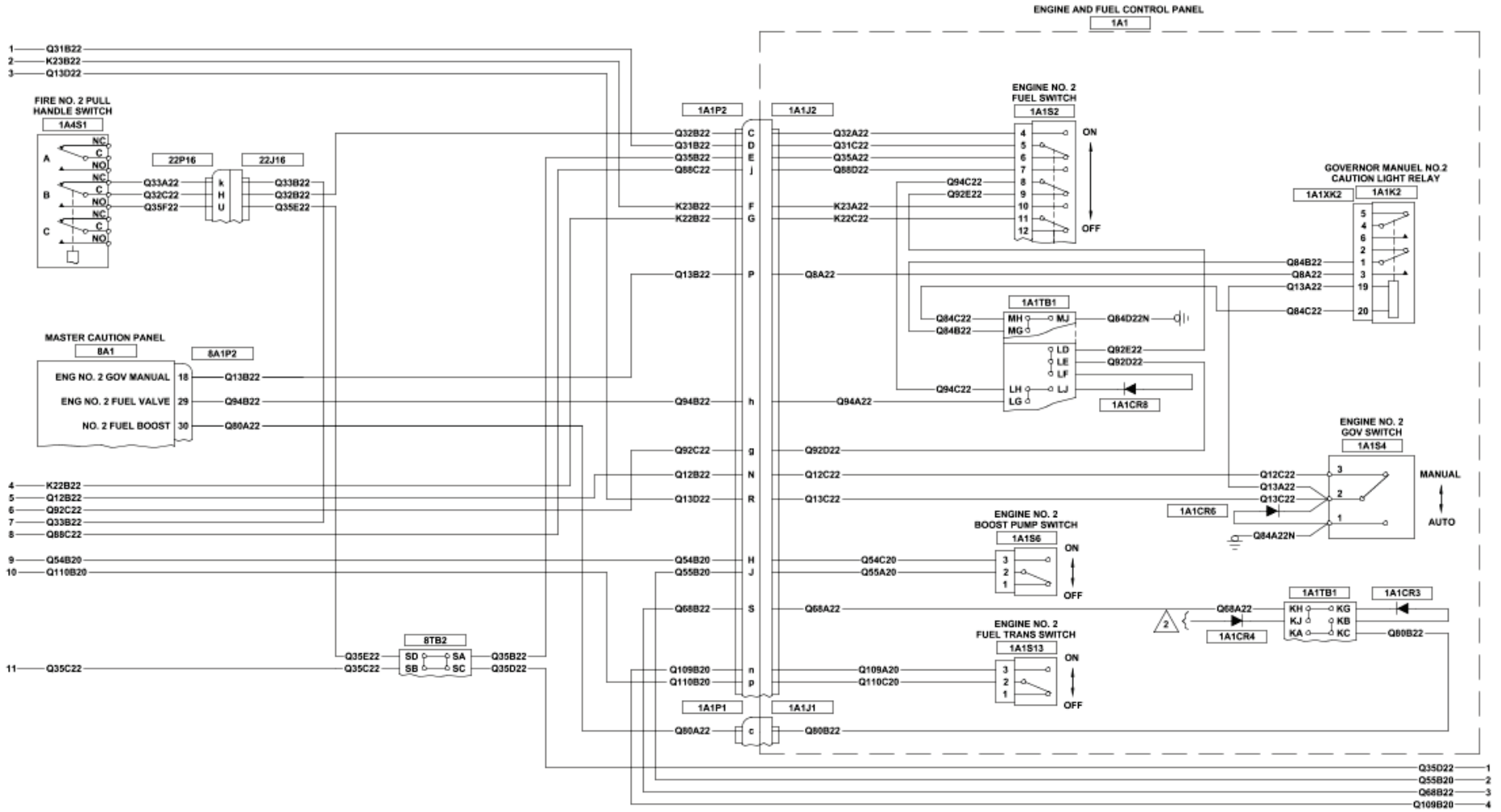
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Figure 1. Fuel Control No. 2 (Sheet 2 of 2)



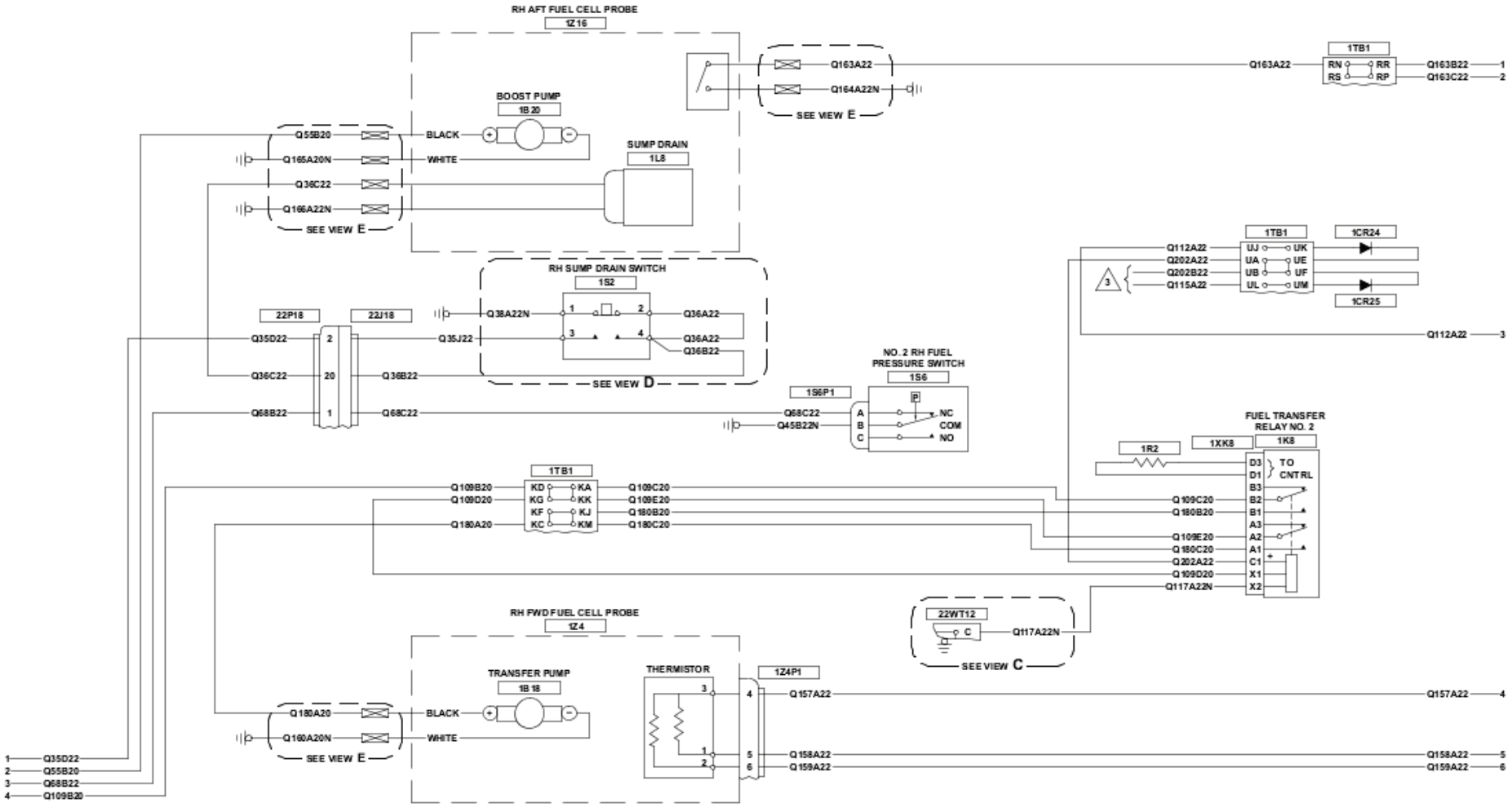
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Figure 1. Fuel Control No. 2 (Sheet 1 of 5)



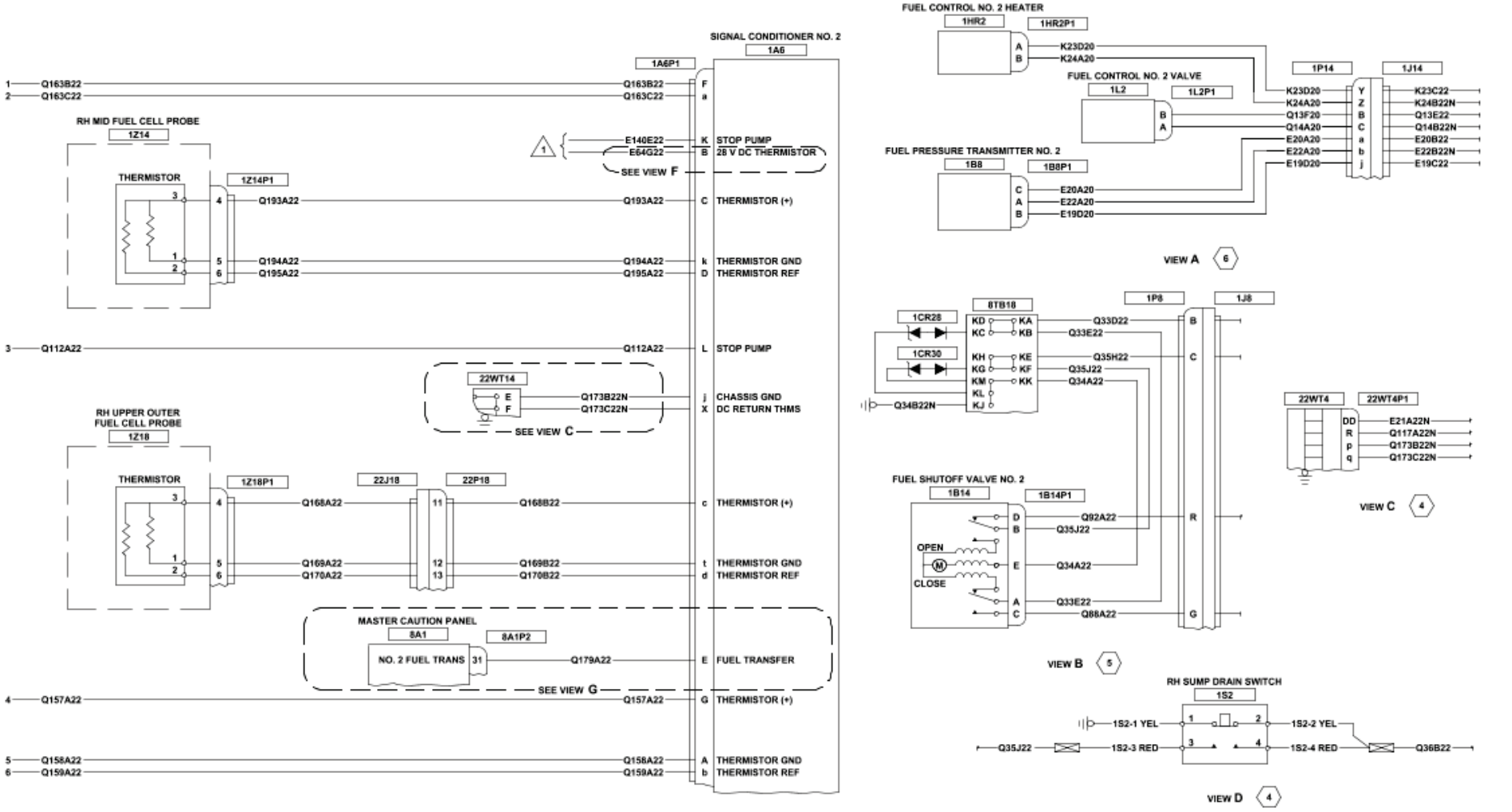
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Figure 1. Fuel Control No. 2 (Sheet 2 of 5)



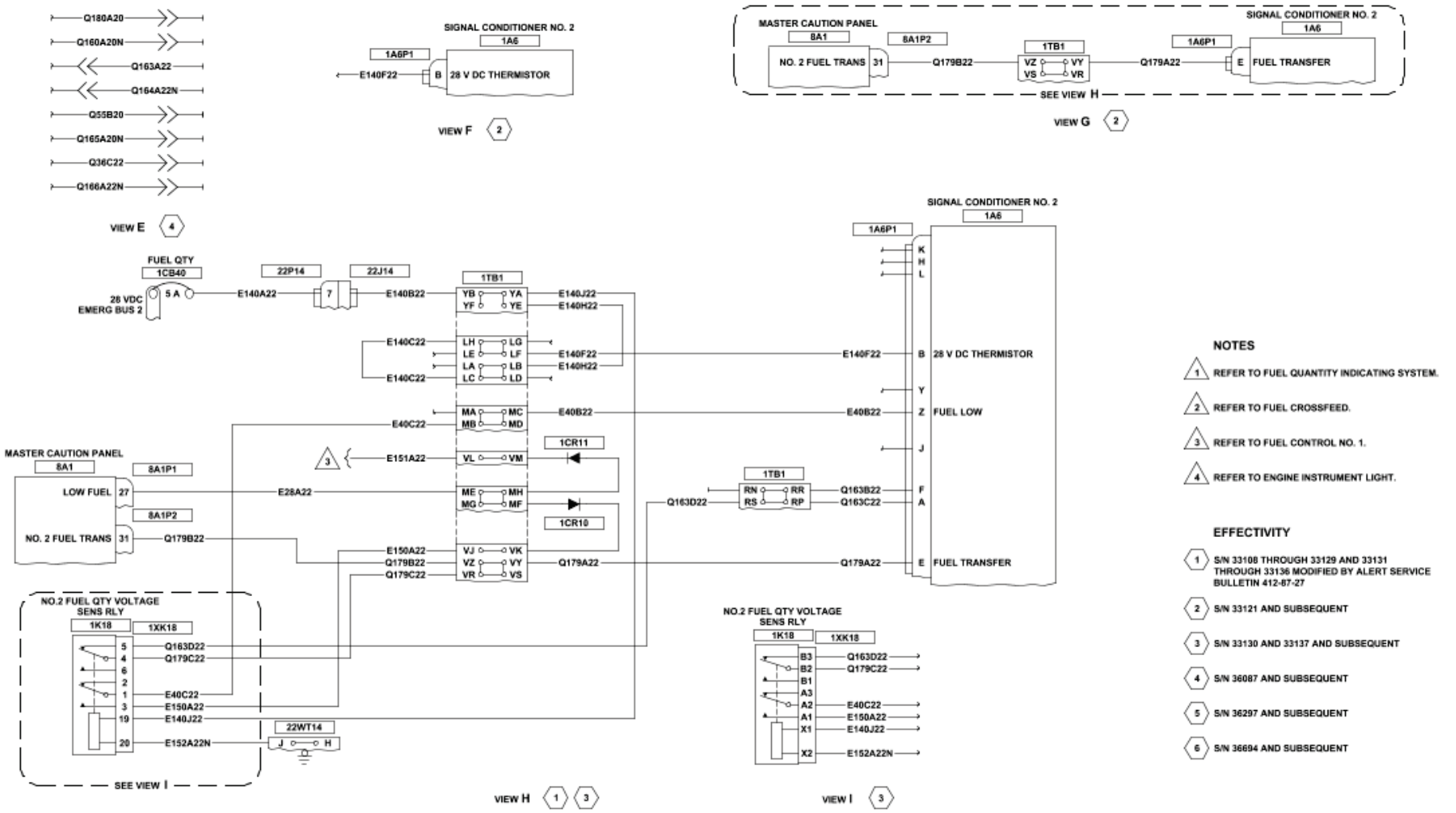
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Figure 1. Fuel Control No. 2 (Sheet 3 of 5)



ICN-412-A-980000-B-97499-00139-A-001-01

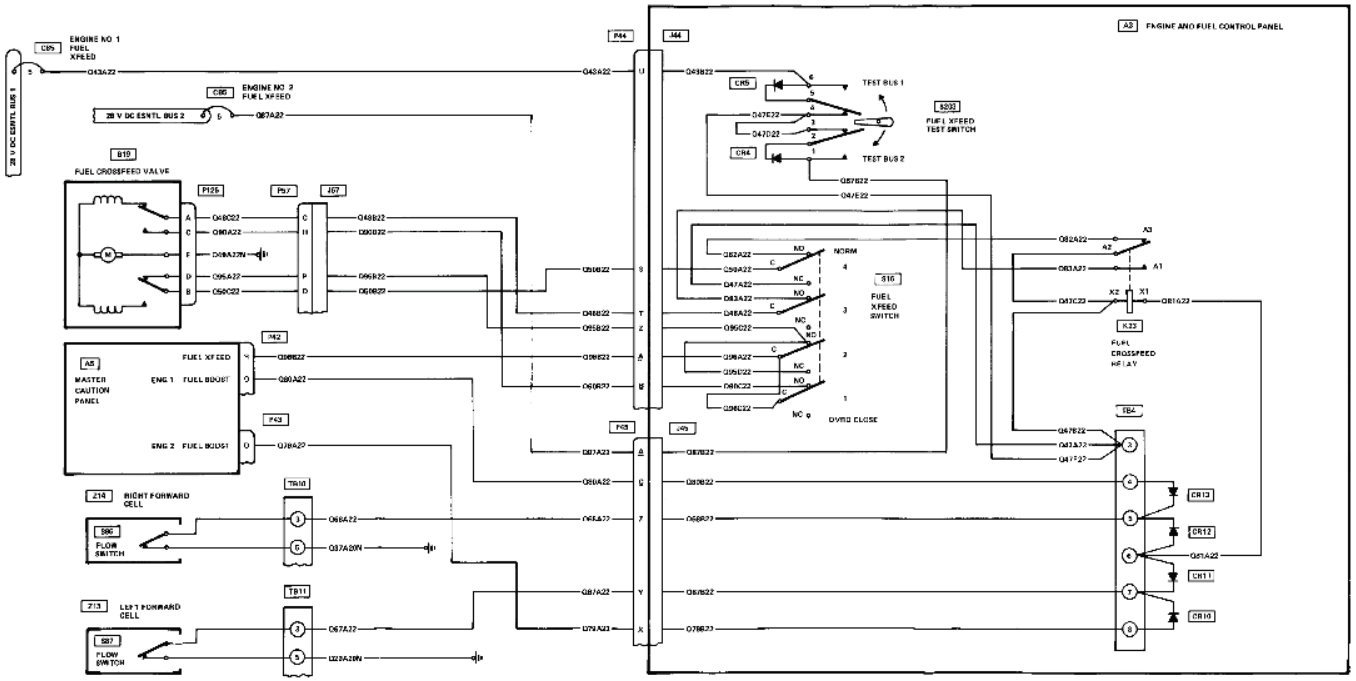
Figure 1. Fuel Control No. 2 (Sheet 4 of 5)



ICN-412-A-980000-B-97499-00140-A-001-01

Figure 1. Fuel Control No. 2 (Sheet 5 of 5)

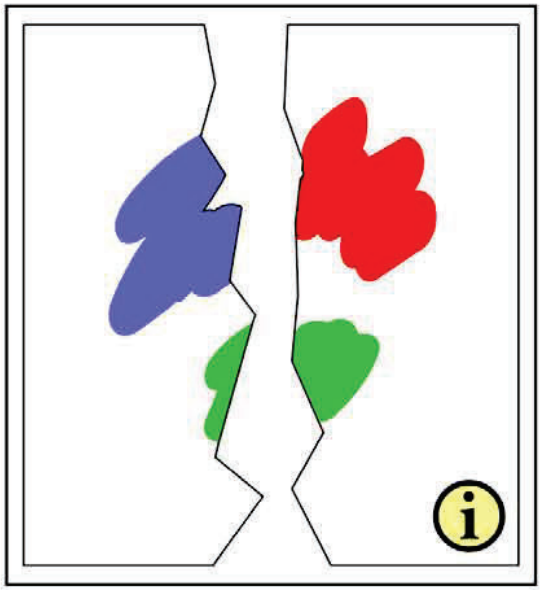
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Figure 1. Fuel Crossfeed (Sheet 1 of 1)

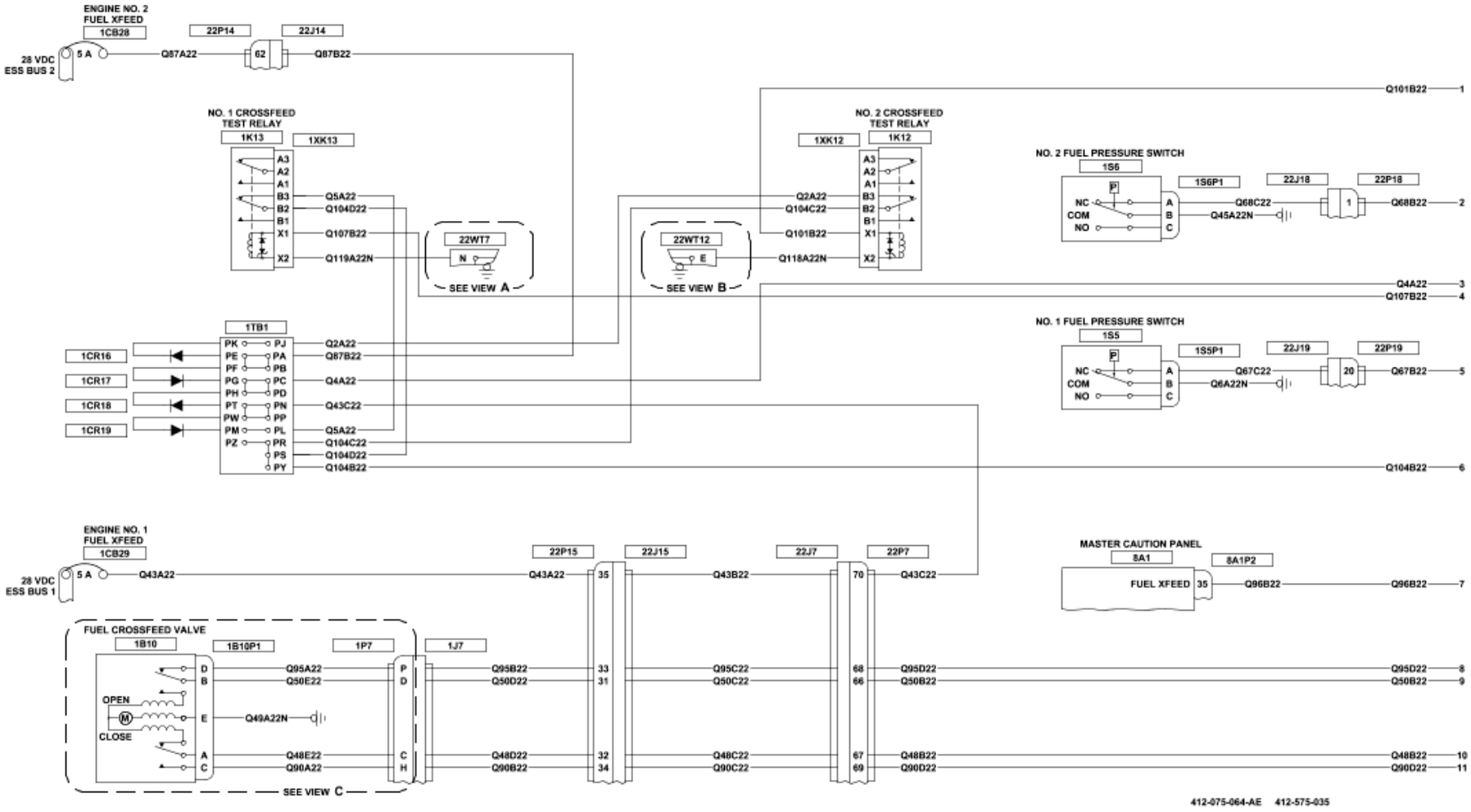
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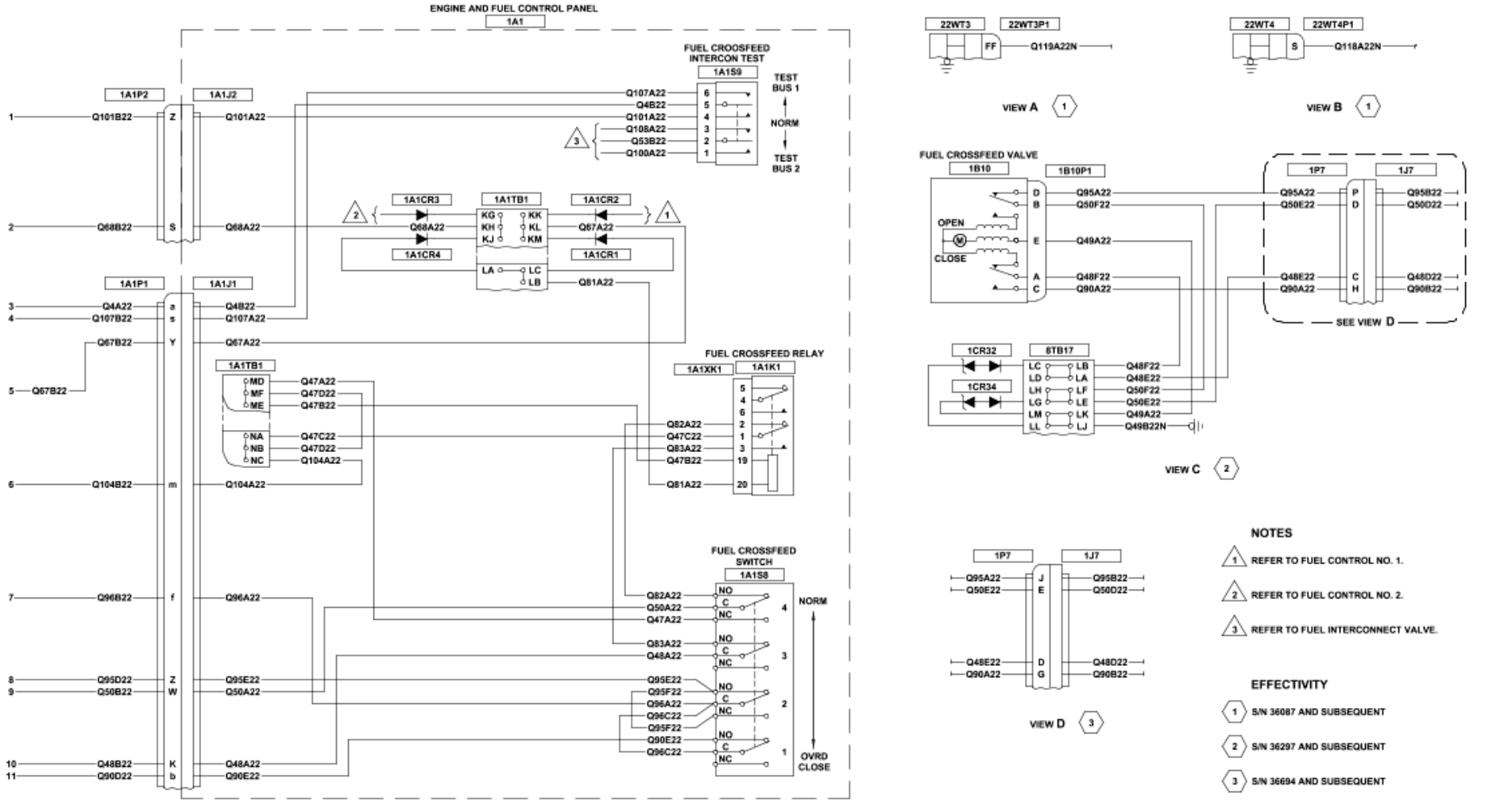
Figure 1. Fuel Crossfeed (Sheet 1 of 1)

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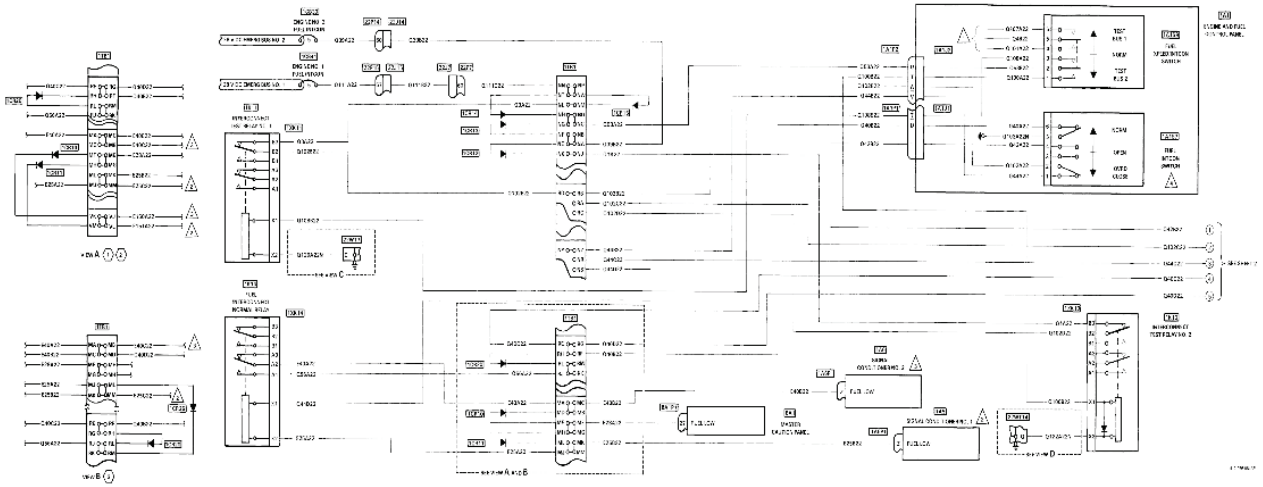
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Figure 1. Fuel Crossfeed (Sheet 1 of 2)



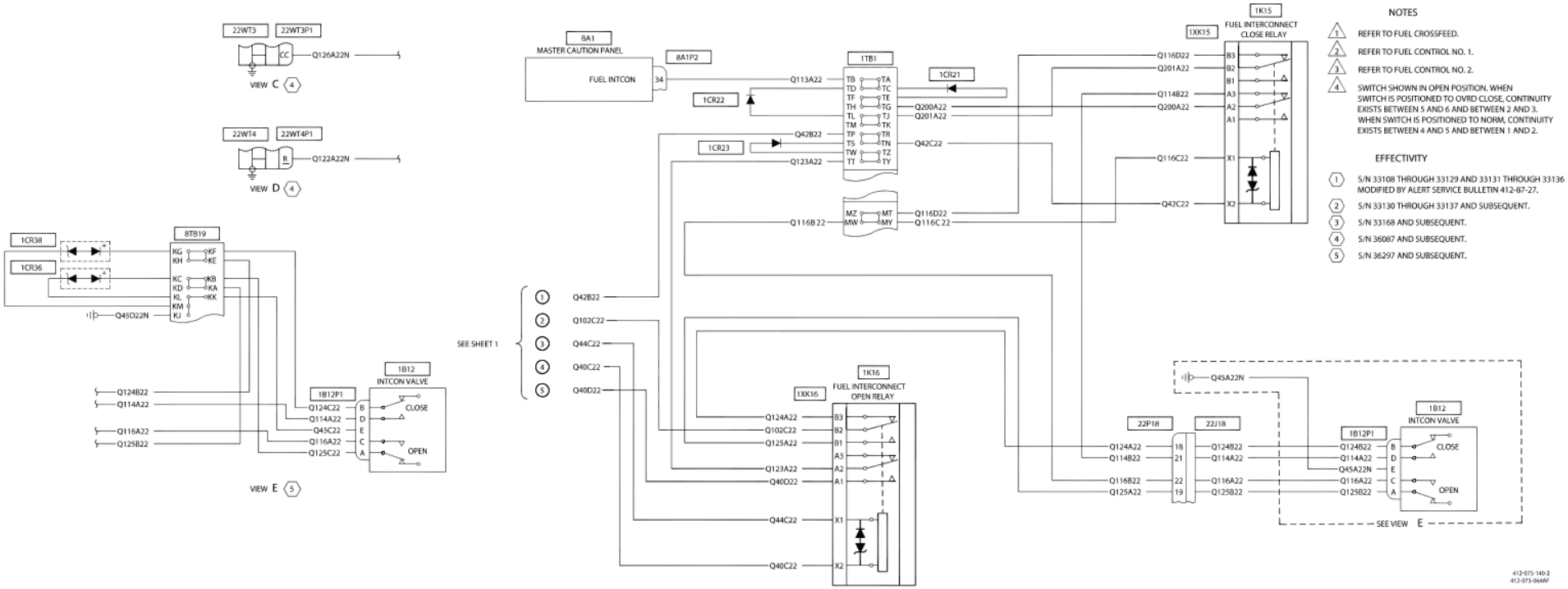
ICN-412-A-980000-B-97499-00144-A-001-01

Figure 1. Fuel Crossfeed (Sheet 2 of 2)



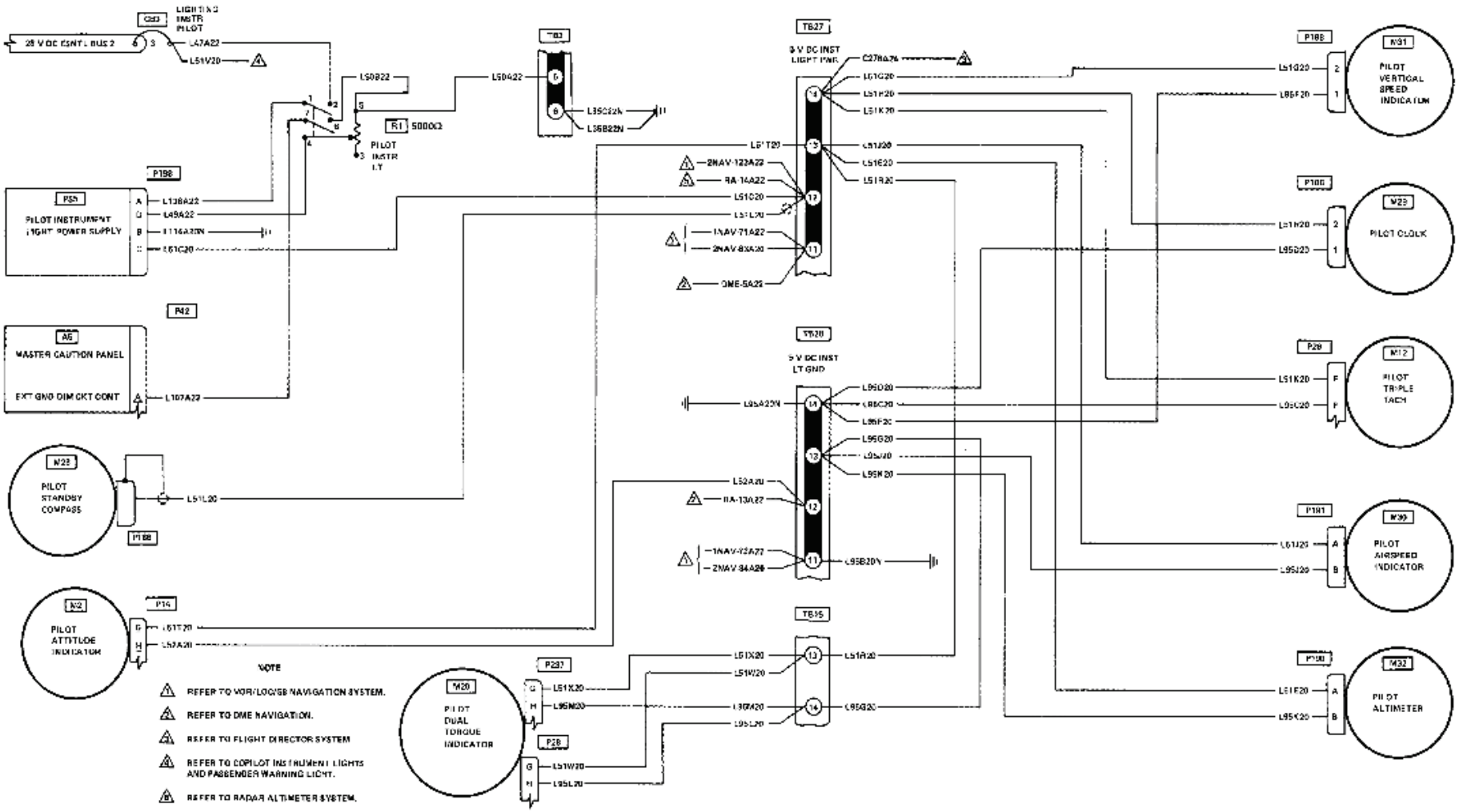
ICN-412-A-980000-B-97499-00145-A-001-01

Figure 1. Fuel Interconnect Valve (Sheet 1 of 2)



ICN-412-A-980000-B-97499-00146-A-001-01

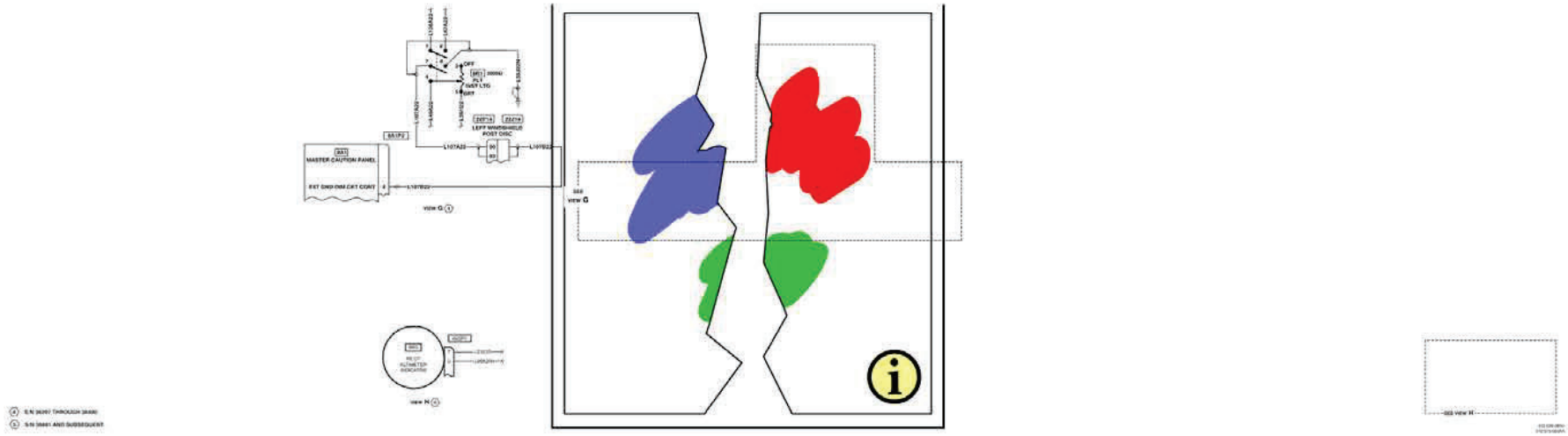
Figure 1. Fuel Interconnect Valve (Sheet 2 of 2)



ICN-412-A-980000-B-97499-00147-A-001-01

Figure 1. Pilot Instrument Lights (Sheet 1 of 1)

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Figure 1. Pilot Instrument Lights (Sheet 1 of 1)

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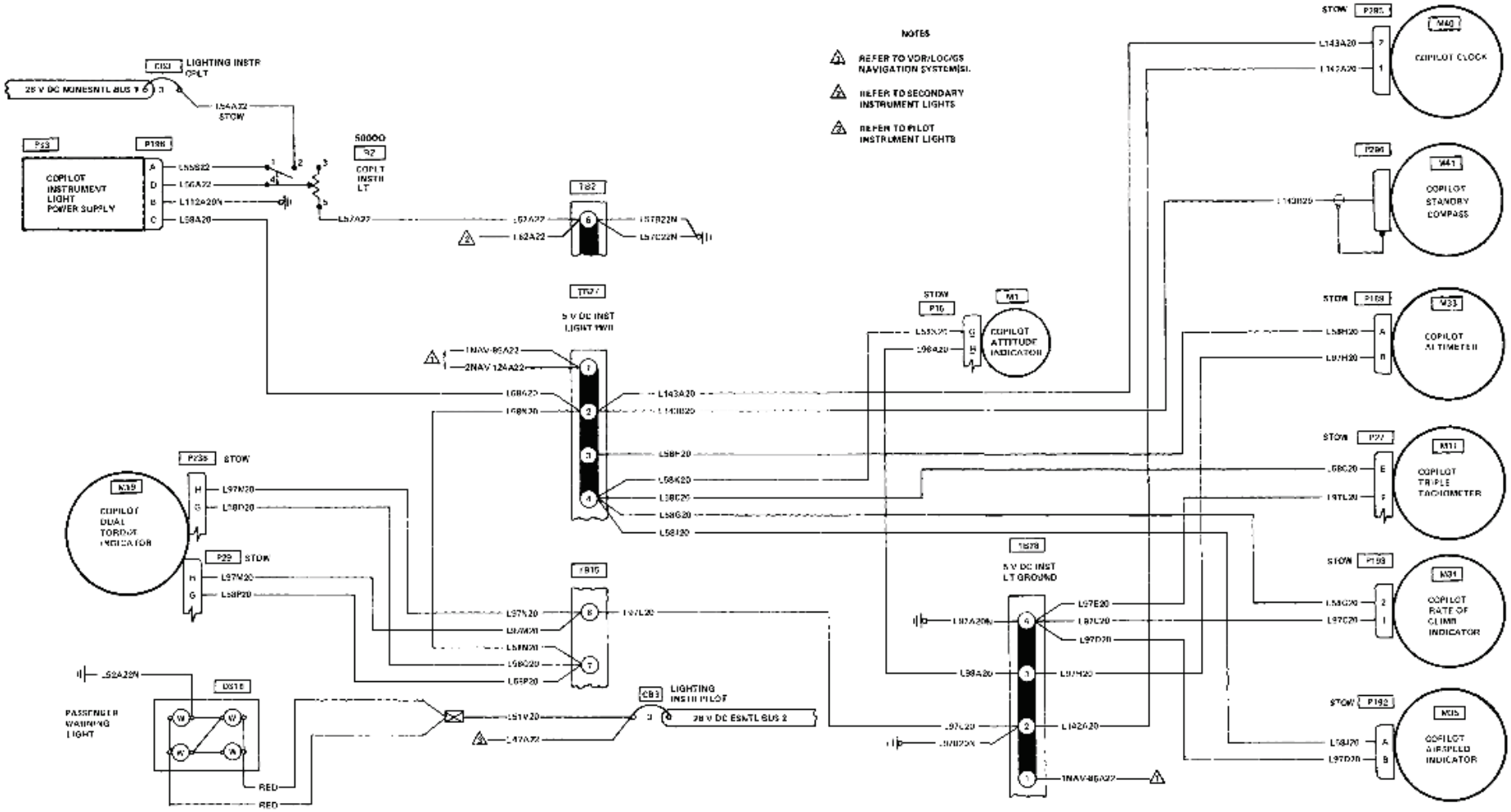
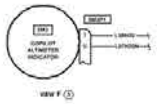
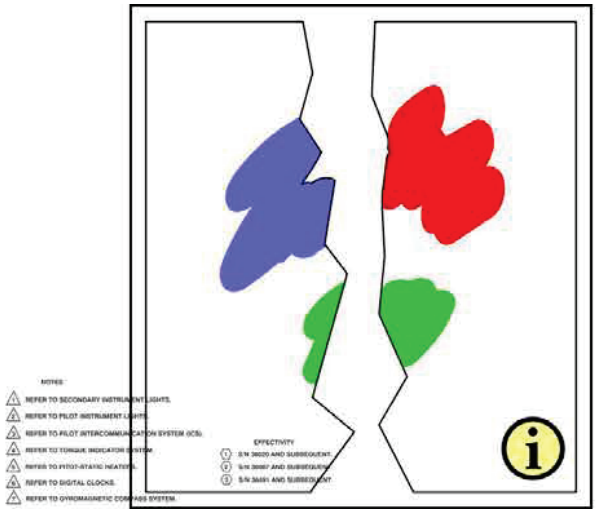


Figure 1. Copilot Instrument Lights and Passenger Warning Light (Sheet 1 of 1)

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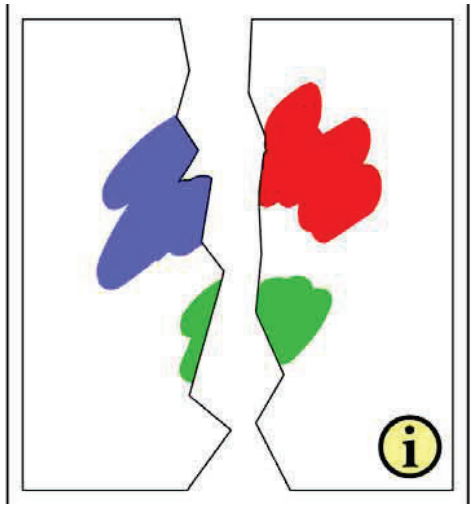




ICN-412-A-980000-B-97499-00150-A-001-01

Figure 1. Copilot Instrument Lights and Passenger Warning Light (Sheet 1 of 1)

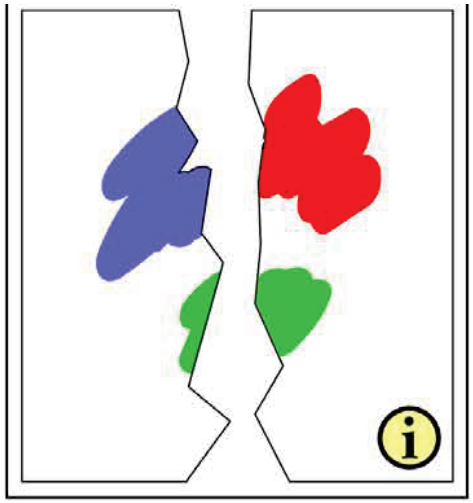
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ICN-412-A-980000-B-97499-00151-A-001-01

Figure 1. Engine Instrument Lights (Sheet 1 of 1)

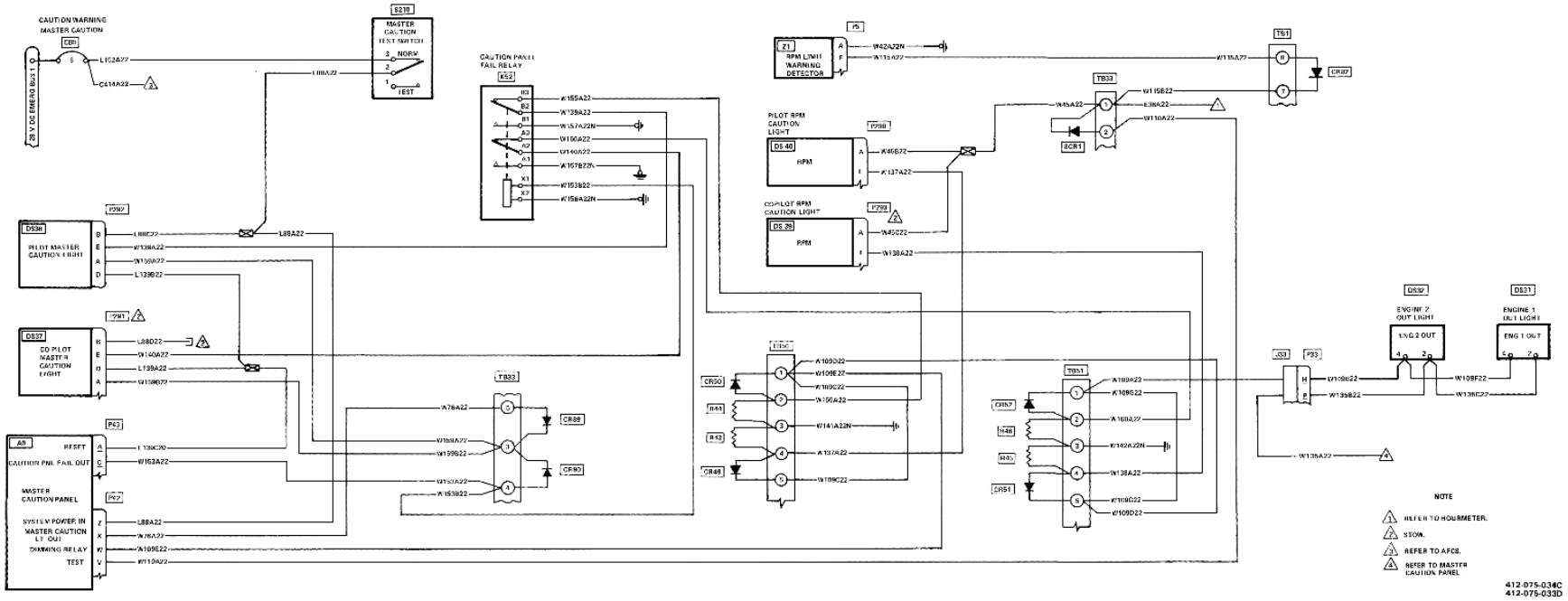
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ICN-412-A-980000-B-97499-00152-A-001-01

Figure 1. Engine Instrument Lights (Sheet 1 of 1)

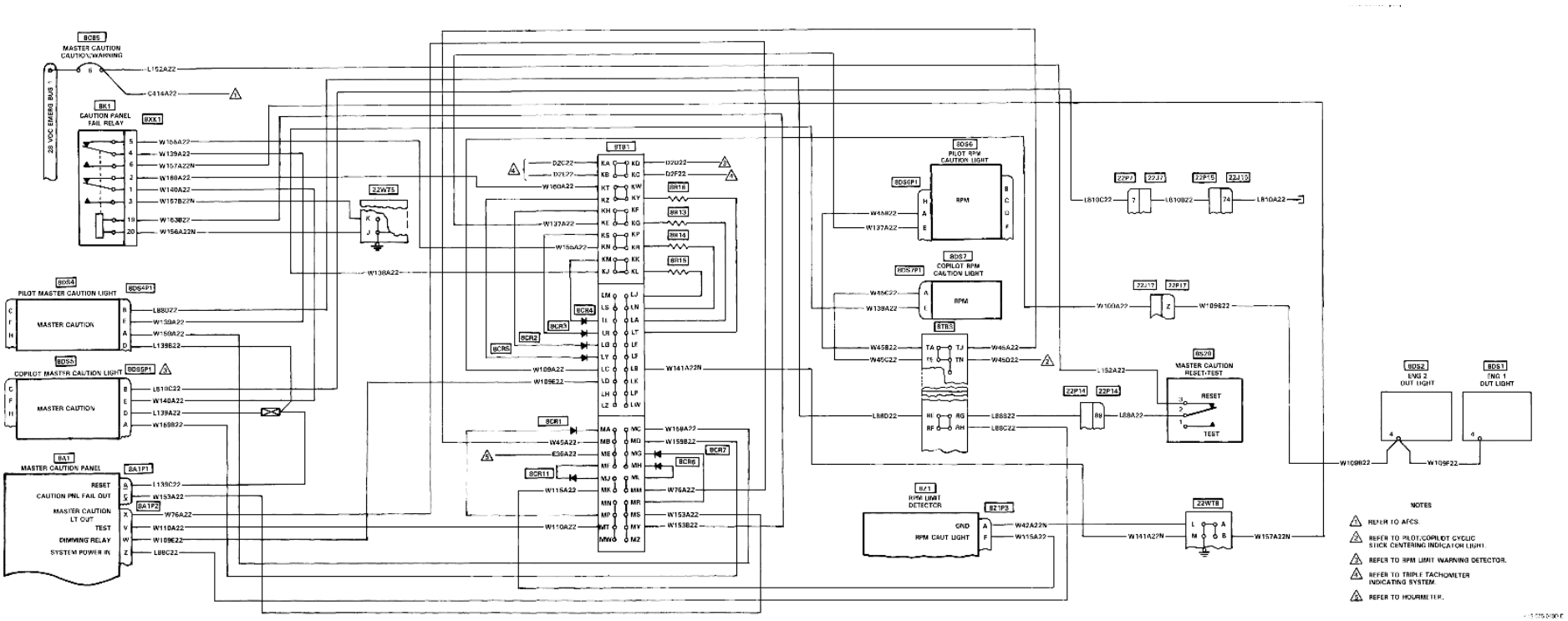
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Figure 1. Master Caution and RPM Caution Light (Sheet 1 of 1)

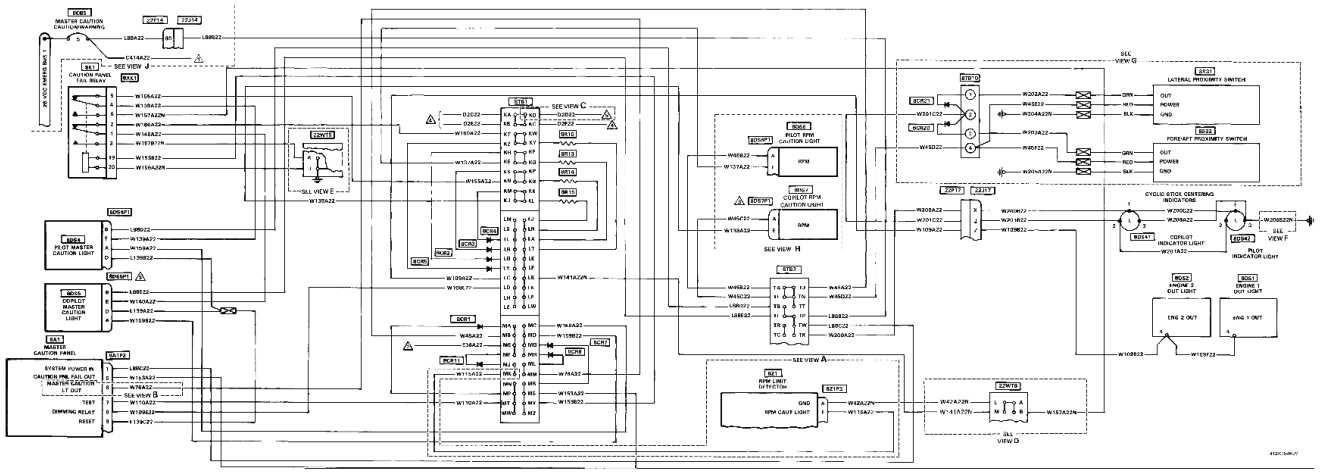
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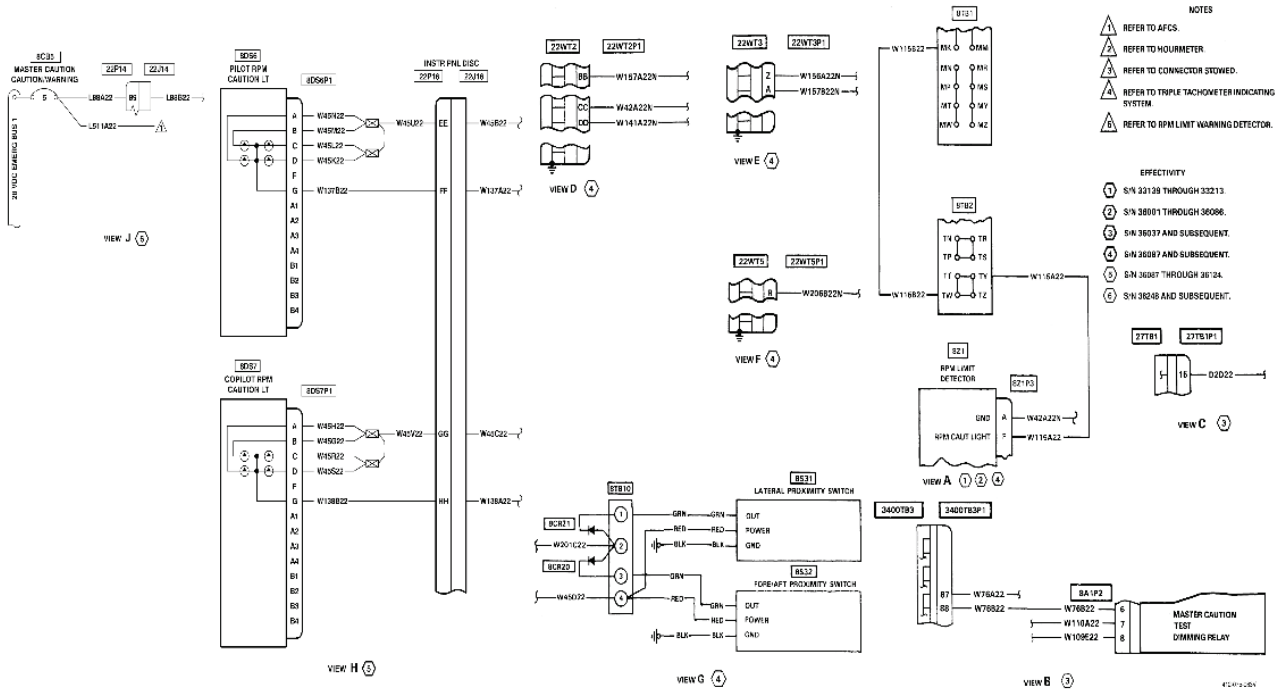
Figure 1. Master Caution and RPM Caution Lights (Sheet 1 of 1)

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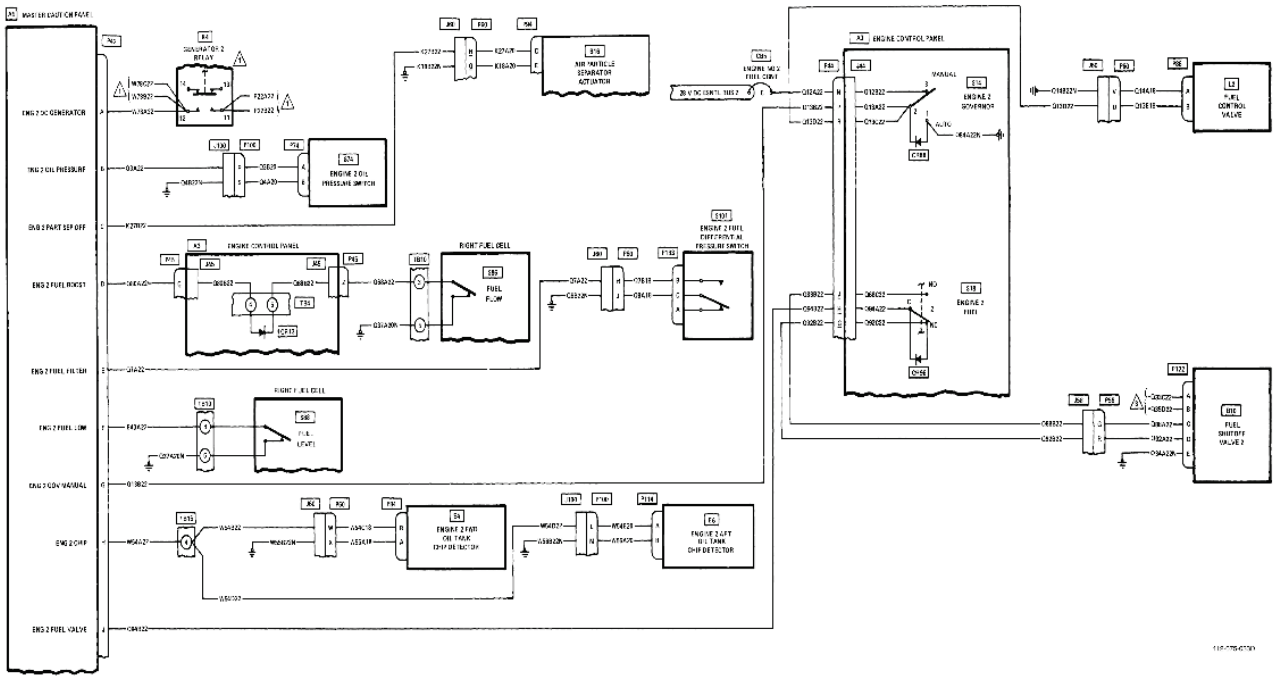
ICN-412-A-980000-B-97499-00155-A-001-01

Figure 1. Master Caution, RPM Caution, and Cyclic Stick Centering Indicator Lights (Sheet 1 of 2)



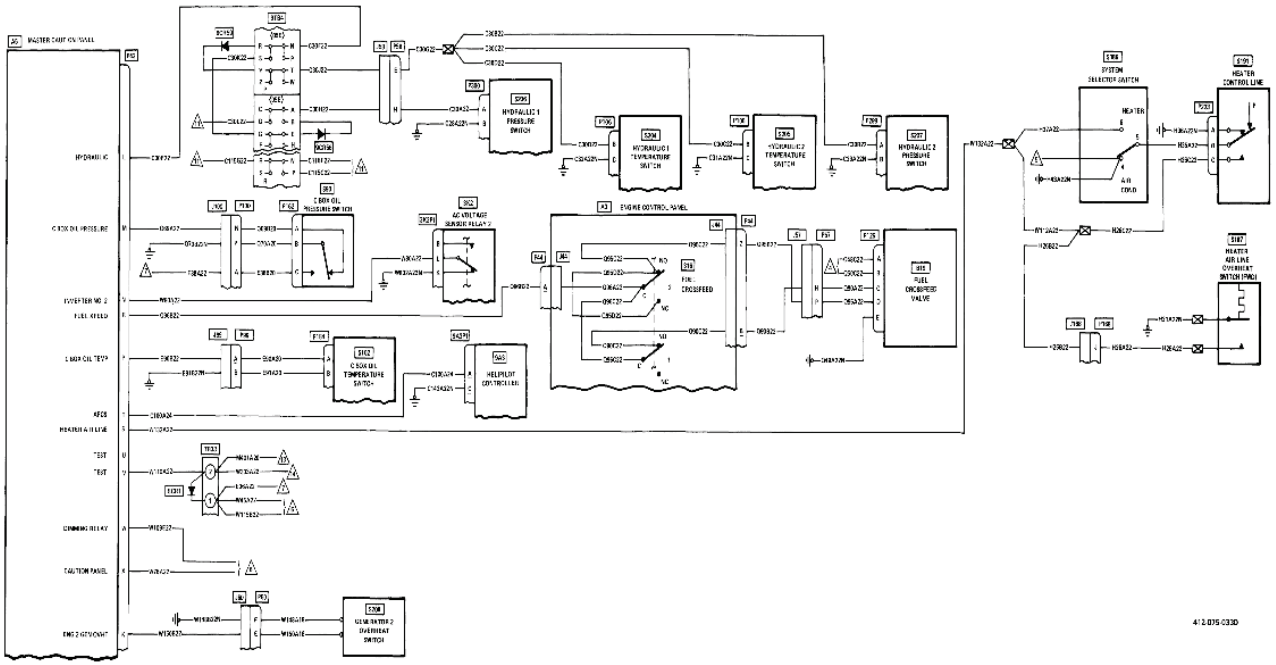
ICN-412-A-980000-B-97499-00156-A-001-01

Figure 1. Master Caution, RPM Caution, and Cyclic Stick Centering Indicator Lights (Sheet 2 of 2)



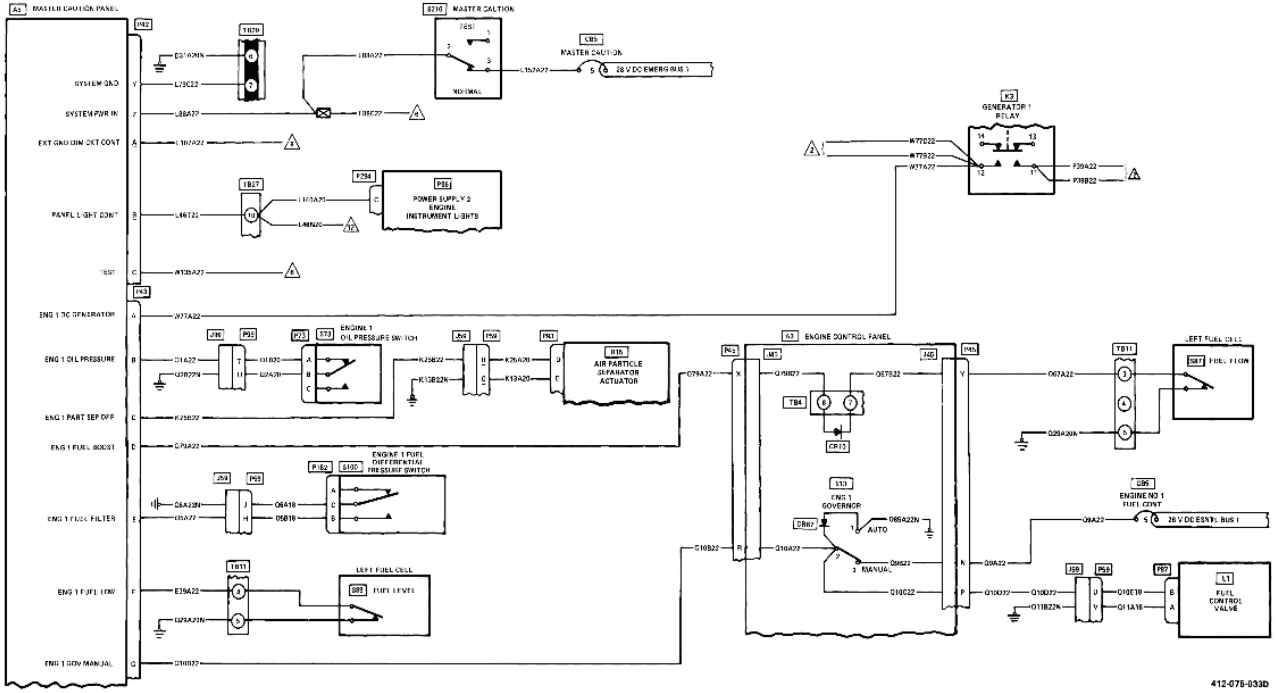
ICN-412-A-980000-B-97499-00157-A-001-01

Figure 1. Master Caution Panel (Sheet 1 of 5)



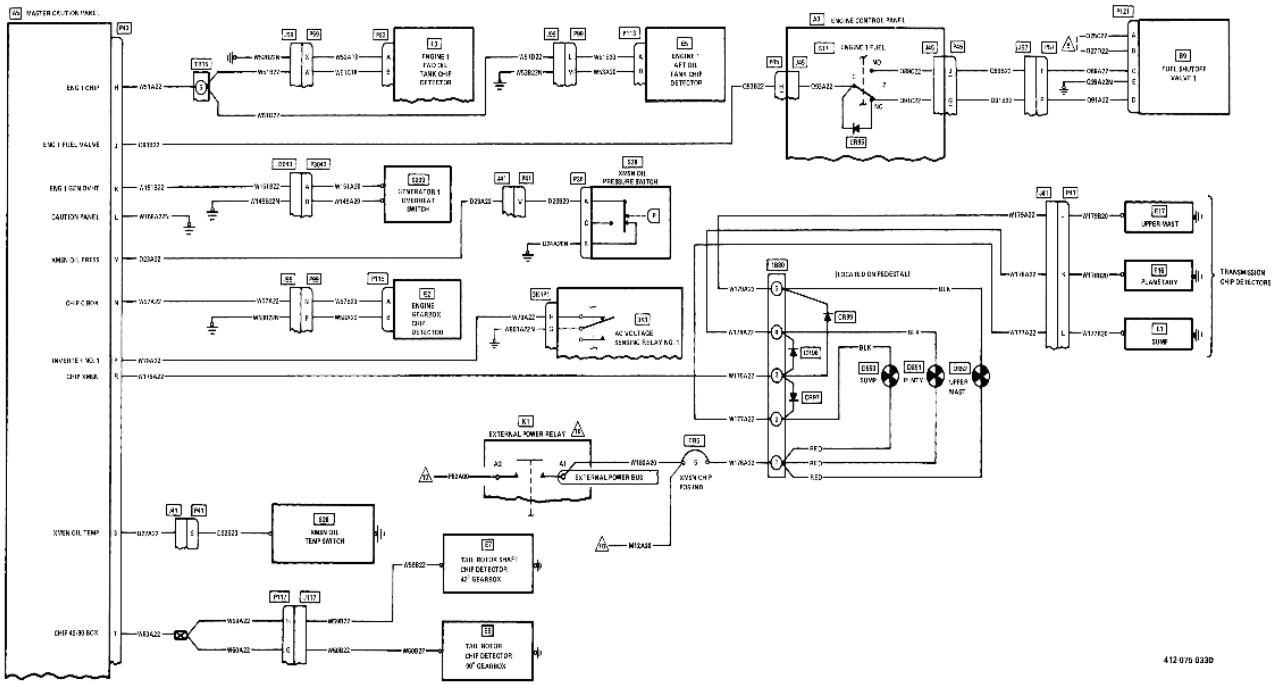
ICN-412-A-980000-B-97499-00158-A-001-01

Figure 1. Master Caution Panel (Sheet 2 of 5)



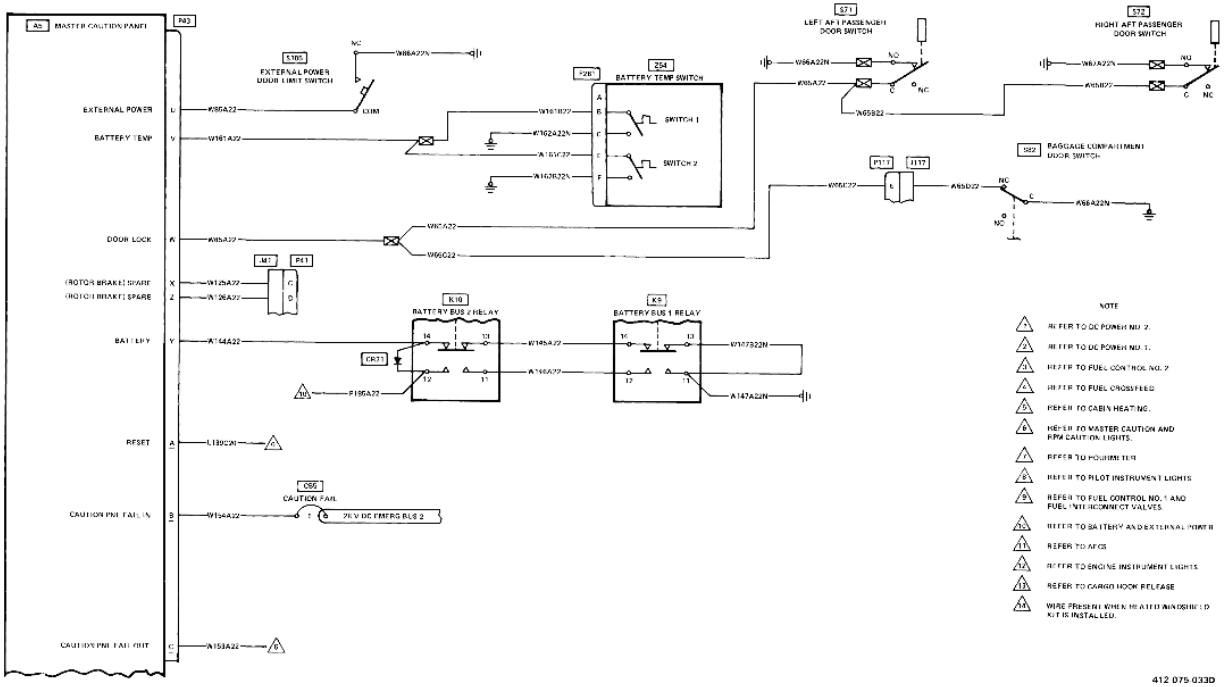
ICN-412-A-980000-B-97499-00159-A-001-01

Figure 1. Master Caution Panel (Sheet 3 of 5)



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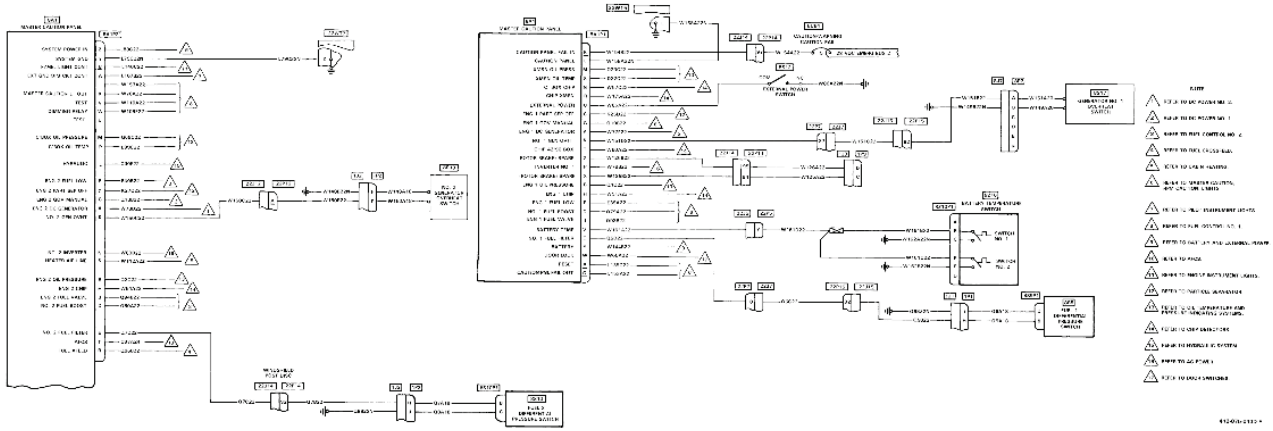
Figure 1. Master Caution Panel (Sheet 4 of 5)



ICN-412-A-980000-B-97499-00161-A-001-01

Figure 1. Master Caution Panel (Sheet 5 of 5)

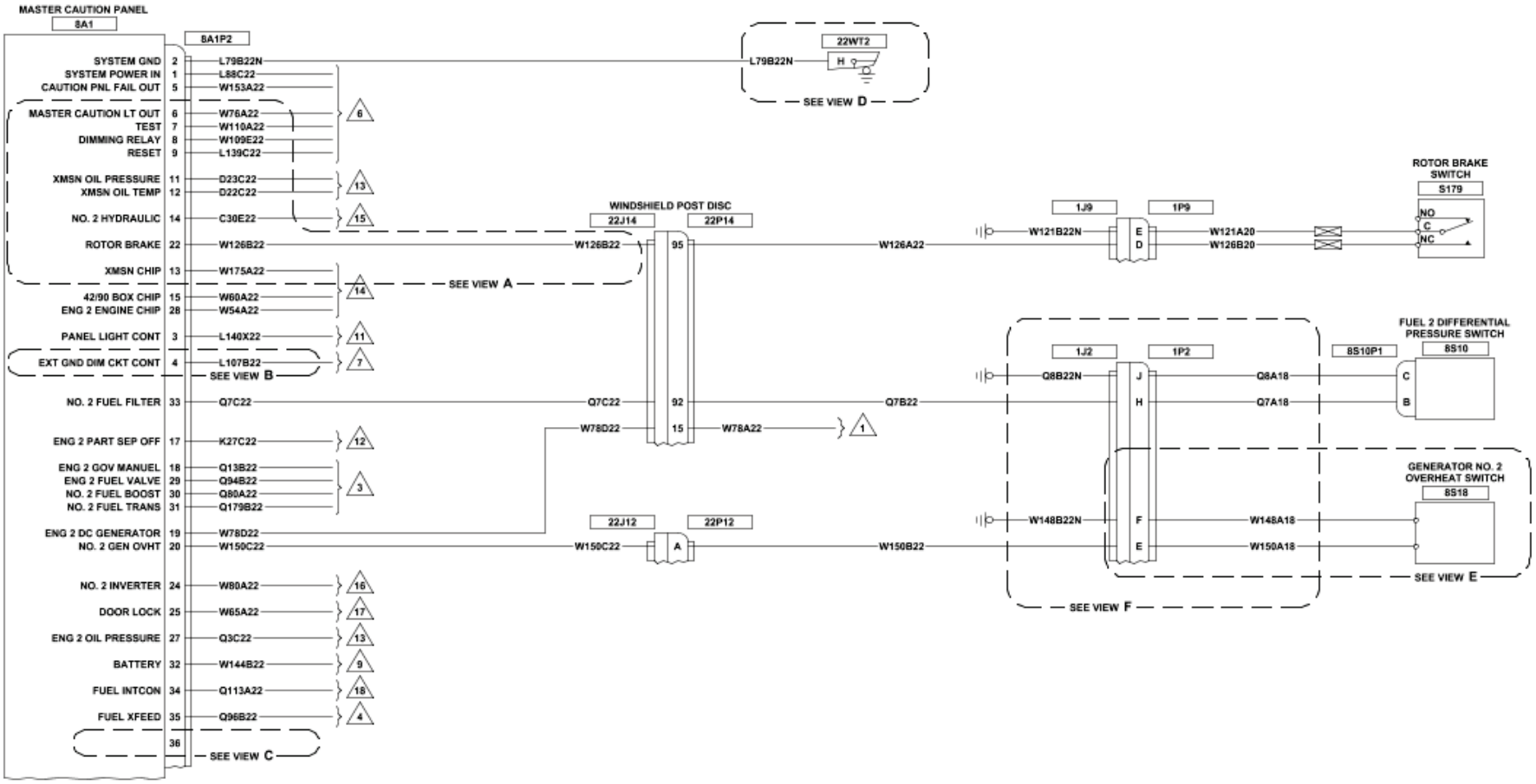
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Figure 1. Master Caution Panel (Sheet 1 of 1)

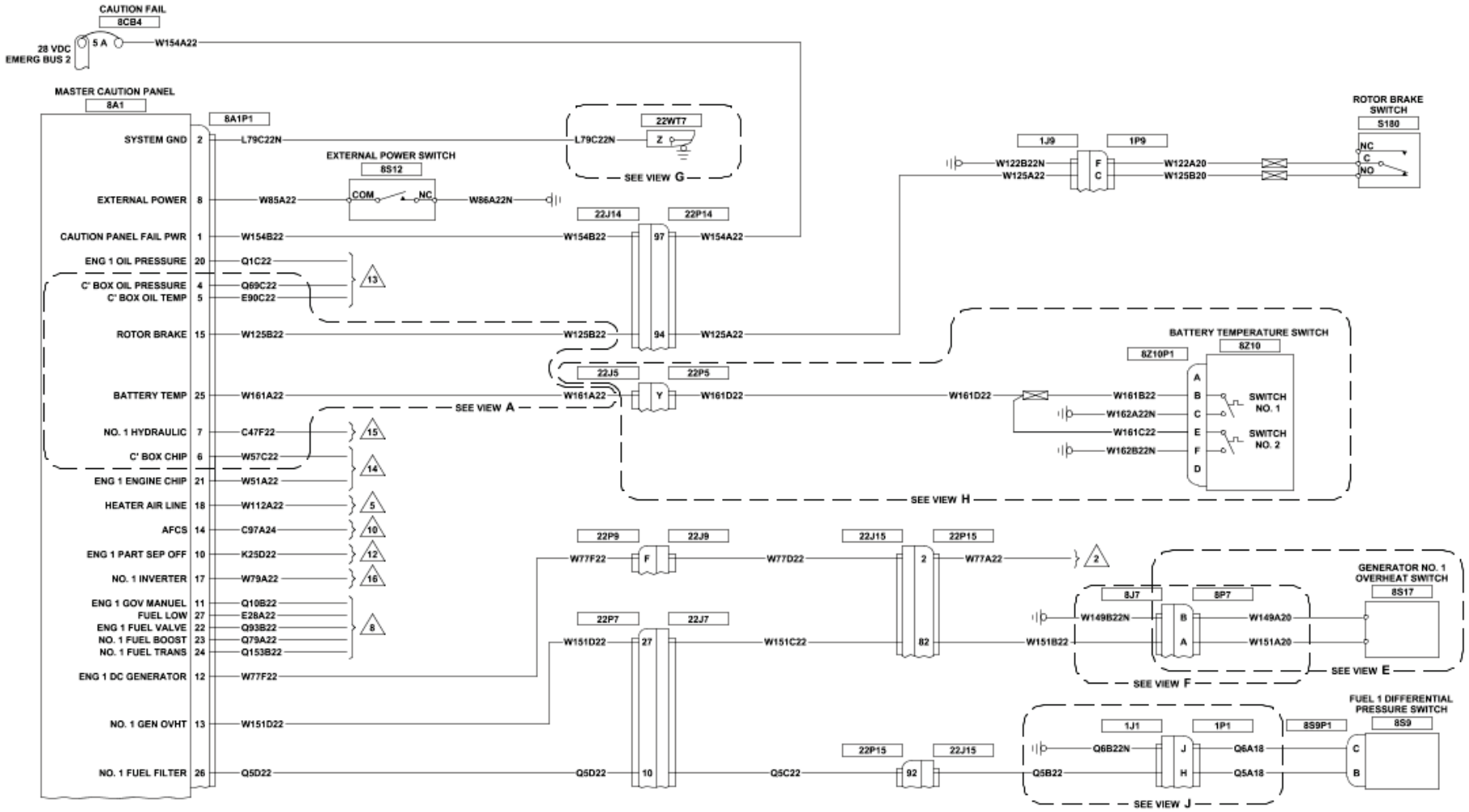
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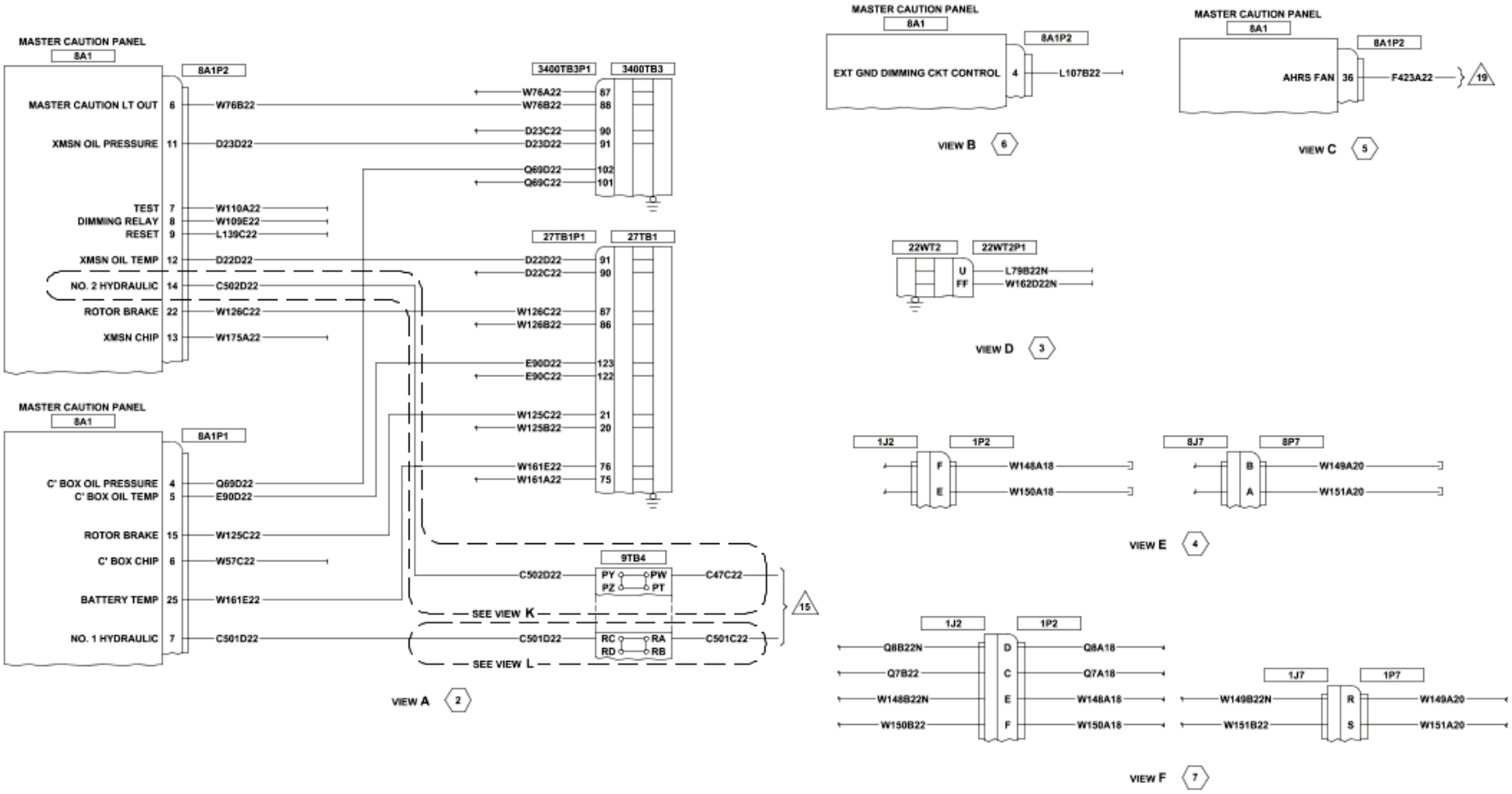
ICN-412-A-980000-B-97499-00163-A-001-01

Figure 1. Master Caution Panel (Sheet 1 of 4)



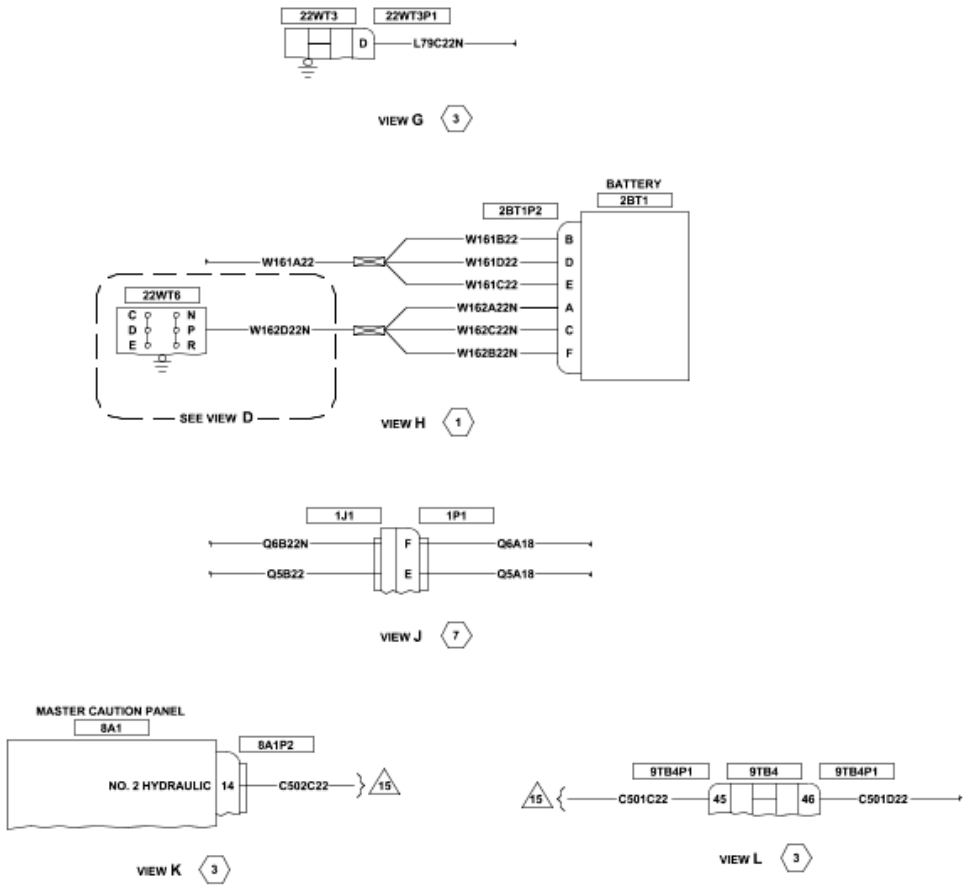
ICN-412-A-980000-B-97499-00164-A-001-01

Figure 1. Master Caution Panel (Sheet 2 of 4)



ICN-412-A-980000-B-97499-00165-A-001-01

Figure 1. Master Caution Panel (Sheet 3 of 4)



NOTES

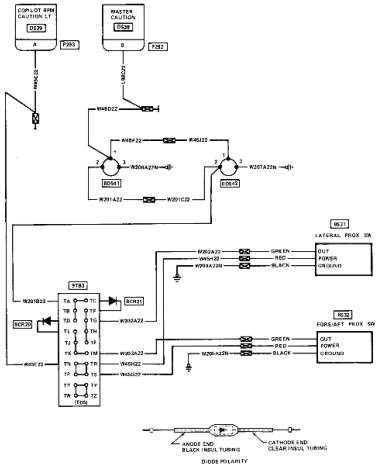
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2. REFER TO DC POWER NO. 1.
3. REFER TO FUEL CONTROL NO. 2.
4. REFER TO FUEL CROSSFEED.
5. REFER TO CABIN HEATING.
6. REFER TO MASTER CAUTION, RPM CAUTION, AND CYCLE STICK CENTERING INDICATOR LIGHTS.
7. REFER TO PILOT INSTRUMENTS LIGHTS.
8. REFER TO FUEL CONTROL NO. 1.
9. REFER TO BATERY AND EXTERNAL POWER.
10. REFER TO AFCS.
11. REFER TO ENGINE INSTRUMENT LIGHTS.
12. REFER TO PARTICLE SEPARATOR.
13. REFER TO OIL TEMPERATURE AND PRESSURE INDICATING SYSTEMS.
14. REFER TO CHIP DETECTORS.
15. REFER TO HYDRAULIC SYSTEM.
16. REFER TO AC POWER.
17. REFER TO DOOR SWITCHES.
18. REFER TO FUEL INTERCONNECT VALVE.
19. REFER TO AHRS 1.

EFFECTIVITY

1. S/N 33121 AND SUBSEQUENT
2. S/N 36037 AND SUBSEQUENT
3. S/N 36087 AND SUBSEQUENT
4. S/N 36139 AND SUBSEQUENT
5. S/N 36248 AND SUBSEQUENT
6. S/N 36297 AND SUBSEQUENT
7. S/N 36694 AND SUBSEQUENT

ICN-412-A-980000-B-97499-00166-A-001-01

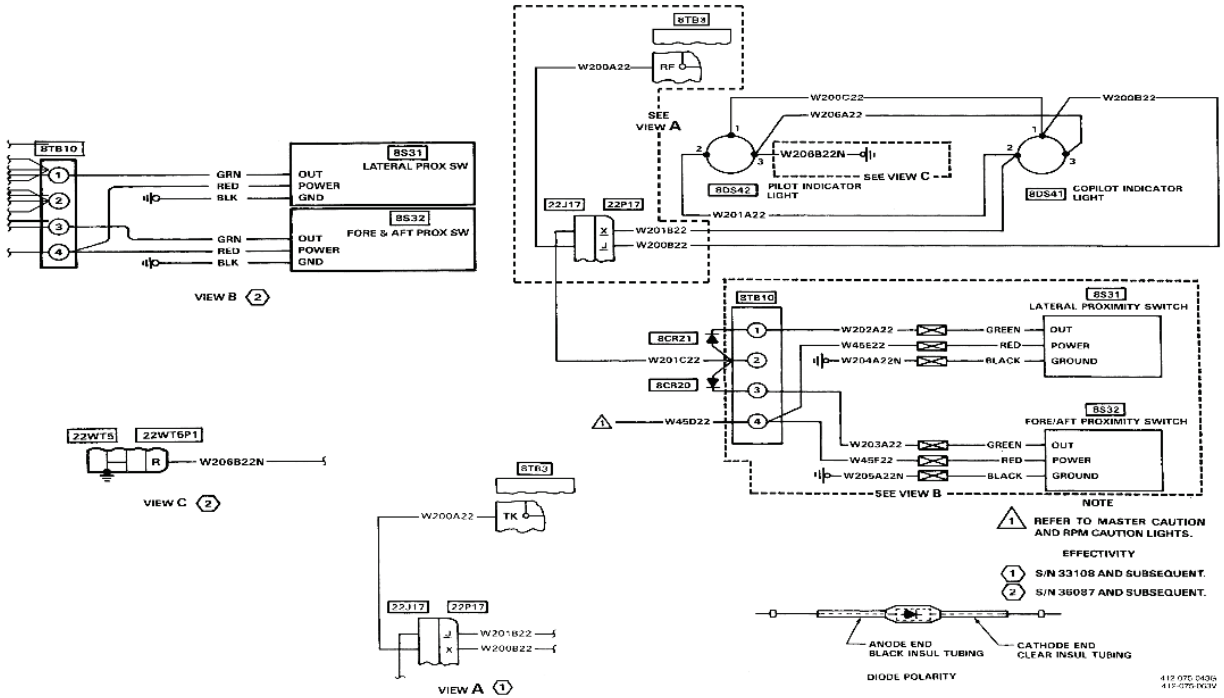
Figure 1. Master Caution Panel (Sheet 4 of 4)



ICN-412-A-980000-B-97499-00167-A-001-01

Figure 1. Pilot/Copilot Cyclic Centering Indicator Light (Sheet 1 of 1)

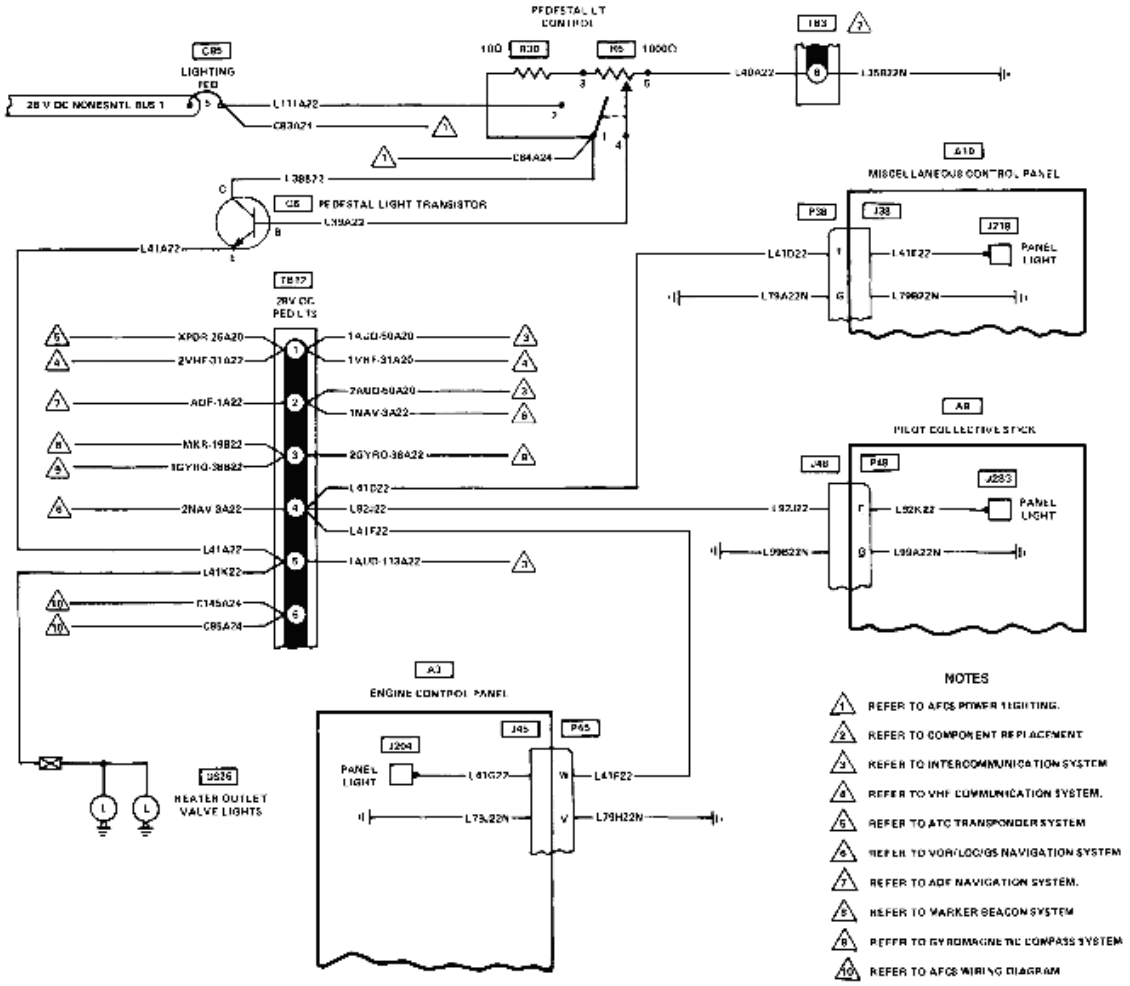
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Figure 1. Pilot/Copilot Cyclic Centering Indicator Light (Sheet 1 of 1)

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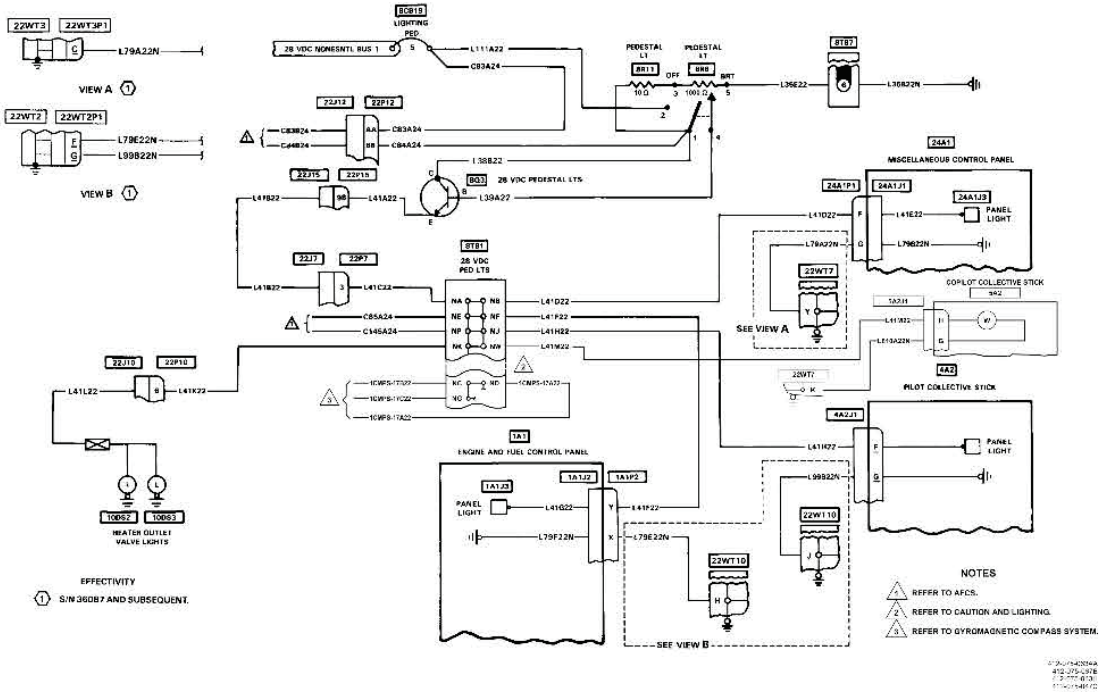


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ICN-412-A-980000-B-97499-00169-A-001-01

Figure 1. Pedestal Lights (Sheet 1 of 1)

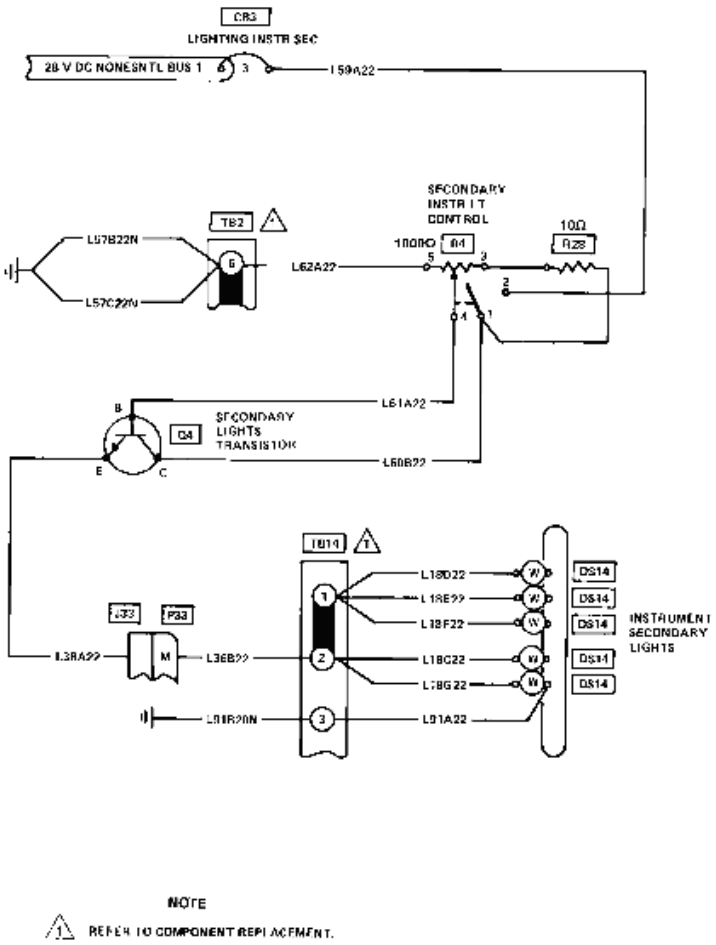
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Figure 1. Pedestal Lights (Sheet 1 of 1)

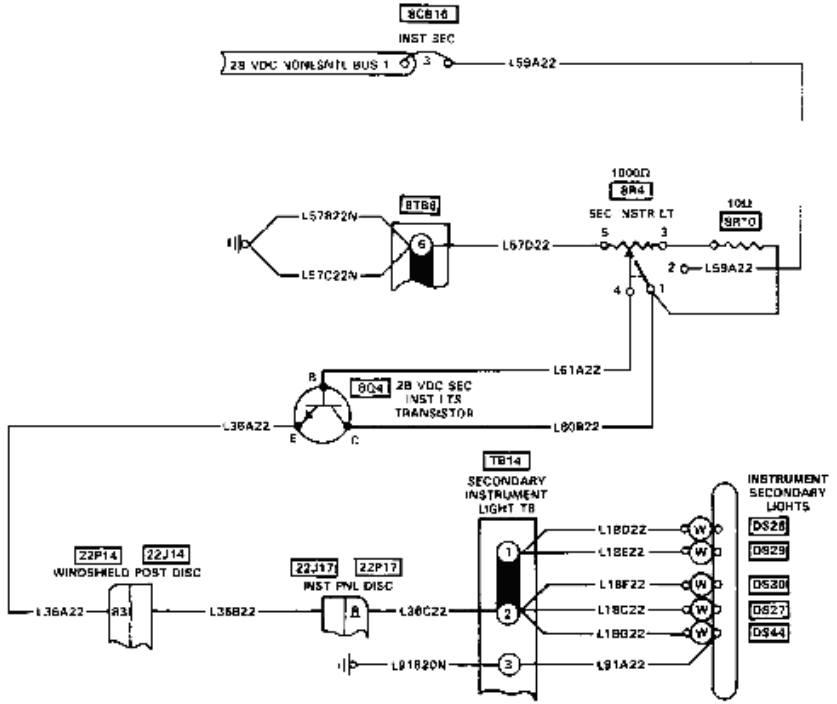
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Figure 1. Secondary Instrument Lights (Sheet 1 of 1)

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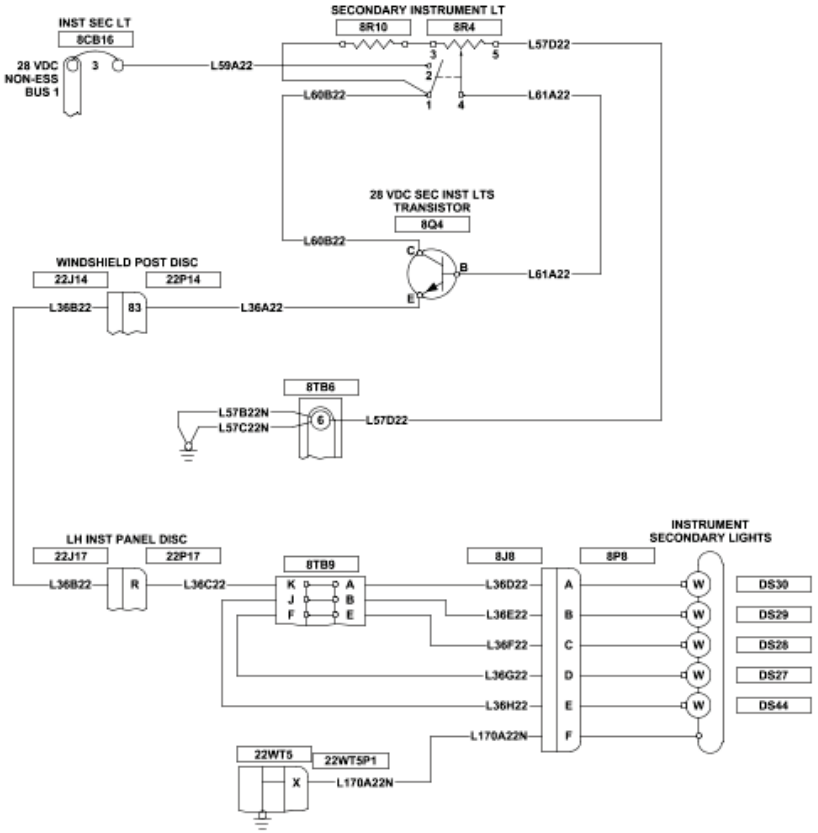


412-075-063G
212-075-064F
412-075-043D
212-075-640F

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Figure 1. Secondary Instrument Lights (Sheet 1 of 1)

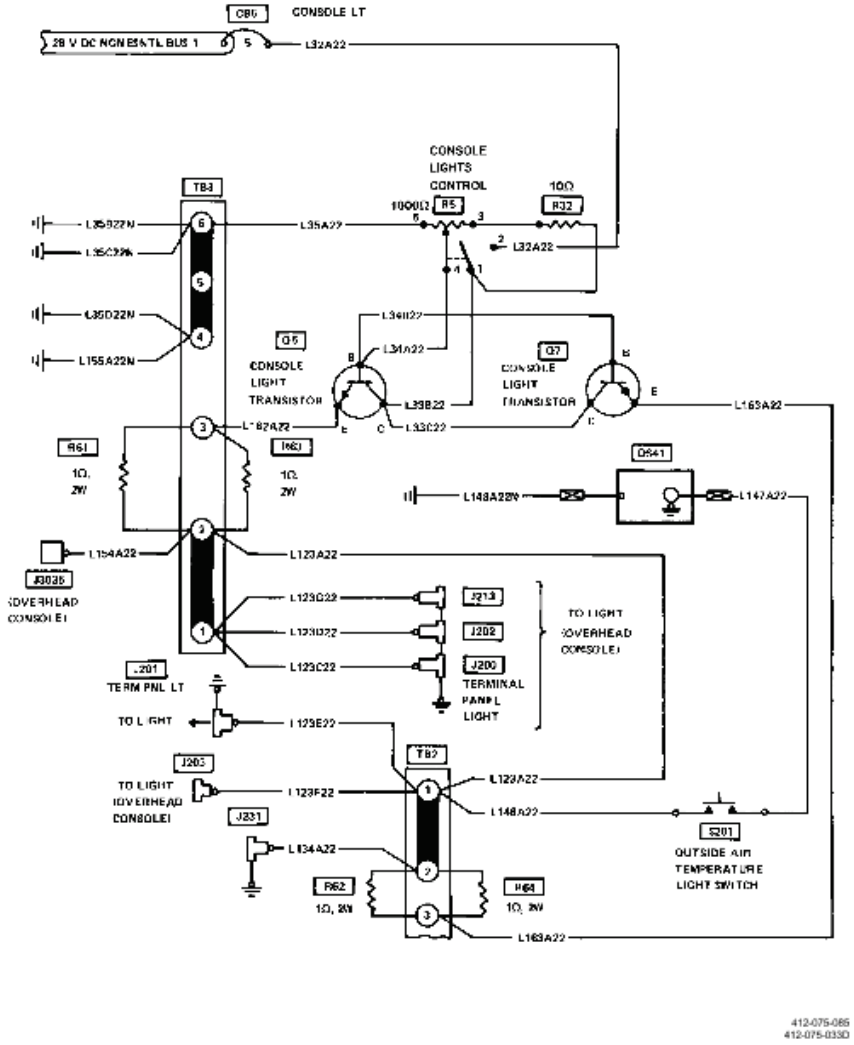
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Figure 1. Secondary Instrument Lights (Sheet 1 of 1)

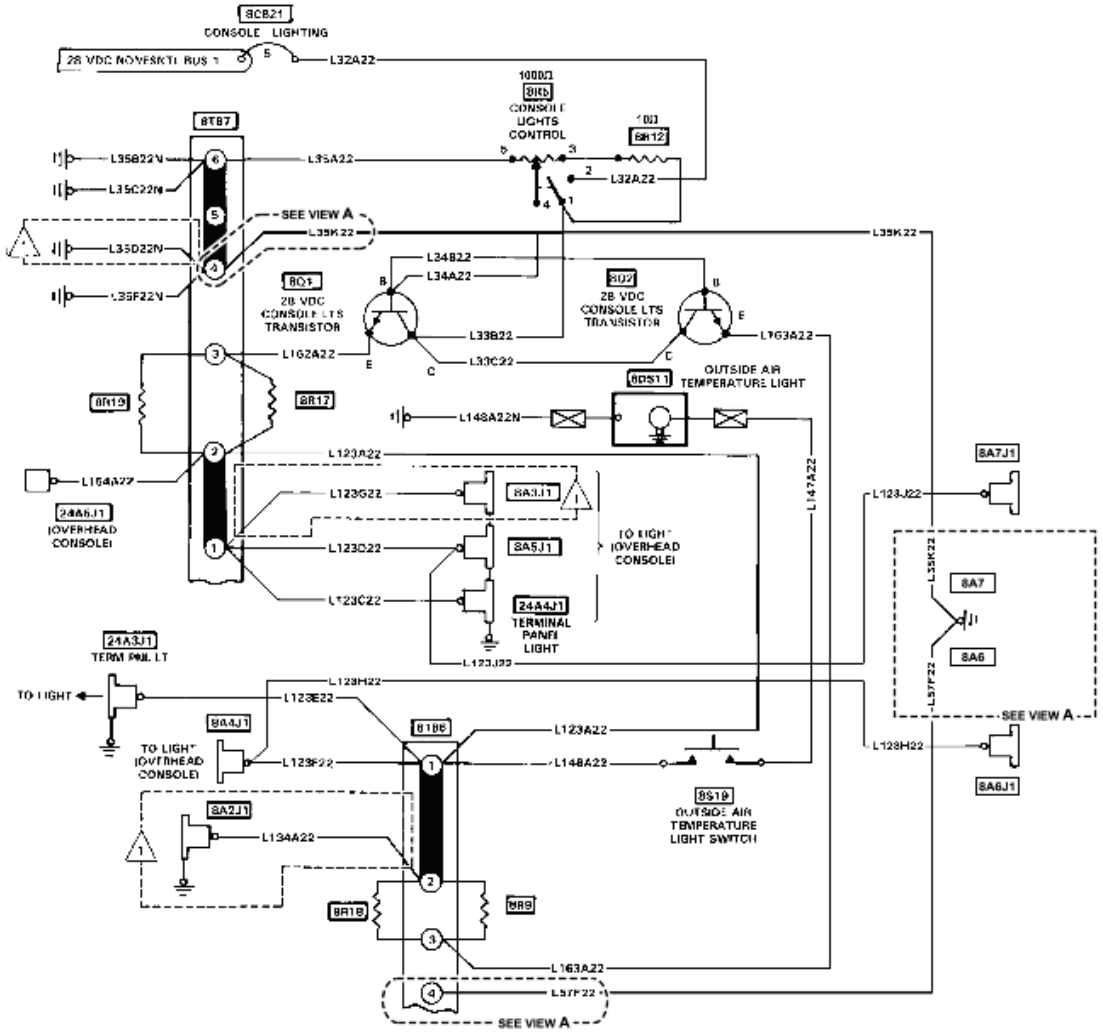
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Figure 1. Overhead Console Lights (Sheet 1 of 1)

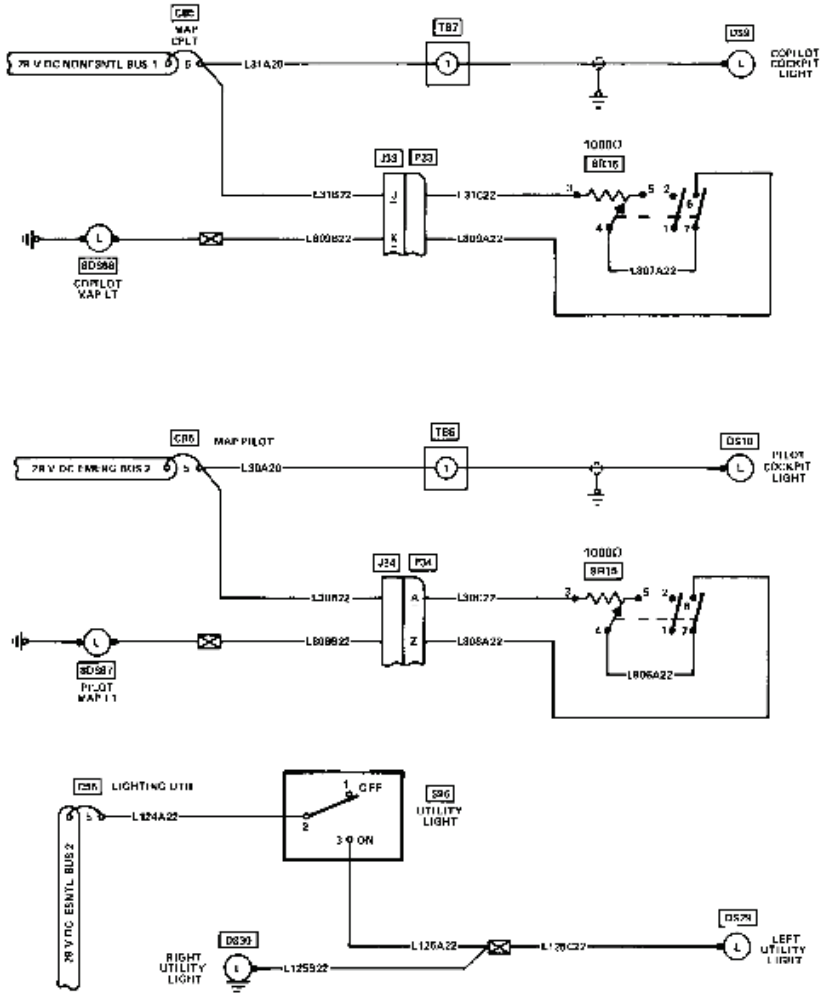
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Figure 1. Overhead Console Lights (Sheet 1 of 1)

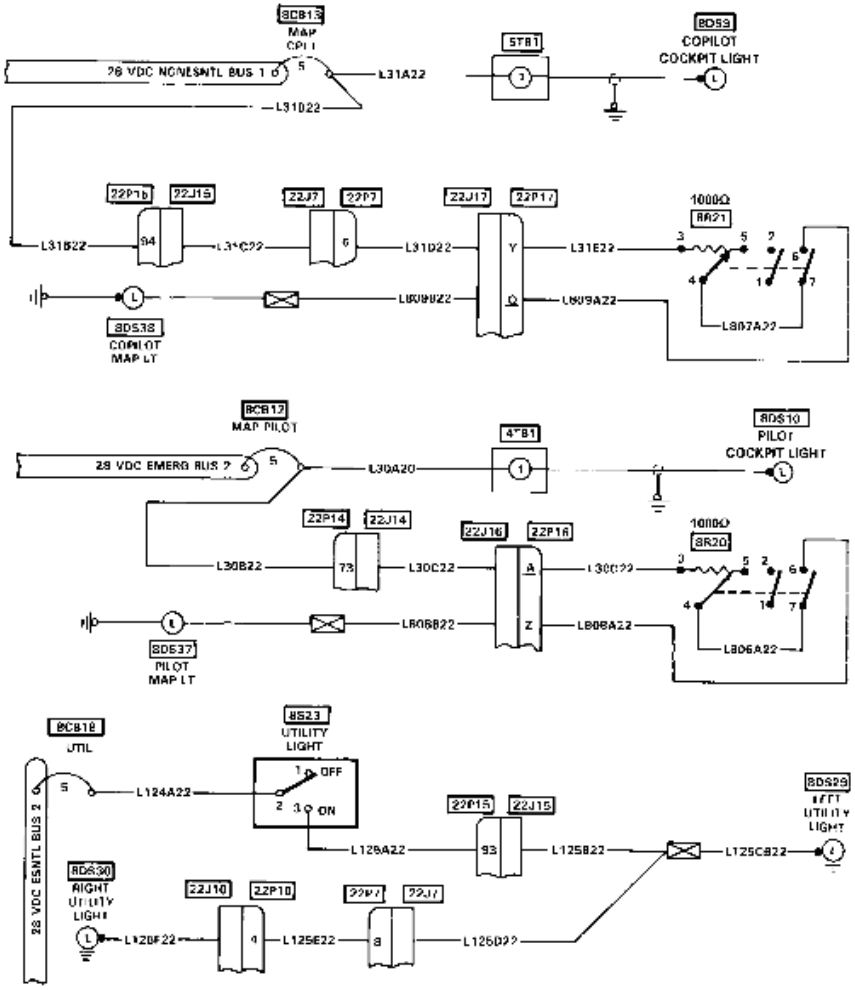
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Figure 1. Cockpit, Map, and Utility Light (Sheet 1 of 1)

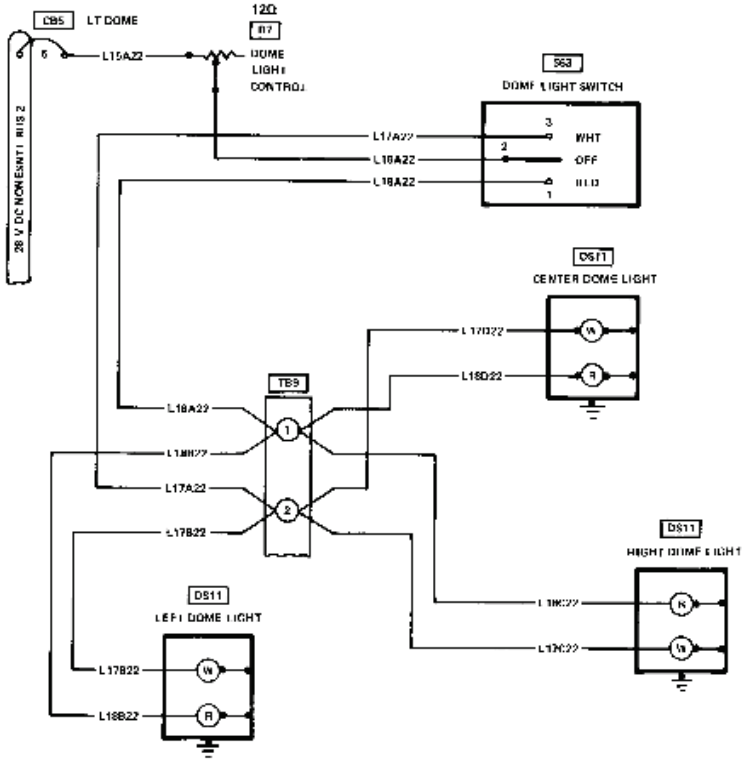
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Figure 1. Cockpit, Map, and Utility Lights (Sheet 1 of 1)

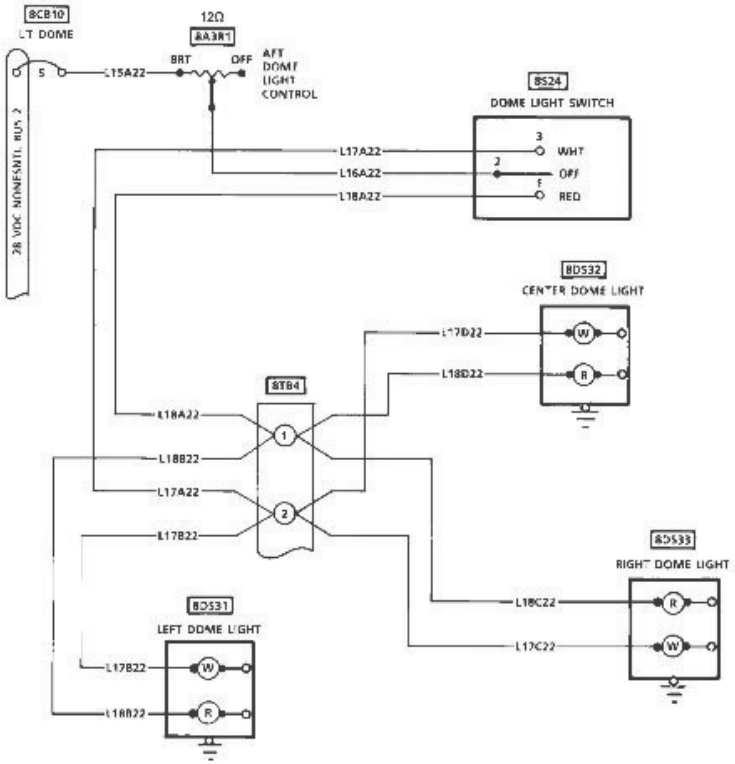
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ICN-412-A-980000-B-97499-00178-A-001-01

Figure 1. Cabin Dome Lights (Sheet 1 of 1)

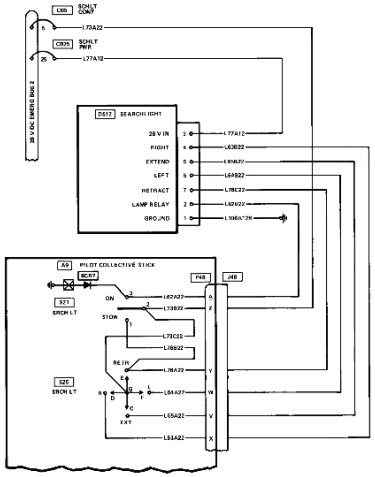
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Figure 1. Cabin Dome Lights (Sheet 1 of 1)

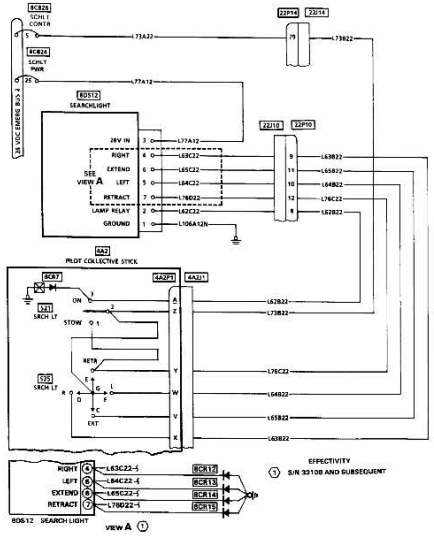
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Figure 1. Searchlight (Sheet 1 of 1)

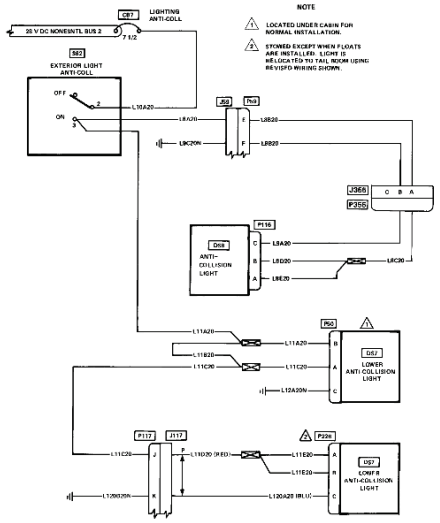
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Figure 1. Searchlight (Sheet 1 of 1)

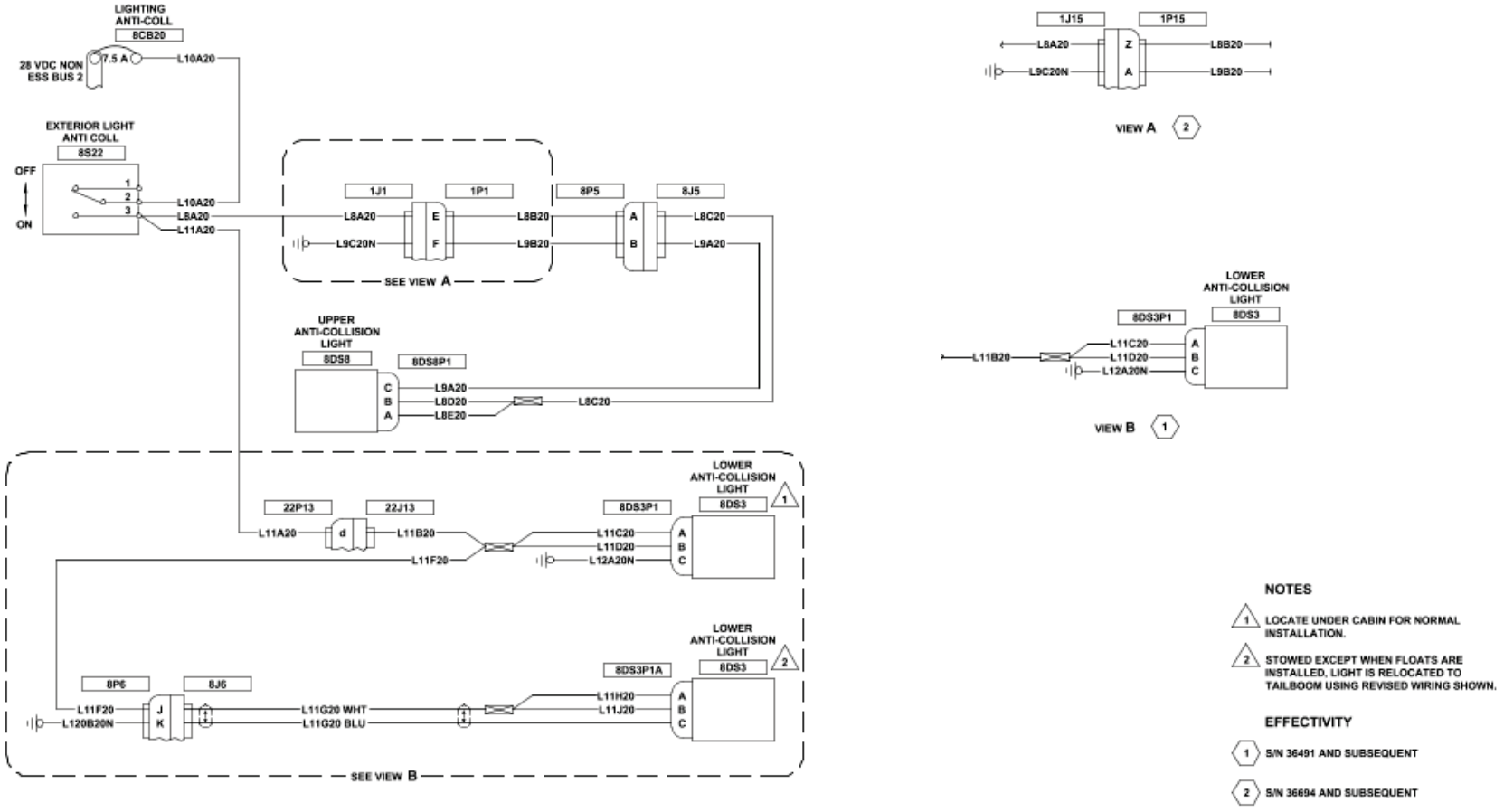
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Figure 1. Anticollision Lights (Sheet 1 of 1)

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NOTES

1. LOCATE UNDER CABIN FOR NORMAL INSTALLATION.
2. STOWED EXCEPT WHEN FLOATS ARE INSTALLED, LIGHT IS RELOCATED TO TAILBOOM USING REVISED WIRING SHOWN.

EFFECTIVITY

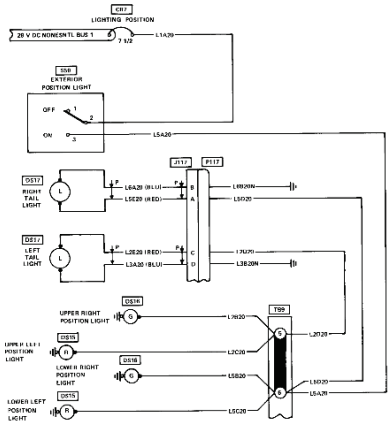
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2. S/N 36694 AND SUBSEQUENT

412-575-035 412-075-063-W

ICN-412-A-980000-B-97499-00183-A-001-01

Figure 1. Anti-collision Lights (Sheet 1 of 1)

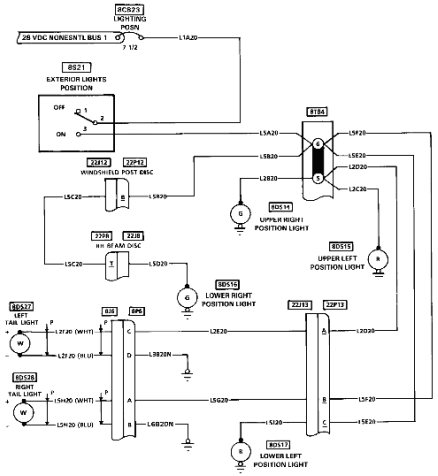
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Figure 1. Position Lights (Sheet 1 of 1)

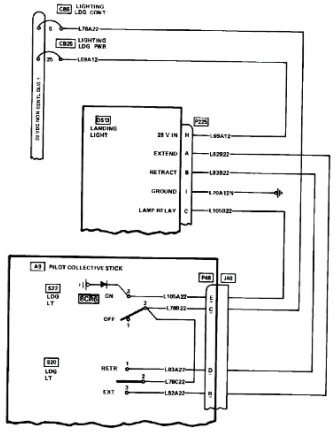
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Figure 1. Position Lights (Sheet 1 of 1)

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Figure 1. Landing Lights (Sheet 1 of 1)

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